26016 Concourse Viewing Platform
Model v21 - combined results

Model 26016_check_v21: v20 updated to drawings 970027-S01-ST (issue 13.11.2026): stair stringers RHS 160x80x5, Level 2 radials UB 457x191x98, external stair landing supports SHS 80x80x6.3. SOFiSTiK 2027 run 13.11.2026 (exit 0, 0 errors, 212 warnings). IDEA StatiCa Connection 26.0.
Prepared by CaSE, 13.11.2026.

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26016 v21 results: model updated to drawings 970027-S01-ST (stair RHS 160x80x5, L2 radials UB 457x191x98, stair landing supports SHS 80x80x6.3)

v21 = v20 updated to drawings 970027-S01-ST (issue 13.11.2026), three section changes only (13 Nov 2026): (1) new SECT 10 RHS 160x80x5 for the external stair stringers, stringer seats and landing edge/middle beams (group 62 SNO 5 -> 10, 34 SLN) and the internal stair sloped stringers (group 60 SLN 79 90 136 137 461 472 489 490); group 60 landing beams/posts stay RHS 200x100x6; (2) SECT 4 L2 radials UB 457x191x89 -> UB 457x191x98; (3) external stair landing supports on the L1 ring (SLN 724 725 769 770) SHS 100x100x5 -> SHS 80x80x6.3 (SECT 9), releases unchanged. Everything else as v20 (qp 0.5 kPa, 12 stringer My hinges, pinned feet, loads, combinations). Picture: v21_changed_members_reading.png.

Run checks

item	v20	v21
exit code / errors	0 / 0	0 / 0
warnings (erg)	212 (tool count)	212 (tool count)
warning difference	-	no new warning type; BDK no. 156: 19 -> 20, no. 205: 6 -> 4, SOFIMSHC no. 12149: 34 -> 35 (one more section line)
mechanism / singularity / error 197	none	none
run time	34.0 min (2041 s), peak RAM 10.1 GB	35.7 min (2139 s), peak RAM 10.1 GB
programs	AQUA SOFIMSHC SOFILOAD ASE MAXIMA BDK x2 ASE x2 (eigen) AQB	same, all finished

Global

item	v20	v21	change
alpha_cr PLC 102 (lowest mode, LC 9201)	6.95	6.95	+0.00
alpha_cr PLC 119 (lowest mode, LC 9221)	7.32	7.15	-0.17
sum reactions LC 102 px, py, pz (kN)	[-15.81, 5.71, 9581.59]	[-15.81, 5.71, 9595.11]	
sum reactions LC 201 pz (kN)	6579.74	6589.75	+10.0
SOFiSTiK stored sum LC 102 / 201	[-15.805157, 5.707161, 9581.59668] / [-10.536701, 3.804818, 6579.741211]	[-15.80513, 5.707152, 9595.100586] / [-10.536813, 3.804839, 6589.751953]	
ULS STR max pz (kN, node, LC)	[427.8, 2, 146]	[428.3, 2, 146]	
ULS STR min pz (kN, node, LC)	[-45.6, 12, 109]	[-45.1, 12, 109]	
ULS STR max horizontal (kN, node, LC)	[113.5, 27, 105]	[112.7, 27, 105]	

Basic load case reaction sums, sum Fz (kN):

LC	v20	v21
1	872.3	882.3
2	842.9	842.9
3	28.9	28.9
4	176.0	176.0
11	2160.7	2160.7
12	259.3	259.3
13	1915.3	1915.3
14	324.3	324.3
71	-650.6	-650.6
72	-607.5	-607.5
73	-636.4	-636.4
74	-622.2	-622.2

Maximum BDK and AQB per group

group	BDK v20 (SLN, LC)	BDK v21 (SLN, LC)	change	AQB v20 (SLN, LC)	AQB v21 (SLN, LC)	change
11	0.383 (36, 147)	0.383 (25, 147)	+0.000	0.231 (24, 152)	0.236 (24, 152)	+0.005
12	0.096 (448, 121)	0.094 (448, 121)	-0.002	0.064 (448, 105)	0.063 (448, 105)	-0.001
13	0.254 (41, 121)	0.253 (41, 121)	-0.001	0.216 (39, 152)	0.219 (39, 152)	+0.003
14	0.360 (399, 145)	0.365 (399, 145)	+0.005	0.286 (399, 145)	0.289 (399, 135)	+0.003
15	0.672 (223, 153)	0.663 (223, 153)	-0.009	0.550 (1545, 152)	0.572 (1539, 152)	+0.022
21	0.398 (17, 145)	0.389 (17, 145)	-0.009	0.387 (9, 146)	0.389 (9, 146)	+0.002
23	0.141 (96, 119)	0.139 (96, 119)	-0.002	0.182 (96, 152)	0.180 (96, 152)	-0.002
24	0.290 (417, 122)	0.292 (417, 122)	+0.002	0.297 (417, 153)	0.295 (417, 153)	-0.002
25	1.060 (56, 120)	1.063 (56, 120)	+0.003	1.061 (56, 120)	1.066 (56, 120)	+0.005
31	0.239 (380, 147)	0.237 (380, 147)	-0.002	0.351 (376, 145)	0.348 (376, 145)	-0.003
35	0.377 (318, 107)	0.377 (318, 107)	+0.000	0.287 (345, 107)	0.287 (345, 107)	+0.000
60	0.314 (490, 122)	0.428 (490, 122)	+0.114	0.698 (458, None)	0.720 (458, None)	+0.022
61	0.111 (602, 120)	0.110 (602, 120)	-0.001	0.101 (602, 120)	0.101 (602, 120)	+0.000
62	0.975 (722, 122)	0.900 (767, 121)	-0.075	0.516 (727, None)	0.570 (728, None)	+0.054

L1 ring r 8.5 (group 15, UB 254, 20 deg segments)

Max |My|, |Mz| over ULS 101-154 (kNm, LC). BDK = max(nm-y, nm-z) per input SLN (radials: BDK id per member as in v20, '-' = no BDK id). AQB = max station utilisation of the SLN's beams. SLS = largest deflection relative to the line between the SLN end nodes, SLS 201-221 (mm, L/x).

SLN	deg	r m	My v20	My v21	Mz v20	Mz v21	BDK v20	BDK v21	AQB v20	AQB v21	SLS v20	SLS v21
185	360-20	8.5-8.5	22.3 (102)	22.3 (102)	2.07 (135)	2.13 (135)	0.194	0.194	0.143	0.143	-1.08 (L/2723)	-1.08 (L/2723)
264	20-40	8.5-8.5	22.3 (102)	22.3 (102)	0.96 (104)	1.00 (135)	0.178	0.178	0.143	0.143	-1.09 (L/2720)	-1.09 (L/2720)
265	40-60	8.5-8.5	22.7 (102)	22.7 (102)	0.96 (119)	0.94 (119)	0.181	0.181	0.146	0.146	-1.11 (L/2670)	-1.11 (L/2670)
189	60-80	8.5-8.5	22.5 (102)	22.5 (102)	0.43 (106)	0.44 (148)	0.173	0.173	0.145	0.145	-1.09 (L/2695)	-1.09 (L/2695)

192	80-100	8.5-8.5	22.3 (102)	22.3 (102)	0.68 (122)	0.69 (122)	0.173	0.173	0.143	0.143	-1.08 (L/2724)	-1.08 (L/2724)
195	100-120	8.5-8.5	22.5 (102)	22.5 (102)	0.63 (148)	0.63 (148)	0.175	0.175	0.144	0.144	-1.09 (L/2698)	-1.09 (L/2698)
198	120-140	8.5-8.5	22.7 (102)	22.7 (102)	0.77 (120)	0.77 (120)	0.178	0.178	0.146	0.146	-1.1 (L/2673)	-1.1 (L/2673)
210	140-160	8.5-8.5	22.3 (102)	22.3 (102)	0.68 (146)	0.68 (146)	0.175	0.175	0.143	0.143	-1.09 (L/2715)	-1.09 (L/2715)
213	160-180	8.5-8.5	22.4 (102)	22.4 (102)	0.80 (104)	0.80 (104)	0.176	0.176	0.144	0.144	-1.09 (L/2714)	-1.09 (L/2714)
214	180-200	8.5-8.5	22.3 (102)	22.3 (102)	0.72 (146)	0.72 (146)	0.176	0.176	0.143	0.143	-1.09 (L/2716)	-1.09 (L/2716)
217	200-220	8.5-8.5	22.3 (102)	22.3 (102)	0.70 (120)	0.69 (120)	0.175	0.175	0.143	0.143	-1.09 (L/2715)	-1.09 (L/2715)
218	240-220	8.5-8.5	22.8 (102)	22.8 (102)	0.72 (121)	0.72 (121)	0.178	0.178	0.146	0.146	-1.11 (L/2662)	-1.11 (L/2662)
221	260-240	8.5-8.5	22.4 (102)	22.4 (102)	0.54 (123)	0.55 (123)	0.174	0.174	0.144	0.144	-1.09 (L/2702)	-1.09 (L/2702)
426	280-260	8.5-8.5	22.2 (102)	22.2 (102)	1.00 (121)	0.99 (121)	0.178	0.177	0.143	0.143	-1.08 (L/2732)	-1.08 (L/2732)
425	300-280	8.5-8.5	22.1 (102)	22.1 (102)	0.82 (135)	0.90 (135)	0.174	0.174	0.142	0.142	-1.08 (L/2737)	-1.08 (L/2737)
225	320-300	8.5-8.5	22.2 (102)	22.2 (102)	2.03 (135)	2.08 (135)	0.192	0.191	0.142	0.142	-1.08 (L/2735)	-1.08 (L/2735)
226	340-320	8.5-8.5	22.3 (102)	22.3 (102)	0.65 (103)	0.65 (103)	0.174	0.174	0.143	0.143	-1.09 (L/2717)	-1.09 (L/2717)
227	340-360	8.5-8.5	22.5 (102)	22.5 (102)	0.58 (121)	0.60 (121)	0.175	0.175	0.144	0.144	-1.1 (L/2697)	-1.1 (L/2697)

L1 radials (group 15)

Max |My|, |Mz| over ULS 101-154 (kNm, LC). BDK = max(nm-y, nm-z) per input SLN (radials: BDK id per member as in v20, '-' = no BDK id). AQB = max station utilisation of the SLN's beams. SLS = largest deflection relative to the line between the SLN end nodes, SLS 201-221 (mm, L/x).

SLN	deg	r m	My v20	My v21	Mz v20	Mz v21	BDK v20	BDK v21	AQB v20	AQB v21	SLS v20	SLS v21
169	0-360	12.0-8.5	53.5 (153)	53.7 (153)	0.99 (106)	0.97 (106)	0.578	0.578	0.344	0.345	-0.79 (L/4446)	-0.81 (L/4321)
180	360-0	8.5-3.0	56.9 (153)	57.0 (153)	0.29 (126)	0.30 (114)	-	-	0.365	0.366	-4.58 (L/1201)	-4.57 (L/1203)
181	0-0	3.0-1.9	5.2 (154)	5.2 (154)	0.68 (114)	0.62 (114)	-	-	0.039	0.039	0.01 (L/100272)	0.01 (L/99968)
182	20-20	1.9-3.0	4.8 (152)	4.8 (152)	0.71 (112)	0.63 (112)	0.055	0.054	0.031	0.031	0.01 (L/183277)	0.01 (L/184301)
183	20-20	3.0-8.5	70.7 (153)	71.8 (153)	1.02 (135)	1.06 (135)	0.538	0.542	0.454	0.461	-1.43 (L/3837)	-1.37 (L/4024)
184	20-20	8.5-12.0	77.5 (153)	79.0 (153)	11.22 (120)	9.45 (120)	0.636	0.627	0.501	0.508	0.83 (L/4243)	0.99 (L/3535)
246	40-40	3.0-1.9	4.9 (154)	4.9 (154)	0.75 (114)	0.81 (114)	-	-	0.044	0.043	0.01 (L/100263)	0.01 (L/100569)
262	40-40	8.5-3.0	65.0 (153)	65.8 (153)	0.35 (111)	0.35 (111)	-	-	0.417	0.422	-3.92 (L/1402)	-3.86 (L/1425)
263	40-40	12.0-8.5	62.7 (153)	64.5 (153)	0.78 (122)	0.77 (122)	0.645	0.654	0.403	0.414	-1.87 (L/1871)	-1.99 (L/1758)
188	60-60	8.5-3.0	66.2 (153)	66.0 (153)	0.39 (111)	0.38 (111)	0.520	0.518	0.425	0.424	-3.75 (L/1467)	-3.76 (L/1462)
245	60-60	3.0-1.9	4.9 (154)	4.9 (154)	1.51 (113)	1.45 (113)	0.065	0.064	0.044	0.043	0.01 (L/104068)	0.01 (L/103415)
266	60-60	12.0-8.5	67.3 (153)	67.1 (153)	0.27 (122)	0.27 (122)	0.435	0.433	0.432	0.431	1.14 (L/3083)	1.12 (L/3113)
187	80-80	8.5-3.0	47.6 (153)	47.6 (153)	0.40 (111)	0.38 (111)	-	-	0.305	0.306	-2.94 (L/1871)	-2.95 (L/1865)
190	80-80	12.0-8.5	45.9 (153)	45.9 (153)	1.25 (121)	1.19 (121)	0.321	0.320	0.294	0.295	-0.96 (L/3652)	-0.96 (L/3652)
244	80-80	3.0-1.9	4.9 (152)	4.9 (152)	0.88 (111)	0.89 (111)	-	-	0.039	0.039	0.01 (L/113366)	0.01 (L/112592)
191	100-100	8.5-3.0	67.6 (153)	67.6 (153)	0.44 (111)	0.44 (111)	0.517	0.517	0.434	0.434	-3.66 (L/1503)	-3.66 (L/1503)
193	100-100	12.0-8.5	68.8 (153)	68.8 (153)	0.43 (123)	0.45 (123)	0.458	0.458	0.442	0.442	1.18 (L/2975)	1.18 (L/2972)
243	100-100	3.0-1.9	4.9 (154)	4.9 (154)	1.33 (121)	1.33 (121)	0.065	0.065	0.040	0.041	0.01 (L/103092)	0.01 (L/103092)
194	120-120	8.5-3.0	56.6 (153)	56.7 (153)	0.34 (111)	0.34 (111)	-	-	0.364	0.364	-4.58 (L/1200)	-4.58 (L/1201)
196	120-120	12.0-8.5	52.9 (153)	52.9 (153)	0.50 (122)	0.48 (122)	0.569	0.569	0.339	0.340	-0.65 (L/5422)	-0.65 (L/5351)
242	120-120	3.0-1.9	4.9 (154)	4.9 (154)	0.56 (112)	0.55 (112)	-	-	0.045	0.045	0.01 (L/100278)	0.01 (L/101201)
197	140-140	8.5-3.0	60.3 (153)	60.4 (153)	0.45 (111)	0.45 (111)	0.482	0.482	0.387	0.388	-2.1 (L/2617)	-2.11 (L/2606)
209	140-140	12.0-8.5	63.4 (153)	63.5 (153)	0.42 (125)	0.46 (125)	0.426	0.427	0.407	0.408	1.0 (L/3488)	1.01 (L/3481)
241	140-140	3.0-1.9	4.8 (152)	4.8 (152)	0.67 (113)	0.67 (113)	0.059	0.059	0.031	0.031	0.01 (L/118244)	0.01 (L/118669)
199	160-160	8.5-3.0	57.3 (153)	57.4 (153)	0.32 (111)	0.31 (111)	-	-	0.368	0.368	-4.55 (L/1209)	-4.54 (L/1210)
211	160-160	12.0-8.5	53.5 (153)	53.6 (153)	0.40 (125)	0.42 (125)	0.573	0.573	0.344	0.344	-0.67 (L/5187)	-0.68 (L/5122)
240	160-160	3.0-1.9	4.8 (154)	4.8 (154)	0.47 (134)	0.47 (134)	-	-	0.046	0.046	0.01 (L/98483)	0.01 (L/99074)
200	180-180	8.5-3.0	68.8 (153)	68.8 (153)	0.33 (123)	0.33 (124)	0.546	0.546	0.442	0.442	-3.59 (L/1531)	-3.59 (L/1532)
212	180-180	12.0-8.5	70.3 (153)	70.3 (153)	0.42 (125)	0.46 (125)	0.470	0.470	0.451	0.452	1.21 (L/2887)	1.21 (L/2885)
239	180-180	3.0-1.9	5.2 (154)	5.2 (154)	0.52 (114)	0.52 (114)	0.062	0.062	0.033	0.033	0.01 (L/104397)	0.01 (L/104397)
202	200-200	8.5-3.0	47.1 (153)	47.2 (153)	0.32 (124)	0.33 (124)	-	-	0.302	0.303	-2.99 (L/1838)	-3.0 (L/1836)
215	200-200	12.0-8.5	45.0 (153)	45.1 (153)	0.25 (125)	0.28 (125)	0.304	0.305	0.289	0.290	-0.81 (L/4339)	-0.82 (L/4294)
238	200-200	3.0-1.9	4.9 (152)	4.9 (152)	0.50 (113)	0.50 (113)	-	-	0.045	0.045	0.01 (L/112980)	0.01 (L/113368)
201	220-220	8.5-3.0	67.1 (153)	67.1 (153)	0.43 (112)	0.43 (112)	0.521	0.521	0.431	0.431	-3.68 (L/1496)	-3.68 (L/1497)
216	220-220	12.0-8.5	68.3 (153)	68.3 (153)	0.38 (125)	0.40 (125)	0.456	0.456	0.439	0.439	1.18 (L/2974)	1.18 (L/2970)
237	220-220	3.0-1.9	4.9 (154)	4.9 (154)	1.02 (114)	1.03 (114)	0.063	0.063	0.032	0.032	0.01 (L/103093)	0.01 (L/102772)
204	240-240	3.0-8.5	56.4 (153)	56.4 (153)	0.31 (122)	0.32 (126)	-	-	0.362	0.362	-4.56 (L/1206)	-4.56 (L/1206)
219	240-240	8.5-12.0	53.0 (153)	53.1 (153)	1.28 (103)	1.20 (103)	-	-	0.341	0.341	-0.77 (L/4559)	-0.77 (L/4547)
236	240-240	1.9-3.0	4.9 (154)	4.9 (154)	0.51 (113)	0.53 (113)	0.408	0.407	0.041	0.041	0.01 (L/101195)	0.01 (L/100578)
203	260-260	3.0-8.5	57.8 (153)	57.6 (153)	0.38 (112)	0.39 (112)	0.472	0.470	0.371	0.370	-2.37 (L/2323)	-2.38 (L/2310)
220	260-260	8.5-12.0	63.1 (153)	62.8 (153)	0.31 (125)	0.34 (125)	0.407	0.405	0.405	0.403	0.98 (L/3577)	0.97 (L/3622)
235	260-260	1.9-3.0	4.8 (152)	4.8 (152)	1.13 (103)	1.06 (103)	0.066	0.065	0.031	0.031	0.01 (L/175477)	0.01 (L/176415)
206	280-280	3.0-8.5	64.5 (153)	65.2 (153)	0.43 (112)	0.43 (112)	-	-	0.414	0.419	-3.91 (L/1407)	-3.85 (L/1427)
222	280-280	8.5-12.0	62.3 (153)	64.0 (153)	1.01 (121)	0.99 (121)	-	-	0.400	0.411	-1.9 (L/1846)	-2.02 (L/1736)
234	280-280	1.9-3.0	4.9 (154)	4.9 (154)	0.47 (104)	0.55 (104)	0.425	0.429	0.046	0.045	0.01 (L/100272)	0.01 (L/99666)
205	300-300	3.0-8.5	77.3 (153)	78.4 (153)	1.13 (119)	1.13 (135)	0.584	0.587	0.496	0.503	-2.84 (L/1935)	-2.75 (L/2001)
223	300-300	8.5-12.0	82.4 (153)	83.9 (153)	11.64 (120)	9.70 (120)	0.672	0.663	0.542	0.548	1.0 (L/3499)	1.17 (L/2981)

232	300-300	1.9-3.0	4.9 (154)	4.9 (154)	1.43 (112)	1.34 (112)	0.064	0.062	0.037	0.034	0.01 (L/104733)	0.01 (L/105066)
208	320-320	3.0-8.5	47.4 (153)	47.6 (153)	0.39 (112)	0.39 (112)	0.312	0.313	0.304	0.305	-2.99 (L/1842)	-2.98 (L/1845)
224	320-320	12.0-8.5	45.7 (153)	45.9 (153)	0.76 (121)	0.77 (121)	0.301	0.302	0.293	0.295	-0.93 (L/3781)	-0.95 (L/3687)
231	320-320	1.9-3.0	4.9 (152)	4.9 (152)	0.67 (113)	0.61 (113)	0.036	0.035	0.038	0.038	0.01 (L/169181)	0.01 (L/169181)
207	340-340	3.0-8.5	64.5 (153)	63.9 (153)	0.39 (112)	0.39 (112)	0.484	0.485	0.414	0.411	-3.95 (L/1391)	-4.01 (L/1373)
228	340-340	8.5-12.0	64.5 (153)	63.7 (153)	0.35 (129)	0.39 (129)	0.433	0.428	0.414	0.409	1.0 (L/3517)	0.96 (L/3645)
233	340-340	1.9-3.0	4.9 (154)	4.9 (154)	0.87 (114)	0.88 (114)	0.052	0.053	0.031	0.031	0.01 (L/102777)	0.01 (L/102458)

Ring r 8.5: Mz at the ring ends at each joint (max |Mz| over ULS 101-154, kNm, local ring axes)

joint deg	joint	ring SLN	v20 Mz (LC)	v21 Mz (LC)
000	A, released	185	+0.00 (101)	+0.00 (101)
000	A, released	227	+0.00 (101)	+0.00 (101)
020	S/T, unchanged	185	+2.07 (135)	+2.13 (135)
020	S/T, unchanged	264	-0.96 (104)	-1.00 (135)
040	B, released	264	+0.00 (101)	+0.00 (101)
040	B, released	265	+0.00 (101)	+0.00 (101)
060	S/T, unchanged	189	+0.43 (106)	+0.44 (148)
060	S/T, unchanged	265	+0.96 (119)	+0.94 (119)
080	A, released	189	+0.00 (101)	+0.00 (101)
080	A, released	192	+0.00 (101)	+0.00 (101)
100	S/T, unchanged	192	+0.68 (122)	+0.69 (122)
100	S/T, unchanged	195	+0.63 (148)	+0.63 (148)
120	A, released	195	+0.00 (101)	+0.00 (101)
120	A, released	198	+0.00 (101)	+0.00 (101)
140	S/T, unchanged	198	+0.77 (120)	+0.77 (120)
140	S/T, unchanged	210	+0.68 (146)	+0.68 (146)
160	A, released	210	+0.00 (101)	+0.00 (101)
160	A, released	213	+0.00 (101)	+0.00 (101)
180	S/T, unchanged	213	+0.80 (104)	+0.80 (104)
180	S/T, unchanged	214	+0.72 (146)	+0.72 (146)
200	A, released	214	+0.00 (101)	+0.00 (101)
200	A, released	217	+0.00 (101)	+0.00 (101)
220	S/T, unchanged	217	+0.70 (120)	+0.69 (120)
220	S/T, unchanged	218	-0.72 (121)	-0.72 (121)
240	A, released	218	+0.00 (101)	+0.00 (101)
240	A, released	221	+0.00 (101)	+0.00 (101)
260	S/T, unchanged	221	-0.54 (123)	-0.55 (123)
260	S/T, unchanged	426	-1.00 (121)	-0.99 (121)
280	B, released	425	+0.00 (101)	+0.00 (101)
280	B, released	426	+0.00 (101)	+0.00 (101)
300	S/T, unchanged	225	-2.03 (135)	-2.08 (135)
300	S/T, unchanged	425	+0.82 (135)	+0.90 (135)
320	A, released	225	+0.00 (101)	+0.00 (101)
320	A, released	226	+0.00 (101)	+0.00 (101)
340	S/T, unchanged	226	-0.65 (103)	-0.65 (103)
340	S/T, unchanged	227	+0.58 (121)	+0.60 (121)

L2 outer ring r 12, SLN 56 / 57 / 62

SLN	BDK v20 (LC)	BDK v21 (LC)	AQB v20 (LC)	AQB v21 (LC)	node-end My v20	node-end My v21	My at 300 mm v20	My at 300 mm v21
56	1.060 (120)	1.063 (120)	1.061 (120)	1.066 (120)	-163.2893	-163.8246	-136.054	-136.615
57	1.011 (120)	1.013 (120)	1.007 (120)	1.011 (120)	-156.1116	-156.4369	-130.885	-131.263
62	1.003 (122)	1.008 (122)	1.000 (153)	1.005 (122)	-155.0457	-155.8506	-128.331	-129.127

Flags: utilisation worse by more than 0.02 (whole model: BDK per SLN, AQB per beam)

621 entries (full list: compare_v20_v21.csv):

check	id	SLN	v20 (LC)	v21 (LC)	change
AQB	190173	763	0.214 (135)	0.395 (152)	+0.181
AQB	190178	763	0.219 (121)	0.397 (121)	+0.178
AQB	190028	718	0.207 (135)	0.383 (122)	+0.176
AQB	170410	1600	0.179 (120)	0.352 (120)	+0.173
AQB	190033	718	0.213 (122)	0.385 (122)	+0.172
BDK	746	746	0.211 (135)	0.382 (135)	+0.171
BDK	701	701	0.211 (135)	0.381 (135)	+0.170
AQB	170340	1747	0.185 (152)	0.354 (120)	+0.169
AQB	190016	711	0.259 (135)	0.413 (135)	+0.154
AQB	170402	1608	0.332 (122)	0.479 (122)	+0.147
AQB	170331	1684	0.311 (120)	0.457 (120)	+0.146
AQB	170282	1470	0.204 (121)	0.350 (121)	+0.146
AQB	170332	1721	0.331 (120)	0.477 (120)	+0.146
AQB	170401	1697	0.311 (122)	0.456 (122)	+0.145
BDK	713	713	0.166 (122)	0.310 (135)	+0.144
BDK	758	758	0.168 (121)	0.312 (135)	+0.144
AQB	190161	756	0.274 (135)	0.418 (135)	+0.144
AQB	170403	1607	0.343 (122)	0.484 (122)	+0.141
AQB	170283	1479	0.198 (121)	0.338 (121)	+0.140

BDK	759	759	0.168 (135)	0.308 (135)	+0.140
AQB	170271	1454	0.132 (152)	0.271 (152)	+0.139
BDK	489	489	0.220 (122)	0.359 (135)	+0.139
AQB	170330	1537	0.277 (120)	0.416 (120)	+0.139
AQB	170333	1736	0.340 (120)	0.479 (120)	+0.139
BDK	702	702	0.176 (119)	0.314 (119)	+0.138
AQB	170352	1564	0.173 (119)	0.310 (122)	+0.137
AQB	170400	1609	0.276 (122)	0.413 (122)	+0.137
BDK	136	136	0.219 (135)	0.355 (135)	+0.136
AQB	190418	1729	0.203 (105)	0.338 (105)	+0.135
AQB	190419	1743	0.199 (105)	0.334 (105)	+0.135
BDK	747	747	0.167 (121)	0.302 (121)	+0.135
AQB	170272	1463	0.130 (152)	0.263 (152)	+0.133
AQB	190417	1707	0.197 (105)	0.329 (105)	+0.132
AQB	190420	1753	0.191 (105)	0.323 (105)	+0.132
AQB	170353	1573	0.168 (119)	0.300 (122)	+0.132
AQB	190326	1749	0.205 (103)	0.336 (103)	+0.131
AQB	190179	764	0.164 (121)	0.295 (121)	+0.131
AQB	170404	1606	0.343 (122)	0.474 (122)	+0.131
AQB	190325	1739	0.198 (103)	0.328 (103)	+0.130
AQB	170319	1534	0.186 (135)	0.315 (135)	+0.129

For information: 252 entries went down by more than 0.02: BDK 92 (SLN 92) 0.707 -> 0.626, BDK 115 (SLN 115) 0.698 -> 0.617, BDK 127 (SLN 127) 0.708 -> 0.627, BDK 111 (SLN 111) 0.692 -> 0.612, BDK 722 (SLN 722) 0.975 -> 0.897, BDK 95 (SLN 95) 0.708 -> 0.631, BDK 110 (SLN 110) 0.701 -> 0.625, BDK 118 (SLN 118) 0.720 -> 0.645, BDK 73 (SLN 73) 0.725 -> 0.651, AQB 100859 (SLN 1551) 0.424 -> 0.353, BDK 122 (SLN 122) 0.599 -> 0.529, BDK 767 (SLN 767) 0.966 -> 0.900, AQB 100841 (SLN 1685) 0.409 -> 0.345, AQB 140834 (SLN 1520) 0.556 -> 0.506, AQB 140442 (SLN 1467) 0.124 -> 0.074.

Members above 0.90 in v21 (summary_v21.json):

- AQB SLN 56 group 25: 1.066 LC 120 (Line)
- BDK SLN 56 group 25: 1.063 LC 120 (Line)
- BDK SLN 57 group 25: 1.013 LC 120 (Line)
- AQB SLN 57 group 25: 1.011 LC 120 (Line)
- BDK SLN 62 group 25: 1.008 LC 122 (Line)
- AQB SLN 62 group 25: 1.005 LC 122 (Line)
- BDK SLN 69 group 25: 0.978 LC 119 (Line)
- AQB SLN 69 group 25: 0.970 LC 153 (Line)
- BDK SLN 68 group 25: 0.962 LC 121 (Line)
- BDK SLN 66 group 25: 0.960 LC 121 (Line)
- AQB SLN 68 group 25: 0.957 LC 153 (Line)
- AQB SLN 66 group 25: 0.950 LC 153 (Line)
- BDK SLN 67 group 25: 0.943 LC 121 (Line)
- AQB SLN 67 group 25: 0.933 LC 153 (Line)
- BDK SLN 70 group 25: 0.909 LC 119 (Line)
- BDK SLN 61 group 25: 0.904 LC 122 (Line)

Status changes v20 -> v21 (utilisation crossing 1.00, BDK per SLN and AQB per beam)

check	id	SLN	v20 (LC)	v21 (LC)	status
AQB	140080	62	1.000 (153)	1.005 (122)	PASS -> FAIL

Members above 1.00 in v21: AQB SLN 56 1.066 LC 120, BDK SLN 56 1.063 LC 120, BDK SLN 57 1.013 LC 120, AQB SLN 57 1.011 LC 120, BDK SLN 62 1.008 LC 122, AQB SLN 62 1.005 LC 122

Governing L2 ring check, SLN 56 (L2 outer ring r 12, UB 254, LC 120)

item	v20	v21
BDK (LC)	1.06 (120)	1.063 (120) Not passed
AQB (LC)	1.061 (120)	1.066 (120)
node end N (node 1460)	18.36	16.48
node end Vz (node 1460)	-91.02	-90.93
node end My (node 1460)	-163.29	-163.82
My at 300 mm from the node	-136.05	-136.62

Units kN, kNm; SOFiSTiK local axes. LC 120 = ULS G+Q+psiW -X down (wind as the accompanying action, so the qp change hardly moves it).

External stair (group 62)

member	v20 BDK / AQB (LC)	v21 BDK / AQB (LC)
SLN 701 right flight 1 inner stringer	0.211 (135) / 0.205 (103)	0.381 (135) / 0.336 (103)
SLN 702 right flight 1 outer stringer	0.176 (119) / 0.166 (135)	0.314 (119) / 0.284 (135)
SLN 713 right flight 2 inner stringer	0.166 (122) / 0.225 (122)	0.31 (135) / 0.343 (122)
SLN 714 right flight 2 outer stringer	0.171 (135) / 0.201 (135)	0.287 (135) / 0.307 (135)
SLN 722 right landing 2 leg outer-low	0.975 (122) / 0.21 (152)	0.897 (122) / 0.216 (152)
SLN 746 left flight 1 inner stringer	0.211 (135) / 0.203 (105)	0.382 (135) / 0.338 (105)
SLN 747 left flight 1 outer stringer	0.167 (121) / 0.164 (135)	0.302 (121) / 0.282 (135)
SLN 758 left flight 2 inner stringer	0.168 (121) / 0.225 (121)	0.312 (135) / 0.343 (121)
SLN 759 left flight 2 outer stringer	0.168 (135) / 0.198 (121)	0.308 (135) / 0.303 (135)

SLN 767 left landing 2 leg outer-low	0.966 (121) / 0.214 (152)	0.9 (121) / 0.222 (152)

Landing horizontal movement (mm):

landing	LC 201 v20	LC 201 v21	worst wind SLS v20 (LC)	worst wind SLS v21 (LC)
right landing 1	2.9	4.0	5.9 (202)	7.8 (202)
right landing 2	18.8	23.4	21.0 (218)	26.7 (218)
left landing 1	2.9	3.9	5.4 (202)	7.0 (202)
left landing 2	18.3	22.8	19.9 (220)	25.3 (220)

Stair support reactions (foot / leg bases, ULS envelope): see stair_ext.json 'supports' and env_support_ULS.json.

alpha_cr (lowest buckling factors) and mode description [v20 in brackets]

LC 9201 (Buckling 1 fact 6.95) [v20: Buckling 1 fact 6.95]: max node 2369 r 11.82 at 171.4 deg z 1.74 [v20 node 2369]; groups with > 10 % of the max: 11 1.0, 15 0.168, 21 0.228
LC 9220 (Buckling 20 fact 9.50) [v20: Buckling 20 fact 9.51]: max node 2332 r 11.82 at 108.6 deg z 1.74 [v20 node 2332]; groups with > 10 % of the max: 11 1.0, 12 0.272, 15 0.111, 21 0.139, 62 0.117
LC 9221 (Buckling 1 fact 7.15) [v20: Buckling 1 fact 7.32]: max node 3267 r 4.57 at 300.0 deg z 4.05 [v20 node 3267]; groups with > 10 % of the max: 12 0.409, 13 0.426, 14 0.832, 15 1.0, 23 0.404, 24 0.798, 60 0.186, 61 0.238
LC 9240 (Buckling 20 fact 10.07) [v20: Buckling 20 fact 10.11]: max node 2343 r 11.82 at 131.4 deg z 1.74 [v20 node 2343]; groups with > 10 % of the max: 11 1.0, 12 0.342, 13 0.134, 14 0.31, 21 0.171

PLC 102: lowest mode = local ground-floor raking column (group 11) at about 171 deg, as in v20. PLC 119: lowest mode = Level 1 pinwheel (inner core turns, radials r 3-8.5 bend about the weak axis), as in v20. Both < 10 (same class as before).

Max SLS deflections [v20 in brackets]

(a) L1 ring r 12, LC 201: rel -14.28 at node 1558 / abs -15.61 (span 0-40, L 8.335, LC 201) [-13.31 / -14.64]
(a) L1 ring r 12, G+Q+0.5W worst: rel -14.54 at node 1558 / abs -15.79 (span 0-40, L 8.335, LC 221) [-13.53 / -14.78]
(a) L1 ring r 12, LC 11 only: rel -7.6 at node 1559 / abs -7.85 (span 320-0, L 8.335, LC 11) [-7.77 / -8.03]
(b) L2 radial, LC 201: rel -16.36 at node 5532 / abs -18.49 (span 220, L 11.34, LC 201) [-17.85 / -19.99]
(b) L2 radial, LC 11+12+13: rel -12.33 at node 5532 / abs -13.78 (span 220, L 11.34, LC (11, 12, 13)) [-13.53 / -14.99]
(b) L2 radial largest abs, LC 201: rel -9.83 at node 5564 / abs -29.59 (span 240, L 11.34, LC 201) [-11.25 / -30.97]
(c) roof radial, LC 201: rel -12.34 at node 3990 / abs -15.6 (span 340, L 11.34, LC 201) [-12.34 / -15.61]
(c) roof radial, LC 12: rel -8.08 at node 3765 / abs -8.35 (span 80, L 11.34, LC 12) [-8.08 / -8.35]
(c) roof radial upward G+W: rel 16.55 at node 3892 / abs 18.11 (span 180, L 11.34, LC 202) [16.55 / 18.11]
(d) max horizontal L1: 1.45 mm node 1712 LC 216 [1.35 node 1712 LC 216]
(d) max horizontal L2: 1.32 mm node 5301 LC 202 [1.37 node 5301 LC 210]
(d) max horizontal RF: 1.3 mm node 3651 LC 204 [1.32 node 3651 LC 204]
(e) L1 radial 280 deg r 8.5-12 under the landing posts, LC 201: rel -1.93 at node 5722 / abs -2.98 (span r8.50-12.00, L 3.5, LC None) [-1.82 / -2.87]
(e) L1 radial 40 deg r 8.5-12, LC 201: rel -1.91 at node 5733 / abs -2.92 (span r8.50-12.00, L 3.501, LC None) [-1.79 / -2.8]
(f) internal stair stringer, SLS envelope worst: rel 0.0 at node 1494 / abs 1.83 (span None, L 0.239, LC 201) [-4.43 / -14.31]

Max reactions [v20 in brackets]

max pz: 428.3 node 2 LC 146 [427.8 node 2 LC 146]
min pz: -45.1 node 12 LC 109 [-45.6 node 12 LC 109]
max horizontal resultant: 112.7 node 27 LC 105 [113.5 node 27 LC 105]

Sliding and overturning from the real v21 reactions (unfactored, mu 0.40, as the original report)

FoS sliding = 0.40 x sum Fz / sum H. Overturning: no support may lift off (min support Fz > 0), as the original report; plus the EQU 0.9G + 1.5W uplift per support for the anchors (reactions_EQU.json).

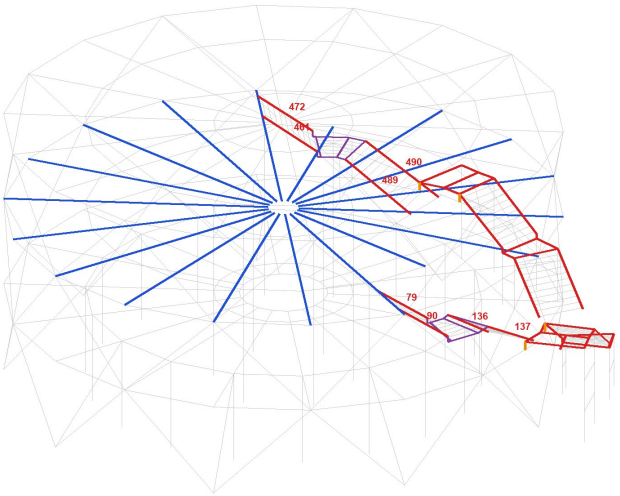
combination	wind	sum Fz (kN)	H (kN)	FoS sliding	min support Fz (kN, node)
DL + WL	+X	1279.5	201.0	2.55	-7.4 (901)
DL + WL	-X	1322.7	196.0	2.70	-7.9 (902)
DL + WL	-Y	1293.7	207.5	2.49	-5.4 (909)
DL + WL	+Y	1307.9	205.5	2.55	-3.2 (911)
DL + 0.7LL + 0.5WL	+X	4685.0	100.5	18.65	2.9 (901)
DL + 0.7LL + 0.5WL	-X	4706.6	98.0	19.21	0.6 (902)
DL + 0.7LL + 0.5WL	-Y	4692.1	103.7	18.09	3.9 (909)
DL + 0.7LL + 0.5WL	+Y	4699.3	102.8	18.29	3.4 (910)
DL + LL + 0.5WL	+X	6005.1	100.5	23.90	5.2 (901)
DL + LL + 0.5WL	-X	6026.7	98.0	24.59	2.2 (902)
DL + LL + 0.5WL	-Y	6012.2	103.7	23.18	6.2 (909)
DL + LL + 0.5WL	+Y	6019.4	102.8	23.43	5.0 (910)

Minimum sliding FoS (DL + WL): 2.49 [v20 2.47].
Overturning (unfactored): supports with uplift: DL + WL +X node 901 -7.4 kN; DL + WL -X node 902 -7.9 kN; DL + WL -Y node 909 -5.4 kN; DL + WL +Y node 911 -3.2 kN (anchors needed, see EQU).
EQU 0.9G + 1.5W (131-134): LC 131 max uplift -47.9 kN at node 2 (20 supports in uplift); LC 132 max uplift -44.8 kN at node 34 (17 supports in uplift); LC 133 max uplift -53.2 kN at node 12 (18 supports in uplift); LC 134 max uplift -45.0 kN at node 27 (18 supports in uplift)
[v20: LC 131 -48.7 kN node 2; LC 132 -44.9 kN node 34; LC 133 -53.6 kN node 12; LC 134 -45.3 kN node 27]

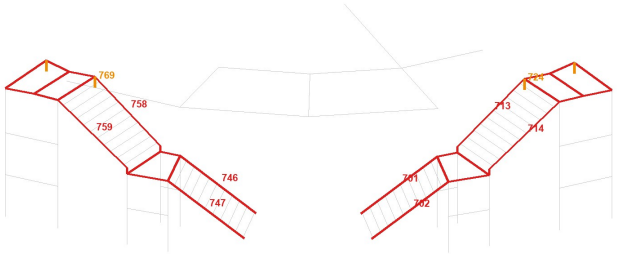
v21: members changed to drawings 970027-S01-ST (stair RHS 160x80x5, L2 radials UB 457x191x98, landing supports SHS 80x80x6.3)

26016 v21 - members changed to drawings 970027-S01-ST (issue 13.11.2026), coloured by NEW section

Whole model (v21 geometry = v20)

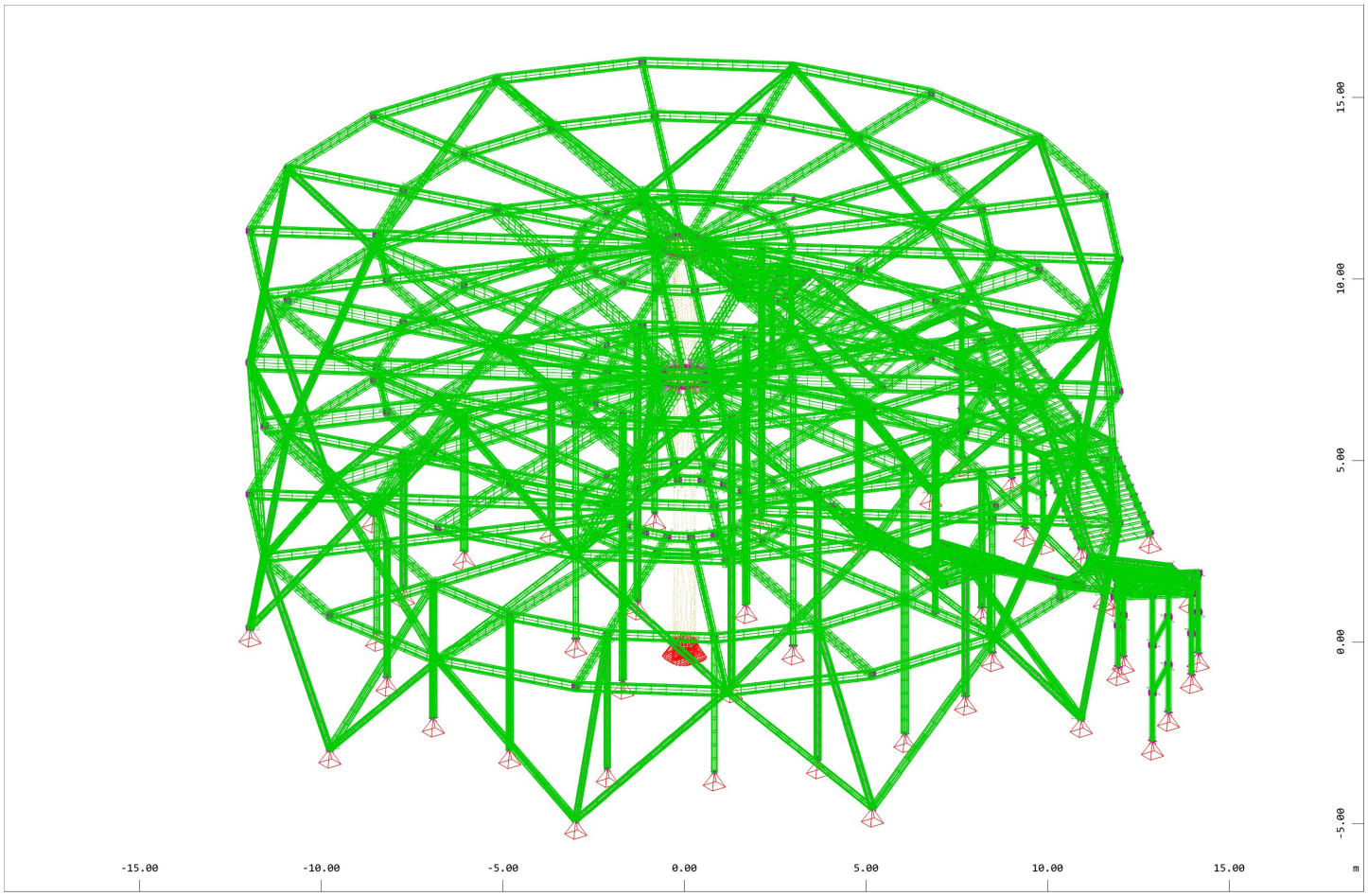


External stair (group 62), from outside

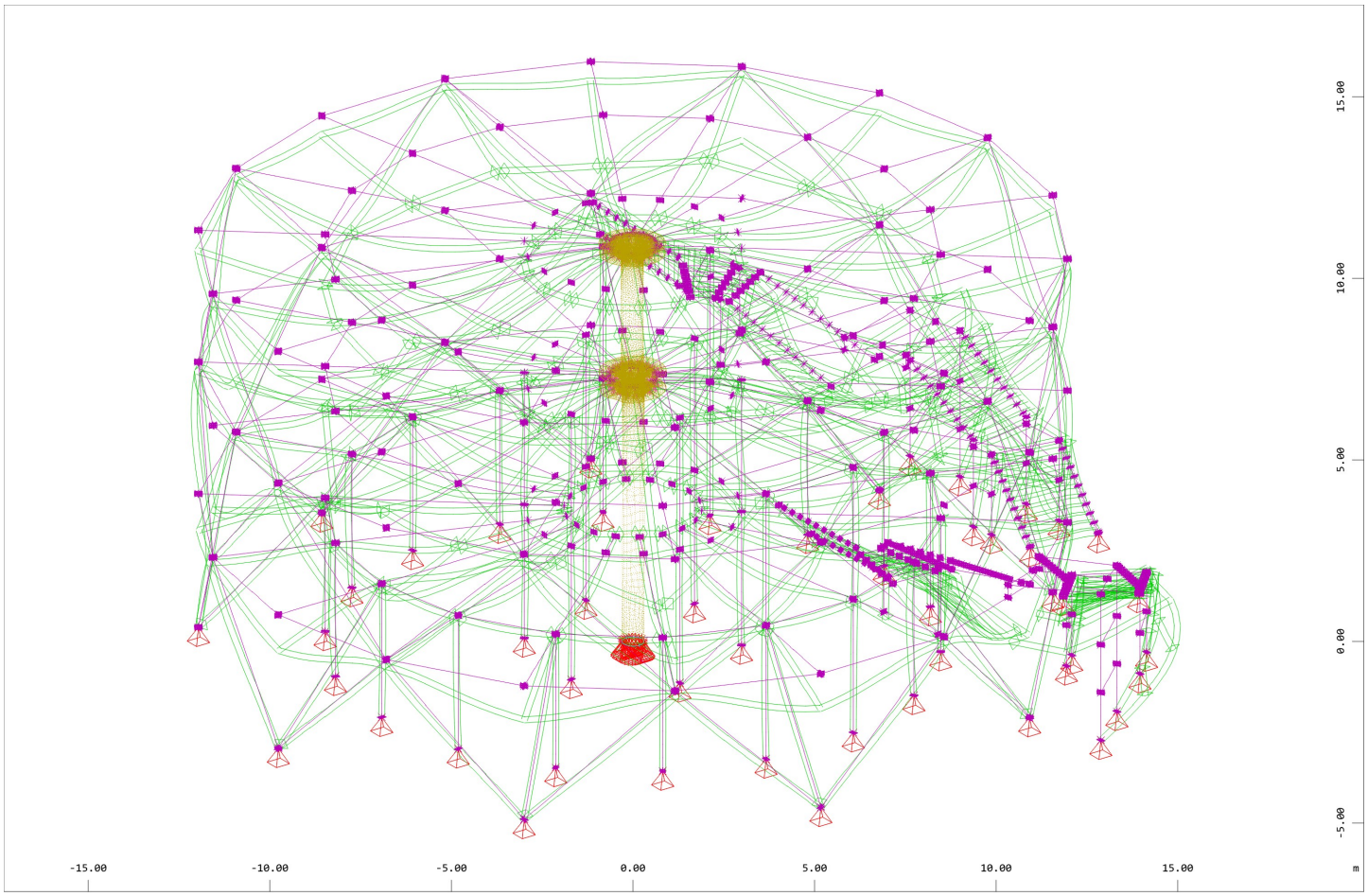


- UB 457x191x98: 18 Level 2 radials (section 4, was UB 457x191x89; drg 1101/1102)
- RHS 160x80x5 (new section 10, was RHS 200x100x6): external stair stringers 701 702 713 714 746 747 758 759, seats 711 712 756 757, landing edge / middle beams 703-706 715-721 748-751 760-766; internal stair stringers 79 90 136 137 461 472 489 490 (drg 1201/1302/1303)
- SHS 80x80x6.3 (section 9, was SHS 100x100x5): external stair landing supports on the L1 ring 724 725 769 770 (drg 1304), Z-only as before
- kept RHS 200x100x6: internal stair landing beams and short posts at +6.04 (only the stringers change per the brief)
- unchanged

W1 model v21, 3D with rendered cross sections



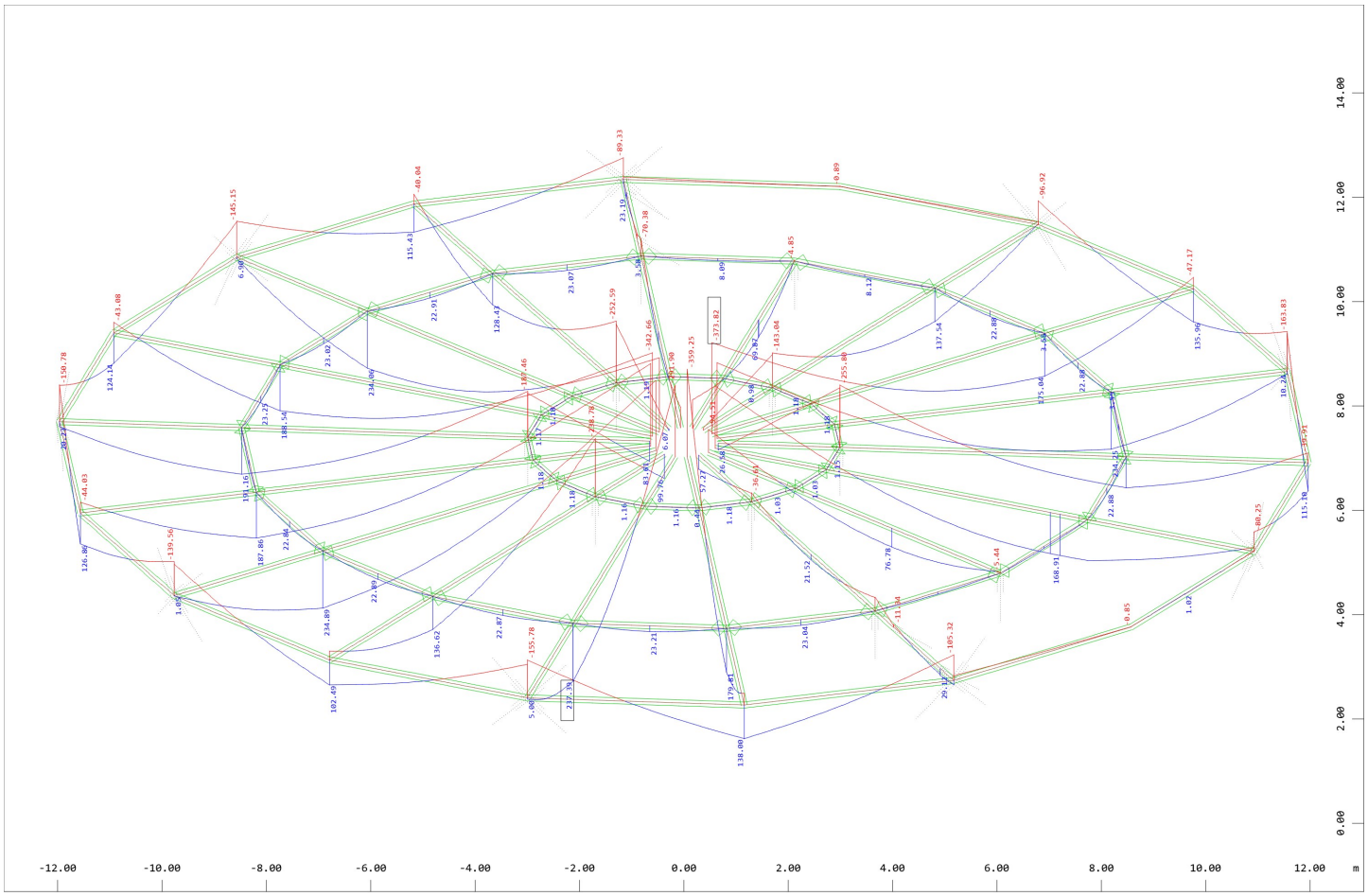
W2 deformed shape, SLS LC 201 (G + Q), x100



Deformed Structure from LC 201 201 SLS G+Q Enlarged by 100.00

M 1 : 148
X * 0.850
Y * 0.675
Z * 0.907

W3 Level 2 framing, My, ULS LC 119



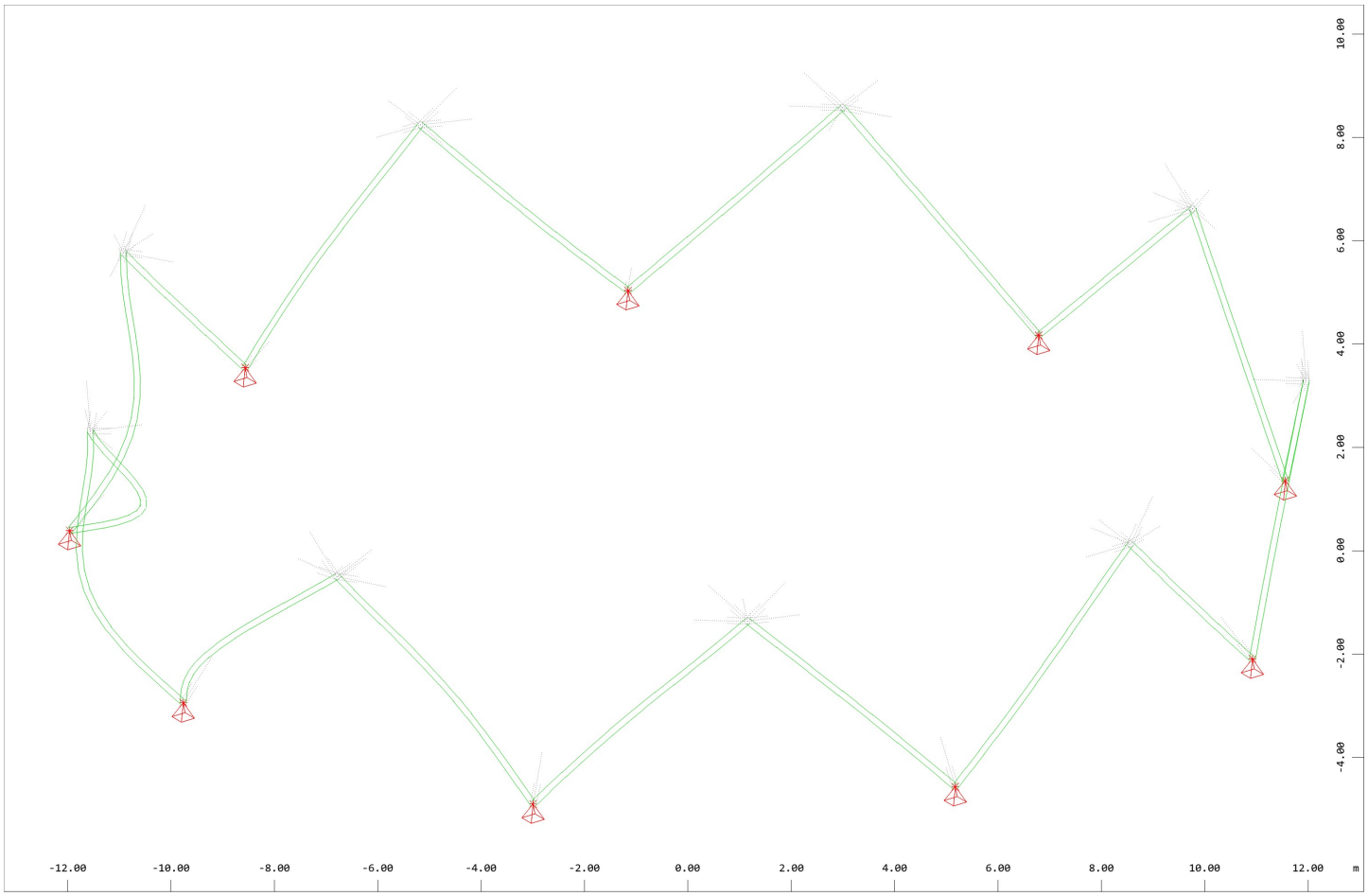
Sector of system Group 25

Z
Y X Beam Elements , Bending moment My, Loadcase 119 119 ULS G+Q+psiW +X down , 1 cm 3D = 200.00 kNm (Min=-373.82) (Max=237.39)

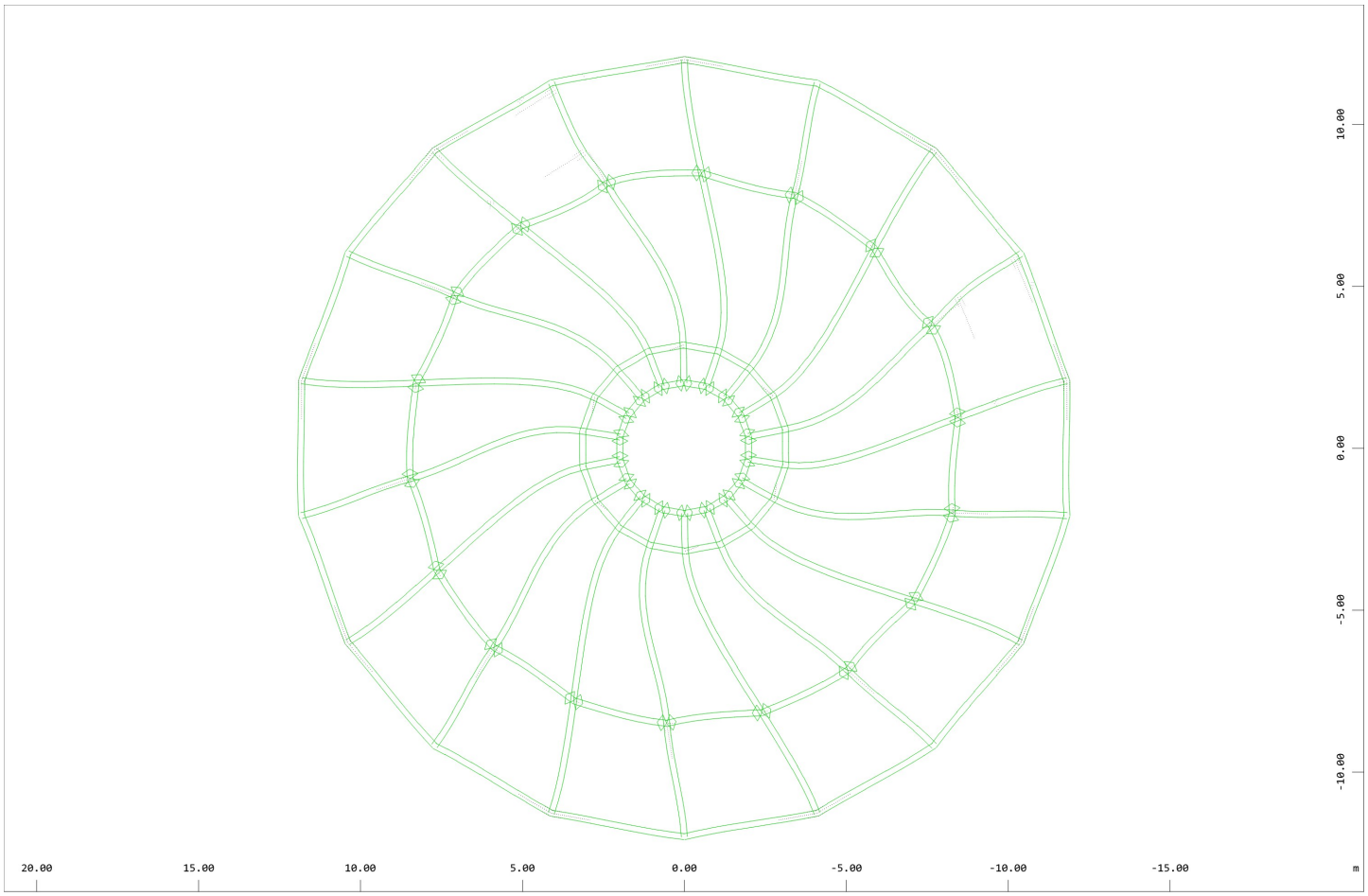
M 1 : 103

X * 0.850
Y * 0.675
Z * 0.907

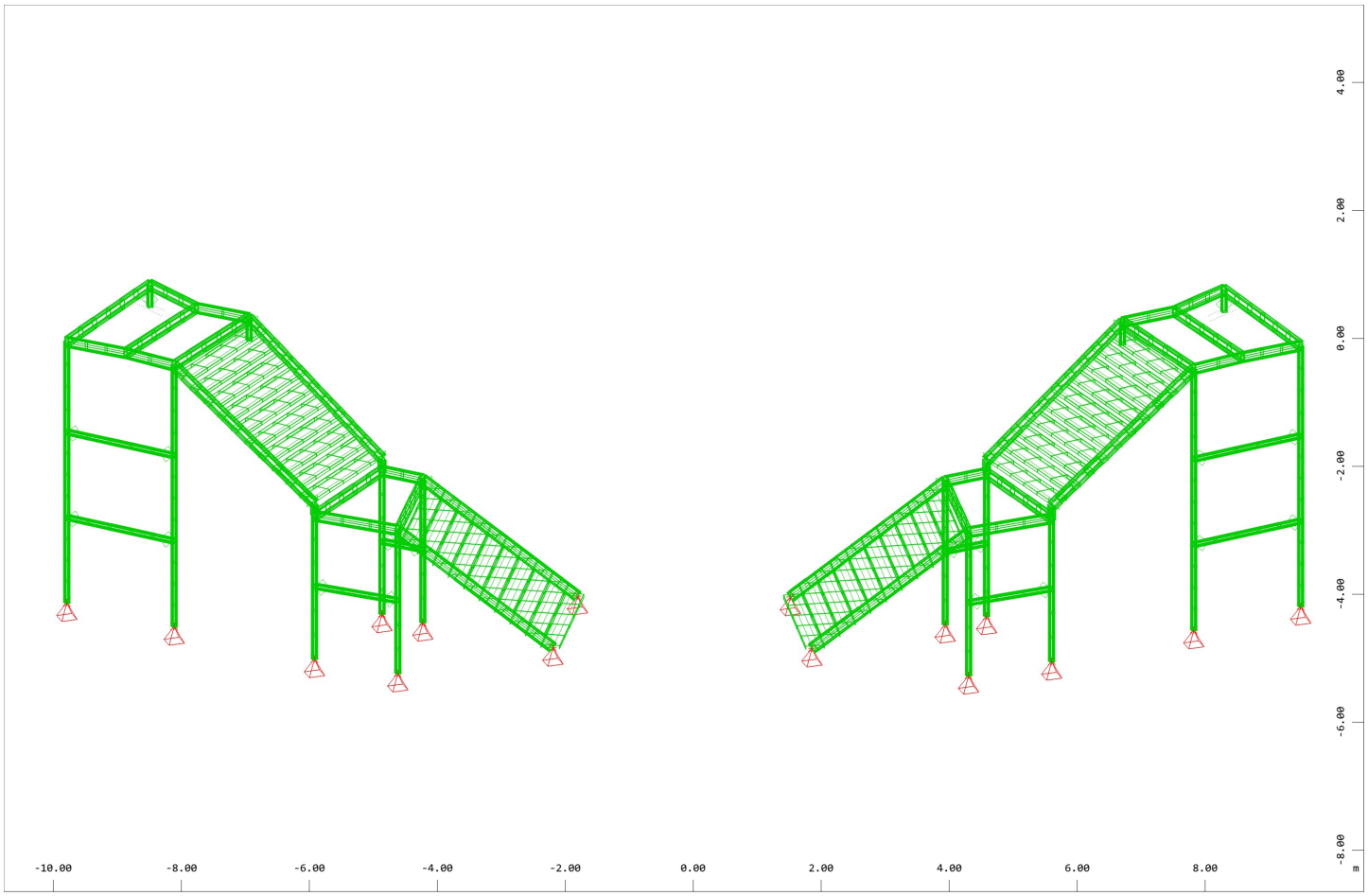
W4 buckling mode 1 of PLC 102 (alpha_cr 6.95)



W5 buckling mode 1 of PLC 119, Level 1 plan (alpha_cr 7.15)



W8 external stair, sections to scale



26016 Concourse viewing platform
Model v21 - ULS member force envelopes (SOFiSTiK WinGRAF)

Prepared 13 Nov 2026 from 26016_check_v21 (v20 + drawing 970027-S01-ST sections: stair RHS 160x80x5, L2 radials UB 457x191x98, stair landing supports SHS 80x80x6.3), read from a copy of the database.

Envelope: the model's own MAXIMA superposition COMB 3 (EXTR NONL, TYPE DESI, BASE 4300): each of the 43 ULS STR combinations 101-130, 135, 141-148, 151-154 enters as an exclusive alternative (TYPE AG1, factor 1.00), so every envelope value is the value of one complete combination. EQU combinations 131-134 are not included. The combinations are linear (ASE SYST PROB LINE) COPY combinations.

Beam envelope load cases: N 4321/4322, Vy 4323/4324, Vz 4325/4326, Mt 4327/4328, My 4329/4330, Mz 4331/4332 (max/min). SLS characteristic 201-221: 4421-4432 (same order).

Check: at every result station of every beam the MAXIMA value equals the largest / smallest value read from the 43 combinations (largest difference over all stations: N 0.0e+00, My 4.0e-09 kNm).

Sign convention: SOFiSTiK local beam axes, N > 0 tension, My about the local y (strong) axis, Mz about the local z (weak) axis. Red = max load case, blue = min load case. Units kN, kNm.

Floor framing is drawn in a steep isometric view because My, Vz act in the vertical plane and cannot be read in a true top view (tested: diagrams rotated into the plan flip side with the member direction). Mz of Level 1 is drawn in a true plan view (X right, Y up).

Kingpost / central mast: modelled with shell elements (QUAD), no beam group, so no beam N / My diagram is given here.

ULS envelope per member group - forces N, Vy, Vz

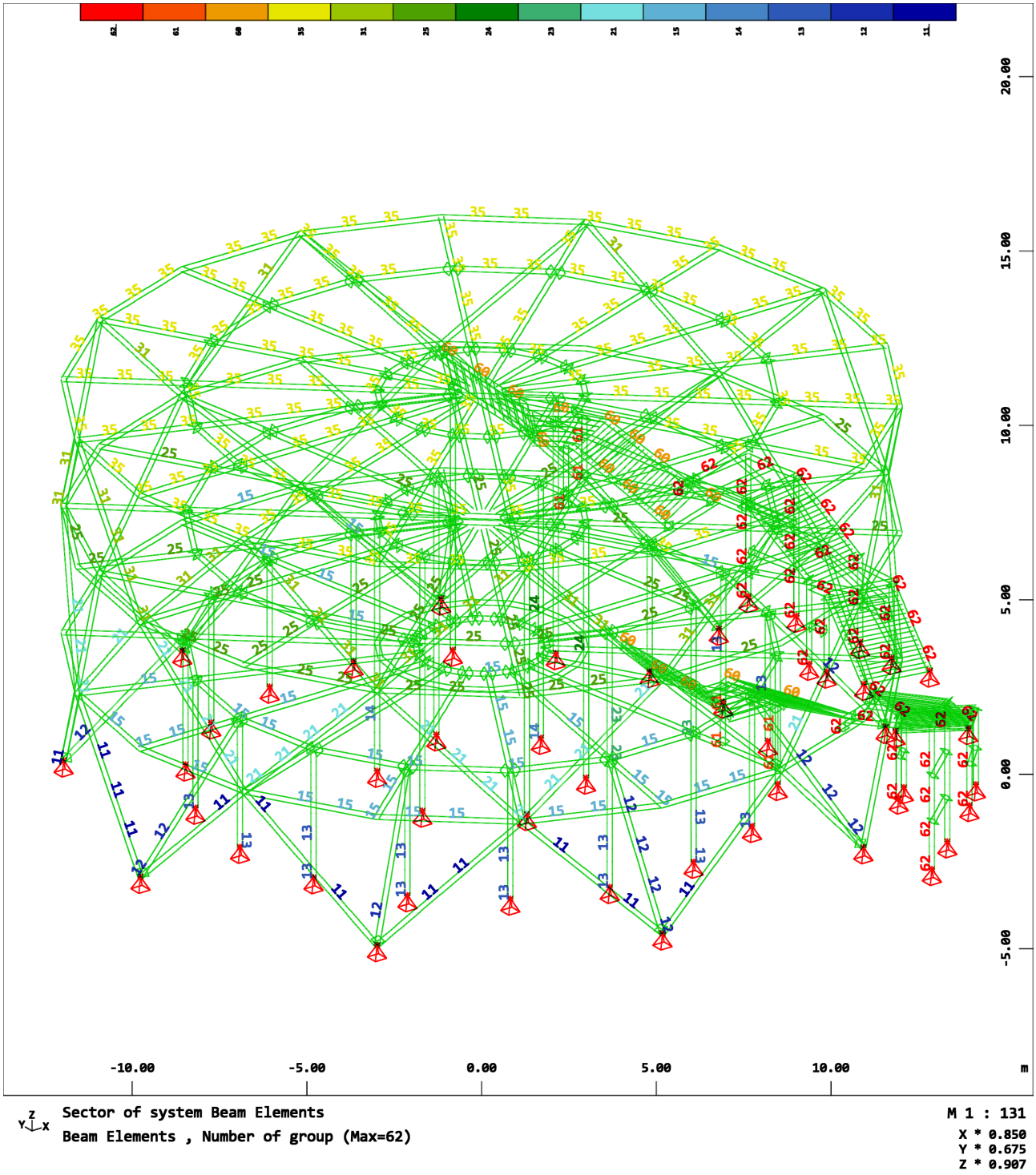
Each cell: envelope value, governing combination (LC), beam element and x [m from beam start]. Full data incl. SLN: env_table_ULS.csv / .json.

Group							
11	V-column rakers, ground to L1 SHS 150x150x10	18.2 (LC 109) b 60196 x 0.42	-290.7 (LC 147) b 60239 x 0.00	1.5 (LC 120) b 60141 x 0.00	-1.6 (LC 126) b 60084 x 0.42	2.8 (LC 120) b 60071 x 0.00	-2.8 (LC 122) b 60154 x 0.42
12	rakers, ground to L1 SHS 150x150x10	37.4 (LC 116) b 70104 x 0.41	-75.5 (LC 105) b 70053 x 0.00	0.8 (LC 130) b 70079 x 0.00	-1.1 (LC 125) b 70092 x 0.00	1.4 (LC 115) b 70001 x 0.00	-1.4 (LC 123) b 70013 x 0.41
13	vertical columns (ground-L1, L1-L2) SHS 150x150x10	10.7 (LC 110) b 80061 x 0.00	-263.7 (LC 121) b 80050 x 0.41	6.1 (LC 124) b 80011 x 0.00	-4.8 (LC 124) b 80030 x 0.41	5.1 (LC 123) b 80141 x 0.00	-5.0 (LC 128) b 80151 x 0.00
14	vertical columns, ground to L1 SHS 150x150x10	-21.2 (LC 109) b 90011 x 0.00	-403.7 (LC 146) b 90030 x 0.41	0.9 (LC 152) b 90051 x 0.00	-1.4 (LC 151) b 90011 x 0.00	0.3 (LC 113) b 90010 x 0.41	-0.2 (LC 112) b 90060 x 0.41
15	Level 1 framing (radials + rings) UB 254x146x43	108.8 (LC 135) b 100507 x 0.00	-57.8 (LC 104) b 100333 x 0.00	48.0 (LC 122) b 100841 x 0.25	-49.4 (LC 152) b 100859 x 0.00	79.8 (LC 153) b 100842 x 0.00	-79.6 (LC 153) b 100868 x 0.39
21	V-column rakers, L1 to L2 SHS 150x150x10	29.4 (LC 109) b 110085 x 0.00	-228.4 (LC 145) b 110238 x 0.41	4.2 (LC 120) b 110168 x 0.41	-4.1 (LC 120) b 110014 x 0.41	7.3 (LC 121) b 110113 x 0.00	-6.6 (LC 119) b 110196 x 0.41
23	vertical columns, L1 to L2 SHS 150x150x10	-6.5 (LC 109) b 120001 x 0.00	-145.0 (LC 121) b 120020 x 0.40	3.3 (LC 154) b 120001 x 0.00	-4.0 (LC 152) b 120020 x 0.40	0.6 (LC 128) b 120010 x 0.40	-0.4 (LC 123) b 120020 x 0.40
24	vertical columns, L1 to L2 SHS 150x150x10	-5.8 (LC 109) b 130021 x 0.00	-268.8 (LC 146) b 130040 x 0.40	4.6 (LC 110) b 130021 x 0.00	-8.3 (LC 125) b 130060 x 0.40	5.2 (LC 110) b 130040 x 0.40	-5.6 (LC 124) b 130060 x 0.40
25	Level 2 framing (radials + rings) UB 254x146x43; UB 457x191x98	62.7 (LC 145) b 140632 x 0.00	-33.3 (LC 110) b 140748 x 0.00	25.1 (LC 152) b 140775 x 0.00	-24.9 (LC 152) b 140769 x 0.00	149.7 (LC 146) b 140684 x 0.00	-111.1 (LC 145) b 140389 x 0.39
31	V-column rakers, L2 to roof SHS 150x150x10	45.7 (LC 112) b 150085 x 0.00	-60.0 (LC 147) b 150099 x 0.00	5.5 (LC 145) b 150196 x 0.41	-5.6 (LC 145) b 150169 x 0.00	7.3 (LC 153) b 150127 x 0.00	-6.9 (LC 145) b 150042 x 0.41
35	roof framing (radials + rings) UB 254x146x43	55.6 (LC 113) b 160070 x 0.42	-54.1 (LC 112) b 160030 x 0.42	1.1 (LC 125) b 160129 x 0.42	-1.1 (LC 126) b 160070 x 0.42	19.1 (LC 110) b 160627 x 0.39	-19.3 (LC 108) b 160081 x 0.00
60	internal stair (stringers, landing, treads) RHS 200x100x6; SHS 50x50x5; H = 5 mm; RHS 160x80x5 s	59.2 (LC 152) b 170329 x 0.34	-24.5 (LC 135) b 170386 x 0.00	21.9 (LC 122) b 170330 x 0.30	-27.8 (LC 152) b 170138 x 0.00	36.0 (LC 122) b 170376 x 0.00	-28.9 (LC 152) b 170008 x 0.00
61	internal stair landing posts SHS 100x100x5	0.7 (LC 107) b 180001 x 0.00	-50.8 (LC 120) b 180010 x 0.40	0.0 (LC 125) b 180006 x 0.00	-0.0 (LC 135) b 180002 x 0.00	0.0 (LC 135) b 180012 x 0.00	-0.0 (LC 107) b 180018 x 0.00
62	external stair ground to L1 H = 5 mm; SHS 80x80x6.3 external stair legs; RHS 160	48.5 (LC 111) b 190321 x 0.05	-54.3 (LC 135) b 190321 x 0.00	27.1 (LC 135) b 190016 x 0.00	-28.9 (LC 135) b 190161 x 0.00	12.5 (LC 121) b 190459 x 0.00	-15.1 (LC 135) b 190016 x 0.00

ULS envelope per member group - moments Mt, My, Mz

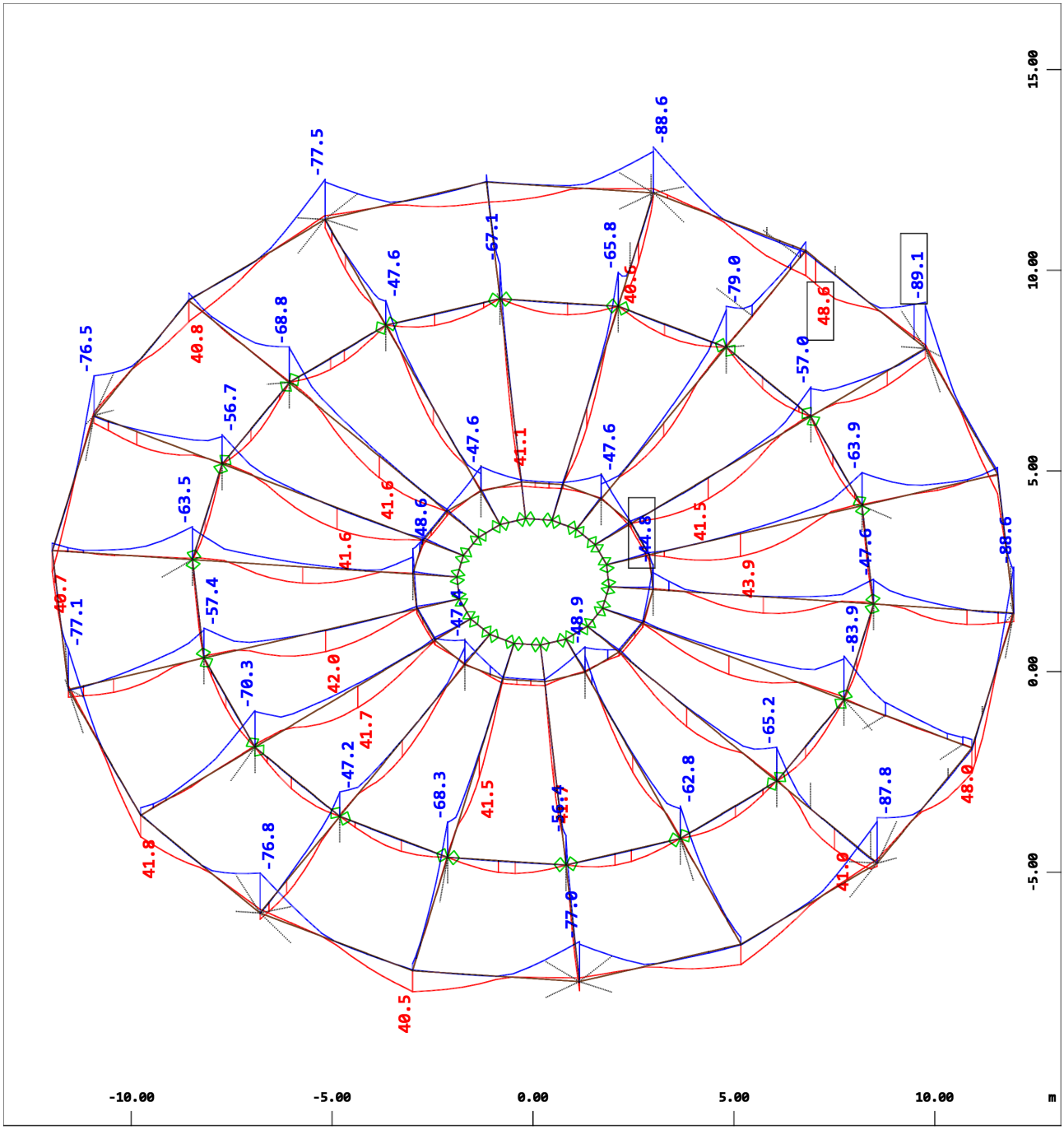
Each cell: envelope value, governing combination (LC), beam element and x [m from beam start]. Full data incl. SLN: env_table_ULS.csv / .json.

Group							
11	V-column rakers, ground to L1 SHS 150x150x10	1.1 (LC 152) b 60167 x 0.00	-1.1 (LC 152) b 60058 x 0.00	4.8 (LC 105) b 60085 x 0.00	-6.0 (LC 151) b 60113 x 0.00	6.0 (LC 152) b 60084 x 0.42	-2.3 (LC 152) b 60071 x 0.00
12	rakers, ground to L1 SHS 150x150x10	0.4 (LC 124) b 70074 x 0.00	-0.4 (LC 125) b 70010 x 0.00	1.8 (LC 115) b 70007 x 0.41	-0.0 (LC 118) b 70078 x 0.41	2.3 (LC 125) b 70104 x 0.41	-1.8 (LC 126) b 70078 x 0.41
13	vertical columns (ground-L1, L1-L2) SHS 150x150x10	0.1 (LC 112) b 80183 x 0.00	-0.6 (LC 152) b 80200 x 0.00	4.6 (LC 118) b 80015 x 0.41	-4.6 (LC 125) b 80015 x 0.41	9.4 (LC 152) b 80200 x 0.40	-12.6 (LC 152) b 80021 x 0.00
14	vertical columns, ground to L1 SHS 150x150x10	0.0 (LC 151) b 90005 x 0.00	-0.0 (LC 151) b 90023 x 0.00	0.7 (LC 112) b 90051 x 0.00	-1.1 (LC 113) b 90001 x 0.00	3.6 (LC 152) b 90051 x 0.00	-5.5 (LC 151) b 90011 x 0.00
15	Level 1 framing (radials + rings) UB 254x146x43	0.9 (LC 135) b 100859 x 0.00	-0.8 (LC 135) b 100841 x 0.00	48.6 (LC 152) b 100790 x 0.00	-89.1 (LC 152) b 100799 x 0.39	9.6 (LC 105) b 100769 x 0.00	-9.8 (LC 105) b 100770 x 0.33
21	V-column rakers, L1 to L2 SHS 150x150x10	2.0 (LC 122) b 110011 x 0.00	-2.0 (LC 121) b 110251 x 0.00	20.5 (LC 120) b 110224 x 0.41	-22.7 (LC 121) b 110113 x 0.00	9.1 (LC 121) b 110183 x 0.00	-14.7 (LC 119) b 110252 x 0.41
23	vertical columns, L1 to L2 SHS 150x150x10	0.7 (LC 103) b 120015 x 0.00	-0.1 (LC 125) b 120001 x 0.00	2.2 (LC 128) b 120010 x 0.40	-1.6 (LC 123) b 120020 x 0.40	9.9 (LC 152) b 120020 x 0.40	-6.5 (LC 151) b 120020 x 0.40
24	vertical columns, L1 to L2 SHS 150x150x10	0.1 (LC 129) b 130011 x 0.00	-0.1 (LC 123) b 130001 x 0.00	3.9 (LC 130) b 130040 x 0.40	-4.0 (LC 123) b 130030 x 0.40	10.5 (LC 153) b 130020 x 0.40	-13.0 (LC 153) b 130051 x 0.00
25	Level 2 framing (radials + rings) UB 254x146x43; UB 457x191x98	1.1 (LC 120) b 140490 x 0.00	-1.2 (LC 148) b 140489 x 0.00	237.7 (LC 120) b 140698 x 0.39	-374.4 (LC 145) b 140389 x 0.39	11.1 (LC 152) b 140769 x 0.41	-9.5 (LC 152) b 140775 x 0.60
31	V-column rakers, L2 to roof SHS 150x150x10	3.7 (LC 111) b 150194 x 0.00	-3.7 (LC 111) b 150172 x 0.00	21.0 (LC 104) b 150070 x 0.41	-23.8 (LC 153) b 150127 x 0.00	17.9 (LC 145) b 150182 x 0.41	-14.5 (LC 145) b 150169 x 0.00
35	roof framing (radials + rings) UB 254x146x43	0.0 (LC 112) b 160815 x 0.00	-0.0 (LC 108) b 160782 x 0.00	44.7 (LC 107) b 160735 x 0.39	-41.1 (LC 135) b 160735 x 0.39	1.6 (LC 122) b 160070 x 0.42	-1.6 (LC 121) b 160119 x 0.42
60	internal stair (stringers, landing, treads) RHS 200x100x6; SHS 50x50x5; H = 5 mm; RHS 160x80x5 s	5.3 (LC 135) b 170307 x 0.32	-5.3 (LC 135) b 170377 x 0.32	15.4 (LC 121) b 170002 x 0.00	-26.2 (LC 120) b 170305 x 0.16	5.0 (LC 120) b 170328 x 0.00	-14.0 (LC 122) b 170137 x 0.00
61	internal stair landing posts SHS 100x100x5	0.3 (LC 121) b 180006 x 0.00	-0.3 (LC 122) b 180016 x 0.00	0.0 (LC 135) b 180011 x 0.40	-0.0 (LC 121) b 180007 x 0.00	0.0 (LC 135) b 180004 x 0.40	-0.0 (LC 122) b 180017 x 0.00
62	external stair ground to L1 H = 5 mm; SHS 80x80x6.3 external stair legs; RHS 160	7.1 (LC 135) b 190173 x 0.00	-6.8 (LC 135) b 190028 x 0.00	8.9 (LC 106) b 190373 x 0.00	-7.6 (LC 119) b 190022 x 0.44	6.2 (LC 135) b 190016 x 0.00	-7.5 (LC 135) b 190161 x 0.00



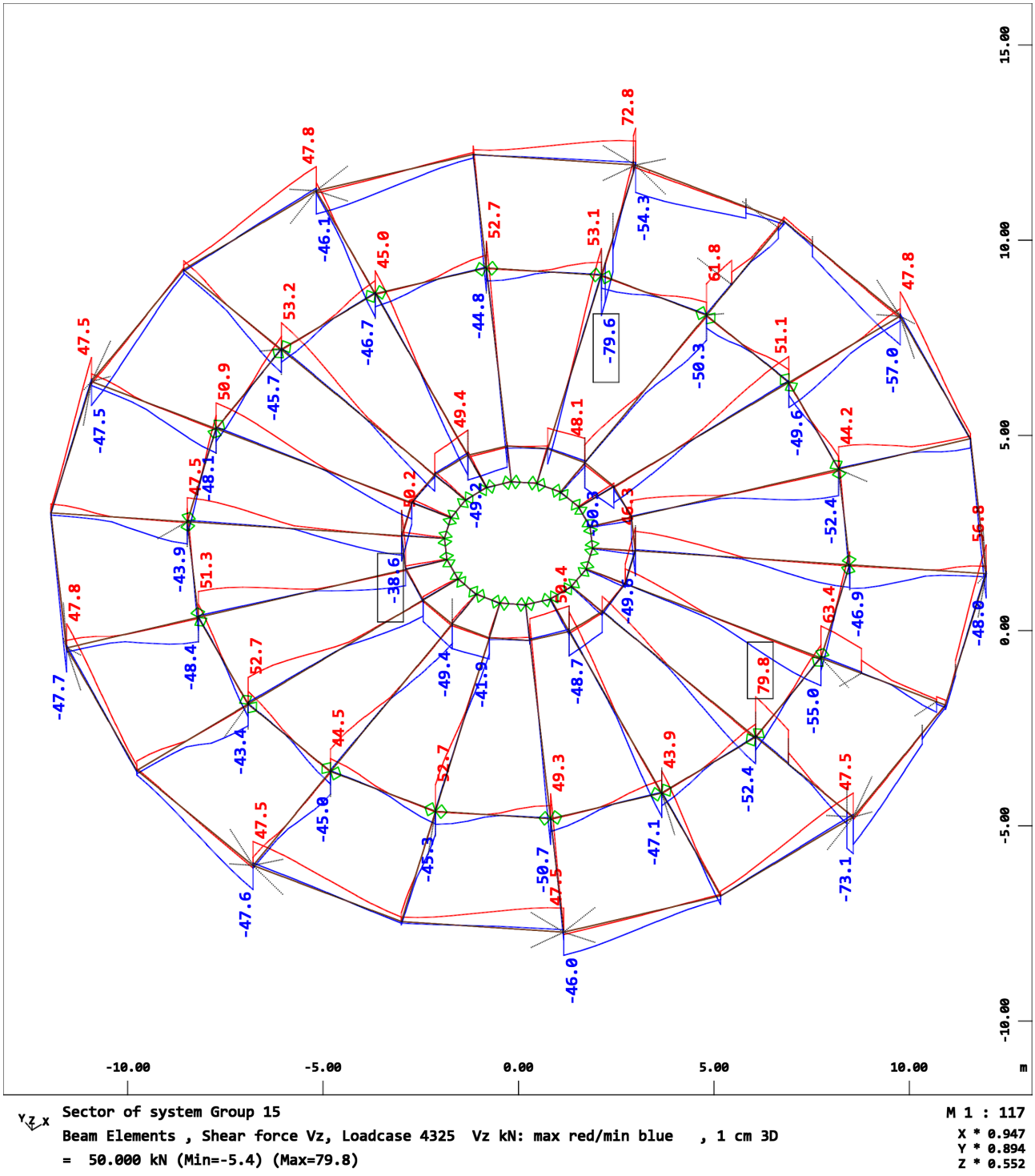
P00_key_beam_groups

Key: beam groups of model v21 (group number on each member, colour legend at top). Groups 1, 2, 6, 7 are shell (QUAD) groups and are not shown.



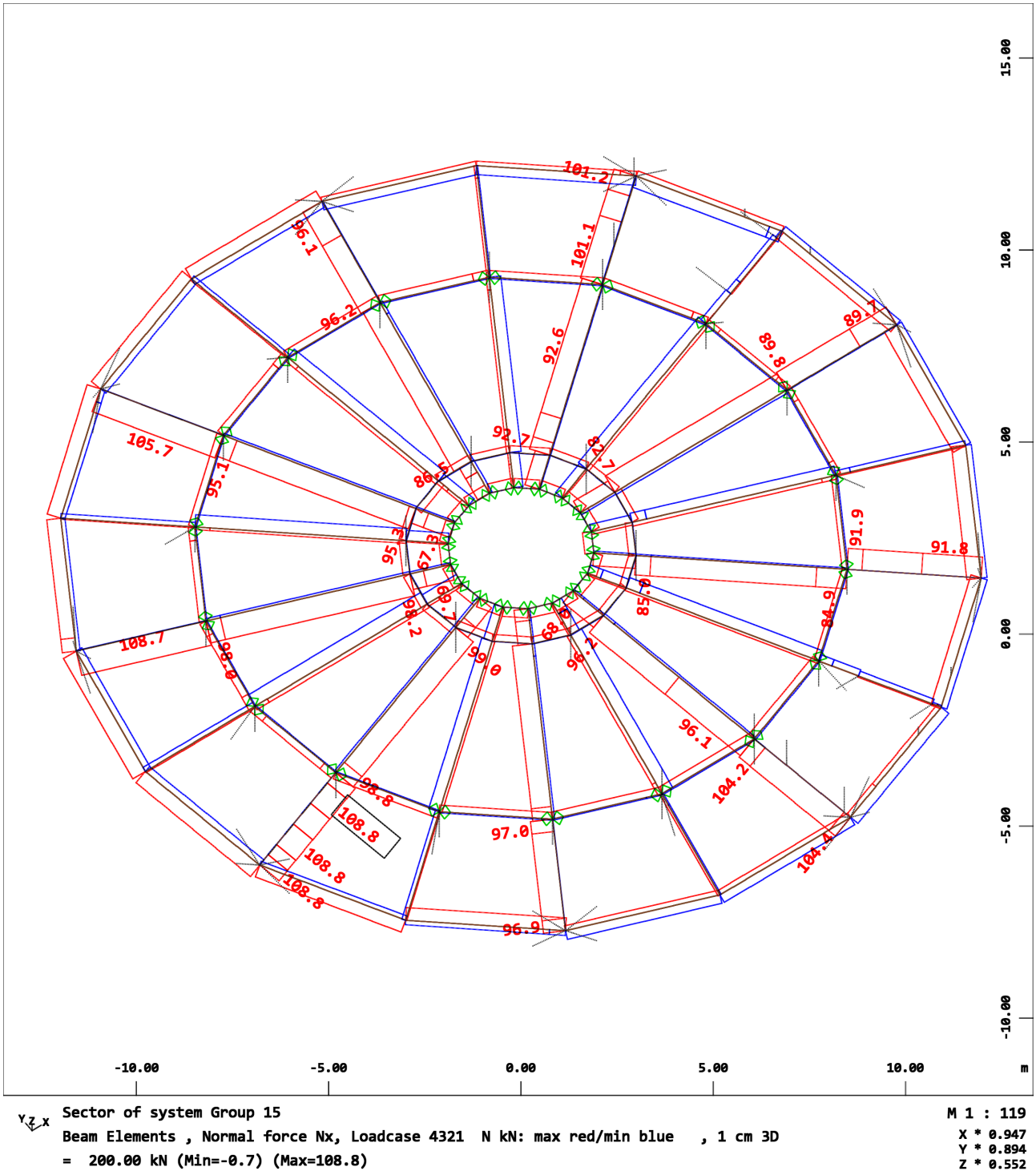
P01_L1_My

My [kNm], ULS envelope: max LC 4329 (red) and min LC 4330 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Level 1 framing (group 15) My. Diagram scale 1 cm = 50 kNm (on the 25.5 cm sheet); labels only where |value| > 40 kNm. Envelope extremes: max 48.6, min -89.11 kNm.



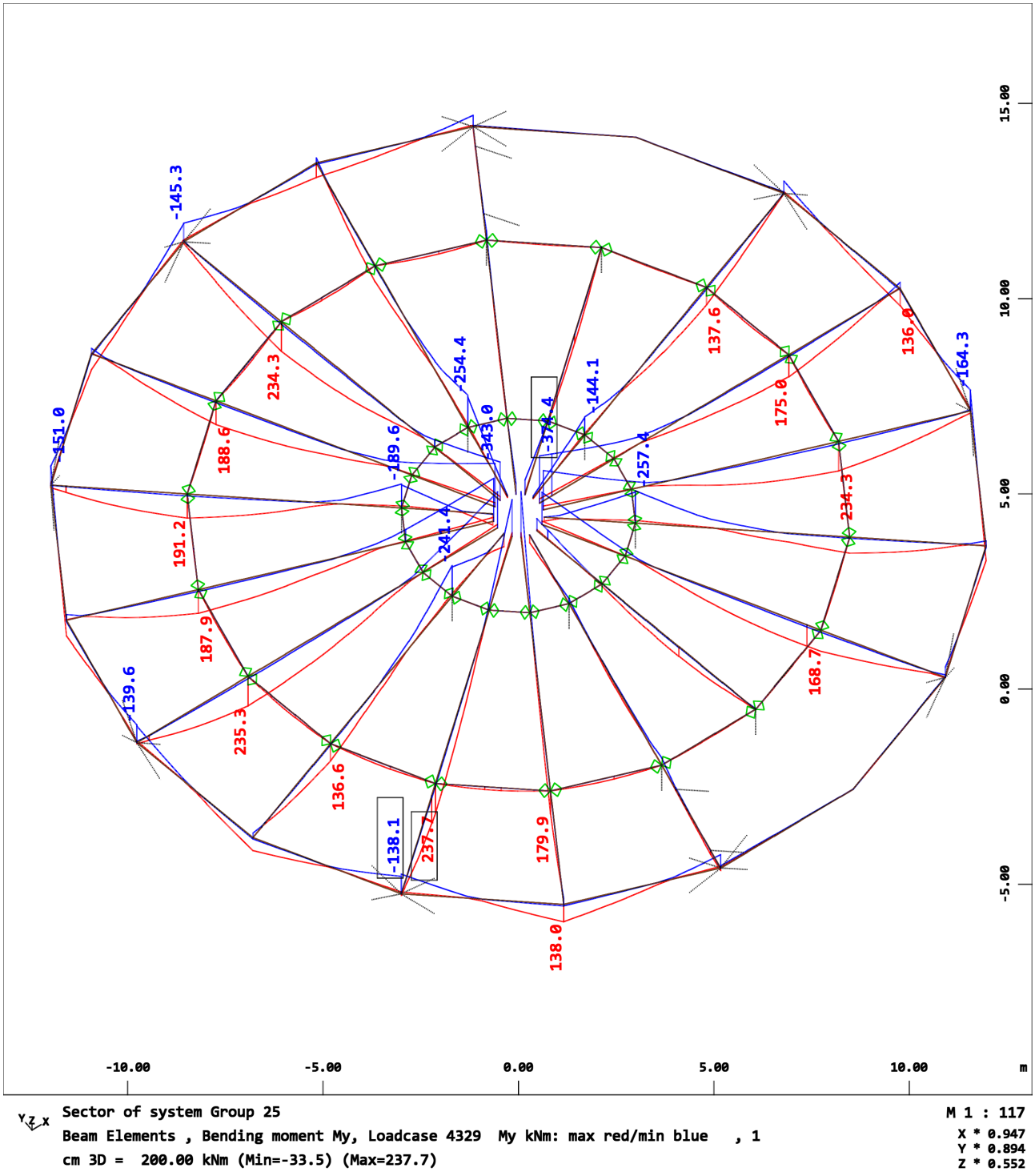
P02_L1_Vz

Vz [kN], ULS envelope: max LC 4325 (red) and min LC 4326 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Level 1 framing (group 15) Vz. Diagram scale 1 cm = 50 kN (on the 25.5 cm sheet); labels only where |value| > 36 kN. Envelope extremes: max 79.83, min -79.63 kN.



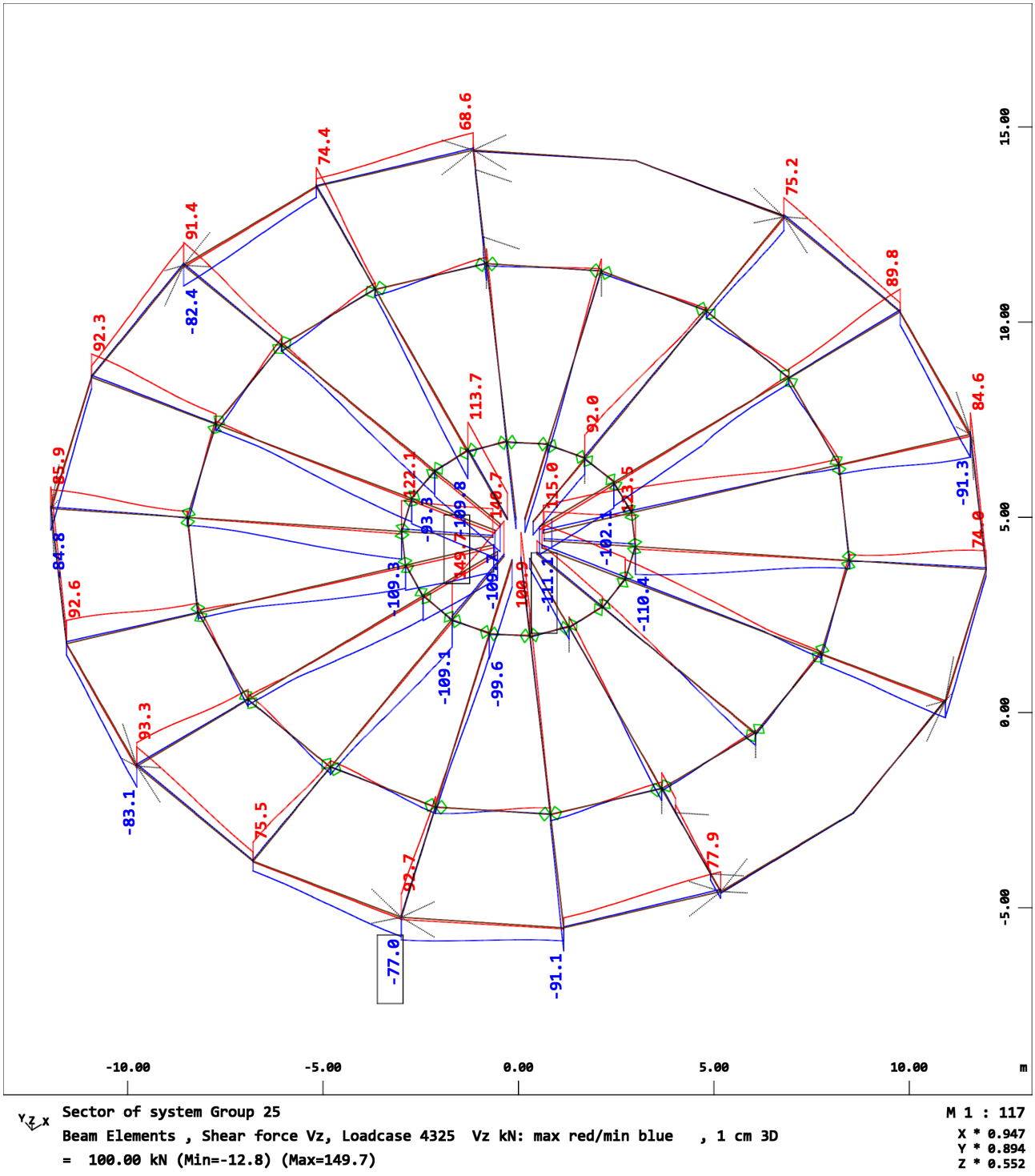
P03_L1_N

N [kN], ULS envelope: max LC 4321 (red) and min LC 4322 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Level 1 framing (group 15) N. Diagram scale 1 cm = 200 kN (on the 25.5 cm sheet); labels only where |value| > 65 kN. Envelope extremes: max 108.83, min -57.77 kN.



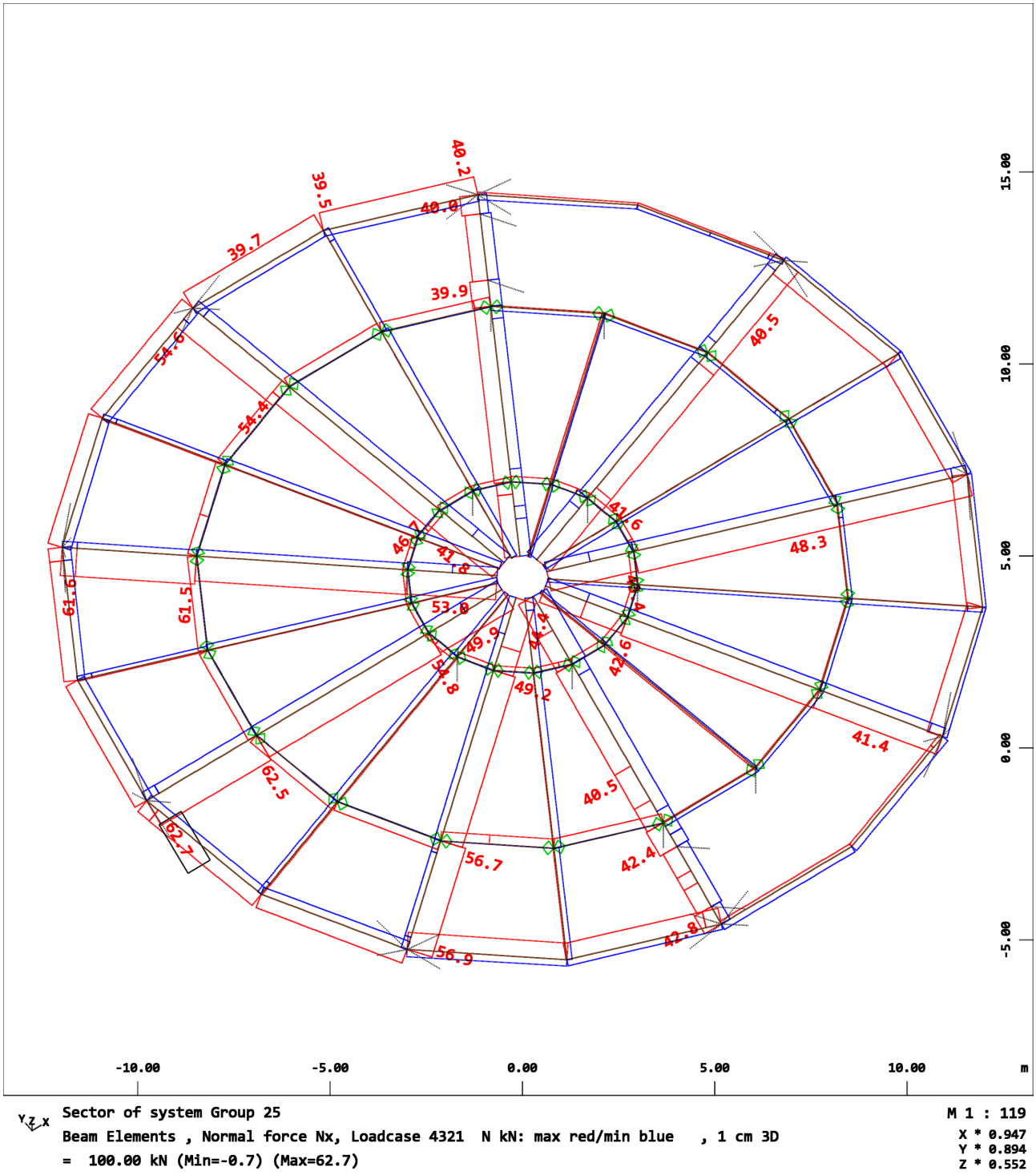
P04_L2_My

My [kNm], ULS envelope: max LC 4329 (red) and min LC 4330 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Level 2 framing (group 25) My. Diagram scale 1 cm = 200 kNm (on the 25.5 cm sheet); labels only where |value| > 131 kNm. Envelope extremes: max 237.68, min -374.44 kNm.



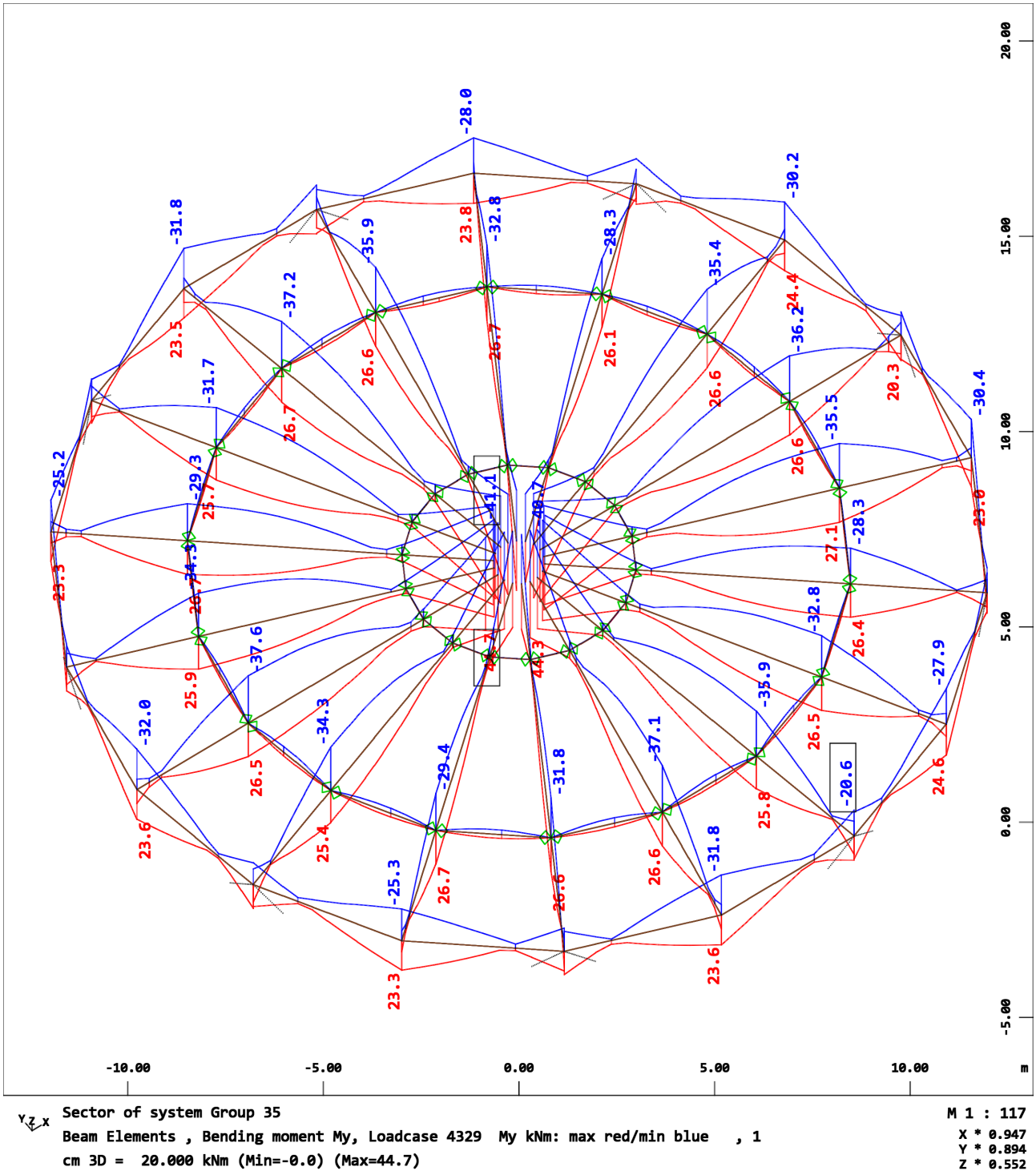
P05_L2_Vz

Vz [kN], ULS envelope: max LC 4325 (red) and min LC 4326 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Level 2 framing (group 25) Vz. Diagram scale 1 cm = 100 kN (on the 25.5 cm sheet); labels only where |value| > 67 kN. Envelope extremes: max 149.68, min -111.1 kN.



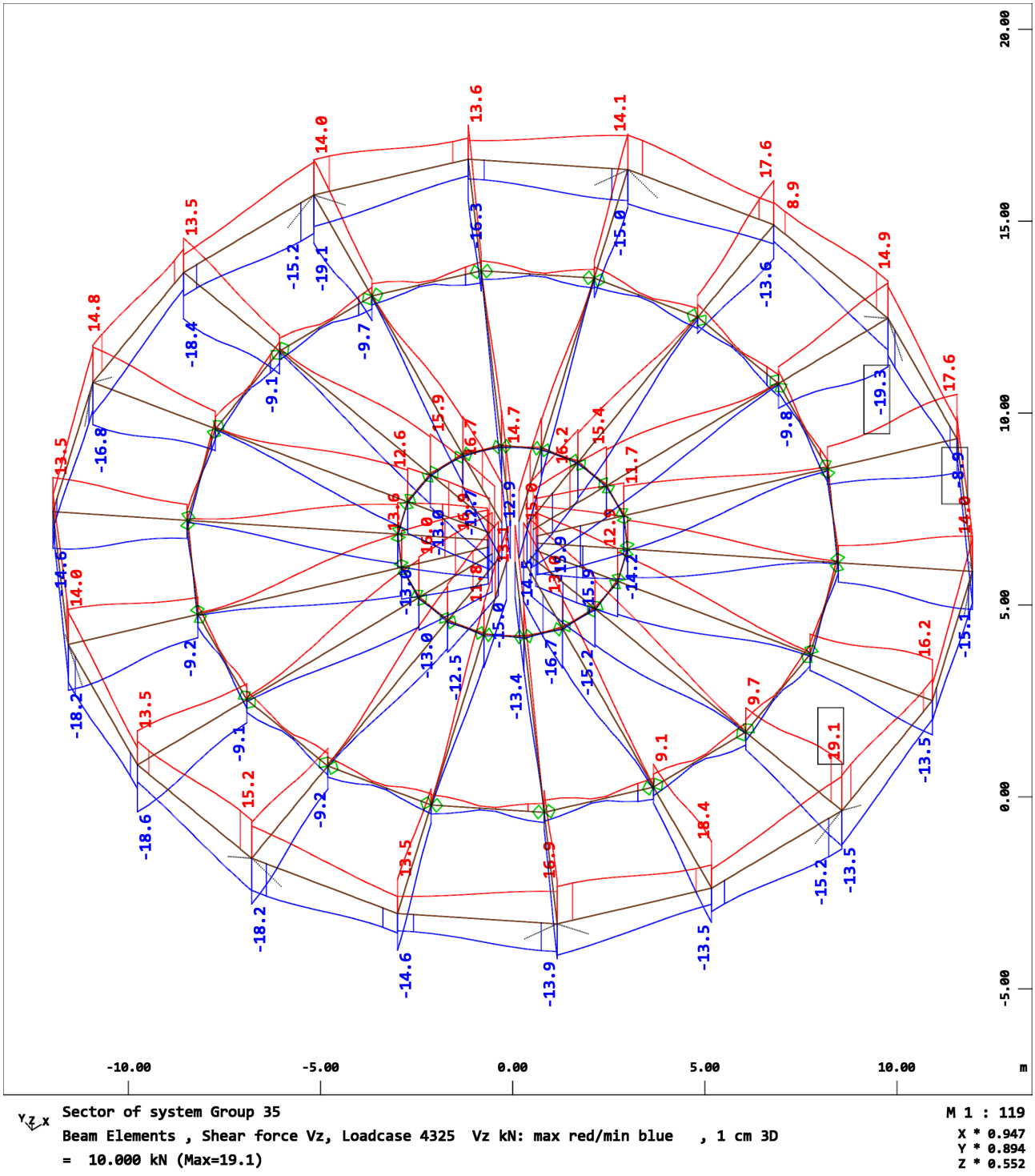
P06_L2_N

N [kN], ULS envelope: max LC 4321 (red) and min LC 4322 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Level 2 framing (group 25) N. Diagram scale 1 cm = 100 kN (on the 25.5 cm sheet); labels only where |value| > 38 kN. Envelope extremes: max 62.7, min -33.33 kN.



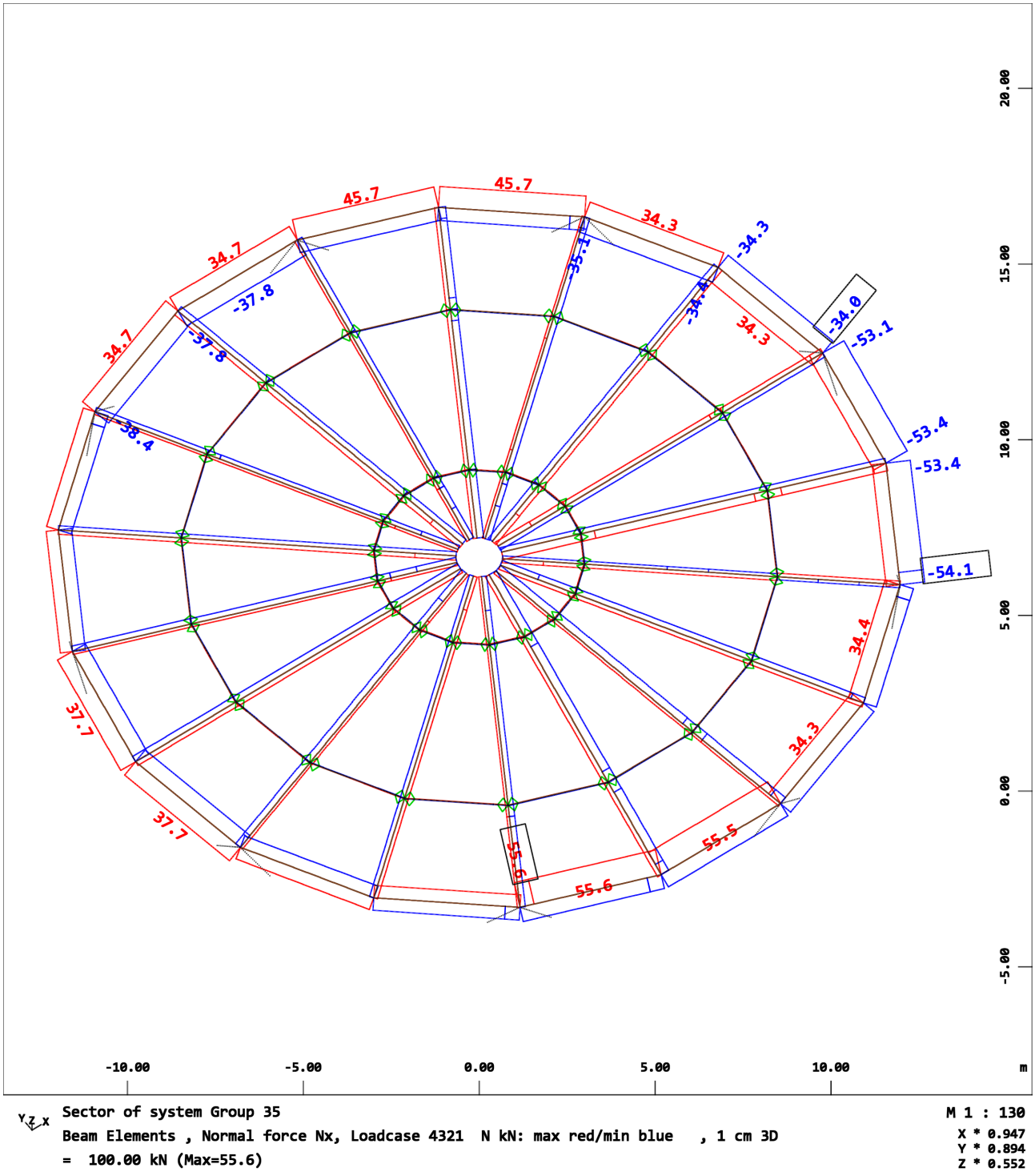
P07_RF_My

My [kNm], ULS envelope: max LC 4329 (red) and min LC 4330 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Roof framing (group 35) My. Diagram scale 1 cm = 20 kNm (on the 25.5 cm sheet); labels only where |value| > 20 kNm. Envelope extremes: max 44.67, min -41.13 kNm.



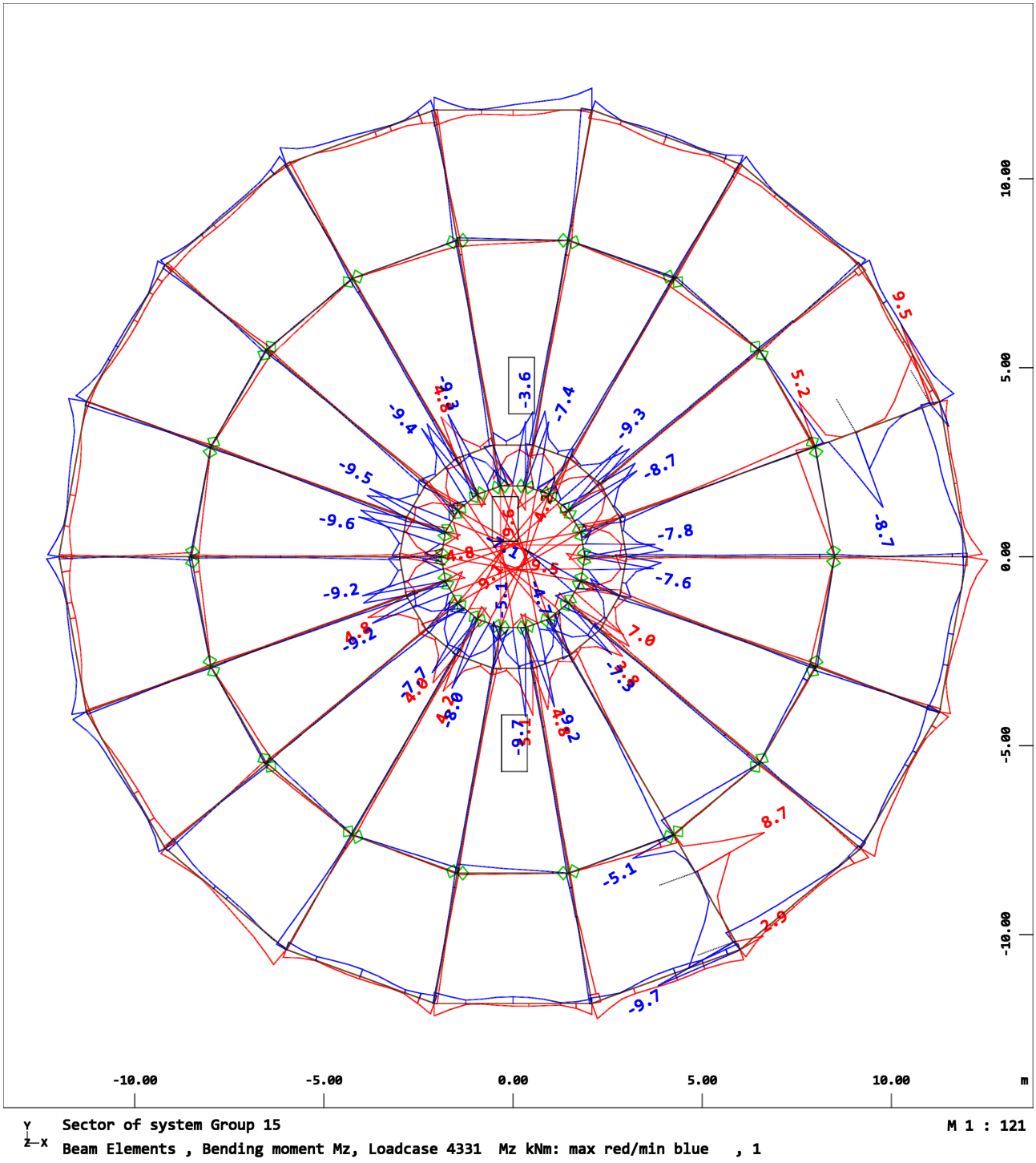
P08_RF_Vz

Vz [kN], ULS envelope: max LC 4325 (red) and min LC 4326 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Roof framing (group 35) Vz. Diagram scale 1 cm = 10 kN (on the 25.5 cm sheet); labels only where |value| > 8.7 kN. Envelope extremes: max 19.08, min -19.29 kN.



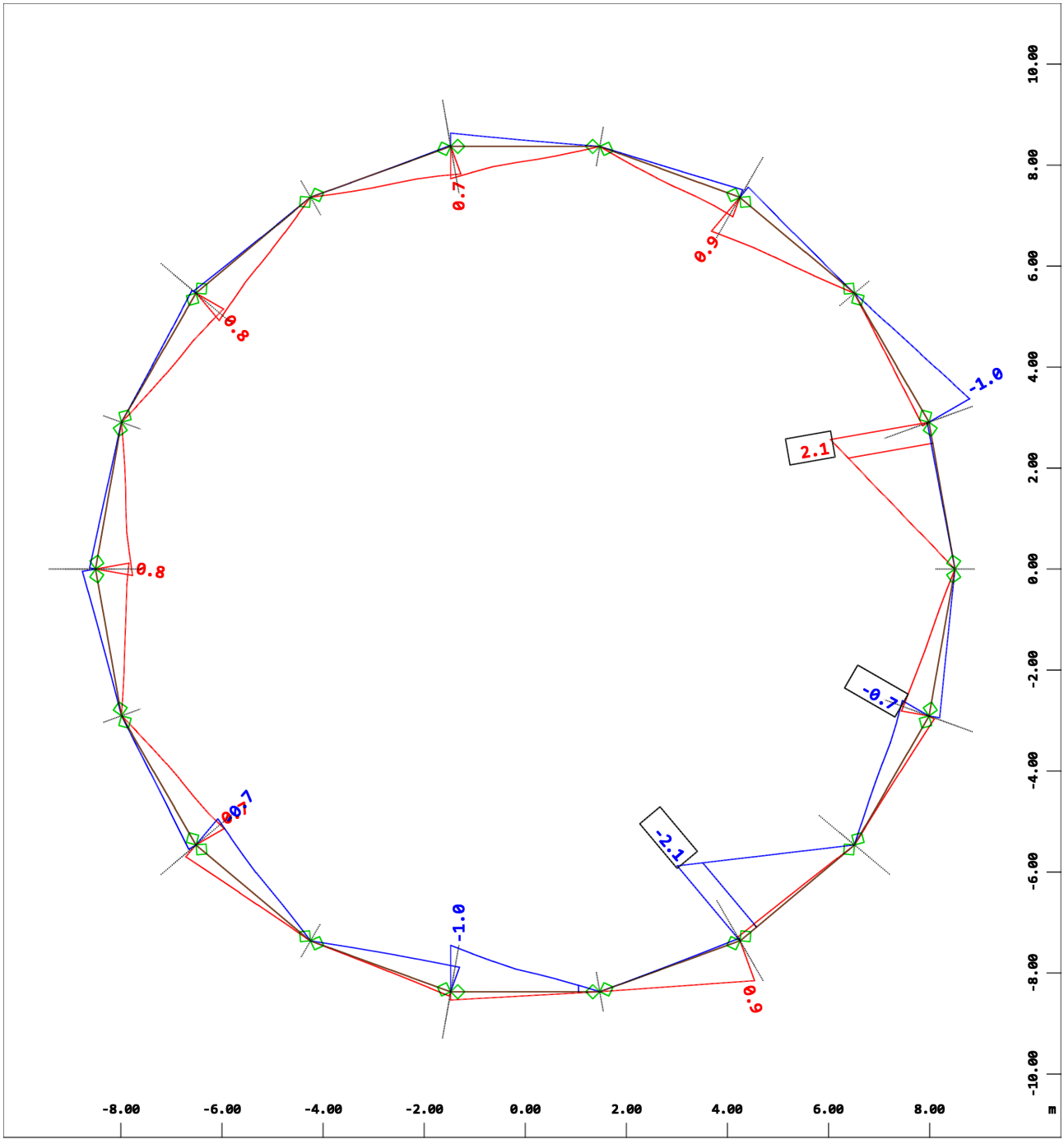
P09_RF_N

N [kN], ULS envelope: max LC 4321 (red) and min LC 4322 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Roof framing (group 35) N. Diagram scale 1 cm = 100 kN (on the 25.5 cm sheet); labels only where |value| > 33 kN. Envelope extremes: max 55.58, min -54.11 kN.



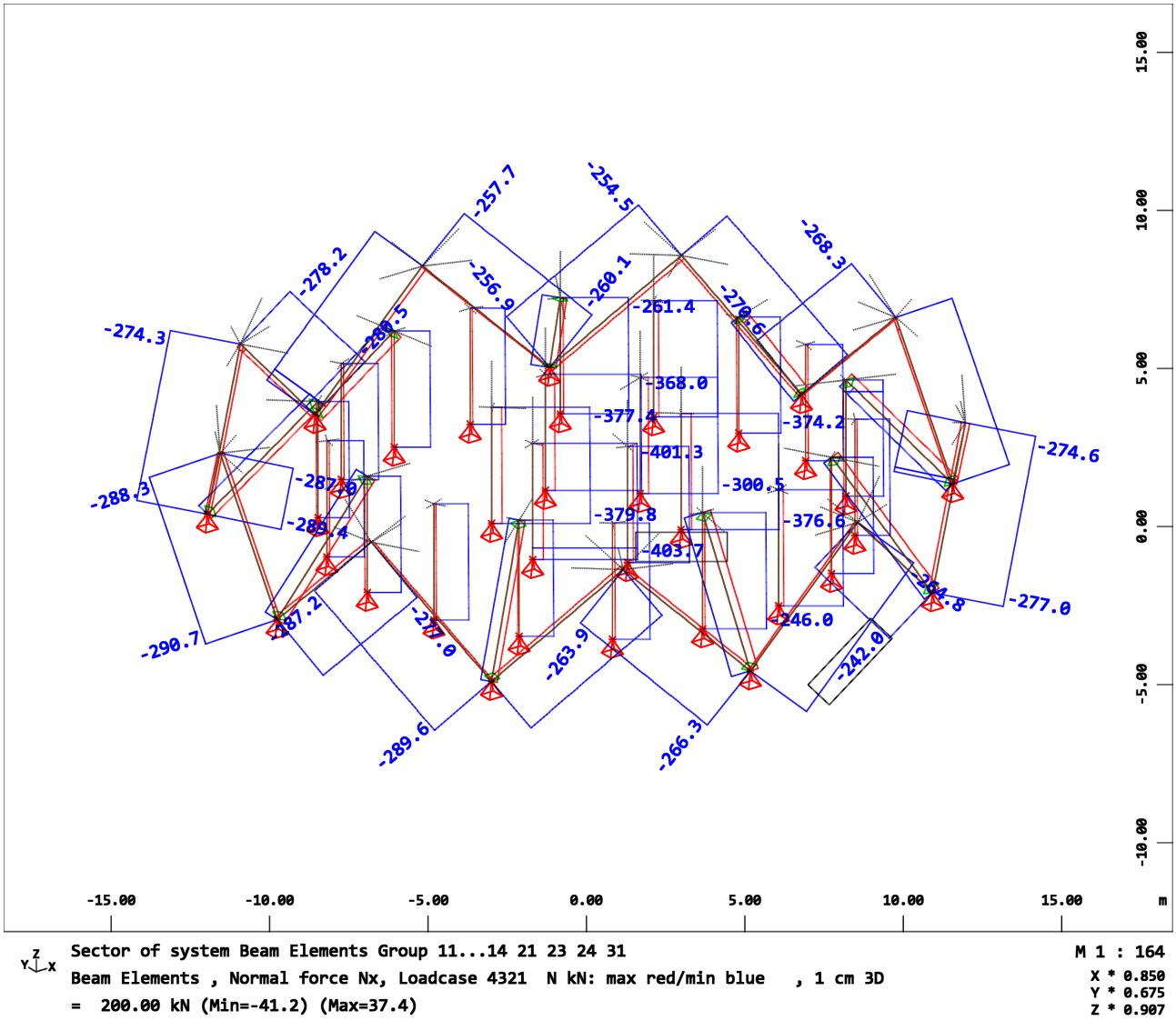
P10_L1_Mz_plan

Mz [kNm], ULS envelope: max LC 4331 (red) and min LC 4332 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Level 1 framing (group 15) Mz, plan. Diagram scale 1 cm = 5 kNm (on the 25.5 cm sheet); labels only where |value| > 2.9 kNm. Envelope extremes: max 9.61, min -9.75 kNm.



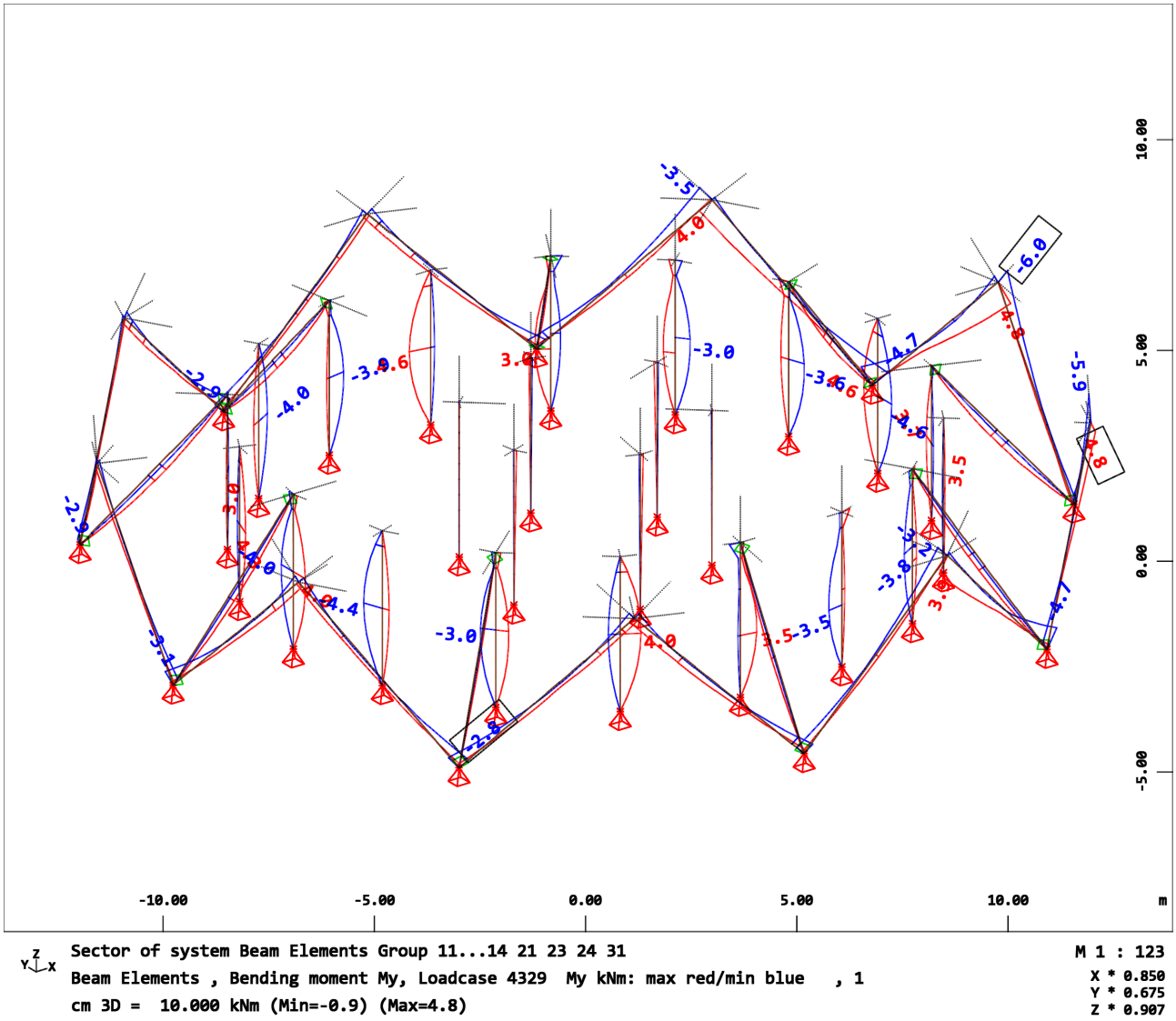
P11_L1_ring85_Mz_plan

Mz [kNm], ULS envelope: max LC 4331 (red) and min LC 4332 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Level 1 ring r 8.5 only (group 15) Mz, plan. Diagram scale 1 cm = 1 kNm (on the 25.5 cm sheet); labels only where |value| > 0.6 kNm. Envelope extremes: max 2.13, min -2.08 kNm.



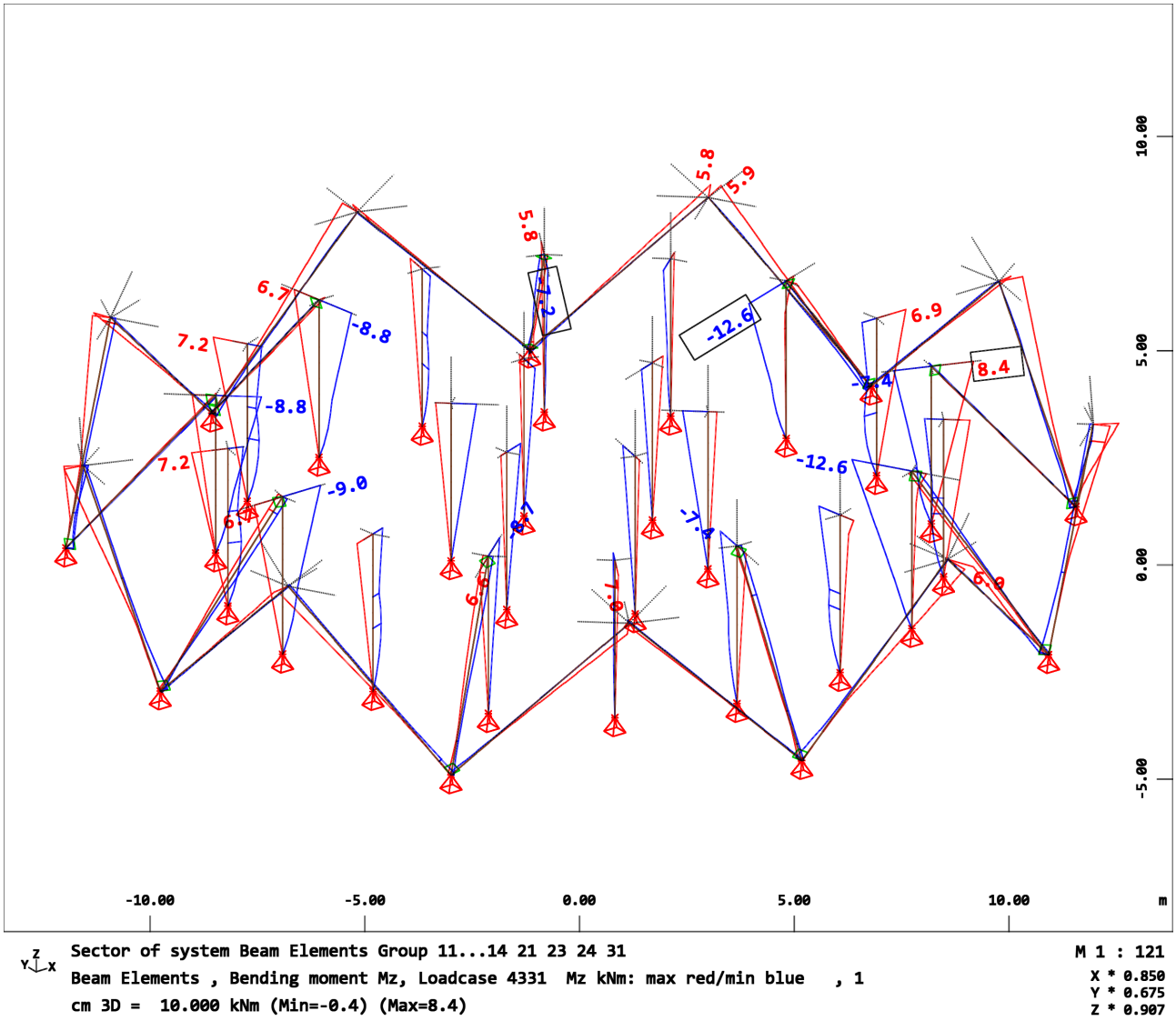
P12_V_G-L1_N

N [kN], ULS envelope: max LC 4321 (red) and min LC 4322 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers ground to Level 1 (groups 11, 12, 13, 14, 21, 23, 24, 31) N. Diagram scale 1 cm = 200 kN (on the 25.5 cm sheet); labels only where |value| > 242 kN. Envelope extremes: max 37.4, min -403.71 kN.



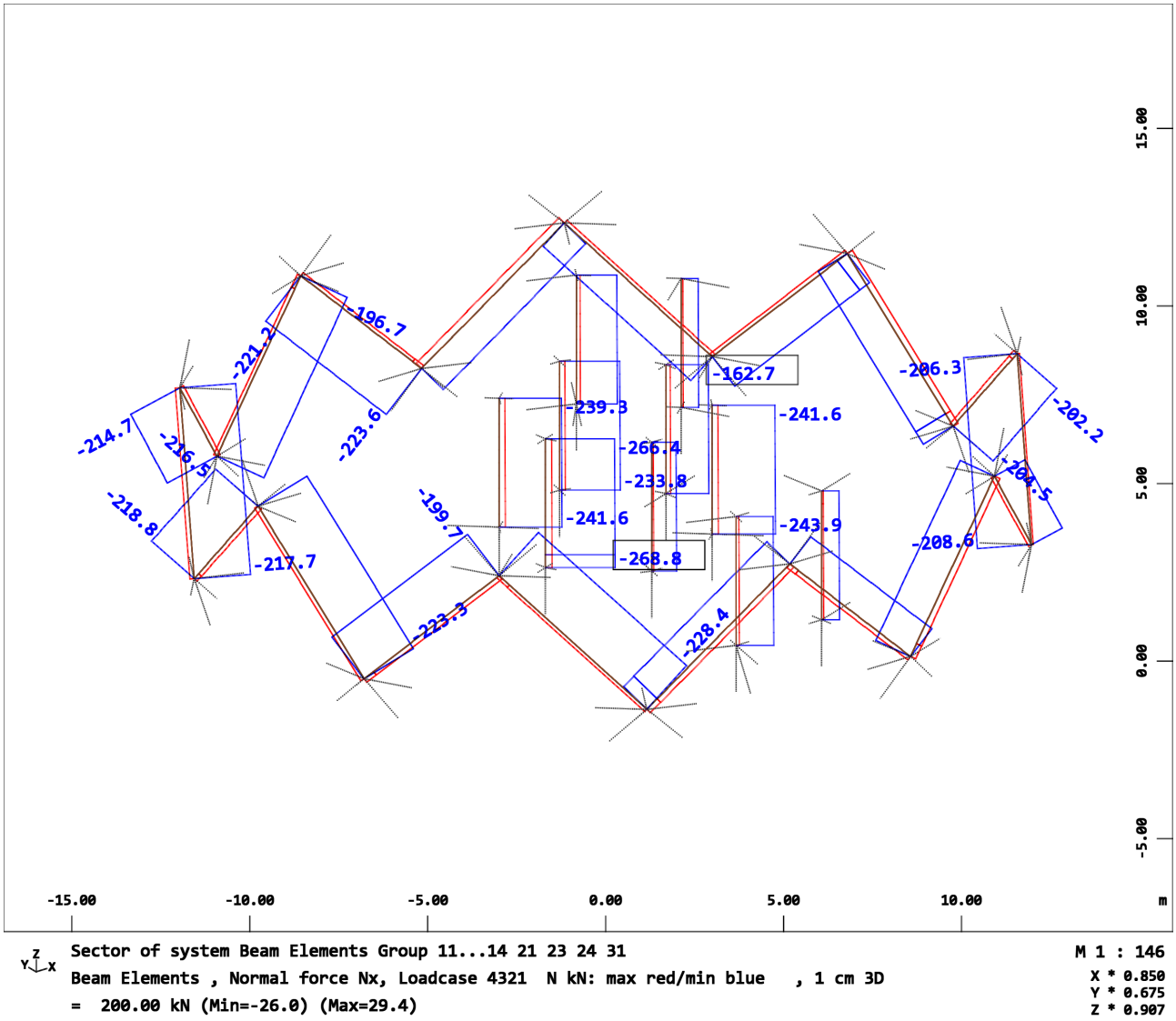
P13_V_G-L1_My

My [kNm], ULS envelope: max LC 4329 (red) and min LC 4330 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers ground to Level 1 (groups 11, 12, 13, 14, 21, 23, 24, 31) My. Diagram scale 1 cm = 10 kNm (on the 25.5 cm sheet); labels only where |value| > 2.7 kNm. Envelope extremes: max 4.84, min -6.04 kNm.



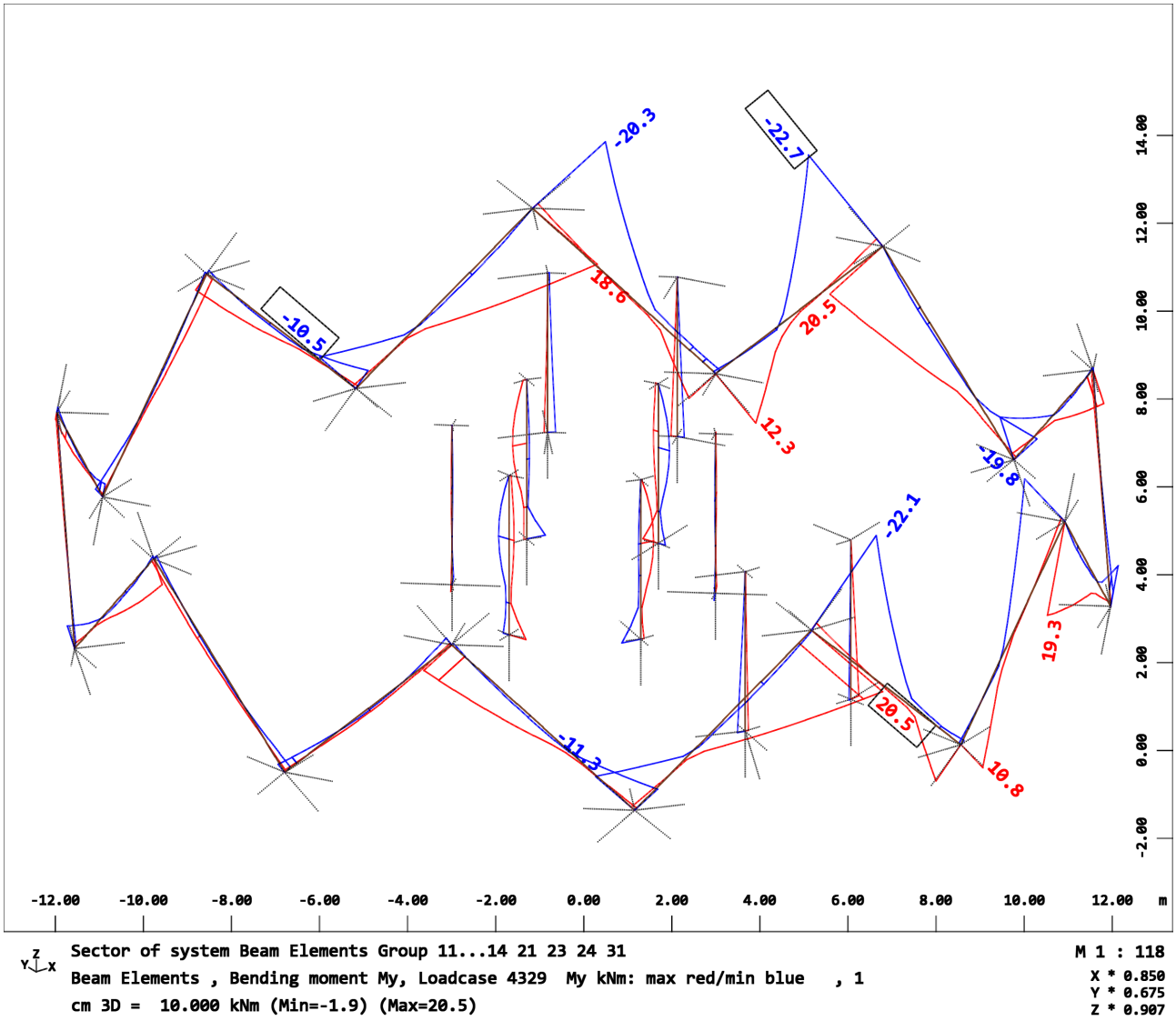
P14_V_G-L1_Mz

Mz [kNm], ULS envelope: max LC 4331 (red) and min LC 4332 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers ground to Level 1 (groups 11, 12, 13, 14, 21, 23, 24, 31) Mz. Diagram scale 1 cm = 10 kNm (on the 25.5 cm sheet); labels only where |value| > 5.7 kNm. Envelope extremes: max 8.44, min -12.63 kNm.



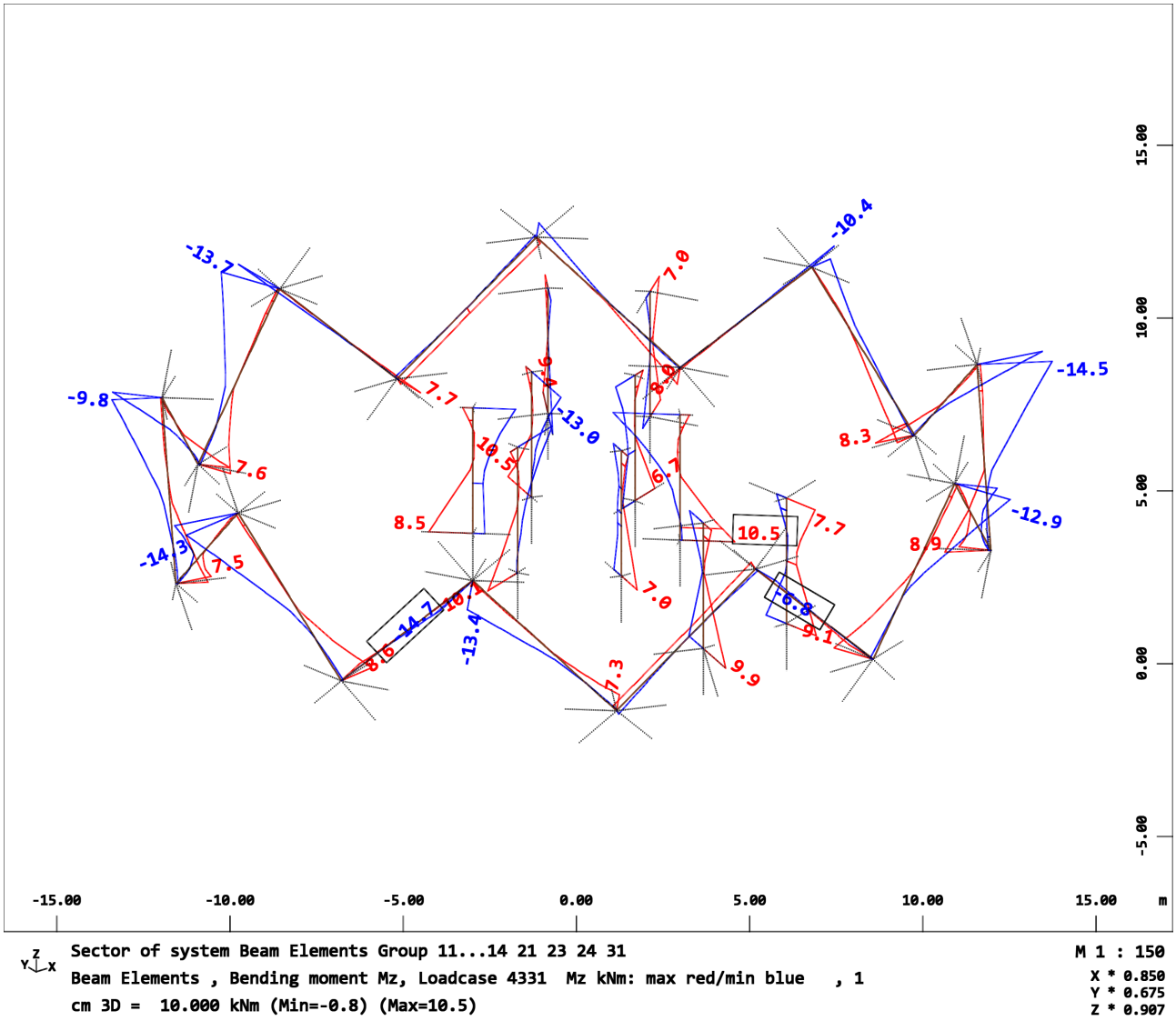
P15_V_L1-L2_N

N [kN], ULS envelope: max LC 4321 (red) and min LC 4322 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers Level 1 to Level 2 (groups 11, 12, 13, 14, 21, 23, 24, 31) N. Diagram scale 1 cm = 200 kN (on the 25.5 cm sheet); labels only where |value| > 161 kN. Envelope extremes: max 29.45, min -268.76 kN.



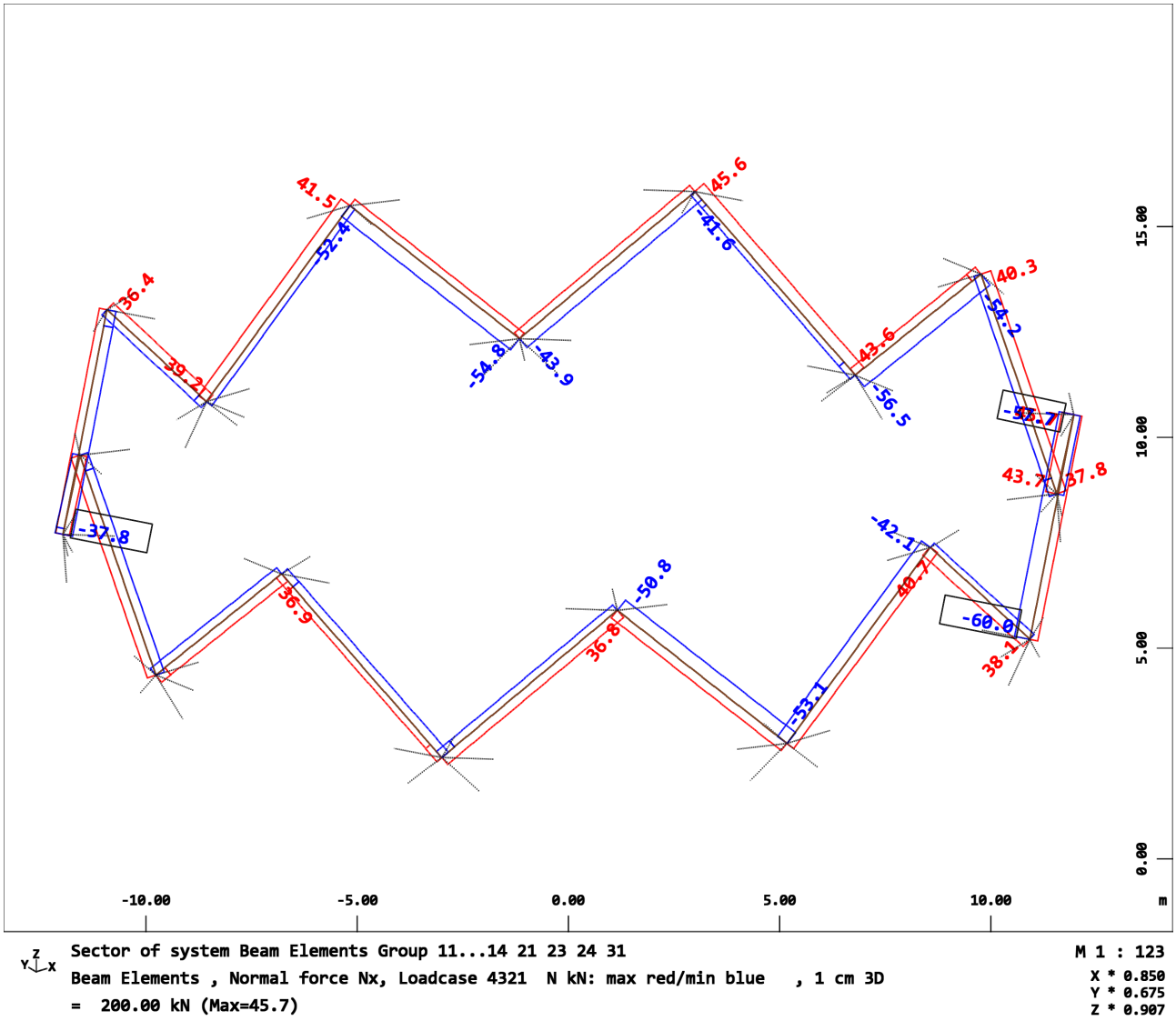
P16_V_L1-L2_My

My [kNm], ULS envelope: max LC 4329 (red) and min LC 4330 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers Level 1 to Level 2 (groups 11, 12, 13, 14, 21, 23, 24, 31) My. Diagram scale 1 cm = 10 kNm (on the 25.5 cm sheet); labels only where |value| > 10 kNm. Envelope extremes: max 20.53, min -22.7 kNm.



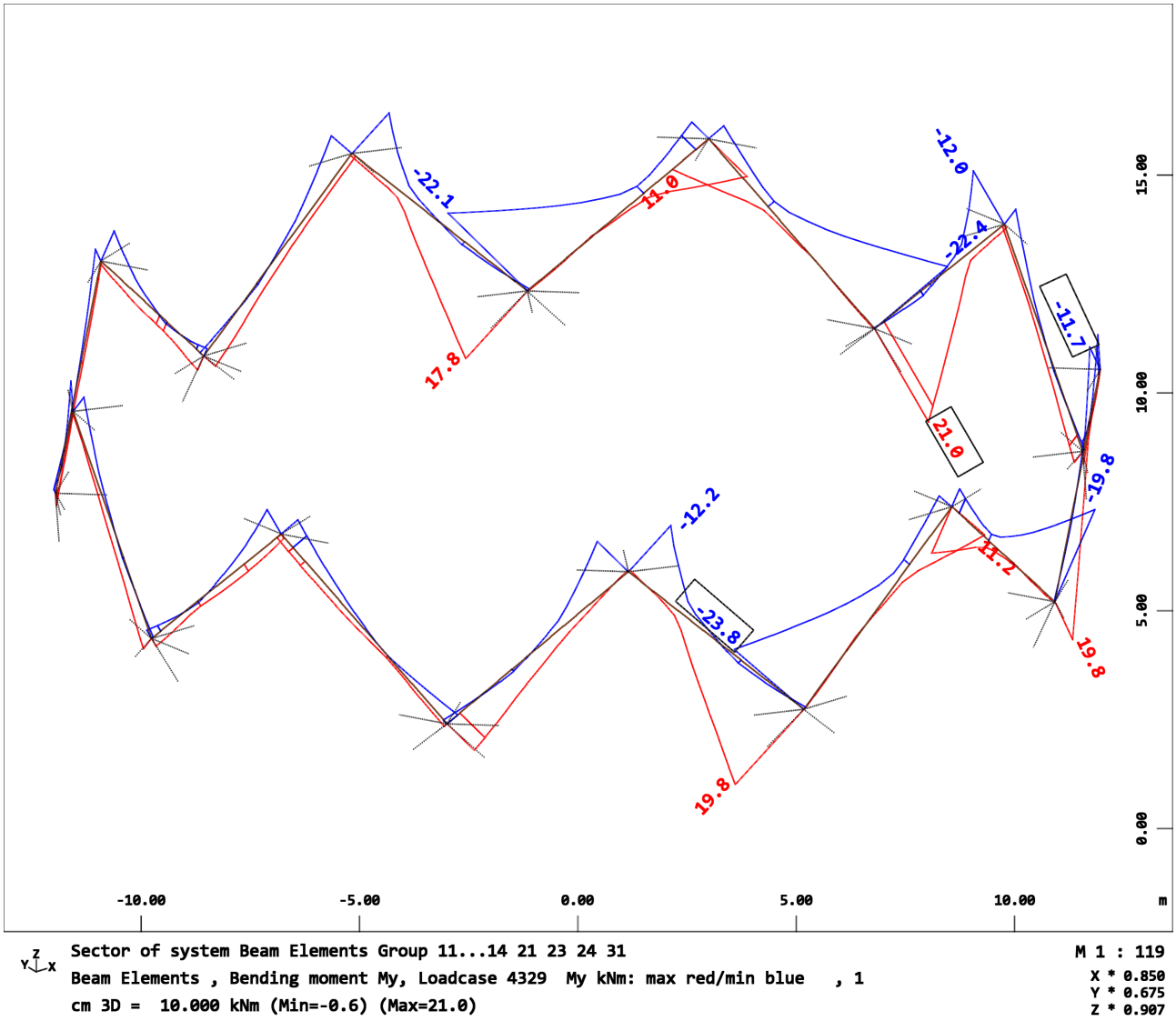
P17_V_L1-L2_Mz

Mz [kNm], ULS envelope: max LC 4331 (red) and min LC 4332 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers Level 1 to Level 2 (groups 11, 12, 13, 14, 21, 23, 24, 31) Mz. Diagram scale 1 cm = 10 kNm (on the 25.5 cm sheet); labels only where |value| > 6.6 kNm. Envelope extremes: max 10.51, min -14.67 kNm.



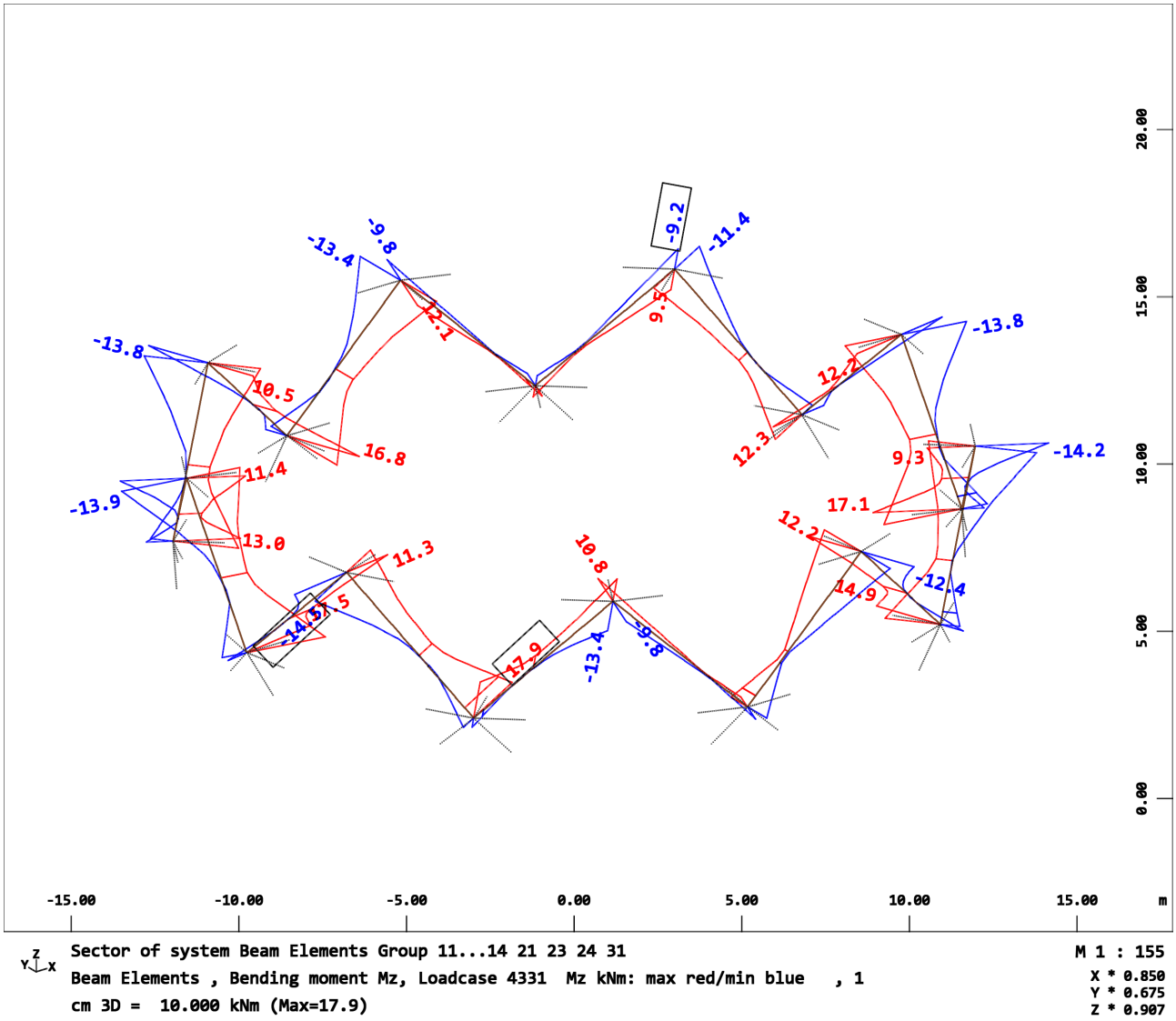
P18_V_L2-RF_N

N [kN], ULS envelope: max LC 4321 (red) and min LC 4322 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers Level 2 to roof (groups 11, 12, 13, 14, 21, 23, 24, 31) N. Diagram scale 1 cm = 200 kN (on the 25.5 cm sheet); labels only where |value| > 36 kN. Envelope extremes: max 45.7, min -60.05 kN.



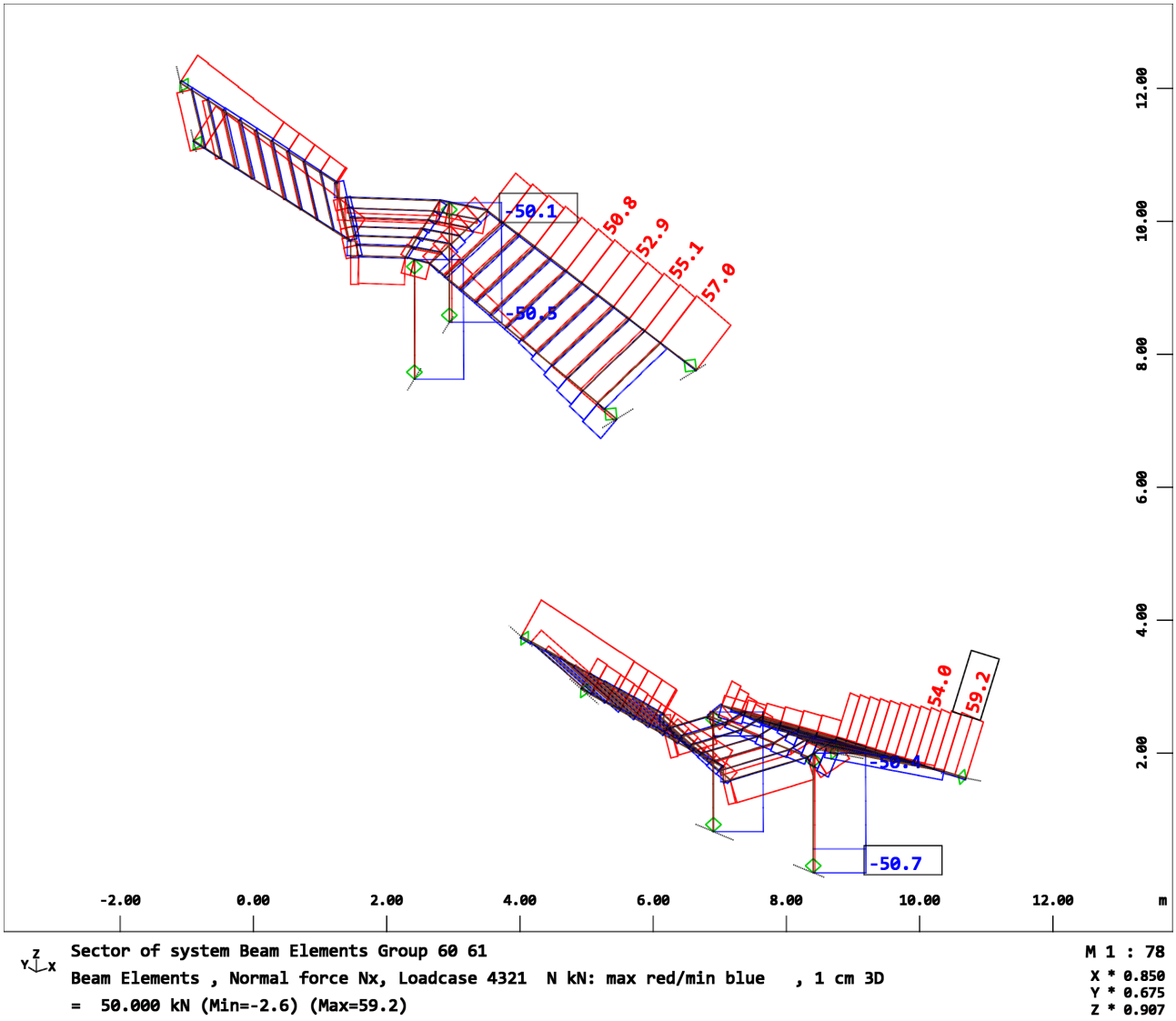
P19_V_L2-RF_My

My [kNm], ULS envelope: max LC 4329 (red) and min LC 4330 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers Level 2 to roof (groups 11, 12, 13, 14, 21, 23, 24, 31) My. Diagram scale 1 cm = 10 kNm (on the 25.5 cm sheet); labels only where |value| > 11 kNm. Envelope extremes: max 21, min -23.75 kNm.



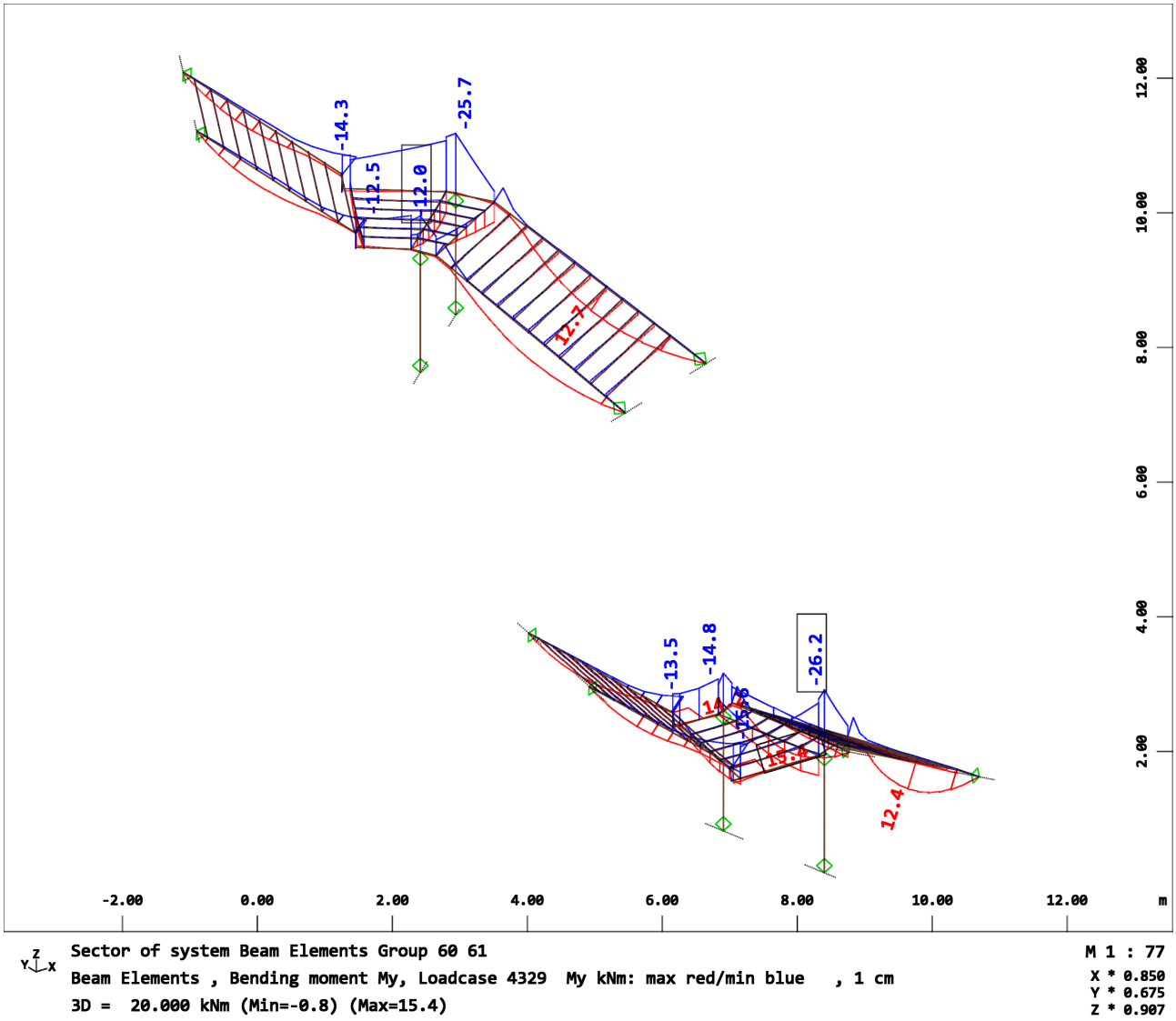
P20_V_L2-RF_Mz

Mz [kNm], ULS envelope: max LC 4331 (red) and min LC 4332 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers Level 2 to roof (groups 11, 12, 13, 14, 21, 23, 24, 31) Mz. Diagram scale 1 cm = 10 kNm (on the 25.5 cm sheet); labels only where |value| > 8.1 kNm. Envelope extremes: max 17.9, min -14.48 kNm.



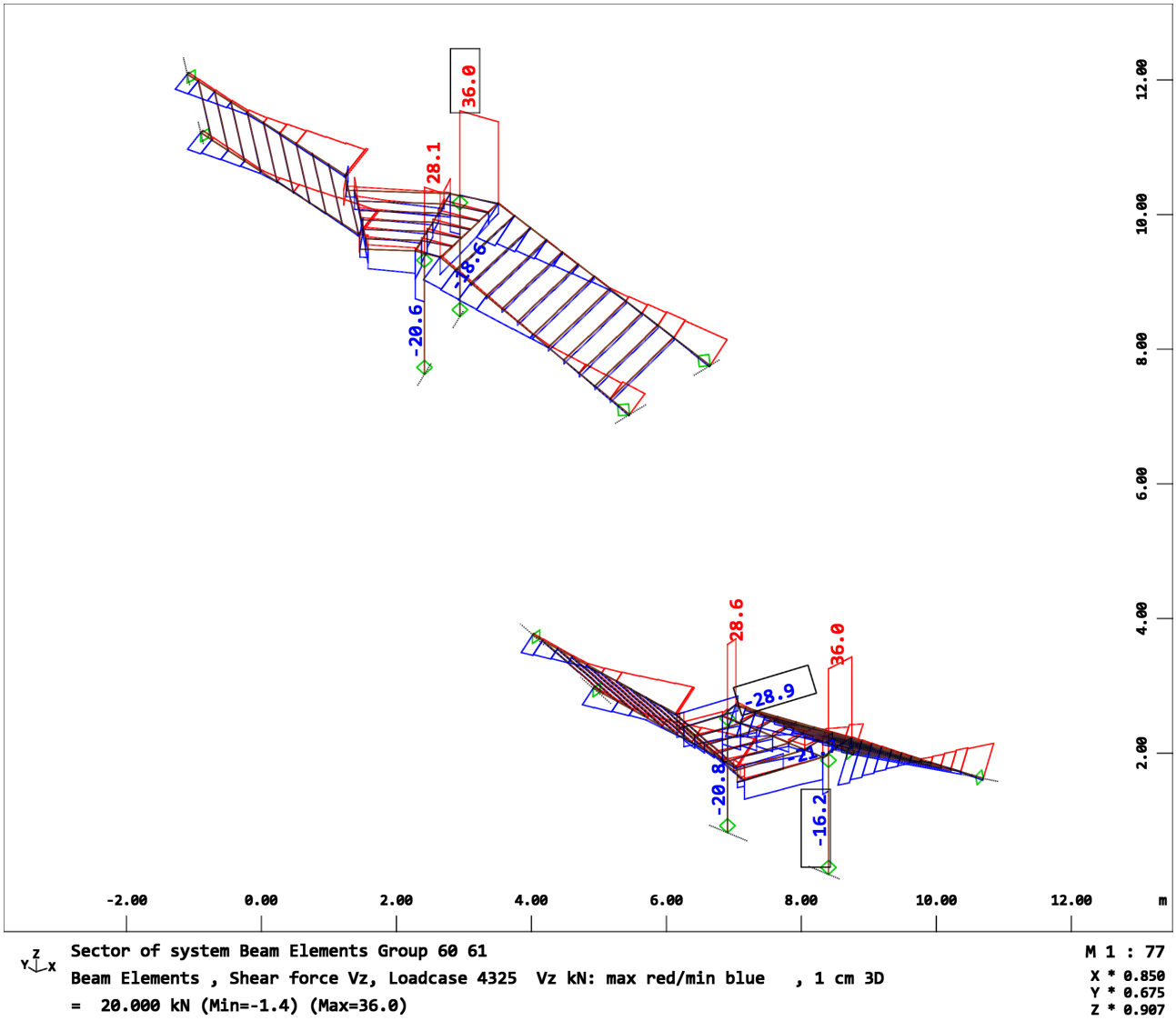
P21_S_int_N

N [kN], ULS envelope: max LC 4321 (red) and min LC 4322 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. internal stairs (groups 60, 61) N. Diagram scale 1 cm = 50 kN (on the 25.5 cm sheet); labels only where |value| > 50 kN. Envelope extremes: max 59.18, min -50.75 kN.



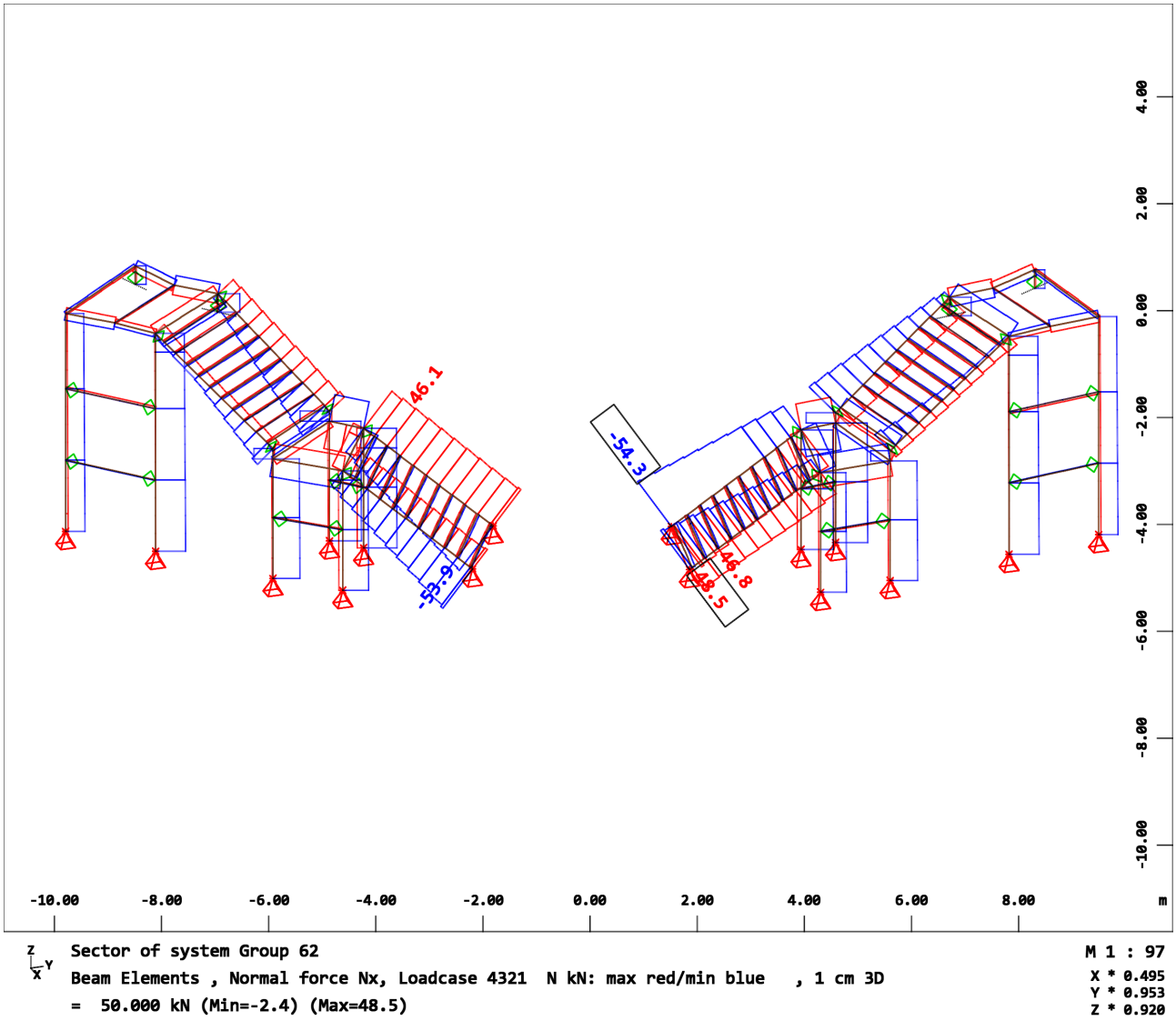
P22_S_int_My

My [kNm], ULS envelope: max LC 4329 (red) and min LC 4330 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. internal stairs (groups 60, 61) My. Diagram scale 1 cm = 20 kNm (on the 25.5 cm sheet); labels only where |value| > 12 kNm. Envelope extremes: max 15.42, min -26.23 kNm.



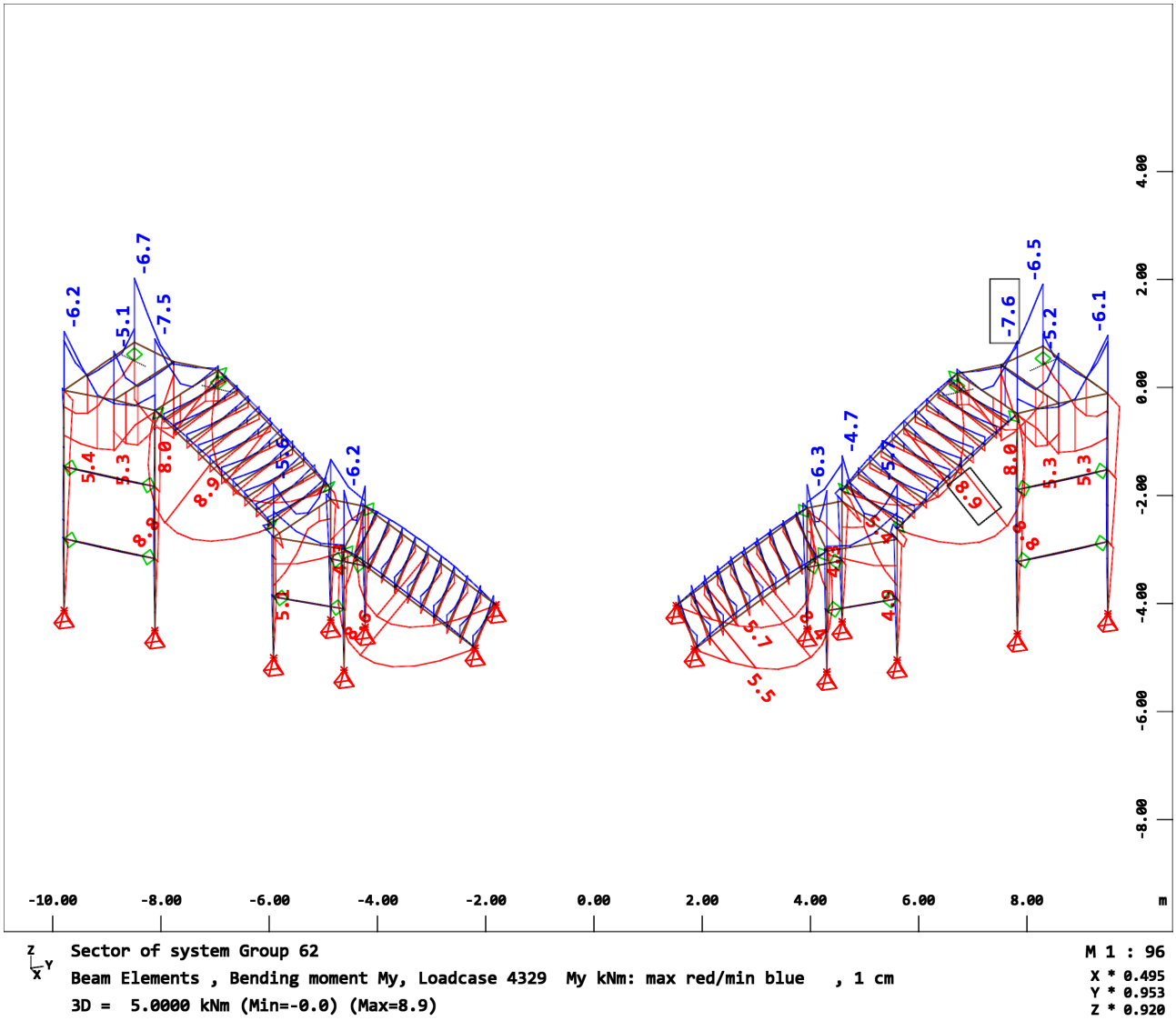
P23_S_int_Vz

Vz [kN], ULS envelope: max LC 4325 (red) and min LC 4326 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. internal stairs (groups 60, 61) Vz. Diagram scale 1 cm = 20 kN (on the 25.5 cm sheet); labels only where |value| > 16 kN. Envelope extremes: max 36.04, min -28.88 kN.



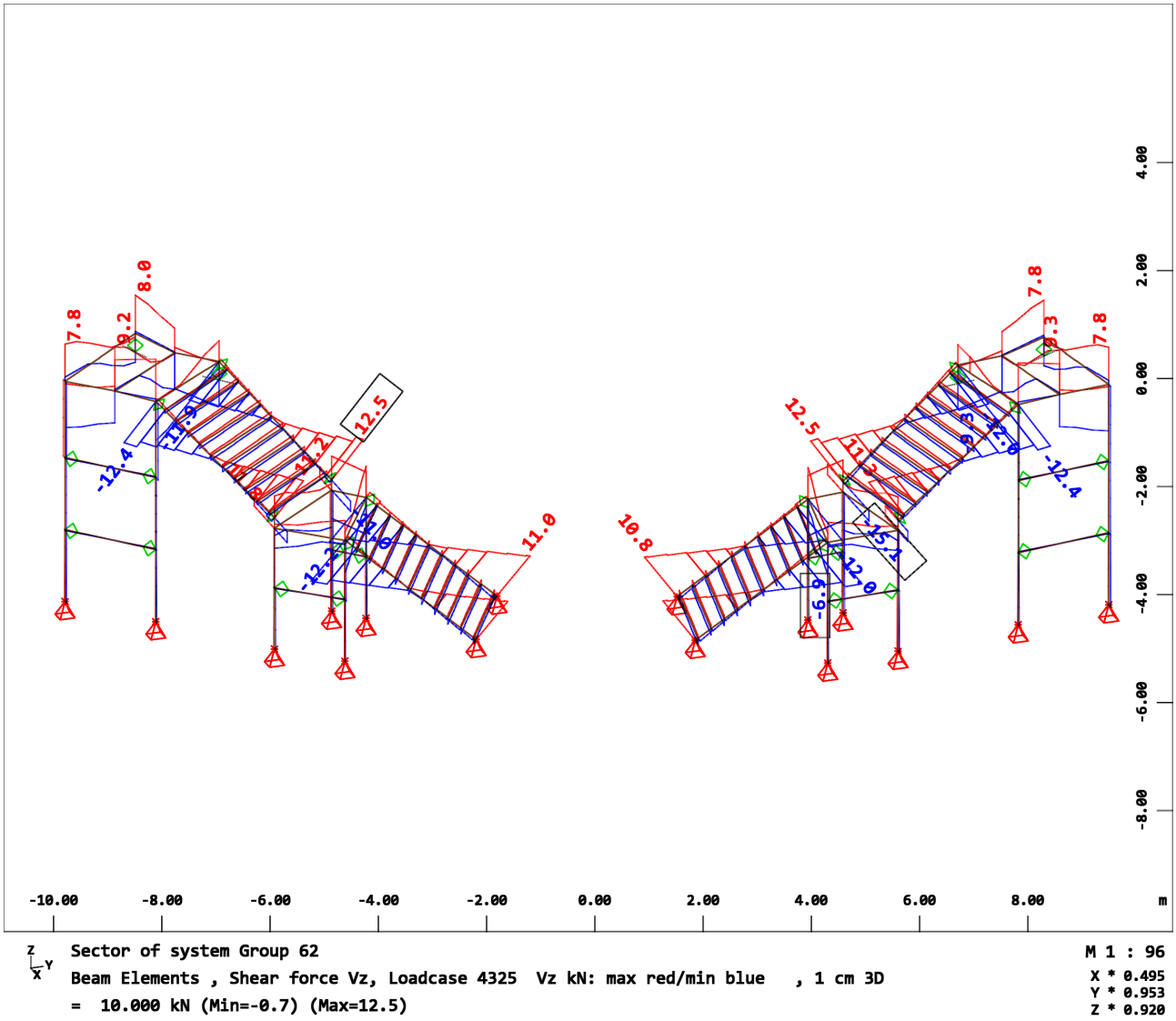
P24_S_ext_N

N [kN], ULS envelope: max LC 4321 (red) and min LC 4322 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. external stair (group 62) N. Diagram scale 1 cm = 50 kN (on the 25.5 cm sheet); labels only where |value| > 46 kN. Envelope extremes: max 48.5, min -54.34 kN.



P25_S_ext_My

My [kNm], ULS envelope: max LC 4329 (red) and min LC 4330 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. external stair (group 62) My. Diagram scale 1 cm = 5 kNm (on the 25.5 cm sheet); labels only where |value| > 4 kNm. Envelope extremes: max 8.93, min -7.58 kNm.



P26_S_ext_Vz

Vz [kN], ULS envelope: max LC 4325 (red) and min LC 4326 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. external stair (group 62) Vz. Diagram scale 1 cm = 10 kN (on the 25.5 cm sheet); labels only where |value| > 6.8 kN. Envelope extremes: max 12.54, min -15.12 kN.

26016 v21 kingpost - von Mises check of the shell model

Model 26016_check_v21 (run 13 Nov 2026), read only from the copy 26016_env_v21.cdb; nothing re-analysed. Reading picture: kingpost_reading_v21.png.

Kingpost shells, all t = 20 mm, S275 (fy = 275 MPa, t <= 40 mm, gamma_M0 = 1.0): tube wall groups 6 + 7 (r 0.305, z 0-12.2, about CHS 610 x 20, 12,117 elements); horizontal collar / cap plates group 1 (r <= 0.916; Level 2 z 7.81 and 8.30, roof z 11.91 and 12.20; 18,055 elements); vertical radial fins group 2 (r 0.304-0.506, 18 per level; 5,328 elements). Base: 20 support nodes on the tube end. Each Level 2 / roof radial beam end is tied by kinematic constraints to 4 plate nodes (288 tied nodes); no beam shares a node with the shells.

Method: ASE shell forces of the 43 ULS STR combinations (101-130, 135, 141-148, 151-154) and SLS 201 are in the cdb (key 210/LC). Per element: sigma = n/t +- 6 m/t^2 at top / bottom, mid = n/t; sigma_v = sqrt(sx^2 + sy^2 - sx sy + 3 txy^2). 'Element' = element-centre forces (averaged over the element); 'gauss' = largest Gauss point (raw). Scripts: scripts/kp_vonmises.py, scripts/kp_peaks.py.

Results, ULS (element centre, MPa, utilisation on 275)

zone	max sigma_v	util	element	z / r / angle	LC	surface
horizontal collar / cap plates, Level 2	609.3	2.216	23531	7.811 / 0.602 / 334.8	122	bottom
horizontal collar / cap plates, roof	156.6	0.57	17671	11.914 / 0.51 / 179.6	107	top
tube wall, Level 2 connection zone (z 7.4-8.7)	137.5	0.5	50812	7.615 / 0.305 / 239.1	122	bottom
tube wall, base zone (z 0-0.5)	34.3	0.125	41481	0.028 / 0.301 / 9.1	145	top
tube wall, between levels	45.9	0.167	42798	11.418 / 0.305 / 275.1	110	bottom
tube wall, roof connection zone (z > 11.5)	83.1	0.302	54454	11.708 / 0.305 / 259.4	110	bottom
vertical radial fin plates, Level 2	298.8	1.087	34073	7.803 / 0.498 / 240.1	122	top
vertical radial fin plates, roof	111.6	0.406	31557	11.906 / 0.498 / 280.0	110	top

Local peaks at the radial ties (ULS, element centre, excluding elements closer than the distance to a tied or support node)

zone	0 mm	50 mm	100 mm	150 mm
collar / cap plates, Level 2	609.3 (2.216)	360.2 (1.31)	360.2 (1.31)	160.1 (0.582)
collar / cap plates, roof	156.6 (0.569)	156.6 (0.569)	156.6 (0.569)	62.2 (0.226)
radial fin plates, Level 2	298.8 (1.087)	298.8 (1.087)	298.8 (1.087)	135.1 (0.491)
radial fin plates, roof	111.6 (0.406)	111.6 (0.406)	111.6 (0.406)	62.4 (0.227)
tube wall, Level 2 zone	137.5 (0.5)	137.5 (0.5)	137.5 (0.5)	137.5 (0.5)
tube wall, base	34.3 (0.125)	34.3 (0.125)	32.5 (0.118)	31.2 (0.113)
tube wall, between levels	45.9 (0.167)	45.9 (0.167)	45.9 (0.167)	45.9 (0.167)
tube wall, roof zone	83.1 (0.302)	83.1 (0.302)	83.1 (0.302)	83.1 (0.302)

Raw Gauss-point peak (Level 2 bottom collar, LC 122): see kingpost_von_mises_v21.json ('gauss' rows). SLS 201 for reference: Level 2 collar 415 MPa at the same tie, tube wall 94 MPa.

Reading and recommendation (for decision)

- All 87 elements above fy are on the Level 2 bottom collar plate (z 7.81), next to the 4-node ties of the radial ends (max 609 MPa, element 23531, LC 122, 12 mm from a tied node). There the plate moment (32 kNm/m) exceeds the plastic moment of a 20 mm plate (27.5 kNm/m) and the transverse shear is 1.8 MN/m: a point-load artefact of tying a 457 deep beam end to 4 nodes, not the real bolted / welded end connection.
- 50-100 mm from the ties the peak is 360 MPa at the fin tip at 240 deg (re-entrant corner of fin and collar, a shell-model singularity); 150 mm or more away the Level 2 collar is 160 MPa (0.58) and the fins 135 MPa (0.49).
- Tube wall: 138 MPa (0.50) at Level 2, 83 (0.30) at the roof, 46 between levels, 34 at the base. Roof plates 157 (0.57), roof fins 112 (0.41).
- Recommended for the report: tube wall max 0.50 and plates 0.58 (at >= 150 mm from the ties), with a note that the beam-to-kingpost load introduction is not represented by the 4-node ties and is checked separately (connection design / IDEA). Alternative if a more conservative check is wanted it conservative: refine the tie (couple the full beam end depth to the fins) and rerun.

Pictures (WinGRAF, MPa, see pictures.json)

WinGRAF cannot envelope SV over load cases, so each picture shows the governing combination of its zone: LC 122 (G+Q+psiW +Y down) for Level 2 and the whole kingpost, LC 110 for the roof, LC 145 for the base. The plots average SV to the nodes (legend max 383 MPa at Level 2 for LC 122, against 609 MPa at the element centre in the table).

- K1_kingpost_vonMises_LC122_iso.png: Kingpost, von Mises SV, whole height, isometric (LC 122, legend max 382.6 MPa) - whole kingpost; peak on the Level 2 bottom collar plate at the radial ties
- K2_kingpost_vonMises_LC122_opposite.png: Kingpost, von Mises SV, opposite side (LC 122, legend max 382.6 MPa) - same as K1 from the opposite side
- K3_L2_zone_vonMises_LC122.png: Level 2 connection zone (z 7.5-8.6), von Mises (LC 122, legend max 382.6 MPa) - collar plates z 7.81 / 8.30, fins, tube; hot spots at the 4-node radial ties on the bottom collar
- K3b_L2_bottom_collar_from_below_LC122.png: Level 2 bottom collar plate (z 7.81) seen from below (LC 122, legend max 382.6 MPa) - ring of hot spots at r 0.6-0.7 = the tied nodes of the 18 radial ends; highest near 240 / 340 deg
- K4_roof_zone_vonMises_LC110.png: Roof connection zone (z > 11.5), von Mises (LC 110, legend max 120.9 MPa) - roof plates and fins, max well below fy

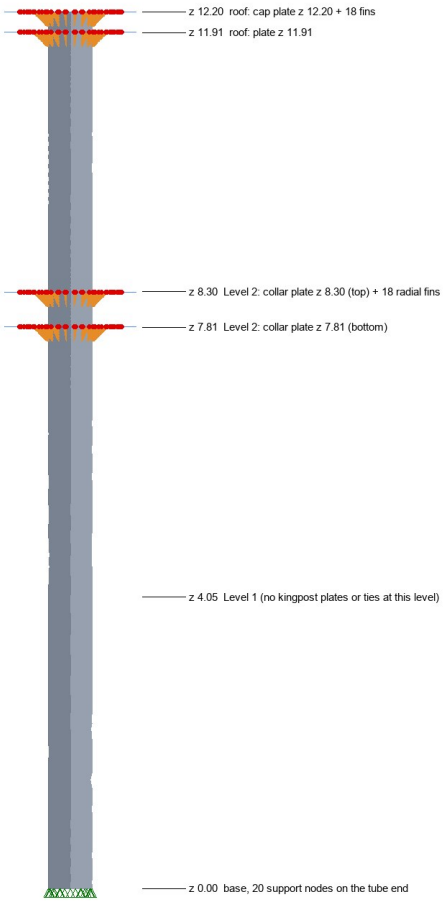
- K5_base_zone_vonMises_LC145.png: Base zone (z < 1.0), von Mises (LC 145, legend max 34.3 MPa) - tube wall at the 20 support nodes, low stress
- K6_L2_zone_deformed_vonMises_LC122.png: Level 2 zone, deformed shape x50 with von Mises (LC 122, legend max 382.6 MPa) - governing LC 122, local plate bending of the bottom collar under the radial ties

Recipe: 26016_kp_wing.dat (WinGRAF on the copy cdb, PLBCONVERTER to PNG).

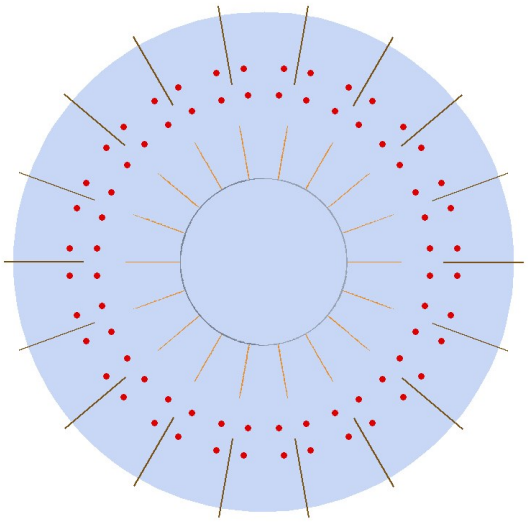
Kingpost shell model as read from the cdb

26016 v21 kingpost: shell model as read from the cdb (to be confirmed)

ELEVATION (x horizontal, z up; 1 m = 100 px)



PLAN, Level 2 (z 7.6-8.4): collar plates, fins, tube



tube wall, groups 6 + 7: r 0.305, t 20, z 0-12.2 (about CHS 610 x 20), 12,117 elements

horizontal collar / cap plates, group 1: r <= 0.916, t 20, Level 2 z 7.81 / 8.30, roof z 11.91 / 12.20

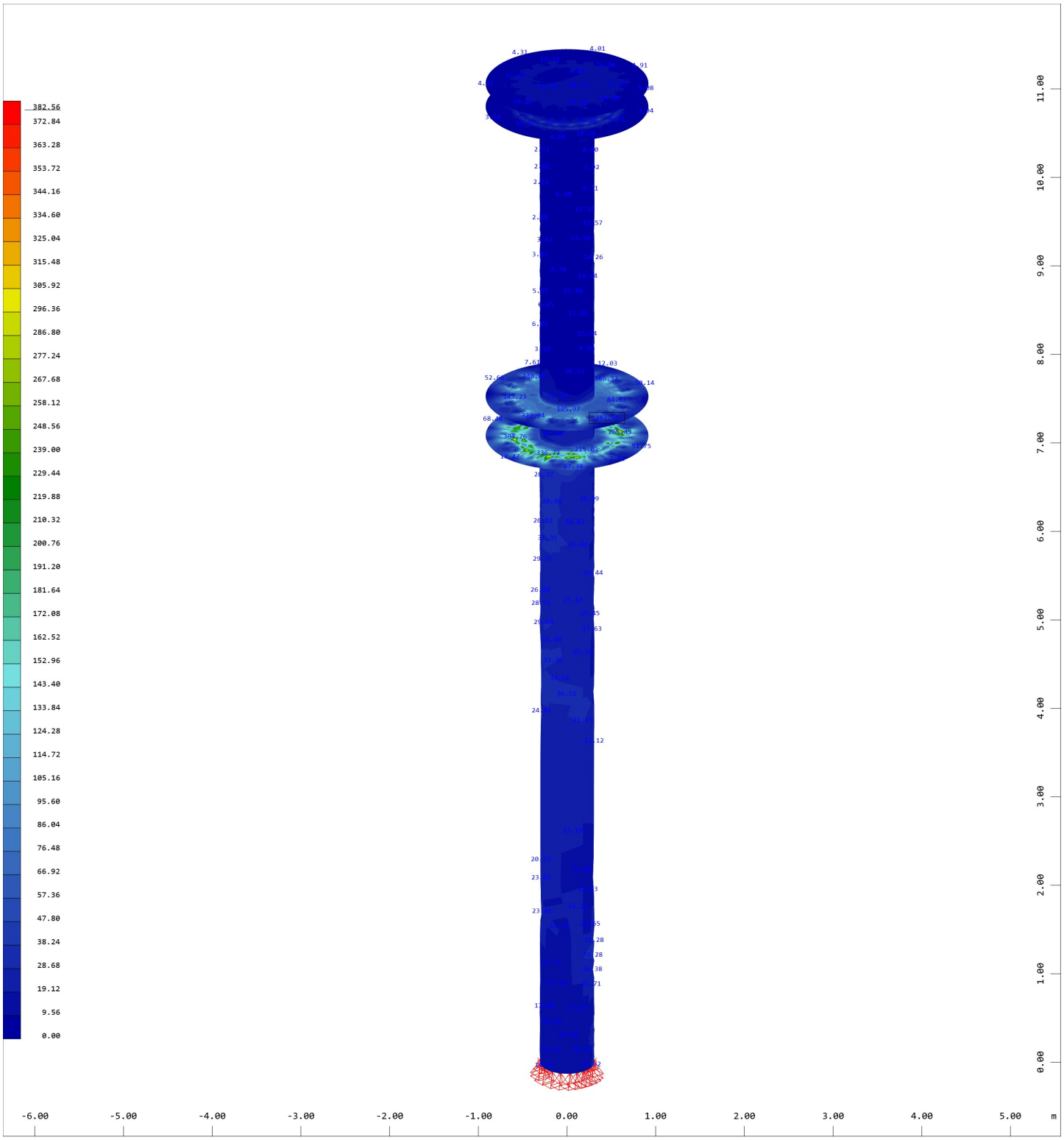
vertical radial fin plates, group 2: r 0.304-0.506, t 20, 18 per level (every 20 deg)

red dots: shell nodes tied by kinematic constraints to a radial beam end (4 nodes per beam end, 288 in total)

brown lines (plan): radial beams, ending at r 0.66 (beam end node at the beam axis, tied to the 4 plate nodes)

Material S275 (MNO 3), fy = 275 MPa for t = 20 <= 40 mm, gamma_M0 = 1.0.
No beam shares a node with the shells: each radial end is tied to 4 nodes
(point-like load introduction; the source of the local peaks).

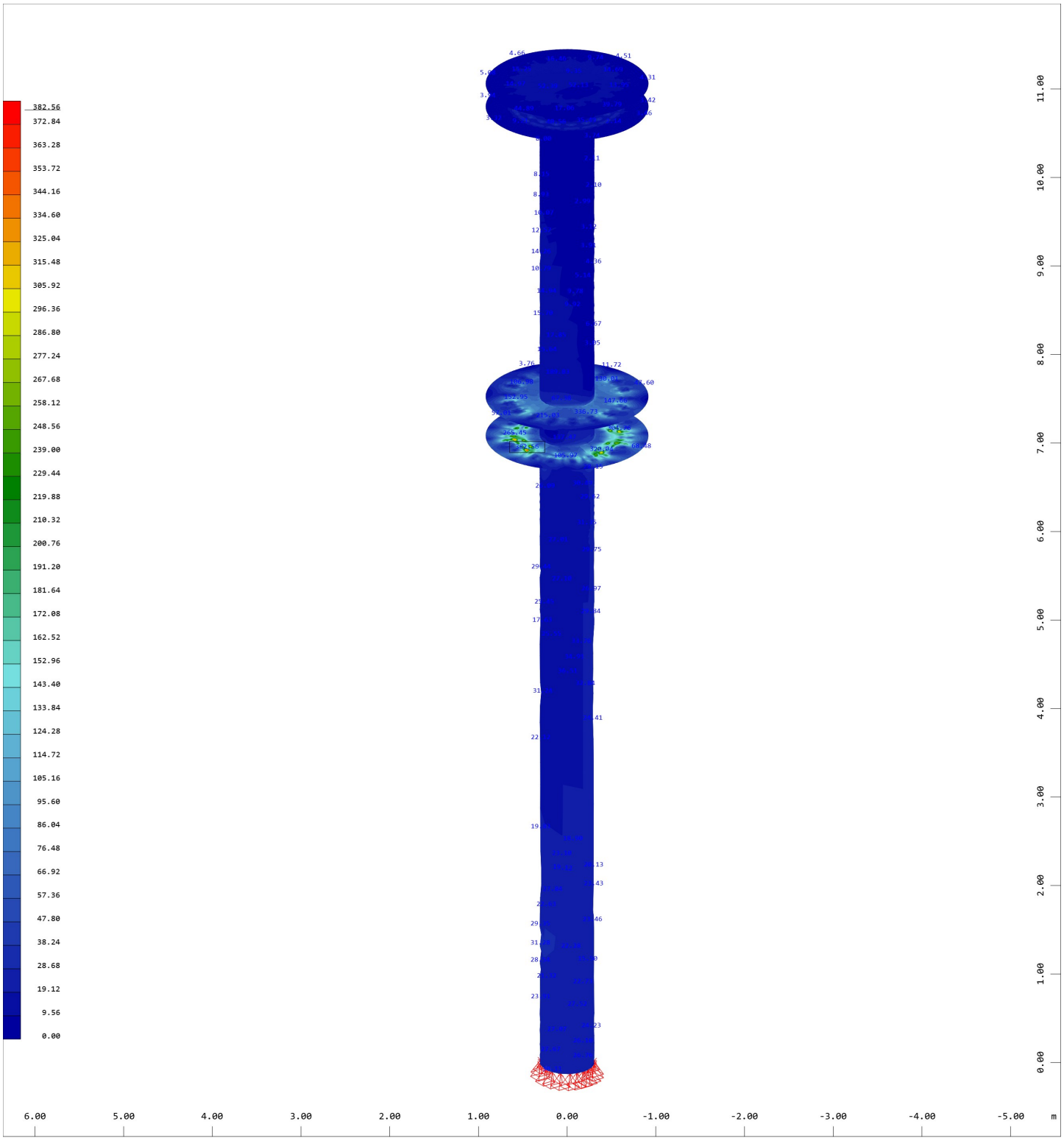
Kingpost, von Mises SV, whole height, isometric - LC 122, legend max 382.6 MPa (node-averaged). whole kingpost; peak on the Level 2 bottom collar plate at the radial ties



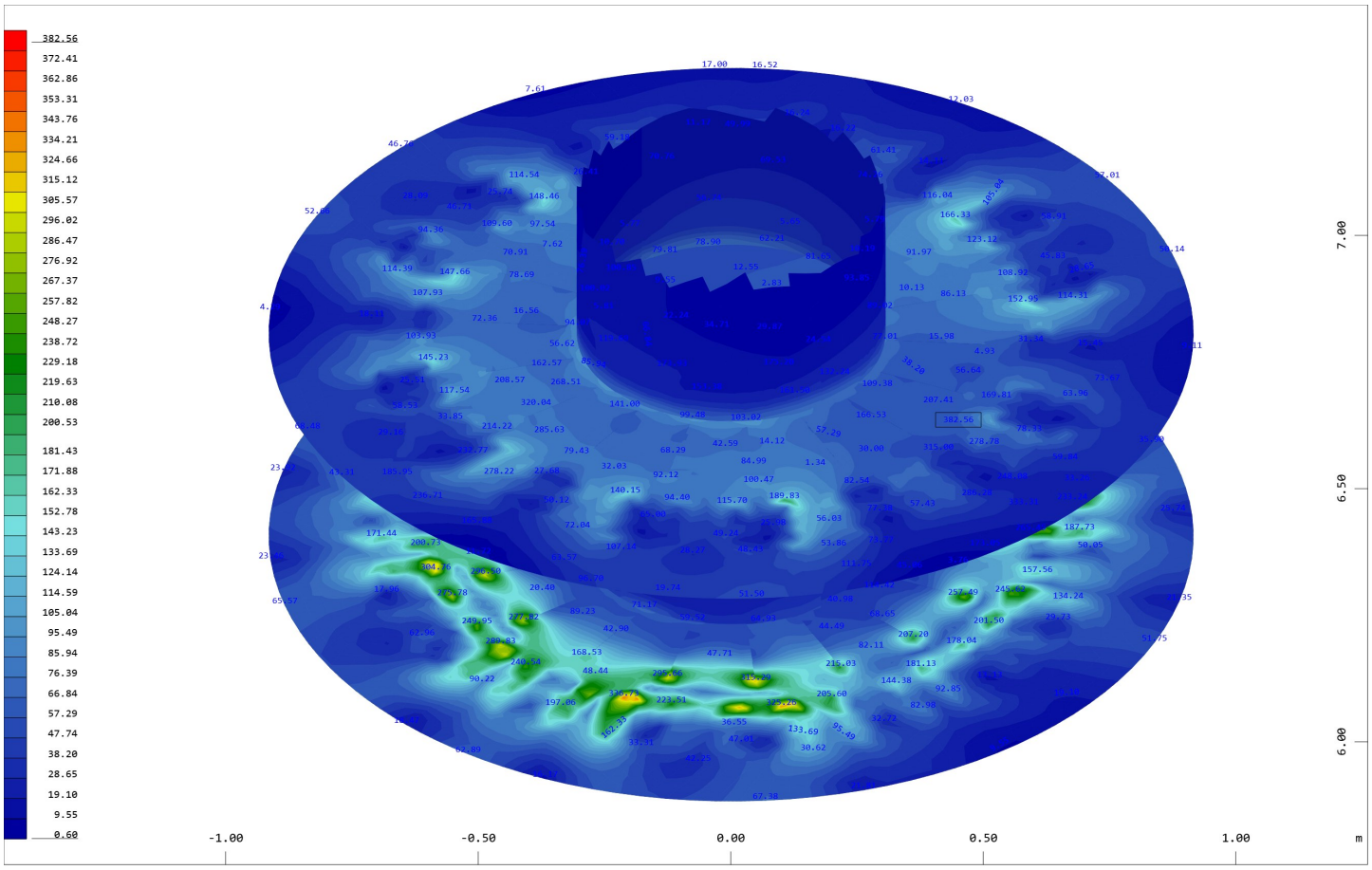
Sector of system Quadrilateral Elements
Maximum v.Mises stress from middle of element, Loadcase 122 122 ULS G+Q+psiW +Y down , from 0.16 to 382.56 step 9.56
MPa

M 1 : 53
X * 0.850
Y * 0.675
Z * 0.907

Kingpost, von Mises SV, opposite side - LC 122, legend max 382.6 MPa (node-averaged). same as K1 from the opposite side



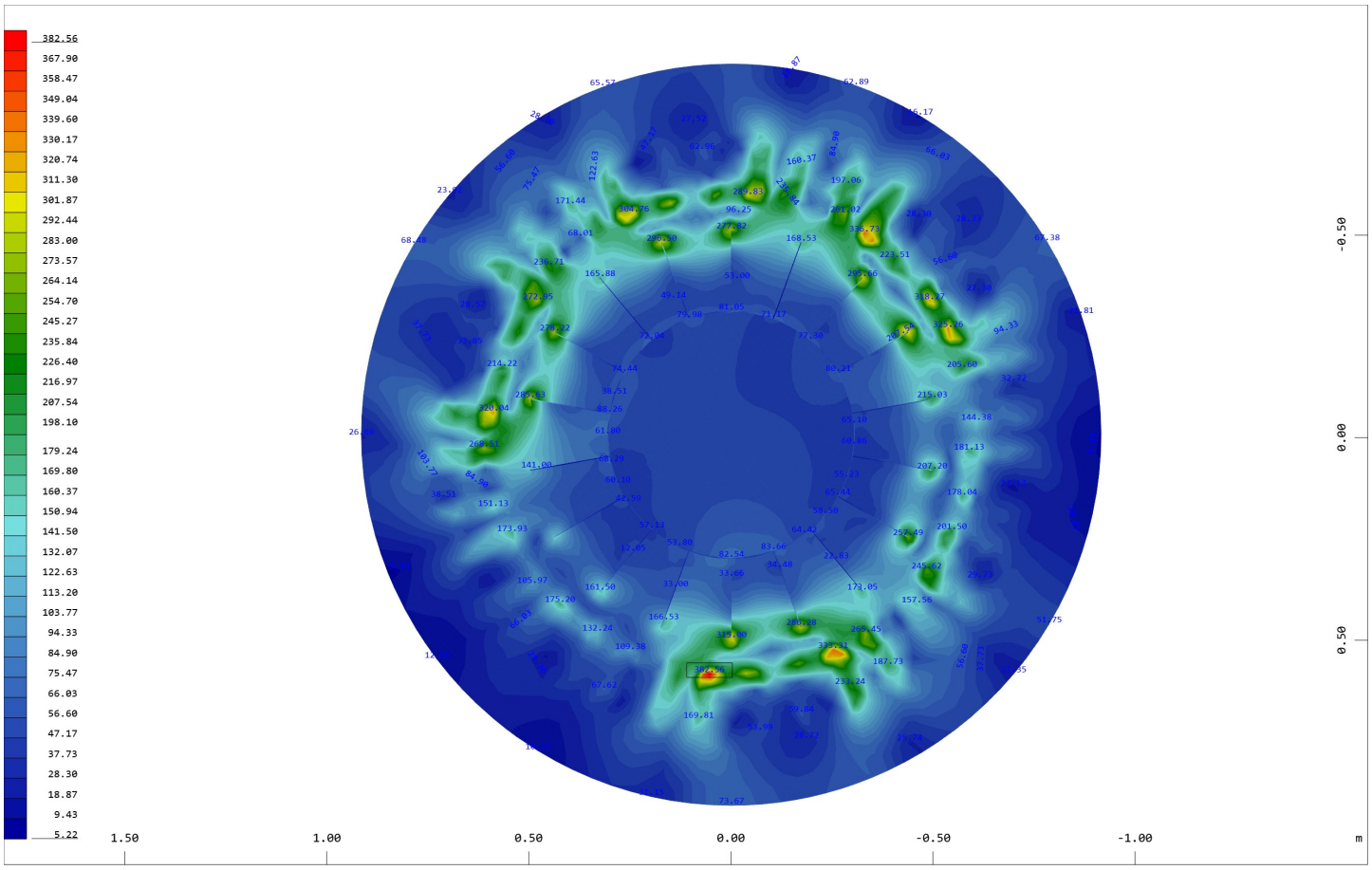
Level 2 connection zone (z 7.5-8.6), von Mises - LC 122, legend max 382.6 MPa (node-averaged). collar plates z 7.81 / 8.30, fins, tube; hot spots at the 4-node radial ties on the bottom collar



Sector of system Quadrilateral Elements
Maximum v.Mises stress from middle of element, Loadcase 122 122 ULS G+Q+psiW +Y down , from 0.60 to 382.56 step 9.55
MPa

M 1 : 12
X * 0.879
Y * 0.745
Z * 0.820

Level 2 bottom collar plate (z 7.81) seen from below - LC 122, legend max 382.6 MPa (node-averaged). ring of hot spots at r 0.6-0.7 = the tied nodes of the 18 radial ends; highest near 240 / 340 deg



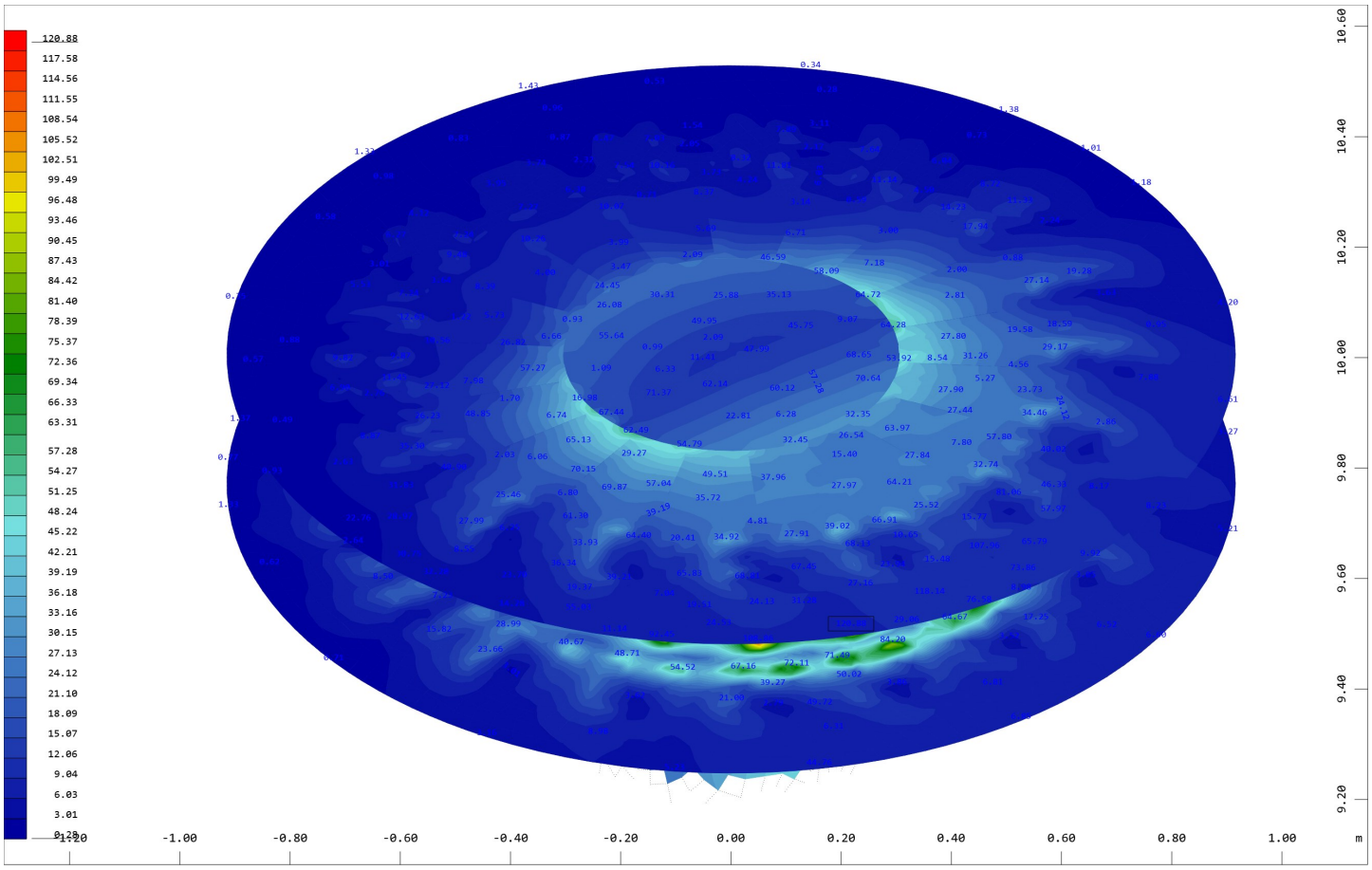
Sector of system Quadrilateral Elements

M 1 : 15

Maximum v.Mises stress from middle of element, Loadcase 122 122 ULS G+Q+psiW +Y down , from 5.22 to 382.56 step 9.43

MPa

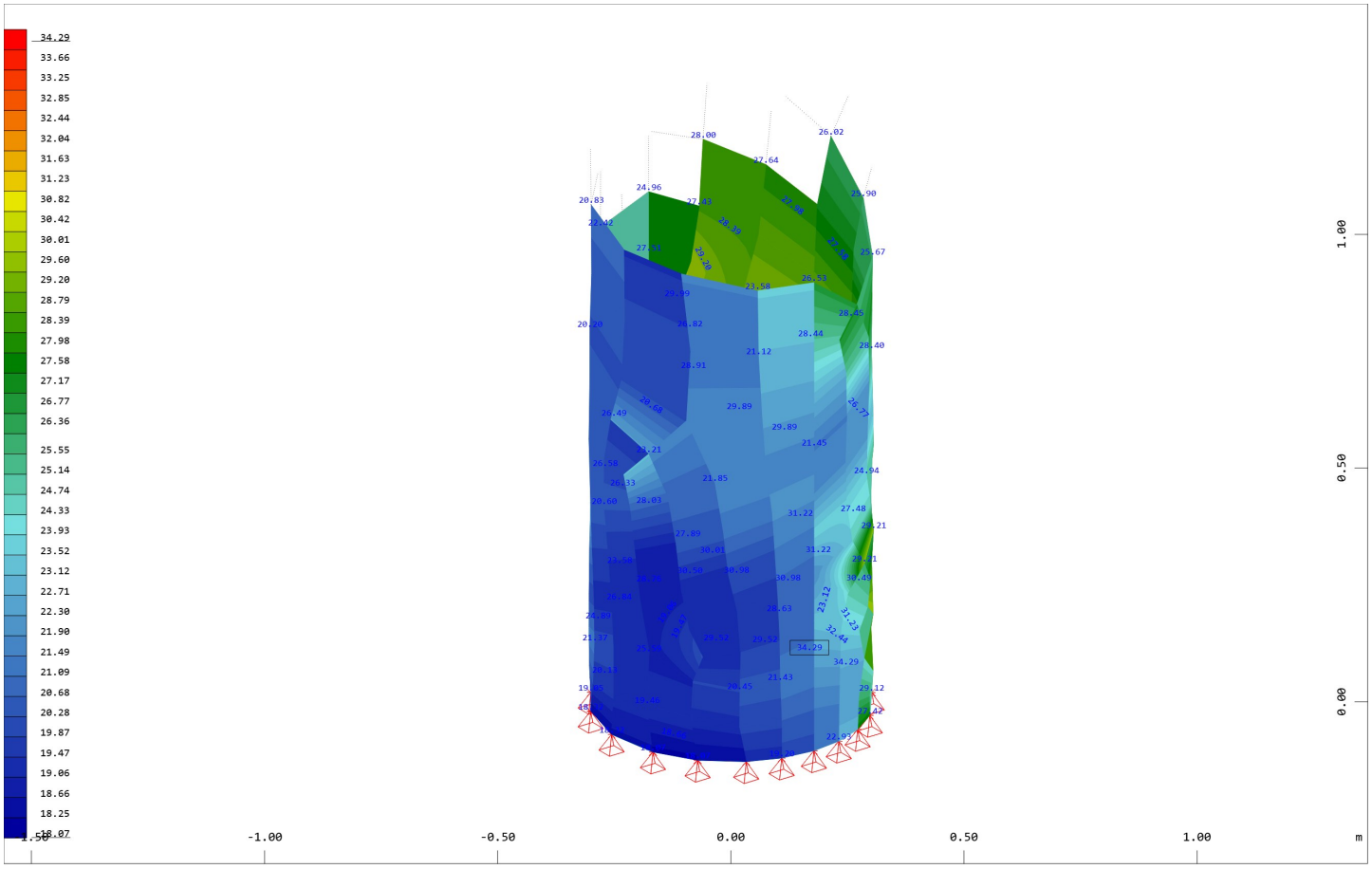
Roof connection zone (z > 11.5), von Mises - LC 110, legend max 120.9 MPa (node-averaged). roof plates and fins, max well below fy



Sector of system Quadrilateral Elements
Maximum v.Mises stress from middle of element, Loadcase 110 110 ULS 1.0G+W +Y uplift , from 0.28 to 120.88 step 3.01
MPa

M 1 : 11
X * 0.879
Y * 0.745
Z * 0.820

Base zone (z < 1.0), von Mises - LC 145, legend max 34.3 MPa (node-averaged). tube wall at the 20 support nodes, low stress



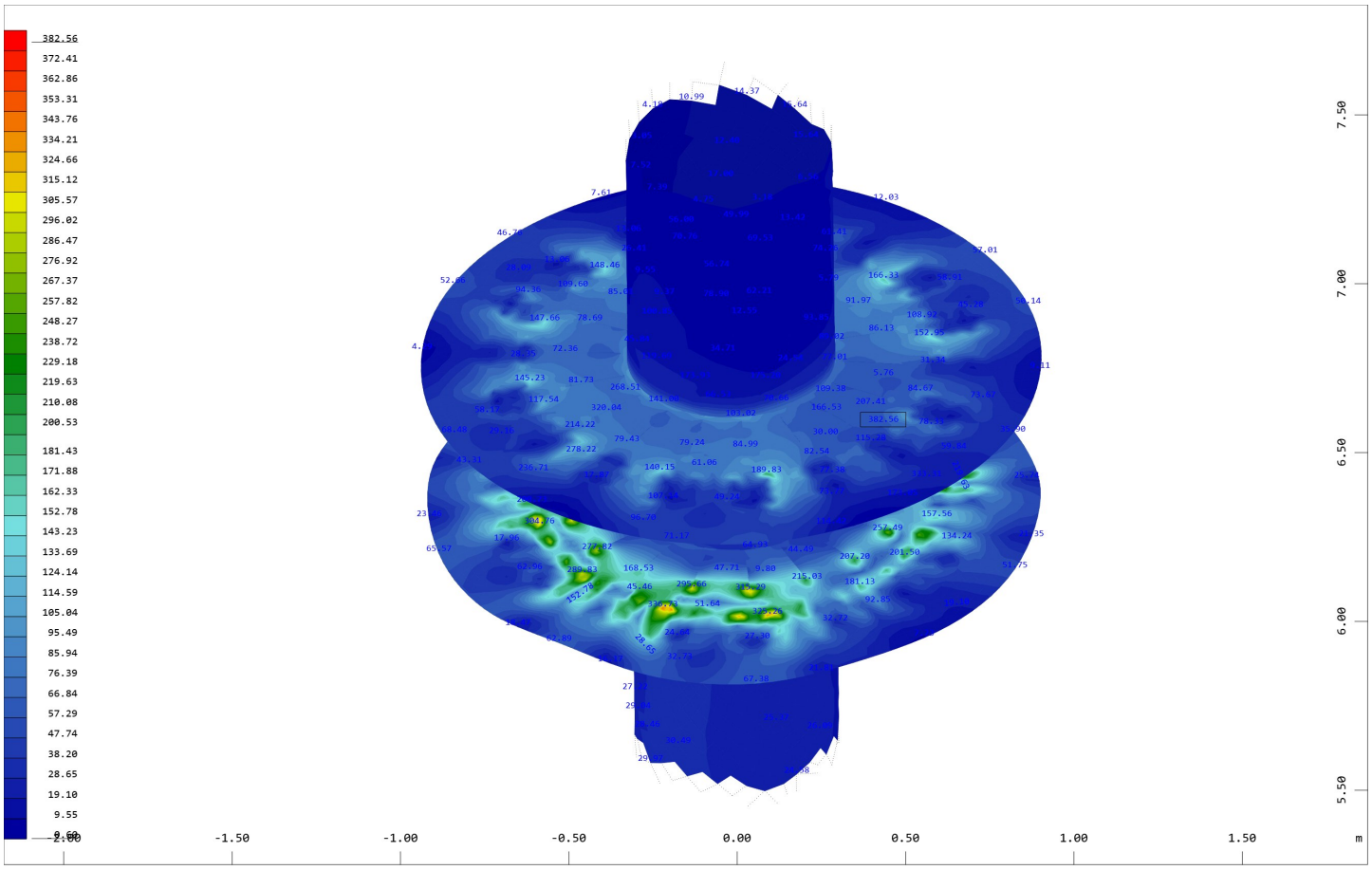
Sector of system Quadrilateral Elements

Maximum v.Mises stress from middle of element, Loadcase 145 145 ULS 1.35G+1.5Q + EHF +X , from 18.07 to 34.29 step 0.41 MPa

M 1 : 13

X * 0.850
Y * 0.675
Z * 0.907

Level 2 zone, deformed shape x50 with von Mises - LC 122, legend max 382.6 MPa (node-averaged). governing LC 122, local plate bending of the bottom collar under the radial ties



Sector of system Quadrilateral Elements
Deformed Structure from LC 122 122 ULS G+Q+psiW +Y down Enlarged by 50.000
Maximum v.Mises stress from middle of element, Loadcase 122 122 ULS G+Q+psiW +Y down , from 0.60 to 382.56 step 9.55

M 1 : 18
X * 0.879
Y * 0.745
Z * 0.820

26016 L2-R12-E1 (Level 2 ring to radial joint at r 12), rebuilt to drawing 970027-S01-ST-1103 rev 04, v21 forces

Joint: node 1485 (r 12.0, 0 deg, z 8.055), radial SLN 73 (UB 457x191x98) ends on the continuous L2 outer ring SLN 55 / 56 (UB 254x146x43).
Forces: 26016_check_v21.cdb (run 13 Nov 2026), 43 ULS STR combinations (101-130, 135, 141-148, 151-154), one IDEA load effect each.
IDEA StatiCa Connection 26.0, CBFEM, EN 1993-1-8. Report: L2-R12-E1_000.pdf (author CaSE, 13.11.2026). Model picture: L2-R12-E1_as_built_v21.png.

Result: FAIL (for decision)

check	max	load effect	where
Analysis	100 %	all	load reached in all 43
Plates (plastic strain, limit 5 %)	6.6 %	LC 153	existing ring UB 254 bottom flange next to the joint
	4.9 %	LC 119	ring top flange
	2.8 % / 2.7 %	LC 121 / 122	ring web / 20 ring stiffener at 315 (SLN 55 side); new plates <= 1.3 %
Bolts (6 x M24 10.9)	60.7 %	LC 122	
Welds	109.1 %	LC 121	20 ring stiffener at 315 to the ring flange (a 7.0)
	107.6 / 106.1 / 104.5 / 103.5 %	LC 122 / 121	395x145x20 fin lower pieces to the ring web (a 4.2, 6 mm leg)
	107.4 %	LC 122	20 inclined stiffener to the ring top flange (a 7.0)
	98-100 %	LC 153 / 135 / 122	bracket web / inclined stiffeners / top plates (SLN 56 side)

Forces at the joint (IDEA axes, kN / kNm), LC 121: radial N -8.5, Vz -89.8, My -47.2; ring each side N +24.8, My +135.9, Vz +36.9 / -52.9.
The ring moment 135.9 kNm is close to the elastic moment of UB 254x146x43 (W_el 504 cm3 x 275 = 138.6 kNm; M_pl 155.7 kNm). In the member check the same ring line is already above 1.0 (SOFiSTiK BDK SLN 56 1.063, AQB 1.066, LC 120). So the strain and the weld overloads at the joint come mainly
from the existing ring at its limit, not from the new plates or the bolts.

How the model was built (assumptions to check)

- Reading of D1 / E as in L2-R12-E1_reading_v21.png, option (a): radial web passes the ring web through a slot. In IDEA the ring web stays whole and overlaps the radial web (slightly stiff). Radial bottom flange cut back over the ring (to 105 mm inside the ring centre line, covering the 10 deg ring chord).
- 395x145x20 plates each side of the radial web, centred on the ring web, from 33 to 428 mm above the bottom of steel; split around the ring top flange and web (upper piece + 2 lower halves each side), every piece welded to the ring.
- Bolts M24 10.9, holes 26, bearing, through plate + radial web + plate: rows 105 / 230 / 330 mm below the plate top (98 mm above the bottom of steel), columns +-39 mm from the ring web (35 mm edges).
- Bracket in the ring web plane each side: 10 mm web plate, 20 mm top plate 145 wide under the radial top flange out to 155 mm, 20 mm inclined stiffener from 155 mm at the top to 315 mm on the ring top flange.
- 20 mm ring stiffener at 315 mm each side, modelled full width and welded to both ring flanges (a half-plate on one side never connected in IDEA 26.0 and made the model a mechanism).
- Welds: throat a = 7.0 mm (10 mm leg) to flanges, a = 4.2 mm (6 mm leg) to webs (drawing note), double fillets. Plates S275 (as the other 26016 joints).
- Left out: the two 20 x 295 plates above the ring bottom flange (with them IDEA reported a mechanism; they stiffen the ring bottom flange, so leaving them out is on the safe side for the ring), the 205x70x20 stiffener and the 185x70x20 plate of section E.
- No contact between the radial top flange and the bracket top plate (IDEA has no weld or contact there) - the bracket stiffens the ring but does not carry the radial flange directly.
- Joint at 0 deg built (the v14 location): radial end Vz 89.8 / My 47.2 kNm. Other r 12 radial ends: 120 deg 92.3 / 43.1, 160 deg 92.6 / 44.0, 240 deg 91.1 / 47.9, 80 / 200 / 320 deg about 75 / 40. Ring end moments at the other r 12 nodes were not compared; a joint at 160 or 240 deg could govern by a few percent.

Recommendation

1. The ring UB 254x146x43 at r 12 is at its limit near the radial joints in both the member check (BDK 1.06) and IDEA (6.6 % strain). A local reinforcement of the ring (for example the drawn 20 mm plates above the bottom flange made to work, or a heavier ring section) is needed before the welds can be sized; increasing only the weld sizes does not fix the flange strain.
2. If the ring is reinforced, rerun this joint; the bolts (61 %) and the new plates (<= 1.3 % strain) have reserve.
3. Please confirm the reading of the radial web at the ring (slot vs cope) and the role of the 20 x 295 plates.

Where it fails (added 13 Nov 2026, see the lower half of `L2-R12-E1_as_built_v21.png`)

- 6.57 % plastic strain (LC 153) sits in the EXISTING ring bottom flange directly below the radial web / lower fin plates
  (the 20 x 295 plates that the drawing shows above the ring bottom flange are not in the model there).
- Welds above 100 %: ring stiffener at 315 (SLN 55 side) to ring top flange 109.1 % (a 7.0, LC 121) and to bottom flange 103.8 %;
  fin lower plates to ring web 107.6 / 106.1 % (SLN 55 side) and 104.5 / 103.5 % (SLN 56 side) (a 4.2, LC 122 / 121);
  inclined stiffener b (SLN 55 side) to ring top flange 107.4 % (a 7.0, LC 122); bracket web (SLN 56 side) 100.2 % (a 4.2, LC 153).

Governing forces at the joint (IDEA axes, kN / kNm)

LC	radial SLN 73 N / Vz / My	ring SLN 55 N / Vz / My	ring SLN 56 N / Vz / My
121	-8.5 / -89.8 / -47.2	+24.9 / +36.9 / +135.9	+24.8 / -52.9 / +136.0
122	-7.6 / -89.8 / -47.2	+22.2 / +36.9 / +135.9	+22.2 / -52.9 / +135.9
153	-9.2 / -89.4 / -47.2	+27.0 / +36.6 / +135.9	+27.0 / -52.8 / +135.9

Sensitivity to the web weld size (6 mm leg read from the drawing note)

Rerun with 10 mm leg (a 7.0) to the webs: ring-web welds 107.6 -> 99.2 %, but the maximum weld stays 108.3 % (flange welds to the ring), plate-to-plate welds 108.7 -> 107.9 %, plastic strain 6.6 % unchanged, bolts 60.3 %. The result is not governed by reading the web welds as 6 mm.

Variants with the drawing's extra bottom plates (13 Nov 2026, picture `L2-R12-E1_variants_v21.png`)

variant	what is added	connects	max plate strain (where, LC)	max weld (which, LC)	bolts
0	none (current model)	yes	6.6 % ring bottom flange below the radial web / lower fins (LC 153)	109.1 % 315 stiffener to ring top flange, a 7.0 (LC 121)	60.7 %
1	20 x 295 plates on the ring bottom flange, 140 wide (2 x 70 + web), from the radial web face to the 315 stiffener; long edges welded to the flange face (a 7.0), end to the stiffener, lower fins welded onto them	yes	7.1 % ring bottom flange at the end of the new plate (LC 120)	102.8 % 295 plate to ring bottom flange, SLN 55 side (LC 119)	59.1 %
2	section E 185x70x20 plates under the fins (2 x 70 + web, 185 along the ring), edges welded to the flange face; radial web cut to sit on it	yes	8.0 % ring bottom flange at the plate ends (LC 119)	105.3 % ring top flange welds; 185 plate to flange 104.9 % (LC 120)	60.4 %

The earlier mechanism with the 20 x 295 plates came from modelling them as two halves either side of the ring web; as one plate welded to the flange face along its long edges they connect.
Both variants move the peak to the end of the new plate / weld on the existing ring bottom flange and raise it; neither brings it under 5 %. The 205x70x20 stiffener (top of section E) is away from the peak and was not tried.
Two variants were enough to show the trend (not the 4 allowed). Reading: the ring flange is near yield over the joint, so the peak is a local concentration at every attachment end; the IDEA result does not support "add the drawn plates" as the fix on its own.

Review comments (13 Nov 2026) and the corrected model (variant 4)

Findings in variant 0:
1. "Missing weld between end plate and bottom flange": TRUE. The lower 395x145x20 fin plates stopped 20 mm above the ring bottom flange (a gap left for the 20 mm plates that were not modelled). They were welded only to the ring web (a 4.2) and to the underside of the ring top flange (a 7.0).
   The upper fin piece was welded to the top of the ring top flange (a 7.0). The drawing (section E) shows the plates going down to the bottom flange zone, welded.
2. "Missing stiffener": the SLN 56 side 315 stiffener WAS in the model (one full-width plate, both faces of the web). But it sat off the IDEA ring, see 3.
3. "UB straight but haunch not": IDEA 26.0 builds a continuous member (ring SLN 55 + 56) as ONE straight line. It took the SLN 55 chord and ran it on through the node, 20 deg off the real SLN 56 chord. My bracket and stiffener parts followed the real chords (+-10 deg from the tangent), so on the SLN 56 side they were up to about 110 mm off the IDEA ring. The ring forces on that side were also applied along the wrong line.

Correction (variant 4, var_V4s): the ring is modelled straight along the tangent at the node (the bisector of the two chords). Every bracket and stiffener part sits on it: 10 web plate in the ring web plane, top plates and inclined stiffeners centred on the ring top flange, both sides. Both 315 stiffeners are full width (both faces), welded to both flanges. The lower fin plates go down to the ring bottom flange and are welded to it (a 7.0).
Limitation: the 10 deg kink of each chord is not modelled; at the 315 stiffeners the real ring web is about 55 mm inward of the model's.
A kinked continuous ring cannot be modelled in IDEA 26.0 (a continuous member is straight).

model	strain (where, LC)	max weld (which, LC)	bolts
v0 (chords, as first reported)	6.6 % ring bottom flange (153)	109.1 %	60.7 %
v0s = v0 on the straight ring	3.5 % (122)	101.4 % (153)	70.2 %
V4 = v0s + fin feet welded to the bottom flange (corrected model)	3.0 % ring top flange (122)	100.6 % 315 stiffener SLN 55 side to ring flange, a 7.0, L 146 (153); bracket top plates / inclined stiffeners 98-99.5 %	69.8 % (122)
V5 = V4 with the 20 x 295 plates (fins stand on them)	7.1 % ring bottom flange at the plate end (119)	99.2 % 295 plates to flange (153 / 122)	67.0 %

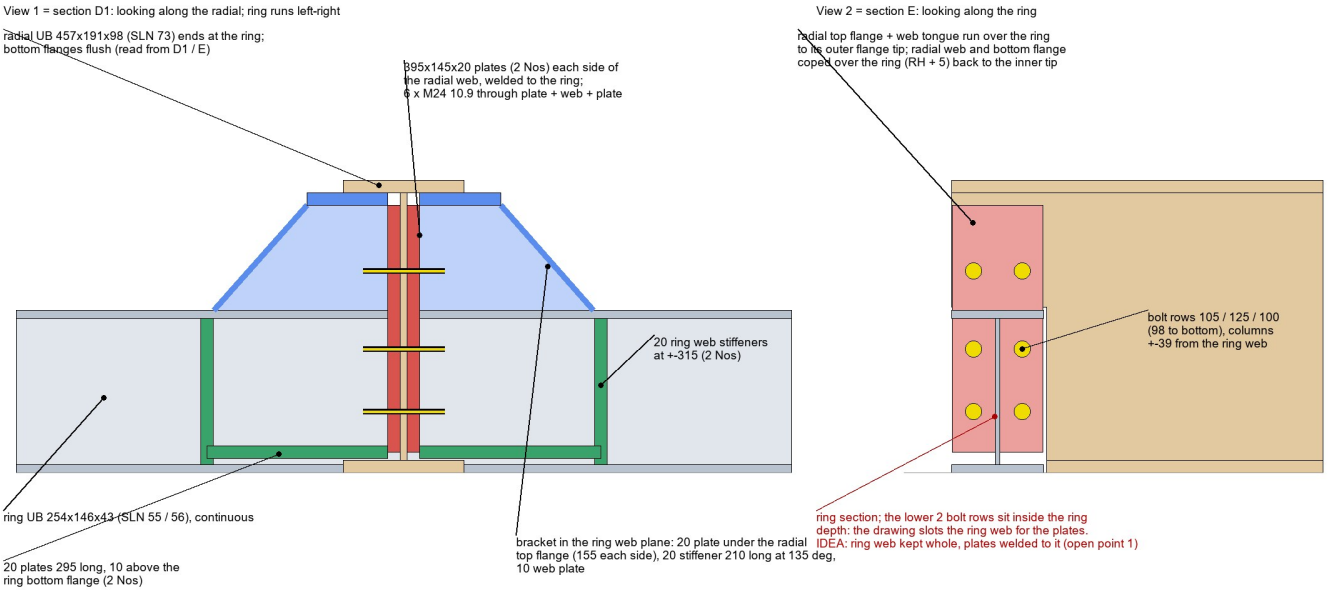
The large strain in v0 to v3 came mainly from the ring geometry error (point 3), not from the ring capacity.
Corrected model V4: strain passes (3.0 % < 5 %), bolts 70 %, welds just over at 100.6 % (one 315 stiffener weld, a 7.0), many welds at 98-99 %.
V4 report: L2-R12-E1_000_V4.pdf (author CaSE). The superseded statements above about the ring being at its limit apply to the faulty geometry.

V6 = V4 with the 315 stiffener to ring flange welds 12 mm leg (a 8.5), nothing else changed	**3.6 % (122)**	**99.8 % (119); bracket inclined stiffeners / top plates to the bracket web / fin 99.4-99.5 % (a 7.0 / a 4.2)**	**69.9 % (122)**

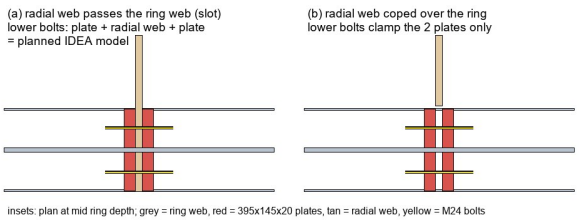
V6 PASSES all checks (IDEA summary: analysis 100 %, plates 3.6 % < 5 %, bolts 69.9 %, welds 99.8 %). Report L2-R12-E1_000_V6.pdf (author CaSE, 13.11.2026).
Note: the bracket welds are at 99.4-99.5 % in both V4 and V6, so they have no reserve. A small change in loads or geometry would take them over 100 %.

Reading of drawing 970027-S01-ST-1103 rev 04, sections D1 / E

26016 L2-R12-E1 (v21): reading of drawing 970027-S01-ST-1103 rev 04, sections D1 and E, and the IDEA idealisation (to be confirmed)



Welds: 10 mm fillets to flanges, 6 mm to webs (drawing note). 10 mm where shown. Plates S275 (as the other 26016 joints). Bolts M24 10.9, bearing.
Forces: v21 database, 43 ULS STR cases at the SLN 73 end node (r 12). Open points: (1) the radial web / ring web crossing (slot vs cope);
(2) the 205x70x20 stiffener and 185x70x20 plate at the ring flanges (E) are left out unless they carry load in your reading.

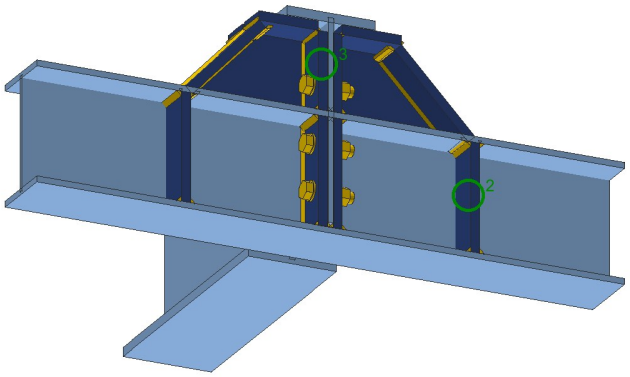


insets: plan at mid ring depth; grey = ring web, red = 395x145x20 plates, tan = radial web, yellow = M24 bolts

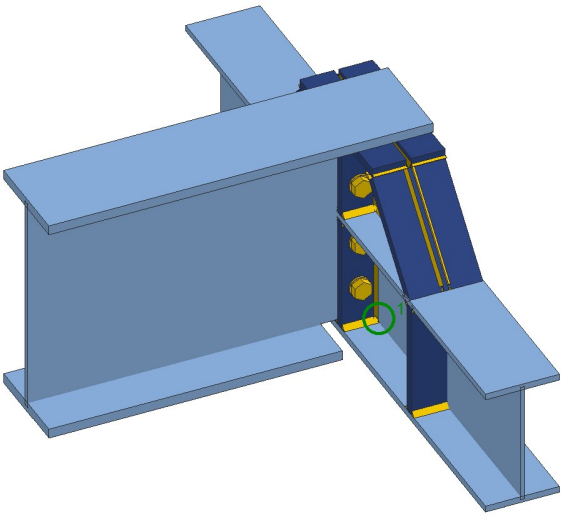
Corrected model (variant 4) after the review comments: views, plan, strain

26016 L2-R12-E1_000 (node 1485, r 12, 0 deg) on v21 forces - CORRECTED model (variant 4) after the 3 review comments

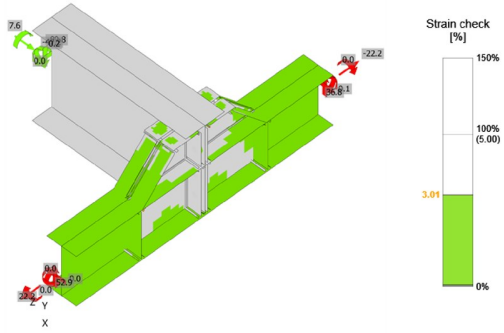
view from below / outside



view from above

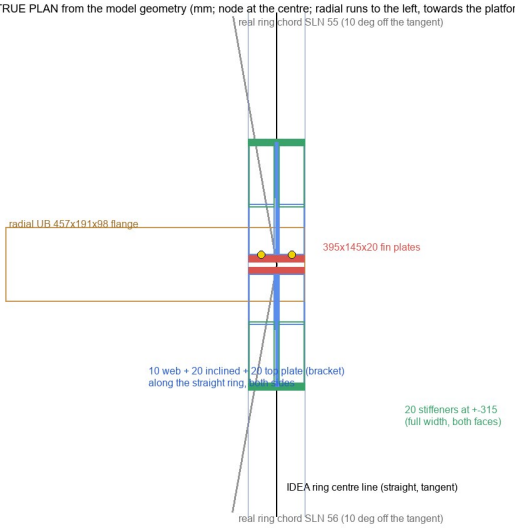


IDEA plastic strain, corrected model: 3.0 % (LC 122) < 5 %



Strain check, C LC_122

TRUE PLAN from the model geometry (mm; node at the centre; radial runs to the left, towards the platform centre)



Review comments and what was found / changed:

1 'missing weld between end plate and bottom flange': TRUE. The lower 395x145x20 fin plates stopped 20 mm above the ring bottom flange (gap left for the 20 mm plates that were not modelled) and were welded only to the ring web (a 4.2) and the top flange underside (a 7.0). Now taken down to the bottom flange and welded to it (a 7.0) - yellow welds at the plate feet in the view from above.

2 'missing stiffener': the SLN 56 side 315 stiffener WAS in the model (full width, both faces), but IDEA drew the ring as ONE straight line along the SLN 55 chord, 20 deg off the SLN 56 chord, so that stiffener and the SLN 56 bracket sat up to ~110 mm off the IDEA ring (hidden in the view).

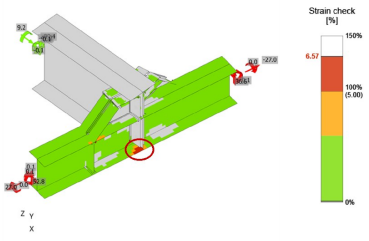
3 'UB looks straight but the haunch is not': same cause. IDEA models a continuous member straight; my bracket followed the real chords (+-10 deg). Fix: the ring is now modelled straight along the tangent at the node (bisector of the two chords) and every bracket / stiffener part sits on it (plan above). Error: the real chords are 10 deg off the tangent; at the 315 stiffeners the real ring web is 55 mm inward of the model's. Drawing D1 shows the bracket in elevation only; the plan (sheet set) shows straight ring chords between radials. A straight local model is the usual CBFEM idealisation; the 10 deg kink is not modelled.

Results (43 ULS cases): v0 = first model, v0s = first model on the straight ring, V4 = v0s + fin feet welded to the bottom flange, V5 = V4 with 20x295 plates
v0 6.6 % / welds 109.1 % / bolts 60.7 % v0s 3.5 % / 101.4 % / 70.2 % V4 3.0 % / 100.6 % / 69.8 % V5 7.1 % / 99.2 % / 67.0 %
V4 max weld: 315 stiffener (SLN 55 side) to ring flange, a 7.0, 100.6 % LC 153; bracket top plates / inclined stiffeners 98-99.5 %. V4 is the corrected model.

Earlier variants (before the straight-ring correction): strain with the drawing's extra plates

26016 L2-R12-E1 (v21 forces): plastic strain in IDEA without and with the extra plates of drawing 1103 (limit 5 %)

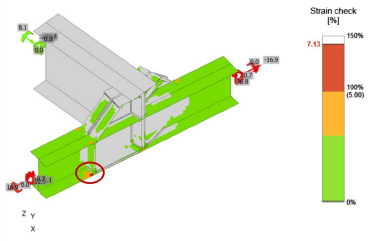
Variant 0 (current): no 20 x 295 plates



Strain check, C LC_ 153

plate strain 6.6 % (LC 153), ring bottom flange below the radial web / lower fins
max weld 109.1 % stiffener at 315 to ring top flange (a 7.0, LC 121)
bolts 60.7 %

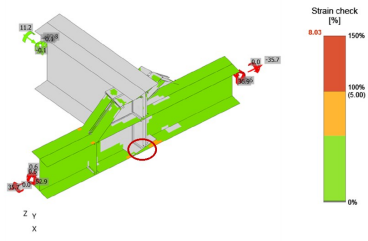
Variant 1: + 20 x 295 plates on the ring bottom flange



Strain check, C LC_ 120

(140 wide, radial web face to the 315 stiffener; edges welded to the flange,
 end to the stiffener, lower fins welded onto them)
plate strain 7.1 % (LC 120), ring bottom flange at the end of the new plate
max weld 102.8 % 295 plate to ring bottom flange (SLN 55 side, a 7.0, LC 119)
bolts 59.1 %

Variant 2: + E-section 185x70x20 plates under the fins



Strain check, C LC_ 119

(2 x 70 + web, 185 along the ring, edges welded to the flange; radial web sits on it)
plate strain 8.0 % (LC 119), ring bottom flange at the plate ends (under the plate)
max weld 105.3 % ring top flange welds; 185 plate to flange 104.9 % (LC 120)
bolts 60.4 %

In both variants the extra plates connect and work, but the strain peak moves to the end of the new plate / weld on the existing ring bottom flange and rises (7.1 % / 8.0 %). The ring flange is close to yield over the joint (ring My 136 kNm vs M_el 138.6 kNm), so every plate or weld end on it becomes a local peak. Red circles: location of the peak in each plot.

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R12-E1_000
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN73 radial (ends at the ring) - ring SLN55 + SLN56 (continuous)

Name	SLN73 radial (ends at the ring) - ring SLN55 + SLN56 (continuous)
Description	
Analysis	Stress, strain/ simplified loading

Design

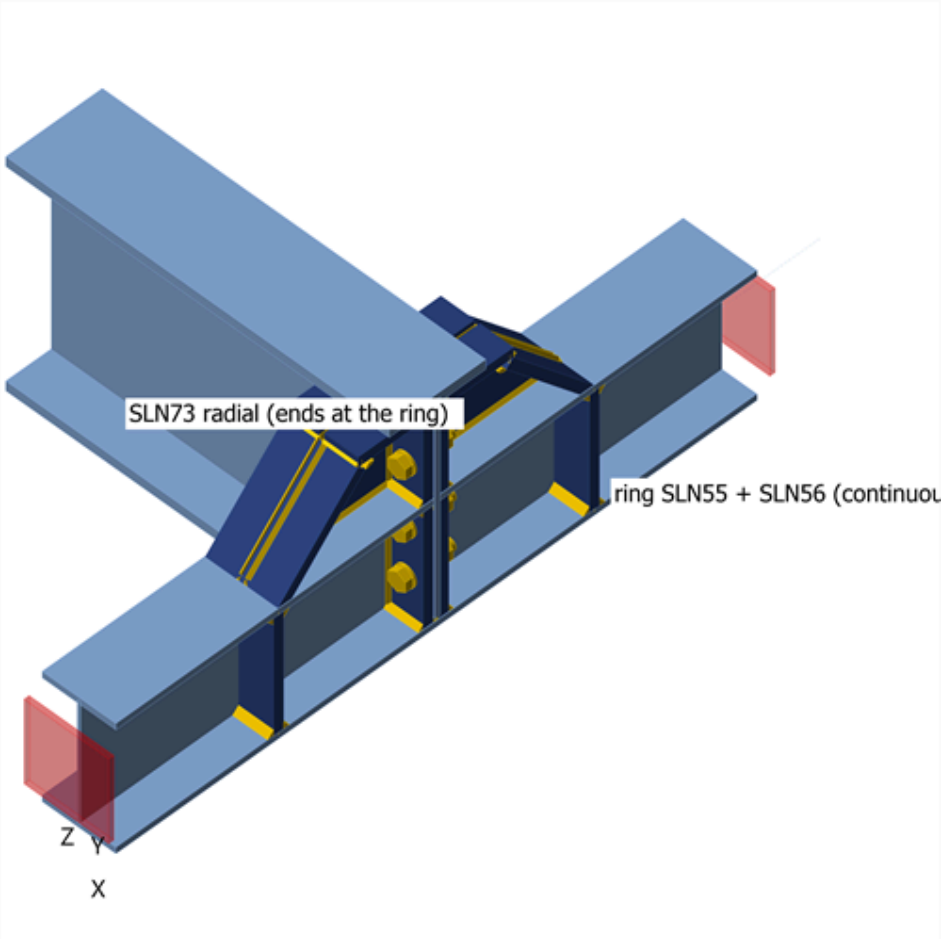
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN73 radial (ends at the ring)	1 - UB457x191x98	0.0	0.0	0.0	76	0	0
ring SLN55 + SLN56 (continuous)	2 - UB254x146x43	-90.0	0.0	0.0	0	0	-104

Supports and forces

Name	Support	Forces in	X [mm]
SLN73 radial (ends at the ring) / begin		Position	0
ring SLN55 + SLN56 (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
ring SLN55 + SLN56 (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0



SLN73 radial (ends at the ring) - ring SLN55 + SLN56 (continuous), Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M24 10.9	24	900.0	1000.0	452

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN73 radial (ends at the ring) / Begin	2.1	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_102	SLN73 radial (ends at the ring) / Begin	9.2	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_103	SLN73 radial (ends at the ring) / Begin	9.6	-0.1	-86.2	-0.1	-45.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_104	SLN73 radial (ends at the ring) / Begin	7.9	0.0	-86.4	-0.1	-45.4	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_105	SLN73 radial (ends at the ring) / Begin	8.8	-0.1	-86.3	-0.1	-45.4	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_106	SLN73 radial (ends at the ring) / Begin	8.0	0.0	-86.3	-0.1	-45.4	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_107	SLN73 radial (ends at the ring) / Begin	1.8	-0.1	-7.8	0.0	-4.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_108	SLN73 radial (ends at the ring) / Begin	1.7	0.1	-8.4	0.0	-4.6	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_109	SLN73 radial (ends at the ring) / Begin	1.4	0.1	-8.1	0.0	-4.5	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_110	SLN73 radial (ends at the ring) / Begin	0.6	-0.1	-8.1	0.0	-4.4	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_111	SLN73 radial (ends at the ring) / Begin	7.4	-0.1	-61.7	0.0	-32.6	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_112	SLN73 radial (ends at the ring) / Begin	6.3	0.1	-62.3	0.0	-32.8	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_113	SLN73 radial (ends at the ring) / Begin	6.7	0.0	-62.0	0.0	-32.7	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_114	SLN73 radial (ends at the ring) / Begin	5.6	-0.1	-62.0	0.0	-32.7	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_115	SLN73 radial (ends at the ring) / Begin	2.4	-0.1	-13.1	0.0	-7.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_116	SLN73 radial (ends at the ring) / Begin	2.1	0.1	-13.6	0.0	-7.4	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_117	SLN73 radial (ends at the ring) / Begin	2.0	0.0	-13.3	0.0	-7.3	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_118	SLN73 radial (ends at the ring) / Begin	1.1	-0.1	-13.3	0.0	-7.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_119	SLN73 radial (ends at the ring) / Begin	11.2	-0.1	-89.8	-0.1	-47.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_120	SLN73 radial (ends at the ring) / Begin	8.1	0.0	-89.8	-0.1	-47.2	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_121	SLN73 radial (ends at the ring) / Begin	8.5	-0.1	-89.8	-0.1	-47.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_122	SLN73 radial (ends at the ring) / Begin	7.6	0.0	-89.8	-0.1	-47.2	0.2
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_123	SLN73 radial (ends at the ring) / Begin	10.8	-0.1	-68.9	0.0	-36.3	0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_124	SLN73 radial (ends at the ring) / Begin	6.7	0.1	-68.9	0.0	-36.3	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_125	SLN73 radial (ends at the ring) / Begin	6.2	0.0	-68.9	0.0	-36.3	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_126	SLN73 radial (ends at the ring) / Begin	4.8	-0.1	-68.9	0.0	-36.3	0.2
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_127	SLN73 radial (ends at the ring) / Begin	5.7	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_128	SLN73 radial (ends at the ring) / Begin	2.5	0.1	-20.2	0.0	-10.9	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_129	SLN73 radial (ends at the ring) / Begin	1.5	0.0	-20.3	0.0	-10.9	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_130	SLN73 radial (ends at the ring) / Begin	0.3	-0.1	-20.3	0.0	-10.9	0.2
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_135	SLN73 radial (ends at the ring) / Begin	7.7	0.0	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_141	SLN73 radial (ends at the ring) / Begin	2.3	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_142	SLN73 radial (ends at the ring) / Begin	1.8	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_143	SLN73 radial (ends at the ring) / Begin	2.1	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_144	SLN73 radial (ends at the ring) / Begin	2.0	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_145	SLN73 radial (ends at the ring) / Begin	10.1	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_146	SLN73 radial (ends at the ring) / Begin	8.3	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_147	SLN73 radial (ends at the ring) / Begin	9.4	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_148	SLN73 radial (ends at the ring) / Begin	9.0	0.0	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_151	SLN73 radial (ends at the ring) / Begin	6.9	-0.1	-46.4	0.0	-22.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_152	SLN73 radial (ends at the ring) / Begin	9.3	0.0	-63.5	0.0	-35.9	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_153	SLN73 radial (ends at the ring) / Begin	9.2	-0.1	-89.4	0.0	-47.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_154	SLN73 radial (ends at the ring) / Begin	7.0	-0.1	-46.8	0.0	-22.5	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	3.6 < 5.0%	OK
Bolts	69.9 < 100%	OK
Welds	99.8 < 100%	OK
Buckling	Not calculated	

Plates

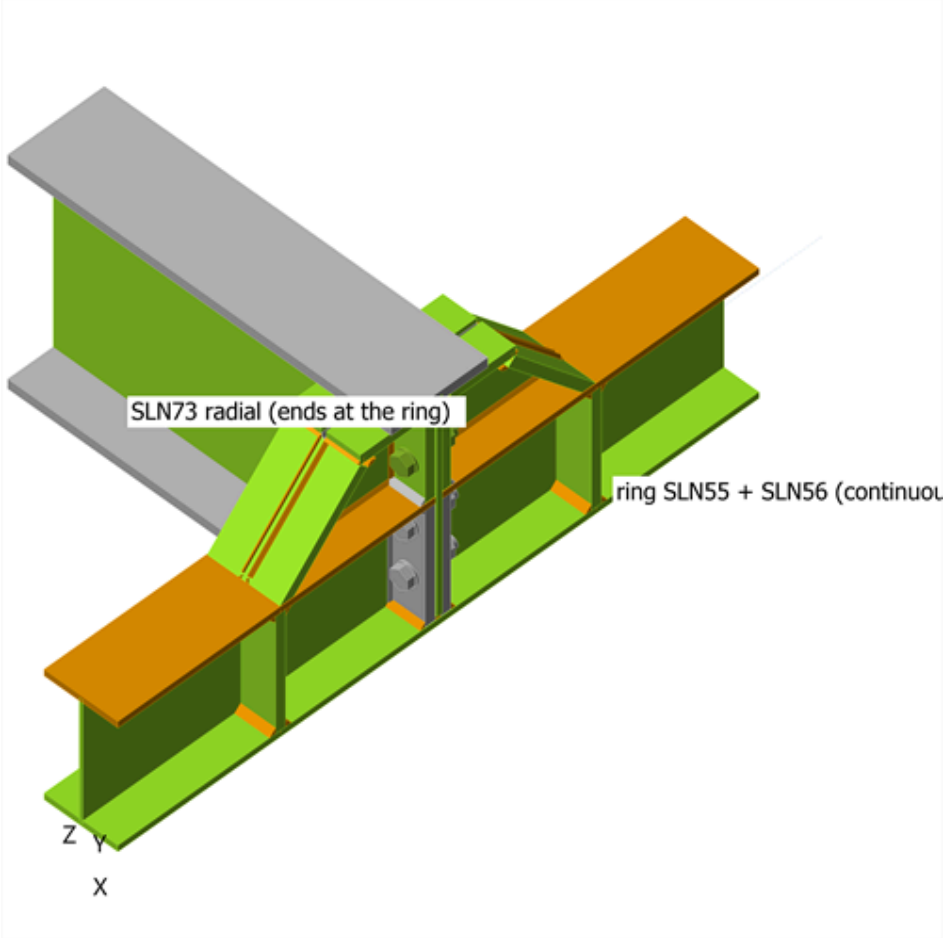
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN73 radial (ends at the ring)-bfl 1	19.6	C LC_122	14.8	0.0	0.0	OK
SLN73 radial (ends at the ring)-tfl 1	19.6	C LC_119	140.0	0.0	5.8	OK
SLN73 radial (ends at the ring)-w 1	11.4	C LC_122	242.8	0.0	71.3	OK
ring SLN55 + SLN56 (continuous)-bfl 1	12.7	C LC_119	280.7	2.7	0.0	OK
ring SLN55 + SLN56 (continuous)-tfl 1	12.7	C LC_122	282.5	3.6	0.0	OK
ring SLN55 + SLN56 (continuous)-w 1	7.2	C LC_122	275.4	0.2	0.0	OK
395x145x20 fin plate upper (SLN55 side)	20.0	C LC_122	275.2	0.1	479.6	OK
395x145x20 fin plate lower outer (SLN55 side)	20.0	C LC_120	203.4	0.0	197.3	OK
395x145x20 fin plate lower inner (SLN55 side)	20.0	C LC_122	205.3	0.0	38.8	OK
395x145x20 fin plate upper (SLN56 side)	20.0	C LC_135	275.1	0.1	476.8	OK
395x145x20 fin plate lower outer (SLN56 side)	20.0	C LC_122	205.4	0.0	198.6	OK
395x145x20 fin plate lower inner (SLN56 side)	20.0	C LC_122	205.1	0.0	38.5	OK
20 ring stiffener at 315 (SLN55 side)	20.0	C LC_121	279.1	1.9	0.0	OK
10 bracket web plate (SLN55 side)	10.0	C LC_122	275.6	0.3	0.0	OK
20 bracket top plate a (SLN55 side)	20.0	C LC_119	275.0	0.0	104.0	OK
20 inclined stiffener a (SLN55 side)	20.0	C LC_122	275.4	0.2	0.0	OK
20 bracket top plate b (SLN55 side)	20.0	C LC_122	275.9	0.4	0.0	OK
20 inclined stiffener b (SLN55 side)	20.0	C LC_122	275.4	0.2	0.0	OK
20 ring stiffener at 315 (SLN56 side)	20.0	C LC_122	279.1	1.9	0.0	OK
10 bracket web plate (SLN56 side)	10.0	C LC_122	275.6	0.3	0.0	OK
20 bracket top plate a (SLN56 side)	20.0	C LC_120	275.9	0.4	0.0	OK
20 inclined stiffener a (SLN56 side)	20.0	C LC_122	275.4	0.2	0.0	OK
20 bracket top plate b (SLN56 side)	20.0	C LC_119	275.0	0.0	104.0	OK
20 inclined stiffener b (SLN56 side)	20.0	C LC_122	275.4	0.2	0.0	OK

Design data

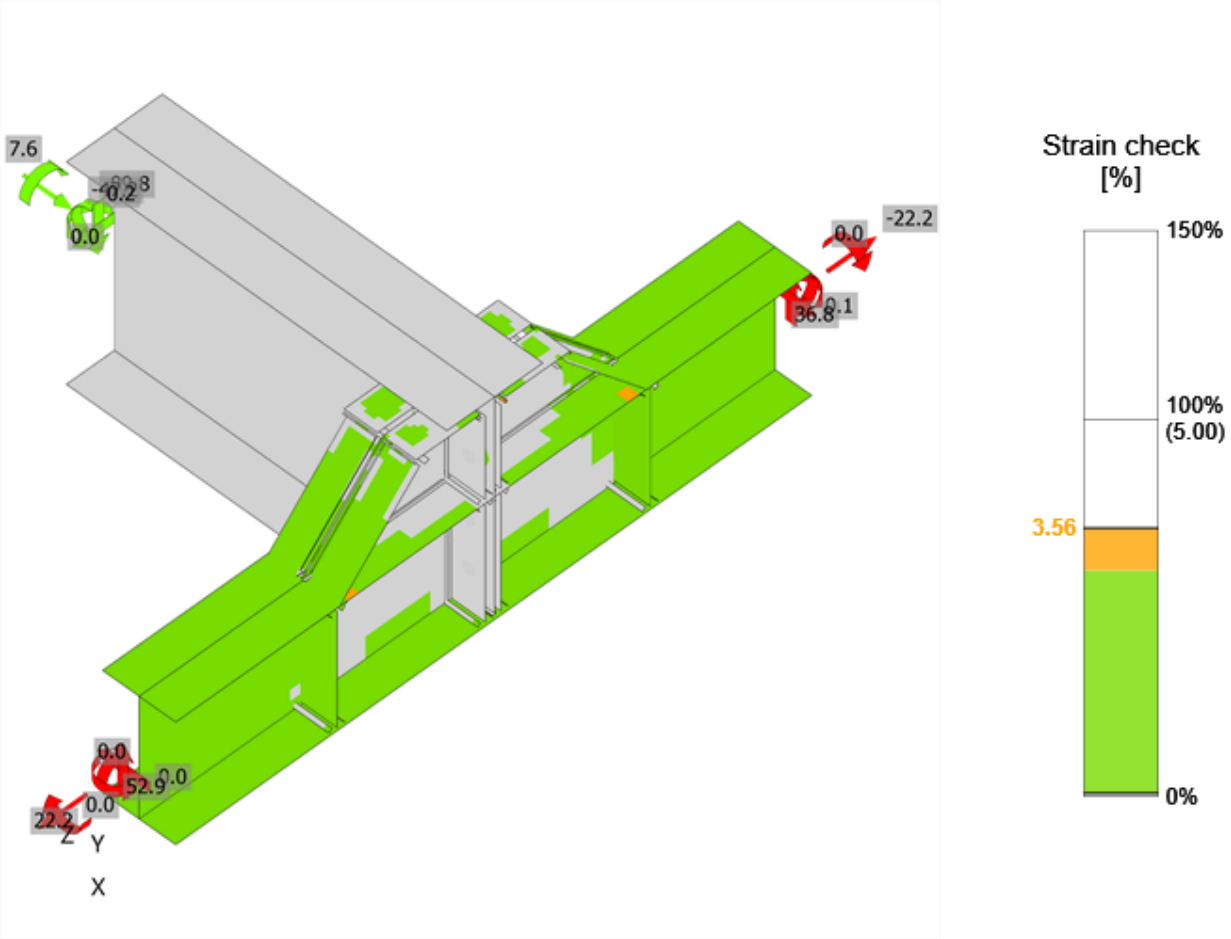
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

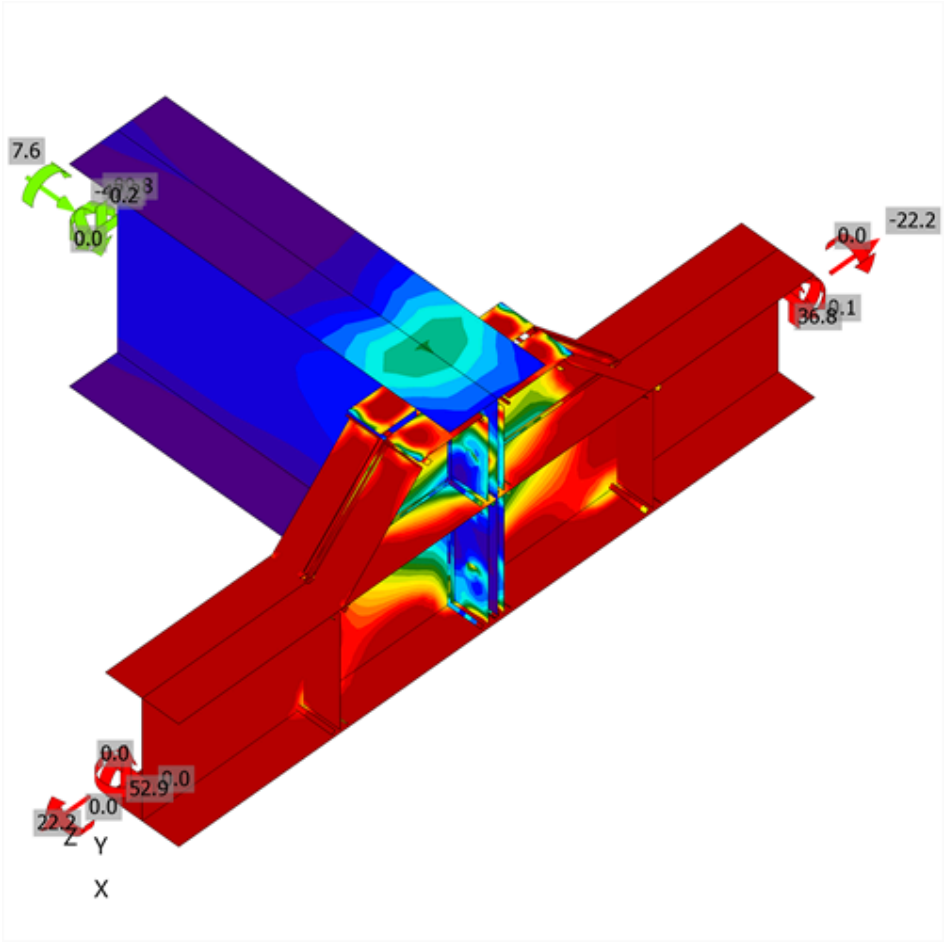
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

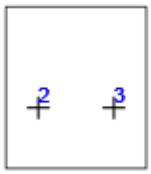
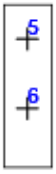
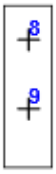


Strain check, C LC_122



Equivalent stress, C LC_122

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M24 10.9 - 1	C LC_122	46.7	34.5	235.3	18.4	29.2	37.5	OK	OK
	B3	M24 10.9 - 1	C LC_122	6.7	35.2	100.7	2.6	69.9	26.8	OK	OK
	B5	M24 10.9 - 1	C LC_119	3.5	10.1	235.3	1.4	8.6	8.2	OK	OK
	B6	M24 10.9 - 1	C LC_119	1.4	32.2	235.3	0.5	27.3	23.2	OK	OK
	B8	M24 10.9 - 1	C LC_153	0.0	12.7	225.3	0.0	11.3	9.0	OK	OK
	B9	M24 10.9 - 1	C LC_119	11.7	30.8	235.3	4.6	26.2	25.1	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M24 10.9 - 1	254.2	492.7	141.2

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
10 bracket web plate (SLN55 side)	20 bracket top plate a (SLN55 side)	-	129	-	-	-	-	-	-	-	-	OK	OK
		-	129	-	-	-	-	-	-	-	-	OK	OK
10 bracket web plate (SLN55 side)	20 bracket top plate b (SLN55 side)	-	129	-	-	-	-	-	-	-	-	OK	OK
		-	129	-	-	-	-	-	-	-	-	OK	OK
10 bracket web plate (SLN56 side)	20 bracket top plate a (SLN56 side)	-	129	-	-	-	-	-	-	-	-	OK	OK
		-	129	-	-	-	-	-	-	-	-	OK	OK
10 bracket web plate (SLN56 side)	20 bracket top plate b (SLN56 side)	-	129	-	-	-	-	-	-	-	-	OK	OK
		-	129	-	-	-	-	-	-	-	-	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate upper (SLN55 side)	◢ 7.0 ◣	144	C LC_119	171.4	0.0	78.6	79.6	37.3	42.3	29.5	OK	OK
		◢ 7.0 ◣	144	C LC_119	146.7	0.0	-77.8	63.2	-34.2	36.3	24.0	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	395x145x20 fin plate lower outer (SLN55 side)	◢ 7.0 ◣	69	C LC_122	255.7	0.0	6.4	-112.0	96.1	63.2	45.3	OK	OK
		◢ 7.0 ◣	69	C LC_122	396.9	0.2	-57.1	-4.2	-226.7	98.1	63.8	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower outer (SLN55 side)	◢ 7.0 ◣	68	C LC_119	118.3	0.0	47.8	61.8	-8.9	29.2	22.0	OK	OK
		◢ 7.0 ◣	69	C LC_119	151.3	0.0	87.9	-71.1	-2.6	37.4	27.7	OK	OK
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower outer (SLN55 side)	◢ 4.2 ◣	233	C LC_119	118.5	0.0	1.5	-1.5	68.4	29.3	20.5	OK	OK
		◢ 4.2 ◣	232	C LC_122	114.0	0.0	-29.7	8.3	63.0	28.2	19.6	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	395x145x20 fin plate lower inner (SLN55 side)	◢ 7.0 ◣	69	C LC_122	384.7	0.0	37.1	73.1	208.6	95.1	49.4	OK	OK
		◢ 7.0 ◣	69	C LC_122	397.0	0.3	49.0	-33.9	-224.9	98.1	55.0	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower inner (SLN55 side)	◢ 7.0 ◣	69	C LC_119	124.8	0.0	-21.1	-71.0	0.5	30.8	22.9	OK	OK
		◢ 7.0 ◣	69	C LC_119	151.8	0.0	-110.2	60.2	2.9	37.5	26.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower inner (SLN55 side)	◢ 4.2 ◣	232	C LC_119	97.2	0.0	-5.9	-2.7	55.9	24.0	16.9	OK	OK
		◢ 4.2 ◣	232	C LC_122	148.9	0.0	-17.5	17.7	83.5	36.8	23.8	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate upper (SLN56 side)	◢ 7.0 ◣	144	C LC_119	148.0	0.0	-77.6	-63.7	35.1	36.6	24.1	OK	OK
		◢ 7.0 ◣	144	C LC_119	173.8	0.0	80.3	-80.4	-38.1	42.9	29.9	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	395x145x20 fin plate lower outer (SLN56 side)	◢ 7.0 ◣	69	C LC_122	396.9	0.2	-56.9	4.1	226.7	98.1	63.8	OK	OK
		◢ 7.0 ◣	69	C LC_122	255.5	0.0	6.3	112.4	-95.4	63.1	45.3	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower outer (SLN56 side)	◢ 7.0 ◣	69	C LC_119	151.5	0.0	87.5	71.4	1.9	37.4	27.8	OK	OK
		◢ 7.0 ◣	68	C LC_119	120.9	0.0	49.6	-63.0	9.1	29.9	22.5	OK	OK
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower outer (SLN56 side)	◢ 4.2 ◣	232	C LC_122	113.6	0.0	-30.2	-8.9	-62.6	28.1	19.5	OK	OK
		◢ 4.2 ◣	233	C LC_119	118.5	0.0	1.6	1.3	-68.4	29.3	20.4	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	395x145x20 fin plate lower inner (SLN56 side)	◢ 7.0 ◣	69	C LC_122	397.0	0.3	49.0	33.7	225.0	98.1	55.1	OK	OK
		◢ 7.0 ◣	69	C LC_122	386.3	0.0	37.8	-74.1	-209.2	95.4	49.7	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower inner (SLN56 side)	◢ 7.0 ◣	69	C LC_119	152.2	0.0	-109.9	-60.7	-3.2	37.6	26.1	OK	OK
		◢ 7.0 ◣	69	C LC_119	127.1	0.0	-23.0	72.2	-0.1	31.4	23.3	OK	OK
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower inner (SLN56 side)	◢ 4.2 ◣	232	C LC_122	148.2	0.0	-17.9	-18.1	-83.0	36.6	23.9	OK	OK
		◢ 4.2 ◣	232	C LC_119	97.2	0.0	-5.9	2.7	-55.9	24.0	16.9	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	20 ring stiffener at 315 (SLN55 side)	◢ 8.5 ◣	147	C LC_119	396.6	0.0	74.7	145.5	-171.5	98.0	80.2	OK	OK
		◢ 8.5 ◣	146	C LC_119	285.5	0.0	-85.7	31.4	-154.1	70.6	49.3	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
ring SLN55 + SLN56 (continuous)-tfl 1	20 ring stiffener at 315 (SLN55 side)	◢ 8.5 ◣	146	C LC_122	240.3	0.0	-0.4	22.3	136.9	59.4	45.8	OK	OK
		◢ 8.5 ◣	146	C LC_153	397.3	0.4	-192.3	159.9	121.3	98.2	88.2	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	10 bracket web plate (SLN55 side)	◢ 4.2 ◣	288	C LC_122	397.3	0.4	-27.0	-33.1	-226.5	98.2	62.1	OK	OK
		◢ 4.2 ◣	288	C LC_122	397.4	0.5	4.0	-9.6	-229.2	98.2	69.1	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener a (SLN55 side)	◢ 7.0 ◣	67	C LC_119	403.8	4.4	132.0	-199.9	92.5	99.8	98.2	OK	OK
		◢ 7.0 ◣	67	C LC_122	402.4	3.6	146.9	207.9	-59.6	99.4	91.0	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener b (SLN55 side)	◢ 7.0 ◣	67	C LC_119	402.4	3.6	-113.7	196.4	105.2	99.4	93.6	OK	OK
		◢ 7.0 ◣	67	C LC_121	402.7	3.8	-137.3	-208.8	-64.6	99.5	88.2	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	20 ring stiffener at 315 (SLN56 side)	◢ 8.5 ◣	147	C LC_119	396.7	0.0	74.8	145.1	171.8	98.0	77.2	OK	OK
		◢ 8.5 ◣	146	C LC_119	286.4	0.0	-85.8	31.5	154.6	70.8	50.5	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 ring stiffener at 315 (SLN56 side)	◢ 8.5 ◣	146	C LC_122	240.1	0.0	-0.5	22.1	-136.9	59.3	46.7	OK	OK
		◢ 8.5 ◣	146	C LC_153	397.3	0.4	-192.4	160.0	-121.1	98.2	88.4	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	10 bracket web plate (SLN56 side)	◢ 4.2 ◣	288	C LC_122	397.4	0.5	4.1	9.3	229.3	98.2	69.1	OK	OK
		◢ 4.2 ◣	288	C LC_122	397.3	0.4	-26.7	32.7	226.5	98.2	62.0	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener a (SLN56 side)	◢ 7.0 ◣	67	C LC_119	402.4	3.6	-114.4	196.6	-104.6	99.4	94.5	OK	OK
		◢ 7.0 ◣	67	C LC_122	402.7	3.8	-137.2	-208.9	64.5	99.5	87.2	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener b (SLN56 side)	◢ 7.0 ◣	67	C LC_119	403.8	4.5	132.7	-200.1	-92.0	99.8	97.4	OK	OK
		◢ 7.0 ◣	67	C LC_122	402.4	3.6	147.5	207.8	59.4	99.4	92.7	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
395x145x20 fin plate upper (SLN55 side)	10 bracket web plate (SLN55 side)	◢ 4.2 ◣	167	C LC_119	399.4	1.7	-302.0	110.2	103.1	98.7	77.6	OK	OK
		◢ 4.2 ◣	167	C LC_119	400.3	2.3	134.9	187.7	110.1	98.9	80.8	OK	OK
395x145x20 fin plate upper (SLN56 side)	10 bracket web plate (SLN56 side)	◢ 4.2 ◣	167	C LC_119	400.3	2.3	135.4	-187.8	-109.8	98.9	80.5	OK	OK
		◢ 4.2 ◣	167	C LC_119	399.5	1.8	-301.2	-110.8	-103.4	98.7	77.7	OK	OK
20 bracket top plate a (SLN55 side)	20 inclined stiffener a (SLN55 side)	◢ 7.0 ◣	67	C LC_122	397.5	0.5	-48.7	2.4	227.7	98.2	79.0	OK	OK
		◢ 7.0 ◣	67	C LC_122	396.7	0.0	-55.3	16.9	226.1	98.0	67.8	OK	OK
20 bracket top plate b (SLN55 side)	20 inclined stiffener b (SLN55 side)	◢ 7.0 ◣	67	C LC_122	397.2	0.4	7.9	-31.9	227.1	98.2	77.4	OK	OK
		◢ 7.0 ◣	67	C LC_122	384.2	0.0	-68.0	31.7	216.0	94.9	67.5	OK	OK
20 bracket top plate a (SLN56 side)	20 inclined stiffener a (SLN56 side)	◢ 7.0 ◣	67	C LC_122	397.2	0.4	7.6	-31.8	-227.1	98.2	74.6	OK	OK
		◢ 7.0 ◣	67	C LC_122	384.2	0.0	-67.1	30.8	-216.2	94.9	62.2	OK	OK
20 bracket top plate b (SLN56 side)	20 inclined stiffener b (SLN56 side)	◢ 7.0 ◣	67	C LC_122	397.5	0.5	-48.8	2.2	-227.7	98.2	77.9	OK	OK
		◢ 7.0 ◣	67	C LC_122	396.7	0.0	-56.4	17.4	-226.0	98.0	69.9	OK	OK
10 bracket web plate (SLN55 side)	20 inclined stiffener a (SLN55 side)	◢ 4.2 ◣	222	C LC_119	398.5	1.2	-261.5	-147.1	-92.1	98.5	92.4	OK	OK
		◢ 4.2 ◣	222	C LC_119	398.6	1.2	-81.4	189.5	121.9	98.5	78.2	OK	OK
10 bracket web plate (SLN55 side)	20 inclined stiffener b (SLN55 side)	◢ 4.2 ◣	222	C LC_119	398.9	1.4	129.6	122.3	-180.2	98.6	93.5	OK	OK
		◢ 4.2 ◣	222	C LC_119	398.2	1.0	164.8	-174.8	115.1	98.4	71.2	OK	OK
10 bracket web plate (SLN56 side)	20 inclined stiffener a (SLN56 side)	◢ 4.2 ◣	222	C LC_119	398.9	1.4	130.0	122.5	180.0	98.6	93.3	OK	OK
		◢ 4.2 ◣	222	C LC_119	398.2	1.0	164.7	-174.8	-115.2	98.4	75.7	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
10 bracket web plate (SLN56 side)	20 inclined stiffener b (SLN56 side)	◢ 4.2 ◣	222	C LC_119	398.5	1.2	-261.2	-147.2	92.3	98.5	91.1	OK	OK
		◢ 4.2 ◣	222	C LC_119	398.6	1.2	-81.7	189.3	-122.0	98.5	76.8	OK	OK
395x145x20 fin plate upper (SLN55 side)	20 bracket top plate a (SLN55 side)	◢ 4.2 ◣	67	C LC_122	396.9	0.2	-199.1	-188.1	-62.6	98.1	71.0	OK	OK
		◢ 4.2 ◣	67	C LC_122	397.0	0.2	231.5	-164.0	-88.1	98.1	78.8	OK	OK
395x145x20 fin plate upper (SLN55 side)	20 bracket top plate b (SLN55 side)	◢ 4.2 ◣	67	C LC_122	402.8	3.8	-233.8	189.3	4.0	99.5	96.8	OK	OK
		◢ 4.2 ◣	67	C LC_122	402.8	3.8	-224.2	188.9	-40.6	99.5	95.2	OK	OK
395x145x20 fin plate upper (SLN56 side)	20 bracket top plate a (SLN56 side)	◢ 4.2 ◣	67	C LC_120	402.5	3.6	-234.3	188.9	-3.0	99.4	96.7	OK	OK
		◢ 4.2 ◣	67	C LC_120	402.5	3.6	-224.8	188.3	41.4	99.5	93.0	OK	OK
395x145x20 fin plate upper (SLN56 side)	20 bracket top plate b (SLN56 side)	◢ 4.2 ◣	67	C LC_122	396.9	0.2	-200.6	-187.3	63.2	98.1	71.8	OK	OK
		◢ 4.2 ◣	67	C LC_122	397.0	0.2	232.1	-163.6	88.4	98.1	76.3	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{C}	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R12-E1_000
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN73 radial (ends at the ring) - ring SLN55 + SLN56 (continuous)

Name	SLN73 radial (ends at the ring) - ring SLN55 + SLN56 (continuous)
Description	
Analysis	Stress, strain/ simplified loading

Design

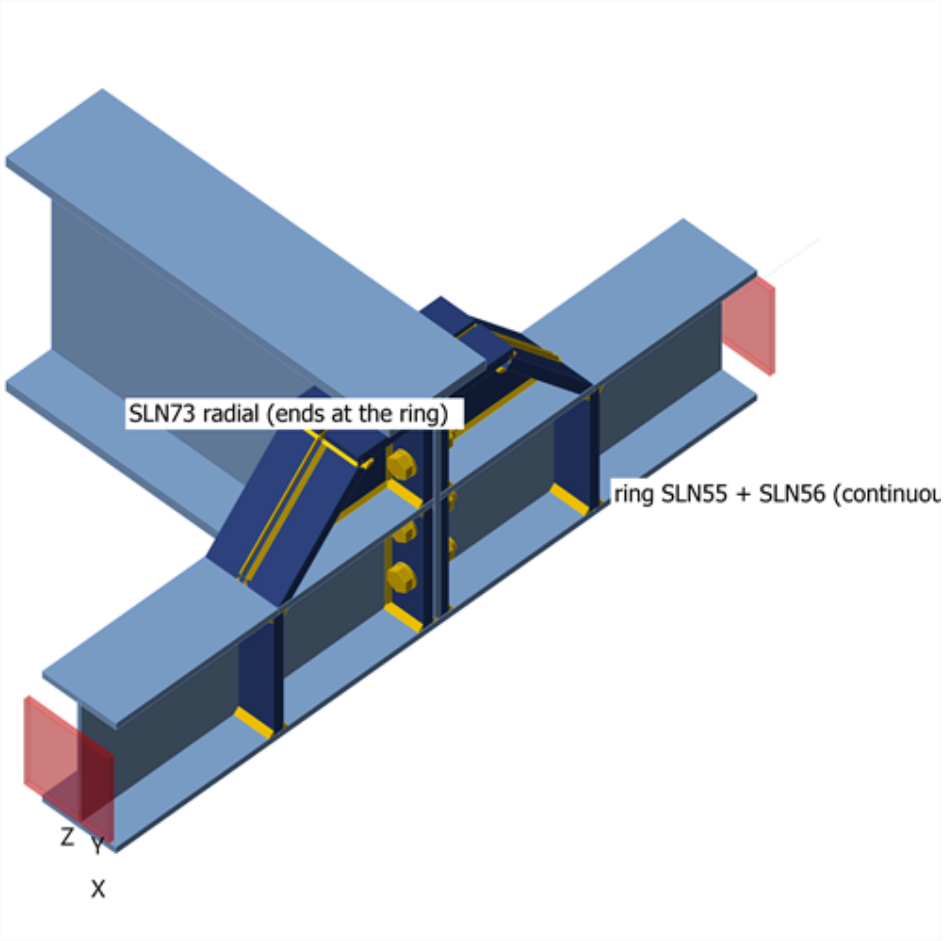
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN73 radial (ends at the ring)	1 - UB457x191x98	0.0	0.0	0.0	76	0	0
ring SLN55 + SLN56 (continuous)	2 - UB254x146x43	-90.0	0.0	0.0	0	0	-104

Supports and forces

Name	Support	Forces in	X [mm]
SLN73 radial (ends at the ring) / begin		Position	0
ring SLN55 + SLN56 (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
ring SLN55 + SLN56 (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0



SLN73 radial (ends at the ring) - ring SLN55 + SLN56 (continuous), Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M24 10.9	24	900.0	1000.0	452

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN73 radial (ends at the ring) / Begin	2.1	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_102	SLN73 radial (ends at the ring) / Begin	9.2	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_103	SLN73 radial (ends at the ring) / Begin	9.6	-0.1	-86.2	-0.1	-45.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_104	SLN73 radial (ends at the ring) / Begin	7.9	0.0	-86.4	-0.1	-45.4	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_105	SLN73 radial (ends at the ring) / Begin	8.8	-0.1	-86.3	-0.1	-45.4	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_106	SLN73 radial (ends at the ring) / Begin	8.0	0.0	-86.3	-0.1	-45.4	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_107	SLN73 radial (ends at the ring) / Begin	1.8	-0.1	-7.8	0.0	-4.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_108	SLN73 radial (ends at the ring) / Begin	1.7	0.1	-8.4	0.0	-4.6	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_109	SLN73 radial (ends at the ring) / Begin	1.4	0.1	-8.1	0.0	-4.5	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_110	SLN73 radial (ends at the ring) / Begin	0.6	-0.1	-8.1	0.0	-4.4	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_111	SLN73 radial (ends at the ring) / Begin	7.4	-0.1	-61.7	0.0	-32.6	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_112	SLN73 radial (ends at the ring) / Begin	6.3	0.1	-62.3	0.0	-32.8	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_113	SLN73 radial (ends at the ring) / Begin	6.7	0.0	-62.0	0.0	-32.7	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_114	SLN73 radial (ends at the ring) / Begin	5.6	-0.1	-62.0	0.0	-32.7	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_115	SLN73 radial (ends at the ring) / Begin	2.4	-0.1	-13.1	0.0	-7.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_116	SLN73 radial (ends at the ring) / Begin	2.1	0.1	-13.6	0.0	-7.4	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_117	SLN73 radial (ends at the ring) / Begin	2.0	0.0	-13.3	0.0	-7.3	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_118	SLN73 radial (ends at the ring) / Begin	1.1	-0.1	-13.3	0.0	-7.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_119	SLN73 radial (ends at the ring) / Begin	11.2	-0.1	-89.8	-0.1	-47.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_120	SLN73 radial (ends at the ring) / Begin	8.1	0.0	-89.8	-0.1	-47.2	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_121	SLN73 radial (ends at the ring) / Begin	8.5	-0.1	-89.8	-0.1	-47.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_122	SLN73 radial (ends at the ring) / Begin	7.6	0.0	-89.8	-0.1	-47.2	0.2
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_123	SLN73 radial (ends at the ring) / Begin	10.8	-0.1	-68.9	0.0	-36.3	0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_124	SLN73 radial (ends at the ring) / Begin	6.7	0.1	-68.9	0.0	-36.3	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_125	SLN73 radial (ends at the ring) / Begin	6.2	0.0	-68.9	0.0	-36.3	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_126	SLN73 radial (ends at the ring) / Begin	4.8	-0.1	-68.9	0.0	-36.3	0.2
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_127	SLN73 radial (ends at the ring) / Begin	5.7	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_128	SLN73 radial (ends at the ring) / Begin	2.5	0.1	-20.2	0.0	-10.9	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_129	SLN73 radial (ends at the ring) / Begin	1.5	0.0	-20.3	0.0	-10.9	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_130	SLN73 radial (ends at the ring) / Begin	0.3	-0.1	-20.3	0.0	-10.9	0.2
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_135	SLN73 radial (ends at the ring) / Begin	7.7	0.0	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_141	SLN73 radial (ends at the ring) / Begin	2.3	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_142	SLN73 radial (ends at the ring) / Begin	1.8	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_143	SLN73 radial (ends at the ring) / Begin	2.1	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_144	SLN73 radial (ends at the ring) / Begin	2.0	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_145	SLN73 radial (ends at the ring) / Begin	10.1	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_146	SLN73 radial (ends at the ring) / Begin	8.3	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_147	SLN73 radial (ends at the ring) / Begin	9.4	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_148	SLN73 radial (ends at the ring) / Begin	9.0	0.0	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_151	SLN73 radial (ends at the ring) / Begin	6.9	-0.1	-46.4	0.0	-22.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_152	SLN73 radial (ends at the ring) / Begin	9.3	0.0	-63.5	0.0	-35.9	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_153	SLN73 radial (ends at the ring) / Begin	9.2	-0.1	-89.4	0.0	-47.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_154	SLN73 radial (ends at the ring) / Begin	7.0	-0.1	-46.8	0.0	-22.5	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	3.0 < 5.0%	OK
Bolts	69.8 < 100%	OK
Welds	100.6 > 100%	**Not OK!**
Buckling	Not calculated	

Plates

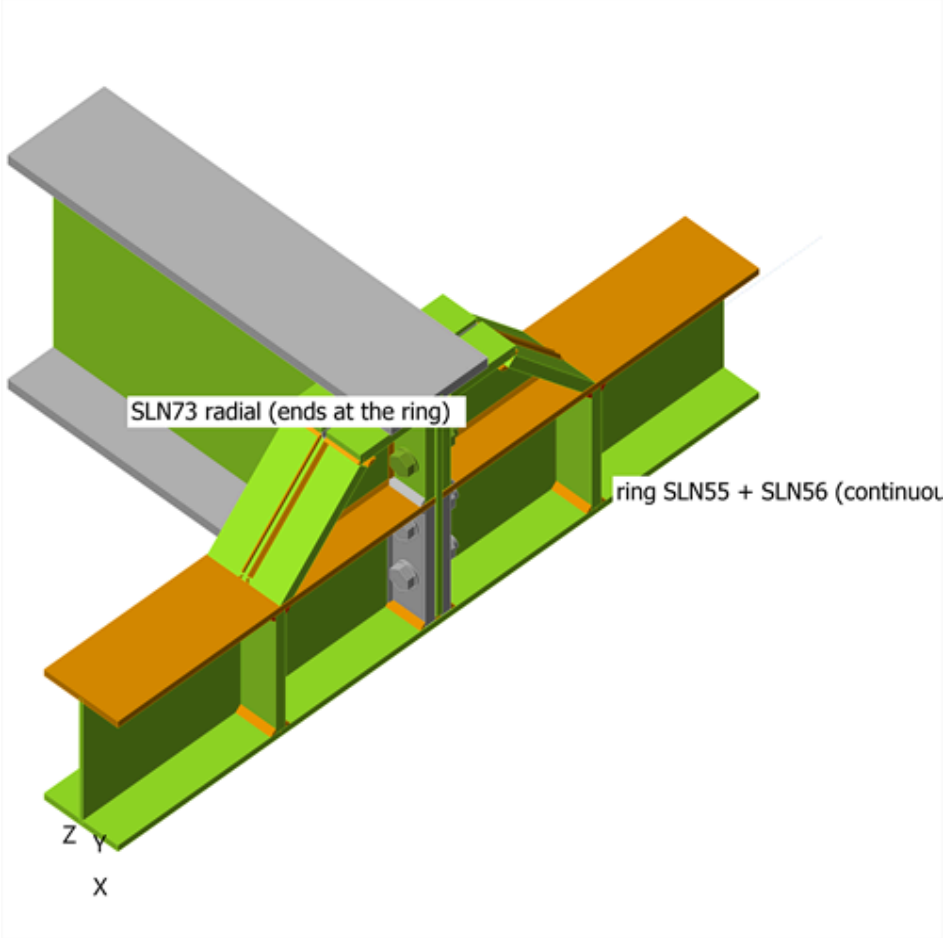
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN73 radial (ends at the ring)-bfl 1	19.6	C LC_122	14.8	0.0	0.0	OK
SLN73 radial (ends at the ring)-tfl 1	19.6	C LC_119	136.9	0.0	5.9	OK
SLN73 radial (ends at the ring)-w 1	11.4	C LC_122	242.8	0.0	70.4	OK
ring SLN55 + SLN56 (continuous)-bfl 1	12.7	C LC_122	280.9	2.8	0.0	OK
ring SLN55 + SLN56 (continuous)-tfl 1	12.7	C LC_122	281.3	3.0	0.0	OK
ring SLN55 + SLN56 (continuous)-w 1	7.2	C LC_122	275.4	0.2	0.0	OK
395x145x20 fin plate upper (SLN55 side)	20.0	C LC_122	275.1	0.1	473.8	OK
395x145x20 fin plate lower outer (SLN55 side)	20.0	C LC_120	211.1	0.0	203.5	OK
395x145x20 fin plate lower inner (SLN55 side)	20.0	C LC_122	211.3	0.0	40.0	OK
395x145x20 fin plate upper (SLN56 side)	20.0	C LC_135	275.1	0.1	470.9	OK
395x145x20 fin plate lower outer (SLN56 side)	20.0	C LC_122	213.1	0.0	204.8	OK
395x145x20 fin plate lower inner (SLN56 side)	20.0	C LC_122	211.2	0.0	39.6	OK
20 ring stiffener at 315 (SLN55 side)	20.0	C LC_122	275.9	0.4	0.0	OK
10 bracket web plate (SLN55 side)	10.0	C LC_122	275.5	0.3	0.0	OK
20 bracket top plate a (SLN55 side)	20.0	C LC_119	275.0	0.0	105.1	OK
20 inclined stiffener a (SLN55 side)	20.0	C LC_122	275.4	0.2	0.0	OK
20 bracket top plate b (SLN55 side)	20.0	C LC_122	275.9	0.4	0.0	OK
20 inclined stiffener b (SLN55 side)	20.0	C LC_122	275.5	0.2	0.0	OK
20 ring stiffener at 315 (SLN56 side)	20.0	C LC_122	275.9	0.4	0.0	OK
10 bracket web plate (SLN56 side)	10.0	C LC_122	275.5	0.3	0.0	OK
20 bracket top plate a (SLN56 side)	20.0	C LC_120	275.9	0.4	0.0	OK
20 inclined stiffener a (SLN56 side)	20.0	C LC_122	275.5	0.2	0.0	OK
20 bracket top plate b (SLN56 side)	20.0	C LC_119	275.0	0.0	105.1	OK
20 inclined stiffener b (SLN56 side)	20.0	C LC_122	275.4	0.2	0.0	OK

Design data

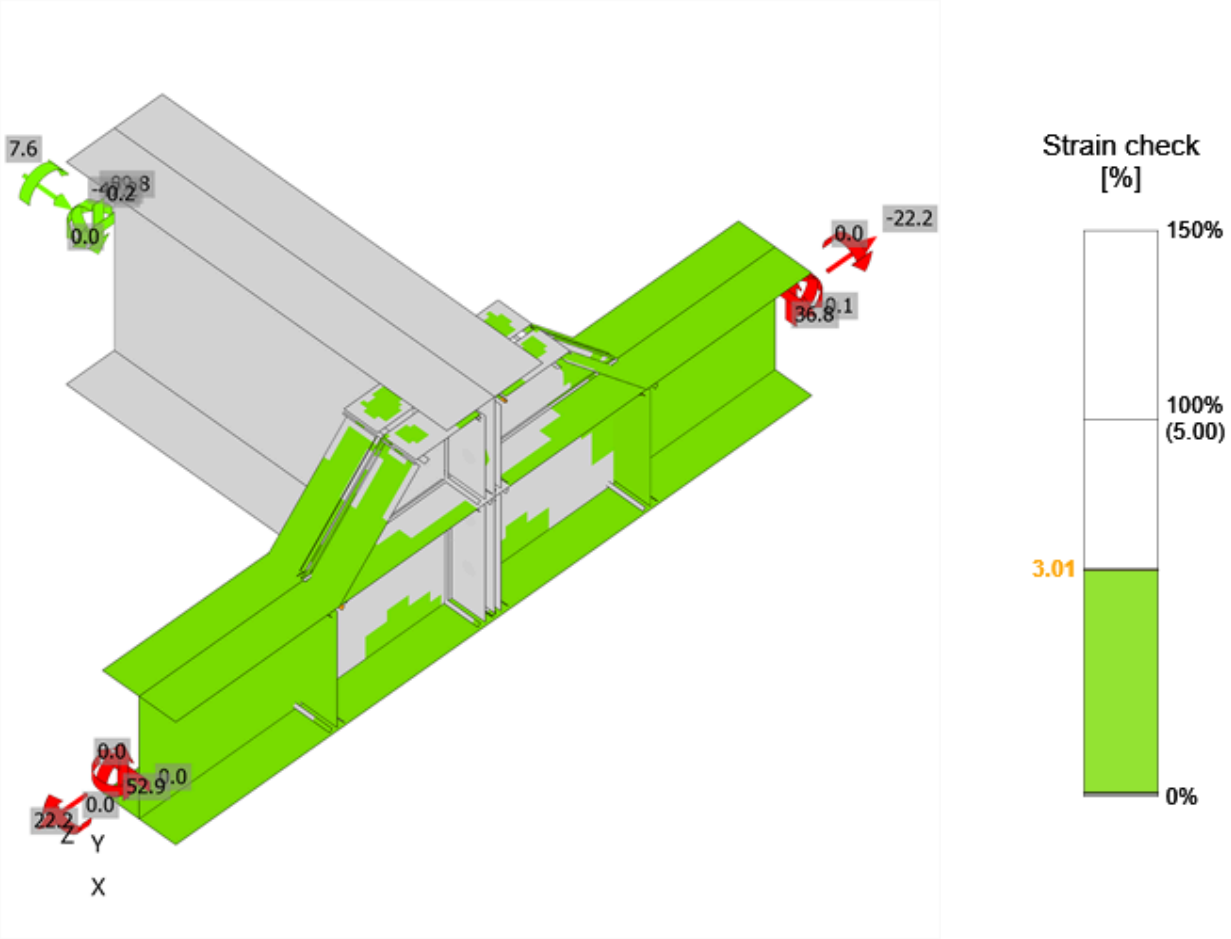
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

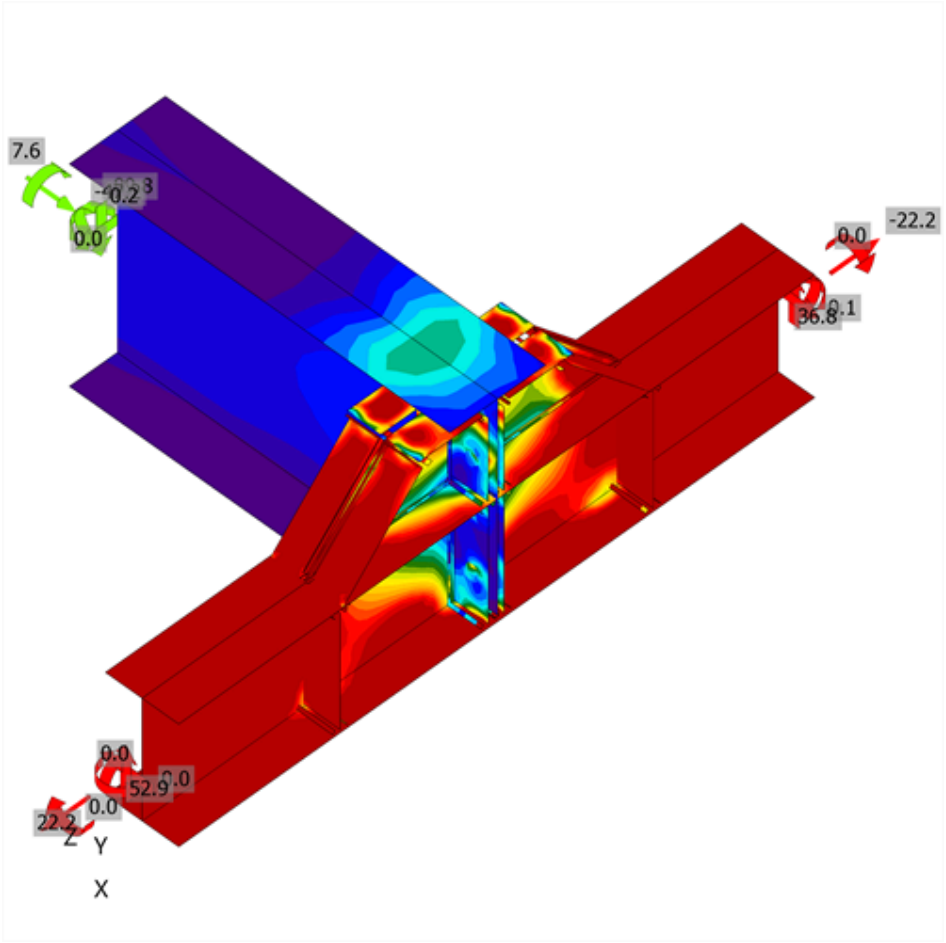
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

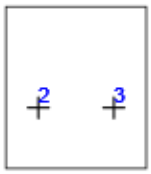
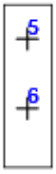
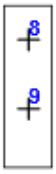


Strain check, C LC_122



Equivalent stress, C LC_122

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M24 10.9 - 1	C LC_122	46.0	34.5	235.3	18.1	29.2	37.4	OK	OK
	B3	M24 10.9 - 1	C LC_122	6.6	35.2	100.7	2.6	69.8	26.8	OK	OK
	B5	M24 10.9 - 1	C LC_119	3.6	10.2	235.3	1.4	8.7	8.2	OK	OK
	B6	M24 10.9 - 1	C LC_119	1.4	32.3	235.3	0.5	27.4	23.3	OK	OK
	B8	M24 10.9 - 1	C LC_153	0.0	12.7	224.9	0.0	11.2	9.0	OK	OK
	B9	M24 10.9 - 1	C LC_119	12.1	30.9	235.3	4.7	26.2	25.3	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M24 10.9 - 1	254.2	492.7	141.2

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
10 bracket web plate (SLN55 side)	20 bracket top plate a (SLN55 side)	-	129	-	-	-	-	-	-	-	-	OK	OK
		-	129	-	-	-	-	-	-	-	-	OK	OK
10 bracket web plate (SLN55 side)	20 bracket top plate b (SLN55 side)	-	129	-	-	-	-	-	-	-	-	OK	OK
		-	129	-	-	-	-	-	-	-	-	OK	OK
10 bracket web plate (SLN56 side)	20 bracket top plate a (SLN56 side)	-	129	-	-	-	-	-	-	-	-	OK	OK
		-	129	-	-	-	-	-	-	-	-	OK	OK
10 bracket web plate (SLN56 side)	20 bracket top plate b (SLN56 side)	-	129	-	-	-	-	-	-	-	-	OK	OK
		-	129	-	-	-	-	-	-	-	-	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate upper (SLN55 side)	◢ 7.0 ◣	144	C LC_119	163.6	0.0	75.2	76.5	34.3	40.4	28.3	OK	OK
		◢ 7.0 ◣	144	C LC_119	139.7	0.0	-76.6	59.9	-31.0	34.5	23.1	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	395x145x20 fin plate lower outer (SLN55 side)	◢ 7.0 ◣	69	C LC_122	268.5	0.0	9.4	-120.3	97.6	66.4	47.2	OK	OK
		◢ 7.0 ◣	69	C LC_120	397.0	0.2	-56.7	-4.4	-226.8	98.1	64.3	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower outer (SLN55 side)	◢ 7.0 ◣	68	C LC_119	113.2	0.0	45.3	59.3	-8.5	28.0	21.2	OK	OK
		◢ 7.0 ◣	69	C LC_119	145.9	0.0	85.1	-68.3	-4.0	36.1	26.8	OK	OK
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower outer (SLN55 side)	◢ 4.2 ◣	233	C LC_119	116.1	0.0	1.5	-1.4	67.0	28.7	20.7	OK	OK
		◢ 4.2 ◣	232	C LC_122	118.8	0.0	-31.7	9.1	65.4	29.3	20.4	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	395x145x20 fin plate lower inner (SLN55 side)	◢ 7.0 ◣	69	C LC_120	396.6	0.0	36.7	80.9	213.2	98.0	51.4	OK	OK
		◢ 7.0 ◣	69	C LC_122	397.1	0.3	50.2	-32.8	-225.1	98.1	55.6	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower inner (SLN55 side)	◢ 7.0 ◣	69	C LC_119	120.1	0.0	-17.5	-68.6	1.6	29.7	22.2	OK	OK
		◢ 7.0 ◣	69	C LC_119	146.3	0.0	-108.0	56.9	1.2	36.1	25.2	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower inner (SLN55 side)	◢ 4.2 ◣	232	C LC_119	95.3	0.0	-5.6	-2.7	54.8	23.5	16.8	OK	OK
		◢ 4.2 ◣	232	C LC_122	154.2	0.0	-17.9	18.4	86.5	38.1	24.4	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate upper (SLN56 side)	◢ 7.0 ◣	144	C LC_119	140.9	0.0	-76.4	-60.4	31.9	34.8	23.2	OK	OK
		◢ 7.0 ◣	144	C LC_119	166.0	0.0	76.9	-77.4	-35.1	41.0	28.7	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	395x145x20 fin plate lower outer (SLN56 side)	◢ 7.0 ◣	69	C LC_122	397.0	0.2	-56.6	4.4	226.8	98.1	64.4	OK	OK
		◢ 7.0 ◣	69	C LC_122	268.4	0.0	9.4	120.8	-96.9	66.3	47.2	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower outer (SLN56 side)	◢ 7.0 ◣	69	C LC_119	146.1	0.0	84.7	68.6	3.3	36.1	26.9	OK	OK
		◢ 7.0 ◣	68	C LC_119	115.9	0.0	47.2	-60.5	8.7	28.6	21.7	OK	OK
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower outer (SLN56 side)	◢ 4.2 ◣	232	C LC_122	118.5	0.0	-32.2	-9.7	-65.1	29.3	20.3	OK	OK
		◢ 4.2 ◣	233	C LC_119	116.1	0.0	1.6	1.3	-67.0	28.7	20.6	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	395x145x20 fin plate lower inner (SLN56 side)	◢ 7.0 ◣	69	C LC_122	397.1	0.3	50.2	32.7	225.1	98.1	55.6	OK	OK
		◢ 7.0 ◣	69	C LC_122	396.6	0.0	37.1	-81.4	-213.0	98.0	51.5	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower inner (SLN56 side)	◢ 7.0 ◣	69	C LC_119	146.7	0.0	-107.7	-57.4	-1.6	36.2	25.3	OK	OK
		◢ 7.0 ◣	69	C LC_119	122.3	0.0	-19.4	69.7	-1.3	30.2	22.6	OK	OK
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower inner (SLN56 side)	◢ 4.2 ◣	232	C LC_122	153.5	0.0	-18.3	-18.8	-86.0	37.9	24.5	OK	OK
		◢ 4.2 ◣	232	C LC_122	95.6	0.0	18.5	-19.1	50.6	23.6	16.6	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	20 ring stiffener at 315 (SLN55 side)	◢ 7.0 ◣	146	C LC_119	396.9	0.2	91.6	129.1	-181.8	98.1	85.9	OK	OK
		◢ 7.0 ◣	146	C LC_120	338.4	0.0	-108.3	41.1	-180.5	83.6	57.7	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
ring SLN55 + SLN56 (continuous)-tfl 1	20 ring stiffener at 315 (SLN55 side)	◢ 7.0 ◣	146	C LC_153	396.8	0.1	153.0	-209.5	-27.8	98.0	85.4	OK	OK
		◢ 7.0 ◣	146	C LC_153	407.2	6.5	-203.4	199.6	40.3	100.6	99.3	OK	**Not OK!**
ring SLN55 + SLN56 (continuous)-tfl 1	10 bracket web plate (SLN55 side)	◢ 4.2 ◣	288	C LC_122	397.3	0.4	-47.9	-37.1	-224.6	98.2	60.8	OK	OK
		◢ 4.2 ◣	288	C LC_122	397.4	0.5	18.4	-8.6	-229.0	98.2	68.8	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener a (SLN55 side)	◢ 7.0 ◣	67	C LC_119	402.8	3.8	171.9	-184.4	101.2	99.5	96.9	OK	OK
		◢ 7.0 ◣	67	C LC_122	399.9	2.0	148.6	193.9	-91.3	98.8	86.0	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener b (SLN55 side)	◢ 7.0 ◣	67	C LC_119	402.6	3.7	-142.6	189.5	106.7	99.5	92.8	OK	OK
		◢ 7.0 ◣	67	C LC_122	401.2	2.8	-141.1	-200.2	-83.3	99.1	84.2	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	20 ring stiffener at 315 (SLN56 side)	◢ 7.0 ◣	146	C LC_119	396.9	0.2	91.1	128.4	182.4	98.1	85.5	OK	OK
		◢ 7.0 ◣	146	C LC_122	339.2	0.0	-108.7	41.3	180.9	83.8	59.9	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 ring stiffener at 315 (SLN56 side)	◢ 7.0 ◣	146	C LC_153	396.8	0.1	151.7	-209.8	28.0	98.0	85.5	OK	OK
		◢ 7.0 ◣	146	C LC_119	407.2	6.5	-203.8	199.6	-39.7	100.6	99.3	OK	**Not OK!**
ring SLN55 + SLN56 (continuous)-tfl 1	10 bracket web plate (SLN56 side)	◢ 4.2 ◣	288	C LC_122	397.4	0.5	18.4	8.4	229.0	98.2	68.7	OK	OK
		◢ 4.2 ◣	288	C LC_122	397.3	0.4	-47.8	36.8	224.7	98.2	60.8	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener a (SLN56 side)	◢ 7.0 ◣	67	C LC_119	402.7	3.7	-143.4	189.5	-106.2	99.5	94.4	OK	OK
		◢ 7.0 ◣	67	C LC_122	401.2	2.8	-141.5	-200.0	83.4	99.1	80.7	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener b (SLN56 side)	◢ 7.0 ◣	67	C LC_119	402.9	3.9	172.9	-184.4	-100.7	99.5	96.5	OK	OK
		◢ 7.0 ◣	67	C LC_122	399.9	2.0	149.2	193.8	91.3	98.8	88.6	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
395x145x20 fin plate upper (SLN55 side)	10 bracket web plate (SLN55 side)	◢ 4.2 ◣	167	C LC_119	399.3	1.7	-302.5	109.5	103.2	98.7	76.5	OK	OK
		◢ 4.2 ◣	167	C LC_119	400.2	2.2	133.2	187.4	111.1	98.9	80.6	OK	OK
395x145x20 fin plate upper (SLN56 side)	10 bracket web plate (SLN56 side)	◢ 4.2 ◣	167	C LC_119	400.2	2.2	133.6	-187.5	-110.8	98.9	80.3	OK	OK
		◢ 4.2 ◣	167	C LC_119	399.3	1.7	-301.7	-110.1	-103.5	98.7	76.6	OK	OK
20 bracket top plate a (SLN55 side)	20 inclined stiffener a (SLN55 side)	◢ 7.0 ◣	67	C LC_122	397.4	0.5	-48.4	3.0	227.7	98.2	78.1	OK	OK
		◢ 7.0 ◣	67	C LC_122	396.6	0.0	-55.7	16.1	226.2	98.0	67.4	OK	OK
20 bracket top plate b (SLN55 side)	20 inclined stiffener b (SLN55 side)	◢ 7.0 ◣	67	C LC_122	397.1	0.3	6.6	-33.4	226.8	98.1	76.9	OK	OK
		◢ 7.0 ◣	67	C LC_122	379.1	0.0	-66.8	30.6	213.3	93.7	66.2	OK	OK
20 bracket top plate a (SLN56 side)	20 inclined stiffener a (SLN56 side)	◢ 7.0 ◣	67	C LC_122	397.1	0.3	6.3	-33.4	-226.8	98.1	73.3	OK	OK
		◢ 7.0 ◣	67	C LC_122	379.0	0.0	-65.9	29.8	-213.4	93.6	60.9	OK	OK
20 bracket top plate b (SLN56 side)	20 inclined stiffener b (SLN56 side)	◢ 7.0 ◣	67	C LC_122	397.4	0.5	-48.5	2.9	-227.7	98.2	76.6	OK	OK
		◢ 7.0 ◣	67	C LC_122	396.6	0.0	-56.8	16.7	-226.0	98.0	69.7	OK	OK
10 bracket web plate (SLN55 side)	20 inclined stiffener a (SLN55 side)	◢ 4.2 ◣	222	C LC_119	398.4	1.1	-262.2	-147.0	-91.5	98.4	92.3	OK	OK
		◢ 4.2 ◣	222	C LC_119	398.5	1.2	-80.7	189.2	122.3	98.5	78.9	OK	OK
10 bracket web plate (SLN55 side)	20 inclined stiffener b (SLN55 side)	◢ 4.2 ◣	222	C LC_119	398.8	1.3	128.1	121.5	-181.0	98.5	93.5	OK	OK
		◢ 4.2 ◣	222	C LC_119	398.1	0.9	165.7	-174.7	114.8	98.4	69.6	OK	OK
10 bracket web plate (SLN56 side)	20 inclined stiffener a (SLN56 side)	◢ 4.2 ◣	222	C LC_119	398.8	1.4	128.5	121.7	180.8	98.5	93.3	OK	OK
		◢ 4.2 ◣	222	C LC_119	398.1	0.9	165.5	-174.6	-114.9	98.4	75.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
10 bracket web plate (SLN56 side)	20 inclined stiffener b (SLN56 side)	◢ 4.2 ◣	222	C LC_119	398.4	1.1	-261.9	-147.1	91.6	98.4	90.9	OK	OK
		◢ 4.2 ◣	222	C LC_119	398.5	1.2	-80.9	189.1	-122.4	98.5	77.4	OK	OK
395x145x20 fin plate upper (SLN55 side)	20 bracket top plate a (SLN55 side)	◢ 4.2 ◣	67	C LC_122	396.9	0.2	-206.2	-185.7	-62.0	98.1	70.6	OK	OK
		◢ 4.2 ◣	67	C LC_122	396.9	0.2	228.2	-165.7	-87.8	98.1	78.5	OK	OK
395x145x20 fin plate upper (SLN55 side)	20 bracket top plate b (SLN55 side)	◢ 4.2 ◣	67	C LC_122	402.5	3.6	-231.7	190.0	3.5	99.5	96.1	OK	OK
		◢ 4.2 ◣	67	C LC_122	402.5	3.6	-221.5	189.7	-40.9	99.5	94.3	OK	OK
395x145x20 fin plate upper (SLN56 side)	20 bracket top plate a (SLN56 side)	◢ 4.2 ◣	67	C LC_120	402.2	3.4	-232.1	189.6	-2.4	99.4	95.9	OK	OK
		◢ 4.2 ◣	67	C LC_120	402.2	3.5	-222.1	189.0	41.7	99.4	92.3	OK	OK
395x145x20 fin plate upper (SLN56 side)	20 bracket top plate b (SLN56 side)	◢ 4.2 ◣	67	C LC_122	396.9	0.2	-207.7	-184.9	62.6	98.1	71.1	OK	OK
		◢ 4.2 ◣	67	C LC_122	396.9	0.2	228.7	-165.3	88.1	98.1	76.3	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{C}	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

IDEA StatiCa Connection Reports

26016 Concourse Viewing Platform Date: 13.11.2026

The steel connections at radius 8.5 m were verified with IDEA StatiCa Connection 26.0 (CBFEM) for the ULS combinations of this report and were found to be satisfactory.

Joint	Location	Plates (strain %, limit 5)	Bolts %	Welds %	Status	Page
L1-R8-A_000	Level 1, radius 8.5 m, 0°	0.0	7.5	69.2	OK	2
L1-R8-A_080	Level 1, radius 8.5 m, 80°	0.0	7.2	58.9	OK	13
L1-R8-A_120	Level 1, radius 8.5 m, 120°	0.0	7.2	71.4	OK	24
L1-R8-A_160	Level 1, radius 8.5 m, 160°	0.0	7.6	72.7	OK	35
L1-R8-A_200	Level 1, radius 8.5 m, 200°	0.0	7.2	60.7	OK	46
L1-R8-A_240	Level 1, radius 8.5 m, 240°	0.0	7.1	68.5	OK	57
L1-R8-A_320	Level 1, radius 8.5 m, 320°	0.0	6.1	60.5	OK	68
L1-R8-B_040	Level 1, radius 8.5 m, 40°	0.0	19.3	63.1	OK	79
L1-R8-B_280	Level 1, radius 8.5 m, 280°	0.2	19.1	72.4	OK	91
L1-R8-S_020	Level 1, radius 8.5 m, 20°	0.1	21.3	98.1	OK	103
L1-R8-S_100	Level 1, radius 8.5 m, 100°	0.1	16.9	98.1	OK	118
L1-R8-S_140	Level 1, radius 8.5 m, 140°	0.1	14.6	98.1	OK	133
L1-R8-S_180	Level 1, radius 8.5 m, 180°	0.1	18.1	98.2	OK	148
L1-R8-S_220	Level 1, radius 8.5 m, 220°	0.0	14.4	98.1	OK	163
L1-R8-S_300	Level 1, radius 8.5 m, 300°	0.1	18.5	98.1	OK	178
L1-R8-S_340	Level 1, radius 8.5 m, 340°	0.0	15.5	98.0	OK	193
L1-R8-T_060	Level 1, radius 8.5 m, 60°	0.9	17.1	98.8	OK	208
L1-R8-T_260	Level 1, radius 8.5 m, 260°	1.0	17.1	98.8	OK	224
L2-R8-A_000	Level 2, radius 8.5 m, 0°	0.0	0.9	18.1	OK	240
L2-R8-A_020	Level 2, radius 8.5 m, 20°	0.0	0.6	17.7	OK	253
L2-R8-A_080	Level 2, radius 8.5 m, 80°	0.0	1.3	20.8	OK	266
L2-R8-A_100	Level 2, radius 8.5 m, 100°	0.0	0.9	20.0	OK	279
L2-R8-A_120	Level 2, radius 8.5 m, 120°	0.0	0.9	19.7	OK	292
L2-R8-A_140	Level 2, radius 8.5 m, 140°	0.0	0.9	20.0	OK	305
L2-R8-A_160	Level 2, radius 8.5 m, 160°	0.0	0.9	19.6	OK	318
L2-R8-A_180	Level 2, radius 8.5 m, 180°	0.0	0.9	19.6	OK	331
L2-R8-A_200	Level 2, radius 8.5 m, 200°	0.0	0.9	16.6	OK	344
L2-R8-A_220	Level 2, radius 8.5 m, 220°	0.0	0.9	16.9	OK	357
L2-R8-A_240	Level 2, radius 8.5 m, 240°	0.0	1.5	19.4	OK	370
L2-R8-A_300	Level 2, radius 8.5 m, 300°	0.0	0.6	18.2	OK	383
L2-R8-A_320	Level 2, radius 8.5 m, 320°	0.0	0.9	17.9	OK	396
L2-R8-A_340	Level 2, radius 8.5 m, 340°	0.0	0.9	18.0	OK	409
L2-R8-B_040	Level 2, radius 8.5 m, 40°	0.0	9.4	36.9	OK	422
L2-R8-B_060	Level 2, radius 8.5 m, 60°	0.0	5.1	50.1	OK	438
L2-R8-B_260	Level 2, radius 8.5 m, 260°	0.0	4.5	43.0	OK	453
L2-R8-B_280	Level 2, radius 8.5 m, 280°	0.0	8.9	36.4	OK	468

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-A_000
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN180+SLN169 radial (continuous) - SLN38 column below - SLN185 ring - SLN227 ring

Name	SLN180+SLN169 radial (continuous) - SLN38 column below - SLN185 ring - SLN227 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

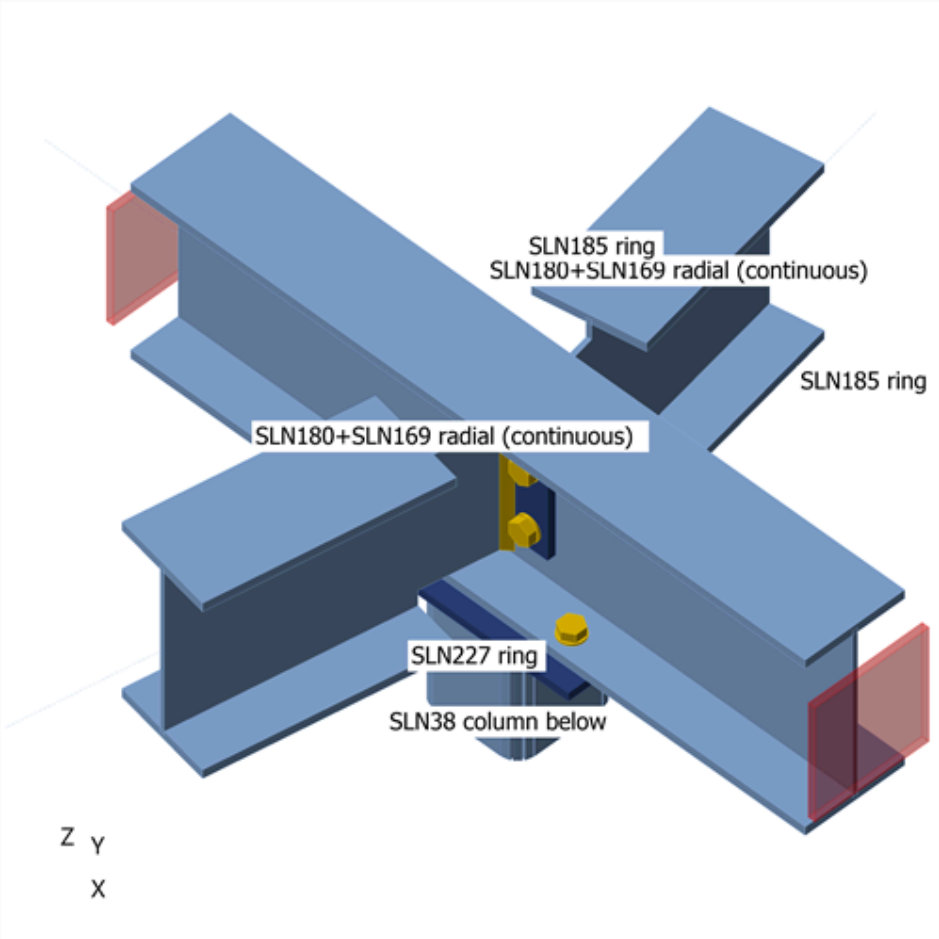
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN180+SLN169 radial (continuous)	1 - UB254x146x43	0.0	0.0	0.0	0	0	0
SLN38 column below	2 - SHS150x150x10	0.0	90.0	180.0	0	0	0
SLN185 ring	1 - UB254x146x43	100.0	0.0	0.0	0	0	0
SLN227 ring	1 - UB254x146x43	-100.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN180+SLN169 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN180+SLN169 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN38 column below / begin		Position	0
SLN185 ring / end		Position	0
SLN227 ring / end		Position	0



SLN180+SLN169 radial (continuous) - SLN38 column below - SLN185 ring - SLN227 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN38 column below / Begin	28.5	0.0	-0.1	0.0	-0.5	0.0
	SLN185 ring / End	6.3	0.2	-4.9	0.0	0.0	0.0
	SLN227 ring / End	6.2	0.0	-4.9	0.0	0.0	0.0
C LC_102	SLN38 column below / Begin	142.8	0.0	-0.8	0.0	-3.1	0.0
	SLN185 ring / End	18.2	0.6	-25.6	0.0	0.0	0.0
	SLN227 ring / End	17.8	-0.2	-25.8	0.0	0.0	0.0
C LC_103	SLN38 column below / Begin	137.0	0.0	-0.8	0.0	-3.1	0.0
	SLN185 ring / End	9.4	0.6	-24.5	0.0	0.0	0.0
	SLN227 ring / End	9.0	-0.1	-24.7	0.0	0.0	0.0
C LC_104	SLN38 column below / Begin	136.5	0.0	-3.5	0.0	-4.8	0.0
	SLN185 ring / End	18.9	0.7	-24.5	0.0	0.0	0.0
	SLN227 ring / End	18.6	-0.2	-24.7	0.0	0.0	0.0
C LC_105	SLN38 column below / Begin	136.9	2.3	-0.7	0.0	-3.0	0.0
	SLN185 ring / End	12.3	0.6	-24.5	0.0	0.0	0.0
	SLN227 ring / End	9.6	-0.2	-24.7	0.0	0.0	0.0
C LC_106	SLN38 column below / Begin	137.0	-2.3	-0.7	0.0	-3.0	0.0
	SLN185 ring / End	10.1	0.6	-24.5	0.0	0.0	0.0
	SLN227 ring / End	12.2	-0.1	-24.7	0.0	0.0	0.0
C LC_107	SLN38 column below / Begin	10.0	0.0	-0.1	0.0	-0.3	0.0
	SLN185 ring / End	-5.1	0.2	-1.6	0.0	0.0	0.0
	SLN227 ring / End	-5.1	0.0	-1.6	0.0	0.0	0.0
C LC_108	SLN38 column below / Begin	8.8	0.0	-5.5	0.0	-3.9	0.0
	SLN185 ring / End	9.7	0.1	-1.6	0.0	0.0	0.0
	SLN227 ring / End	9.8	0.0	-1.6	0.0	0.0	0.0
C LC_109	SLN38 column below / Begin	9.6	4.6	-0.1	0.0	-0.2	0.0
	SLN185 ring / End	-1.5	0.0	-1.6	0.0	0.0	0.0
	SLN227 ring / End	-6.0	-0.1	-1.6	0.0	0.0	0.0
C LC_110	SLN38 column below / Begin	9.9	-4.6	0.0	0.0	-0.2	0.0
	SLN185 ring / End	-5.5	0.0	-1.6	0.0	0.0	0.0
	SLN227 ring / End	-1.2	0.1	-1.6	0.0	0.0	0.0
C LC_111	SLN38 column below / Begin	97.3	0.0	-0.6	0.0	-2.3	0.0
	SLN185 ring / End	2.2	0.5	-17.3	0.0	0.0	0.0
	SLN227 ring / End	1.9	-0.1	-17.4	0.0	0.0	0.0
C LC_112	SLN38 column below / Begin	96.1	0.0	-6.0	0.0	-5.8	0.0
	SLN185 ring / End	18.4	0.5	-17.3	0.0	0.0	0.0
	SLN227 ring / End	18.2	-0.1	-17.4	0.0	0.0	0.0
C LC_113	SLN38 column below / Begin	96.9	4.6	-0.5	0.0	-2.2	0.0
	SLN185 ring / End	6.5	0.4	-17.3	0.0	0.0	0.0
	SLN227 ring / End	1.7	-0.2	-17.4	0.0	0.0	0.0
C LC_114	SLN38 column below / Begin	97.2	-4.6	-0.5	0.0	-2.1	0.0
	SLN185 ring / End	2.3	0.4	-17.3	0.0	0.0	0.0
	SLN227 ring / End	6.6	0.0	-17.4	0.0	0.0	0.0
C LC_115	SLN38 column below / Begin	17.4	0.0	-0.1	0.0	-0.4	0.0
	SLN185 ring / End	-3.5	0.2	-2.8	0.0	0.0	0.0
	SLN227 ring / End	-3.6	0.0	-2.9	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN38 column below / Begin	16.2	0.0	-5.6	0.0	-4.0	0.0
	SLN185 ring / End	11.4	0.2	-2.8	0.0	0.0	0.0
	SLN227 ring / End	11.4	0.0	-2.9	0.0	0.0	0.0
C LC_117	SLN38 column below / Begin	17.0	4.6	-0.1	0.0	-0.4	0.0
	SLN185 ring / End	0.1	0.1	-2.8	0.0	0.0	0.0
	SLN227 ring / End	-4.4	-0.1	-2.9	0.0	0.0	0.0
C LC_118	SLN38 column below / Begin	17.3	-4.6	-0.1	0.0	-0.3	0.0
	SLN185 ring / End	-3.9	0.1	-2.8	0.0	0.0	0.0
	SLN227 ring / End	0.4	0.1	-2.9	0.0	0.0	0.0
C LC_119	SLN38 column below / Begin	142.6	0.0	-0.8	0.0	-3.2	0.0
	SLN185 ring / End	11.3	0.6	-25.6	0.0	0.0	0.0
	SLN227 ring / End	11.0	-0.1	-25.8	0.0	0.0	0.0
C LC_120	SLN38 column below / Begin	142.3	0.0	-3.5	0.0	-5.0	0.0
	SLN185 ring / End	20.9	0.7	-25.6	0.0	0.0	0.0
	SLN227 ring / End	20.6	-0.2	-25.8	0.0	0.0	0.0
C LC_121	SLN38 column below / Begin	142.6	2.3	-0.8	0.0	-3.2	0.0
	SLN185 ring / End	14.3	0.6	-25.6	0.0	0.0	0.0
	SLN227 ring / End	11.5	-0.2	-25.8	0.0	0.0	0.0
C LC_122	SLN38 column below / Begin	142.7	-2.3	-0.8	0.0	-3.1	0.0
	SLN185 ring / End	12.1	0.6	-25.6	0.0	0.0	0.0
	SLN227 ring / End	14.1	-0.1	-25.8	0.0	0.0	0.0
C LC_123	SLN38 column below / Begin	108.4	0.0	-0.6	0.0	-2.5	0.0
	SLN185 ring / End	6.1	0.5	-19.4	0.0	0.0	0.0
	SLN227 ring / End	5.8	-0.1	-19.5	0.0	0.0	0.0
C LC_124	SLN38 column below / Begin	107.8	0.0	-6.1	0.0	-6.1	0.0
	SLN185 ring / End	22.4	0.6	-19.4	0.0	0.0	0.0
	SLN227 ring / End	22.2	-0.1	-19.5	0.0	0.0	0.0
C LC_125	SLN38 column below / Begin	108.4	4.6	-0.6	0.0	-2.4	0.0
	SLN185 ring / End	10.5	0.4	-19.4	0.0	0.0	0.0
	SLN227 ring / End	5.6	-0.2	-19.5	0.0	0.0	0.0
C LC_126	SLN38 column below / Begin	108.7	-4.6	-0.6	0.0	-2.3	0.0
	SLN185 ring / End	6.3	0.4	-19.4	0.0	0.0	0.0
	SLN227 ring / End	10.6	0.0	-19.5	0.0	0.0	0.0
C LC_127	SLN38 column below / Begin	28.5	0.0	-0.2	0.0	-0.7	0.0
	SLN185 ring / End	0.4	0.2	-4.9	0.0	0.0	0.0
	SLN227 ring / End	0.3	0.0	-4.9	0.0	0.0	0.0
C LC_128	SLN38 column below / Begin	27.9	0.0	-5.6	0.0	-4.3	0.0
	SLN185 ring / End	15.5	0.2	-4.9	0.0	0.0	0.0
	SLN227 ring / End	15.5	0.0	-4.9	0.0	0.0	0.0
C LC_129	SLN38 column below / Begin	28.5	4.6	-0.2	0.0	-0.6	0.0
	SLN185 ring / End	4.1	0.1	-4.9	0.0	0.0	0.0
	SLN227 ring / End	-0.5	-0.1	-4.9	0.0	0.0	0.0
C LC_130	SLN38 column below / Begin	28.8	-4.6	-0.1	0.0	-0.5	0.0
	SLN185 ring / End	0.0	0.1	-4.9	0.0	0.0	0.0
	SLN227 ring / End	4.4	0.0	-4.9	0.0	0.0	0.0
C LC_135	SLN38 column below / Begin	142.6	0.0	-0.8	0.0	-3.1	0.0
	SLN185 ring / End	15.4	0.7	-25.6	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN227 ring / End	15.0	-0.2	-25.8	0.0	0.0	0.0
C LC_141	SLN38 column below / Begin	28.5	0.0	-0.1	0.0	-0.5	0.0
	SLN185 ring / End	6.0	0.2	-4.9	0.0	0.0	0.0
	SLN227 ring / End	5.9	0.0	-4.9	0.0	0.0	0.0
C LC_142	SLN38 column below / Begin	28.5	0.0	-0.1	0.0	-0.5	0.0
	SLN185 ring / End	6.5	0.2	-4.9	0.0	0.0	0.0
	SLN227 ring / End	6.4	-0.1	-4.9	0.0	0.0	0.0
C LC_143	SLN38 column below / Begin	28.5	0.0	-0.1	0.0	-0.5	0.0
	SLN185 ring / End	6.3	0.2	-4.9	0.0	0.0	0.0
	SLN227 ring / End	6.2	0.0	-4.9	0.0	0.0	0.0
C LC_144	SLN38 column below / Begin	28.5	0.0	-0.1	0.0	-0.5	0.0
	SLN185 ring / End	6.2	0.2	-4.9	0.0	0.0	0.0
	SLN227 ring / End	6.2	0.0	-4.9	0.0	0.0	0.0
C LC_145	SLN38 column below / Begin	142.8	0.0	-0.8	0.0	-3.1	0.0
	SLN185 ring / End	17.1	0.6	-25.6	0.0	0.0	0.0
	SLN227 ring / End	16.7	-0.1	-25.8	0.0	0.0	0.0
C LC_146	SLN38 column below / Begin	142.8	0.0	-0.8	0.0	-3.0	0.0
	SLN185 ring / End	19.3	0.7	-25.6	0.0	0.0	0.0
	SLN227 ring / End	19.0	-0.2	-25.8	0.0	0.0	0.0
C LC_147	SLN38 column below / Begin	142.8	0.0	-0.8	0.0	-3.1	0.0
	SLN185 ring / End	18.3	0.6	-25.6	0.0	0.0	0.0
	SLN227 ring / End	17.7	-0.2	-25.8	0.0	0.0	0.0
C LC_148	SLN38 column below / Begin	142.8	0.0	-0.8	0.0	-3.1	0.0
	SLN185 ring / End	18.1	0.6	-25.6	0.0	0.0	0.0
	SLN227 ring / End	17.9	-0.2	-25.8	0.0	0.0	0.0
C LC_151	SLN38 column below / Begin	91.7	0.0	-1.7	0.0	-6.9	0.0
	SLN185 ring / End	6.2	0.4	-11.7	0.0	0.0	0.0
	SLN227 ring / End	6.0	-0.1	-11.5	0.0	0.0	0.0
C LC_152	SLN38 column below / Begin	80.2	0.0	0.8	0.0	3.4	0.0
	SLN185 ring / End	19.7	0.5	-14.2	0.0	0.0	0.0
	SLN227 ring / End	19.4	-0.1	-14.2	0.0	0.0	0.0
C LC_153	SLN38 column below / Begin	142.4	0.0	-0.8	0.0	-3.3	0.0
	SLN185 ring / End	18.2	0.6	-21.0	0.0	0.0	0.0
	SLN227 ring / End	17.9	-0.2	-20.7	0.0	0.0	0.0
C LC_154	SLN38 column below / Begin	91.0	0.0	-1.7	0.0	-6.8	0.0
	SLN185 ring / End	6.3	0.4	-11.7	0.0	0.0	0.0
	SLN227 ring / End	6.1	-0.1	-11.5	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	7.5 < 100%	OK
Welds	69.2 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN180+SLN169 radial (continuous)-bfl 1	12.7	C LC_120	144.4	0.0	42.7	OK
SLN180+SLN169 radial (continuous)-tfl 1	12.7	C LC_153	72.8	0.0	0.0	OK
SLN180+SLN169 radial (continuous)-w 1	7.2	C LC_125	191.0	0.0	29.5	OK
SLN38 column below	10.0	C LC_120	157.3	0.0	0.0	OK
SLN185 ring-bfl 1	12.7	C LC_120	28.4	0.0	0.0	OK
SLN185 ring-tfl 1	12.7	C LC_135	25.4	0.0	0.0	OK
SLN185 ring-w 1	7.2	C LC_120	168.4	0.0	0.0	OK
SLN227 ring-bfl 1	12.7	C LC_120	26.8	0.0	0.0	OK
SLN227 ring-tfl 1	12.7	C LC_121	21.5	0.0	0.0	OK
SLN227 ring-w 1	7.2	C LC_146	122.3	0.0	0.0	OK
End plate column below SLN38 column below	12.0	C LC_120	113.4	0.0	39.0	OK
End plate SLN227 ring	12.0	C LC_126	26.9	0.0	2.9	OK
End plate SLN185 ring	12.0	C LC_125	28.1	0.0	3.2	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

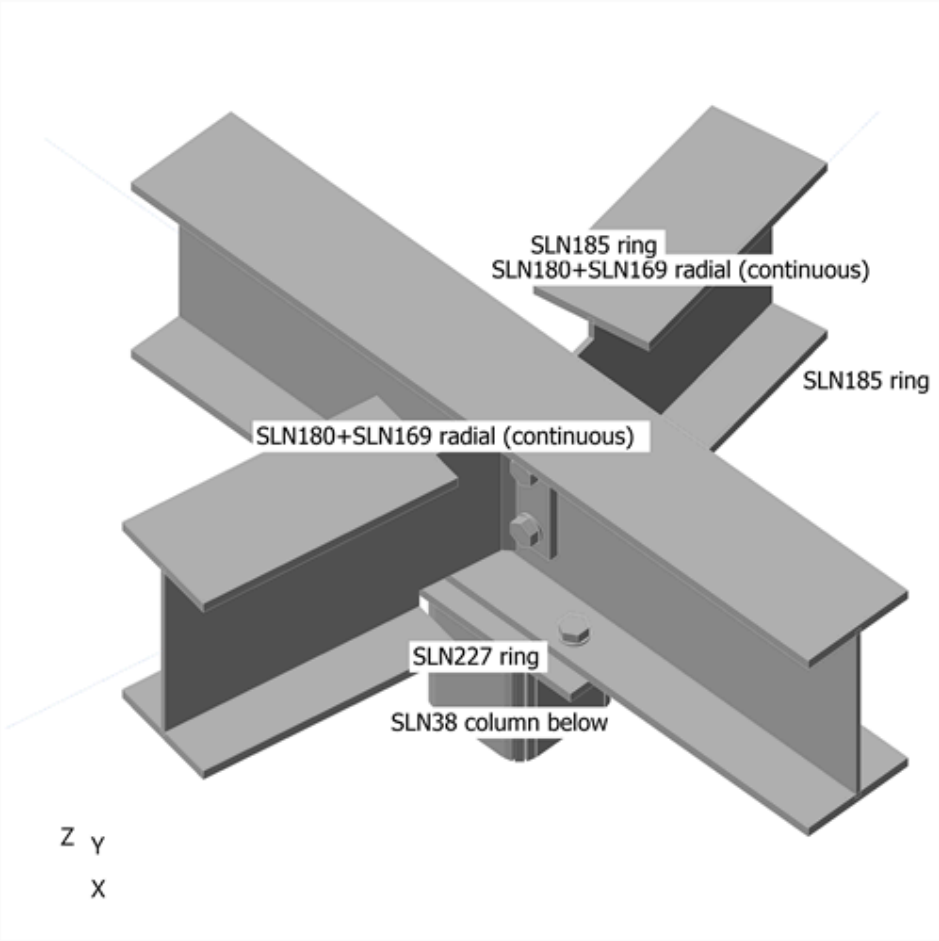
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

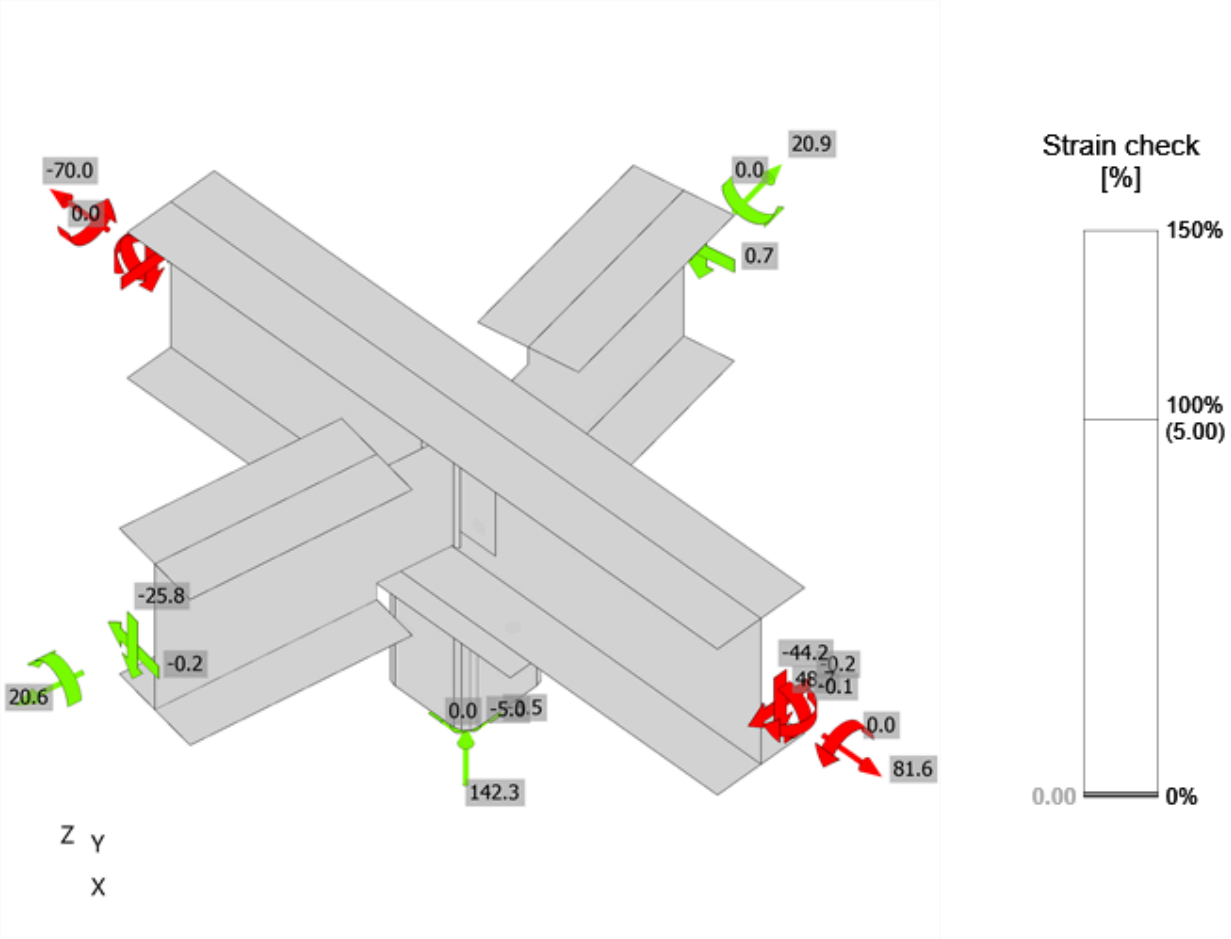
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN38 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

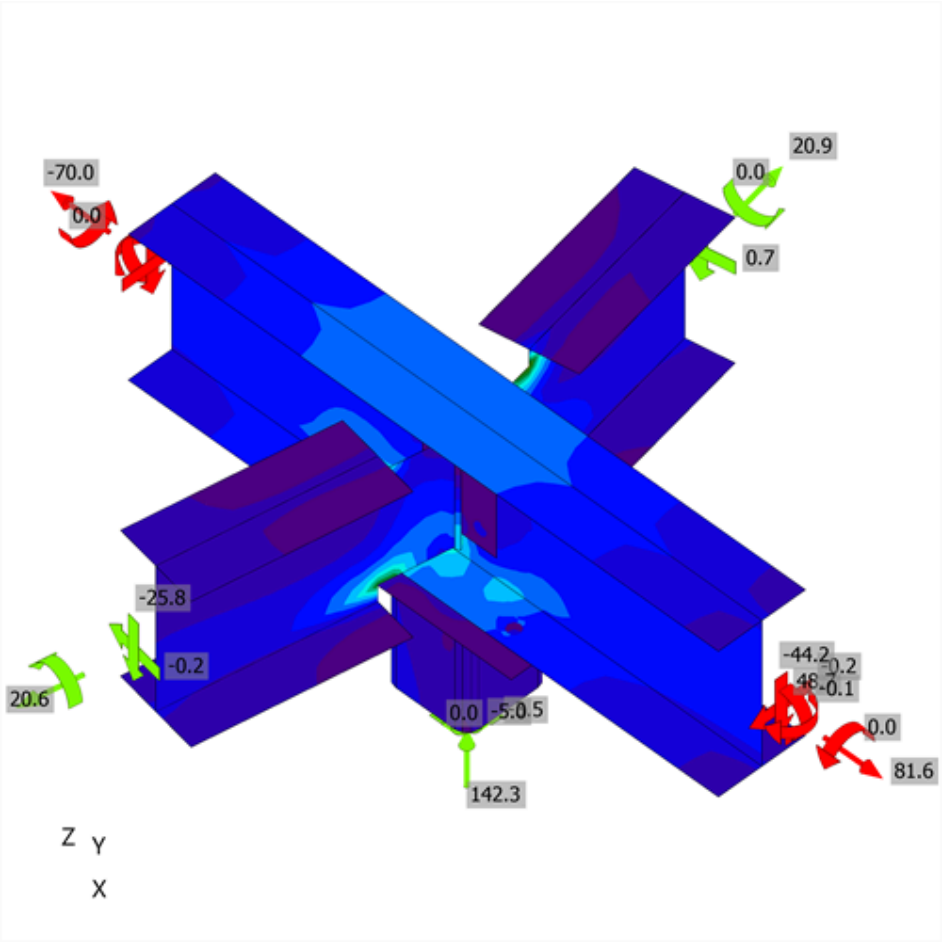
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

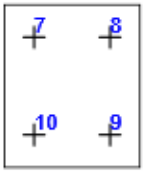


Strain check, C LC_120



Equivalent stress, C LC_120

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_120	0.5	4.3	176.0	0.4	4.6	4.9	OK	OK
	B3	M20 8.8 - 1	C LC_108	10.6	1.4	99.6	7.5	1.5	6.9	OK	OK
	B4	M20 8.8 - 1	C LC_108	10.6	1.4	99.5	7.5	1.5	6.9	OK	OK
	B5	M20 8.8 - 1	C LC_120	0.5	4.3	176.1	0.4	4.6	4.8	OK	OK
	B7	M20 8.8 - 1	C LC_151	0.1	0.3	123.8	0.0	0.4	0.3	OK	OK
	B8	M20 8.8 - 1	C LC_151	0.1	0.5	123.8	0.0	0.7	0.5	OK	OK
	B9	M20 8.8 - 1	C LC_125	1.2	0.4	123.8	0.9	0.4	1.0	OK	OK
	B10	M20 8.8 - 1	C LC_151	0.3	0.9	123.8	0.2	1.4	1.0	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN38 column below	SLN38 column below	◢ 4.2 ◣	521	C LC_120	259.7	0.0	-128.9	-130.2	-0.1	64.2	35.4	OK	OK
		◢ 4.2 ◣	521	C LC_120	280.0	0.0	-140.9	139.7	0.6	69.2	38.7	OK	OK
SLN180+SLN169 radial (continuous)-w 1	SLN185 ring-w 1	◢ 7.2	189	C LC_121	90.8	0.0	5.4	7.6	-51.8	22.4	13.5	OK	OK
SLN180+SLN169 radial (continuous)-w 1	SLN227 ring-w 1	◢ 7.2	189	C LC_122	89.1	0.0	4.3	2.6	-51.3	22.0	13.0	OK	OK
End plate SLN227 ring	SLN227 ring-w 1	◢ 4.2 ◣	149	C LC_122	79.3	0.0	-21.2	-17.0	40.7	19.6	12.0	OK	OK
		◢ 4.2 ◣	149	C LC_122	95.2	0.0	-18.4	24.4	-48.1	23.5	14.7	OK	OK
End plate SLN185 ring	SLN185 ring-w 1	◢ 4.2 ◣	149	C LC_121	96.0	0.0	-17.8	-21.7	49.9	23.7	15.3	OK	OK
		◢ 4.2 ◣	149	C LC_121	79.5	0.0	-21.9	19.9	-39.4	19.6	11.5	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-A_080
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN187+SLN190 radial (continuous) - SLN42 column below - SLN189 ring - SLN192 ring

Name	SLN187+SLN190 radial (continuous) - SLN42 column below - SLN189 ring - SLN192 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

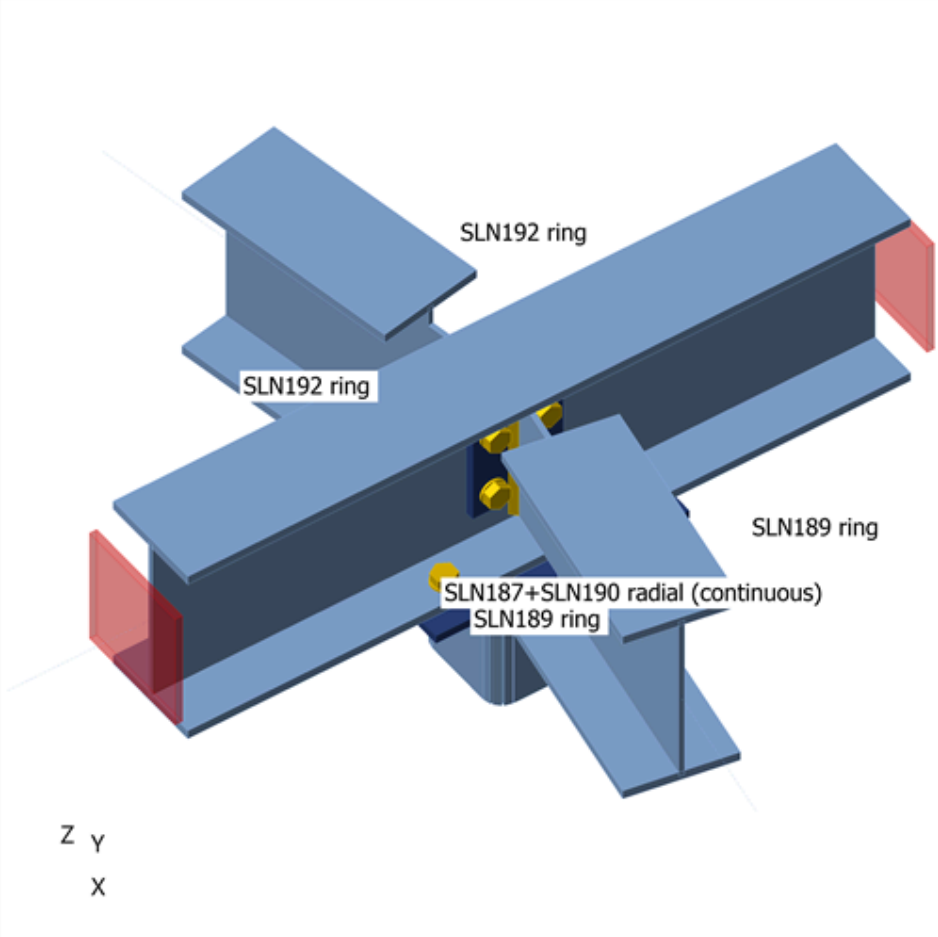
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN187+SLN190 radial (continuous)	1 - UB254x146x43	80.0	0.0	0.0	0	0	0
SLN42 column below	2 - SHS150x150x10	0.0	90.0	-100.0	0	0	0
SLN189 ring	1 - UB254x146x43	-20.0	0.0	0.0	0	0	0
SLN192 ring	1 - UB254x146x43	180.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN187+SLN190 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN187+SLN190 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN42 column below / begin		Position	0
SLN189 ring / end		Position	0
SLN192 ring / end		Position	0



SLN187+SLN190 radial (continuous) - SLN42 column below - SLN189 ring - SLN192 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN42 column below / Begin	26.6	0.0	0.0	0.0	-0.1	0.0
	SLN189 ring / End	8.9	0.0	-5.0	0.0	0.0	0.0
	SLN192 ring / End	8.8	0.1	-4.9	0.0	0.0	0.0
C LC_102	SLN42 column below / Begin	133.1	0.0	-0.4	0.0	-1.4	0.0
	SLN189 ring / End	28.2	-0.1	-25.8	0.0	0.0	0.0
	SLN192 ring / End	27.8	0.2	-25.6	0.0	0.0	0.0
C LC_103	SLN42 column below / Begin	127.5	0.0	-0.4	0.0	-1.4	0.0
	SLN189 ring / End	20.2	-0.1	-24.8	0.0	0.0	0.0
	SLN192 ring / End	20.0	0.1	-24.6	0.0	0.0	0.0
C LC_104	SLN42 column below / Begin	127.4	-2.2	-0.8	0.0	-1.7	0.0
	SLN189 ring / End	22.8	-0.1	-24.8	0.0	0.0	0.0
	SLN192 ring / End	20.0	0.2	-24.6	0.0	0.0	0.0
C LC_105	SLN42 column below / Begin	126.9	0.4	-3.0	0.0	-3.2	0.0
	SLN189 ring / End	29.0	-0.2	-24.8	0.0	0.0	0.0
	SLN192 ring / End	29.0	0.2	-24.6	0.0	0.0	0.0
C LC_106	SLN42 column below / Begin	127.7	0.0	-0.4	0.0	-1.5	0.0
	SLN189 ring / End	18.1	-0.1	-24.8	0.0	0.0	0.0
	SLN192 ring / End	17.8	0.1	-24.6	0.0	0.0	0.0
C LC_107	SLN42 column below / Begin	9.0	0.0	0.0	0.0	-0.1	0.0
	SLN189 ring / End	-2.8	0.0	-1.6	0.0	0.0	0.0
	SLN192 ring / End	-2.9	0.0	-1.6	0.0	0.0	0.0
C LC_108	SLN42 column below / Begin	8.7	-4.5	-1.0	0.0	-0.7	0.0
	SLN189 ring / End	1.4	0.1	-1.6	0.0	0.0	0.0
	SLN192 ring / End	-3.2	0.0	-1.6	0.0	0.0	0.0
C LC_109	SLN42 column below / Begin	7.7	0.8	-5.4	0.0	-3.6	0.0
	SLN189 ring / End	12.2	0.0	-1.6	0.0	0.0	0.0
	SLN192 ring / End	12.8	-0.1	-1.6	0.0	0.0	0.0
C LC_110	SLN42 column below / Begin	9.3	0.0	-0.1	0.0	-0.2	0.0
	SLN189 ring / End	-5.4	0.0	-1.6	0.0	0.0	0.0
	SLN192 ring / End	-5.2	0.1	-1.6	0.0	0.0	0.0
C LC_111	SLN42 column below / Begin	90.3	0.0	-0.3	0.0	-1.0	0.0
	SLN189 ring / End	10.7	0.0	-17.5	0.0	0.0	0.0
	SLN192 ring / End	10.5	0.1	-17.3	0.0	0.0	0.0
C LC_112	SLN42 column below / Begin	90.0	-4.5	-1.2	0.0	-1.6	0.0
	SLN189 ring / End	15.2	0.0	-17.5	0.0	0.0	0.0
	SLN192 ring / End	10.3	0.2	-17.3	0.0	0.0	0.0
C LC_113	SLN42 column below / Begin	89.0	0.8	-5.6	0.0	-4.5	0.0
	SLN189 ring / End	26.5	-0.1	-17.5	0.0	0.0	0.0
	SLN192 ring / End	27.0	0.1	-17.3	0.0	0.0	0.0
C LC_114	SLN42 column below / Begin	90.6	0.0	-0.3	0.0	-1.2	0.0
	SLN189 ring / End	7.6	-0.1	-17.5	0.0	0.0	0.0
	SLN192 ring / End	7.5	0.1	-17.3	0.0	0.0	0.0
C LC_115	SLN42 column below / Begin	15.9	0.0	0.0	0.0	-0.1	0.0
	SLN189 ring / End	-0.5	0.0	-2.9	0.0	0.0	0.0
	SLN192 ring / End	-0.6	0.0	-2.8	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN42 column below / Begin	15.6	-4.5	-1.0	0.0	-0.7	0.0
	SLN189 ring / End	3.7	0.0	-2.9	0.0	0.0	0.0
	SLN192 ring / End	-0.9	0.1	-2.8	0.0	0.0	0.0
C LC_117	SLN42 column below / Begin	14.6	0.8	-5.4	0.0	-3.6	0.0
	SLN189 ring / End	14.5	0.0	-2.9	0.0	0.0	0.0
	SLN192 ring / End	15.1	-0.1	-2.8	0.0	0.0	0.0
C LC_118	SLN42 column below / Begin	16.2	0.0	-0.1	0.0	-0.3	0.0
	SLN189 ring / End	-3.1	-0.1	-2.9	0.0	0.0	0.0
	SLN192 ring / End	-3.0	0.1	-2.8	0.0	0.0	0.0
C LC_119	SLN42 column below / Begin	133.0	0.0	-0.4	0.0	-1.5	0.0
	SLN189 ring / End	22.4	-0.1	-25.8	0.0	0.0	0.0
	SLN192 ring / End	22.2	0.2	-25.6	0.0	0.0	0.0
C LC_120	SLN42 column below / Begin	132.8	-2.2	-0.8	0.0	-1.8	0.0
	SLN189 ring / End	25.1	-0.1	-25.8	0.0	0.0	0.0
	SLN192 ring / End	22.2	0.2	-25.6	0.0	0.0	0.0
C LC_121	SLN42 column below / Begin	132.4	0.4	-3.0	0.0	-3.2	0.0
	SLN189 ring / End	31.4	-0.2	-25.8	0.0	0.0	0.0
	SLN192 ring / End	31.4	0.2	-25.6	0.0	0.0	0.0
C LC_122	SLN42 column below / Begin	133.0	0.0	-0.4	0.0	-1.6	0.0
	SLN189 ring / End	20.2	-0.1	-25.8	0.0	0.0	0.0
	SLN192 ring / End	19.9	0.1	-25.6	0.0	0.0	0.0
C LC_123	SLN42 column below / Begin	101.1	0.0	-0.3	0.0	-1.1	0.0
	SLN189 ring / End	15.1	0.0	-19.6	0.0	0.0	0.0
	SLN192 ring / End	15.0	0.1	-19.4	0.0	0.0	0.0
C LC_124	SLN42 column below / Begin	100.9	-4.5	-1.2	0.0	-1.7	0.0
	SLN189 ring / End	19.8	-0.1	-19.6	0.0	0.0	0.0
	SLN192 ring / End	14.7	0.2	-19.4	0.0	0.0	0.0
C LC_125	SLN42 column below / Begin	100.1	0.8	-5.6	0.0	-4.6	0.0
	SLN189 ring / End	31.4	-0.2	-19.6	0.0	0.0	0.0
	SLN192 ring / End	31.7	0.1	-19.4	0.0	0.0	0.0
C LC_126	SLN42 column below / Begin	101.2	0.0	-0.3	0.0	-1.3	0.0
	SLN189 ring / End	11.9	-0.1	-19.6	0.0	0.0	0.0
	SLN192 ring / End	11.7	0.1	-19.4	0.0	0.0	0.0
C LC_127	SLN42 column below / Begin	26.7	0.0	0.0	0.0	-0.2	0.0
	SLN189 ring / End	3.9	0.0	-5.0	0.0	0.0	0.0
	SLN192 ring / End	3.9	0.0	-4.9	0.0	0.0	0.0
C LC_128	SLN42 column below / Begin	26.4	-4.5	-1.0	0.0	-0.8	0.0
	SLN189 ring / End	8.3	0.0	-5.0	0.0	0.0	0.0
	SLN192 ring / End	3.6	0.1	-4.9	0.0	0.0	0.0
C LC_129	SLN42 column below / Begin	25.7	0.8	-5.4	0.0	-3.7	0.0
	SLN189 ring / End	19.3	-0.1	-5.0	0.0	0.0	0.0
	SLN192 ring / End	19.9	0.0	-4.9	0.0	0.0	0.0
C LC_130	SLN42 column below / Begin	26.7	0.0	-0.1	0.0	-0.3	0.0
	SLN189 ring / End	1.2	0.0	-5.0	0.0	0.0	0.0
	SLN192 ring / End	1.2	0.1	-4.9	0.0	0.0	0.0
C LC_135	SLN42 column below / Begin	133.1	0.0	-0.4	0.0	-1.4	0.0
	SLN189 ring / End	28.6	-0.1	-25.8	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN192 ring / End	28.3	0.2	-25.6	0.0	0.0	0.0
C LC_141	SLN42 column below / Begin	26.6	0.0	0.0	0.0	-0.1	0.0
	SLN189 ring / End	8.9	0.0	-5.0	0.0	0.0	0.0
	SLN192 ring / End	8.8	0.1	-4.9	0.0	0.0	0.0
C LC_142	SLN42 column below / Begin	26.6	0.0	0.0	0.0	-0.1	0.0
	SLN189 ring / End	9.0	0.0	-5.0	0.0	0.0	0.0
	SLN192 ring / End	8.9	0.1	-4.9	0.0	0.0	0.0
C LC_143	SLN42 column below / Begin	26.6	0.0	0.0	0.0	-0.1	0.0
	SLN189 ring / End	9.2	-0.1	-5.0	0.0	0.0	0.0
	SLN192 ring / End	9.1	0.1	-4.9	0.0	0.0	0.0
C LC_144	SLN42 column below / Begin	26.6	0.0	0.0	0.0	-0.1	0.0
	SLN189 ring / End	8.7	0.0	-5.0	0.0	0.0	0.0
	SLN192 ring / End	8.6	0.0	-4.9	0.0	0.0	0.0
C LC_145	SLN42 column below / Begin	133.1	0.0	-0.4	0.0	-1.4	0.0
	SLN189 ring / End	27.9	-0.1	-25.8	0.0	0.0	0.0
	SLN192 ring / End	27.7	0.2	-25.6	0.0	0.0	0.0
C LC_146	SLN42 column below / Begin	133.1	0.0	-0.4	0.0	-1.4	0.0
	SLN189 ring / End	28.5	-0.1	-25.8	0.0	0.0	0.0
	SLN192 ring / End	27.9	0.2	-25.6	0.0	0.0	0.0
C LC_147	SLN42 column below / Begin	133.1	0.0	-0.3	0.0	-1.4	0.0
	SLN189 ring / End	29.3	-0.2	-25.8	0.0	0.0	0.0
	SLN192 ring / End	29.0	0.2	-25.6	0.0	0.0	0.0
C LC_148	SLN42 column below / Begin	133.1	0.0	-0.4	0.0	-1.5	0.0
	SLN189 ring / End	27.0	-0.1	-25.8	0.0	0.0	0.0
	SLN192 ring / End	26.6	0.1	-25.6	0.0	0.0	0.0
C LC_151	SLN42 column below / Begin	81.2	0.0	-1.3	0.0	-5.3	0.0
	SLN189 ring / End	12.6	-0.1	-11.4	0.0	0.0	0.0
	SLN192 ring / End	12.4	0.1	-11.9	0.0	0.0	0.0
C LC_152	SLN42 column below / Begin	78.3	0.0	0.9	0.0	3.6	0.0
	SLN189 ring / End	28.9	-0.1	-14.1	0.0	0.0	0.0
	SLN192 ring / End	28.6	0.2	-14.2	0.0	0.0	0.0
C LC_153	SLN42 column below / Begin	133.5	0.0	-0.4	0.0	-1.7	0.0
	SLN189 ring / End	28.3	-0.1	-20.6	0.0	0.0	0.0
	SLN192 ring / End	27.9	0.2	-21.1	0.0	0.0	0.0
C LC_154	SLN42 column below / Begin	80.3	0.0	-1.3	0.0	-5.1	0.0
	SLN189 ring / End	12.8	-0.1	-11.4	0.0	0.0	0.0
	SLN192 ring / End	12.5	0.1	-11.9	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	7.2 < 100%	OK
Welds	58.9 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN187+SLN190 radial (continuous)-bfl 1	12.7	C LC_121	122.8	0.0	36.1	OK
SLN187+SLN190 radial (continuous)-tfl 1	12.7	C LC_153	65.5	0.0	0.0	OK
SLN187+SLN190 radial (continuous)-w 1	7.2	C LC_124	176.8	0.0	26.0	OK
SLN42 column below	10.0	C LC_121	130.7	0.0	0.0	OK
SLN189 ring-bfl 1	12.7	C LC_121	28.8	0.0	0.0	OK
SLN189 ring-tfl 1	12.7	C LC_122	19.7	0.0	0.0	OK
SLN189 ring-w 1	7.2	C LC_121	132.3	0.0	0.0	OK
SLN192 ring-bfl 1	12.7	C LC_121	28.6	0.0	0.0	OK
SLN192 ring-tfl 1	12.7	C LC_122	19.7	0.0	0.0	OK
SLN192 ring-w 1	7.2	C LC_147	134.2	0.0	0.0	OK
End plate column below SLN42 column below	12.0	C LC_121	96.6	0.0	40.3	OK
End plate SLN189 ring	12.0	C LC_124	26.9	0.0	1.1	OK
End plate SLN192 ring	12.0	C LC_124	23.3	0.0	3.8	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

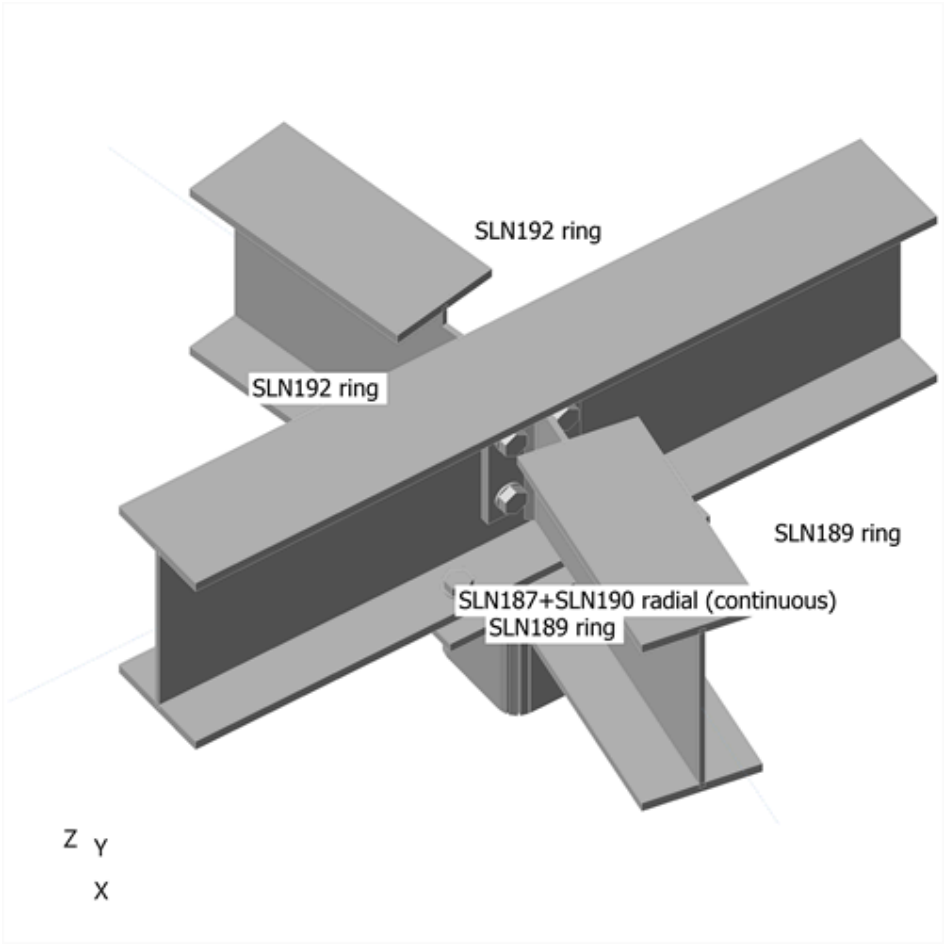
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

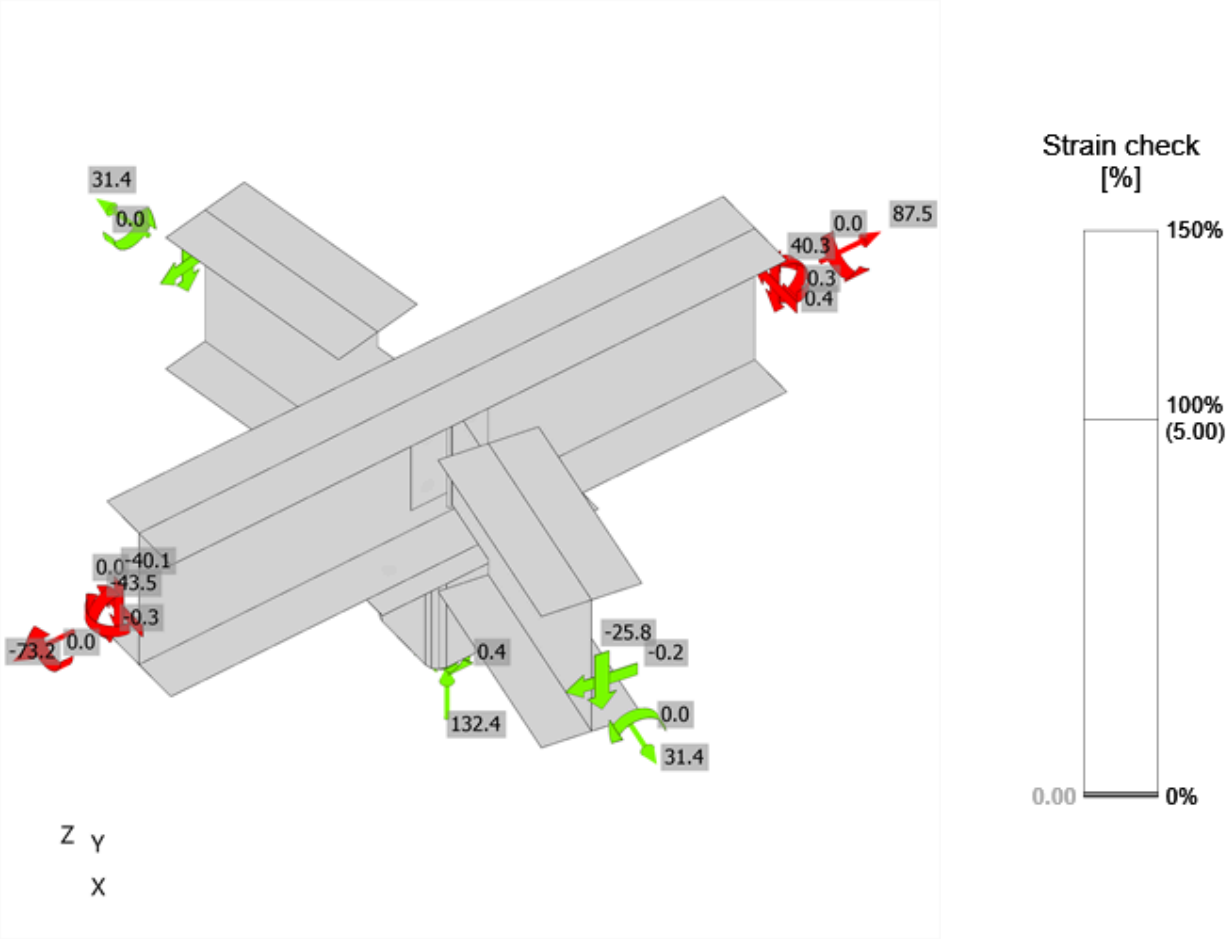
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN42 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

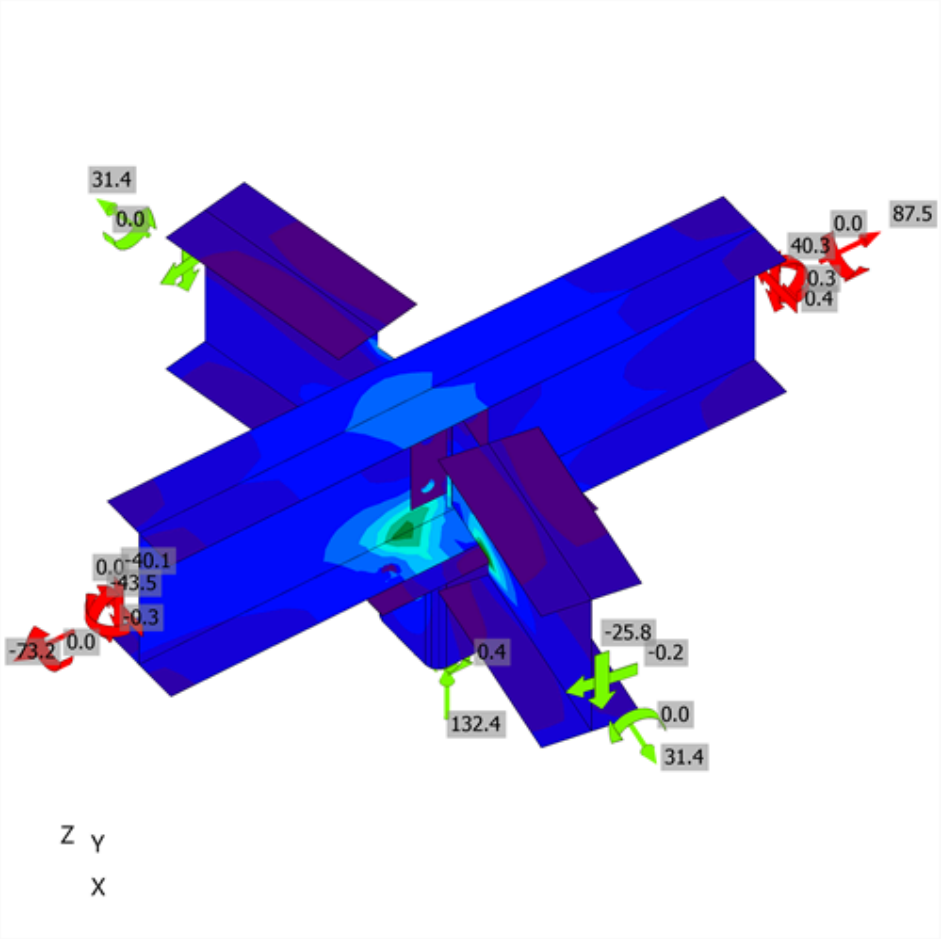
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

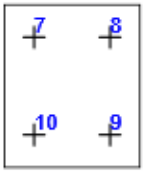


Strain check, C LC_121



Equivalent stress, C LC_121

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_121	0.4	3.7	174.0	0.3	4.0	4.2	OK	OK
	B3	M20 8.8 - 1	C LC_109	10.2	1.5	92.4	7.2	1.7	6.8	OK	OK
	B4	M20 8.8 - 1	C LC_109	9.4	1.3	116.5	6.6	1.4	6.1	OK	OK
	B5	M20 8.8 - 1	C LC_121	0.4	3.9	176.3	0.3	4.2	4.4	OK	OK
	B7	M20 8.8 - 1	C LC_152	0.2	0.2	123.8	0.2	0.4	0.4	OK	OK
	B8	M20 8.8 - 1	C LC_151	0.1	0.4	123.8	0.1	0.6	0.4	OK	OK
	B9	M20 8.8 - 1	C LC_124	1.2	0.4	123.8	0.9	0.4	1.0	OK	OK
	B10	M20 8.8 - 1	C LC_124	1.1	0.6	123.8	0.8	0.7	1.2	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN42 column below	SLN42 column below	◢ 4.2 ◣	521	C LC_121	225.4	0.0	-110.4	-108.2	-34.2	55.7	30.6	OK	OK
		◢ 4.2 ◣	521	C LC_121	238.3	0.0	-114.0	116.4	32.3	58.9	32.6	OK	OK
SLN187+SLN190 radial (continuous)-w 1	SLN189 ring-w 1	◢ 7.2	189	C LC_121	89.2	0.0	27.7	-34.5	34.7	22.0	14.2	OK	OK
SLN187+SLN190 radial (continuous)-w 1	SLN192 ring-w 1	◢ 7.2	189	C LC_121	87.3	0.0	26.8	33.7	-34.2	21.6	14.4	OK	OK
End plate SLN189 ring	SLN189 ring-w 1	◢ 4.2 ◣	149	C LC_120	77.0	0.0	-20.3	-17.0	39.4	19.0	11.6	OK	OK
		◢ 4.2 ◣	149	C LC_120	91.0	0.0	-18.2	23.1	-46.0	22.5	14.0	OK	OK
End plate SLN192 ring	SLN192 ring-w 1	◢ 4.2 ◣	149	C LC_121	89.6	0.0	-17.7	-23.3	45.0	22.1	13.8	OK	OK
		◢ 4.2 ◣	149	C LC_121	75.9	0.0	-20.6	16.7	-38.7	18.8	11.2	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-A_120
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN194+SLN196 radial (continuous) - SLN44 column below - SLN195 ring - SLN198 ring

Name	SLN194+SLN196 radial (continuous) - SLN44 column below - SLN195 ring - SLN198 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

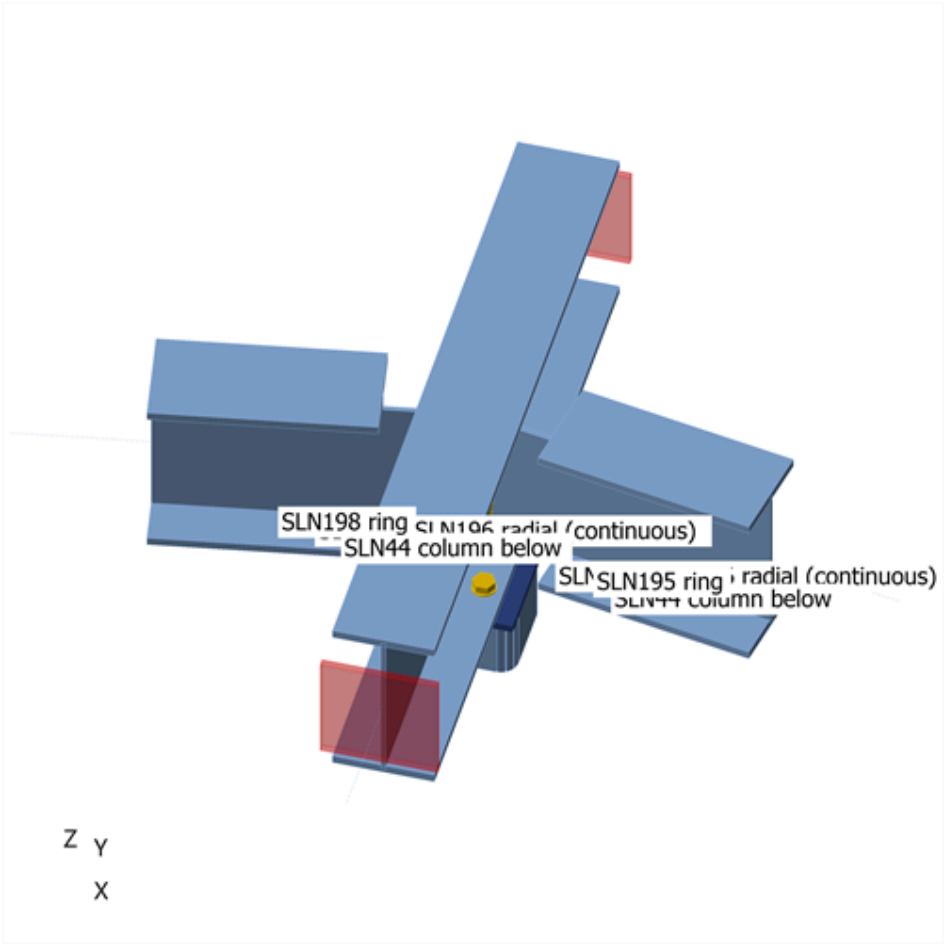
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN194+SLN196 radial (continuous)	1 - UB254x146x43	120.0	0.0	0.0	0	0	0
SLN44 column below	2 - SHS150x150x10	0.0	90.0	-60.0	0	0	0
SLN195 ring	1 - UB254x146x43	20.0	0.0	0.0	0	0	0
SLN198 ring	1 - UB254x146x43	-140.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN194+SLN196 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN194+SLN196 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN44 column below / begin		Position	0
SLN195 ring / end		Position	0
SLN198 ring / end		Position	0



SLN194+SLN196 radial (continuous) - SLN44 column below - SLN195 ring - SLN198 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN44 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN195 ring / End	8.9	-0.1	-4.9	0.0	0.0	0.0
	SLN198 ring / End	8.9	0.1	-5.0	0.0	0.0	0.0
C LC_102	SLN44 column below / Begin	141.6	0.0	-0.9	0.0	-3.5	0.0
	SLN195 ring / End	27.9	-0.2	-25.8	0.0	0.0	0.0
	SLN198 ring / End	28.0	0.2	-26.0	0.0	0.0	0.0
C LC_103	SLN44 column below / Begin	135.6	2.0	-2.2	0.0	-4.4	0.0
	SLN195 ring / End	23.1	-0.2	-24.7	0.0	0.0	0.0
	SLN198 ring / End	25.3	0.2	-24.9	0.0	0.0	0.0
C LC_104	SLN44 column below / Begin	135.8	0.0	-0.9	0.0	-3.5	0.0
	SLN195 ring / End	19.3	-0.1	-24.7	0.0	0.0	0.0
	SLN198 ring / End	19.2	0.2	-24.9	0.0	0.0	0.0
C LC_105	SLN44 column below / Begin	135.3	-1.2	-3.2	0.0	-5.0	0.0
	SLN195 ring / End	28.5	-0.2	-24.7	0.0	0.0	0.0
	SLN198 ring / End	27.2	0.2	-24.9	0.0	0.0	0.0
C LC_106	SLN44 column below / Begin	135.8	0.0	-0.9	0.0	-3.5	0.0
	SLN195 ring / End	18.1	-0.1	-24.7	0.0	0.0	0.0
	SLN198 ring / End	18.4	0.2	-24.9	0.0	0.0	0.0
C LC_107	SLN44 column below / Begin	9.3	4.0	-2.8	0.0	-2.1	0.0
	SLN195 ring / End	2.1	0.0	-1.6	0.0	0.0	0.0
	SLN198 ring / End	6.1	0.0	-1.6	0.0	0.0	0.0
C LC_108	SLN44 column below / Begin	9.7	0.0	-0.1	0.0	-0.3	0.0
	SLN195 ring / End	-3.7	0.0	-1.6	0.0	0.0	0.0
	SLN198 ring / End	-3.8	0.1	-1.6	0.0	0.0	0.0
C LC_109	SLN44 column below / Begin	8.8	-2.3	-4.8	0.0	-3.4	0.0
	SLN195 ring / End	11.7	0.0	-1.6	0.0	0.0	0.0
	SLN198 ring / End	9.3	0.0	-1.6	0.0	0.0	0.0
C LC_110	SLN44 column below / Begin	9.8	0.0	-0.1	0.0	-0.3	0.0
	SLN195 ring / End	-5.0	0.0	-1.6	0.0	0.0	0.0
	SLN198 ring / End	-4.8	0.1	-1.6	0.0	0.0	0.0
C LC_111	SLN44 column below / Begin	95.8	3.9	-3.4	0.0	-4.4	0.0
	SLN195 ring / End	15.9	-0.2	-17.4	0.0	0.0	0.0
	SLN198 ring / End	20.0	0.1	-17.6	0.0	0.0	0.0
C LC_112	SLN44 column below / Begin	96.2	0.0	-0.6	0.0	-2.5	0.0
	SLN195 ring / End	9.5	-0.1	-17.4	0.0	0.0	0.0
	SLN198 ring / End	9.4	0.2	-17.6	0.0	0.0	0.0
C LC_113	SLN44 column below / Begin	95.3	-2.3	-5.3	0.0	-5.6	0.0
	SLN195 ring / End	25.9	-0.1	-17.4	0.0	0.0	0.0
	SLN198 ring / End	23.5	0.1	-17.6	0.0	0.0	0.0
C LC_114	SLN44 column below / Begin	96.3	0.0	-0.6	0.0	-2.6	0.0
	SLN195 ring / End	7.9	-0.1	-17.4	0.0	0.0	0.0
	SLN198 ring / End	8.1	0.2	-17.6	0.0	0.0	0.0
C LC_115	SLN44 column below / Begin	16.6	4.0	-2.8	0.0	-2.3	0.0
	SLN195 ring / End	4.4	-0.1	-2.9	0.0	0.0	0.0
	SLN198 ring / End	8.4	0.0	-2.9	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN44 column below / Begin	17.1	0.0	-0.1	0.0	-0.4	0.0
	SLN195 ring / End	-1.4	0.0	-2.9	0.0	0.0	0.0
	SLN198 ring / End	-1.5	0.1	-2.9	0.0	0.0	0.0
C LC_117	SLN44 column below / Begin	16.1	-2.3	-4.8	0.0	-3.6	0.0
	SLN195 ring / End	14.1	0.0	-2.9	0.0	0.0	0.0
	SLN198 ring / End	11.7	0.0	-2.9	0.0	0.0	0.0
C LC_118	SLN44 column below / Begin	17.1	0.0	-0.1	0.0	-0.5	0.0
	SLN195 ring / End	-2.7	-0.1	-2.9	0.0	0.0	0.0
	SLN198 ring / End	-2.6	0.1	-2.9	0.0	0.0	0.0
C LC_119	SLN44 column below / Begin	141.3	2.0	-2.3	0.0	-4.5	0.0
	SLN195 ring / End	25.4	-0.2	-25.8	0.0	0.0	0.0
	SLN198 ring / End	27.6	0.3	-26.0	0.0	0.0	0.0
C LC_120	SLN44 column below / Begin	141.4	0.0	-0.9	0.0	-3.6	0.0
	SLN195 ring / End	21.6	-0.1	-25.8	0.0	0.0	0.0
	SLN198 ring / End	21.4	0.2	-26.0	0.0	0.0	0.0
C LC_121	SLN44 column below / Begin	141.2	-1.2	-3.3	0.0	-5.2	0.0
	SLN195 ring / End	30.8	-0.2	-25.8	0.0	0.0	0.0
	SLN198 ring / End	29.6	0.3	-26.0	0.0	0.0	0.0
C LC_122	SLN44 column below / Begin	141.4	0.0	-0.9	0.0	-3.6	0.0
	SLN195 ring / End	20.3	-0.1	-25.8	0.0	0.0	0.0
	SLN198 ring / End	20.5	0.2	-26.0	0.0	0.0	0.0
C LC_123	SLN44 column below / Begin	107.4	3.9	-3.4	0.0	-4.6	0.0
	SLN195 ring / End	20.5	-0.2	-19.5	0.0	0.0	0.0
	SLN198 ring / End	24.6	0.2	-19.7	0.0	0.0	0.0
C LC_124	SLN44 column below / Begin	107.6	0.0	-0.7	0.0	-2.8	0.0
	SLN195 ring / End	14.0	-0.1	-19.5	0.0	0.0	0.0
	SLN198 ring / End	13.8	0.2	-19.7	0.0	0.0	0.0
C LC_125	SLN44 column below / Begin	107.0	-2.3	-5.4	0.0	-5.9	0.0
	SLN195 ring / End	30.6	-0.2	-19.5	0.0	0.0	0.0
	SLN198 ring / End	28.2	0.2	-19.7	0.0	0.0	0.0
C LC_126	SLN44 column below / Begin	107.5	0.0	-0.7	0.0	-2.8	0.0
	SLN195 ring / End	12.2	-0.1	-19.5	0.0	0.0	0.0
	SLN198 ring / End	12.5	0.2	-19.7	0.0	0.0	0.0
C LC_127	SLN44 column below / Begin	28.2	4.0	-2.9	0.0	-2.6	0.0
	SLN195 ring / End	9.0	-0.1	-4.9	0.0	0.0	0.0
	SLN198 ring / End	13.0	0.1	-5.0	0.0	0.0	0.0
C LC_128	SLN44 column below / Begin	28.4	0.0	-0.2	0.0	-0.7	0.0
	SLN195 ring / End	3.0	0.0	-4.9	0.0	0.0	0.0
	SLN198 ring / End	2.9	0.1	-5.0	0.0	0.0	0.0
C LC_129	SLN44 column below / Begin	27.9	-2.3	-4.9	0.0	-3.8	0.0
	SLN195 ring / End	18.8	-0.1	-4.9	0.0	0.0	0.0
	SLN198 ring / End	16.4	0.0	-5.0	0.0	0.0	0.0
C LC_130	SLN44 column below / Begin	28.4	0.0	-0.2	0.0	-0.7	0.0
	SLN195 ring / End	1.6	0.0	-4.9	0.0	0.0	0.0
	SLN198 ring / End	1.8	0.1	-5.0	0.0	0.0	0.0
C LC_135	SLN44 column below / Begin	141.6	0.0	-0.9	0.0	-3.5	0.0
	SLN195 ring / End	28.4	-0.2	-25.8	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN198 ring / End	28.5	0.2	-26.0	0.0	0.0	0.0
C LC_141	SLN44 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN195 ring / End	9.0	-0.1	-4.9	0.0	0.0	0.0
	SLN198 ring / End	9.0	0.1	-5.0	0.0	0.0	0.0
C LC_142	SLN44 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN195 ring / End	8.8	0.0	-4.9	0.0	0.0	0.0
	SLN198 ring / End	8.7	0.1	-5.0	0.0	0.0	0.0
C LC_143	SLN44 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN195 ring / End	9.1	-0.1	-4.9	0.0	0.0	0.0
	SLN198 ring / End	9.1	0.1	-5.0	0.0	0.0	0.0
C LC_144	SLN44 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN195 ring / End	8.6	0.0	-4.9	0.0	0.0	0.0
	SLN198 ring / End	8.7	0.1	-5.0	0.0	0.0	0.0
C LC_145	SLN44 column below / Begin	141.6	0.0	-0.9	0.0	-3.5	0.0
	SLN195 ring / End	28.4	-0.2	-25.8	0.0	0.0	0.0
	SLN198 ring / End	28.7	0.3	-26.0	0.0	0.0	0.0
C LC_146	SLN44 column below / Begin	141.6	0.0	-0.9	0.0	-3.5	0.0
	SLN195 ring / End	27.4	-0.2	-25.8	0.0	0.0	0.0
	SLN198 ring / End	27.3	0.2	-26.0	0.0	0.0	0.0
C LC_147	SLN44 column below / Begin	141.6	0.0	-0.9	0.0	-3.5	0.0
	SLN195 ring / End	29.0	-0.2	-25.8	0.0	0.0	0.0
	SLN198 ring / End	28.9	0.3	-26.0	0.0	0.0	0.0
C LC_148	SLN44 column below / Begin	141.6	0.0	-0.9	0.0	-3.5	0.0
	SLN195 ring / End	26.8	-0.1	-25.8	0.0	0.0	0.0
	SLN198 ring / End	27.0	0.2	-26.0	0.0	0.0	0.0
C LC_151	SLN44 column below / Begin	90.3	0.0	-1.8	0.0	-7.2	0.0
	SLN195 ring / End	12.5	-0.1	-11.9	0.0	0.0	0.0
	SLN198 ring / End	12.5	0.1	-11.9	0.0	0.0	0.0
C LC_152	SLN44 column below / Begin	78.7	0.0	0.8	0.0	3.0	0.0
	SLN195 ring / End	28.7	-0.2	-14.1	0.0	0.0	0.0
	SLN198 ring / End	28.8	0.2	-14.0	0.0	0.0	0.0
C LC_153	SLN44 column below / Begin	141.0	0.0	-0.9	0.0	-3.7	0.0
	SLN195 ring / End	28.1	-0.2	-21.1	0.0	0.0	0.0
	SLN198 ring / End	28.1	0.2	-20.9	0.0	0.0	0.0
C LC_154	SLN44 column below / Begin	89.7	0.0	-1.7	0.0	-7.1	0.0
	SLN195 ring / End	12.6	-0.1	-11.9	0.0	0.0	0.0
	SLN198 ring / End	12.6	0.1	-11.9	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	7.2 < 100%	OK
Welds	71.4 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN194+SLN196 radial (continuous)-bfl 1	12.7	C LC_119	143.6	0.0	41.3	OK
SLN194+SLN196 radial (continuous)-tfl 1	12.7	C LC_153	71.9	0.0	0.0	OK
SLN194+SLN196 radial (continuous)-w 1	7.2	C LC_123	198.3	0.0	34.1	OK
SLN44 column below	10.0	C LC_121	159.0	0.0	0.0	OK
SLN195 ring-bfl 1	12.7	C LC_121	28.6	0.0	0.0	OK
SLN195 ring-tfl 1	12.7	C LC_122	19.8	0.0	0.0	OK
SLN195 ring-w 1	7.2	C LC_121	135.7	0.0	0.0	OK
SLN198 ring-bfl 1	12.7	C LC_121	28.6	0.0	0.0	OK
SLN198 ring-tfl 1	12.7	C LC_122	20.3	0.0	0.0	OK
SLN198 ring-w 1	7.2	C LC_147	138.0	0.0	0.0	OK
End plate column below SLN44 column below	12.0	C LC_121	114.4	0.0	36.5	OK
End plate SLN195 ring	12.0	C LC_121	22.5	0.0	0.8	OK
End plate SLN198 ring	12.0	C LC_123	27.5	0.0	2.1	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

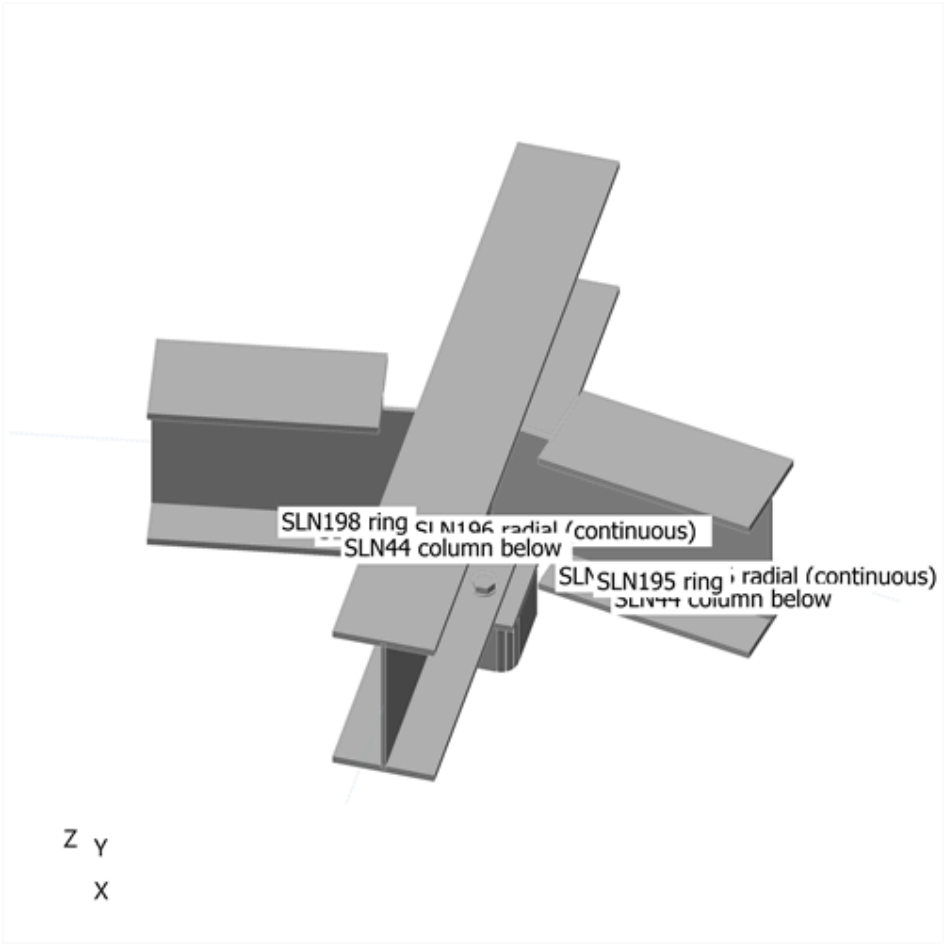
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

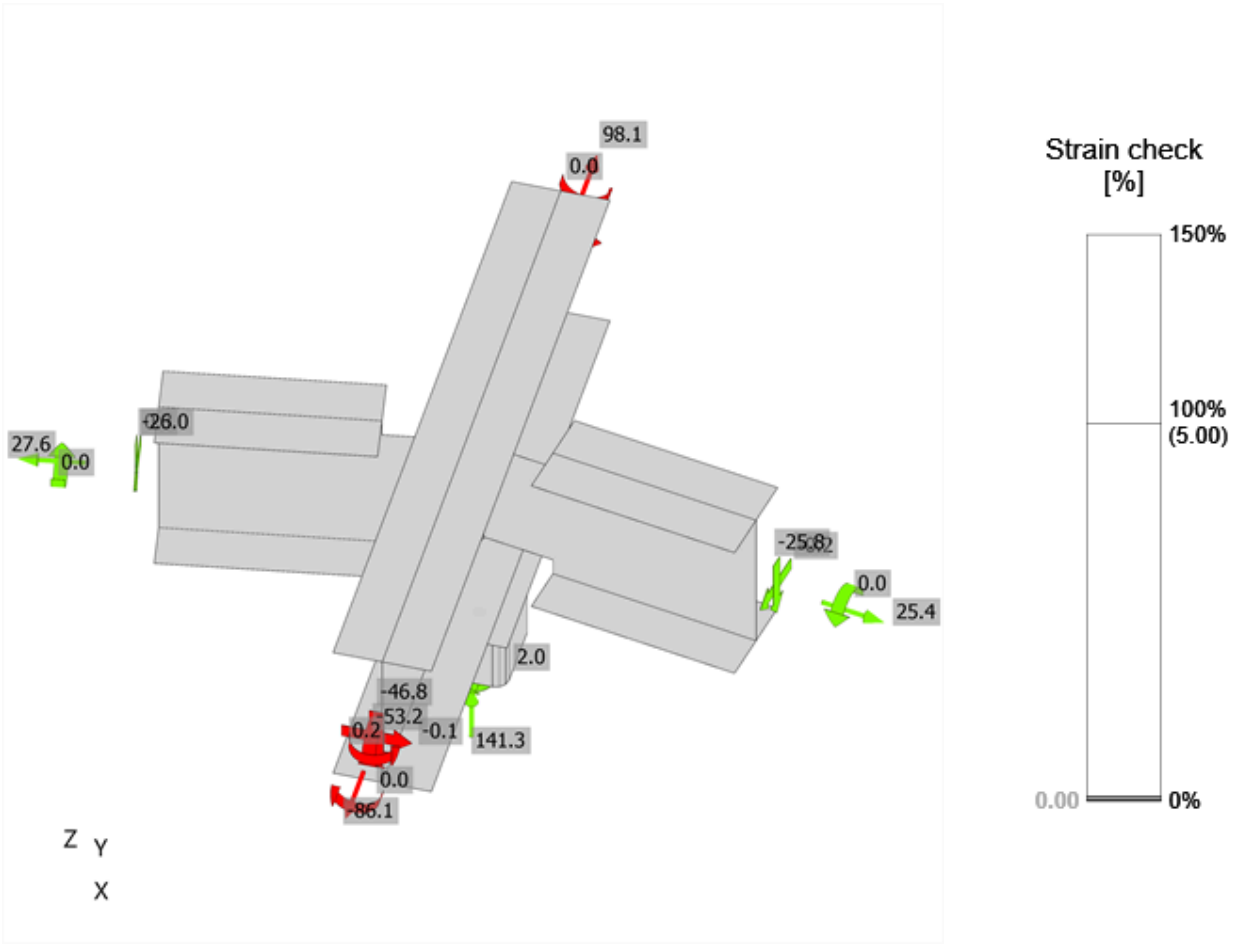
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN44 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

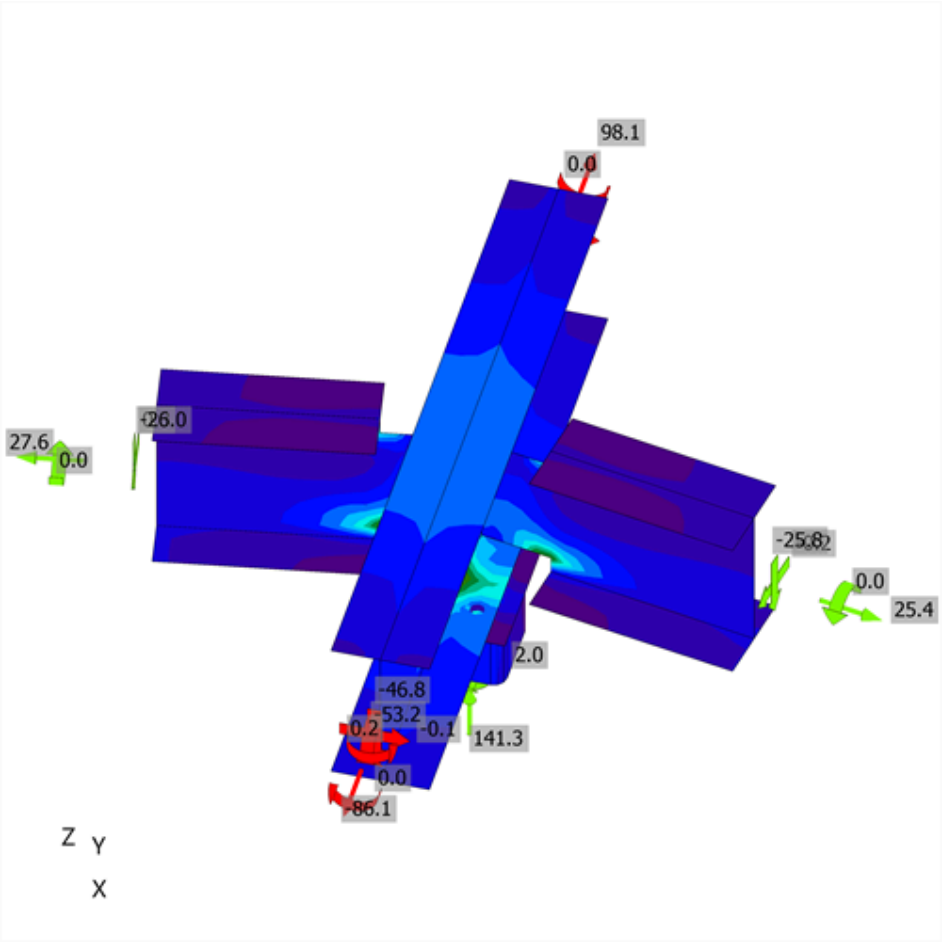
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

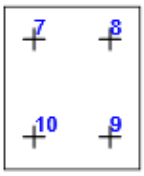


Strain check, C LC_119



Equivalent stress, C LC_119

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_121	0.5	4.4	181.2	0.3	4.6	4.9	OK	OK
	B3	M20 8.8 - 1	C LC_109	7.9	1.0	156.4	5.6	1.1	5.1	OK	OK
	B4	M20 8.8 - 1	C LC_109	10.1	1.6	81.1	7.2	2.0	6.8	OK	OK
	B5	M20 8.8 - 1	C LC_119	0.5	4.3	186.2	0.3	4.5	4.8	OK	OK
	B7	M20 8.8 - 1	C LC_151	0.0	0.3	123.8	0.0	0.5	0.3	OK	OK
	B8	M20 8.8 - 1	C LC_151	0.1	0.5	123.8	0.1	0.8	0.6	OK	OK
	B9	M20 8.8 - 1	C LC_123	1.1	0.3	67.6	0.8	0.3	0.9	OK	OK
	B10	M20 8.8 - 1	C LC_123	1.2	0.8	123.8	0.8	1.1	1.4	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN44 column below	SLN44 column below	◢ 4.2 ◣	521	C LC_121	267.6	0.0	-132.1	-133.2	18.1	66.1	35.3	OK	OK
		◢ 4.2 ◣	521	C LC_121	289.0	0.0	-144.4	143.1	-20.0	71.4	38.8	OK	OK
SLN194+SLN196 radial (continuous)-w 1	SLN195 ring-w 1	◢ 7.2	189	C LC_119	90.7	0.0	26.4	-33.5	37.2	22.4	13.6	OK	OK
SLN194+SLN196 radial (continuous)-w 1	SLN198 ring-w 1	◢ 7.2	189	C LC_119	94.2	0.0	11.5	14.2	-52.1	23.3	15.0	OK	OK
End plate SLN195 ring	SLN195 ring-w 1	◢ 4.2 ◣	149	C LC_121	77.5	0.0	-21.9	-17.9	39.0	19.2	11.4	OK	OK
		◢ 4.2 ◣	149	C LC_121	92.6	0.0	-17.6	23.6	-46.9	22.9	14.3	OK	OK
End plate SLN198 ring	SLN198 ring-w 1	◢ 4.2 ◣	149	C LC_119	98.4	0.0	-18.0	-23.9	50.5	24.3	15.3	OK	OK
		◢ 4.2 ◣	149	C LC_119	83.6	0.0	-23.5	19.8	-41.9	20.7	12.3	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-A_160
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN199+SLN211 radial (continuous) - SLN46 column below - SLN210 ring - SLN213 ring

Name	SLN199+SLN211 radial (continuous) - SLN46 column below - SLN210 ring - SLN213 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

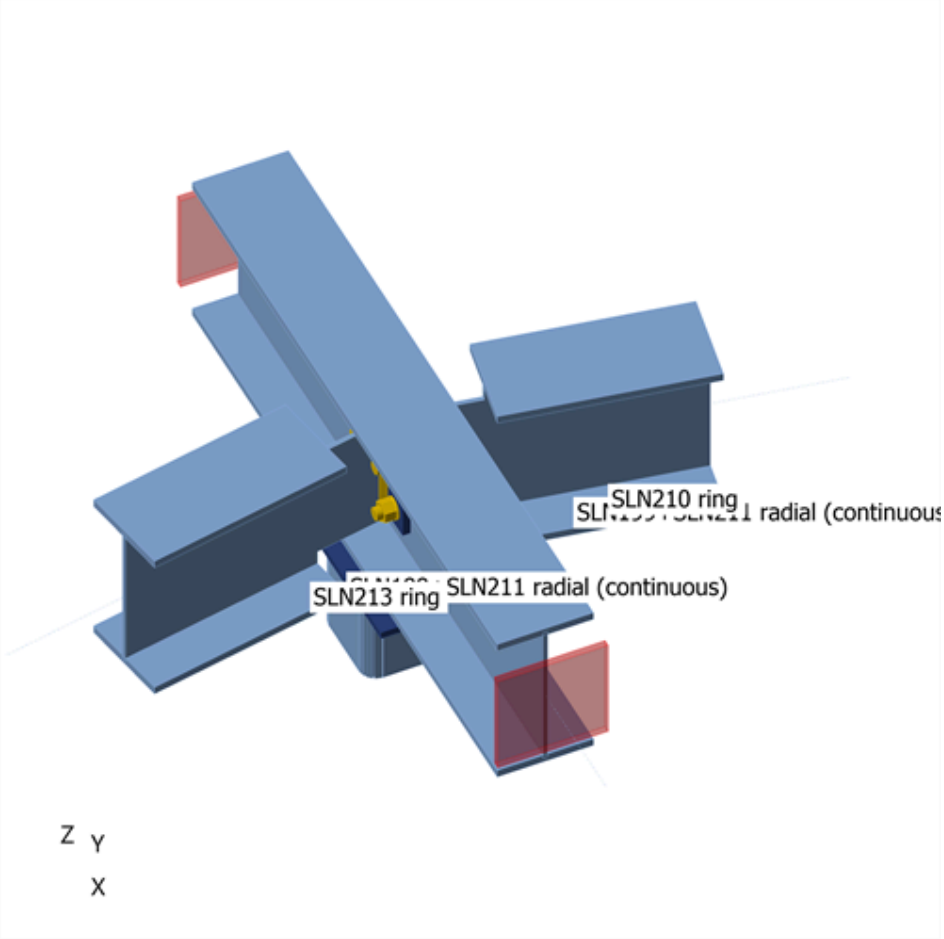
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN199+SLN211 radial (continuous)	1 - UB254x146x43	160.0	0.0	0.0	0	0	0
SLN46 column below	2 - SHS150x150x10	0.0	90.0	-20.0	0	0	0
SLN210 ring	1 - UB254x146x43	60.0	0.0	0.0	0	0	0
SLN213 ring	1 - UB254x146x43	-100.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN199+SLN211 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN199+SLN211 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN46 column below / begin		Position	0
SLN210 ring / end		Position	0
SLN213 ring / end		Position	0



SLN199+SLN211 radial (continuous) - SLN46 column below - SLN210 ring - SLN213 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN46 column below / Begin	28.4	0.0	-0.1	0.0	-0.6	0.0
	SLN210 ring / End	8.9	-0.1	-4.9	0.0	0.0	0.0
	SLN213 ring / End	8.8	0.1	-4.9	0.0	0.0	0.0
C LC_102	SLN46 column below / Begin	141.6	0.0	-0.9	0.0	-3.5	0.0
	SLN210 ring / End	28.0	-0.2	-25.6	0.0	0.0	0.0
	SLN213 ring / End	27.8	0.2	-25.6	0.0	0.0	0.0
C LC_103	SLN46 column below / Begin	135.3	0.8	-3.4	0.0	-5.1	0.0
	SLN210 ring / End	27.4	-0.2	-24.6	0.0	0.0	0.0
	SLN213 ring / End	28.0	0.3	-24.6	0.0	0.0	0.0
C LC_104	SLN46 column below / Begin	135.8	0.0	-0.9	0.0	-3.5	0.0
	SLN210 ring / End	18.7	-0.1	-24.6	0.0	0.0	0.0
	SLN213 ring / End	18.3	0.2	-24.6	0.0	0.0	0.0
C LC_105	SLN46 column below / Begin	135.6	-2.2	-1.8	0.0	-4.1	0.0
	SLN210 ring / End	24.8	-0.2	-24.6	0.0	0.0	0.0
	SLN213 ring / End	22.2	0.2	-24.6	0.0	0.0	0.0
C LC_106	SLN46 column below / Begin	135.8	0.0	-0.8	0.0	-3.4	0.0
	SLN210 ring / End	19.1	-0.2	-24.6	0.0	0.0	0.0
	SLN213 ring / End	19.1	0.2	-24.6	0.0	0.0	0.0
C LC_107	SLN46 column below / Begin	8.8	1.6	-5.2	0.0	-3.7	0.0
	SLN210 ring / End	9.4	0.0	-1.6	0.0	0.0	0.0
	SLN213 ring / End	10.9	0.0	-1.6	0.0	0.0	0.0
C LC_108	SLN46 column below / Begin	9.8	0.0	-0.1	0.0	-0.3	0.0
	SLN210 ring / End	-4.1	0.0	-1.6	0.0	0.0	0.0
	SLN213 ring / End	-4.2	0.1	-1.6	0.0	0.0	0.0
C LC_109	SLN46 column below / Begin	9.4	-4.3	-1.9	0.0	-1.5	0.0
	SLN210 ring / End	5.2	0.0	-1.6	0.0	0.0	0.0
	SLN213 ring / End	0.8	0.0	-1.6	0.0	0.0	0.0
C LC_110	SLN46 column below / Begin	9.7	0.0	-0.1	0.0	-0.3	0.0
	SLN210 ring / End	-4.3	0.0	-1.6	0.0	0.0	0.0
	SLN213 ring / End	-4.2	0.0	-1.6	0.0	0.0	0.0
C LC_111	SLN46 column below / Begin	95.3	1.6	-5.7	0.0	-5.9	0.0
	SLN210 ring / End	23.6	-0.1	-17.3	0.0	0.0	0.0
	SLN213 ring / End	25.0	0.2	-17.4	0.0	0.0	0.0
C LC_112	SLN46 column below / Begin	96.3	0.0	-0.6	0.0	-2.5	0.0
	SLN210 ring / End	8.8	-0.1	-17.3	0.0	0.0	0.0
	SLN213 ring / End	8.5	0.2	-17.4	0.0	0.0	0.0
C LC_113	SLN46 column below / Begin	95.9	-4.3	-2.5	0.0	-3.7	0.0
	SLN210 ring / End	19.1	-0.1	-17.3	0.0	0.0	0.0
	SLN213 ring / End	14.4	0.2	-17.4	0.0	0.0	0.0
C LC_114	SLN46 column below / Begin	96.2	0.0	-0.6	0.0	-2.5	0.0
	SLN210 ring / End	9.0	-0.1	-17.3	0.0	0.0	0.0
	SLN213 ring / End	9.0	0.1	-17.4	0.0	0.0	0.0
C LC_115	SLN46 column below / Begin	16.1	1.6	-5.2	0.0	-3.8	0.0
	SLN210 ring / End	11.7	0.0	-2.8	0.0	0.0	0.0
	SLN213 ring / End	13.3	0.0	-2.8	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN46 column below / Begin	17.2	0.0	-0.1	0.0	-0.5	0.0
	SLN210 ring / End	-1.8	0.0	-2.8	0.0	0.0	0.0
	SLN213 ring / End	-2.0	0.1	-2.8	0.0	0.0	0.0
C LC_117	SLN46 column below / Begin	16.8	-4.3	-2.0	0.0	-1.7	0.0
	SLN210 ring / End	7.6	0.0	-2.8	0.0	0.0	0.0
	SLN213 ring / End	3.1	0.1	-2.8	0.0	0.0	0.0
C LC_118	SLN46 column below / Begin	17.1	0.0	-0.1	0.0	-0.4	0.0
	SLN210 ring / End	-2.0	0.0	-2.8	0.0	0.0	0.0
	SLN213 ring / End	-2.0	0.0	-2.8	0.0	0.0	0.0
C LC_119	SLN46 column below / Begin	141.2	0.8	-3.4	0.0	-5.3	0.0
	SLN210 ring / End	29.7	-0.2	-25.6	0.0	0.0	0.0
	SLN213 ring / End	30.3	0.3	-25.6	0.0	0.0	0.0
C LC_120	SLN46 column below / Begin	141.4	0.0	-0.9	0.0	-3.6	0.0
	SLN210 ring / End	20.9	-0.1	-25.6	0.0	0.0	0.0
	SLN213 ring / End	20.5	0.2	-25.6	0.0	0.0	0.0
C LC_121	SLN46 column below / Begin	141.4	-2.2	-1.8	0.0	-4.2	0.1
	SLN210 ring / End	27.1	-0.2	-25.6	0.0	0.0	0.0
	SLN213 ring / End	24.4	0.2	-25.6	0.0	0.0	0.0
C LC_122	SLN46 column below / Begin	141.5	0.0	-0.9	0.0	-3.6	0.0
	SLN210 ring / End	21.3	-0.2	-25.6	0.0	0.0	0.0
	SLN213 ring / End	21.3	0.2	-25.6	0.0	0.0	0.0
C LC_123	SLN46 column below / Begin	107.0	1.6	-5.8	0.0	-6.2	0.0
	SLN210 ring / End	28.2	-0.2	-19.4	0.0	0.0	0.0
	SLN213 ring / End	29.7	0.2	-19.4	0.0	0.0	0.0
C LC_124	SLN46 column below / Begin	107.5	0.0	-0.7	0.0	-2.8	0.0
	SLN210 ring / End	13.2	-0.1	-19.4	0.0	0.0	0.0
	SLN213 ring / End	12.9	0.2	-19.4	0.0	0.0	0.0
C LC_125	SLN46 column below / Begin	107.5	-4.3	-2.5	0.0	-4.0	0.0
	SLN210 ring / End	23.8	-0.2	-19.4	0.0	0.0	0.0
	SLN213 ring / End	19.0	0.2	-19.4	0.0	0.0	0.0
C LC_126	SLN46 column below / Begin	107.6	0.0	-0.7	0.0	-2.7	0.0
	SLN210 ring / End	13.4	-0.1	-19.4	0.0	0.0	0.0
	SLN213 ring / End	13.5	0.1	-19.4	0.0	0.0	0.0
C LC_127	SLN46 column below / Begin	27.8	1.6	-5.3	0.0	-4.1	0.0
	SLN210 ring / End	16.4	0.0	-4.9	0.0	0.0	0.0
	SLN213 ring / End	17.9	0.1	-4.9	0.0	0.0	0.0
C LC_128	SLN46 column below / Begin	28.4	0.0	-0.2	0.0	-0.7	0.0
	SLN210 ring / End	2.6	-0.1	-4.9	0.0	0.0	0.0
	SLN213 ring / End	2.4	0.1	-4.9	0.0	0.0	0.0
C LC_129	SLN46 column below / Begin	28.4	-4.3	-2.0	0.0	-1.9	0.0
	SLN210 ring / End	12.2	-0.1	-4.9	0.0	0.0	0.0
	SLN213 ring / End	7.7	0.1	-4.9	0.0	0.0	0.0
C LC_130	SLN46 column below / Begin	28.5	0.0	-0.2	0.0	-0.7	0.0
	SLN210 ring / End	2.4	0.0	-4.9	0.0	0.0	0.0
	SLN213 ring / End	2.5	0.0	-4.9	0.0	0.0	0.0
C LC_135	SLN46 column below / Begin	141.6	0.0	-0.9	0.0	-3.5	0.0
	SLN210 ring / End	28.5	-0.2	-25.6	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN213 ring / End	28.2	0.2	-25.6	0.0	0.0	0.0
C LC_141	SLN46 column below / Begin	28.4	0.0	-0.1	0.0	-0.6	0.0
	SLN210 ring / End	9.1	-0.1	-4.9	0.0	0.0	0.0
	SLN213 ring / End	9.1	0.1	-4.9	0.0	0.0	0.0
C LC_142	SLN46 column below / Begin	28.4	0.0	-0.1	0.0	-0.6	0.0
	SLN210 ring / End	8.7	0.0	-4.9	0.0	0.0	0.0
	SLN213 ring / End	8.6	0.1	-4.9	0.0	0.0	0.0
C LC_143	SLN46 column below / Begin	28.4	0.0	-0.1	0.0	-0.6	0.0
	SLN210 ring / End	9.0	-0.1	-4.9	0.0	0.0	0.0
	SLN213 ring / End	8.9	0.1	-4.9	0.0	0.0	0.0
C LC_144	SLN46 column below / Begin	28.4	0.0	-0.1	0.0	-0.6	0.0
	SLN210 ring / End	8.8	-0.1	-4.9	0.0	0.0	0.0
	SLN213 ring / End	8.8	0.1	-4.9	0.0	0.0	0.0
C LC_145	SLN46 column below / Begin	141.6	0.0	-0.9	0.0	-3.5	0.0
	SLN210 ring / End	29.1	-0.3	-25.6	0.0	0.0	0.0
	SLN213 ring / End	28.9	0.3	-25.6	0.0	0.0	0.0
C LC_146	SLN46 column below / Begin	141.6	0.0	-0.9	0.0	-3.5	0.0
	SLN210 ring / End	27.0	-0.2	-25.6	0.0	0.0	0.0
	SLN213 ring / End	26.6	0.2	-25.6	0.0	0.0	0.0
C LC_147	SLN46 column below / Begin	141.6	0.0	-0.9	0.0	-3.5	0.0
	SLN210 ring / End	28.5	-0.2	-25.6	0.0	0.0	0.0
	SLN213 ring / End	28.0	0.2	-25.6	0.0	0.0	0.0
C LC_148	SLN46 column below / Begin	141.6	0.0	-0.9	0.0	-3.5	0.0
	SLN210 ring / End	27.5	-0.2	-25.6	0.0	0.0	0.0
	SLN213 ring / End	27.5	0.2	-25.6	0.0	0.0	0.0
C LC_151	SLN46 column below / Begin	90.2	0.0	-1.8	0.0	-7.2	0.0
	SLN210 ring / End	12.5	-0.1	-11.6	0.0	0.0	0.0
	SLN213 ring / End	12.4	0.1	-11.6	0.0	0.0	0.0
C LC_152	SLN46 column below / Begin	79.1	0.0	0.8	0.0	3.1	0.0
	SLN210 ring / End	28.8	-0.2	-14.1	0.0	0.0	0.0
	SLN213 ring / End	28.6	0.2	-14.2	0.0	0.0	0.0
C LC_153	SLN46 column below / Begin	141.3	0.0	-0.9	0.0	-3.8	0.0
	SLN210 ring / End	28.2	-0.2	-20.8	0.0	0.0	0.0
	SLN213 ring / End	27.9	0.2	-20.8	0.0	0.0	0.0
C LC_154	SLN46 column below / Begin	89.6	0.0	-1.8	0.0	-7.2	0.0
	SLN210 ring / End	12.6	-0.1	-11.6	0.0	0.0	0.0
	SLN213 ring / End	12.5	0.1	-11.6	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	7.6 < 100%	OK
Welds	72.7 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN199+SLN211 radial (continuous)-bfl 1	12.7	C LC_119	146.0	0.0	43.8	OK
SLN199+SLN211 radial (continuous)-tfl 1	12.7	C LC_153	72.7	0.0	0.0	OK
SLN199+SLN211 radial (continuous)-w 1	7.2	C LC_125	190.9	0.0	33.3	OK
SLN46 column below	10.0	C LC_119	164.2	0.0	0.0	OK
SLN210 ring-bfl 1	12.7	C LC_119	28.3	0.0	0.0	OK
SLN210 ring-tfl 1	12.7	C LC_122	19.8	0.0	0.0	OK
SLN210 ring-w 1	7.2	C LC_145	135.2	0.0	0.0	OK
SLN213 ring-bfl 1	12.7	C LC_119	28.6	0.0	0.0	OK
SLN213 ring-tfl 1	12.7	C LC_120	19.9	0.0	0.0	OK
SLN213 ring-w 1	7.2	C LC_119	138.9	0.0	0.0	OK
End plate column below SLN46 column below	12.0	C LC_119	114.6	0.0	48.6	OK
End plate SLN210 ring	12.0	C LC_125	26.1	0.0	1.1	OK
End plate SLN213 ring	12.0	C LC_119	23.8	0.0	0.5	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

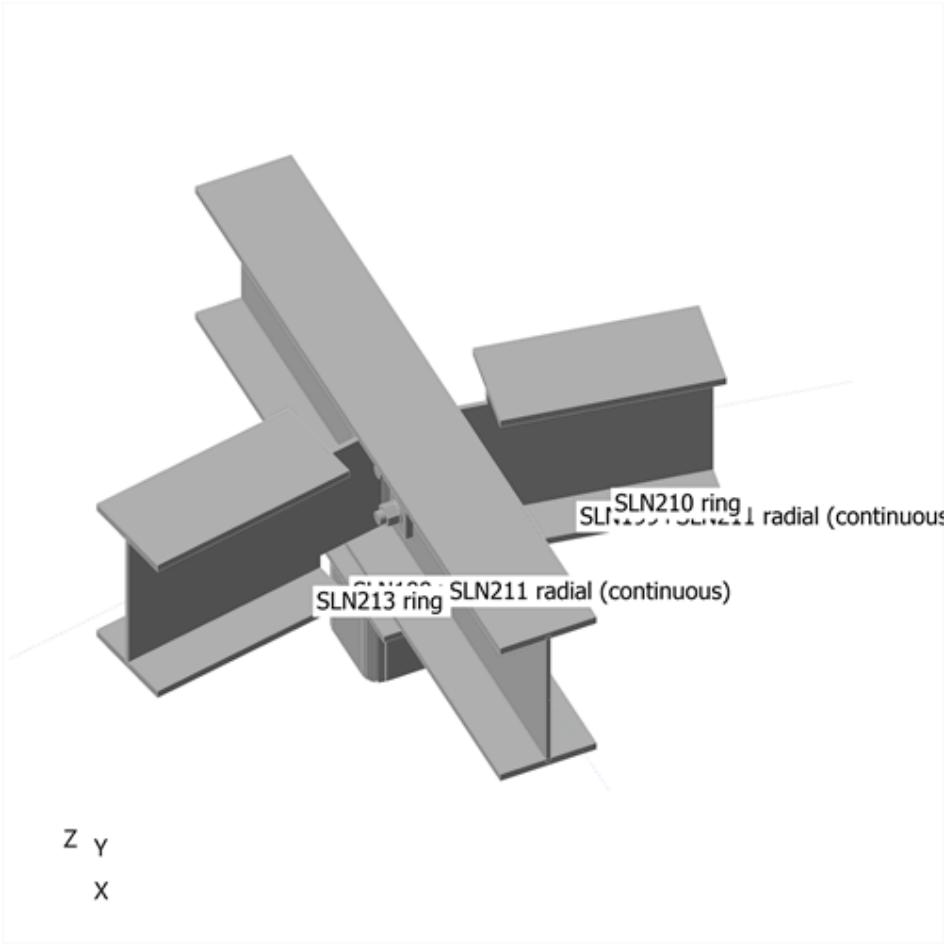
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

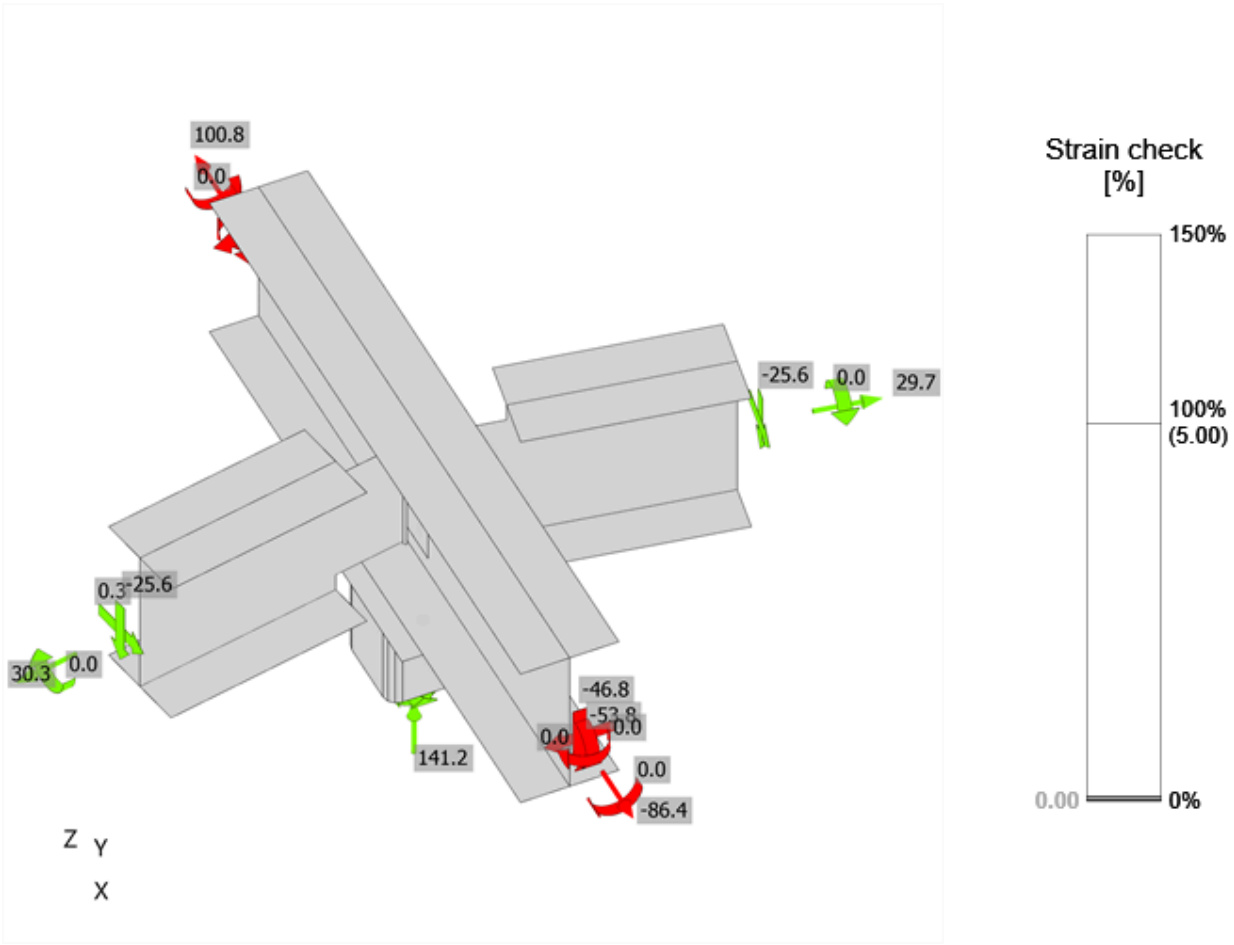
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN46 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

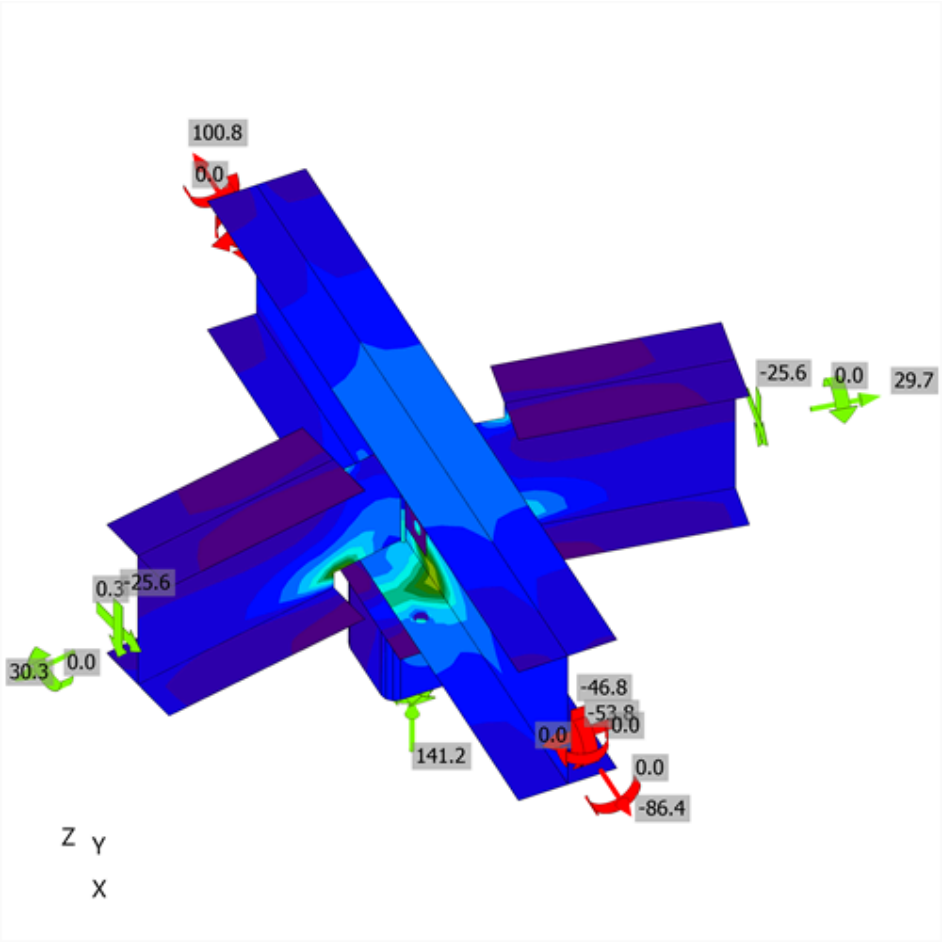
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

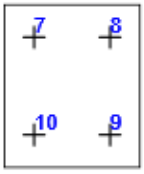


Strain check, C LC_119



Equivalent stress, C LC_119

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_119	0.6	4.1	174.1	0.4	4.4	4.7	OK	OK
	B3	M20 8.8 - 1	C LC_107	10.8	1.6	85.3	7.6	1.9	7.2	OK	OK
	B4	M20 8.8 - 1	C LC_107	9.2	1.2	156.4	6.6	1.2	5.9	OK	OK
	B5	M20 8.8 - 1	C LC_119	0.5	4.4	179.0	0.4	4.7	4.9	OK	OK
	B7	M20 8.8 - 1	C LC_151	0.0	0.3	123.8	0.0	0.5	0.3	OK	OK
	B8	M20 8.8 - 1	C LC_151	0.1	0.5	123.8	0.1	0.8	0.6	OK	OK
	B9	M20 8.8 - 1	C LC_125	1.1	0.3	141.8	0.8	0.3	0.9	OK	OK
	B10	M20 8.8 - 1	C LC_151	0.3	0.9	123.8	0.2	1.4	1.1	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN46 column below	SLN46 column below	◢ 4.2 ◣	521	C LC_119	273.1	0.0	-133.3	-131.1	-41.8	67.5	37.1	OK	OK
		◢ 4.2 ◣	521	C LC_119	294.2	0.0	-141.2	143.4	40.3	72.7	40.4	OK	OK
SLN199+SLN211 radial (continuous)-w 1	SLN210 ring-w 1	◢ 7.2	189	C LC_119	91.9	0.0	26.5	-34.2	37.6	22.7	14.3	OK	OK
SLN199+SLN211 radial (continuous)-w 1	SLN213 ring-w 1	◢ 7.2	189	C LC_119	90.7	0.0	10.6	14.1	-50.1	22.4	14.8	OK	OK
End plate SLN210 ring	SLN210 ring-w 1	◢ 4.2 ◣	149	C LC_121	78.8	0.0	-22.1	-18.8	39.4	19.5	11.7	OK	OK
		◢ 4.2 ◣	149	C LC_121	92.6	0.0	-17.0	22.4	-47.5	22.9	14.4	OK	OK
End plate SLN213 ring	SLN213 ring-w 1	◢ 4.2 ◣	149	C LC_119	96.0	0.0	-18.1	-24.2	48.8	23.7	14.9	OK	OK
		◢ 4.2 ◣	149	C LC_119	79.6	0.0	-22.3	18.2	-40.2	19.7	11.8	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-A_200
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN202+SLN215 radial (continuous) - SLN48 column below - SLN214 ring - SLN217 ring

Name	SLN202+SLN215 radial (continuous) - SLN48 column below - SLN214 ring - SLN217 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

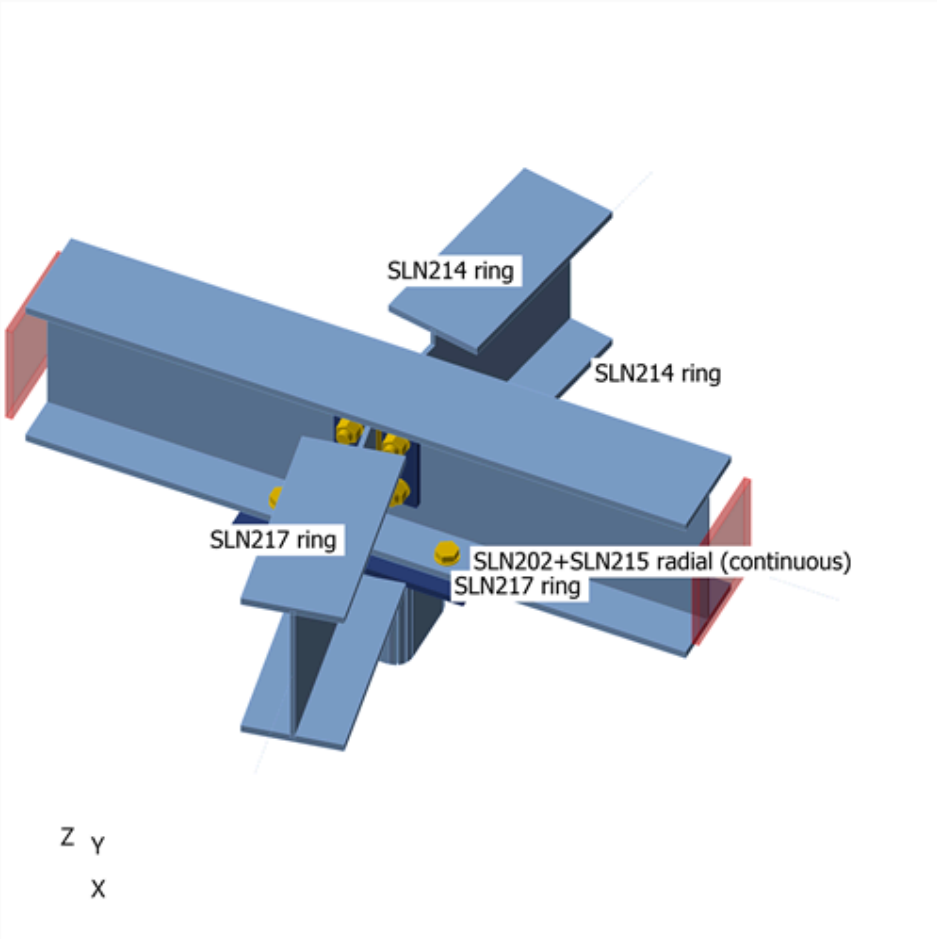
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN202+SLN215 radial (continuous)	1 - UB254x146x43	-160.0	0.0	0.0	0	0	0
SLN48 column below	2 - SHS150x150x10	0.0	90.0	20.0	0	0	0
SLN214 ring	1 - UB254x146x43	100.0	0.0	0.0	0	0	0
SLN217 ring	1 - UB254x146x43	-60.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN202+SLN215 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN202+SLN215 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN48 column below / begin		Position	0
SLN214 ring / end		Position	0
SLN217 ring / end		Position	0



SLN202+SLN215 radial (continuous) - SLN48 column below - SLN214 ring - SLN217 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN48 column below / Begin	26.3	0.0	-0.1	0.0	-0.2	0.0
	SLN214 ring / End	8.9	-0.1	-4.9	0.0	0.0	0.0
	SLN217 ring / End	8.8	0.1	-4.9	0.0	0.0	0.0
C LC_102	SLN48 column below / Begin	131.9	0.0	-0.5	0.0	-1.8	0.0
	SLN214 ring / End	27.9	-0.2	-25.5	0.0	0.0	0.0
	SLN217 ring / End	27.7	0.2	-25.6	0.0	0.0	0.0
C LC_103	SLN48 column below / Begin	125.7	-0.8	-3.0	0.0	-3.5	0.0
	SLN214 ring / End	28.1	-0.2	-24.5	0.0	0.0	0.0
	SLN217 ring / End	27.1	0.3	-24.6	0.0	0.0	0.0
C LC_104	SLN48 column below / Begin	126.5	0.0	-0.5	0.0	-1.9	0.0
	SLN214 ring / End	18.4	-0.1	-24.5	0.0	0.0	0.0
	SLN217 ring / End	18.2	0.2	-24.6	0.0	0.0	0.0
C LC_105	SLN48 column below / Begin	126.4	0.0	-0.5	0.0	-1.9	0.0
	SLN214 ring / End	19.1	-0.2	-24.5	0.0	0.0	0.0
	SLN217 ring / End	18.7	0.1	-24.6	0.0	0.0	0.0
C LC_106	SLN48 column below / Begin	126.1	2.1	-1.4	0.0	-2.4	0.0
	SLN214 ring / End	22.3	-0.2	-24.5	0.0	0.0	0.0
	SLN217 ring / End	24.5	0.2	-24.6	0.0	0.0	0.0
C LC_107	SLN48 column below / Begin	7.6	-1.6	-5.1	0.0	-3.5	0.0
	SLN214 ring / End	10.9	0.1	-1.6	0.0	0.0	0.0
	SLN217 ring / End	9.4	0.1	-1.6	0.0	0.0	0.0
C LC_108	SLN48 column below / Begin	9.2	0.0	-0.1	0.0	-0.2	0.0
	SLN214 ring / End	-4.2	0.0	-1.6	0.0	0.0	0.0
	SLN217 ring / End	-4.3	0.1	-1.6	0.0	0.0	0.0
C LC_109	SLN48 column below / Begin	8.9	0.0	0.0	0.0	-0.1	0.0
	SLN214 ring / End	-4.3	0.0	-1.6	0.0	0.0	0.0
	SLN217 ring / End	-4.4	0.0	-1.6	0.0	0.0	0.0
C LC_110	SLN48 column below / Begin	8.3	4.3	-1.9	0.0	-1.3	0.0
	SLN214 ring / End	0.9	-0.1	-1.6	0.0	0.0	0.0
	SLN217 ring / End	5.3	-0.1	-1.6	0.0	0.0	0.0
C LC_111	SLN48 column below / Begin	88.2	-1.6	-5.4	0.0	-4.7	0.0
	SLN214 ring / End	25.1	-0.1	-17.3	0.0	0.0	0.0
	SLN217 ring / End	23.4	0.2	-17.3	0.0	0.0	0.0
C LC_112	SLN48 column below / Begin	89.8	0.0	-0.4	0.0	-1.4	0.0
	SLN214 ring / End	8.6	-0.1	-17.3	0.0	0.0	0.0
	SLN217 ring / End	8.4	0.2	-17.3	0.0	0.0	0.0
C LC_113	SLN48 column below / Begin	89.5	0.0	-0.3	0.0	-1.4	0.0
	SLN214 ring / End	9.0	-0.1	-17.3	0.0	0.0	0.0
	SLN217 ring / End	8.6	0.1	-17.3	0.0	0.0	0.0
C LC_114	SLN48 column below / Begin	88.9	4.3	-2.2	0.0	-2.5	0.0
	SLN214 ring / End	14.6	-0.2	-17.3	0.0	0.0	0.0
	SLN217 ring / End	19.0	0.1	-17.3	0.0	0.0	0.0
C LC_115	SLN48 column below / Begin	14.5	-1.6	-5.2	0.0	-3.5	0.0
	SLN214 ring / End	13.3	0.1	-2.8	0.0	0.0	0.0
	SLN217 ring / End	11.7	0.1	-2.8	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN48 column below / Begin	16.0	0.0	-0.1	0.0	-0.3	0.0
	SLN214 ring / End	-2.0	0.0	-2.8	0.0	0.0	0.0
	SLN217 ring / End	-2.1	0.1	-2.8	0.0	0.0	0.0
C LC_117	SLN48 column below / Begin	15.8	0.0	0.0	0.0	-0.2	0.0
	SLN214 ring / End	-2.1	0.0	-2.8	0.0	0.0	0.0
	SLN217 ring / End	-2.2	0.0	-2.8	0.0	0.0	0.0
C LC_118	SLN48 column below / Begin	15.2	4.3	-1.9	0.0	-1.4	0.0
	SLN214 ring / End	3.2	-0.1	-2.8	0.0	0.0	0.0
	SLN217 ring / End	7.6	0.0	-2.8	0.0	0.0	0.0
C LC_119	SLN48 column below / Begin	131.2	-0.8	-3.0	0.0	-3.6	0.0
	SLN214 ring / End	30.4	-0.2	-25.5	0.0	0.0	0.0
	SLN217 ring / End	29.4	0.3	-25.6	0.0	0.0	0.0
C LC_120	SLN48 column below / Begin	131.7	0.0	-0.5	0.0	-2.0	0.0
	SLN214 ring / End	20.6	-0.1	-25.5	0.0	0.0	0.0
	SLN217 ring / End	20.4	0.2	-25.6	0.0	0.0	0.0
C LC_121	SLN48 column below / Begin	131.7	0.0	-0.5	0.0	-1.9	0.0
	SLN214 ring / End	21.4	-0.2	-25.5	0.0	0.0	0.0
	SLN217 ring / End	20.9	0.1	-25.6	0.0	0.0	0.0
C LC_122	SLN48 column below / Begin	131.5	2.1	-1.4	0.0	-2.5	0.0
	SLN214 ring / End	24.6	-0.2	-25.5	0.0	0.0	0.0
	SLN217 ring / End	26.8	0.2	-25.6	0.0	0.0	0.0
C LC_123	SLN48 column below / Begin	99.2	-1.6	-5.5	0.0	-4.8	0.0
	SLN214 ring / End	29.8	-0.1	-19.3	0.0	0.0	0.0
	SLN217 ring / End	28.1	0.3	-19.4	0.0	0.0	0.0
C LC_124	SLN48 column below / Begin	100.2	0.0	-0.4	0.0	-1.6	0.0
	SLN214 ring / End	12.9	-0.1	-19.3	0.0	0.0	0.0
	SLN217 ring / End	12.7	0.2	-19.4	0.0	0.0	0.0
C LC_125	SLN48 column below / Begin	100.2	0.0	-0.4	0.0	-1.5	0.0
	SLN214 ring / End	13.5	-0.1	-19.3	0.0	0.0	0.0
	SLN217 ring / End	13.0	0.1	-19.4	0.0	0.0	0.0
C LC_126	SLN48 column below / Begin	99.7	4.3	-2.2	0.0	-2.7	0.0
	SLN214 ring / End	19.1	-0.2	-19.3	0.0	0.0	0.0
	SLN217 ring / End	23.6	0.1	-19.4	0.0	0.0	0.0
C LC_127	SLN48 column below / Begin	25.5	-1.6	-5.2	0.0	-3.7	0.0
	SLN214 ring / End	18.0	0.0	-4.9	0.0	0.0	0.0
	SLN217 ring / End	16.4	0.1	-4.9	0.0	0.0	0.0
C LC_128	SLN48 column below / Begin	26.5	0.0	-0.1	0.0	-0.4	0.0
	SLN214 ring / End	2.3	0.0	-4.9	0.0	0.0	0.0
	SLN217 ring / End	2.2	0.1	-4.9	0.0	0.0	0.0
C LC_129	SLN48 column below / Begin	26.5	0.0	-0.1	0.0	-0.3	0.0
	SLN214 ring / End	2.4	0.0	-4.9	0.0	0.0	0.0
	SLN217 ring / End	2.2	0.0	-4.9	0.0	0.0	0.0
C LC_130	SLN48 column below / Begin	26.0	4.3	-1.9	0.0	-1.5	0.0
	SLN214 ring / End	7.8	-0.1	-4.9	0.0	0.0	0.0
	SLN217 ring / End	12.2	0.0	-4.9	0.0	0.0	0.0
C LC_135	SLN48 column below / Begin	131.9	0.0	-0.5	0.0	-1.9	0.0
	SLN214 ring / End	28.3	-0.2	-25.5	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN217 ring / End	28.2	0.2	-25.6	0.0	0.0	0.0
C LC_141	SLN48 column below / Begin	26.3	0.0	-0.1	0.0	-0.2	0.0
	SLN214 ring / End	9.1	-0.1	-4.9	0.0	0.0	0.0
	SLN217 ring / End	9.0	0.1	-4.9	0.0	0.0	0.0
C LC_142	SLN48 column below / Begin	26.4	0.0	-0.1	0.0	-0.2	0.0
	SLN214 ring / End	8.6	0.0	-4.9	0.0	0.0	0.0
	SLN217 ring / End	8.6	0.0	-4.9	0.0	0.0	0.0
C LC_143	SLN48 column below / Begin	26.3	0.0	-0.1	0.0	-0.2	0.0
	SLN214 ring / End	8.8	-0.1	-4.9	0.0	0.0	0.0
	SLN217 ring / End	8.7	0.1	-4.9	0.0	0.0	0.0
C LC_144	SLN48 column below / Begin	26.3	0.0	-0.1	0.0	-0.2	0.0
	SLN214 ring / End	8.9	-0.1	-4.9	0.0	0.0	0.0
	SLN217 ring / End	8.9	0.1	-4.9	0.0	0.0	0.0
C LC_145	SLN48 column below / Begin	131.8	0.0	-0.5	0.0	-1.8	0.0
	SLN214 ring / End	29.0	-0.3	-25.5	0.0	0.0	0.0
	SLN217 ring / End	28.8	0.3	-25.6	0.0	0.0	0.0
C LC_146	SLN48 column below / Begin	131.9	0.0	-0.5	0.0	-1.9	0.0
	SLN214 ring / End	26.7	-0.2	-25.5	0.0	0.0	0.0
	SLN217 ring / End	26.6	0.1	-25.6	0.0	0.0	0.0
C LC_147	SLN48 column below / Begin	131.9	0.0	-0.5	0.0	-1.9	0.0
	SLN214 ring / End	27.6	-0.2	-25.5	0.0	0.0	0.0
	SLN217 ring / End	27.2	0.2	-25.6	0.0	0.0	0.0
C LC_148	SLN48 column below / Begin	131.9	0.0	-0.5	0.0	-1.8	0.0
	SLN214 ring / End	28.2	-0.2	-25.5	0.0	0.0	0.0
	SLN217 ring / End	28.2	0.2	-25.6	0.0	0.0	0.0
C LC_151	SLN48 column below / Begin	80.2	0.0	-1.3	0.0	-5.4	0.0
	SLN214 ring / End	12.4	-0.1	-11.9	0.0	0.0	0.0
	SLN217 ring / End	12.3	0.1	-11.8	0.0	0.0	0.0
C LC_152	SLN48 column below / Begin	77.4	0.0	0.8	0.0	3.3	0.0
	SLN214 ring / End	28.7	-0.2	-14.5	0.0	0.0	0.0
	SLN217 ring / End	28.5	0.2	-14.1	0.0	0.0	0.0
C LC_153	SLN48 column below / Begin	131.9	0.0	-0.5	0.0	-2.1	0.0
	SLN214 ring / End	28.0	-0.2	-21.5	0.0	0.0	0.0
	SLN217 ring / End	27.8	0.2	-21.0	0.0	0.0	0.0
C LC_154	SLN48 column below / Begin	79.4	0.0	-1.3	0.0	-5.3	0.0
	SLN214 ring / End	12.5	-0.1	-11.9	0.0	0.0	0.0
	SLN217 ring / End	12.5	0.1	-11.8	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	7.2 < 100%	OK
Welds	60.7 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN202+SLN215 radial (continuous)-bfl 1	12.7	C LC_119	123.3	0.0	37.0	OK
SLN202+SLN215 radial (continuous)-tfl 1	12.7	C LC_153	64.1	0.0	0.0	OK
SLN202+SLN215 radial (continuous)-w 1	7.2	C LC_126	181.9	0.0	27.9	OK
SLN48 column below	10.0	C LC_119	134.0	0.0	0.0	OK
SLN214 ring-bfl 1	12.7	C LC_119	28.4	0.0	0.0	OK
SLN214 ring-tfl 1	12.7	C LC_120	19.6	0.0	0.0	OK
SLN214 ring-w 1	7.2	C LC_145	135.9	0.0	0.0	OK
SLN217 ring-bfl 1	12.7	C LC_119	28.5	0.0	0.0	OK
SLN217 ring-tfl 1	12.7	C LC_120	19.7	0.0	0.0	OK
SLN217 ring-w 1	7.2	C LC_119	137.5	0.0	0.0	OK
End plate column below SLN48 column below	12.0	C LC_119	98.0	0.0	34.7	OK
End plate SLN214 ring	12.0	C LC_126	23.1	0.0	5.4	OK
End plate SLN217 ring	12.0	C LC_126	27.4	0.0	0.8	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

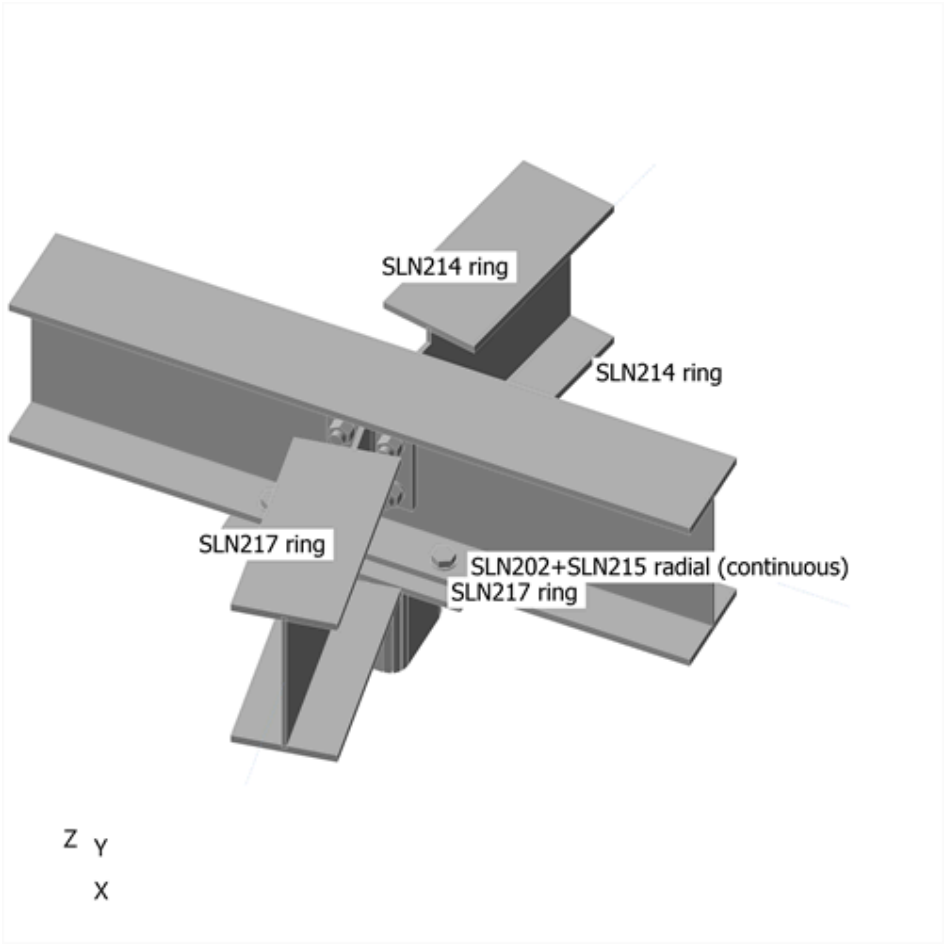
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

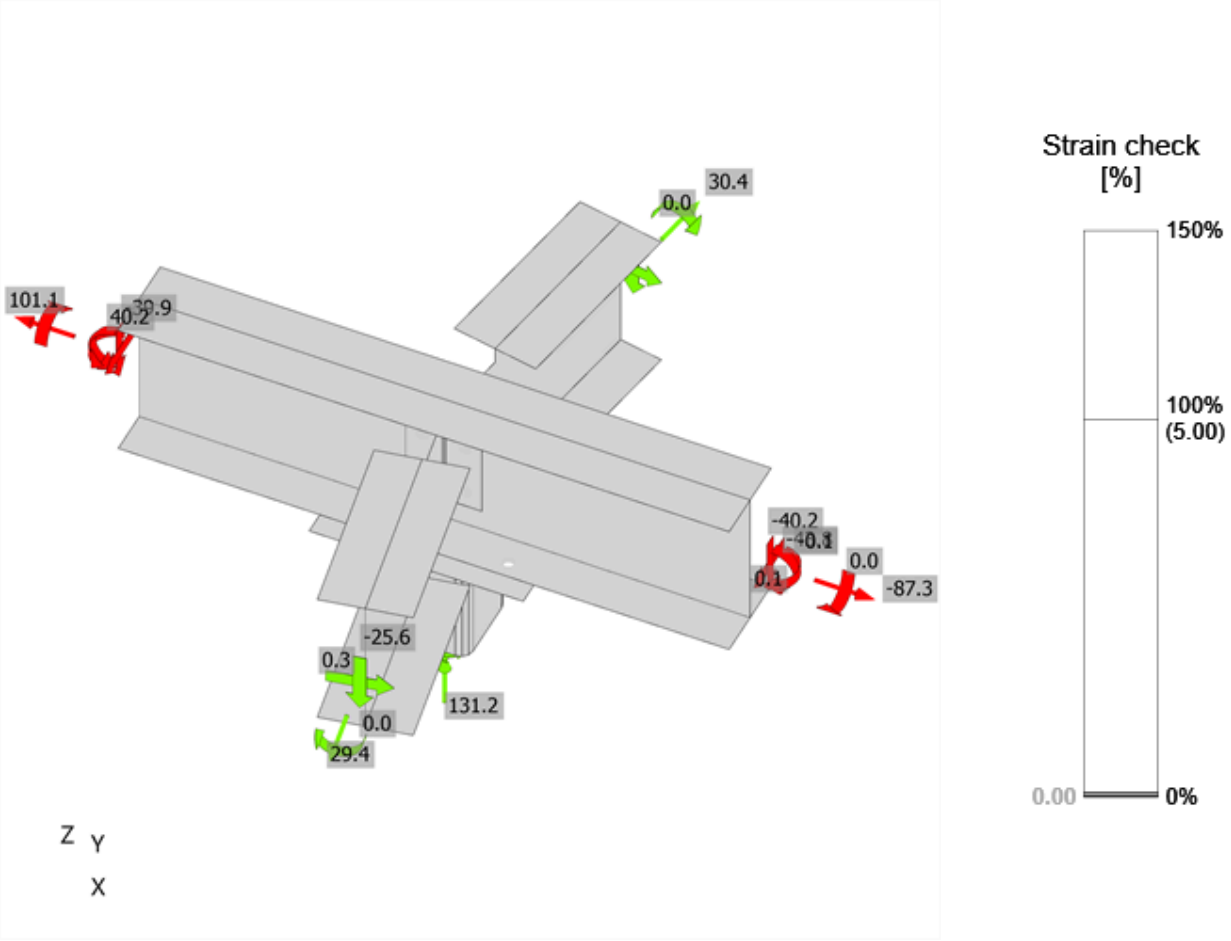
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN48 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

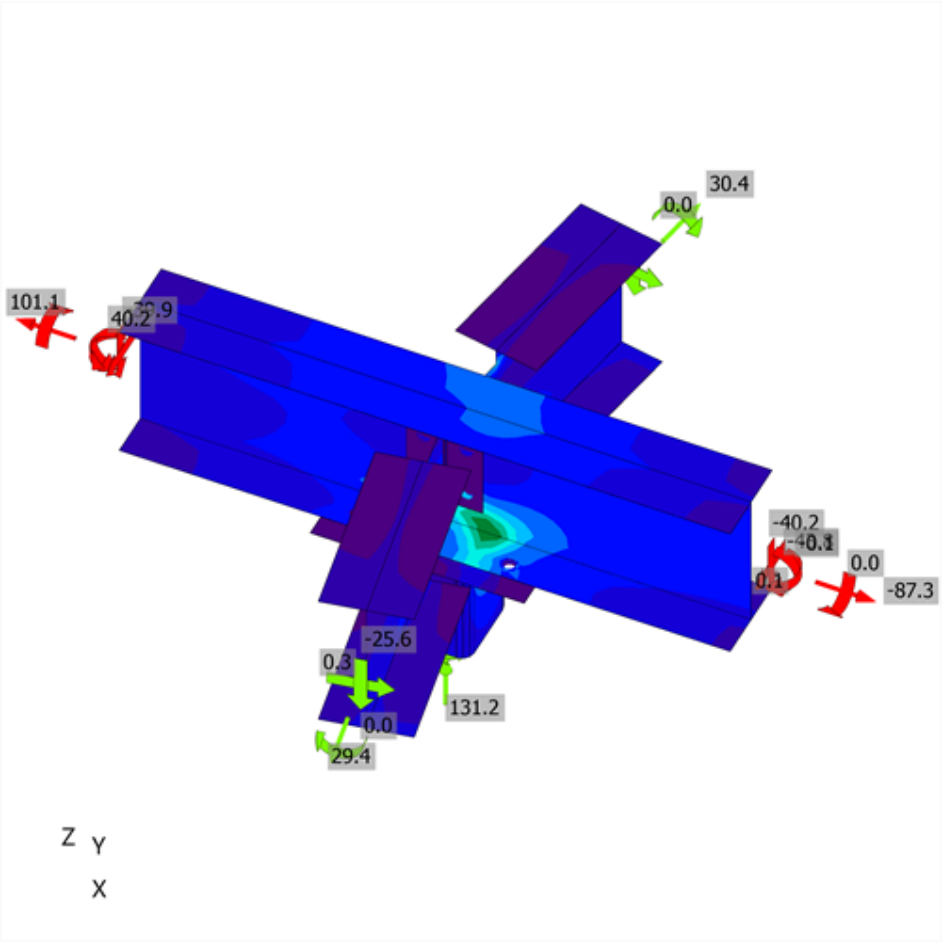
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

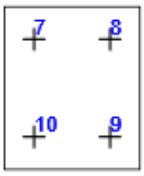


Strain check, C LC_119



Equivalent stress, C LC_119

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_119	0.4	3.9	178.9	0.3	4.1	4.3	OK	OK
	B3	M20 8.8 - 1	C LC_107	8.7	1.2	156.4	6.2	1.3	5.7	OK	OK
	B4	M20 8.8 - 1	C LC_107	10.2	1.6	85.9	7.2	1.9	6.9	OK	OK
	B5	M20 8.8 - 1	C LC_119	0.4	3.7	173.1	0.3	4.0	4.2	OK	OK
	B7	M20 8.8 - 1	C LC_152	0.3	0.2	123.8	0.2	0.4	0.4	OK	OK
	B8	M20 8.8 - 1	C LC_151	0.1	0.4	123.8	0.1	0.6	0.4	OK	OK
	B9	M20 8.8 - 1	C LC_126	1.2	0.3	123.8	0.9	0.4	1.0	OK	OK
	B10	M20 8.8 - 1	C LC_126	1.2	0.7	123.8	0.8	0.9	1.3	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN48 column below	SLN48 column below	◢ 4.2 ◣	521	C LC_119	232.2	0.0	-112.4	-115.7	19.4	57.4	30.6	OK	OK
		◢ 4.2 ◣	521	C LC_119	245.8	0.0	-124.2	120.9	-19.4	60.7	32.8	OK	OK
SLN202+SLN215 radial (continuous)-w 1	SLN214 ring-w 1	◢ 7.2	189	C LC_122	84.7	0.0	31.0	25.5	-37.7	20.9	12.3	OK	OK
SLN202+SLN215 radial (continuous)-w 1	SLN217 ring-w 1	◢ 7.2	189	C LC_122	84.8	0.0	11.4	-8.5	47.7	20.9	13.4	OK	OK
End plate SLN214 ring	SLN214 ring-w 1	◢ 4.2 ◣	149	C LC_119	70.5	0.0	-18.9	-14.9	36.3	17.4	10.6	OK	OK
		◢ 4.2 ◣	149	C LC_119	83.3	0.0	-16.7	22.4	-41.5	20.6	13.1	OK	OK
End plate SLN217 ring	SLN217 ring-w 1	◢ 4.2 ◣	149	C LC_122	91.4	0.0	-18.5	-24.0	45.8	22.6	14.3	OK	OK
		◢ 4.2 ◣	149	C LC_122	76.6	0.0	-19.9	15.9	-39.6	18.9	11.6	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-A_240
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN204+SLN219 radial (continuous) - SLN50 column below - SLN218 ring - SLN221 ring

Name	SLN204+SLN219 radial (continuous) - SLN50 column below - SLN218 ring - SLN221 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

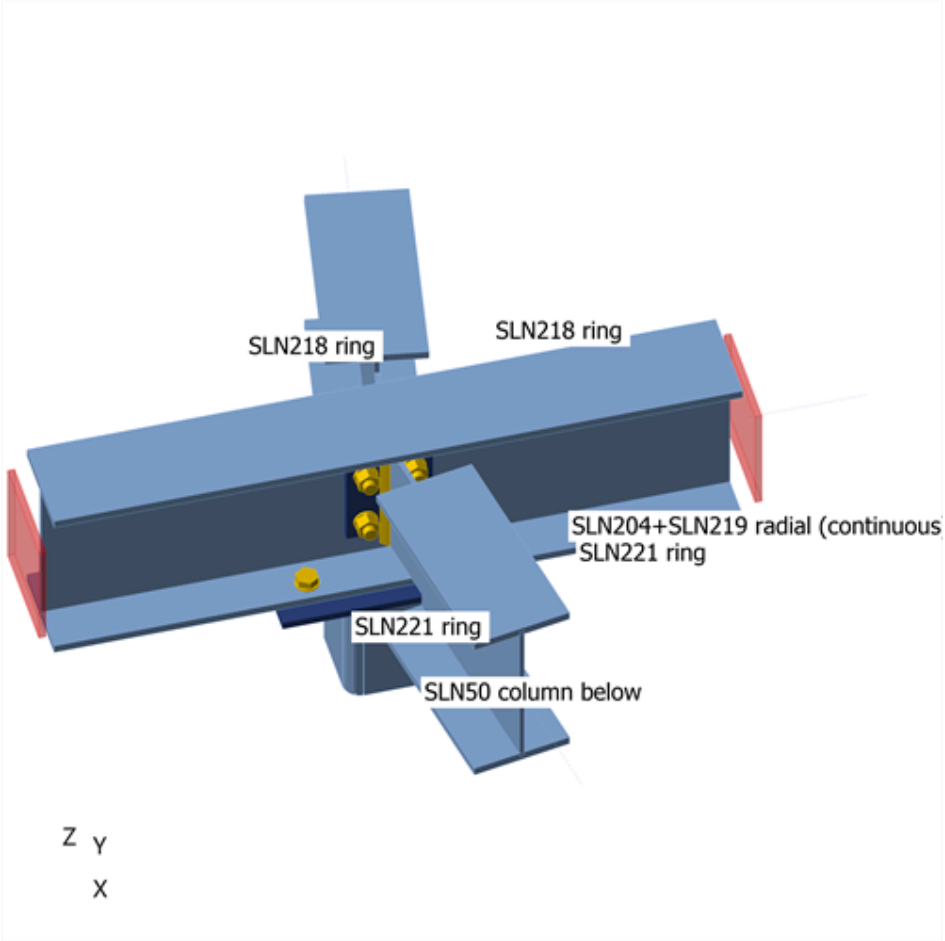
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN204+SLN219 radial (continuous)	1 - UB254x146x43	-120.0	0.0	0.0	0	0	0
SLN50 column below	2 - SHS150x150x10	0.0	90.0	60.0	0	0	0
SLN218 ring	1 - UB254x146x43	140.0	0.0	0.0	0	0	0
SLN221 ring	1 - UB254x146x43	-20.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN204+SLN219 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN204+SLN219 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN50 column below / begin		Position	0
SLN218 ring / end		Position	0
SLN221 ring / end		Position	0



SLN204+SLN219 radial (continuous) - SLN50 column below - SLN218 ring - SLN221 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN50 column below / Begin	28.5	0.0	-0.1	0.0	-0.5	0.0
	SLN218 ring / End	8.8	-0.1	-5.0	0.0	0.0	0.0
	SLN221 ring / End	8.9	0.0	-4.9	0.0	0.0	0.0
C LC_102	SLN50 column below / Begin	142.4	0.0	-0.8	0.0	-3.2	0.0
	SLN218 ring / End	27.6	-0.2	-26.1	0.0	0.0	0.0
	SLN221 ring / End	28.0	0.2	-25.7	0.0	0.0	0.0
C LC_103	SLN50 column below / Begin	136.2	-2.0	-2.1	0.0	-4.0	0.0
	SLN218 ring / End	25.0	-0.1	-25.0	0.0	0.0	0.0
	SLN221 ring / End	23.2	0.2	-24.7	0.0	0.0	0.0
C LC_104	SLN50 column below / Begin	136.5	0.0	-0.8	0.0	-3.1	0.0
	SLN218 ring / End	18.6	-0.1	-25.0	0.0	0.0	0.0
	SLN221 ring / End	19.1	0.1	-24.7	0.0	0.0	0.0
C LC_105	SLN50 column below / Begin	136.6	0.0	-0.8	0.0	-3.2	0.0
	SLN218 ring / End	17.9	-0.2	-25.0	0.0	0.0	0.0
	SLN221 ring / End	18.1	0.1	-24.7	0.0	0.0	0.0
C LC_106	SLN50 column below / Begin	136.2	1.1	-3.1	0.0	-4.7	0.0
	SLN218 ring / End	26.8	-0.2	-25.0	0.0	0.0	0.0
	SLN221 ring / End	28.6	0.2	-24.7	0.0	0.0	0.0
C LC_107	SLN50 column below / Begin	9.1	-4.0	-2.8	0.0	-2.0	0.0
	SLN218 ring / End	6.1	0.1	-1.6	0.0	0.0	0.0
	SLN221 ring / End	2.2	0.1	-1.6	0.0	0.0	0.0
C LC_108	SLN50 column below / Begin	9.7	0.0	-0.1	0.0	-0.2	0.0
	SLN218 ring / End	-4.1	0.0	-1.6	0.0	0.0	0.0
	SLN221 ring / End	-4.2	0.1	-1.6	0.0	0.0	0.0
C LC_109	SLN50 column below / Begin	9.8	0.0	-0.1	0.0	-0.3	0.0
	SLN218 ring / End	-4.9	-0.1	-1.6	0.0	0.0	0.0
	SLN221 ring / End	-5.0	0.0	-1.6	0.0	0.0	0.0
C LC_110	SLN50 column below / Begin	9.0	2.3	-4.8	0.0	-3.4	0.0
	SLN218 ring / End	9.3	0.0	-1.6	0.0	0.0	0.0
	SLN221 ring / End	11.8	0.0	-1.6	0.0	0.0	0.0
C LC_111	SLN50 column below / Begin	96.1	-4.0	-3.3	0.0	-4.0	0.0
	SLN218 ring / End	19.8	0.0	-17.6	0.0	0.0	0.0
	SLN221 ring / End	16.1	0.2	-17.4	0.0	0.0	0.0
C LC_112	SLN50 column below / Begin	96.7	0.0	-0.6	0.0	-2.3	0.0
	SLN218 ring / End	8.8	-0.1	-17.6	0.0	0.0	0.0
	SLN221 ring / End	9.1	0.1	-17.4	0.0	0.0	0.0
C LC_113	SLN50 column below / Begin	96.8	0.0	-0.6	0.0	-2.3	0.0
	SLN218 ring / End	7.8	-0.2	-17.6	0.0	0.0	0.0
	SLN221 ring / End	7.8	0.0	-17.4	0.0	0.0	0.0
C LC_114	SLN50 column below / Begin	96.0	2.3	-5.3	0.0	-5.4	0.0
	SLN218 ring / End	23.2	-0.2	-17.6	0.0	0.0	0.0
	SLN221 ring / End	26.0	0.1	-17.4	0.0	0.0	0.0
C LC_115	SLN50 column below / Begin	16.5	-4.0	-2.8	0.0	-2.2	0.0
	SLN218 ring / End	8.4	0.1	-2.9	0.0	0.0	0.0
	SLN221 ring / End	4.6	0.1	-2.9	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN50 column below / Begin	17.1	0.0	-0.1	0.0	-0.4	0.0
	SLN218 ring / End	-1.9	0.0	-2.9	0.0	0.0	0.0
	SLN221 ring / End	-1.9	0.1	-2.9	0.0	0.0	0.0
C LC_117	SLN50 column below / Begin	17.2	0.0	-0.1	0.0	-0.4	0.0
	SLN218 ring / End	-2.7	-0.1	-2.9	0.0	0.0	0.0
	SLN221 ring / End	-2.8	0.0	-2.9	0.0	0.0	0.0
C LC_118	SLN50 column below / Begin	16.4	2.3	-4.8	0.0	-3.5	0.0
	SLN218 ring / End	11.7	0.0	-2.9	0.0	0.0	0.0
	SLN221 ring / End	14.1	0.0	-2.9	0.0	0.0	0.0
C LC_119	SLN50 column below / Begin	142.0	-2.0	-2.2	0.0	-4.1	0.0
	SLN218 ring / End	27.3	-0.2	-26.1	0.0	0.0	0.0
	SLN221 ring / End	25.5	0.2	-25.7	0.0	0.0	0.0
C LC_120	SLN50 column below / Begin	142.2	0.0	-0.8	0.0	-3.3	0.0
	SLN218 ring / End	20.8	-0.1	-26.1	0.0	0.0	0.0
	SLN221 ring / End	21.3	0.1	-25.7	0.0	0.0	0.0
C LC_121	SLN50 column below / Begin	142.2	0.0	-0.8	0.0	-3.3	0.0
	SLN218 ring / End	20.1	-0.2	-26.1	0.0	0.0	0.0
	SLN221 ring / End	20.3	0.1	-25.7	0.0	0.0	0.0
C LC_122	SLN50 column below / Begin	142.0	1.1	-3.2	0.0	-4.8	0.0
	SLN218 ring / End	29.2	-0.2	-26.1	0.0	0.0	0.0
	SLN221 ring / End	30.9	0.2	-25.7	0.0	0.0	0.0
C LC_123	SLN50 column below / Begin	107.8	-4.0	-3.3	0.0	-4.3	0.0
	SLN218 ring / End	24.5	-0.1	-19.7	0.0	0.0	0.0
	SLN221 ring / End	20.7	0.2	-19.5	0.0	0.0	0.0
C LC_124	SLN50 column below / Begin	108.1	0.0	-0.6	0.0	-2.5	0.0
	SLN218 ring / End	13.1	-0.1	-19.7	0.0	0.0	0.0
	SLN221 ring / End	13.5	0.1	-19.5	0.0	0.0	0.0
C LC_125	SLN50 column below / Begin	108.1	0.0	-0.6	0.0	-2.5	0.0
	SLN218 ring / End	12.1	-0.2	-19.7	0.0	0.0	0.0
	SLN221 ring / End	12.2	0.0	-19.5	0.0	0.0	0.0
C LC_126	SLN50 column below / Begin	107.7	2.3	-5.3	0.0	-5.7	0.0
	SLN218 ring / End	27.8	-0.2	-19.7	0.0	0.0	0.0
	SLN221 ring / End	30.7	0.2	-19.5	0.0	0.0	0.0
C LC_127	SLN50 column below / Begin	28.2	-4.0	-2.9	0.0	-2.4	0.0
	SLN218 ring / End	13.1	0.0	-5.0	0.0	0.0	0.0
	SLN221 ring / End	9.2	0.1	-4.9	0.0	0.0	0.0
C LC_128	SLN50 column below / Begin	28.5	0.0	-0.1	0.0	-0.6	0.0
	SLN218 ring / End	2.5	0.0	-5.0	0.0	0.0	0.0
	SLN221 ring / End	2.5	0.1	-4.9	0.0	0.0	0.0
C LC_129	SLN50 column below / Begin	28.5	0.0	-0.2	0.0	-0.6	0.0
	SLN218 ring / End	1.6	-0.1	-5.0	0.0	0.0	0.0
	SLN221 ring / End	1.5	0.0	-4.9	0.0	0.0	0.0
C LC_130	SLN50 column below / Begin	28.1	2.3	-4.9	0.0	-3.8	0.0
	SLN218 ring / End	16.3	-0.1	-5.0	0.0	0.0	0.0
	SLN221 ring / End	18.9	0.1	-4.9	0.0	0.0	0.0
C LC_135	SLN50 column below / Begin	142.4	0.0	-0.8	0.0	-3.2	0.0
	SLN218 ring / End	28.0	-0.2	-26.1	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN221 ring / End	28.5	0.2	-25.7	0.0	0.0	0.0
C LC_141	SLN50 column below / Begin	28.5	0.0	-0.1	0.0	-0.5	0.0
	SLN218 ring / End	8.9	-0.1	-5.0	0.0	0.0	0.0
	SLN221 ring / End	9.0	0.1	-4.9	0.0	0.0	0.0
C LC_142	SLN50 column below / Begin	28.5	0.0	-0.1	0.0	-0.5	0.0
	SLN218 ring / End	8.6	0.0	-5.0	0.0	0.0	0.0
	SLN221 ring / End	8.8	0.0	-4.9	0.0	0.0	0.0
C LC_143	SLN50 column below / Begin	28.5	0.0	-0.1	0.0	-0.5	0.0
	SLN218 ring / End	8.6	0.0	-5.0	0.0	0.0	0.0
	SLN221 ring / End	8.7	0.0	-4.9	0.0	0.0	0.0
C LC_144	SLN50 column below / Begin	28.5	0.0	-0.1	0.0	-0.5	0.0
	SLN218 ring / End	9.0	-0.1	-5.0	0.0	0.0	0.0
	SLN221 ring / End	9.1	0.1	-4.9	0.0	0.0	0.0
C LC_145	SLN50 column below / Begin	142.4	0.0	-0.8	0.0	-3.1	0.0
	SLN218 ring / End	28.3	-0.2	-26.1	0.0	0.0	0.0
	SLN221 ring / End	28.5	0.2	-25.7	0.0	0.0	0.0
C LC_146	SLN50 column below / Begin	142.4	0.0	-0.8	0.0	-3.2	0.0
	SLN218 ring / End	26.9	-0.2	-26.1	0.0	0.0	0.0
	SLN221 ring / End	27.5	0.1	-25.7	0.0	0.0	0.0
C LC_147	SLN50 column below / Begin	142.4	0.0	-0.8	0.0	-3.2	0.0
	SLN218 ring / End	26.6	-0.1	-26.1	0.0	0.0	0.0
	SLN221 ring / End	26.9	0.1	-25.7	0.0	0.0	0.0
C LC_148	SLN50 column below / Begin	142.4	0.0	-0.8	0.0	-3.1	0.0
	SLN218 ring / End	28.5	-0.2	-26.1	0.0	0.0	0.0
	SLN221 ring / End	29.1	0.2	-25.7	0.0	0.0	0.0
C LC_151	SLN50 column below / Begin	90.1	0.0	-1.7	0.0	-7.0	0.0
	SLN218 ring / End	12.3	-0.1	-11.6	0.0	0.0	0.0
	SLN221 ring / End	12.5	0.1	-11.7	0.0	0.0	0.0
C LC_152	SLN50 column below / Begin	79.3	0.0	0.8	0.0	3.3	0.0
	SLN218 ring / End	28.4	-0.2	-14.0	0.0	0.0	0.0
	SLN221 ring / End	28.6	0.1	-13.9	0.0	0.0	0.0
C LC_153	SLN50 column below / Begin	141.3	0.0	-0.8	0.0	-3.3	0.0
	SLN218 ring / End	27.7	-0.2	-20.7	0.0	0.0	0.0
	SLN221 ring / End	28.1	0.2	-20.6	0.0	0.0	0.0
C LC_154	SLN50 column below / Begin	89.5	0.0	-1.7	0.0	-6.9	0.0
	SLN218 ring / End	12.4	-0.1	-11.6	0.0	0.0	0.0
	SLN221 ring / End	12.7	0.1	-11.7	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	7.1 < 100%	OK
Welds	68.5 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN204+SLN219 radial (continuous)-bfl 1	12.7	C LC_122	144.4	0.0	42.8	OK
SLN204+SLN219 radial (continuous)-tfl 1	12.7	C LC_153	72.5	0.0	0.0	OK
SLN204+SLN219 radial (continuous)-w 1	7.2	C LC_123	191.5	0.0	34.0	OK
SLN50 column below	10.0	C LC_122	155.9	0.0	0.0	OK
SLN218 ring-bfl 1	12.7	C LC_122	28.6	0.0	0.0	OK
SLN218 ring-tfl 1	12.7	C LC_121	20.3	0.0	0.0	OK
SLN218 ring-w 1	7.2	C LC_122	134.4	0.0	0.0	OK
SLN221 ring-bfl 1	12.7	C LC_122	28.6	0.0	0.0	OK
SLN221 ring-tfl 1	12.7	C LC_121	19.7	0.0	0.0	OK
SLN221 ring-w 1	7.2	C LC_122	130.9	0.0	0.0	OK
End plate column below SLN50 column below	12.0	C LC_122	114.0	0.0	38.8	OK
End plate SLN218 ring	12.0	C LC_123	26.9	0.0	0.9	OK
End plate SLN221 ring	12.0	C LC_122	22.8	0.0	0.7	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

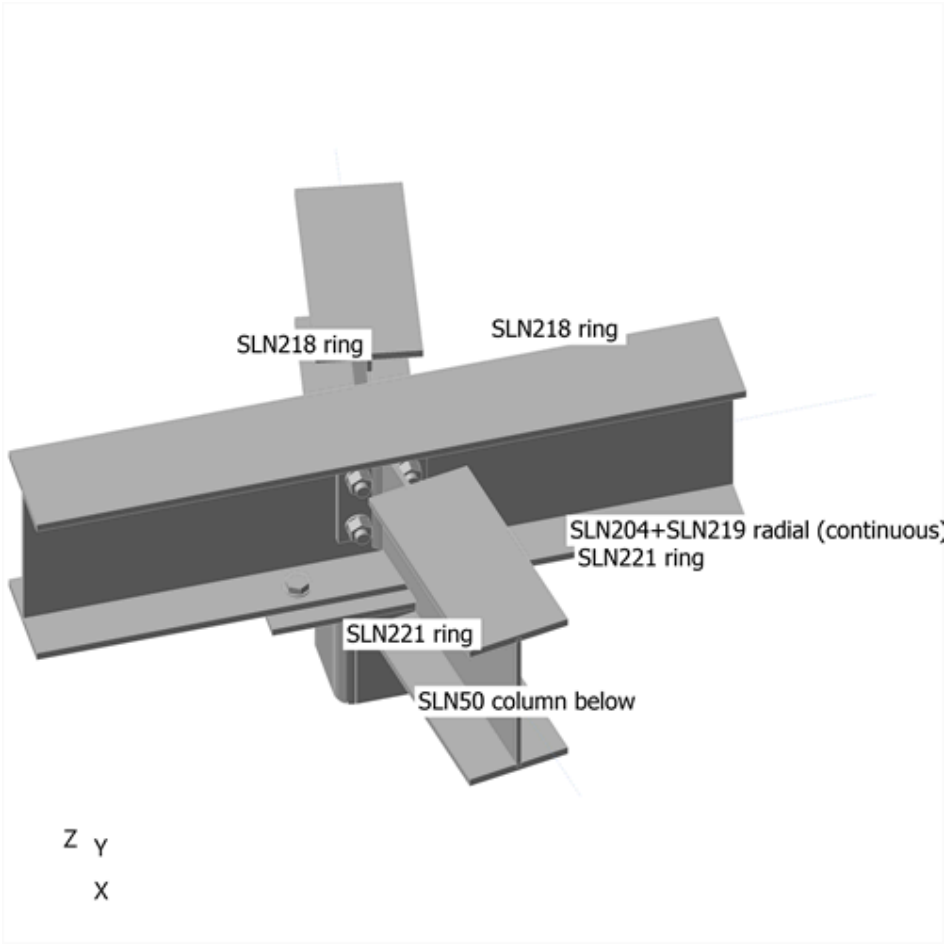
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

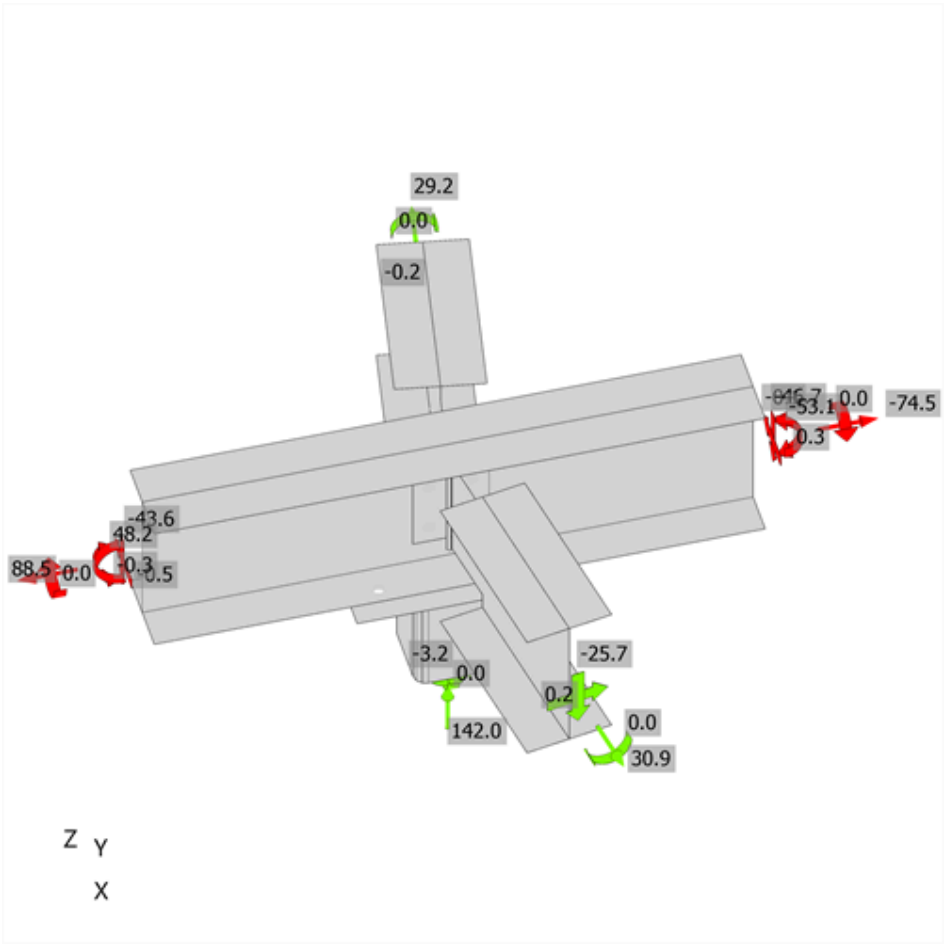
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN50 column below	150	C LC_154	0	5	0.0	OK

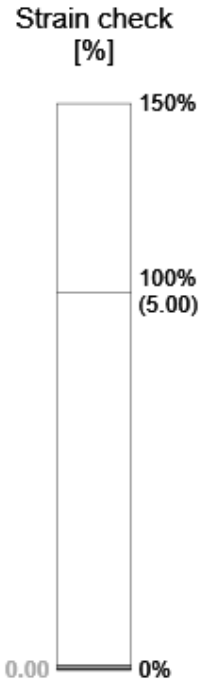
Symbol explanation

d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation

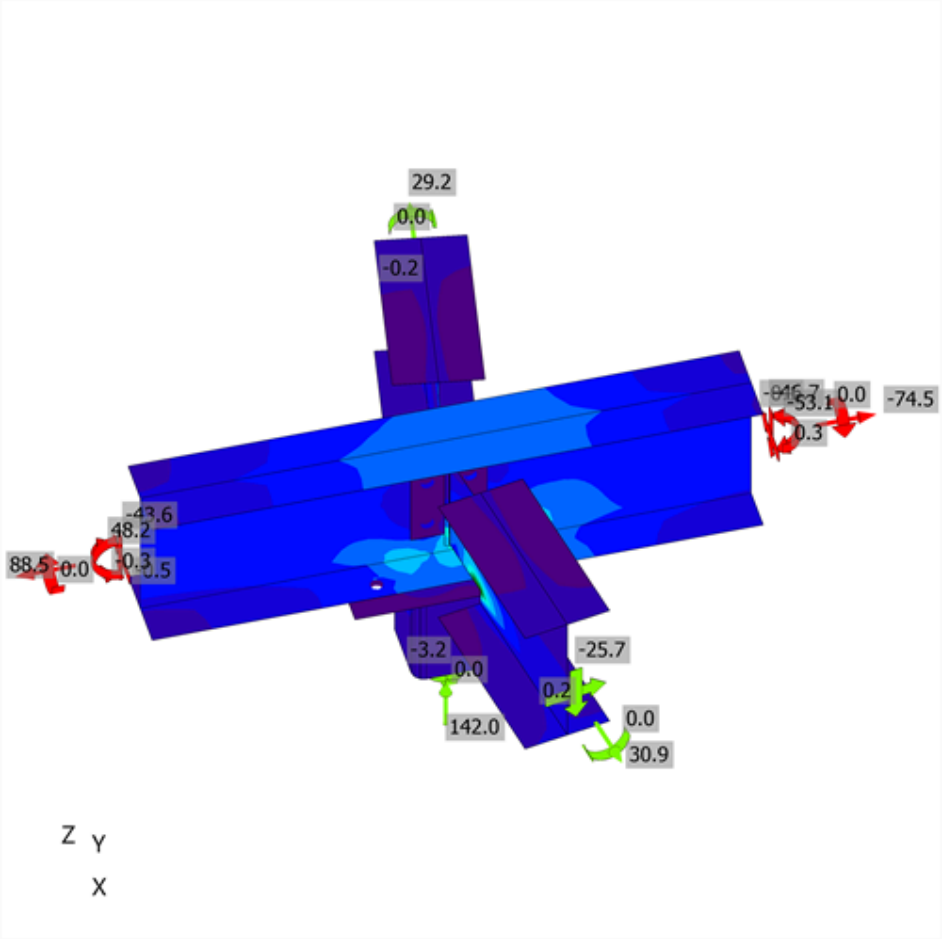


Overall check



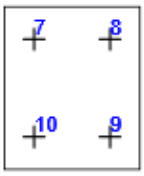


Strain check, C LC_122



Equivalent stress, C LC_122

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_119	0.4	4.3	185.3	0.3	4.6	4.8	OK	OK
	B3	M20 8.8 - 1	C LC_110	10.1	1.6	81.0	7.1	2.0	6.8	OK	OK
	B4	M20 8.8 - 1	C LC_110	7.8	1.0	156.4	5.5	1.1	5.0	OK	OK
	B5	M20 8.8 - 1	C LC_122	0.5	4.3	181.0	0.4	4.6	4.8	OK	OK
	B7	M20 8.8 - 1	C LC_151	0.0	0.3	123.8	0.0	0.5	0.3	OK	OK
	B8	M20 8.8 - 1	C LC_151	0.1	0.5	123.8	0.1	0.8	0.6	OK	OK
	B9	M20 8.8 - 1	C LC_152	0.4	0.6	123.8	0.3	0.9	0.8	OK	OK
	B10	M20 8.8 - 1	C LC_123	1.1	0.8	123.8	0.8	1.1	1.4	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN50 column below	SLN50 column below	◢ 4.2 ◣	521	C LC_122	256.9	0.0	-127.9	-128.6	-0.3	63.5	35.2	OK	OK
		◢ 4.2 ◣	521	C LC_122	277.1	0.0	-139.1	138.4	1.9	68.5	38.4	OK	OK
SLN204+SLN219 radial (continuous)-w 1	SLN218 ring-w 1	◢ 7.2	189	C LC_122	87.8	0.0	33.2	25.9	-39.2	21.7	13.3	OK	OK
SLN204+SLN219 radial (continuous)-w 1	SLN221 ring-w 1	◢ 7.2	189	C LC_119	87.8	0.0	30.8	-24.3	40.8	21.7	12.8	OK	OK
End plate SLN218 ring	SLN218 ring-w 1	◢ 4.2 ◣	149	C LC_119	79.7	0.0	-21.4	-17.4	40.8	19.7	12.0	OK	OK
		◢ 4.2 ◣	149	C LC_119	94.0	0.0	-17.9	23.8	-47.7	23.2	14.6	OK	OK
End plate SLN221 ring	SLN221 ring-w 1	◢ 4.2 ◣	149	C LC_122	87.6	0.0	-16.4	-22.4	44.3	21.6	13.7	OK	OK
		◢ 4.2 ◣	149	C LC_122	76.0	0.0	-21.3	17.4	-38.4	18.8	11.4	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-A_320
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN208+SLN224 radial (continuous) - SLN54 column below - SLN225 ring - SLN226 ring

Name	SLN208+SLN224 radial (continuous) - SLN54 column below - SLN225 ring - SLN226 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

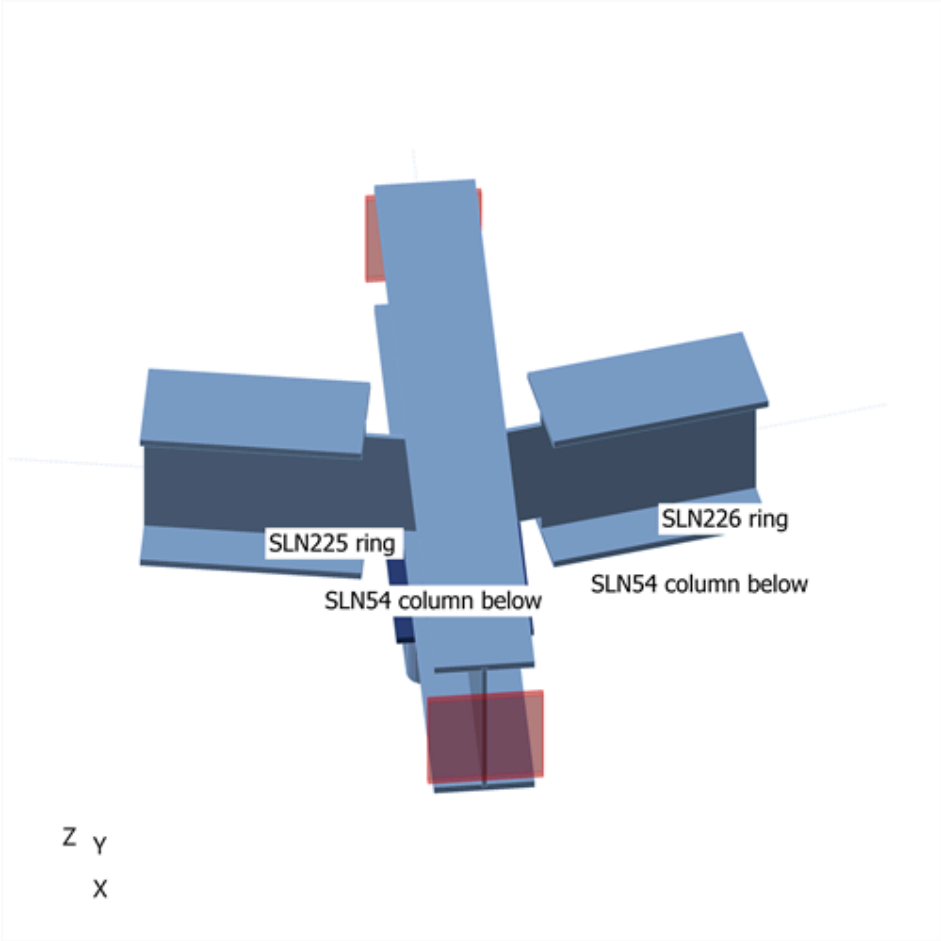
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN208+SLN224 radial (continuous)	1 - UB254x146x43	-40.0	0.0	0.0	0	0	0
SLN54 column below	2 - SHS150x150x10	0.0	90.0	140.0	0	0	0
SLN225 ring	1 - UB254x146x43	-140.0	0.0	0.0	0	0	0
SLN226 ring	1 - UB254x146x43	60.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN208+SLN224 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN208+SLN224 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN54 column below / begin		Position	0
SLN225 ring / end		Position	0
SLN226 ring / end		Position	0



SLN208+SLN224 radial (continuous) - SLN54 column below - SLN225 ring - SLN226 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN54 column below / Begin	26.5	0.0	0.0	0.0	-0.2	0.0
	SLN225 ring / End	6.2	-0.2	-4.9	0.0	0.0	0.0
	SLN226 ring / End	6.2	0.0	-4.9	0.0	0.0	0.0
C LC_102	SLN54 column below / Begin	133.4	0.0	-0.4	0.0	-1.4	0.0
	SLN225 ring / End	18.1	-0.6	-25.4	0.0	0.0	0.0
	SLN226 ring / End	17.8	0.2	-25.6	0.0	0.0	0.0
C LC_103	SLN54 column below / Begin	127.9	0.0	-0.4	0.0	-1.5	0.0
	SLN225 ring / End	9.7	-0.5	-24.4	0.0	0.0	0.0
	SLN226 ring / End	9.2	0.1	-24.6	0.0	0.0	0.0
C LC_104	SLN54 column below / Begin	127.3	1.5	-2.4	0.0	-2.8	0.0
	SLN225 ring / End	16.3	-0.6	-24.4	0.0	0.0	0.0
	SLN226 ring / End	17.5	0.2	-24.6	0.0	0.0	0.0
C LC_105	SLN54 column below / Begin	128.1	0.0	-0.4	0.0	-1.5	0.0
	SLN225 ring / End	9.1	-0.5	-24.4	0.0	0.0	0.0
	SLN226 ring / End	8.9	0.1	-24.6	0.0	0.0	0.0
C LC_106	SLN54 column below / Begin	127.2	-1.8	-2.1	0.0	-2.6	0.0
	SLN225 ring / End	17.0	-0.6	-24.4	0.0	0.0	0.0
	SLN226 ring / End	14.9	0.2	-24.6	0.0	0.0	0.0
C LC_107	SLN54 column below / Begin	9.1	0.0	0.0	0.0	-0.2	0.0
	SLN225 ring / End	-5.0	-0.1	-1.6	0.0	0.0	0.0
	SLN226 ring / End	-5.0	0.1	-1.6	0.0	0.0	0.0
C LC_108	SLN54 column below / Begin	8.1	2.9	-4.2	0.0	-2.8	0.0
	SLN225 ring / End	5.3	0.0	-1.6	0.0	0.0	0.0
	SLN226 ring / End	8.1	0.0	-1.6	0.0	0.0	0.0
C LC_109	SLN54 column below / Begin	9.4	0.0	0.0	0.0	-0.1	0.0
	SLN225 ring / End	-6.0	-0.1	-1.6	0.0	0.0	0.0
	SLN226 ring / End	-6.2	0.0	-1.6	0.0	0.0	0.0
C LC_110	SLN54 column below / Begin	7.9	-3.5	-3.5	0.0	-2.4	0.0
	SLN225 ring / End	6.8	-0.1	-1.6	0.0	0.0	0.0
	SLN226 ring / End	3.4	0.1	-1.6	0.0	0.0	0.0
C LC_111	SLN54 column below / Begin	90.7	0.0	-0.3	0.0	-1.2	0.0
	SLN225 ring / End	2.4	-0.4	-17.2	0.0	0.0	0.0
	SLN226 ring / End	2.1	0.2	-17.3	0.0	0.0	0.0
C LC_112	SLN54 column below / Begin	89.6	2.9	-4.4	0.0	-3.7	0.0
	SLN225 ring / End	13.7	-0.4	-17.2	0.0	0.0	0.0
	SLN226 ring / End	16.4	0.1	-17.3	0.0	0.0	0.0
C LC_113	SLN54 column below / Begin	91.0	0.0	-0.3	0.0	-1.1	0.0
	SLN225 ring / End	1.4	-0.4	-17.2	0.0	0.0	0.0
	SLN226 ring / End	1.1	0.1	-17.3	0.0	0.0	0.0
C LC_114	SLN54 column below / Begin	89.4	-3.5	-3.8	0.0	-3.4	0.0
	SLN225 ring / End	15.2	-0.4	-17.2	0.0	0.0	0.0
	SLN226 ring / End	11.5	0.2	-17.3	0.0	0.0	0.0
C LC_115	SLN54 column below / Begin	16.0	0.0	-0.1	0.0	-0.2	0.0
	SLN225 ring / End	-3.4	-0.1	-2.8	0.0	0.0	0.0
	SLN226 ring / End	-3.5	0.1	-2.8	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN54 column below / Begin	15.0	2.9	-4.2	0.0	-2.9	0.0
	SLN225 ring / End	7.0	-0.1	-2.8	0.0	0.0	0.0
	SLN226 ring / End	9.8	0.0	-2.8	0.0	0.0	0.0
C LC_117	SLN54 column below / Begin	16.3	0.0	0.0	0.0	-0.2	0.0
	SLN225 ring / End	-4.5	-0.1	-2.8	0.0	0.0	0.0
	SLN226 ring / End	-4.6	0.0	-2.8	0.0	0.0	0.0
C LC_118	SLN54 column below / Begin	14.7	-3.5	-3.5	0.0	-2.5	0.0
	SLN225 ring / End	8.4	-0.1	-2.8	0.0	0.0	0.0
	SLN226 ring / End	5.1	0.1	-2.8	0.0	0.0	0.0
C LC_119	SLN54 column below / Begin	133.2	0.0	-0.4	0.0	-1.6	0.0
	SLN225 ring / End	11.6	-0.5	-25.4	0.0	0.0	0.0
	SLN226 ring / End	11.2	0.1	-25.6	0.0	0.0	0.0
C LC_120	SLN54 column below / Begin	132.8	1.5	-2.5	0.0	-2.9	0.0
	SLN225 ring / End	18.3	-0.6	-25.4	0.0	0.0	0.0
	SLN226 ring / End	19.6	0.2	-25.6	0.0	0.0	0.0
C LC_121	SLN54 column below / Begin	133.3	0.0	-0.4	0.0	-1.5	0.0
	SLN225 ring / End	11.1	-0.5	-25.4	0.0	0.0	0.0
	SLN226 ring / End	10.9	0.1	-25.6	0.0	0.0	0.0
C LC_122	SLN54 column below / Begin	132.7	-1.8	-2.1	0.0	-2.7	0.0
	SLN225 ring / End	19.0	-0.6	-25.4	0.0	0.0	0.0
	SLN226 ring / End	16.9	0.2	-25.6	0.0	0.0	0.0
C LC_123	SLN54 column below / Begin	101.1	0.0	-0.3	0.0	-1.3	0.0
	SLN225 ring / End	6.3	-0.4	-19.3	0.0	0.0	0.0
	SLN226 ring / End	6.0	0.2	-19.4	0.0	0.0	0.0
C LC_124	SLN54 column below / Begin	100.5	2.9	-4.5	0.0	-3.9	0.0
	SLN225 ring / End	17.7	-0.5	-19.3	0.0	0.0	0.0
	SLN226 ring / End	20.4	0.1	-19.4	0.0	0.0	0.0
C LC_125	SLN54 column below / Begin	101.5	0.0	-0.3	0.0	-1.2	0.0
	SLN225 ring / End	5.3	-0.4	-19.3	0.0	0.0	0.0
	SLN226 ring / End	5.0	0.1	-19.4	0.0	0.0	0.0
C LC_126	SLN54 column below / Begin	100.2	-3.5	-3.8	0.0	-3.5	0.0
	SLN225 ring / End	19.2	-0.5	-19.3	0.0	0.0	0.0
	SLN226 ring / End	15.5	0.2	-19.4	0.0	0.0	0.0
C LC_127	SLN54 column below / Begin	26.5	0.0	-0.1	0.0	-0.3	0.0
	SLN225 ring / End	0.5	-0.1	-4.9	0.0	0.0	0.0
	SLN226 ring / End	0.4	0.1	-4.9	0.0	0.0	0.0
C LC_128	SLN54 column below / Begin	25.9	2.9	-4.2	0.0	-3.0	0.0
	SLN225 ring / End	11.0	-0.1	-4.9	0.0	0.0	0.0
	SLN226 ring / End	13.8	0.0	-4.9	0.0	0.0	0.0
C LC_129	SLN54 column below / Begin	26.9	0.0	-0.1	0.0	-0.3	0.0
	SLN225 ring / End	-0.5	-0.2	-4.9	0.0	0.0	0.0
	SLN226 ring / End	-0.7	0.0	-4.9	0.0	0.0	0.0
C LC_130	SLN54 column below / Begin	25.6	-3.5	-3.6	0.0	-2.6	0.0
	SLN225 ring / End	12.4	-0.2	-4.9	0.0	0.0	0.0
	SLN226 ring / End	9.0	0.1	-4.9	0.0	0.0	0.0
C LC_135	SLN54 column below / Begin	133.2	0.0	-0.4	0.0	-1.5	0.0
	SLN225 ring / End	15.3	-0.7	-25.4	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN226 ring / End	15.0	0.2	-25.6	0.0	0.0	0.0
C LC_141	SLN54 column below / Begin	26.5	0.0	0.0	0.0	-0.2	0.0
	SLN225 ring / End	6.1	-0.2	-4.9	0.0	0.0	0.0
	SLN226 ring / End	6.0	0.0	-4.9	0.0	0.0	0.0
C LC_142	SLN54 column below / Begin	26.5	0.0	0.0	0.0	-0.2	0.0
	SLN225 ring / End	6.4	-0.2	-4.9	0.0	0.0	0.0
	SLN226 ring / End	6.4	0.1	-4.9	0.0	0.0	0.0
C LC_143	SLN54 column below / Begin	26.5	0.0	0.0	0.0	-0.2	0.0
	SLN225 ring / End	6.1	-0.2	-4.9	0.0	0.0	0.0
	SLN226 ring / End	6.0	0.0	-4.9	0.0	0.0	0.0
C LC_144	SLN54 column below / Begin	26.5	0.0	0.0	0.0	-0.2	0.0
	SLN225 ring / End	6.4	-0.2	-4.9	0.0	0.0	0.0
	SLN226 ring / End	6.3	0.1	-4.9	0.0	0.0	0.0
C LC_145	SLN54 column below / Begin	133.4	0.0	-0.4	0.0	-1.5	0.0
	SLN225 ring / End	17.3	-0.6	-25.4	0.0	0.0	0.0
	SLN226 ring / End	16.9	0.1	-25.6	0.0	0.0	0.0
C LC_146	SLN54 column below / Begin	133.3	0.0	-0.4	0.0	-1.4	0.0
	SLN225 ring / End	18.9	-0.6	-25.4	0.0	0.0	0.0
	SLN226 ring / End	18.8	0.2	-25.6	0.0	0.0	0.0
C LC_147	SLN54 column below / Begin	133.4	0.0	-0.4	0.0	-1.5	0.0
	SLN225 ring / End	17.3	-0.6	-25.4	0.0	0.0	0.0
	SLN226 ring / End	17.2	0.1	-25.6	0.0	0.0	0.0
C LC_148	SLN54 column below / Begin	133.3	0.0	-0.4	0.0	-1.4	0.0
	SLN225 ring / End	18.9	-0.6	-25.4	0.0	0.0	0.0
	SLN226 ring / End	18.5	0.2	-25.6	0.0	0.0	0.0
C LC_151	SLN54 column below / Begin	82.1	0.0	-1.3	0.0	-5.1	0.0
	SLN225 ring / End	6.1	-0.4	-11.7	0.0	0.0	0.0
	SLN226 ring / End	6.0	0.1	-11.9	0.0	0.0	0.0
C LC_152	SLN54 column below / Begin	78.6	0.0	0.9	0.0	3.7	0.0
	SLN225 ring / End	19.7	-0.5	-13.9	0.0	0.0	0.0
	SLN226 ring / End	19.4	0.1	-14.1	0.0	0.0	0.0
C LC_153	SLN54 column below / Begin	133.3	0.0	-0.4	0.0	-1.6	0.0
	SLN225 ring / End	18.2	-0.6	-20.6	0.0	0.0	0.0
	SLN226 ring / End	17.8	0.2	-21.0	0.0	0.0	0.0
C LC_154	SLN54 column below / Begin	81.3	0.0	-1.2	0.0	-5.0	0.0
	SLN225 ring / End	6.3	-0.4	-11.7	0.0	0.0	0.0
	SLN226 ring / End	6.1	0.1	-11.9	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	6.1 < 100%	OK
Welds	60.5 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN208+SLN224 radial (continuous)-bfl 1	12.7	C LC_120	122.1	0.0	35.3	OK
SLN208+SLN224 radial (continuous)-tfl 1	12.7	C LC_153	65.3	0.0	0.0	OK
SLN208+SLN224 radial (continuous)-w 1	7.2	C LC_126	167.0	0.0	29.7	OK
SLN54 column below	10.0	C LC_120	129.7	0.0	0.0	OK
SLN225 ring-bfl 1	12.7	C LC_122	28.0	0.0	0.0	OK
SLN225 ring-tfl 1	12.7	C LC_135	24.6	0.0	0.0	OK
SLN225 ring-w 1	7.2	C LC_135	164.2	0.0	0.0	OK
SLN226 ring-bfl 1	12.7	C LC_120	26.5	0.0	0.0	OK
SLN226 ring-tfl 1	12.7	C LC_121	21.1	0.0	0.0	OK
SLN226 ring-w 1	7.2	C LC_146	120.8	0.0	0.0	OK
End plate column below SLN54 column below	12.0	C LC_120	94.6	0.0	41.7	OK
End plate SLN225 ring	12.0	C LC_126	24.0	0.0	0.8	OK
End plate SLN226 ring	12.0	C LC_120	22.9	0.0	0.4	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

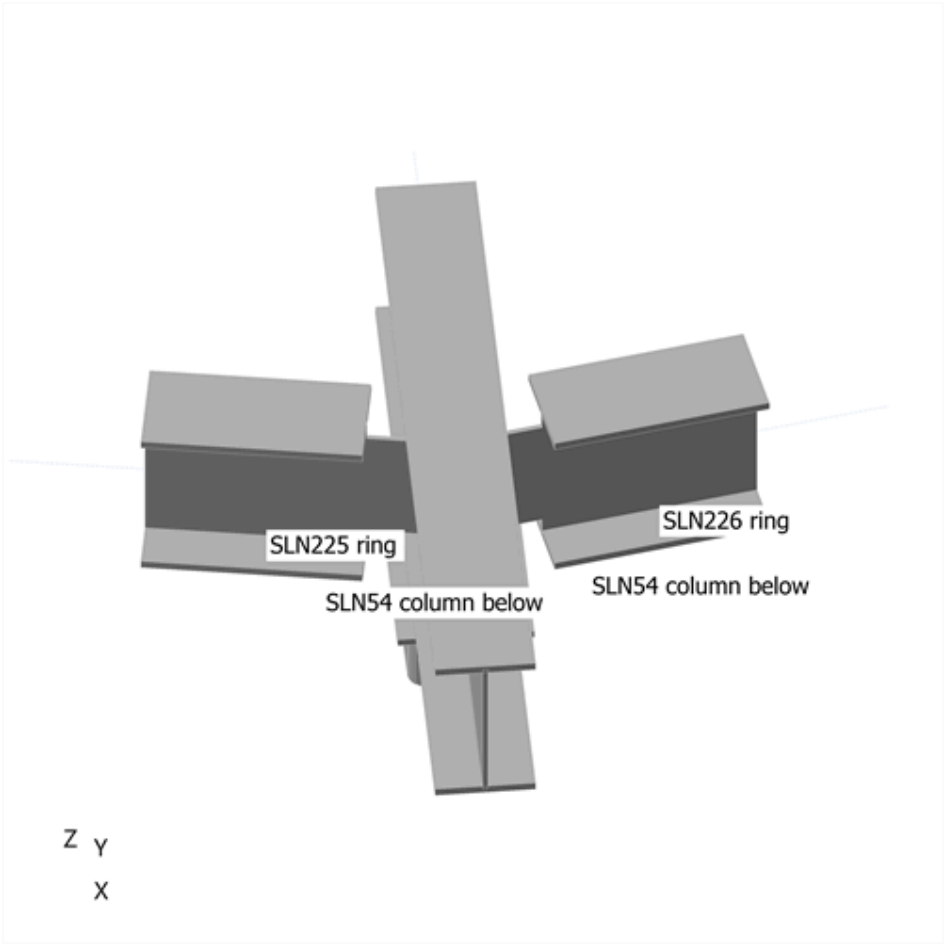
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

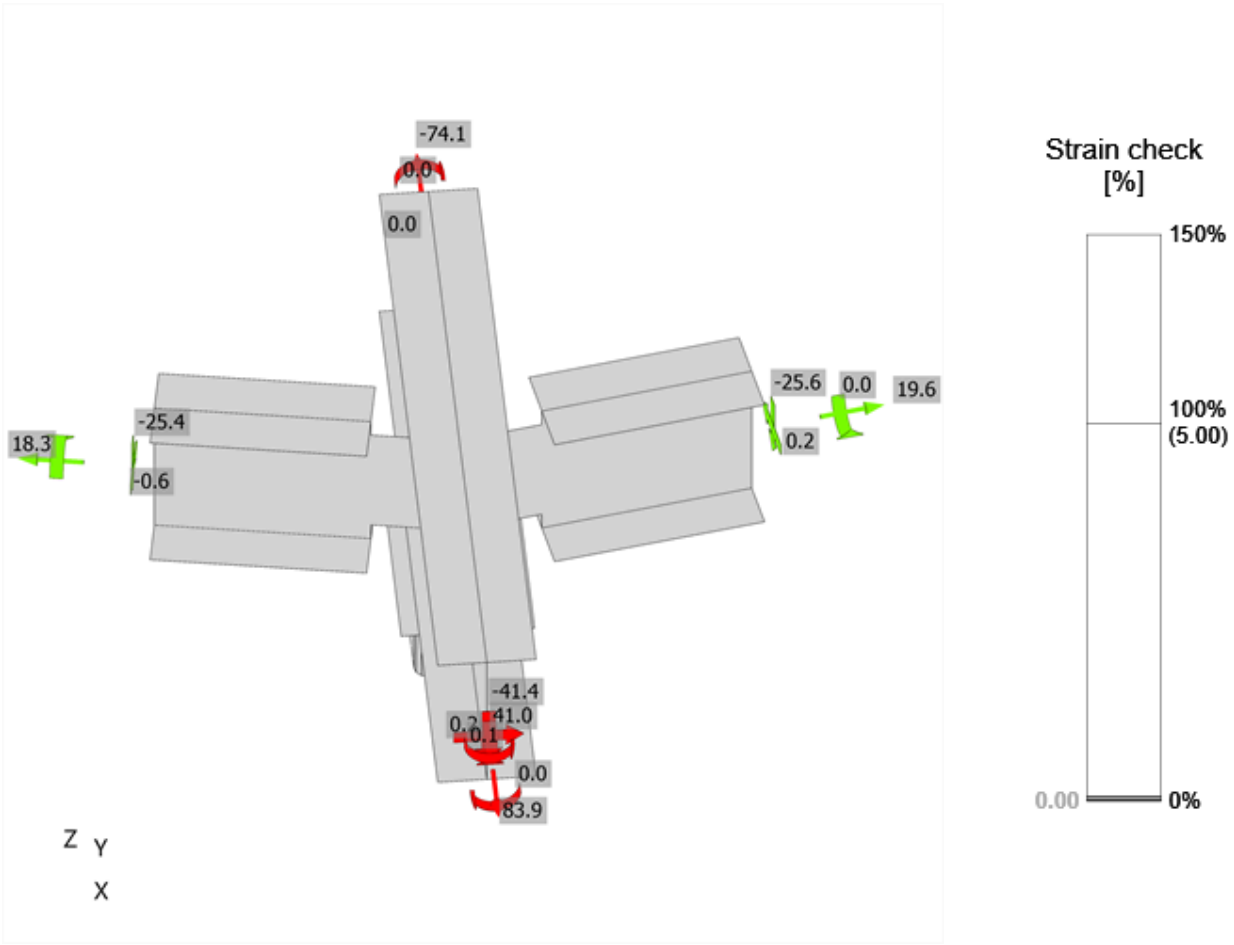
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN54 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

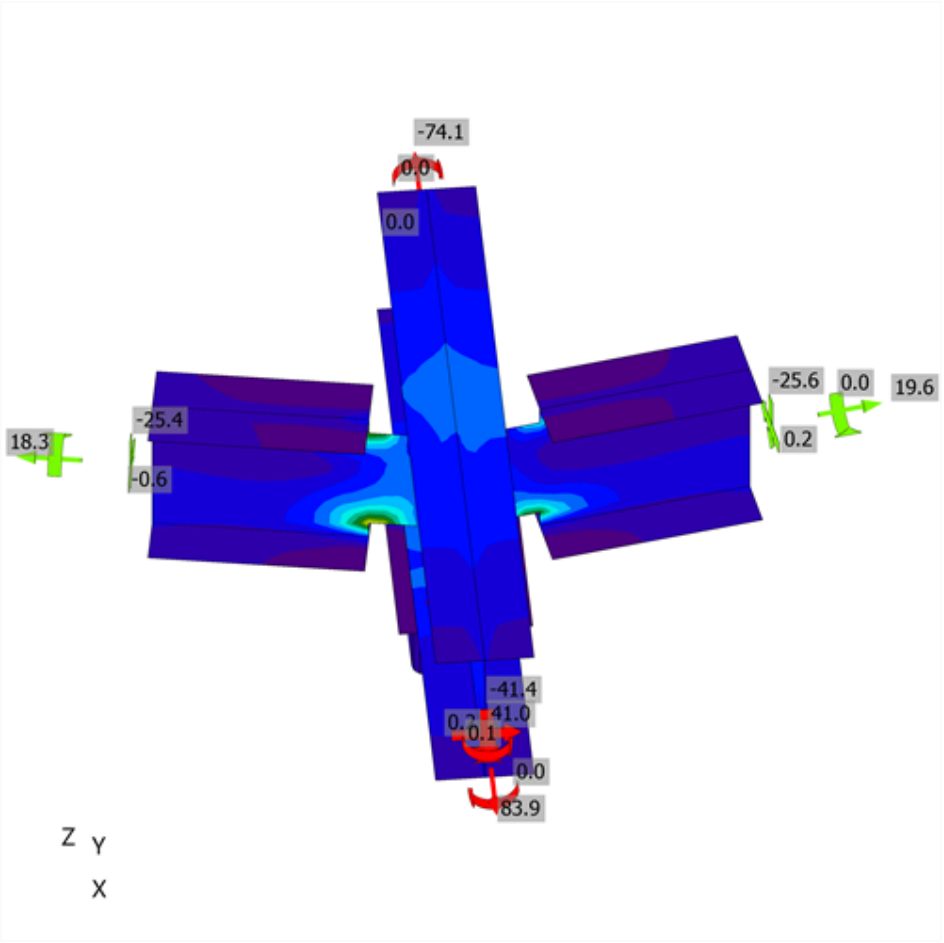
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

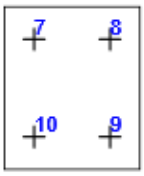


Strain check, C LC_120



Equivalent stress, C LC_120

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_122	0.3	3.8	183.7	0.2	4.1	4.2	OK	OK
	B3	M20 8.8 - 1	C LC_108	8.5	1.6	78.6	6.1	2.0	6.0	OK	OK
	B4	M20 8.8 - 1	C LC_110	7.7	1.5	76.2	5.4	2.0	5.5	OK	OK
	B5	M20 8.8 - 1	C LC_120	0.4	3.9	181.2	0.3	4.2	4.4	OK	OK
	B7	M20 8.8 - 1	C LC_152	0.2	0.3	123.8	0.1	0.4	0.4	OK	OK
	B8	M20 8.8 - 1	C LC_151	0.0	0.4	123.8	0.0	0.6	0.4	OK	OK
	B9	M20 8.8 - 1	C LC_152	0.3	0.6	123.8	0.2	1.0	0.8	OK	OK
	B10	M20 8.8 - 1	C LC_126	1.0	0.7	123.8	0.7	1.0	1.2	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN54 column below	SLN54 column below	◢ 4.2 ◣	521	C LC_120	232.7	0.0	-114.1	-110.6	-38.2	57.5	31.0	OK	OK
		◢ 4.2 ◣	521	C LC_120	244.7	0.0	-115.2	118.7	37.9	60.5	33.0	OK	OK
SLN208+SLN224 radial (continuous)-w 1	SLN225 ring-w 1	◢ 7.2	189	C LC_122	79.7	0.0	7.0	3.5	-45.7	19.7	12.2	OK	OK
SLN208+SLN224 radial (continuous)-w 1	SLN226 ring-w 1	◢ 7.2	189	C LC_120	81.2	0.0	6.7	-3.8	46.6	20.1	12.4	OK	OK
End plate SLN225 ring	SLN225 ring-w 1	◢ 4.2 ◣	149	C LC_122	70.1	0.0	-18.1	-15.2	36.1	17.3	10.3	OK	OK
		◢ 4.2 ◣	149	C LC_122	88.1	0.0	-18.1	22.2	-44.6	21.8	14.2	OK	OK
End plate SLN226 ring	SLN226 ring-w 1	◢ 4.2 ◣	149	C LC_120	88.9	0.0	-18.1	-23.8	44.3	22.0	13.8	OK	OK
		◢ 4.2 ◣	149	C LC_120	72.5	0.0	-18.7	14.5	-37.8	17.9	11.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-B_040
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN262+SLN263 radial (continuous) - SLN40 column below - SLN121 column above - SLN264 ring - SLN265 ring

Name	SLN262+SLN263 radial (continuous) - SLN40 column below - SLN121 column above - SLN264 ring - SLN265 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

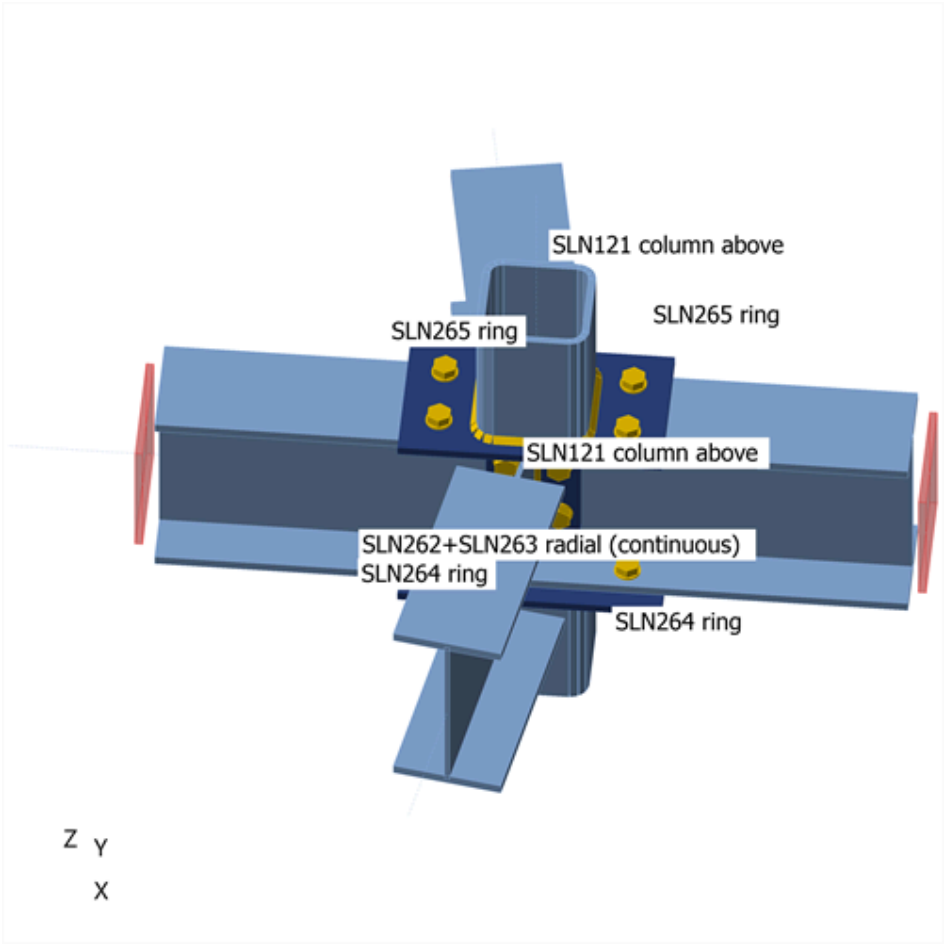
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN262+SLN263 radial (continuous)	1 - UB254x146x43	40.0	0.0	0.0	0	0	0
SLN40 column below	2 - SHS150x150x10	0.0	90.0	-140.0	0	0	0
SLN121 column above	2 - SHS150x150x10	0.0	90.0	-140.0	0	0	0
SLN264 ring	1 - UB254x146x43	-60.0	0.0	0.0	0	0	0
SLN265 ring	1 - UB254x146x43	140.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN262+SLN263 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN262+SLN263 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN40 column below / begin		Position	0
SLN121 column above / end		Position	0
SLN264 ring / end		Position	0
SLN265 ring / end		Position	0



SLN262+SLN263 radial (continuous) - SLN40 column below - SLN121 column above - SLN264 ring - SLN265 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN40 column below / Begin	53.6	0.0	0.1	0.0	0.2	0.0
	SLN121 column above / End	-17.7	0.0	0.3	0.0	0.0	0.0
	SLN264 ring / End	8.8	0.1	-4.9	0.0	0.0	0.0
	SLN265 ring / End	8.9	0.1	-5.0	0.0	0.0	0.0
C LC_102	SLN40 column below / Begin	240.3	0.0	0.0	0.0	0.2	-0.1
	SLN121 column above / End	-65.9	0.0	1.4	-0.1	-1.0	0.1
	SLN264 ring / End	27.8	0.2	-25.6	0.0	0.0	0.0
	SLN265 ring / End	28.0	0.3	-26.0	0.0	0.0	0.0
C LC_103	SLN40 column below / Begin	229.5	0.1	0.0	0.0	0.0	-0.2
	SLN121 column above / End	-63.4	0.1	1.4	-0.1	-1.2	0.2
	SLN264 ring / End	18.8	0.3	-24.6	0.0	0.0	0.0
	SLN265 ring / End	19.2	0.3	-25.0	0.0	0.0	0.0
C LC_104	SLN40 column below / Begin	233.9	-1.6	-2.0	0.0	-1.0	0.6
	SLN121 column above / End	-63.5	-0.2	1.1	-0.1	-0.5	-0.6
	SLN264 ring / End	27.5	0.3	-24.6	0.0	0.0	0.0
	SLN265 ring / End	25.7	0.3	-25.0	0.0	0.0	0.0
C LC_105	SLN40 column below / Begin	229.5	1.9	-1.8	0.0	-1.2	-0.7
	SLN121 column above / End	-63.6	0.2	1.3	-0.1	-1.0	0.6
	SLN264 ring / End	25.0	0.2	-24.6	0.0	0.0	0.0
	SLN265 ring / End	27.3	0.2	-25.0	0.0	0.0	0.0
C LC_106	SLN40 column below / Begin	233.8	0.1	0.1	0.0	0.4	-0.3
	SLN121 column above / End	-63.3	0.1	1.2	-0.1	-0.7	0.3
	SLN264 ring / End	17.9	0.2	-24.6	0.0	0.0	0.0
	SLN265 ring / End	18.1	0.2	-25.0	0.0	0.0	0.0
C LC_107	SLN40 column below / Begin	18.5	0.1	0.0	0.0	-0.2	-0.2
	SLN121 column above / End	-8.1	0.1	0.1	-0.1	-0.1	0.2
	SLN264 ring / End	-3.5	0.1	-1.6	0.0	0.0	0.0
	SLN265 ring / End	-3.4	0.1	-1.6	0.0	0.0	0.0
C LC_108	SLN40 column below / Begin	27.1	-3.3	-4.0	0.0	-2.2	1.3
	SLN121 column above / End	-8.4	-0.3	-0.3	-0.1	1.0	-1.3
	SLN264 ring / End	10.2	0.2	-1.6	0.0	0.0	0.0
	SLN265 ring / End	6.6	0.0	-1.6	0.0	0.0	0.0
C LC_109	SLN40 column below / Begin	18.5	3.8	-3.5	0.0	-2.5	-1.2
	SLN121 column above / End	-8.5	0.3	0.1	0.1	0.1	1.2
	SLN264 ring / End	5.7	0.0	-1.6	0.0	0.0	0.0
	SLN265 ring / End	9.6	-0.1	-1.6	0.0	0.0	0.0
C LC_110	SLN40 column below / Begin	27.0	0.1	0.1	0.0	0.6	-0.4
	SLN121 column above / End	-8.0	0.1	-0.2	0.0	0.9	0.4
	SLN264 ring / End	-5.8	0.0	-1.6	0.0	0.0	0.0
	SLN265 ring / End	-5.5	0.1	-1.6	0.0	0.0	0.0
C LC_111	SLN40 column below / Begin	162.9	0.1	0.0	0.0	-0.2	-0.3
	SLN121 column above / End	-46.4	0.1	1.0	-0.1	-0.9	0.3
	SLN264 ring / End	9.3	0.2	-17.3	0.0	0.0	0.0
	SLN265 ring / End	9.7	0.2	-17.6	0.0	0.0	0.0
C LC_112	SLN40 column below / Begin	171.6	-3.3	-4.0	0.0	-2.2	1.3

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN121 column above / End	-46.7	-0.3	0.6	-0.1	0.3	-1.3
	SLN264 ring / End	24.2	0.3	-17.3	0.0	0.0	0.0
	SLN265 ring / End	20.6	0.2	-17.6	0.0	0.0	0.0
C LC_113	SLN40 column below / Begin	162.9	3.8	-3.5	0.0	-2.5	-1.2
	SLN121 column above / End	-46.8	0.3	0.9	0.0	-0.6	1.2
	SLN264 ring / End	19.6	0.1	-17.3	0.0	0.0	0.0
	SLN265 ring / End	23.7	0.1	-17.6	0.0	0.0	0.0
C LC_114	SLN40 column below / Begin	171.5	0.1	0.1	0.0	0.6	-0.5
	SLN121 column above / End	-46.3	0.1	0.7	-0.1	0.1	0.4
	SLN264 ring / End	7.2	0.1	-17.3	0.0	0.0	0.0
	SLN265 ring / End	7.6	0.2	-17.6	0.0	0.0	0.0
C LC_115	SLN40 column below / Begin	32.4	0.1	0.0	0.0	-0.1	-0.2
	SLN121 column above / End	-12.7	0.1	0.2	-0.1	-0.1	0.2
	SLN264 ring / End	-1.3	0.1	-2.8	0.0	0.0	0.0
	SLN265 ring / End	-1.1	0.1	-2.9	0.0	0.0	0.0
C LC_116	SLN40 column below / Begin	41.0	-3.3	-4.0	0.0	-2.2	1.3
	SLN121 column above / End	-13.0	-0.3	-0.2	-0.1	1.0	-1.3
	SLN264 ring / End	12.5	0.2	-2.8	0.0	0.0	0.0
	SLN265 ring / End	8.9	0.1	-2.9	0.0	0.0	0.0
C LC_117	SLN40 column below / Begin	32.3	3.8	-3.5	0.0	-2.4	-1.2
	SLN121 column above / End	-13.1	0.3	0.1	0.1	0.1	1.2
	SLN264 ring / End	8.0	0.0	-2.8	0.0	0.0	0.0
	SLN265 ring / End	12.0	-0.1	-2.9	0.0	0.0	0.0
C LC_118	SLN40 column below / Begin	40.9	0.1	0.2	0.0	0.6	-0.4
	SLN121 column above / End	-12.6	0.1	-0.2	0.0	0.9	0.4
	SLN264 ring / End	-3.6	0.0	-2.8	0.0	0.0	0.0
	SLN265 ring / End	-3.2	0.1	-2.9	0.0	0.0	0.0
C LC_119	SLN40 column below / Begin	237.5	0.1	0.0	0.0	-0.1	-0.2
	SLN121 column above / End	-66.0	0.1	1.6	-0.1	-1.6	0.2
	SLN264 ring / End	21.0	0.3	-25.6	0.0	0.0	0.0
	SLN265 ring / End	21.4	0.3	-26.0	0.0	0.0	0.0
C LC_120	SLN40 column below / Begin	241.7	-1.6	-2.0	0.0	-1.2	0.6
	SLN121 column above / End	-65.8	-0.2	1.2	-0.1	-0.5	-0.6
	SLN264 ring / End	29.9	0.2	-25.6	0.0	0.0	0.0
	SLN265 ring / End	28.1	0.3	-26.0	0.0	0.0	0.0
C LC_121	SLN40 column below / Begin	237.3	1.9	-1.8	0.0	-1.3	-0.7
	SLN121 column above / End	-65.8	0.2	1.3	-0.1	-1.0	0.7
	SLN264 ring / End	27.4	0.2	-25.6	0.0	0.0	0.0
	SLN265 ring / End	29.7	0.3	-26.0	0.0	0.0	0.0
C LC_122	SLN40 column below / Begin	241.9	0.1	0.1	0.0	0.3	-0.3
	SLN121 column above / End	-65.9	0.1	1.4	-0.1	-1.1	0.3
	SLN264 ring / End	20.0	0.2	-25.6	0.0	0.0	0.0
	SLN265 ring / End	20.2	0.2	-26.0	0.0	0.0	0.0
C LC_123	SLN40 column below / Begin	179.0	0.1	-0.1	0.0	-0.3	-0.3
	SLN121 column above / End	-51.7	0.1	1.5	-0.1	-1.7	0.3
	SLN264 ring / End	13.6	0.3	-19.4	0.0	0.0	0.0
	SLN265 ring / End	14.0	0.3	-19.7	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_124	SLN40 column below / Begin	187.3	-3.3	-4.1	0.0	-2.5	1.3
	SLN121 column above / End	-51.2	-0.3	0.6	-0.1	0.3	-1.4
	SLN264 ring / End	29.0	0.3	-19.4	0.0	0.0	0.0
	SLN265 ring / End	25.3	0.3	-19.7	0.0	0.0	0.0
C LC_125	SLN40 column below / Begin	178.7	3.8	-3.6	0.0	-2.8	-1.2
	SLN121 column above / End	-51.4	0.3	0.9	0.0	-0.6	1.2
	SLN264 ring / End	24.3	0.1	-19.4	0.0	0.0	0.0
	SLN265 ring / End	28.5	0.1	-19.7	0.0	0.0	0.0
C LC_126	SLN40 column below / Begin	187.6	0.1	0.1	0.0	0.4	-0.4
	SLN121 column above / End	-51.5	0.1	1.0	-0.1	-0.6	0.4
	SLN264 ring / End	11.4	0.1	-19.4	0.0	0.0	0.0
	SLN265 ring / End	11.8	0.2	-19.7	0.0	0.0	0.0
C LC_127	SLN40 column below / Begin	48.5	0.1	-0.1	0.0	-0.3	-0.3
	SLN121 column above / End	-18.0	0.1	0.7	-0.1	-0.9	0.3
	SLN264 ring / End	2.9	0.1	-4.9	0.0	0.0	0.0
	SLN265 ring / End	3.2	0.1	-5.0	0.0	0.0	0.0
C LC_128	SLN40 column below / Begin	56.7	-3.3	-4.1	0.0	-2.5	1.4
	SLN121 column above / End	-17.5	-0.4	-0.1	-0.1	1.0	-1.4
	SLN264 ring / End	17.3	0.1	-4.9	0.0	0.0	0.0
	SLN265 ring / End	13.6	0.1	-5.0	0.0	0.0	0.0
C LC_129	SLN40 column below / Begin	48.1	3.8	-3.6	0.0	-2.8	-1.2
	SLN121 column above / End	-17.7	0.3	0.2	0.0	0.1	1.2
	SLN264 ring / End	12.8	0.0	-4.9	0.0	0.0	0.0
	SLN265 ring / End	16.8	0.0	-5.0	0.0	0.0	0.0
C LC_130	SLN40 column below / Begin	57.0	0.1	0.1	0.0	0.4	-0.4
	SLN121 column above / End	-17.8	0.1	0.2	0.0	0.2	0.4
	SLN264 ring / End	0.7	0.0	-4.9	0.0	0.0	0.0
	SLN265 ring / End	1.0	0.1	-5.0	0.0	0.0	0.0
C LC_135	SLN40 column below / Begin	244.7	0.0	0.1	0.0	0.5	-0.1
	SLN121 column above / End	-65.9	0.0	1.2	-0.1	-0.5	0.1
	SLN264 ring / End	28.2	0.3	-25.6	0.0	0.0	0.0
	SLN265 ring / End	28.5	0.3	-26.0	0.0	0.0	0.0
C LC_141	SLN40 column below / Begin	53.5	0.0	0.1	0.0	0.2	0.0
	SLN121 column above / End	-17.7	0.0	0.3	0.0	0.0	0.0
	SLN264 ring / End	8.6	0.1	-4.9	0.0	0.0	0.0
	SLN265 ring / End	8.7	0.1	-5.0	0.0	0.0	0.0
C LC_142	SLN40 column below / Begin	53.6	0.0	0.1	0.0	0.2	0.0
	SLN121 column above / End	-17.7	0.0	0.2	0.0	0.0	0.0
	SLN264 ring / End	9.0	0.1	-4.9	0.0	0.0	0.0
	SLN265 ring / End	9.1	0.1	-5.0	0.0	0.0	0.0
C LC_143	SLN40 column below / Begin	53.5	0.0	0.1	0.0	0.2	0.0
	SLN121 column above / End	-17.7	0.0	0.2	0.0	0.0	0.0
	SLN264 ring / End	9.0	0.1	-4.9	0.0	0.0	0.0
	SLN265 ring / End	9.1	0.1	-5.0	0.0	0.0	0.0
C LC_144	SLN40 column below / Begin	53.6	0.0	0.1	0.0	0.2	0.0
	SLN121 column above / End	-17.7	0.0	0.3	0.0	0.0	0.0
	SLN264 ring / End	8.7	0.1	-4.9	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN265 ring / End	8.7	0.1	-5.0	0.0	0.0	0.0
C LC_145	SLN40 column below / Begin	240.2	0.0	0.0	0.0	0.2	-0.1
	SLN121 column above / End	-65.9	0.0	1.4	-0.1	-1.1	0.1
	SLN264 ring / End	26.8	0.3	-25.6	0.0	0.0	0.0
	SLN265 ring / End	27.2	0.2	-26.0	0.0	0.0	0.0
C LC_146	SLN40 column below / Begin	240.4	0.0	0.0	0.0	0.2	-0.1
	SLN121 column above / End	-65.9	0.0	1.3	-0.1	-0.9	0.1
	SLN264 ring / End	28.8	0.2	-25.6	0.0	0.0	0.0
	SLN265 ring / End	28.8	0.3	-26.0	0.0	0.0	0.0
C LC_147	SLN40 column below / Begin	240.2	0.0	0.0	0.0	0.2	-0.1
	SLN121 column above / End	-65.9	0.0	1.3	-0.1	-0.9	0.1
	SLN264 ring / End	28.5	0.2	-25.6	0.0	0.0	0.0
	SLN265 ring / End	28.9	0.3	-26.0	0.0	0.0	0.0
C LC_148	SLN40 column below / Begin	240.4	0.0	0.1	0.0	0.2	-0.1
	SLN121 column above / End	-65.9	0.0	1.4	-0.1	-1.1	0.1
	SLN264 ring / End	27.0	0.3	-25.6	0.0	0.0	0.0
	SLN265 ring / End	27.1	0.2	-26.0	0.0	0.0	0.0
C LC_151	SLN40 column below / Begin	192.7	0.0	-0.7	0.0	-2.7	-0.1
	SLN121 column above / End	-66.5	0.0	3.1	-0.1	-5.4	0.0
	SLN264 ring / End	12.4	0.2	-11.3	0.0	0.0	0.0
	SLN265 ring / End	12.6	0.2	-11.3	0.0	0.0	0.0
C LC_152	SLN40 column below / Begin	128.8	0.0	1.4	0.0	5.6	-0.1
	SLN121 column above / End	-16.9	0.0	-2.8	-0.1	8.0	0.1
	SLN264 ring / End	28.6	0.2	-14.0	0.0	0.0	0.0
	SLN265 ring / End	28.7	0.2	-14.0	0.0	0.0	0.0
C LC_153	SLN40 column below / Begin	238.6	0.0	0.0	0.0	-0.1	-0.1
	SLN121 column above / End	-65.2	0.0	1.4	-0.1	-1.2	0.1
	SLN264 ring / End	27.9	0.2	-20.4	0.0	0.0	0.0
	SLN265 ring / End	28.1	0.3	-20.3	0.0	0.0	0.0
C LC_154	SLN40 column below / Begin	192.9	0.0	-0.6	0.0	-2.6	-0.1
	SLN121 column above / End	-67.2	0.0	3.1	-0.1	-5.5	0.1
	SLN264 ring / End	12.5	0.2	-11.3	0.0	0.0	0.0
	SLN265 ring / End	12.7	0.2	-11.3	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	19.3 < 100%	OK
Welds	63.1 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN262+SLN263 radial (continuous)-bfl 1	12.7	C LC_120	183.8	0.0	52.9	OK
SLN262+SLN263 radial (continuous)-tfl 1	12.7	C LC_152	242.8	0.0	24.5	OK
SLN262+SLN263 radial (continuous)-w 1	7.2	C LC_124	275.1	0.0	44.3	OK
SLN40 column below	10.0	C LC_120	176.6	0.0	0.0	OK
SLN121 column above	10.0	C LC_152	205.8	0.0	0.0	OK
SLN264 ring-bfl 1	12.7	C LC_135	29.0	0.0	0.0	OK
SLN264 ring-tfl 1	12.7	C LC_119	19.8	0.0	0.0	OK
SLN264 ring-w 1	7.2	C LC_135	150.8	0.0	0.0	OK
SLN265 ring-bfl 1	12.7	C LC_146	29.1	0.0	0.0	OK
SLN265 ring-tfl 1	12.7	C LC_122	20.3	0.0	0.0	OK
SLN265 ring-w 1	7.2	C LC_146	144.7	0.0	0.0	OK
End plate column below SLN40 column below	12.0	C LC_120	141.3	0.0	46.8	OK
End plate column above SLN121 column above	12.0	C LC_152	197.0	0.0	24.5	OK
End plate SLN264 ring	12.0	C LC_125	61.8	0.0	24.0	OK
End plate SLN265 ring	12.0	C LC_124	67.6	0.0	26.7	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

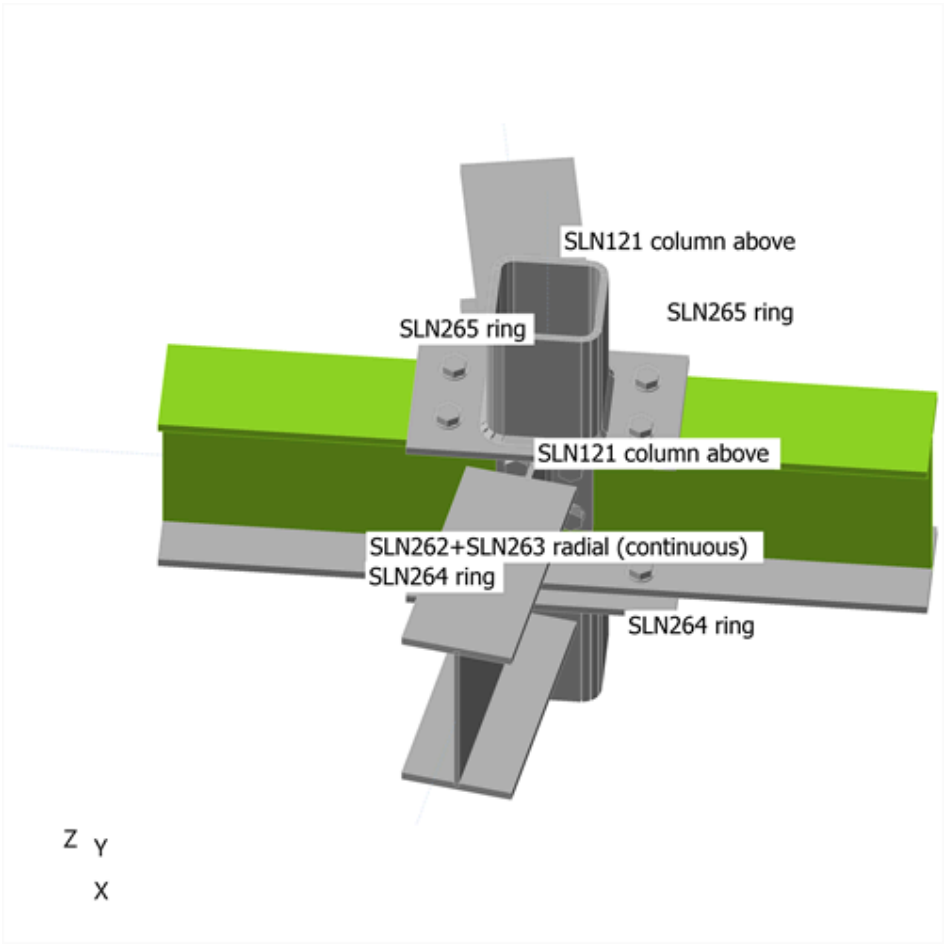
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

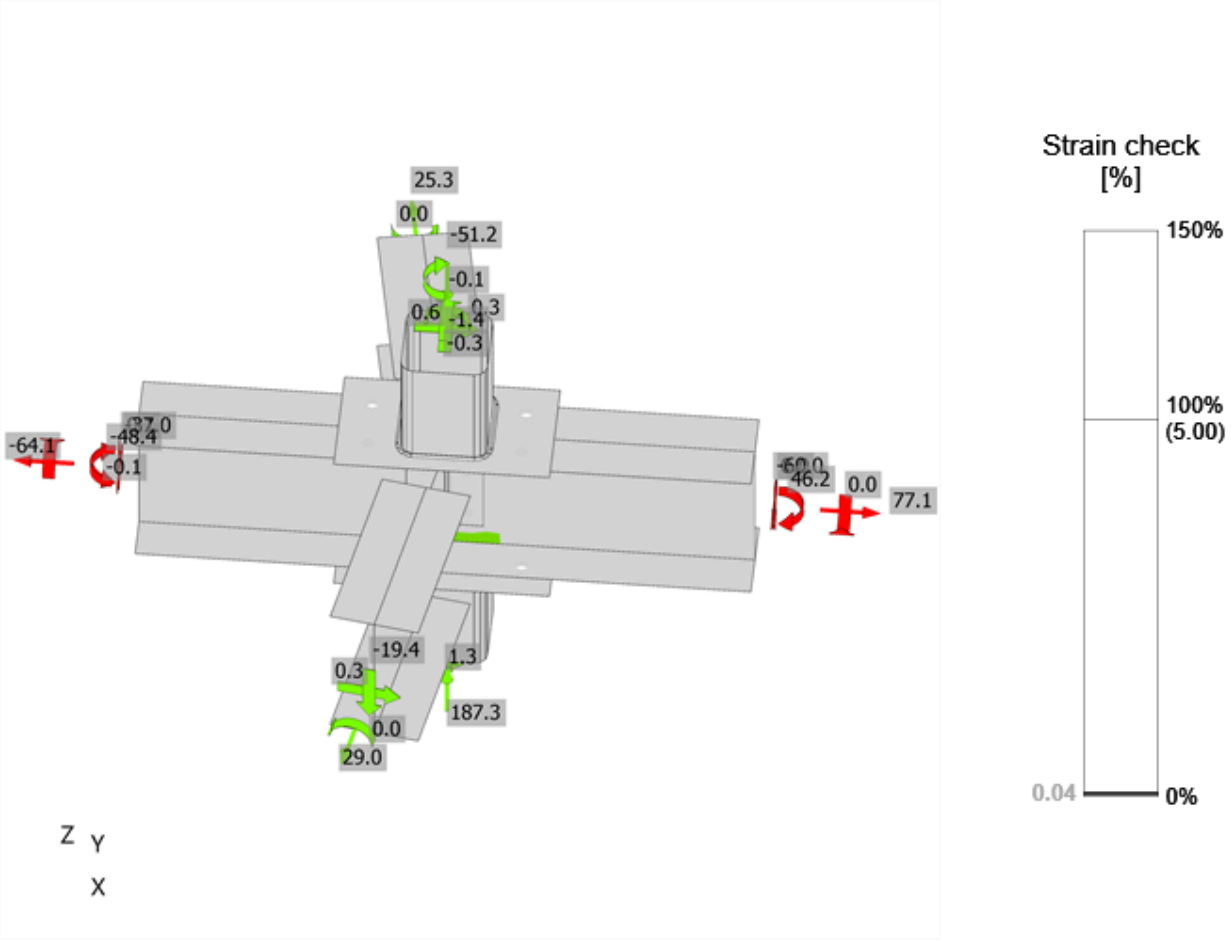
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN40 column below	150	C LC_154	0	5	0.0	OK
SLN121 column above	150	C LC_154	0	5	0.0	OK

Symbol explanation

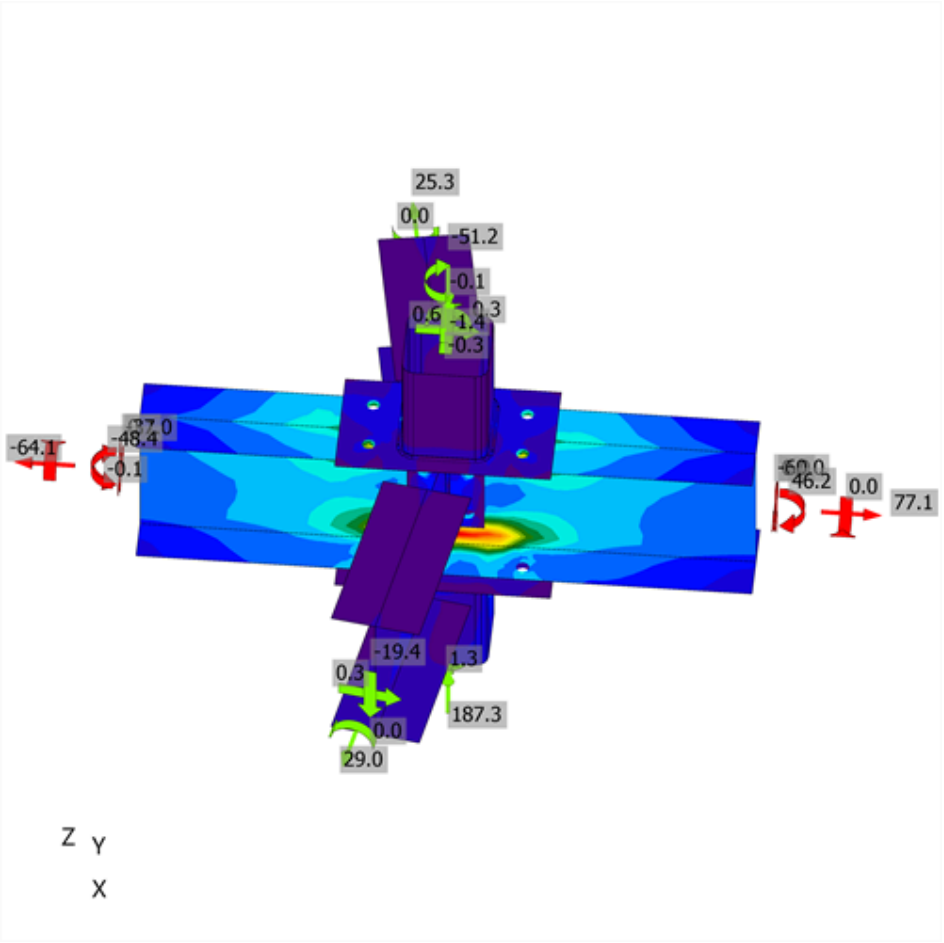
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

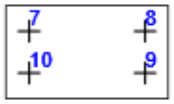
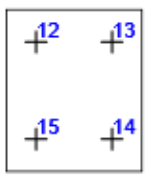


Strain check, C LC_124



Equivalent stress, C LC_124

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_124	2.5	4.6	192.0	1.8	4.9	6.2	OK	OK
	B3	M20 8.8 - 1	C LC_153	0.6	5.2	175.3	0.4	5.5	5.8	OK	OK
	B4	M20 8.8 - 1	C LC_153	0.7	5.1	175.8	0.5	5.4	5.8	OK	OK
	B5	M20 8.8 - 1	C LC_120	0.7	5.6	171.6	0.5	6.0	6.3	OK	OK
	B7	M20 8.8 - 1	C LC_125	4.8	2.8	156.4	3.4	3.0	5.4	OK	OK
	B8	M20 8.8 - 1	C LC_152	27.3	3.5	93.5	19.3	3.7	17.5	OK	OK
	B9	M20 8.8 - 1	C LC_152	26.8	3.6	88.4	19.0	4.1	17.4	OK	OK
	B10	M20 8.8 - 1	C LC_124	5.1	2.9	156.4	3.6	3.1	5.7	OK	OK
	B12	M20 8.8 - 1	C LC_124	3.1	0.7	63.9	2.2	1.1	2.3	OK	OK
	B13	M20 8.8 - 1	C LC_124	3.1	0.8	62.7	2.2	1.3	2.4	OK	OK
	B14	M20 8.8 - 1	C LC_124	2.5	0.8	64.7	1.8	0.8	2.1	OK	OK
	B15	M20 8.8 - 1	C LC_124	2.5	0.9	123.8	1.8	1.1	2.3	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN40 column below	SLN40 column below	◢ 5.6 ◣	520	C LC_120	250.1	0.0	-126.5	-124.5	5.6	61.8	34.3	OK	OK
		◢ 5.6 ◣	521	C LC_120	255.5	0.0	-126.3	128.2	1.3	63.1	35.3	OK	OK
End plate column above SLN121 column above	SLN121 column above	◢ 5.6 ◣	521	C LC_154	129.6	0.0	-64.7	-64.0	-10.7	32.0	17.2	OK	OK
		◢ 5.6 ◣	521	C LC_152	156.3	0.0	110.7	-55.3	31.8	38.6	24.0	OK	OK
SLN262+SLN263 radial (continuous)-w 1	SLN264 ring-w 1	◢ 7.2	189	C LC_125	217.8	0.0	-3.3	3.5	125.7	53.8	36.9	OK	OK
SLN262+SLN263 radial (continuous)-w 1	SLN265 ring-w 1	◢ 7.2	189	C LC_124	235.9	0.0	-5.1	-3.9	-136.1	58.3	39.8	OK	OK
End plate SLN264 ring	SLN264 ring-w 1	◢ 4.2 ◣	149	C LC_121	173.3	0.0	-47.0	-41.0	87.2	42.8	26.5	OK	OK
		◢ 4.2 ◣	149	C LC_125	195.6	0.0	-36.2	47.0	-100.5	48.3	29.6	OK	OK
End plate SLN265 ring	SLN265 ring-w 1	◢ 4.2 ◣	149	C LC_124	215.0	0.0	-44.5	-53.8	108.9	53.1	31.9	OK	OK
		◢ 4.2 ◣	149	C LC_124	175.5	0.0	-45.1	38.7	-89.9	43.4	26.7	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-B_280
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN206+SLN222 radial (continuous) - SLN52 column below - SLN85 column above - SLN425 ring - SLN426 ring

Name	SLN206+SLN222 radial (continuous) - SLN52 column below - SLN85 column above - SLN425 ring - SLN426 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

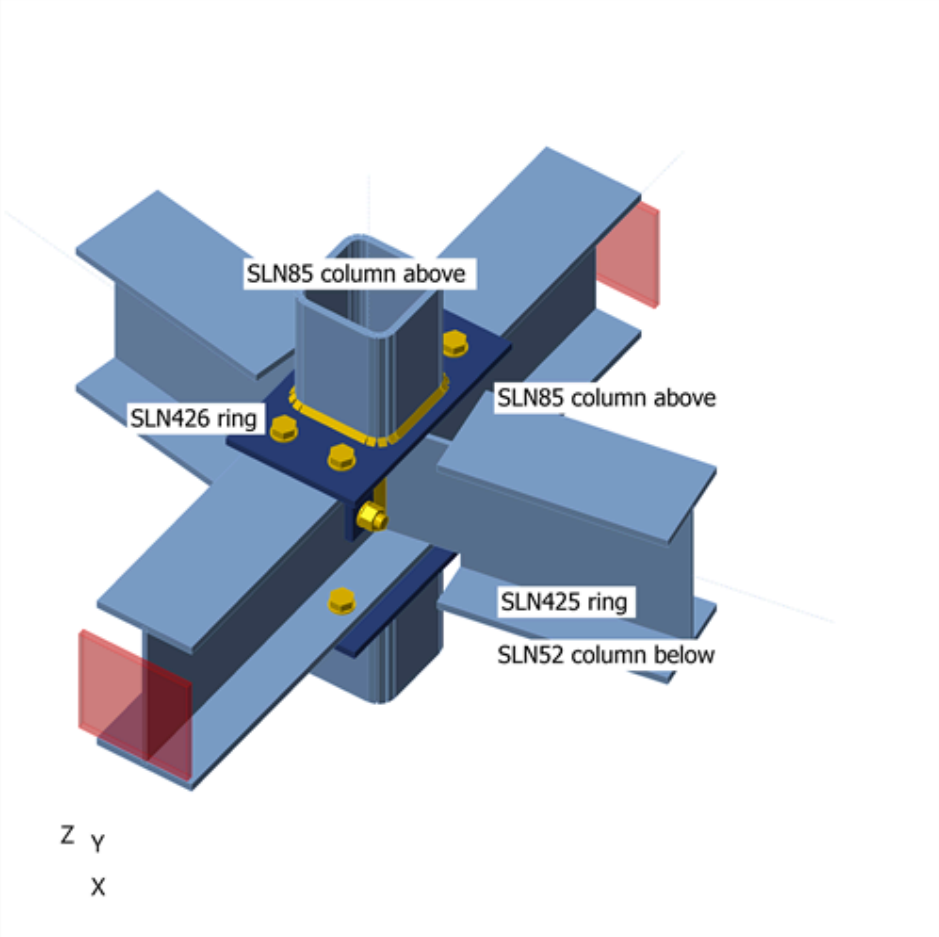
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN206+SLN222 radial (continuous)	1 - UB254x146x43	-80.0	0.0	0.0	0	0	0
SLN52 column below	2 - SHS150x150x10	0.0	90.0	100.0	0	0	0
SLN85 column above	2 - SHS150x150x10	0.0	90.0	100.0	0	0	0
SLN425 ring	1 - UB254x146x43	20.0	0.0	0.0	0	0	0
SLN426 ring	1 - UB254x146x43	-180.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN206+SLN222 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN206+SLN222 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN52 column below / begin		Position	0
SLN85 column above / end		Position	0
SLN425 ring / end		Position	0
SLN426 ring / end		Position	0



SLN206+SLN222 radial (continuous) - SLN52 column below - SLN85 column above - SLN425 ring - SLN426 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN52 column below / Begin	54.1	0.0	0.1	0.0	0.2	0.0
	SLN85 column above / End	-18.4	0.0	0.3	0.0	0.0	0.0
	SLN425 ring / End	8.8	-0.1	-4.9	0.0	0.0	0.0
	SLN426 ring / End	8.8	-0.1	-4.9	0.0	0.0	0.0
C LC_102	SLN52 column below / Begin	242.6	0.0	0.1	0.0	0.3	0.1
	SLN85 column above / End	-68.7	0.0	1.6	0.1	-1.2	0.0
	SLN425 ring / End	27.5	-0.2	-25.5	0.0	0.0	0.0
	SLN426 ring / End	27.7	-0.3	-25.5	0.0	0.0	0.0
C LC_103	SLN52 column below / Begin	231.3	0.0	0.0	0.0	0.0	-0.1
	SLN85 column above / End	-66.1	0.0	1.6	0.0	-1.4	0.2
	SLN425 ring / End	19.4	-0.2	-24.5	0.0	0.0	0.0
	SLN426 ring / End	19.7	-0.2	-24.5	0.0	0.0	0.0
C LC_104	SLN52 column below / Begin	236.7	2.5	-0.3	0.0	0.2	-1.0
	SLN85 column above / End	-66.0	0.3	1.4	0.1	-0.7	1.0
	SLN425 ring / End	22.3	-0.2	-24.5	0.0	0.0	0.0
	SLN426 ring / End	19.6	-0.2	-24.5	0.0	0.0	0.0
C LC_105	SLN52 column below / Begin	234.2	-0.1	0.1	0.0	0.3	0.4
	SLN85 column above / End	-65.9	-0.1	1.5	0.1	-1.1	-0.3
	SLN425 ring / End	17.1	-0.2	-24.5	0.0	0.0	0.0
	SLN426 ring / End	17.5	-0.2	-24.5	0.0	0.0	0.0
C LC_106	SLN52 column below / Begin	233.6	-0.4	-2.6	0.0	-1.5	0.1
	SLN85 column above / End	-66.2	0.0	1.5	0.1	-0.9	0.0
	SLN425 ring / End	28.5	-0.2	-24.5	0.0	0.0	0.0
	SLN426 ring / End	29.1	-0.3	-24.5	0.0	0.0	0.0
C LC_107	SLN52 column below / Begin	18.0	0.1	-0.1	0.0	-0.2	-0.4
	SLN85 column above / End	-8.6	0.1	0.2	-0.1	-0.3	0.4
	SLN425 ring / End	-3.2	0.0	-1.6	0.0	0.0	0.0
	SLN426 ring / End	-3.4	0.1	-1.6	0.0	0.0	0.0
C LC_108	SLN52 column below / Begin	28.5	5.0	-0.8	0.0	0.1	-2.1
	SLN85 column above / End	-8.5	0.5	-0.3	0.0	1.1	2.1
	SLN425 ring / End	1.6	0.0	-1.6	0.0	0.0	0.0
	SLN426 ring / End	-3.8	0.0	-1.6	0.0	0.0	0.0
C LC_109	SLN52 column below / Begin	23.7	-0.1	0.1	0.0	0.2	0.5
	SLN85 column above / End	-8.3	-0.1	0.0	0.1	0.5	-0.5
	SLN425 ring / End	-6.0	0.0	-1.6	0.0	0.0	0.0
	SLN426 ring / End	-5.6	-0.1	-1.6	0.0	0.0	0.0
C LC_110	SLN52 column below / Begin	22.6	-0.8	-5.3	0.0	-3.4	0.1
	SLN85 column above / End	-8.8	0.0	0.0	0.0	0.5	-0.1
	SLN425 ring / End	12.4	-0.1	-1.6	0.0	0.0	0.0
	SLN426 ring / End	13.1	0.0	-1.6	0.0	0.0	0.0
C LC_111	SLN52 column below / Begin	163.7	0.1	0.0	0.0	-0.2	-0.3
	SLN85 column above / End	-48.5	0.1	1.2	0.0	-1.1	0.4
	SLN425 ring / End	9.9	-0.1	-17.3	0.0	0.0	0.0
	SLN426 ring / End	9.9	-0.1	-17.3	0.0	0.0	0.0
C LC_112	SLN52 column below / Begin	174.4	5.0	-0.8	0.0	0.2	-2.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN85 column above / End	-48.4	0.5	0.7	0.1	0.2	2.1
	SLN425 ring / End	15.0	-0.2	-17.3	0.0	0.0	0.0
	SLN426 ring / End	9.6	-0.2	-17.3	0.0	0.0	0.0
C LC_113	SLN52 column below / Begin	169.5	-0.1	0.1	0.0	0.3	0.6
	SLN85 column above / End	-48.2	-0.1	1.0	0.1	-0.4	-0.5
	SLN425 ring / End	6.5	-0.2	-17.3	0.0	0.0	0.0
	SLN426 ring / End	7.1	-0.2	-17.3	0.0	0.0	0.0
C LC_114	SLN52 column below / Begin	168.4	-0.8	-5.3	0.0	-3.3	0.1
	SLN85 column above / End	-48.7	0.0	1.0	0.1	-0.3	-0.1
	SLN425 ring / End	26.4	-0.2	-17.3	0.0	0.0	0.0
	SLN426 ring / End	27.3	-0.1	-17.3	0.0	0.0	0.0
C LC_115	SLN52 column below / Begin	32.0	0.1	0.0	0.0	-0.2	-0.4
	SLN85 column above / End	-13.3	0.1	0.3	-0.1	-0.3	0.4
	SLN425 ring / End	-0.9	0.0	-2.8	0.0	0.0	0.0
	SLN426 ring / End	-1.2	0.0	-2.8	0.0	0.0	0.0
C LC_116	SLN52 column below / Begin	42.6	5.0	-0.8	0.0	0.1	-2.1
	SLN85 column above / End	-13.3	0.5	-0.2	0.0	1.1	2.1
	SLN425 ring / End	3.9	-0.1	-2.8	0.0	0.0	0.0
	SLN426 ring / End	-1.5	0.0	-2.8	0.0	0.0	0.0
C LC_117	SLN52 column below / Begin	37.7	-0.1	0.1	0.0	0.3	0.5
	SLN85 column above / End	-13.0	-0.1	0.0	0.1	0.5	-0.5
	SLN425 ring / End	-3.8	0.0	-2.8	0.0	0.0	0.0
	SLN426 ring / End	-3.4	-0.1	-2.8	0.0	0.0	0.0
C LC_118	SLN52 column below / Begin	36.7	-0.8	-5.3	0.0	-3.3	0.1
	SLN85 column above / End	-13.6	0.0	0.1	0.0	0.5	-0.1
	SLN425 ring / End	14.7	-0.1	-2.8	0.0	0.0	0.0
	SLN426 ring / End	15.5	0.0	-2.8	0.0	0.0	0.0
C LC_119	SLN52 column below / Begin	239.3	0.0	0.0	0.0	-0.1	-0.1
	SLN85 column above / End	-68.7	0.0	1.8	0.1	-1.6	0.2
	SLN425 ring / End	21.6	-0.2	-25.5	0.0	0.0	0.0
	SLN426 ring / End	22.0	-0.2	-25.5	0.0	0.0	0.0
C LC_120	SLN52 column below / Begin	244.7	2.5	-0.4	0.0	0.1	-1.0
	SLN85 column above / End	-68.5	0.3	1.5	0.1	-0.8	1.1
	SLN425 ring / End	24.5	-0.2	-25.5	0.0	0.0	0.0
	SLN426 ring / End	21.8	-0.3	-25.5	0.0	0.0	0.0
C LC_121	SLN52 column below / Begin	242.4	-0.1	0.1	0.0	0.2	0.4
	SLN85 column above / End	-68.7	-0.1	1.8	0.1	-1.6	-0.3
	SLN425 ring / End	19.2	-0.3	-25.5	0.0	0.0	0.0
	SLN426 ring / End	19.6	-0.2	-25.5	0.0	0.0	0.0
C LC_122	SLN52 column below / Begin	241.6	-0.4	-2.7	0.0	-1.7	0.1
	SLN85 column above / End	-68.5	0.0	1.5	0.1	-0.9	0.0
	SLN425 ring / End	30.9	-0.2	-25.5	0.0	0.0	0.0
	SLN426 ring / End	31.5	-0.3	-25.5	0.0	0.0	0.0
C LC_123	SLN52 column below / Begin	179.9	0.1	-0.1	0.0	-0.4	-0.3
	SLN85 column above / End	-53.7	0.1	1.5	0.0	-1.5	0.3
	SLN425 ring / End	14.4	-0.1	-19.3	0.0	0.0	0.0
	SLN426 ring / End	14.5	-0.1	-19.3	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_124	SLN52 column below / Begin	190.5	5.0	-0.8	0.0	-0.1	-2.1
	SLN85 column above / End	-53.4	0.5	0.9	0.0	0.0	2.2
	SLN425 ring / End	19.6	-0.1	-19.3	0.0	0.0	0.0
	SLN426 ring / End	14.1	-0.2	-19.3	0.0	0.0	0.0
C LC_125	SLN52 column below / Begin	185.9	-0.1	0.0	0.0	0.2	0.6
	SLN85 column above / End	-53.8	-0.1	1.5	0.1	-1.4	-0.5
	SLN425 ring / End	10.7	-0.2	-19.3	0.0	0.0	0.0
	SLN426 ring / End	11.3	-0.3	-19.3	0.0	0.0	0.0
C LC_126	SLN52 column below / Begin	184.3	-0.8	-5.4	0.0	-3.7	0.1
	SLN85 column above / End	-53.4	0.0	1.0	0.1	-0.2	-0.1
	SLN425 ring / End	31.1	-0.2	-19.3	0.0	0.0	0.0
	SLN426 ring / End	32.0	-0.2	-19.3	0.0	0.0	0.0
C LC_127	SLN52 column below / Begin	48.2	0.1	-0.1	0.0	-0.4	-0.3
	SLN85 column above / End	-18.5	0.1	0.5	-0.1	-0.7	0.3
	SLN425 ring / End	3.5	0.0	-4.9	0.0	0.0	0.0
	SLN426 ring / End	3.4	0.0	-4.9	0.0	0.0	0.0
C LC_128	SLN52 column below / Begin	58.7	5.0	-0.8	0.0	-0.1	-2.2
	SLN85 column above / End	-18.3	0.6	0.0	0.0	0.8	2.2
	SLN425 ring / End	8.4	0.0	-4.9	0.0	0.0	0.0
	SLN426 ring / End	2.9	0.0	-4.9	0.0	0.0	0.0
C LC_129	SLN52 column below / Begin	54.1	-0.1	0.0	0.0	0.2	0.5
	SLN85 column above / End	-18.6	-0.1	0.6	0.1	-0.5	-0.5
	SLN425 ring / End	0.4	0.0	-4.9	0.0	0.0	0.0
	SLN426 ring / End	0.9	-0.1	-4.9	0.0	0.0	0.0
C LC_130	SLN52 column below / Begin	52.5	-0.8	-5.4	0.0	-3.7	0.1
	SLN85 column above / End	-18.2	0.0	0.1	0.0	0.6	-0.1
	SLN425 ring / End	19.5	-0.1	-4.9	0.0	0.0	0.0
	SLN426 ring / End	20.3	0.0	-4.9	0.0	0.0	0.0
C LC_135	SLN52 column below / Begin	247.0	0.0	0.2	0.0	0.6	0.1
	SLN85 column above / End	-68.6	0.0	1.4	0.1	-0.7	0.0
	SLN425 ring / End	27.9	-0.3	-25.5	0.0	0.0	0.0
	SLN426 ring / End	28.2	-0.3	-25.5	0.0	0.0	0.0
C LC_141	SLN52 column below / Begin	54.1	0.0	0.1	0.0	0.2	0.0
	SLN85 column above / End	-18.4	0.0	0.3	0.0	0.0	0.0
	SLN425 ring / End	8.7	-0.1	-4.9	0.0	0.0	0.0
	SLN426 ring / End	8.8	-0.1	-4.9	0.0	0.0	0.0
C LC_142	SLN52 column below / Begin	54.2	0.0	0.1	0.0	0.3	0.0
	SLN85 column above / End	-18.4	0.0	0.3	0.0	0.0	0.0
	SLN425 ring / End	8.8	0.0	-4.9	0.0	0.0	0.0
	SLN426 ring / End	8.9	-0.1	-4.9	0.0	0.0	0.0
C LC_143	SLN52 column below / Begin	54.1	0.0	0.1	0.0	0.2	0.0
	SLN85 column above / End	-18.4	0.0	0.3	0.0	-0.1	0.0
	SLN425 ring / End	8.5	-0.1	-4.9	0.0	0.0	0.0
	SLN426 ring / End	8.6	-0.1	-4.9	0.0	0.0	0.0
C LC_144	SLN52 column below / Begin	54.1	0.0	0.1	0.0	0.2	0.0
	SLN85 column above / End	-18.4	0.0	0.3	0.0	0.0	0.0
	SLN425 ring / End	9.0	0.0	-4.9	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN426 ring / End	9.1	-0.1	-4.9	0.0	0.0	0.0
C LC_145	SLN52 column below / Begin	242.5	0.0	0.1	0.0	0.3	0.1
	SLN85 column above / End	-68.7	0.0	1.6	0.1	-1.2	0.0
	SLN425 ring / End	27.2	-0.2	-25.5	0.0	0.0	0.0
	SLN426 ring / End	27.7	-0.3	-25.5	0.0	0.0	0.0
C LC_146	SLN52 column below / Begin	242.7	0.0	0.1	0.0	0.3	0.1
	SLN85 column above / End	-68.7	0.0	1.6	0.1	-1.2	0.0
	SLN425 ring / End	27.7	-0.2	-25.5	0.0	0.0	0.0
	SLN426 ring / End	27.8	-0.3	-25.5	0.0	0.0	0.0
C LC_147	SLN52 column below / Begin	242.6	0.0	0.1	0.0	0.3	0.1
	SLN85 column above / End	-68.7	0.0	1.7	0.1	-1.3	0.0
	SLN425 ring / End	26.3	-0.3	-25.5	0.0	0.0	0.0
	SLN426 ring / End	26.5	-0.2	-25.5	0.0	0.0	0.0
C LC_148	SLN52 column below / Begin	242.6	0.0	0.1	0.0	0.3	0.1
	SLN85 column above / End	-68.7	0.0	1.6	0.1	-1.1	0.0
	SLN425 ring / End	28.7	-0.1	-25.5	0.0	0.0	0.0
	SLN426 ring / End	29.0	-0.3	-25.5	0.0	0.0	0.0
C LC_151	SLN52 column below / Begin	194.8	0.0	-0.6	0.0	-2.6	0.0
	SLN85 column above / End	-68.5	0.0	3.2	0.1	-5.5	0.0
	SLN425 ring / End	12.3	-0.2	-11.5	0.0	0.0	0.0
	SLN426 ring / End	12.4	-0.2	-11.9	0.0	0.0	0.0
C LC_152	SLN52 column below / Begin	130.8	0.0	1.4	0.0	5.7	0.2
	SLN85 column above / End	-18.3	0.0	-2.7	0.1	8.0	-0.1
	SLN425 ring / End	28.2	-0.2	-14.0	0.0	0.0	0.0
	SLN426 ring / End	28.4	-0.2	-14.1	0.0	0.0	0.0
C LC_153	SLN52 column below / Begin	241.9	0.0	0.0	0.0	0.1	0.1
	SLN85 column above / End	-67.9	0.0	1.6	0.1	-1.3	0.0
	SLN425 ring / End	27.6	-0.2	-20.6	0.0	0.0	0.0
	SLN426 ring / End	27.9	-0.3	-21.1	0.0	0.0	0.0
C LC_154	SLN52 column below / Begin	195.0	0.0	-0.6	0.0	-2.5	0.0
	SLN85 column above / End	-69.3	0.0	3.3	0.1	-5.6	0.0
	SLN425 ring / End	12.4	-0.1	-11.5	0.0	0.0	0.0
	SLN426 ring / End	12.6	-0.2	-11.9	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.2 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	19.1 < 100%	OK
Welds	72.4 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN206+SLN222 radial (continuous)-bfl 1	12.7	C LC_122	180.0	0.0	55.3	OK
SLN206+SLN222 radial (continuous)-tfl 1	12.7	C LC_152	242.1	0.0	24.4	OK
SLN206+SLN222 radial (continuous)-w 1	7.2	C LC_124	275.5	0.2	40.2	OK
SLN52 column below	10.0	C LC_122	185.9	0.0	0.0	OK
SLN85 column above	10.0	C LC_152	204.5	0.0	0.0	OK
SLN425 ring-bfl 1	12.7	C LC_122	28.7	0.0	0.0	OK
SLN425 ring-tfl 1	12.7	C LC_121	19.8	0.0	0.0	OK
SLN425 ring-w 1	7.2	C LC_135	142.5	0.0	0.0	OK
SLN426 ring-bfl 1	12.7	C LC_122	28.9	0.0	0.0	OK
SLN426 ring-tfl 1	12.7	C LC_121	20.1	0.0	0.0	OK
SLN426 ring-w 1	7.2	C LC_148	143.2	0.0	0.0	OK
End plate column below SLN52 column below	12.0	C LC_122	144.1	0.0	50.0	OK
End plate column above SLN85 column above	12.0	C LC_152	195.7	0.0	24.4	OK
End plate SLN426 ring	12.0	C LC_124	94.7	0.0	27.0	OK
End plate SLN425 ring	12.0	C LC_128	89.0	0.0	19.4	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

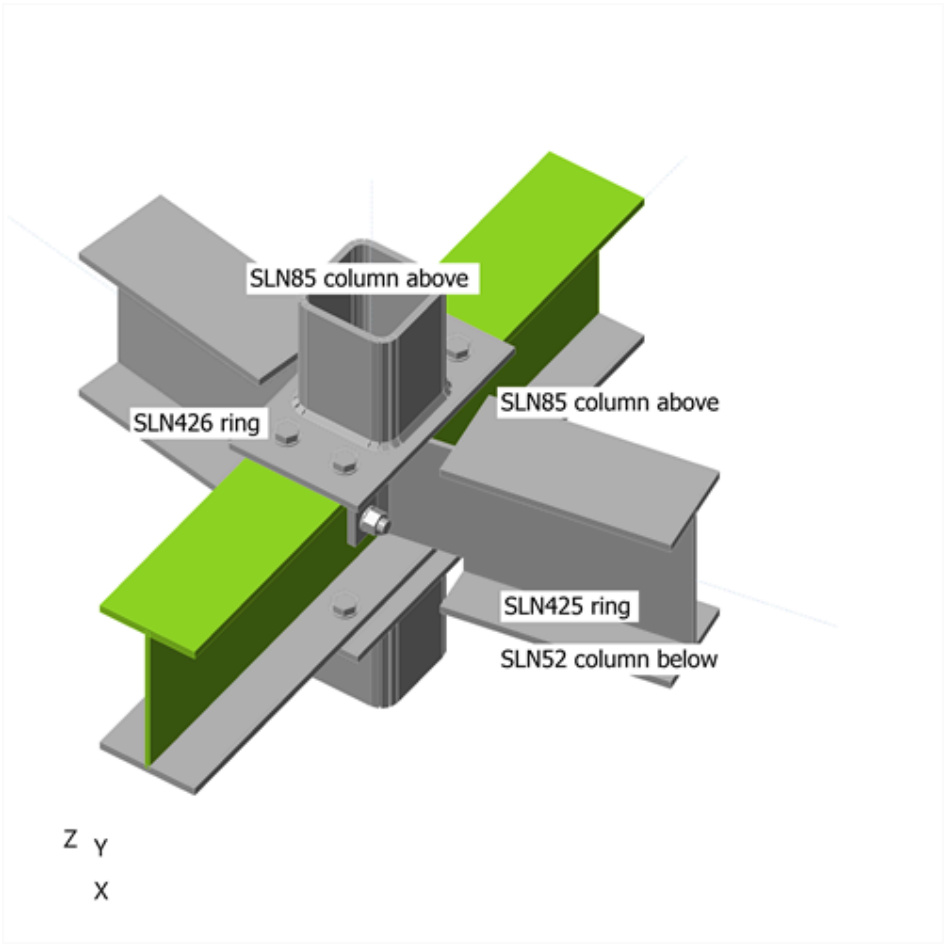
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

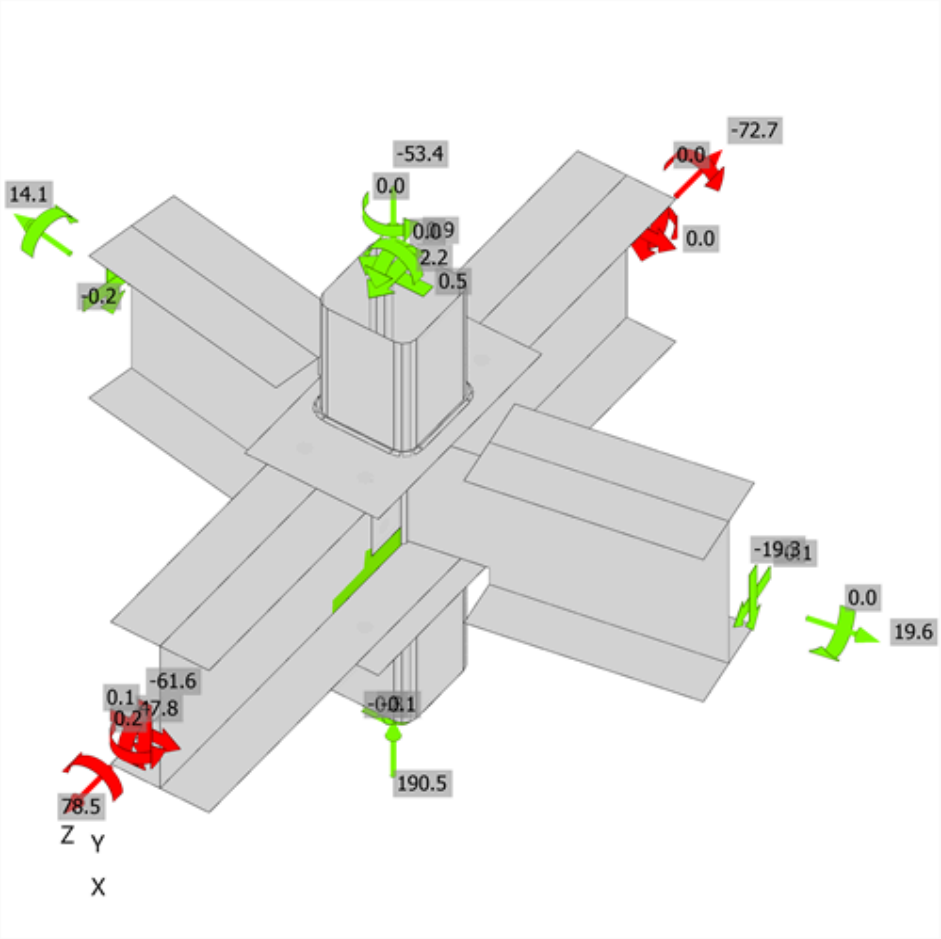
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN52 column below	150	C LC_154	0	5	0.0	OK
SLN85 column above	150	C LC_154	0	5	0.0	OK

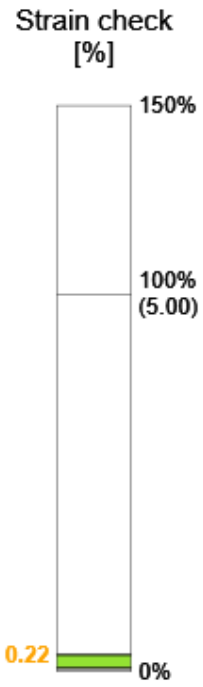
Symbol explanation

d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation

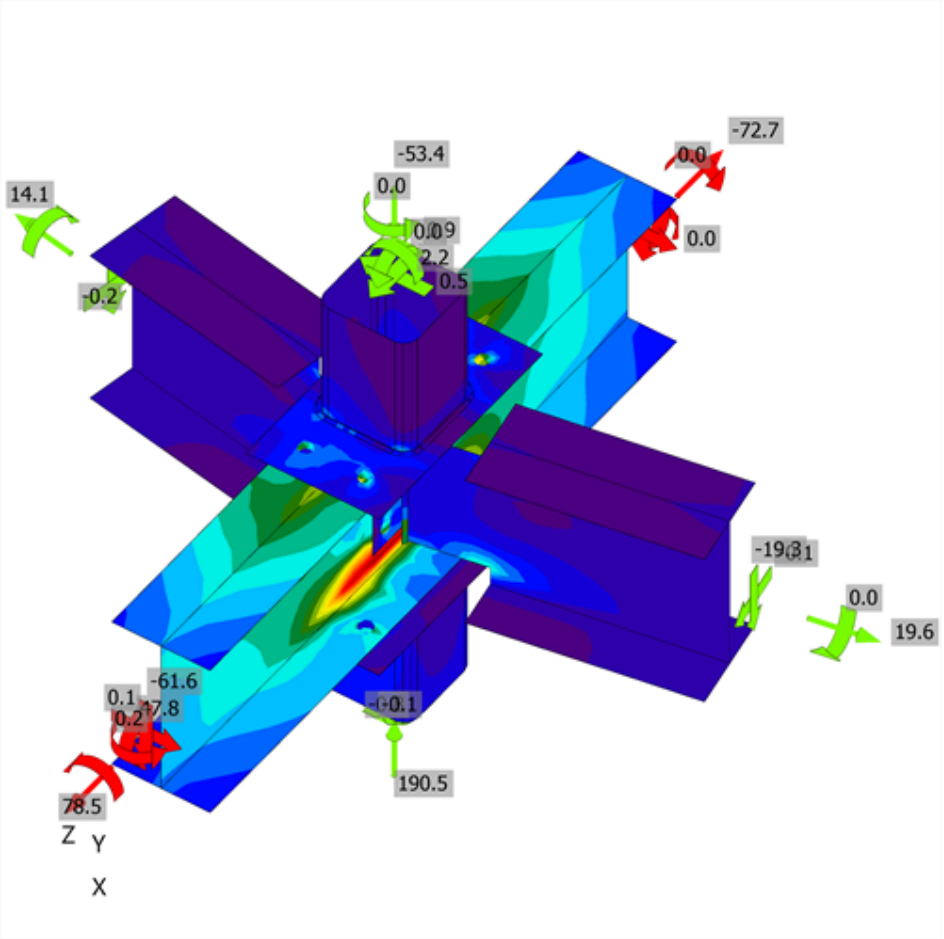


Overall check



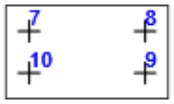
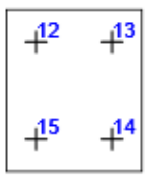


Strain check, C LC_124



Equivalent stress, C LC_124

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_122	0.7	5.6	176.6	0.5	5.9	6.3	OK	OK
	B3	M20 8.8 - 1	C LC_153	0.7	5.1	175.9	0.5	5.5	5.8	OK	OK
	B4	M20 8.8 - 1	C LC_124	4.6	3.5	182.5	3.3	3.8	6.1	OK	OK
	B5	M20 8.8 - 1	C LC_124	4.6	3.9	187.1	3.3	4.1	6.5	OK	OK
	B7	M20 8.8 - 1	C LC_124	8.4	3.6	156.4	6.0	3.8	8.1	OK	OK
	B8	M20 8.8 - 1	C LC_152	26.3	3.5	88.2	18.6	4.0	17.1	OK	OK
	B9	M20 8.8 - 1	C LC_152	26.9	3.5	93.7	19.1	3.7	17.3	OK	OK
	B10	M20 8.8 - 1	C LC_121	2.3	3.5	121.1	1.6	3.7	4.9	OK	OK
	B12	M20 8.8 - 1	C LC_124	5.0	1.0	64.8	3.5	1.6	3.6	OK	OK
	B13	M20 8.8 - 1	C LC_124	5.1	1.1	68.2	3.6	1.6	3.7	OK	OK
	B14	M20 8.8 - 1	C LC_124	4.1	1.1	63.1	2.9	1.3	3.3	OK	OK
	B15	M20 8.8 - 1	C LC_124	4.0	1.1	63.7	2.8	1.2	3.2	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN52 column below	SLN52 column below	◢ 5.6 ◣	520	C LC_122	261.9	0.0	-127.8	-128.5	30.0	64.7	35.0	OK	OK
		◢ 5.6 ◣	521	C LC_122	266.0	0.0	-131.5	130.8	-26.7	65.7	36.1	OK	OK
		◢ 5.6 ◣	521	C LC_152	154.4	0.0	109.7	-54.3	-31.4	38.1	24.1	OK	OK
End plate column above SLN85 column above	SLN85 column above	◢ 5.6 ◣	521	C LC_154	133.6	0.0	-66.1	-67.0	0.8	33.0	17.8	OK	OK
SLN206+SLN222 radial (continuous)-w 1	SLN425 ring-w 1	◢ 7.2	189	C LC_128	234.8	0.0	32.7	-31.6	130.5	58.0	31.1	OK	OK
SLN206+SLN222 radial (continuous)-w 1	SLN426 ring-w 1	◢ 7.2	189	C LC_124	293.0	0.0	-26.5	-26.2	-166.4	72.4	47.3	OK	OK
End plate SLN426 ring	SLN426 ring-w 1	◢ 4.2 ◣	149	C LC_124	204.0	0.0	-50.4	-42.1	106.1	50.4	30.8	OK	OK
		◢ 4.2 ◣	149	C LC_124	242.6	0.0	-48.2	60.1	-123.4	60.0	36.4	OK	OK
End plate SLN425 ring	SLN425 ring-w 1	◢ 4.2 ◣	149	C LC_121	172.1	0.0	-32.5	-41.9	88.2	42.5	26.2	OK	OK
		◢ 4.2 ◣	149	C LC_121	155.9	0.0	-41.5	36.0	-78.9	38.5	23.9	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-S_020
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN184+SLN183 radial (continuous) - SLN39 column below - SLN185 ring - SLN264 ring - SLN451 raking strut

Name	SLN184+SLN183 radial (continuous) - SLN39 column below - SLN185 ring - SLN264 ring - SLN451 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

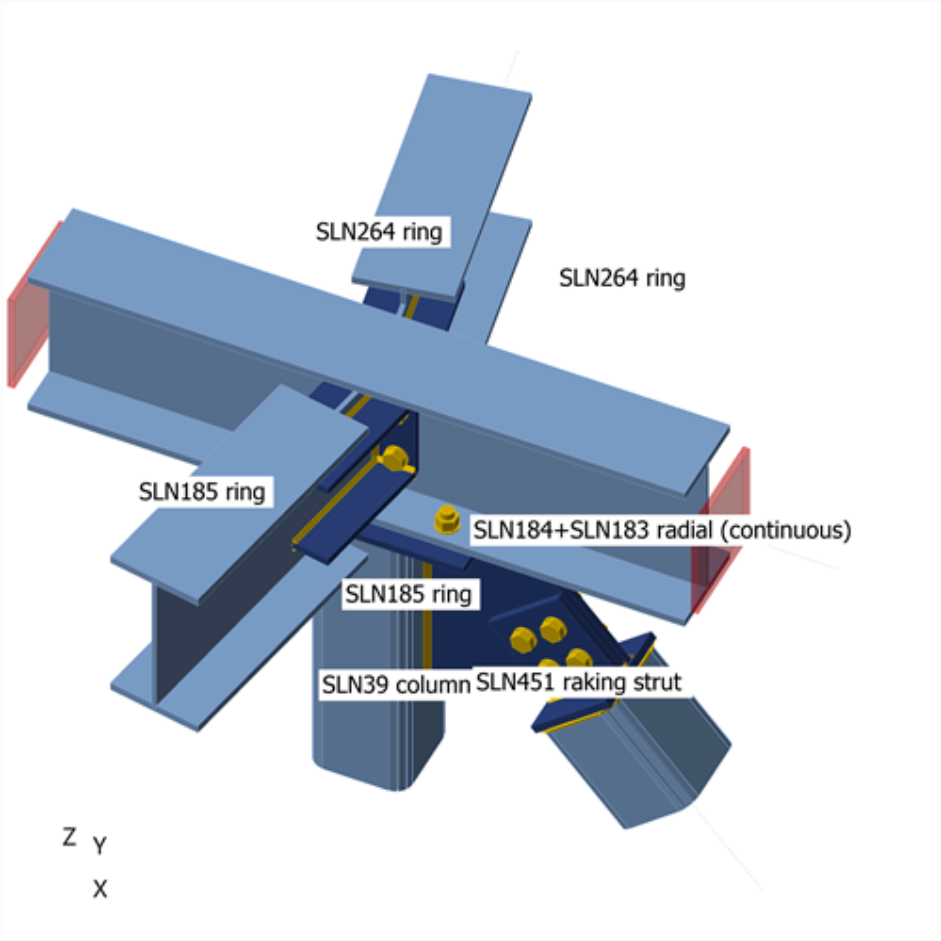
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN184+SLN183 radial (continuous)	1 - UB254x146x43	-160.0	0.0	0.0	0	0	0
SLN39 column below	2 - SHS150x150x10	0.0	90.0	-160.0	0	0	0
SLN185 ring	1 - UB254x146x43	100.0	0.0	0.0	0	0	0
SLN264 ring	1 - UB254x146x43	120.0	0.0	0.0	0	0	0
SLN451 raking strut	2 - SHS150x150x10	20.0	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN184+SLN183 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN184+SLN183 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN39 column below / begin		Position	0
SLN185 ring / begin		Position	0
SLN264 ring / end		Position	0
SLN451 raking strut / end		Position	0



SLN184+SLN183 radial (continuous) - SLN39 column below - SLN185 ring - SLN264 ring - SLN451 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN39 column below / Begin	31.6	0.1	0.4	0.0	1.6	-0.2
	SLN185 ring / Begin	-6.3	-0.2	-4.9	0.0	0.0	0.5
	SLN264 ring / End	8.8	0.1	-4.9	0.0	0.0	0.2
	SLN451 raking strut / End	0.4	0.1	-1.0	0.1	0.0	0.3
C LC_102	SLN39 column below / Begin	147.0	0.3	1.8	0.0	7.3	-1.0
	SLN185 ring / Begin	-18.2	-0.6	-25.6	0.0	0.0	1.8
	SLN264 ring / End	27.8	0.2	-25.6	0.0	0.0	0.7
	SLN451 raking strut / End	-10.2	0.3	-1.0	0.2	0.0	1.0
C LC_103	SLN39 column below / Begin	127.6	0.2	1.8	0.0	7.2	-0.9
	SLN185 ring / Begin	-9.4	-0.7	-24.6	0.0	0.0	1.9
	SLN264 ring / End	18.7	0.2	-24.5	0.0	0.0	0.7
	SLN451 raking strut / End	-27.9	0.2	-0.9	0.2	0.0	0.9
C LC_104	SLN39 column below / Begin	156.2	-0.6	-0.8	0.0	5.4	-0.8
	SLN185 ring / Begin	-18.9	-0.5	-24.6	0.0	0.0	1.7
	SLN264 ring / End	27.6	0.4	-24.5	0.0	0.0	1.0
	SLN451 raking strut / End	9.4	0.3	-1.2	0.2	0.0	0.8
C LC_105	SLN39 column below / Begin	142.1	2.6	0.8	0.0	6.3	-1.7
	SLN185 ring / Begin	-12.5	-0.5	-24.6	0.0	0.0	1.6
	SLN264 ring / End	24.9	0.3	-24.5	0.0	0.0	0.7
	SLN451 raking strut / End	-7.6	0.3	-1.1	0.1	0.0	1.7
C LC_106	SLN39 column below / Begin	135.6	0.3	1.8	0.0	7.3	-1.0
	SLN185 ring / Begin	-9.9	-0.6	-24.6	0.0	0.0	1.7
	SLN264 ring / End	18.0	0.1	-24.5	0.0	0.0	0.6
	SLN451 raking strut / End	-18.4	0.5	-1.0	0.1	0.0	1.0
C LC_107	SLN39 column below / Begin	-8.8	0.0	0.2	0.0	1.0	0.0
	SLN185 ring / Begin	5.1	-0.2	-1.6	0.0	0.0	0.5
	SLN264 ring / End	-3.5	0.1	-1.6	0.0	0.0	0.2
	SLN451 raking strut / End	-26.7	-0.1	-0.4	0.0	0.0	0.1
C LC_108	SLN39 column below / Begin	40.4	-1.6	-4.9	0.0	-2.6	0.2
	SLN185 ring / Begin	-9.7	-0.1	-1.6	0.0	0.0	0.3
	SLN264 ring / End	10.2	0.2	-1.6	0.0	0.0	0.5
	SLN451 raking strut / End	37.1	0.1	-1.1	0.2	0.0	-0.3
C LC_109	SLN39 column below / Begin	15.2	4.7	-1.8	0.0	-0.8	-1.5
	SLN185 ring / Begin	1.5	0.0	-1.6	0.0	0.0	0.0
	SLN264 ring / End	5.7	0.0	-1.6	0.0	0.0	0.0
	SLN451 raking strut / End	7.0	0.1	-0.9	-0.2	0.0	1.7
C LC_110	SLN39 column below / Begin	4.4	0.1	0.3	0.0	1.3	-0.2
	SLN185 ring / Begin	5.6	0.0	-1.6	0.0	0.0	0.1
	SLN264 ring / End	-5.8	-0.1	-1.6	0.0	0.0	-0.2
	SLN451 raking strut / End	-11.8	0.4	-0.6	0.0	0.0	0.3
C LC_111	SLN39 column below / Begin	78.0	0.2	1.3	0.0	5.3	-0.6
	SLN185 ring / Begin	-2.2	-0.6	-17.3	0.0	0.0	1.6
	SLN264 ring / End	9.3	0.1	-17.3	0.0	0.0	0.6
	SLN451 raking strut / End	-36.8	0.1	-0.7	0.1	0.0	0.6
C LC_112	SLN39 column below / Begin	129.7	-1.5	-3.8	0.0	1.8	-0.4

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN185 ring / Begin	-18.3	-0.4	-17.3	0.0	0.0	1.3
	SLN264 ring / End	24.3	0.4	-17.3	0.0	0.0	1.0
	SLN451 raking strut / End	30.5	0.3	-1.3	0.3	0.0	0.3
C LC_113	SLN39 column below / Begin	103.5	4.8	-0.7	0.0	3.6	-2.1
	SLN185 ring / Begin	-6.6	-0.3	-17.3	0.0	0.0	1.0
	SLN264 ring / End	19.5	0.2	-17.3	0.0	0.0	0.5
	SLN451 raking strut / End	-0.8	0.3	-1.1	-0.1	0.0	2.2
C LC_114	SLN39 column below / Begin	92.1	0.2	1.4	0.0	5.7	-0.8
	SLN185 ring / Begin	-2.2	-0.4	-17.3	0.0	0.0	1.1
	SLN264 ring / End	7.3	0.1	-17.3	0.0	0.0	0.3
	SLN451 raking strut / End	-20.6	0.6	-0.9	0.1	0.0	0.8
C LC_115	SLN39 column below / Begin	-0.7	0.0	0.3	0.0	1.4	-0.1
	SLN185 ring / Begin	3.5	-0.2	-2.8	0.0	0.0	0.7
	SLN264 ring / End	-1.3	0.1	-2.8	0.0	0.0	0.2
	SLN451 raking strut / End	-26.8	-0.1	-0.7	0.0	0.0	0.2
C LC_116	SLN39 column below / Begin	48.8	-1.6	-4.8	0.0	-2.2	0.1
	SLN185 ring / Begin	-11.4	-0.1	-2.8	0.0	0.0	0.4
	SLN264 ring / End	12.6	0.2	-2.8	0.0	0.0	0.5
	SLN451 raking strut / End	37.4	0.1	-1.3	0.2	0.0	-0.2
C LC_117	SLN39 column below / Begin	23.4	4.7	-1.7	0.0	-0.4	-1.5
	SLN185 ring / Begin	-0.2	-0.1	-2.8	0.0	0.0	0.2
	SLN264 ring / End	8.0	0.0	-2.8	0.0	0.0	0.1
	SLN451 raking strut / End	7.2	0.1	-1.1	-0.2	0.0	1.7
C LC_118	SLN39 column below / Begin	12.6	0.1	0.4	0.0	1.8	-0.3
	SLN185 ring / Begin	4.0	-0.1	-2.8	0.0	0.0	0.2
	SLN264 ring / End	-3.5	0.0	-2.8	0.0	0.0	-0.1
	SLN451 raking strut / End	-11.8	0.5	-0.9	0.0	0.0	0.3
C LC_119	SLN39 column below / Begin	136.1	0.2	1.8	0.0	7.2	-0.9
	SLN185 ring / Begin	-11.4	-0.7	-25.6	0.0	0.0	2.0
	SLN264 ring / End	20.9	0.2	-25.6	0.0	0.0	0.7
	SLN451 raking strut / End	-23.7	0.2	-0.9	0.2	0.0	0.9
C LC_120	SLN39 column below / Begin	159.5	-0.6	-0.8	0.0	5.5	-0.9
	SLN185 ring / Begin	-20.9	-0.5	-25.6	0.0	0.0	1.8
	SLN264 ring / End	30.0	0.4	-25.6	0.0	0.0	0.9
	SLN451 raking strut / End	6.4	0.3	-1.2	0.3	0.0	0.8
C LC_121	SLN39 column below / Begin	146.5	2.6	0.8	0.0	6.4	-1.7
	SLN185 ring / Begin	-14.5	-0.6	-25.6	0.0	0.0	1.7
	SLN264 ring / End	27.2	0.3	-25.6	0.0	0.0	0.7
	SLN451 raking strut / End	-9.2	0.3	-1.1	0.1	0.0	1.8
C LC_122	SLN39 column below / Begin	141.8	0.3	1.8	0.0	7.4	-1.1
	SLN185 ring / Begin	-11.9	-0.6	-25.6	0.0	0.0	1.7
	SLN264 ring / End	20.2	0.2	-25.6	0.0	0.0	0.6
	SLN451 raking strut / End	-17.4	0.5	-1.0	0.1	0.0	1.0
C LC_123	SLN39 column below / Begin	95.0	0.2	1.3	0.0	5.4	-0.6
	SLN185 ring / Begin	-6.1	-0.6	-19.4	0.0	0.0	1.7
	SLN264 ring / End	13.5	0.2	-19.4	0.0	0.0	0.6
	SLN451 raking strut / End	-28.4	0.1	-0.7	0.1	0.0	0.7

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_124	SLN39 column below / Begin	136.4	-1.4	-3.8	0.0	2.1	-0.5
	SLN185 ring / Begin	-22.4	-0.5	-19.4	0.0	0.0	1.5
	SLN264 ring / End	29.0	0.4	-19.4	0.0	0.0	0.9
	SLN451 raking strut / End	24.4	0.3	-1.3	0.3	0.0	0.4
C LC_125	SLN39 column below / Begin	112.4	4.8	-0.6	0.0	3.8	-2.2
	SLN185 ring / Begin	-10.6	-0.4	-19.4	0.0	0.0	1.2
	SLN264 ring / End	24.2	0.2	-19.4	0.0	0.0	0.5
	SLN451 raking strut / End	-4.0	0.3	-1.1	-0.1	0.0	2.3
C LC_126	SLN39 column below / Begin	104.5	0.2	1.4	0.0	5.9	-0.9
	SLN185 ring / Begin	-6.1	-0.4	-19.4	0.0	0.0	1.2
	SLN264 ring / End	11.5	0.1	-19.4	0.0	0.0	0.3
	SLN451 raking strut / End	-18.6	0.6	-0.9	0.1	0.0	0.8
C LC_127	SLN39 column below / Begin	16.3	0.0	0.4	0.0	1.5	-0.1
	SLN185 ring / Begin	-0.4	-0.3	-4.9	0.0	0.0	0.7
	SLN264 ring / End	2.9	0.1	-4.9	0.0	0.0	0.3
	SLN451 raking strut / End	-18.4	-0.1	-0.7	0.0	0.0	0.2
C LC_128	SLN39 column below / Begin	55.4	-1.6	-4.8	0.0	-1.9	0.1
	SLN185 ring / Begin	-15.5	-0.2	-4.9	0.0	0.0	0.7
	SLN264 ring / End	17.3	0.2	-4.9	0.0	0.0	0.5
	SLN451 raking strut / End	31.2	0.1	-1.3	0.2	0.0	-0.1
C LC_129	SLN39 column below / Begin	32.2	4.7	-1.6	0.0	-0.2	-1.6
	SLN185 ring / Begin	-4.1	-0.1	-4.9	0.0	0.0	0.4
	SLN264 ring / End	12.7	0.0	-4.9	0.0	0.0	0.1
	SLN451 raking strut / End	4.0	0.1	-1.1	-0.1	0.0	1.8
C LC_130	SLN39 column below / Begin	25.0	0.1	0.5	0.0	1.9	-0.3
	SLN185 ring / Begin	0.0	-0.1	-4.9	0.0	0.0	0.3
	SLN264 ring / End	0.7	0.0	-4.9	0.0	0.0	-0.1
	SLN451 raking strut / End	-9.8	0.5	-0.9	0.0	0.0	0.3
C LC_135	SLN39 column below / Begin	151.2	0.3	1.8	0.0	7.2	-1.1
	SLN185 ring / Begin	-15.4	-0.7	-25.6	0.0	0.0	2.1
	SLN264 ring / End	28.2	0.3	-25.6	0.0	0.0	1.0
	SLN451 raking strut / End	-7.8	0.4	-1.0	0.3	0.0	1.3
C LC_141	SLN39 column below / Begin	31.1	0.1	0.4	0.0	1.6	-0.2
	SLN185 ring / Begin	-6.0	-0.2	-4.9	0.0	0.0	0.5
	SLN264 ring / End	8.6	0.0	-4.9	0.0	0.0	0.2
	SLN451 raking strut / End	-0.4	0.1	-1.0	0.1	0.0	0.3
C LC_142	SLN39 column below / Begin	32.2	0.1	0.4	0.0	1.6	-0.2
	SLN185 ring / Begin	-6.5	-0.1	-4.9	0.0	0.0	0.5
	SLN264 ring / End	9.1	0.1	-4.9	0.0	0.0	0.2
	SLN451 raking strut / End	1.1	0.1	-1.0	0.1	0.0	0.3
C LC_143	SLN39 column below / Begin	31.8	0.1	0.4	0.0	1.6	-0.2
	SLN185 ring / Begin	-6.3	-0.2	-4.9	0.0	0.0	0.5
	SLN264 ring / End	9.0	0.1	-4.9	0.0	0.0	0.2
	SLN451 raking strut / End	0.6	0.1	-1.0	0.1	0.0	0.3
C LC_144	SLN39 column below / Begin	31.5	0.1	0.4	0.0	1.6	-0.2
	SLN185 ring / Begin	-6.2	-0.2	-4.9	0.0	0.0	0.5
	SLN264 ring / End	8.7	0.0	-4.9	0.0	0.0	0.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN451 raking strut / End	0.2	0.1	-1.0	0.1	0.0	0.3
C LC_145	SLN39 column below / Begin	144.7	0.3	1.8	0.0	7.3	-1.0
	SLN185 ring / Begin	-17.1	-0.7	-25.6	0.0	0.0	1.9
	SLN264 ring / End	26.7	0.1	-25.6	0.0	0.0	0.6
	SLN451 raking strut / End	-13.2	0.3	-1.0	0.2	0.0	1.0
C LC_146	SLN39 column below / Begin	149.2	0.3	1.8	0.0	7.3	-1.0
	SLN185 ring / Begin	-19.3	-0.5	-25.6	0.0	0.0	1.8
	SLN264 ring / End	28.8	0.3	-25.6	0.0	0.0	0.8
	SLN451 raking strut / End	-7.3	0.3	-1.0	0.2	0.0	1.0
C LC_147	SLN39 column below / Begin	147.6	0.3	1.8	0.0	7.3	-1.0
	SLN185 ring / Begin	-18.5	-0.6	-25.6	0.0	0.0	1.8
	SLN264 ring / End	28.4	0.3	-25.6	0.0	0.0	0.7
	SLN451 raking strut / End	-9.5	0.3	-1.0	0.2	0.0	1.0
C LC_148	SLN39 column below / Begin	146.4	0.2	1.8	0.0	7.3	-1.0
	SLN185 ring / Begin	-17.9	-0.6	-25.6	0.0	0.0	1.9
	SLN264 ring / End	27.2	0.2	-25.6	0.0	0.0	0.7
	SLN451 raking strut / End	-11.0	0.3	-1.0	0.2	0.0	1.0
C LC_151	SLN39 column below / Begin	91.8	0.2	0.2	0.0	0.6	-0.7
	SLN185 ring / Begin	-6.2	-0.4	-11.7	0.0	0.0	1.2
	SLN264 ring / End	12.4	0.2	-11.3	0.0	0.0	0.5
	SLN451 raking strut / End	-11.3	0.3	-1.0	0.1	0.0	0.8
C LC_152	SLN39 column below / Begin	102.1	0.2	3.1	0.0	12.6	-1.0
	SLN185 ring / Begin	-19.7	-0.5	-14.2	0.0	0.0	1.6
	SLN264 ring / End	28.6	0.2	-13.9	0.0	0.0	0.6
	SLN451 raking strut / End	3.6	0.3	-1.0	0.2	0.0	0.8
C LC_153	SLN39 column below / Begin	146.7	0.3	1.7	0.0	7.1	-1.0
	SLN185 ring / Begin	-18.2	-0.6	-21.0	0.0	0.0	1.8
	SLN264 ring / End	27.9	0.2	-20.3	0.0	0.0	0.7
	SLN451 raking strut / End	-9.9	0.3	-1.0	0.2	0.0	1.0
C LC_154	SLN39 column below / Begin	91.1	0.2	0.2	0.0	0.8	-0.7
	SLN185 ring / Begin	-6.3	-0.4	-11.7	0.0	0.0	1.2
	SLN264 ring / End	12.5	0.2	-11.3	0.0	0.0	0.5
	SLN451 raking strut / End	-11.0	0.3	-1.0	0.1	0.0	0.8

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.1 < 5.0%	OK
Loc. deformation	0.1 < 3%	OK
Bolts	21.3 < 100%	OK
Welds	98.1 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN184+SLN183 radial (continuous)-bfl 1	12.7	C LC_119	189.1	0.0	61.7	OK
SLN184+SLN183 radial (continuous)-tfl 1	12.7	C LC_135	146.2	0.0	0.0	OK
SLN184+SLN183 radial (continuous)-w 1	7.2	C LC_135	275.2	0.1	56.2	OK
SLN39 column below	10.0	C LC_111	275.1	0.1	0.0	OK
SLN185 ring-bfl 1	12.7	C LC_135	124.3	0.0	0.0	OK
SLN185 ring-tfl 1	12.7	C LC_135	122.7	0.0	0.0	OK
SLN185 ring-w 1	7.2	C LC_135	275.1	0.0	0.0	OK
SLN264 ring-bfl 1	12.7	C LC_135	66.6	0.0	0.0	OK
SLN264 ring-tfl 1	12.7	C LC_135	60.0	0.0	0.0	OK
SLN264 ring-w 1	7.2	C LC_135	195.9	0.0	0.0	OK
SLN451 raking strut	10.0	C LC_125	54.8	0.0	0.0	OK
End plate column below SLN39 column below	12.0	C LC_152	156.9	0.0	50.9	OK
End plate SLN185 ring	12.0	C LC_135	87.4	0.0	453.2	OK
End plate SLN264 ring	12.0	C LC_135	62.1	0.0	331.2	OK
Cap plate SLN451 raking strut	12.0	C LC_125	128.7	0.0	0.0	OK
Fin plate SLN451 raking strut	12.0	C LC_125	275.1	0.0	77.8	OK
Fin plate 2 SLN451 raking strut	12.0	C LC_125	275.1	0.0	82.4	OK
Gusset SLN451 raking strut	12.0	C LC_125	275.2	0.1	45.4	OK
Flange top a end plate SLN185 ring	12.0	C LC_135	166.7	0.0	0.0	OK
Flange top b end plate SLN185 ring	12.0	C LC_135	144.7	0.0	0.0	OK
Flange bottom a end plate SLN185 ring	12.0	C LC_135	148.9	0.0	0.0	OK
Flange bottom b end plate SLN185 ring	12.0	C LC_135	154.9	0.0	0.0	OK
Flange top a end plate SLN264 ring	12.0	C LC_135	85.2	0.0	0.0	OK
Flange top b end plate SLN264 ring	12.0	C LC_135	69.9	0.0	0.0	OK
Flange bottom a end plate SLN264 ring	12.0	C LC_135	81.3	0.0	0.0	OK
Flange bottom b end plate SLN264 ring	12.0	C LC_135	95.3	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

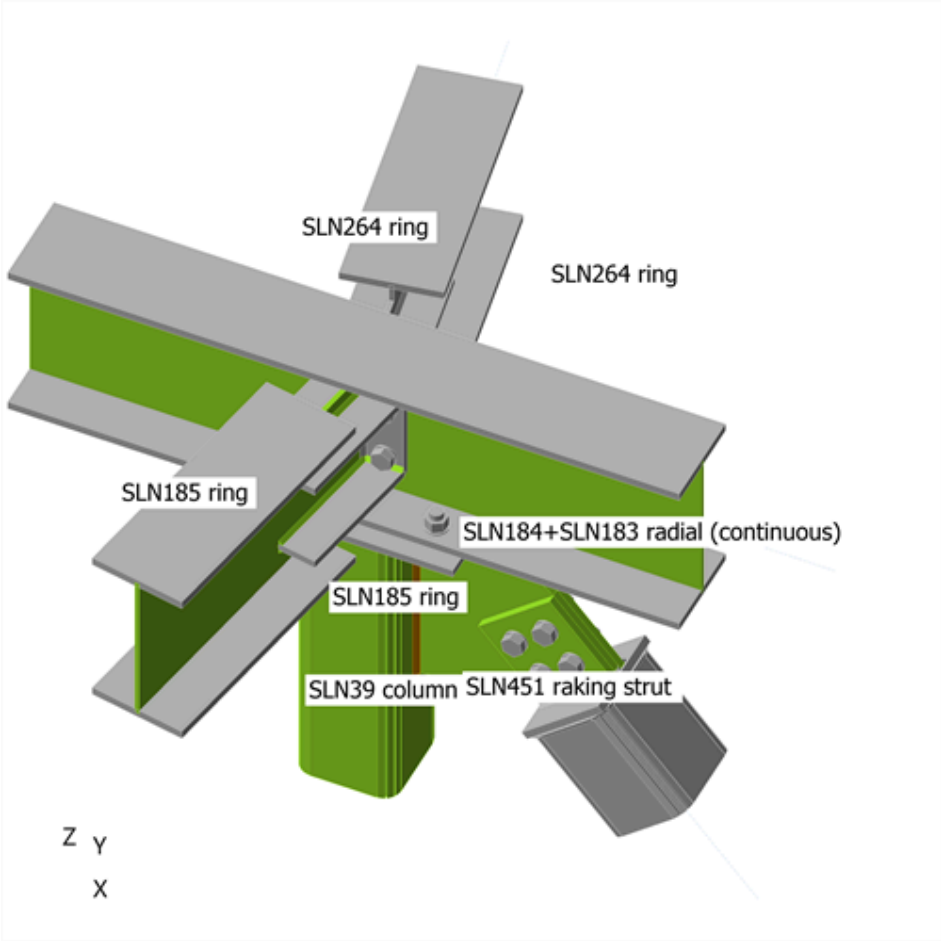
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

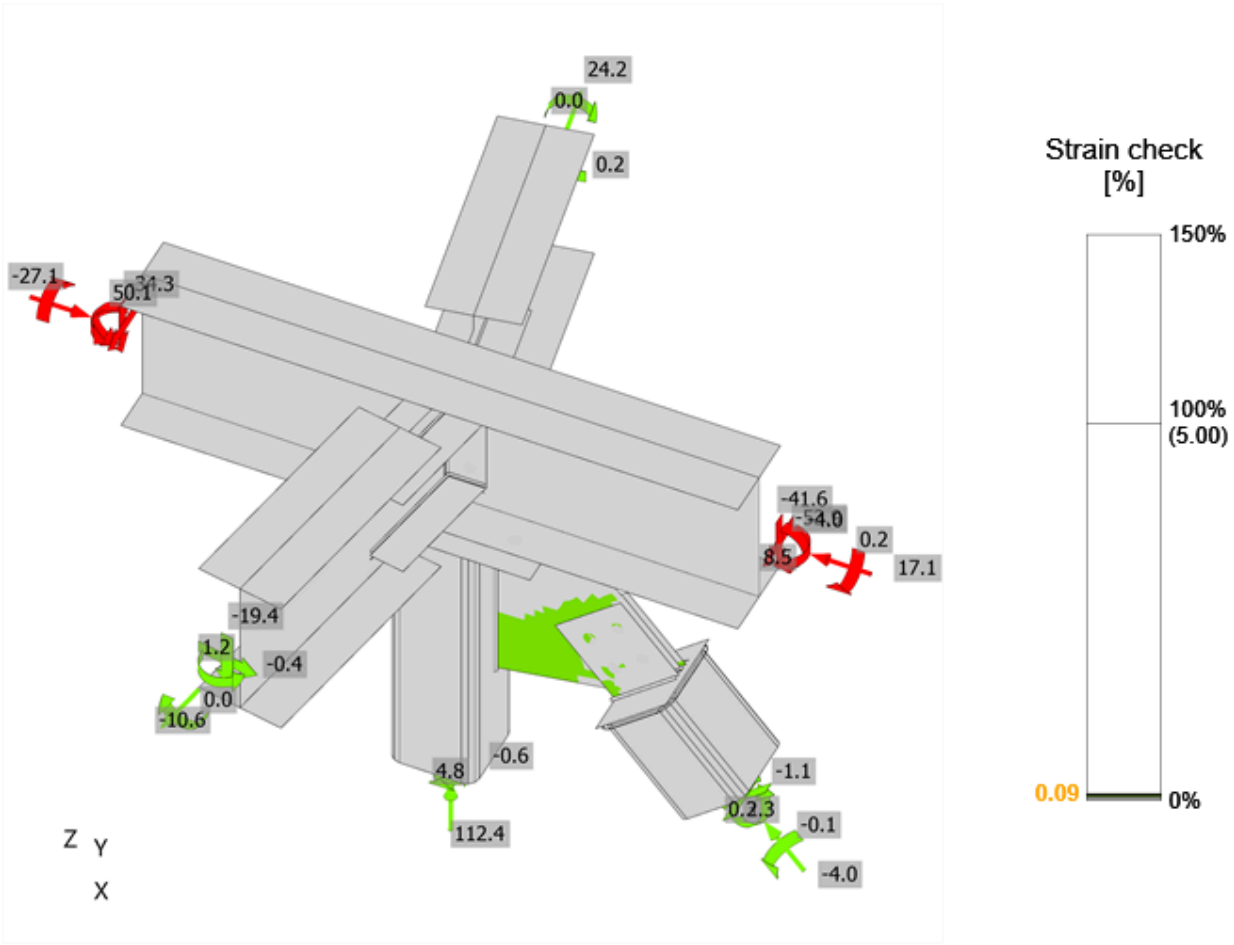
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN39 column below	150	C LC_108	0	5	0.1	OK
SLN451 raking strut	150	C LC_154	0	5	0.0	OK

Symbol explanation

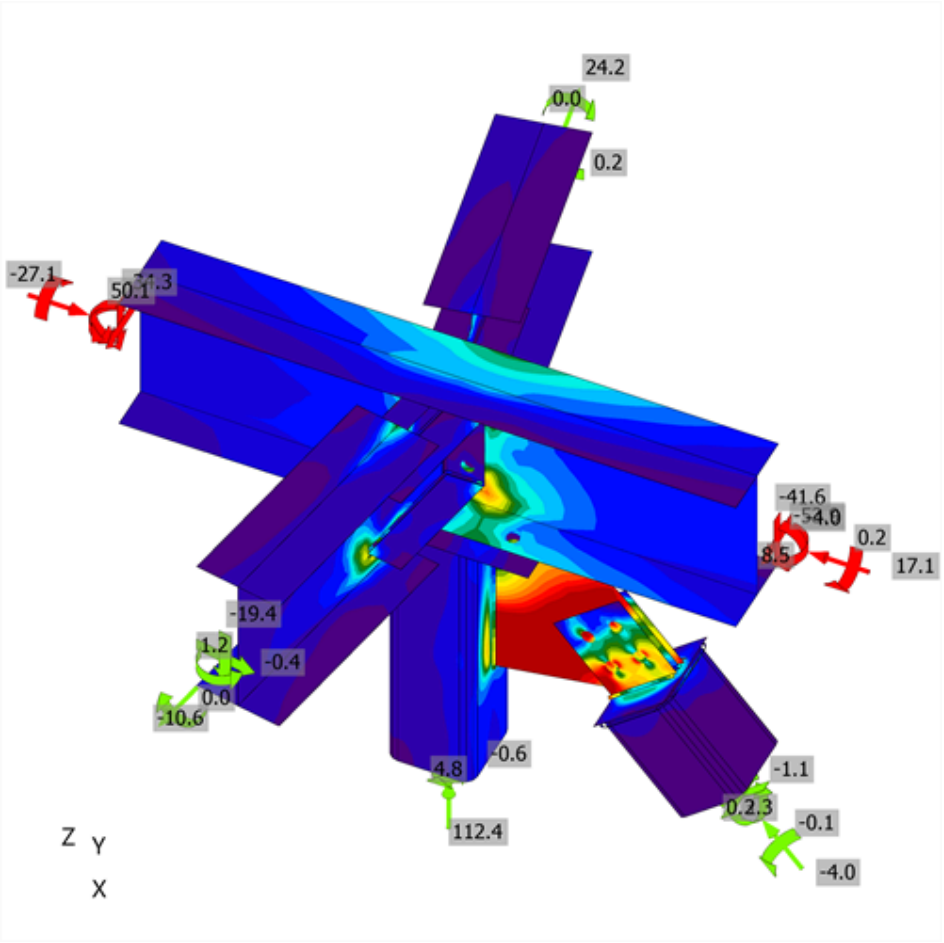
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

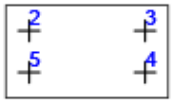
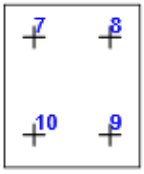
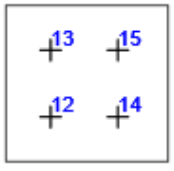


Strain check, C LC_125



Equivalent stress, C LC_125

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_152	17.3	2.0	77.1	12.3	2.5	10.8	OK	OK
	B3	M20 8.8 - 1	C LC_108	19.5	4.9	112.7	13.8	5.2	15.1	OK	OK
	B4	M20 8.8 - 1	C LC_108	18.7	4.6	115.6	13.2	4.8	14.3	OK	OK
	B5	M20 8.8 - 1	C LC_152	22.7	0.5	170.3	16.1	0.5	12.0	OK	OK
	B7	M20 8.8 - 1	C LC_135	4.9	0.7	123.8	3.5	0.9	3.2	OK	OK
	B8	M20 8.8 - 1	C LC_135	3.6	0.5	61.9	2.6	0.7	2.3	OK	OK
	B9	M20 8.8 - 1	C LC_135	5.1	1.4	123.8	3.6	2.1	4.0	OK	OK
	B10	M20 8.8 - 1	C LC_135	7.5	0.6	123.8	5.3	0.8	4.4	OK	OK
	B12	M20 8.8 - 1	C LC_125	15.9	12.5	106.0	11.3	13.3	21.3	OK	OK
	B13	M20 8.8 - 1	C LC_125	8.6	11.6	115.3	6.1	12.4	16.7	OK	OK
	B14	M20 8.8 - 1	C LC_116	1.0	6.8	106.0	0.7	11.2	7.7	OK	OK
	B15	M20 8.8 - 1	C LC_111	0.6	6.5	115.3	0.4	8.8	7.2	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN451 raking strut	SLN451 raking strut	◢ 4.2	519	C LC_125	142.9	0.0	-58.3	72.7	-19.8	35.3	20.6	OK	OK
SLN39 column below-w 3	Gusset SLN451 raking strut	◢ 5.6 ◣	288	C LC_111	396.9	0.2	-129.7	-123.9	-177.6	98.1	59.6	OK	OK
		◢ 5.6 ◣	288	C LC_111	396.9	0.2	-110.7	116.8	186.5	98.1	61.2	OK	OK
SLN185 ring-w 1	Flange top b end plate SLN185 ring	◢ 4.2 ◣	198	C LC_135	200.3	0.0	107.0	88.6	41.3	49.5	30.1	OK	OK
		◢ 4.2 ◣	197	C LC_135	84.8	0.0	-42.7	39.3	15.5	20.9	11.7	OK	OK
SLN185 ring-w 1	Flange bottom b end plate SLN185 ring	◢ 4.2 ◣	198	C LC_135	122.9	0.0	35.9	51.2	44.5	30.4	18.7	OK	OK
		◢ 4.2 ◣	198	C LC_135	255.2	0.0	118.2	-102.9	-80.4	63.1	39.1	OK	OK
SLN264 ring-w 1	Flange top a end plate SLN264 ring	◢ 4.2 ◣	198	C LC_135	129.7	0.0	-58.8	-51.1	42.9	32.0	16.6	OK	OK
		◢ 4.2 ◣	198	C LC_135	65.8	0.0	-18.2	25.9	-25.7	16.2	9.5	OK	OK
SLN264 ring-w 1	Flange bottom a end plate SLN264 ring	◢ 4.2 ◣	197	C LC_135	49.4	0.0	21.8	19.5	-16.6	12.2	9.7	OK	OK
		◢ 4.2 ◣	198	C LC_135	89.7	0.0	4.8	-14.3	-49.7	22.2	13.7	OK	OK
Cap plate SLN451 raking strut	Fin plate SLN451 raking strut	◢ 5.6 ◣	139	C LC_113	217.6	0.0	61.3	97.3	71.1	53.8	42.7	OK	OK
		◢ 5.6 ◣	139	C LC_116	130.3	0.0	64.6	-54.9	-35.4	32.2	19.1	OK	OK
Cap plate SLN451 raking strut	Fin plate 2 SLN451 raking strut	◢ 5.6 ◣	139	C LC_111	113.0	0.0	-41.4	-51.2	32.7	27.9	15.8	OK	OK
		◢ 5.6 ◣	139	C LC_125	215.1	0.0	-106.3	84.5	-67.1	53.1	41.6	OK	OK
End plate column below SLN39 column below	SLN39 column below	◢ 5.6 ◣	521	C LC_103	316.7	0.0	-186.7	-147.6	6.0	78.3	43.2	OK	OK
		◢ 5.6 ◣	521	C LC_152	283.1	0.0	-146.1	139.1	16.2	70.0	41.0	OK	OK
SLN184+SLN183 radial (continuous)-w 1	SLN185 ring-w 1	◢ 7.2	189	C LC_120	135.6	0.0	35.8	7.0	75.2	33.5	21.5	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
SLN184+SLN183 radial (continuous)-w 1	SLN264 ring-w 1	◢ 7.2	189	C LC_120	138.0	0.0	70.5	52.3	-44.2	34.1	20.6	OK	OK
End plate SLN185 ring	SLN185 ring-w 1	◢ 4.2 ◣	149	C LC_135	140.5	0.0	-61.4	-32.1	-65.5	34.7	19.4	OK	OK
		◢ 4.2 ◣	149	C LC_135	93.6	0.0	-19.4	40.8	33.6	23.1	12.6	OK	OK
End plate SLN264 ring	SLN264 ring-w 1	◢ 4.2 ◣	149	C LC_135	88.4	0.0	-21.7	-48.2	10.9	21.8	14.3	OK	OK
		◢ 4.2 ◣	149	C LC_135	69.2	0.0	-38.4	18.6	-27.5	17.1	10.9	OK	OK
SLN185 ring-w 1	Flange top a end plate SLN185 ring	◢ 4.2 ◣	199	C LC_135	293.9	0.0	-24.2	13.4	-168.6	72.6	46.7	OK	OK
		◢ 4.2 ◣	198	C LC_135	215.5	0.0	32.8	35.4	-117.8	53.2	30.0	OK	OK
SLN185 ring-w 1	Flange bottom a end plate SLN185 ring	◢ 4.2 ◣	198	C LC_135	231.3	0.0	42.6	-44.5	123.5	57.2	31.9	OK	OK
		◢ 4.2 ◣	199	C LC_135	243.2	0.0	-36.9	-16.4	137.8	60.1	35.4	OK	OK
SLN264 ring-w 1	Flange top b end plate SLN264 ring	◢ 4.2 ◣	199	C LC_135	118.9	0.0	8.3	-2.9	-68.4	29.4	20.3	OK	OK
		◢ 4.2 ◣	198	C LC_135	103.5	0.0	-16.2	-18.2	-56.2	25.6	14.5	OK	OK
SLN264 ring-w 1	Flange bottom b end plate SLN264 ring	◢ 4.2 ◣	199	C LC_135	98.8	0.0	9.5	-51.0	-24.9	24.4	14.1	OK	OK
		◢ 4.2 ◣	199	C LC_135	167.7	0.0	80.4	38.9	75.5	41.4	29.6	OK	OK
End plate SLN185 ring	Flange top a end plate SLN185 ring	◢ 4.2 ◣	58	C LC_135	273.4	0.0	-129.8	107.9	87.4	67.5	32.3	OK	OK
		◢ 4.2 ◣	58	C LC_135	225.3	0.0	-74.3	-96.2	-76.3	55.7	27.8	OK	OK
End plate SLN185 ring	Flange top b end plate SLN185 ring	◢ 4.2 ◣	58	C LC_135	50.2	0.0	6.3	-0.3	-28.7	12.4	9.2	OK	OK
		◢ 4.2 ◣	58	C LC_135	24.0	0.0	1.1	7.1	11.9	5.9	5.9	OK	OK
End plate SLN185 ring	Flange bottom a end plate SLN185 ring	◢ 4.2 ◣	58	C LC_135	184.1	0.0	-53.9	83.1	58.5	45.5	25.7	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_135	260.5	0.0	-130.7	-101.5	-81.4	64.4	32.8	OK	OK
End plate SLN185 ring	Flange bottom b end plate SLN185 ring	◢ 4.2 ◣	58	C LC_135	70.7	0.0	13.7	-34.1	-21.0	17.5	11.3	OK	OK
		◢ 4.2 ◣	58	C LC_135	63.0	0.0	1.4	-4.7	36.1	15.6	10.0	OK	OK
End plate SLN264 ring	Flange top a end plate SLN264 ring	◢ 4.2 ◣	58	C LC_135	220.4	0.0	-110.8	95.2	55.1	54.5	27.8	OK	OK
		◢ 4.2 ◣	58	C LC_135	185.8	0.0	-70.0	-85.7	-50.3	45.9	24.8	OK	OK
End plate SLN264 ring	Flange top b end plate SLN264 ring	◢ 4.2 ◣	58	C LC_135	25.6	0.0	9.8	-5.3	-12.6	6.3	6.3	OK	OK
		◢ 4.2 ◣	58	C LC_135	41.3	0.0	-4.2	-3.1	23.5	10.2	8.4	OK	OK
End plate SLN264 ring	Flange bottom a end plate SLN264 ring	◢ 4.2 ◣	58	C LC_135	129.5	0.0	-59.6	58.7	30.9	32.0	18.8	OK	OK
		◢ 4.2 ◣	58	C LC_135	178.1	0.0	-85.7	-86.6	-24.9	44.0	23.0	OK	OK
End plate SLN264 ring	Flange bottom b end plate SLN264 ring	◢ 4.2 ◣	58	C LC_135	70.4	0.0	11.2	-24.4	-31.8	17.4	9.8	OK	OK
		◢ 4.2 ◣	58	C LC_104	33.0	0.0	5.6	3.3	18.5	8.2	6.3	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-S_100
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN193+SLN191 radial (continuous) - SLN43 column below - SLN192 ring - SLN195 ring - SLN446 raking strut

Name	SLN193+SLN191 radial (continuous) - SLN43 column below - SLN192 ring - SLN195 ring - SLN446 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

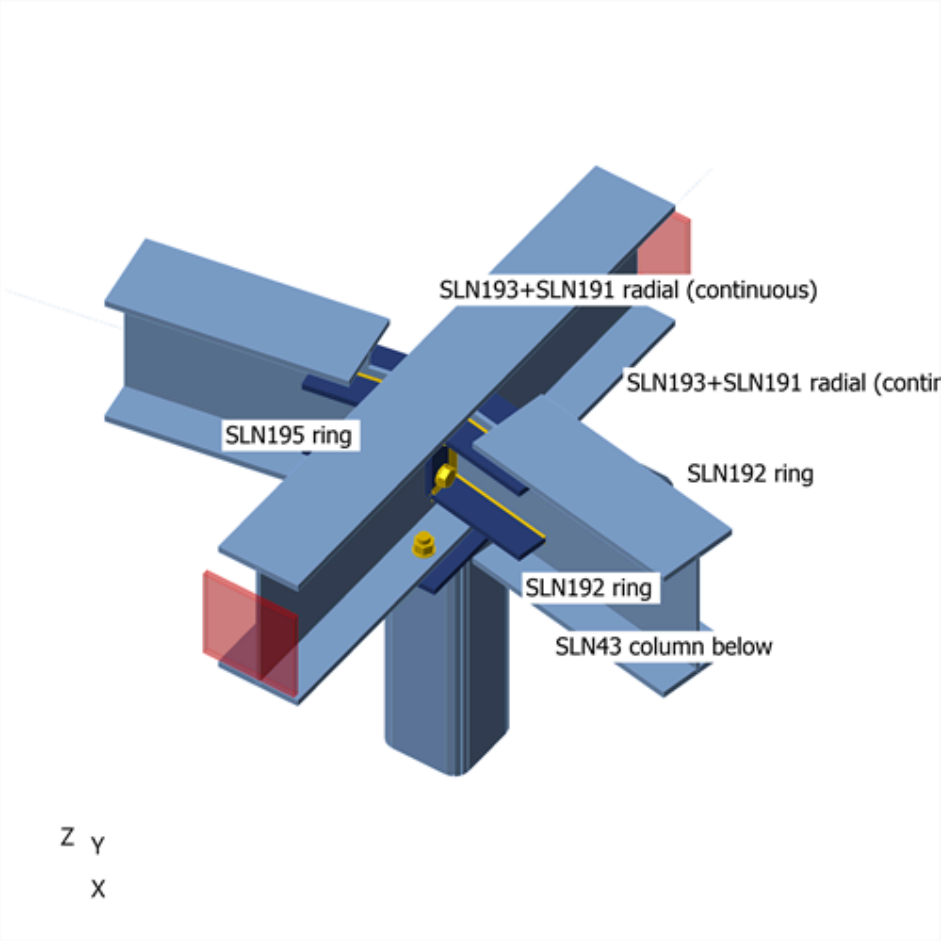
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN193+SLN191 radial (continuous)	1 - UB254x146x43	-80.0	0.0	0.0	0	0	0
SLN43 column below	2 - SHS150x150x10	0.0	90.0	-80.0	0	0	0
SLN192 ring	1 - UB254x146x43	180.0	0.0	0.0	0	0	0
SLN195 ring	1 - UB254x146x43	-160.0	0.0	0.0	0	0	0
SLN446 raking strut	2 - SHS150x150x10	100.1	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN193+SLN191 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN193+SLN191 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN43 column below / begin		Position	0
SLN192 ring / begin		Position	0
SLN195 ring / end		Position	0
SLN446 raking strut / end		Position	0



SLN193+SLN191 radial (continuous) - SLN43 column below - SLN192 ring - SLN195 ring - SLN446 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN43 column below / Begin	26.6	0.0	0.2	0.0	0.6	0.0
	SLN192 ring / Begin	-8.8	-0.1	-4.9	0.0	0.0	0.2
	SLN195 ring / End	8.9	-0.1	-4.9	0.0	0.0	-0.1
	SLN446 raking strut / End	-2.6	0.0	-1.0	0.0	0.0	0.0
C LC_102	SLN43 column below / Begin	124.9	0.0	0.3	0.0	1.3	0.1
	SLN192 ring / Begin	-27.8	-0.2	-25.6	0.0	0.0	0.6
	SLN195 ring / End	27.9	-0.2	-25.7	0.0	0.0	-0.5
	SLN446 raking strut / End	-21.7	0.0	-1.0	0.0	0.0	0.0
C LC_103	SLN43 column below / Begin	119.0	2.4	-0.2	0.0	0.9	-0.6
	SLN192 ring / Begin	-20.1	-0.1	-24.6	0.0	0.0	0.4
	SLN195 ring / End	23.0	-0.1	-24.7	0.0	0.0	-0.5
	SLN446 raking strut / End	-21.8	-0.1	-1.0	-0.1	0.0	0.7
C LC_104	SLN43 column below / Begin	115.2	0.0	0.3	0.0	1.2	-0.1
	SLN192 ring / Begin	-19.8	-0.2	-24.6	0.0	0.0	0.6
	SLN195 ring / End	19.5	-0.2	-24.7	0.0	0.0	-0.5
	SLN446 raking strut / End	-27.0	0.2	-1.0	0.1	0.0	0.2
C LC_105	SLN43 column below / Begin	134.2	-0.4	-2.4	0.0	-0.7	0.2
	SLN192 ring / Begin	-29.0	0.0	-24.6	0.0	0.0	0.3
	SLN195 ring / End	28.5	0.0	-24.7	0.0	0.0	-0.4
	SLN446 raking strut / End	-1.4	0.0	-1.2	0.1	0.0	-0.1
C LC_106	SLN43 column below / Begin	106.3	0.0	0.3	0.0	1.2	0.1
	SLN192 ring / Begin	-17.8	-0.3	-24.6	0.0	0.0	0.7
	SLN195 ring / End	18.1	-0.3	-24.7	0.0	0.0	-0.6
	SLN446 raking strut / End	-38.7	-0.1	-0.9	0.0	0.0	0.0
C LC_107	SLN43 column below / Begin	9.5	4.8	-0.9	0.0	-0.4	-1.3
	SLN192 ring / Begin	2.9	0.0	-1.6	0.0	0.0	-0.1
	SLN195 ring / End	2.1	0.0	-1.6	0.0	0.0	-0.1
	SLN446 raking strut / End	-0.4	0.0	-0.8	-0.3	0.0	1.4
C LC_108	SLN43 column below / Begin	2.8	0.1	0.1	0.0	0.3	-0.3
	SLN192 ring / Begin	3.3	0.0	-1.6	0.0	0.0	0.1
	SLN195 ring / End	-3.7	0.0	-1.6	0.0	0.0	0.0
	SLN446 raking strut / End	-9.4	0.5	-0.7	0.0	0.0	0.4
C LC_109	SLN43 column below / Begin	35.6	-0.8	-5.4	0.0	-3.5	0.2
	SLN192 ring / Begin	-12.8	0.1	-1.6	0.0	0.0	-0.3
	SLN195 ring / End	11.7	0.0	-1.6	0.0	0.0	0.0
	SLN446 raking strut / End	34.7	0.0	-1.1	0.1	0.0	-0.2
C LC_110	SLN43 column below / Begin	-10.7	0.0	0.1	0.0	0.3	0.1
	SLN192 ring / Begin	5.2	-0.1	-1.6	0.0	0.0	0.2
	SLN195 ring / End	-5.0	-0.1	-1.6	0.0	0.0	-0.2
	SLN446 raking strut / End	-27.1	-0.1	-0.4	0.0	0.0	-0.1
C LC_111	SLN43 column below / Begin	84.6	4.8	-0.7	0.0	0.3	-1.3
	SLN192 ring / Begin	-10.6	-0.1	-17.3	0.0	0.0	0.2
	SLN195 ring / End	15.8	-0.1	-17.4	0.0	0.0	-0.4
	SLN446 raking strut / End	-15.1	-0.1	-1.1	-0.3	0.0	1.4
C LC_112	SLN43 column below / Begin	77.7	0.0	0.2	0.0	0.9	-0.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN192 ring / Begin	-10.1	-0.2	-17.3	0.0	0.0	0.5
	SLN195 ring / End	9.7	-0.1	-17.4	0.0	0.0	-0.3
	SLN446 raking strut / End	-24.5	0.4	-1.0	0.1	0.0	0.4
C LC_113	SLN43 column below / Begin	112.1	-0.9	-5.2	0.0	-2.9	0.3
	SLN192 ring / Begin	-27.0	0.1	-17.3	0.0	0.0	0.0
	SLN195 ring / End	25.9	0.0	-17.4	0.0	0.0	-0.2
	SLN446 raking strut / End	21.9	0.0	-1.3	0.1	0.0	-0.2
C LC_114	SLN43 column below / Begin	62.8	0.0	0.2	0.0	1.0	0.1
	SLN192 ring / Begin	-7.5	-0.3	-17.3	0.0	0.0	0.6
	SLN195 ring / End	7.9	-0.2	-17.4	0.0	0.0	-0.5
	SLN446 raking strut / End	-44.1	-0.1	-0.7	0.0	0.0	-0.1
C LC_115	SLN43 column below / Begin	16.4	4.8	-0.8	0.0	-0.2	-1.3
	SLN192 ring / Begin	0.6	0.0	-2.8	0.0	0.0	0.0
	SLN195 ring / End	4.4	0.0	-2.9	0.0	0.0	-0.1
	SLN446 raking strut / End	-1.0	-0.1	-1.1	-0.3	0.0	1.4
C LC_116	SLN43 column below / Begin	9.7	0.1	0.1	0.0	0.4	-0.3
	SLN192 ring / Begin	1.0	-0.1	-2.8	0.0	0.0	0.2
	SLN195 ring / End	-1.4	0.0	-2.9	0.0	0.0	-0.1
	SLN446 raking strut / End	-10.1	0.5	-1.0	0.0	0.0	0.4
C LC_117	SLN43 column below / Begin	42.6	-0.9	-5.3	0.0	-3.3	0.2
	SLN192 ring / Begin	-15.1	0.1	-2.8	0.0	0.0	-0.2
	SLN195 ring / End	14.1	0.0	-2.9	0.0	0.0	0.0
	SLN446 raking strut / End	34.3	0.0	-1.3	0.1	0.0	-0.2
C LC_118	SLN43 column below / Begin	-3.9	0.0	0.1	0.0	0.5	0.1
	SLN192 ring / Begin	3.0	-0.1	-2.8	0.0	0.0	0.3
	SLN195 ring / End	-2.7	-0.1	-2.9	0.0	0.0	-0.2
	SLN446 raking strut / End	-28.0	-0.1	-0.7	0.0	0.0	-0.1
C LC_119	SLN43 column below / Begin	123.3	2.4	-0.2	0.0	0.9	-0.6
	SLN192 ring / Begin	-22.4	-0.2	-25.6	0.0	0.0	0.5
	SLN195 ring / End	25.2	-0.2	-25.7	0.0	0.0	-0.5
	SLN446 raking strut / End	-23.7	-0.1	-1.0	-0.1	0.0	0.7
C LC_120	SLN43 column below / Begin	120.6	0.0	0.3	0.0	1.2	-0.1
	SLN192 ring / Begin	-22.0	-0.2	-25.6	0.0	0.0	0.6
	SLN195 ring / End	21.7	-0.2	-25.7	0.0	0.0	-0.5
	SLN446 raking strut / End	-27.3	0.2	-1.0	0.1	0.0	0.2
C LC_121	SLN43 column below / Begin	137.4	-0.4	-2.4	0.0	-0.6	0.2
	SLN192 ring / Begin	-31.4	0.0	-25.6	0.0	0.0	0.3
	SLN195 ring / End	30.9	-0.1	-25.7	0.0	0.0	-0.5
	SLN446 raking strut / End	-4.8	0.0	-1.2	0.1	0.0	-0.1
C LC_122	SLN43 column below / Begin	114.5	0.0	0.3	0.0	1.2	0.1
	SLN192 ring / Begin	-19.9	-0.3	-25.6	0.0	0.0	0.7
	SLN195 ring / End	20.2	-0.3	-25.7	0.0	0.0	-0.6
	SLN446 raking strut / End	-35.2	-0.1	-0.9	0.0	0.0	0.0
C LC_123	SLN43 column below / Begin	93.3	4.8	-0.7	0.0	0.4	-1.3
	SLN192 ring / Begin	-15.1	-0.1	-19.4	0.0	0.0	0.3
	SLN195 ring / End	20.3	-0.1	-19.5	0.0	0.0	-0.5
	SLN446 raking strut / End	-18.7	-0.1	-1.1	-0.3	0.0	1.5

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_124	SLN43 column below / Begin	88.5	0.0	0.2	0.0	1.0	-0.2
	SLN192 ring / Begin	-14.6	-0.2	-19.4	0.0	0.0	0.5
	SLN195 ring / End	14.1	-0.2	-19.5	0.0	0.0	-0.4
	SLN446 raking strut / End	-25.1	0.4	-1.0	0.1	0.0	0.4
C LC_125	SLN43 column below / Begin	118.5	-0.9	-5.2	0.0	-2.7	0.3
	SLN192 ring / Begin	-31.7	0.0	-19.4	0.0	0.0	0.1
	SLN195 ring / End	30.7	-0.1	-19.5	0.0	0.0	-0.4
	SLN446 raking strut / End	15.1	0.0	-1.3	0.1	0.0	-0.2
C LC_126	SLN43 column below / Begin	79.2	0.0	0.3	0.0	1.0	0.1
	SLN192 ring / Begin	-11.7	-0.3	-19.4	0.0	0.0	0.6
	SLN195 ring / End	12.2	-0.2	-19.5	0.0	0.0	-0.5
	SLN446 raking strut / End	-37.1	-0.1	-0.7	0.0	0.0	-0.1
C LC_127	SLN43 column below / Begin	25.0	4.8	-0.8	0.0	-0.1	-1.3
	SLN192 ring / Begin	-3.9	0.0	-4.9	0.0	0.0	0.1
	SLN195 ring / End	8.9	-0.1	-4.9	0.0	0.0	-0.2
	SLN446 raking strut / End	-4.6	-0.1	-1.1	-0.3	0.0	1.5
C LC_128	SLN43 column below / Begin	20.6	0.1	0.1	0.0	0.5	-0.2
	SLN192 ring / Begin	-3.5	-0.1	-4.9	0.0	0.0	0.2
	SLN195 ring / End	3.1	-0.1	-4.9	0.0	0.0	-0.1
	SLN446 raking strut / End	-10.7	0.5	-1.0	0.0	0.0	0.4
C LC_129	SLN43 column below / Begin	49.0	-0.9	-5.3	0.0	-3.1	0.2
	SLN192 ring / Begin	-19.9	0.1	-4.9	0.0	0.0	-0.1
	SLN195 ring / End	18.8	-0.1	-4.9	0.0	0.0	-0.2
	SLN446 raking strut / End	27.4	0.0	-1.3	0.1	0.0	-0.2
C LC_130	SLN43 column below / Begin	12.5	0.0	0.1	0.0	0.6	0.1
	SLN192 ring / Begin	-1.2	-0.1	-4.9	0.0	0.0	0.3
	SLN195 ring / End	1.6	-0.1	-4.9	0.0	0.0	-0.2
	SLN446 raking strut / End	-21.0	-0.1	-0.7	0.0	0.0	-0.1
C LC_135	SLN43 column below / Begin	124.5	0.0	0.3	0.0	1.3	0.1
	SLN192 ring / Begin	-28.3	-0.2	-25.6	0.0	0.0	0.6
	SLN195 ring / End	28.4	-0.2	-25.7	0.0	0.0	-0.5
	SLN446 raking strut / End	-22.1	0.0	-1.0	0.0	0.0	0.0
C LC_141	SLN43 column below / Begin	26.6	0.0	0.2	0.0	0.6	0.0
	SLN192 ring / Begin	-8.9	-0.1	-4.9	0.0	0.0	0.2
	SLN195 ring / End	8.9	0.0	-4.9	0.0	0.0	-0.1
	SLN446 raking strut / End	-2.5	0.0	-1.0	0.0	0.0	0.0
C LC_142	SLN43 column below / Begin	26.5	0.0	0.2	0.0	0.6	0.0
	SLN192 ring / Begin	-8.8	-0.1	-4.9	0.0	0.0	0.2
	SLN195 ring / End	8.8	-0.1	-4.9	0.0	0.0	-0.2
	SLN446 raking strut / End	-2.7	0.0	-1.0	0.0	0.0	0.0
C LC_143	SLN43 column below / Begin	27.2	0.0	0.1	0.0	0.6	0.0
	SLN192 ring / Begin	-9.1	0.0	-4.9	0.0	0.0	0.1
	SLN195 ring / End	9.1	0.0	-4.9	0.0	0.0	-0.1
	SLN446 raking strut / End	-1.7	0.0	-1.0	0.0	0.0	0.0
C LC_144	SLN43 column below / Begin	25.9	0.0	0.2	0.0	0.6	0.0
	SLN192 ring / Begin	-8.6	-0.1	-4.9	0.0	0.0	0.2
	SLN195 ring / End	8.6	-0.1	-4.9	0.0	0.0	-0.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN446 raking strut / End	-3.5	0.0	-1.0	0.0	0.0	0.0
C LC_145	SLN43 column below / Begin	125.2	0.0	0.3	0.0	1.3	0.1
	SLN192 ring / Begin	-27.9	-0.2	-25.6	0.0	0.0	0.5
	SLN195 ring / End	28.3	-0.2	-25.7	0.0	0.0	-0.5
	SLN446 raking strut / End	-21.3	0.0	-1.0	0.0	0.0	0.0
C LC_146	SLN43 column below / Begin	124.7	0.0	0.3	0.0	1.3	0.1
	SLN192 ring / Begin	-27.7	-0.2	-25.6	0.0	0.0	0.6
	SLN195 ring / End	27.6	-0.2	-25.7	0.0	0.0	-0.6
	SLN446 raking strut / End	-22.0	0.0	-1.0	0.1	0.0	0.0
C LC_147	SLN43 column below / Begin	127.6	0.0	0.3	0.0	1.4	0.1
	SLN192 ring / Begin	-29.0	-0.1	-25.6	0.0	0.0	0.5
	SLN195 ring / End	29.1	-0.1	-25.7	0.0	0.0	-0.5
	SLN446 raking strut / End	-18.1	0.0	-1.0	0.0	0.0	0.0
C LC_148	SLN43 column below / Begin	122.3	0.0	0.3	0.0	1.3	0.1
	SLN192 ring / Begin	-26.6	-0.3	-25.6	0.0	0.0	0.7
	SLN195 ring / End	26.8	-0.3	-25.7	0.0	0.0	-0.6
	SLN446 raking strut / End	-25.2	0.0	-1.0	0.0	0.0	0.0
C LC_151	SLN43 column below / Begin	80.0	0.0	-1.7	0.0	-6.7	0.0
	SLN192 ring / Begin	-12.4	-0.1	-11.9	0.0	0.0	0.3
	SLN195 ring / End	12.5	-0.1	-11.8	0.0	0.0	-0.3
	SLN446 raking strut / End	-13.1	0.0	-1.0	0.0	0.0	0.0
C LC_152	SLN43 column below / Begin	73.6	0.0	2.2	0.0	8.8	0.0
	SLN192 ring / Begin	-28.6	-0.2	-14.2	0.0	0.0	0.5
	SLN195 ring / End	28.7	-0.2	-14.1	0.0	0.0	-0.5
	SLN446 raking strut / End	-8.3	0.0	-1.0	0.0	0.0	0.0
C LC_153	SLN43 column below / Begin	125.3	0.0	0.3	0.0	1.2	0.1
	SLN192 ring / Begin	-27.9	-0.2	-21.2	0.0	0.0	0.6
	SLN195 ring / End	28.1	-0.2	-21.0	0.0	0.0	-0.6
	SLN446 raking strut / End	-21.6	0.0	-1.0	0.0	0.0	0.0
C LC_154	SLN43 column below / Begin	79.6	0.0	-1.6	0.0	-6.6	0.0
	SLN192 ring / Begin	-12.5	-0.1	-11.9	0.0	0.0	0.3
	SLN195 ring / End	12.6	-0.1	-11.8	0.0	0.0	-0.3
	SLN446 raking strut / End	-12.9	0.0	-1.0	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.1 < 5.0%	OK
Loc. deformation	0.1 < 3%	OK
Bolts	16.9 < 100%	OK
Welds	98.1 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN193+SLN191 radial (continuous)-bfl 1	12.7	C LC_122	141.1	0.0	47.2	OK
SLN193+SLN191 radial (continuous)-tfl 1	12.7	C LC_153	72.5	0.0	0.0	OK
SLN193+SLN191 radial (continuous)-w 1	7.2	C LC_122	169.8	0.0	47.2	OK
SLN43 column below	10.0	C LC_114	275.1	0.1	0.0	OK
SLN192 ring-bfl 1	12.7	C LC_122	49.3	0.0	0.0	OK
SLN192 ring-tfl 1	12.7	C LC_122	44.3	0.0	0.0	OK
SLN192 ring-w 1	7.2	C LC_122	150.8	0.0	0.0	OK
SLN195 ring-bfl 1	12.7	C LC_148	46.7	0.0	0.0	OK
SLN195 ring-tfl 1	12.7	C LC_148	40.7	0.0	0.0	OK
SLN195 ring-w 1	7.2	C LC_148	145.6	0.0	0.0	OK
SLN446 raking strut	10.0	C LC_123	45.8	0.0	0.0	OK
End plate column below SLN43 column below	12.0	C LC_109	158.3	0.0	21.4	OK
End plate SLN192 ring	12.0	C LC_122	68.8	0.0	0.0	OK
End plate SLN195 ring	12.0	C LC_148	58.9	0.0	9.3	OK
Cap plate SLN446 raking strut	12.0	C LC_123	94.7	0.0	0.0	OK
Fin plate SLN446 raking strut	12.0	C LC_115	255.1	0.0	85.0	OK
Fin plate 2 SLN446 raking strut	12.0	C LC_123	255.7	0.0	60.0	OK
Gusset SLN446 raking strut	12.0	C LC_123	275.0	0.0	43.7	OK
Flange top a end plate SLN192 ring	12.0	C LC_122	58.4	0.0	0.0	OK
Flange top b end plate SLN192 ring	12.0	C LC_122	46.2	0.0	0.0	OK
Flange bottom a end plate SLN192 ring	12.0	C LC_122	43.3	0.0	0.0	OK
Flange bottom b end plate SLN192 ring	12.0	C LC_148	70.9	0.0	0.0	OK
Flange top a end plate SLN195 ring	12.0	C LC_148	41.7	0.0	0.0	OK
Flange top b end plate SLN195 ring	12.0	C LC_106	53.0	0.0	0.0	OK
Flange bottom a end plate SLN195 ring	12.0	C LC_148	68.7	0.0	0.0	OK
Flange bottom b end plate SLN195 ring	12.0	C LC_148	38.9	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

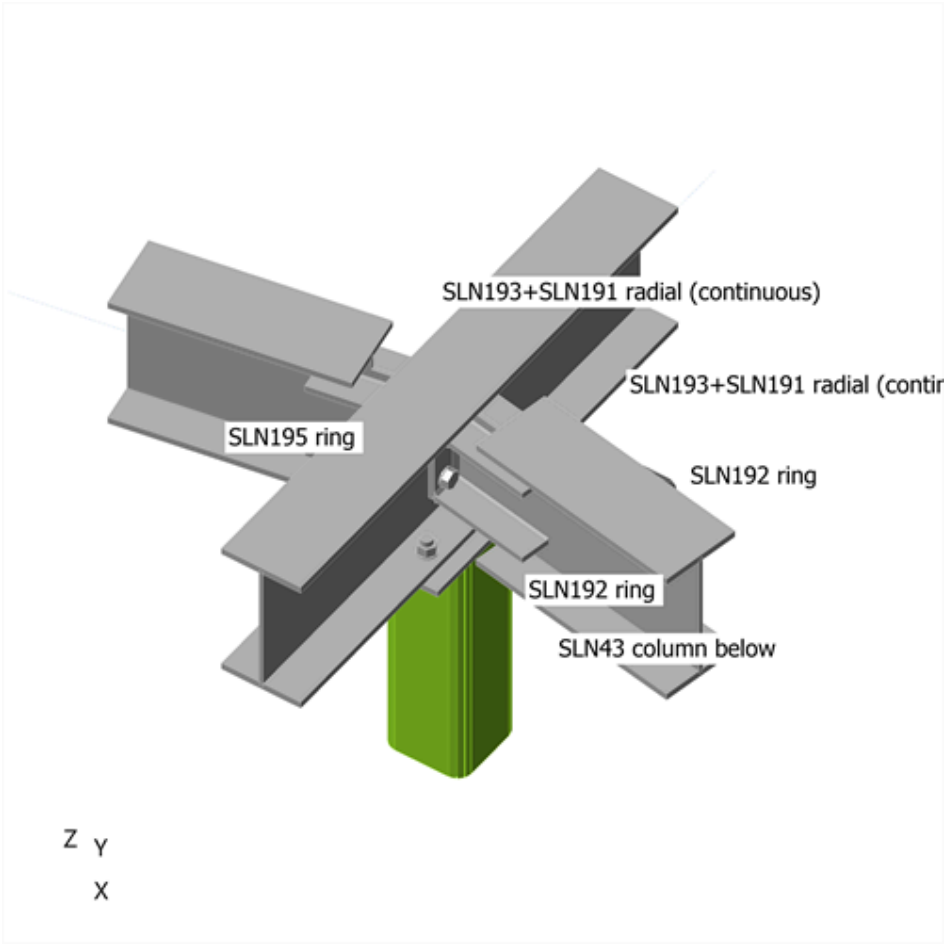
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

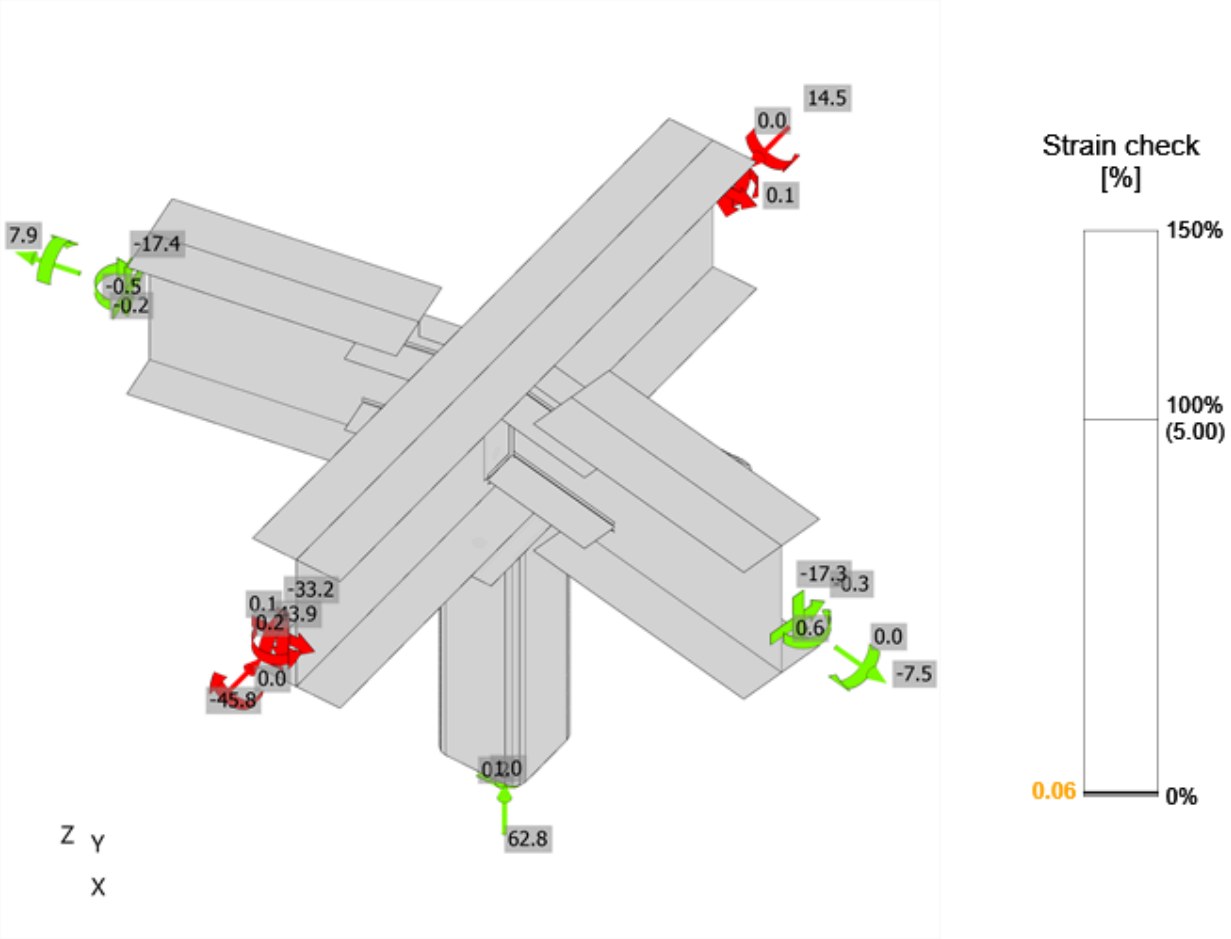
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN43 column below	150	C LC_114	0	5	0.1	OK
SLN446 raking strut	150	C LC_154	0	5	0.0	OK

Symbol explanation

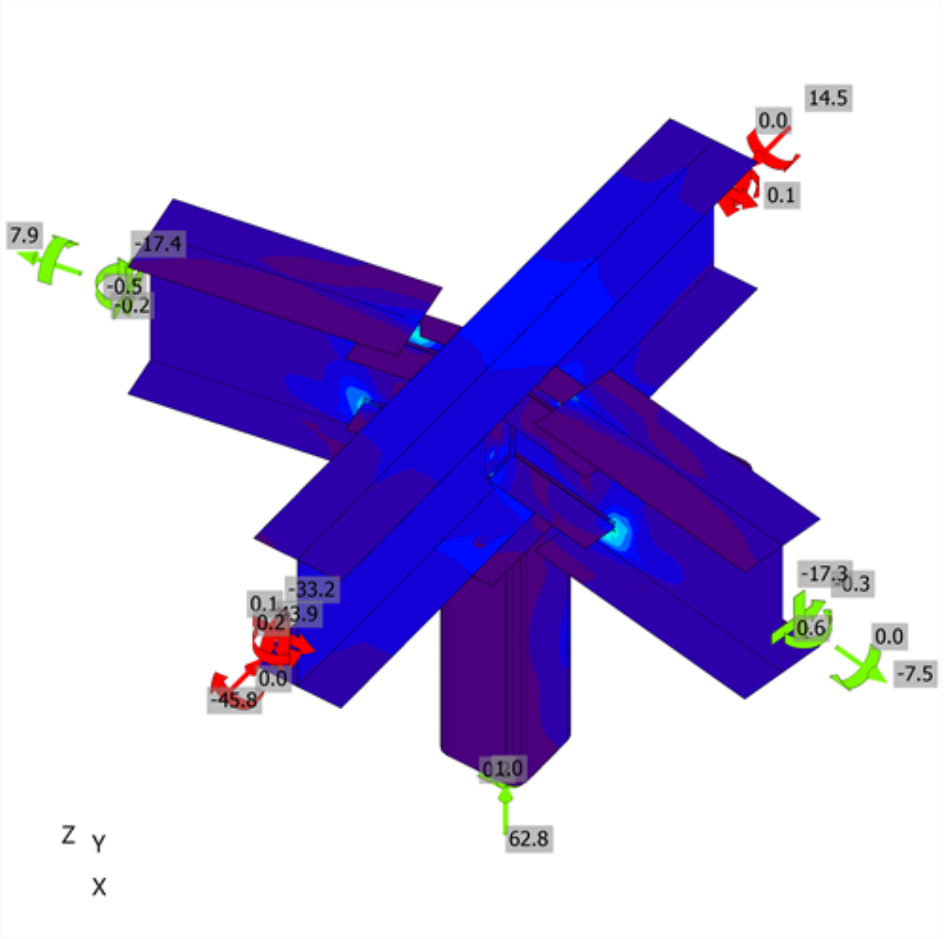
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

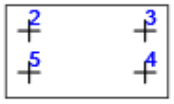
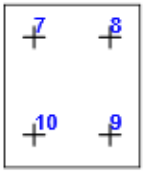
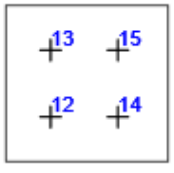


Strain check, C LC_114



Equivalent stress, C LC_114

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_152	15.4	2.6	111.1	10.9	2.7	10.5	OK	OK
	B3	M20 8.8 - 1	C LC_109	22.6	4.0	102.6	16.0	4.2	15.7	OK	OK
	B4	M20 8.8 - 1	C LC_109	23.0	4.2	109.1	16.3	4.4	16.1	OK	OK
	B5	M20 8.8 - 1	C LC_152	15.0	2.6	111.7	10.6	2.7	10.3	OK	OK
	B7	M20 8.8 - 1	C LC_122	9.7	0.7	123.8	6.9	0.8	5.7	OK	OK
	B8	M20 8.8 - 1	C LC_109	3.5	0.5	123.8	2.5	0.6	2.3	OK	OK
	B9	M20 8.8 - 1	C LC_109	3.5	0.5	81.2	2.5	0.6	2.3	OK	OK
	B10	M20 8.8 - 1	C LC_122	10.6	0.9	123.8	7.5	0.9	6.3	OK	OK
	B12	M20 8.8 - 1	C LC_123	11.1	10.6	115.3	7.9	11.2	16.9	OK	OK
	B13	M20 8.8 - 1	C LC_123	5.0	9.9	115.3	3.6	10.5	13.0	OK	OK
	B14	M20 8.8 - 1	C LC_117	0.8	6.3	106.0	0.5	10.5	7.1	OK	OK
	B15	M20 8.8 - 1	C LC_114	0.1	6.0	115.3	0.1	10.4	6.5	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN446 raking strut	SLN446 raking strut	◢ 4.2	519	C LC_123	137.0	0.0	-59.7	70.7	-8.7	33.9	19.0	OK	OK
SLN43 column below-w 3	Gusset SLN446 raking strut	◢ 5.6 ◣	288	C LC_114	397.0	0.2	-130.2	-130.6	-172.7	98.1	60.4	OK	OK
		◢ 5.6 ◣	288	C LC_114	397.0	0.2	-130.0	129.6	173.5	98.1	59.9	OK	OK
SLN192 ring-w 1	Flange top b end plate SLN192 ring	◢ 4.2 ◣	198	C LC_148	54.9	0.0	30.5	24.8	9.0	13.6	9.0	OK	OK
		◢ 4.2 ◣	197	C LC_122	31.9	0.0	-15.3	14.1	7.8	7.9	6.8	OK	OK
SLN192 ring-w 1	Flange bottom b end plate SLN192 ring	◢ 4.2 ◣	198	C LC_148	91.9	0.0	27.1	31.5	39.8	22.7	13.9	OK	OK
		◢ 4.2 ◣	198	C LC_148	131.0	0.0	51.8	-47.4	-50.8	32.4	19.9	OK	OK
SLN195 ring-w 1	Flange top a end plate SLN195 ring	◢ 4.2 ◣	198	C LC_152	52.5	0.0	25.9	22.3	-14.1	13.0	7.6	OK	OK
		◢ 4.2 ◣	197	C LC_106	28.3	0.0	-13.7	12.7	-6.5	7.0	6.2	OK	OK
SLN195 ring-w 1	Flange bottom a end plate SLN195 ring	◢ 4.2 ◣	198	C LC_148	92.1	0.0	27.3	31.0	-40.2	22.8	12.1	OK	OK
		◢ 4.2 ◣	198	C LC_148	128.7	0.0	50.2	-46.4	50.2	31.8	17.3	OK	OK
Cap plate SLN446 raking strut	Fin plate SLN446 raking strut	◢ 5.6 ◣	139	C LC_115	184.9	0.0	61.2	82.1	58.4	45.7	34.5	OK	OK
		◢ 5.6 ◣	139	C LC_114	122.2	0.0	-48.5	52.6	-37.8	30.2	16.9	OK	OK
Cap plate SLN446 raking strut	Fin plate 2 SLN446 raking strut	◢ 5.6 ◣	139	C LC_114	119.7	0.0	-46.8	-51.4	37.5	29.6	16.8	OK	OK
		◢ 5.6 ◣	139	C LC_123	192.3	0.0	-88.2	77.9	-60.5	47.5	36.5	OK	OK
End plate column below SLN43 column below	SLN43 column below	◢ 5.6 ◣	521	C LC_106	275.0	0.0	-175.9	-120.9	-17.1	67.9	37.4	OK	OK
		◢ 5.6 ◣	521	C LC_152	209.8	0.0	-98.4	106.8	-6.7	51.9	30.7	OK	OK
SLN193+SLN191 radial (continuous)-w 1	SLN192 ring-w 1	◢ 7.2	189	C LC_121	94.7	0.0	35.3	34.1	37.6	23.4	14.9	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
SLN193+SLN191 radial (continuous)-w 1	SLN195 ring-w 1	◢ 7.2	189	C LC_120	93.5	0.0	1.3	6.9	-53.5	23.1	14.4	OK	OK
End plate SLN192 ring	SLN192 ring-w 1	◢ 4.2 ◣	149	C LC_121	85.1	0.0	-27.1	-27.3	-37.7	21.0	11.5	OK	OK
		◢ 4.2 ◣	149	C LC_148	67.9	0.0	-13.5	25.9	28.4	16.8	11.1	OK	OK
End plate SLN195 ring	SLN195 ring-w 1	◢ 4.2 ◣	149	C LC_121	94.1	0.0	-30.4	-27.8	43.3	23.3	14.1	OK	OK
		◢ 4.2 ◣	149	C LC_148	72.9	0.0	-19.8	25.0	-31.9	18.0	11.8	OK	OK
SLN192 ring-w 1	Flange top a end plate SLN192 ring	◢ 4.2 ◣	199	C LC_122	98.0	0.0	-11.7	7.5	-55.7	24.2	17.2	OK	OK
		◢ 4.2 ◣	198	C LC_122	63.7	0.0	11.1	12.5	-34.0	15.7	9.4	OK	OK
SLN192 ring-w 1	Flange bottom a end plate SLN192 ring	◢ 4.2 ◣	198	C LC_122	92.3	0.0	16.7	-16.6	49.7	22.8	13.6	OK	OK
		◢ 4.2 ◣	199	C LC_121	73.0	0.0	-9.6	13.2	39.7	18.0	10.0	OK	OK
SLN195 ring-w 1	Flange top b end plate SLN195 ring	◢ 4.2 ◣	199	C LC_106	90.1	0.0	-13.3	8.2	50.8	22.3	14.0	OK	OK
		◢ 4.2 ◣	198	C LC_148	59.5	0.0	10.9	11.8	31.7	14.7	8.8	OK	OK
SLN195 ring-w 1	Flange bottom b end plate SLN195 ring	◢ 4.2 ◣	198	C LC_148	89.4	0.0	15.6	-15.5	-48.4	22.1	13.0	OK	OK
		◢ 4.2 ◣	199	C LC_121	76.7	0.0	-10.4	14.5	-41.4	19.0	11.9	OK	OK
End plate SLN192 ring	Flange top a end plate SLN192 ring	◢ 4.2 ◣	58	C LC_106	29.4	0.0	1.1	1.6	-16.9	7.3	6.6	OK	OK
		◢ 4.2 ◣	58	C LC_109	21.6	0.0	9.3	6.7	9.0	5.3	5.3	OK	OK
End plate SLN192 ring	Flange top b end plate SLN192 ring	◢ 4.2 ◣	58	C LC_122	45.3	0.0	-0.6	-0.6	-26.1	11.2	8.9	OK	OK
		◢ 4.2 ◣	58	C LC_122	55.6	0.0	31.6	21.7	-15.1	13.7	11.4	OK	OK
End plate SLN192 ring	Flange bottom a end plate SLN192 ring	◢ 4.2 ◣	58	C LC_121	56.3	0.0	12.1	-21.9	23.0	13.9	9.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_122	42.3	0.0	-11.0	-17.1	16.3	10.5	8.2	OK	OK
End plate SLN192 ring	Flange bottom b end plate SLN192 ring	◢ 4.2 ◣	58	C LC_122	67.9	0.0	38.5	-29.3	13.6	16.8	13.5	OK	OK
		◢ 4.2 ◣	58	C LC_122	55.5	0.0	-3.5	0.1	31.9	13.7	9.8	OK	OK
End plate SLN195 ring	Flange top a end plate SLN195 ring	◢ 4.2 ◣	58	C LC_148	38.4	0.0	1.0	-2.9	22.0	9.5	8.6	OK	OK
		◢ 4.2 ◣	58	C LC_106	44.7	0.0	5.1	9.0	24.0	11.0	9.2	OK	OK
End plate SLN195 ring	Flange top b end plate SLN195 ring	◢ 4.2 ◣	58	C LC_125	37.2	0.0	-18.6	14.5	-11.6	9.2	7.3	OK	OK
		◢ 4.2 ◣	58	C LC_125	32.7	0.0	-10.3	-14.5	10.6	8.1	6.7	OK	OK
End plate SLN195 ring	Flange bottom a end plate SLN195 ring	◢ 4.2 ◣	58	C LC_148	58.1	0.0	32.3	-25.7	-10.7	14.3	11.7	OK	OK
		◢ 4.2 ◣	58	C LC_148	47.9	0.0	-2.7	1.5	-27.6	11.8	9.5	OK	OK
End plate SLN195 ring	Flange bottom b end plate SLN195 ring	◢ 4.2 ◣	58	C LC_121	59.1	0.0	12.6	-23.0	-24.2	14.6	8.1	OK	OK
		◢ 4.2 ◣	58	C LC_122	42.2	0.0	-11.4	-19.4	-13.2	10.4	7.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-S_140
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN209+SLN197 radial (continuous) - SLN45 column below - SLN198 ring - SLN210 ring - SLN445 raking strut

Name	SLN209+SLN197 radial (continuous) - SLN45 column below - SLN198 ring - SLN210 ring - SLN445 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

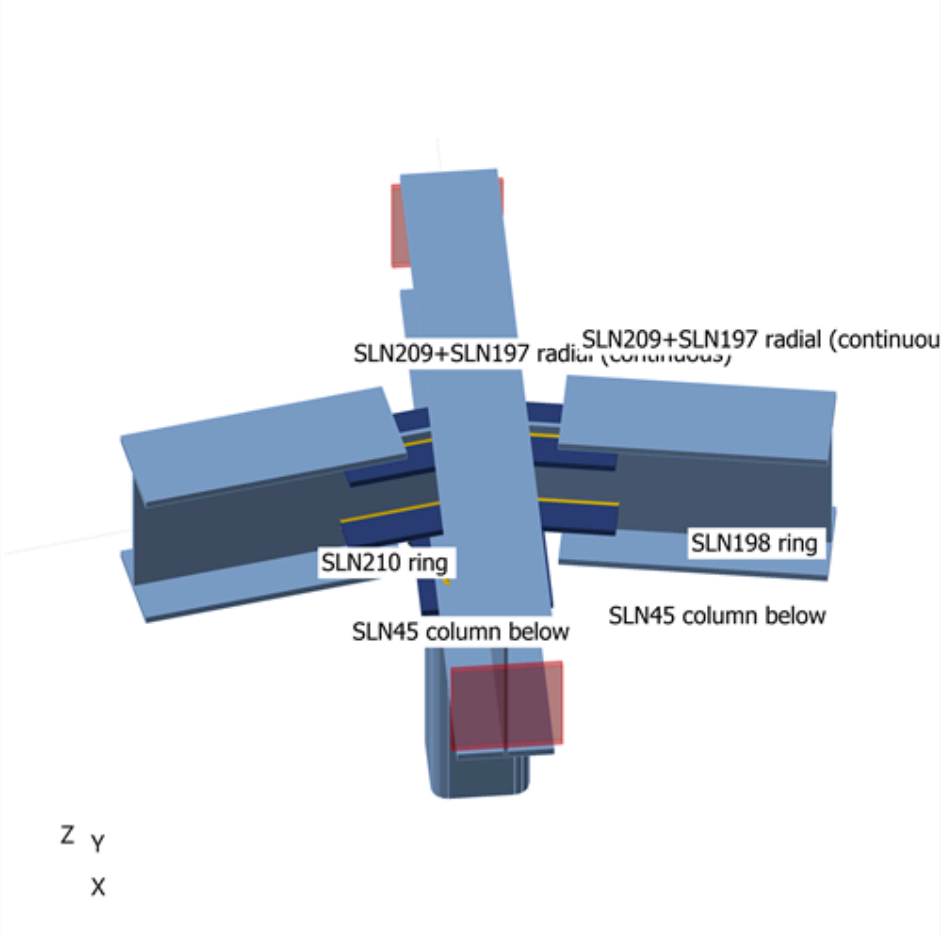
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN209+SLN197 radial (continuous)	1 - UB254x146x43	-40.0	0.0	0.0	0	0	0
SLN45 column below	2 - SHS150x150x10	0.0	90.0	-40.0	0	0	0
SLN198 ring	1 - UB254x146x43	-140.0	0.0	0.0	0	0	0
SLN210 ring	1 - UB254x146x43	-120.0	0.0	0.0	0	0	0
SLN445 raking strut	2 - SHS150x150x10	140.1	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN209+SLN197 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN209+SLN197 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN45 column below / begin		Position	0
SLN198 ring / begin		Position	0
SLN210 ring / end		Position	0
SLN445 raking strut / end		Position	0



SLN209+SLN197 radial (continuous) - SLN45 column below - SLN198 ring - SLN210 ring - SLN445 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN45 column below / Begin	23.8	0.0	0.2	0.0	1.0	0.0
	SLN198 ring / Begin	-8.9	-0.1	-5.0	0.0	0.0	0.2
	SLN210 ring / End	8.9	-0.1	-4.9	0.0	0.0	-0.2
	SLN445 raking strut / End	-4.4	0.0	-1.0	0.0	0.0	0.0
C LC_102	SLN45 column below / Begin	111.5	0.0	0.8	0.0	3.2	0.0
	SLN198 ring / Begin	-28.0	-0.2	-26.0	0.0	0.0	0.7
	SLN210 ring / End	28.0	-0.2	-25.6	0.0	0.0	-0.6
	SLN445 raking strut / End	-30.1	0.0	-1.0	0.0	0.0	0.0
C LC_103	SLN45 column below / Begin	116.1	1.6	-1.4	0.0	1.5	-0.4
	SLN198 ring / Begin	-25.5	-0.1	-25.0	0.0	0.0	0.5
	SLN210 ring / End	27.3	-0.1	-24.6	0.0	0.0	-0.4
	SLN445 raking strut / End	-16.2	0.0	-1.1	-0.1	0.0	0.5
C LC_104	SLN45 column below / Begin	96.6	0.0	0.7	0.0	3.0	-0.1
	SLN198 ring / Begin	-19.0	-0.3	-25.0	0.0	0.0	0.8
	SLN210 ring / End	18.8	-0.3	-24.6	0.0	0.0	-0.6
	SLN445 raking strut / End	-42.6	0.2	-0.9	0.0	0.0	0.2
C LC_105	SLN45 column below / Begin	114.5	-1.9	-1.0	0.0	1.8	0.5
	SLN198 ring / Begin	-27.1	-0.1	-25.0	0.0	0.0	0.5
	SLN210 ring / End	24.9	-0.1	-24.6	0.0	0.0	-0.4
	SLN445 raking strut / End	-18.4	0.0	-1.1	0.1	0.0	-0.5
C LC_106	SLN45 column below / Begin	97.3	0.0	0.7	0.0	3.0	0.1
	SLN198 ring / Begin	-18.5	-0.3	-25.0	0.0	0.0	0.7
	SLN210 ring / End	18.9	-0.3	-24.6	0.0	0.0	-0.6
	SLN445 raking strut / End	-41.6	-0.2	-0.9	0.0	0.0	-0.1
C LC_107	SLN45 column below / Begin	25.8	3.1	-4.1	0.0	-2.4	-0.9
	SLN198 ring / Begin	-6.1	0.0	-1.6	0.0	0.0	-0.1
	SLN210 ring / End	9.4	0.0	-1.6	0.0	0.0	0.1
	SLN445 raking strut / End	22.8	0.0	-1.0	-0.2	0.0	0.9
C LC_108	SLN45 column below / Begin	-6.8	0.0	0.1	0.0	0.5	-0.2
	SLN198 ring / Begin	3.8	-0.1	-1.6	0.0	0.0	0.2
	SLN210 ring / End	-4.1	-0.1	-1.6	0.0	0.0	-0.1
	SLN445 raking strut / End	-21.5	0.3	-0.5	0.0	0.0	0.2
C LC_109	SLN45 column below / Begin	22.8	-3.7	-3.4	0.0	-2.0	1.0
	SLN198 ring / Begin	-9.3	0.0	-1.6	0.0	0.0	-0.1
	SLN210 ring / End	5.3	0.0	-1.6	0.0	0.0	0.0
	SLN445 raking strut / End	18.6	0.1	-1.0	0.3	0.0	-1.1
C LC_110	SLN45 column below / Begin	-5.6	0.0	0.1	0.0	0.4	0.2
	SLN198 ring / Begin	4.8	-0.1	-1.6	0.0	0.0	0.2
	SLN210 ring / End	-4.3	0.0	-1.6	0.0	0.0	-0.1
	SLN445 raking strut / End	-19.8	-0.4	-0.5	0.0	0.0	-0.3
C LC_111	SLN45 column below / Begin	93.7	3.1	-3.7	0.0	-0.7	-0.9
	SLN198 ring / Begin	-20.1	-0.1	-17.6	0.0	0.0	0.3
	SLN210 ring / End	23.5	0.0	-17.3	0.0	0.0	-0.2
	SLN445 raking strut / End	4.2	0.0	-1.3	-0.2	0.0	1.0
C LC_112	SLN45 column below / Begin	58.9	0.0	0.5	0.0	2.2	-0.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN198 ring / Begin	-9.3	-0.3	-17.6	0.0	0.0	0.7
	SLN210 ring / End	8.9	-0.2	-17.3	0.0	0.0	-0.5
	SLN445 raking strut / End	-42.9	0.3	-0.8	0.0	0.0	0.3
C LC_113	SLN45 column below / Begin	90.5	-3.7	-3.0	0.0	-0.2	1.0
	SLN198 ring / Begin	-23.4	0.0	-17.6	0.0	0.0	0.3
	SLN210 ring / End	19.2	-0.1	-17.3	0.0	0.0	-0.3
	SLN445 raking strut / End	-0.1	0.1	-1.2	0.3	0.0	-1.1
C LC_114	SLN45 column below / Begin	60.2	-0.1	0.5	0.0	2.2	0.2
	SLN198 ring / Begin	-8.2	-0.3	-17.6	0.0	0.0	0.6
	SLN210 ring / End	8.9	-0.2	-17.3	0.0	0.0	-0.5
	SLN445 raking strut / End	-41.1	-0.4	-0.8	0.0	0.0	-0.3
C LC_115	SLN45 column below / Begin	32.1	3.1	-4.0	0.0	-2.2	-0.9
	SLN198 ring / Begin	-8.4	0.0	-2.9	0.0	0.0	0.0
	SLN210 ring / End	11.7	0.0	-2.8	0.0	0.0	0.1
	SLN445 raking strut / End	21.8	0.0	-1.3	-0.2	0.0	1.0
C LC_116	SLN45 column below / Begin	-0.8	0.0	0.2	0.0	0.7	-0.2
	SLN198 ring / Begin	1.5	-0.1	-2.9	0.0	0.0	0.3
	SLN210 ring / End	-1.8	-0.1	-2.8	0.0	0.0	-0.2
	SLN445 raking strut / End	-22.8	0.3	-0.8	0.0	0.0	0.2
C LC_117	SLN45 column below / Begin	29.0	-3.7	-3.4	0.0	-1.7	1.0
	SLN198 ring / Begin	-11.7	0.0	-2.9	0.0	0.0	0.0
	SLN210 ring / End	7.6	0.0	-2.8	0.0	0.0	0.0
	SLN445 raking strut / End	17.6	0.1	-1.2	0.3	0.0	-1.1
C LC_118	SLN45 column below / Begin	0.5	0.0	0.2	0.0	0.7	0.2
	SLN198 ring / Begin	2.6	-0.1	-2.9	0.0	0.0	0.3
	SLN210 ring / End	-2.1	-0.1	-2.8	0.0	0.0	-0.2
	SLN445 raking strut / End	-21.1	-0.4	-0.8	0.0	0.0	-0.3
C LC_119	SLN45 column below / Begin	118.8	1.6	-1.3	0.0	1.7	-0.4
	SLN198 ring / Begin	-27.8	-0.1	-26.0	0.0	0.0	0.6
	SLN210 ring / End	29.6	-0.1	-25.6	0.0	0.0	-0.5
	SLN445 raking strut / End	-19.8	0.0	-1.1	-0.1	0.0	0.5
C LC_120	SLN45 column below / Begin	103.5	0.0	0.8	0.0	3.1	-0.1
	SLN198 ring / Begin	-21.2	-0.3	-26.0	0.0	0.0	0.8
	SLN210 ring / End	21.0	-0.3	-25.6	0.0	0.0	-0.6
	SLN445 raking strut / End	-40.6	0.2	-0.9	0.0	0.0	0.2
C LC_121	SLN45 column below / Begin	117.6	-1.9	-1.0	0.0	1.9	0.5
	SLN198 ring / Begin	-29.5	-0.1	-26.0	0.0	0.0	0.6
	SLN210 ring / End	27.3	-0.1	-25.6	0.0	0.0	-0.5
	SLN445 raking strut / End	-21.5	0.0	-1.1	0.1	0.0	-0.5
C LC_122	SLN45 column below / Begin	103.7	0.0	0.8	0.0	3.1	0.1
	SLN198 ring / Begin	-20.7	-0.3	-26.0	0.0	0.0	0.8
	SLN210 ring / End	21.1	-0.3	-25.6	0.0	0.0	-0.6
	SLN445 raking strut / End	-40.2	-0.2	-0.9	0.0	0.0	-0.1
C LC_123	SLN45 column below / Begin	99.1	3.1	-3.6	0.0	-0.4	-0.9
	SLN198 ring / Begin	-24.7	-0.1	-19.7	0.0	0.0	0.4
	SLN210 ring / End	28.2	-0.1	-19.4	0.0	0.0	-0.3
	SLN445 raking strut / End	-3.1	0.0	-1.3	-0.2	0.0	1.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_124	SLN45 column below / Begin	72.7	0.0	0.6	0.0	2.4	-0.2
	SLN198 ring / Begin	-13.7	-0.3	-19.7	0.0	0.0	0.7
	SLN210 ring / End	13.3	-0.2	-19.4	0.0	0.0	-0.5
	SLN445 raking strut / End	-38.8	0.3	-0.8	0.0	0.0	0.3
C LC_125	SLN45 column below / Begin	96.9	-3.8	-2.9	0.0	0.1	1.0
	SLN198 ring / Begin	-28.1	-0.1	-19.7	0.0	0.0	0.4
	SLN210 ring / End	23.9	-0.1	-19.4	0.0	0.0	-0.4
	SLN445 raking strut / End	-6.2	0.1	-1.2	0.3	0.0	-1.1
C LC_126	SLN45 column below / Begin	73.0	0.0	0.6	0.0	2.4	0.2
	SLN198 ring / Begin	-12.6	-0.3	-19.7	0.0	0.0	0.7
	SLN210 ring / End	13.2	-0.2	-19.4	0.0	0.0	-0.5
	SLN445 raking strut / End	-38.4	-0.4	-0.8	-0.1	0.0	-0.3
C LC_127	SLN45 column below / Begin	37.5	3.1	-4.0	0.0	-1.9	-0.9
	SLN198 ring / Begin	-13.0	0.0	-5.0	0.0	0.0	0.2
	SLN210 ring / End	16.4	0.0	-4.9	0.0	0.0	-0.1
	SLN445 raking strut / End	14.5	0.0	-1.3	-0.2	0.0	1.0
C LC_128	SLN45 column below / Begin	13.0	0.0	0.2	0.0	0.9	-0.2
	SLN198 ring / Begin	-2.9	-0.1	-5.0	0.0	0.0	0.3
	SLN210 ring / End	2.6	-0.1	-4.9	0.0	0.0	-0.2
	SLN445 raking strut / End	-18.7	0.3	-0.8	0.0	0.0	0.2
C LC_129	SLN45 column below / Begin	35.4	-3.8	-3.3	0.0	-1.4	1.0
	SLN198 ring / Begin	-16.3	0.0	-5.0	0.0	0.0	0.1
	SLN210 ring / End	12.2	0.0	-4.9	0.0	0.0	-0.2
	SLN445 raking strut / End	11.5	0.1	-1.2	0.3	0.0	-1.1
C LC_130	SLN45 column below / Begin	13.2	0.0	0.2	0.0	0.9	0.2
	SLN198 ring / Begin	-1.8	-0.1	-5.0	0.0	0.0	0.3
	SLN210 ring / End	2.3	-0.1	-4.9	0.0	0.0	-0.2
	SLN445 raking strut / End	-18.4	-0.4	-0.8	0.0	0.0	-0.3
C LC_135	SLN45 column below / Begin	110.9	0.0	0.8	0.0	3.2	0.0
	SLN198 ring / Begin	-28.5	-0.2	-26.0	0.0	0.0	0.7
	SLN210 ring / End	28.5	-0.2	-25.6	0.0	0.0	-0.6
	SLN445 raking strut / End	-30.9	0.0	-1.0	0.0	0.0	0.0
C LC_141	SLN45 column below / Begin	24.2	0.0	0.2	0.0	1.0	0.0
	SLN198 ring / Begin	-9.1	0.0	-5.0	0.0	0.0	0.2
	SLN210 ring / End	9.1	0.0	-4.9	0.0	0.0	-0.2
	SLN445 raking strut / End	-3.8	0.0	-1.0	0.0	0.0	0.0
C LC_142	SLN45 column below / Begin	23.3	0.0	0.2	0.0	1.0	0.0
	SLN198 ring / Begin	-8.7	-0.1	-5.0	0.0	0.0	0.2
	SLN210 ring / End	8.7	-0.1	-4.9	0.0	0.0	-0.2
	SLN445 raking strut / End	-5.0	0.0	-1.0	0.0	0.0	0.0
C LC_143	SLN45 column below / Begin	24.2	0.0	0.2	0.0	1.0	0.0
	SLN198 ring / Begin	-9.1	0.0	-5.0	0.0	0.0	0.2
	SLN210 ring / End	9.0	0.0	-4.9	0.0	0.0	-0.2
	SLN445 raking strut / End	-3.9	0.0	-1.0	0.0	0.0	0.0
C LC_144	SLN45 column below / Begin	23.4	0.0	0.2	0.0	1.0	0.0
	SLN198 ring / Begin	-8.7	-0.1	-5.0	0.0	0.0	0.2
	SLN210 ring / End	8.8	-0.1	-4.9	0.0	0.0	-0.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN445 raking strut / End	-5.0	0.0	-1.0	0.0	0.0	0.0
C LC_145	SLN45 column below / Begin	113.3	0.0	0.8	0.0	3.2	0.0
	SLN198 ring / Begin	-28.8	-0.2	-26.0	0.0	0.0	0.6
	SLN210 ring / End	29.0	-0.1	-25.6	0.0	0.0	-0.6
	SLN445 raking strut / End	-27.7	0.0	-1.0	0.0	0.0	0.0
C LC_146	SLN45 column below / Begin	109.8	0.0	0.8	0.0	3.2	0.0
	SLN198 ring / Begin	-27.2	-0.3	-26.0	0.0	0.0	0.7
	SLN210 ring / End	27.0	-0.3	-25.6	0.0	0.0	-0.7
	SLN445 raking strut / End	-32.4	0.0	-1.0	0.0	0.0	0.0
C LC_147	SLN45 column below / Begin	113.1	0.0	0.8	0.0	3.2	0.0
	SLN198 ring / Begin	-28.8	-0.1	-26.0	0.0	0.0	0.6
	SLN210 ring / End	28.7	-0.1	-25.6	0.0	0.0	-0.6
	SLN445 raking strut / End	-27.9	0.0	-1.0	0.0	0.0	0.0
C LC_148	SLN45 column below / Begin	109.9	0.0	0.8	0.0	3.2	0.0
	SLN198 ring / Begin	-27.1	-0.3	-26.0	0.0	0.0	0.7
	SLN210 ring / End	27.4	-0.3	-25.6	0.0	0.0	-0.7
	SLN445 raking strut / End	-32.3	0.0	-1.0	0.0	0.0	0.0
C LC_151	SLN45 column below / Begin	69.2	0.0	-1.1	0.0	-4.5	0.0
	SLN198 ring / Begin	-12.5	-0.1	-11.9	0.0	0.0	0.3
	SLN210 ring / End	12.5	-0.1	-11.5	0.0	0.0	-0.3
	SLN445 raking strut / End	-14.9	0.0	-1.0	0.0	0.0	0.0
C LC_152	SLN45 column below / Begin	68.3	0.0	2.2	0.0	8.8	0.0
	SLN198 ring / Begin	-28.8	-0.2	-14.0	0.0	0.0	0.5
	SLN210 ring / End	28.8	-0.2	-14.1	0.0	0.0	-0.5
	SLN445 raking strut / End	-14.9	0.0	-1.0	0.0	0.0	0.0
C LC_153	SLN45 column below / Begin	111.2	0.0	0.8	0.0	3.1	0.0
	SLN198 ring / Begin	-28.1	-0.2	-21.0	0.0	0.0	0.7
	SLN210 ring / End	28.2	-0.2	-20.7	0.0	0.0	-0.6
	SLN445 raking strut / End	-29.7	0.0	-1.0	0.0	0.0	0.0
C LC_154	SLN45 column below / Begin	68.5	0.0	-1.1	0.0	-4.4	0.0
	SLN198 ring / Begin	-12.6	-0.1	-11.9	0.0	0.0	0.3
	SLN210 ring / End	12.6	-0.1	-11.5	0.0	0.0	-0.3
	SLN445 raking strut / End	-14.8	0.0	-1.0	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.1 < 5.0%	OK
Loc. deformation	0.1 < 3%	OK
Bolts	14.6 < 100%	OK
Welds	98.1 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN209+SLN197 radial (continuous)-bfl 1	12.7	C LC_120	150.5	0.0	52.5	OK
SLN209+SLN197 radial (continuous)-tfl 1	12.7	C LC_153	69.2	0.0	0.0	OK
SLN209+SLN197 radial (continuous)-w 1	7.2	C LC_122	180.5	0.0	52.4	OK
SLN45 column below	10.0	C LC_104	275.2	0.1	0.0	OK
SLN198 ring-bfl 1	12.7	C LC_120	54.1	0.0	0.0	OK
SLN198 ring-tfl 1	12.7	C LC_120	48.9	0.0	0.0	OK
SLN198 ring-w 1	7.2	C LC_146	163.9	0.0	0.0	OK
SLN210 ring-bfl 1	12.7	C LC_148	49.4	0.0	0.0	OK
SLN210 ring-tfl 1	12.7	C LC_148	43.6	0.0	0.0	OK
SLN210 ring-w 1	7.2	C LC_148	152.2	0.0	0.0	OK
SLN445 raking strut	10.0	C LC_104	38.4	0.0	0.0	OK
End plate column below SLN45 column below	12.0	C LC_152	132.9	0.0	41.7	OK
End plate SLN198 ring	12.0	C LC_120	76.9	0.0	0.0	OK
End plate SLN210 ring	12.0	C LC_146	61.7	0.0	10.7	OK
Cap plate SLN445 raking strut	12.0	C LC_109	71.6	0.0	0.0	OK
Fin plate SLN445 raking strut	12.0	C LC_115	201.3	0.0	56.2	OK
Fin plate 2 SLN445 raking strut	12.0	C LC_117	222.2	0.0	64.3	OK
Gusset SLN445 raking strut	12.0	C LC_109	213.9	0.0	31.7	OK
Flange top a end plate SLN198 ring	12.0	C LC_120	65.4	0.0	0.0	OK
Flange top b end plate SLN198 ring	12.0	C LC_120	53.5	0.0	0.0	OK
Flange bottom a end plate SLN198 ring	12.0	C LC_120	47.6	0.0	0.0	OK
Flange bottom b end plate SLN198 ring	12.0	C LC_146	76.5	0.0	0.0	OK
Flange top a end plate SLN210 ring	12.0	C LC_148	47.0	0.0	0.0	OK
Flange top b end plate SLN210 ring	12.0	C LC_146	56.3	0.0	0.0	OK
Flange bottom a end plate SLN210 ring	12.0	C LC_148	71.4	0.0	0.0	OK
Flange bottom b end plate SLN210 ring	12.0	C LC_148	41.6	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

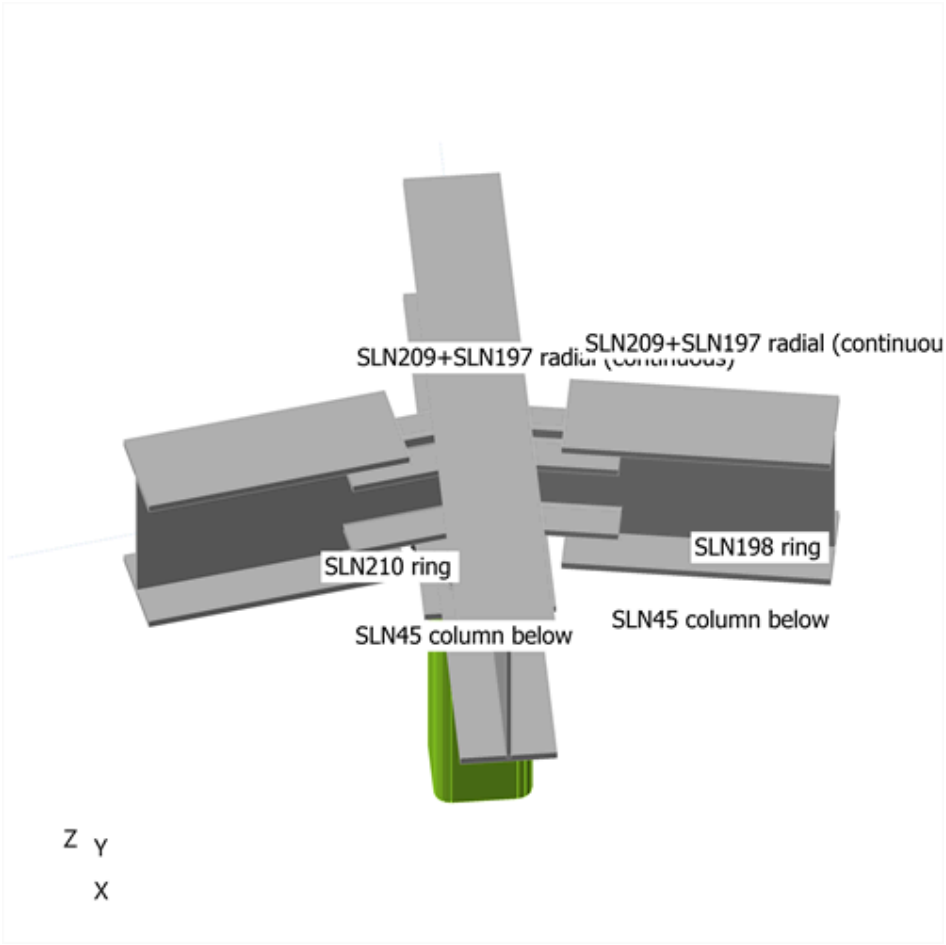
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

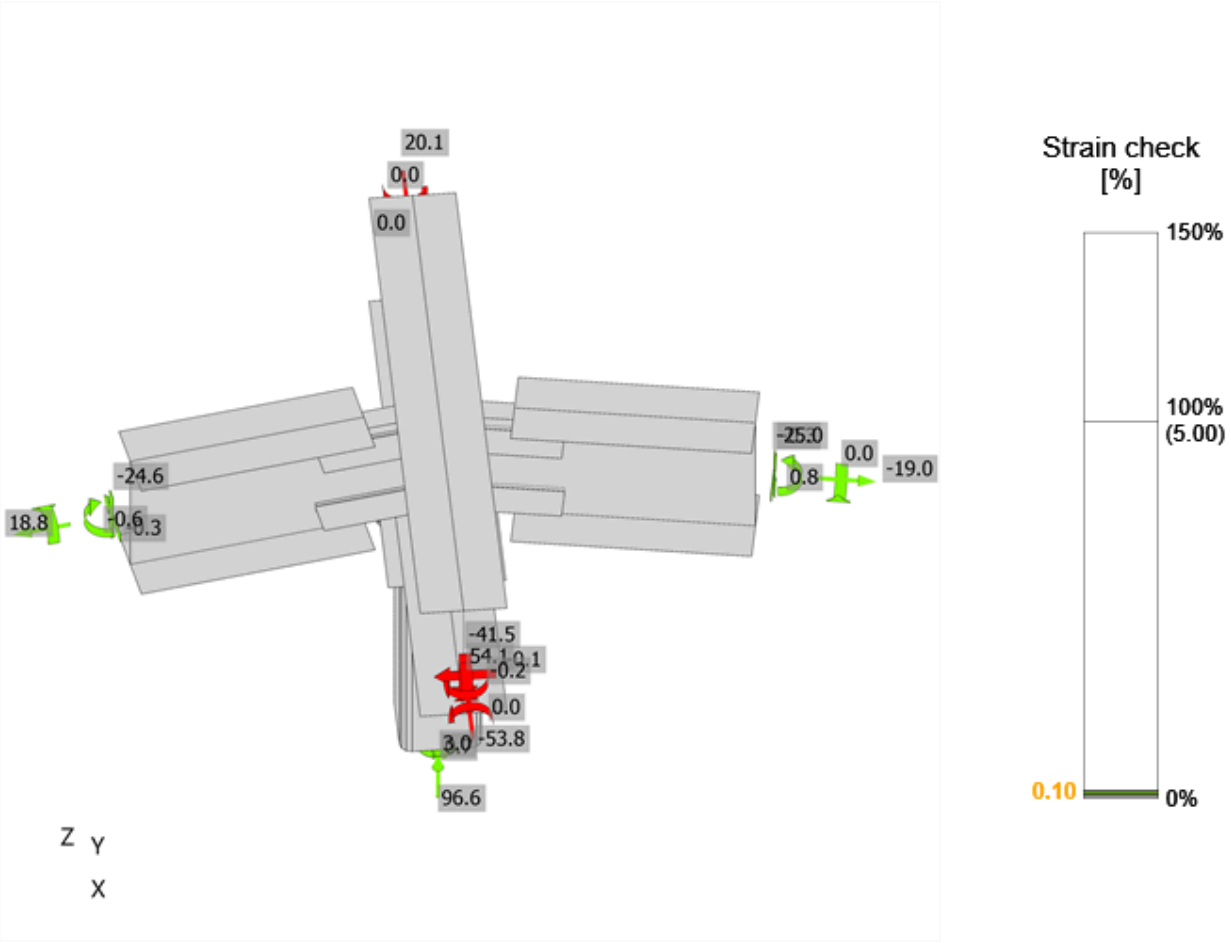
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN45 column below	150	C LC_104	0	5	0.1	OK
SLN445 raking strut	150	C LC_154	0	5	0.0	OK

Symbol explanation

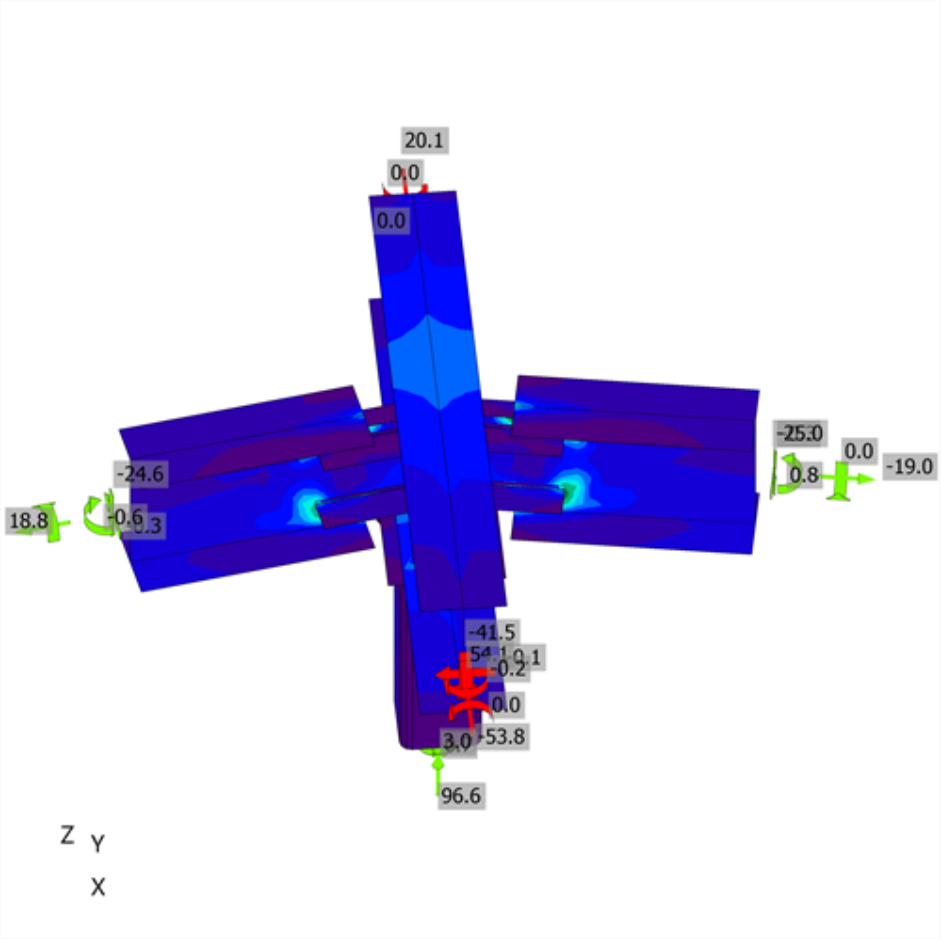
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

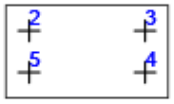
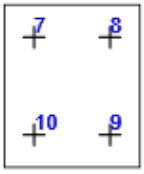
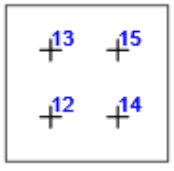


Strain check, C LC_104



Equivalent stress, C LC_104

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_152	18.3	3.5	111.7	13.0	3.8	13.0	OK	OK
	B3	M20 8.8 - 1	C LC_107	14.7	2.9	171.2	10.4	3.1	10.5	OK	OK
	B4	M20 8.8 - 1	C LC_107	14.1	2.3	84.0	10.0	2.8	9.6	OK	OK
	B5	M20 8.8 - 1	C LC_152	18.4	3.6	116.8	13.1	3.8	13.1	OK	OK
	B7	M20 8.8 - 1	C LC_120	11.1	0.9	123.8	7.8	1.0	6.5	OK	OK
	B8	M20 8.8 - 1	C LC_104	0.0	1.0	63.8	0.0	1.5	1.0	OK	OK
	B9	M20 8.8 - 1	C LC_152	0.0	1.2	123.8	0.0	2.0	1.3	OK	OK
	B10	M20 8.8 - 1	C LC_120	11.6	0.9	123.8	8.2	1.0	6.9	OK	OK
	B12	M20 8.8 - 1	C LC_117	9.3	9.3	106.0	6.6	9.9	14.6	OK	OK
	B13	M20 8.8 - 1	C LC_125	3.7	6.9	151.8	2.6	7.3	9.2	OK	OK
	B14	M20 8.8 - 1	C LC_117	1.7	6.8	106.0	1.2	7.3	8.1	OK	OK
	B15	M20 8.8 - 1	C LC_104	0.1	6.2	115.3	0.1	10.4	6.6	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN445 raking strut	SLN445 raking strut	◢ 4.2	519	C LC_104	135.2	0.0	-64.5	68.5	-2.5	33.4	17.8	OK	OK
SLN45 column below-w 3	Gusset SLN445 raking strut	◢ 5.6 ◣	288	C LC_104	397.1	0.3	-125.2	-124.1	-178.7	98.1	61.0	OK	OK
		◢ 5.6 ◣	288	C LC_104	397.1	0.3	-123.7	124.9	178.5	98.1	61.7	OK	OK
SLN198 ring-w 1	Flange top b end plate SLN198 ring	◢ 4.2 ◣	198	C LC_153	68.6	0.0	36.3	30.2	14.7	16.9	10.6	OK	OK
		◢ 4.2 ◣	197	C LC_120	33.4	0.0	-16.1	14.7	8.2	8.3	6.9	OK	OK
SLN198 ring-w 1	Flange bottom b end plate SLN198 ring	◢ 4.2 ◣	198	C LC_146	95.9	0.0	28.4	33.5	40.9	23.7	14.5	OK	OK
		◢ 4.2 ◣	198	C LC_146	140.0	0.0	56.3	-51.2	-53.4	34.6	21.2	OK	OK
SLN210 ring-w 1	Flange top a end plate SLN210 ring	◢ 4.2 ◣	198	C LC_153	60.7	0.0	31.9	27.0	-12.7	15.0	8.5	OK	OK
		◢ 4.2 ◣	197	C LC_146	29.2	0.0	-14.6	13.4	-5.8	7.2	6.6	OK	OK
SLN210 ring-w 1	Flange bottom a end plate SLN210 ring	◢ 4.2 ◣	198	C LC_148	93.3	0.0	27.9	31.8	-40.4	23.0	12.3	OK	OK
		◢ 4.2 ◣	198	C LC_148	132.3	0.0	52.2	-48.3	50.9	32.7	17.8	OK	OK
Cap plate SLN445 raking strut	Fin plate SLN445 raking strut	◢ 5.6 ◣	139	C LC_115	193.7	0.0	71.4	85.2	59.5	47.9	33.4	OK	OK
		◢ 5.6 ◣	139	C LC_106	118.6	0.0	-46.7	51.4	-36.4	29.3	16.1	OK	OK
Cap plate SLN445 raking strut	Fin plate 2 SLN445 raking strut	◢ 5.6 ◣	139	C LC_104	123.8	0.0	-49.5	-53.8	37.3	30.6	16.8	OK	OK
		◢ 5.6 ◣	139	C LC_117	199.2	0.0	72.6	-87.5	-61.8	49.2	34.9	OK	OK
End plate column below SLN45 column below	SLN45 column below	◢ 5.6 ◣	521	C LC_120	298.7	0.0	-188.8	-132.0	-21.1	73.8	40.6	OK	OK
		◢ 5.6 ◣	521	C LC_152	212.4	0.0	-94.0	109.8	-5.6	52.5	31.6	OK	OK
SLN209+SLN197 radial (continuous)-w 1	SLN198 ring-w 1	◢ 7.2	189	C LC_119	84.9	0.0	8.2	5.4	48.5	21.0	13.7	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
SLN209+SLN197 radial (continuous)-w 1	SLN210 ring-w 1	◢ 7.2	189	C LC_121	90.7	0.0	4.5	10.2	-51.3	22.4	14.7	OK	OK
End plate SLN198 ring	SLN198 ring-w 1	◢ 4.2 ◣	149	C LC_119	86.6	0.0	-32.2	-18.7	-42.4	21.4	12.2	OK	OK
		◢ 4.2 ◣	149	C LC_148	70.2	0.0	-12.3	27.5	28.9	17.4	11.3	OK	OK
End plate SLN210 ring	SLN210 ring-w 1	◢ 4.2 ◣	149	C LC_121	85.3	0.0	-26.2	-22.8	41.0	21.1	13.8	OK	OK
		◢ 4.2 ◣	149	C LC_146	69.5	0.0	-17.8	24.0	-30.5	17.2	11.3	OK	OK
SLN198 ring-w 1	Flange top a end plate SLN198 ring	◢ 4.2 ◣	199	C LC_120	112.8	0.0	-10.3	6.2	-64.6	27.9	19.6	OK	OK
		◢ 4.2 ◣	198	C LC_146	74.7	0.0	11.6	13.0	-40.6	18.5	10.6	OK	OK
SLN198 ring-w 1	Flange bottom a end plate SLN198 ring	◢ 4.2 ◣	198	C LC_146	99.9	0.0	18.9	-18.8	53.5	24.7	14.6	OK	OK
		◢ 4.2 ◣	198	C LC_104	77.2	0.0	-10.7	-2.5	44.0	19.1	11.2	OK	OK
SLN210 ring-w 1	Flange top b end plate SLN210 ring	◢ 4.2 ◣	199	C LC_148	98.7	0.0	-10.4	6.1	56.3	24.4	15.0	OK	OK
		◢ 4.2 ◣	198	C LC_148	66.9	0.0	10.8	12.1	36.1	16.5	9.7	OK	OK
SLN210 ring-w 1	Flange bottom b end plate SLN210 ring	◢ 4.2 ◣	198	C LC_148	94.2	0.0	16.5	-16.5	-50.9	23.3	13.6	OK	OK
		◢ 4.2 ◣	199	C LC_121	68.3	0.0	-12.8	14.1	-36.1	16.9	10.8	OK	OK
End plate SLN198 ring	Flange top a end plate SLN198 ring	◢ 4.2 ◣	58	C LC_122	33.0	0.0	1.2	1.8	-19.0	8.2	7.0	OK	OK
		◢ 4.2 ◣	58	C LC_120	21.9	0.0	-6.3	-10.8	-5.6	5.4	5.4	OK	OK
End plate SLN198 ring	Flange top b end plate SLN198 ring	◢ 4.2 ◣	58	C LC_146	51.9	0.0	-0.8	-1.5	-29.9	12.8	9.6	OK	OK
		◢ 4.2 ◣	58	C LC_120	63.6	0.0	36.3	24.9	-16.9	15.7	12.6	OK	OK
End plate SLN198 ring	Flange bottom a end plate SLN198 ring	◢ 4.2 ◣	58	C LC_119	54.4	0.0	11.7	-19.0	24.1	13.4	8.9	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_122	43.2	0.0	-11.7	-16.8	17.2	10.7	8.4	OK	OK
End plate SLN198 ring	Flange bottom b end plate SLN198 ring	◢ 4.2 ◣	58	C LC_120	73.6	0.0	41.6	-31.5	15.4	18.2	14.5	OK	OK
		◢ 4.2 ◣	58	C LC_146	62.2	0.0	-3.9	0.9	35.8	15.4	10.6	OK	OK
End plate SLN210 ring	Flange top a end plate SLN210 ring	◢ 4.2 ◣	58	C LC_148	41.7	0.0	0.7	-3.4	23.8	10.3	9.0	OK	OK
		◢ 4.2 ◣	58	C LC_120	49.1	0.0	6.3	10.1	26.2	12.1	9.7	OK	OK
End plate SLN210 ring	Flange top b end plate SLN210 ring	◢ 4.2 ◣	58	C LC_104	25.5	0.0	-1.5	4.7	13.9	6.3	6.0	OK	OK
		◢ 4.2 ◣	58	C LC_122	26.2	0.0	-9.6	-11.5	8.1	6.5	6.3	OK	OK
End plate SLN210 ring	Flange bottom a end plate SLN210 ring	◢ 4.2 ◣	58	C LC_146	59.9	0.0	33.0	-26.6	-11.3	14.8	12.0	OK	OK
		◢ 4.2 ◣	58	C LC_148	50.7	0.0	-3.0	2.0	-29.2	12.5	9.7	OK	OK
End plate SLN210 ring	Flange bottom b end plate SLN210 ring	◢ 4.2 ◣	58	C LC_121	56.9	0.0	11.3	-20.9	-24.5	14.1	7.8	OK	OK
		◢ 4.2 ◣	58	C LC_120	41.2	0.0	-10.6	-18.4	-13.8	10.2	6.9	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-S_180
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN212+SLN200 radial (continuous) - SLN47 column below - SLN213 ring - SLN214 ring - SLN434 raking strut

Name	SLN212+SLN200 radial (continuous) - SLN47 column below - SLN213 ring - SLN214 ring - SLN434 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

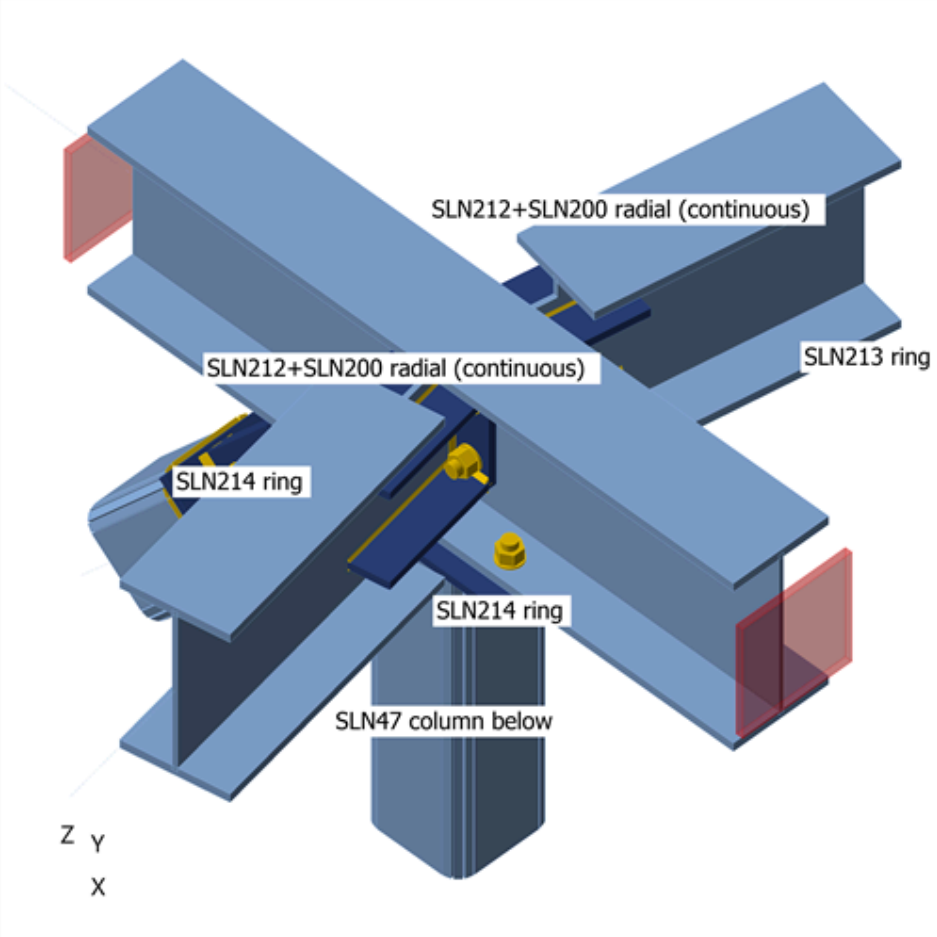
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN212+SLN200 radial (continuous)	1 - UB254x146x43	0.0	0.0	0.0	0	0	0
SLN47 column below	2 - SHS150x150x10	0.0	90.0	0.0	0	0	0
SLN213 ring	1 - UB254x146x43	-100.0	0.0	0.0	0	0	0
SLN214 ring	1 - UB254x146x43	-80.0	0.0	0.0	0	0	0
SLN434 raking strut	2 - SHS150x150x10	-179.9	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN212+SLN200 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN212+SLN200 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN47 column below / begin		Position	0
SLN213 ring / begin		Position	0
SLN214 ring / end		Position	0
SLN434 raking strut / end		Position	0



SLN212+SLN200 radial (continuous) - SLN47 column below - SLN213 ring - SLN214 ring - SLN434 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN47 column below / Begin	24.8	0.0	0.2	0.0	0.7	0.0
	SLN213 ring / Begin	-8.8	-0.1	-4.9	0.0	0.0	0.2
	SLN214 ring / End	8.9	-0.1	-4.9	0.0	0.0	-0.2
	SLN434 raking strut / End	-5.0	0.0	-1.0	0.0	0.0	0.0
C LC_102	SLN47 column below / Begin	116.8	0.0	0.4	0.0	1.5	0.0
	SLN213 ring / Begin	-27.8	-0.2	-25.6	0.0	0.0	0.7
	SLN214 ring / End	27.9	-0.2	-25.6	0.0	0.0	-0.6
	SLN434 raking strut / End	-32.8	0.0	-1.0	0.0	0.0	0.0
C LC_103	SLN47 column below / Begin	125.7	0.0	-2.4	0.0	-0.5	0.0
	SLN213 ring / Begin	-28.0	-0.1	-24.5	0.0	0.0	0.5
	SLN214 ring / End	28.1	0.0	-24.6	0.0	0.0	-0.3
	SLN434 raking strut / End	-13.0	0.0	-1.2	0.0	0.0	0.0
C LC_104	SLN47 column below / Begin	99.2	0.0	0.4	0.0	1.4	0.0
	SLN213 ring / Begin	-18.3	-0.4	-24.5	0.0	0.0	0.8
	SLN214 ring / End	18.4	-0.3	-24.6	0.0	0.0	-0.7
	SLN434 raking strut / End	-48.4	0.0	-0.8	0.0	0.0	0.0
C LC_105	SLN47 column below / Begin	109.0	-2.5	0.3	0.0	1.4	0.7
	SLN213 ring / Begin	-22.0	-0.2	-24.5	0.0	0.0	0.6
	SLN214 ring / End	19.3	-0.2	-24.6	0.0	0.0	-0.5
	SLN434 raking strut / End	-35.5	0.0	-1.0	0.2	0.0	-0.7
C LC_106	SLN47 column below / Begin	109.5	2.4	0.3	0.0	1.4	-0.6
	SLN213 ring / Begin	-19.3	-0.2	-24.5	0.0	0.0	0.5
	SLN214 ring / End	22.1	-0.2	-24.6	0.0	0.0	-0.7
	SLN434 raking strut / End	-34.9	0.0	-1.0	-0.2	0.0	0.7
C LC_107	SLN47 column below / Begin	34.1	0.0	-5.4	0.0	-3.6	0.0
	SLN213 ring / Begin	-10.9	0.0	-1.6	0.0	0.0	0.0
	SLN214 ring / End	11.0	0.1	-1.6	0.0	0.0	0.3
	SLN434 raking strut / End	32.7	0.0	-1.1	0.0	0.0	0.0
C LC_108	SLN47 column below / Begin	-10.2	0.0	0.1	0.0	0.3	0.0
	SLN213 ring / Begin	4.2	-0.1	-1.6	0.0	0.0	0.3
	SLN214 ring / End	-4.2	-0.1	-1.6	0.0	0.0	-0.1
	SLN434 raking strut / End	-26.6	0.0	-0.4	0.0	0.0	0.0
C LC_109	SLN47 column below / Begin	5.1	-4.9	0.1	0.0	0.3	1.3
	SLN213 ring / Begin	-0.8	0.0	-1.6	0.0	0.0	0.1
	SLN214 ring / End	-4.3	0.0	-1.6	0.0	0.0	0.0
	SLN434 raking strut / End	-6.3	0.1	-0.8	0.4	0.0	-1.4
C LC_110	SLN47 column below / Begin	5.7	4.9	0.1	0.0	0.3	-1.3
	SLN213 ring / Begin	4.2	0.0	-1.6	0.0	0.0	-0.1
	SLN214 ring / End	0.9	-0.1	-1.6	0.0	0.0	-0.2
	SLN434 raking strut / End	-5.6	-0.1	-0.8	-0.3	0.0	1.4
C LC_111	SLN47 column below / Begin	105.5	0.0	-5.3	0.0	-2.8	0.0
	SLN213 ring / Begin	-25.1	0.0	-17.3	0.0	0.0	0.3
	SLN214 ring / End	25.2	0.1	-17.3	0.0	0.0	0.0
	SLN434 raking strut / End	12.9	0.0	-1.4	0.0	0.0	0.0
C LC_112	SLN47 column below / Begin	58.4	0.0	0.3	0.0	1.1	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN213 ring / Begin	-8.5	-0.3	-17.3	0.0	0.0	0.7
	SLN214 ring / End	8.5	-0.2	-17.3	0.0	0.0	-0.5
	SLN434 raking strut / End	-50.1	0.0	-0.7	0.0	0.0	0.0
C LC_113	SLN47 column below / Begin	75.1	-4.9	0.3	0.0	1.1	1.3
	SLN213 ring / Begin	-14.3	-0.2	-17.3	0.0	0.0	0.5
	SLN214 ring / End	9.1	-0.1	-17.3	0.0	0.0	-0.3
	SLN434 raking strut / End	-28.0	0.1	-1.0	0.4	0.0	-1.4
C LC_114	SLN47 column below / Begin	75.9	4.9	0.3	0.0	1.0	-1.3
	SLN213 ring / Begin	-9.1	-0.1	-17.3	0.0	0.0	0.2
	SLN214 ring / End	14.5	-0.2	-17.3	0.0	0.0	-0.6
	SLN434 raking strut / End	-27.2	-0.1	-1.0	-0.3	0.0	1.4
C LC_115	SLN47 column below / Begin	40.7	0.0	-5.4	0.0	-3.4	0.0
	SLN213 ring / Begin	-13.3	0.0	-2.8	0.0	0.0	0.0
	SLN214 ring / End	13.3	0.1	-2.8	0.0	0.0	0.2
	SLN434 raking strut / End	31.6	0.0	-1.4	0.0	0.0	0.0
C LC_116	SLN47 column below / Begin	-3.9	0.0	0.1	0.0	0.5	0.0
	SLN213 ring / Begin	2.0	-0.1	-2.8	0.0	0.0	0.3
	SLN214 ring / End	-2.0	-0.1	-2.8	0.0	0.0	-0.2
	SLN434 raking strut / End	-28.1	0.0	-0.7	0.0	0.0	0.0
C LC_117	SLN47 column below / Begin	11.6	-4.9	0.1	0.0	0.5	1.3
	SLN213 ring / Begin	-3.1	-0.1	-2.8	0.0	0.0	0.2
	SLN214 ring / End	-2.0	0.0	-2.8	0.0	0.0	0.0
	SLN434 raking strut / End	-7.6	0.1	-1.0	0.4	0.0	-1.4
C LC_118	SLN47 column below / Begin	12.2	4.9	0.1	0.0	0.5	-1.3
	SLN213 ring / Begin	1.9	0.0	-2.8	0.0	0.0	-0.1
	SLN214 ring / End	3.2	-0.1	-2.8	0.0	0.0	-0.3
	SLN434 raking strut / End	-6.9	-0.1	-1.0	-0.3	0.0	1.4
C LC_119	SLN47 column below / Begin	128.4	0.0	-2.4	0.0	-0.4	0.0
	SLN213 ring / Begin	-30.3	-0.1	-25.6	0.0	0.0	0.6
	SLN214 ring / End	30.5	-0.1	-25.6	0.0	0.0	-0.4
	SLN434 raking strut / End	-17.2	0.0	-1.2	0.0	0.0	0.0
C LC_120	SLN47 column below / Begin	107.3	0.0	0.4	0.0	1.4	0.0
	SLN213 ring / Begin	-20.5	-0.4	-25.6	0.0	0.0	0.8
	SLN214 ring / End	20.5	-0.3	-25.6	0.0	0.0	-0.7
	SLN434 raking strut / End	-45.1	0.0	-0.8	0.0	0.0	0.0
C LC_121	SLN47 column below / Begin	113.5	-2.5	0.4	0.0	1.5	0.7
	SLN213 ring / Begin	-24.3	-0.2	-25.6	0.0	0.0	0.7
	SLN214 ring / End	21.5	-0.2	-25.6	0.0	0.0	-0.6
	SLN434 raking strut / End	-37.0	0.0	-1.0	0.2	0.0	-0.7
C LC_122	SLN47 column below / Begin	114.2	2.4	0.4	0.0	1.5	-0.6
	SLN213 ring / Begin	-21.5	-0.2	-25.6	0.0	0.0	0.6
	SLN214 ring / End	24.4	-0.2	-25.6	0.0	0.0	-0.7
	SLN434 raking strut / End	-36.2	0.0	-1.0	-0.2	0.0	0.7
C LC_123	SLN47 column below / Begin	110.8	0.0	-5.2	0.0	-2.6	0.0
	SLN213 ring / Begin	-29.7	-0.1	-19.4	0.0	0.0	0.5
	SLN214 ring / End	29.8	0.0	-19.4	0.0	0.0	-0.2
	SLN434 raking strut / End	4.5	0.0	-1.4	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_124	SLN47 column below / Begin	74.6	0.0	0.3	0.0	1.2	0.0
	SLN213 ring / Begin	-12.9	-0.3	-19.4	0.0	0.0	0.7
	SLN214 ring / End	12.9	-0.2	-19.4	0.0	0.0	-0.5
	SLN434 raking strut / End	-43.5	0.0	-0.7	0.0	0.0	0.1
C LC_125	SLN47 column below / Begin	84.2	-4.9	0.3	0.0	1.2	1.4
	SLN213 ring / Begin	-18.8	-0.2	-19.4	0.0	0.0	0.6
	SLN214 ring / End	13.6	-0.2	-19.4	0.0	0.0	-0.4
	SLN434 raking strut / End	-31.1	0.1	-1.0	0.4	0.0	-1.4
C LC_126	SLN47 column below / Begin	85.2	4.9	0.3	0.0	1.2	-1.3
	SLN213 ring / Begin	-13.6	-0.1	-19.4	0.0	0.0	0.3
	SLN214 ring / End	19.0	-0.2	-19.4	0.0	0.0	-0.6
	SLN434 raking strut / End	-29.8	-0.1	-1.0	-0.4	0.0	1.4
C LC_127	SLN47 column below / Begin	45.9	0.0	-5.3	0.0	-3.2	0.0
	SLN213 ring / Begin	-17.9	-0.1	-4.9	0.0	0.0	0.2
	SLN214 ring / End	18.0	0.0	-4.9	0.0	0.0	0.1
	SLN434 raking strut / End	23.2	0.0	-1.4	0.0	0.0	0.0
C LC_128	SLN47 column below / Begin	12.3	0.0	0.2	0.0	0.6	0.0
	SLN213 ring / Begin	-2.4	-0.1	-4.9	0.0	0.0	0.3
	SLN214 ring / End	2.3	-0.1	-4.9	0.0	0.0	-0.2
	SLN434 raking strut / End	-21.4	0.0	-0.7	0.0	0.0	0.0
C LC_129	SLN47 column below / Begin	20.6	-4.9	0.2	0.0	0.6	1.3
	SLN213 ring / Begin	-7.6	-0.1	-4.9	0.0	0.0	0.2
	SLN214 ring / End	2.5	0.0	-4.9	0.0	0.0	-0.1
	SLN434 raking strut / End	-10.6	0.1	-1.0	0.4	0.0	-1.4
C LC_130	SLN47 column below / Begin	21.5	4.9	0.1	0.0	0.6	-1.3
	SLN213 ring / Begin	-2.5	0.0	-4.9	0.0	0.0	0.0
	SLN214 ring / End	7.7	-0.1	-4.9	0.0	0.0	-0.3
	SLN434 raking strut / End	-9.6	-0.1	-1.0	-0.4	0.0	1.4
C LC_135	SLN47 column below / Begin	116.2	0.0	0.4	0.0	1.5	0.0
	SLN213 ring / Begin	-28.2	-0.2	-25.6	0.0	0.0	0.7
	SLN214 ring / End	28.3	-0.2	-25.6	0.0	0.0	-0.6
	SLN434 raking strut / End	-33.7	0.0	-1.0	0.0	0.0	0.0
C LC_141	SLN47 column below / Begin	25.4	0.0	0.2	0.0	0.7	0.0
	SLN213 ring / Begin	-9.1	0.0	-4.9	0.0	0.0	0.2
	SLN214 ring / End	9.1	0.0	-4.9	0.0	0.0	-0.2
	SLN434 raking strut / End	-4.2	0.0	-1.0	0.0	0.0	0.0
C LC_142	SLN47 column below / Begin	24.2	0.0	0.2	0.0	0.7	0.0
	SLN213 ring / Begin	-8.6	-0.1	-4.9	0.0	0.0	0.2
	SLN214 ring / End	8.6	-0.1	-4.9	0.0	0.0	-0.2
	SLN434 raking strut / End	-5.8	0.0	-1.0	0.0	0.0	0.0
C LC_143	SLN47 column below / Begin	24.8	0.0	0.2	0.0	0.7	0.0
	SLN213 ring / Begin	-8.9	-0.1	-4.9	0.0	0.0	0.2
	SLN214 ring / End	8.8	-0.1	-4.9	0.0	0.0	-0.2
	SLN434 raking strut / End	-5.0	0.0	-1.0	0.0	0.0	0.0
C LC_144	SLN47 column below / Begin	24.9	0.0	0.2	0.0	0.7	0.0
	SLN213 ring / Begin	-8.8	-0.1	-4.9	0.0	0.0	0.2
	SLN214 ring / End	8.9	-0.1	-4.9	0.0	0.0	-0.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN434 raking strut / End	-5.0	0.0	-1.0	0.0	0.0	0.0
C LC_145	SLN47 column below / Begin	119.2	0.0	0.4	0.0	1.6	0.0
	SLN213 ring / Begin	-28.9	-0.1	-25.6	0.0	0.0	0.6
	SLN214 ring / End	29.0	-0.1	-25.6	0.0	0.0	-0.5
	SLN434 raking strut / End	-29.6	0.0	-1.0	0.0	0.0	0.0
C LC_146	SLN47 column below / Begin	114.4	0.0	0.4	0.0	1.5	0.0
	SLN213 ring / Begin	-26.6	-0.3	-25.6	0.0	0.0	0.8
	SLN214 ring / End	26.7	-0.3	-25.6	0.0	0.0	-0.7
	SLN434 raking strut / End	-36.0	0.0	-1.0	0.0	0.0	0.0
C LC_147	SLN47 column below / Begin	116.7	0.0	0.4	0.0	1.5	0.0
	SLN213 ring / Begin	-27.9	-0.2	-25.6	0.0	0.0	0.7
	SLN214 ring / End	27.7	-0.2	-25.6	0.0	0.0	-0.6
	SLN434 raking strut / End	-32.9	0.0	-1.0	0.0	0.0	0.0
C LC_148	SLN47 column below / Begin	116.9	0.0	0.4	0.0	1.5	0.1
	SLN213 ring / Begin	-27.6	-0.2	-25.6	0.0	0.0	0.7
	SLN214 ring / End	28.0	-0.2	-25.6	0.0	0.0	-0.6
	SLN434 raking strut / End	-32.6	0.0	-1.0	0.0	0.0	0.0
C LC_151	SLN47 column below / Begin	75.9	0.0	-1.6	0.0	-6.7	0.0
	SLN213 ring / Begin	-12.4	-0.1	-11.6	0.0	0.0	0.3
	SLN214 ring / End	12.4	-0.1	-12.3	0.0	0.0	-0.3
	SLN434 raking strut / End	-18.0	0.0	-1.0	0.0	0.0	0.0
C LC_152	SLN47 column below / Begin	68.1	0.0	2.2	0.0	9.0	0.0
	SLN213 ring / Begin	-28.6	-0.2	-14.2	0.0	0.0	0.5
	SLN214 ring / End	28.7	-0.2	-16.2	0.0	0.0	-0.5
	SLN434 raking strut / End	-15.3	0.0	-1.0	0.0	0.0	0.0
C LC_153	SLN47 column below / Begin	116.5	0.0	0.4	0.0	1.5	0.0
	SLN213 ring / Begin	-27.9	-0.2	-20.9	0.0	0.0	0.7
	SLN214 ring / End	28.0	-0.2	-23.6	0.0	0.0	-0.6
	SLN434 raking strut / End	-32.6	0.0	-1.0	0.0	0.0	0.0
C LC_154	SLN47 column below / Begin	75.4	0.0	-1.6	0.0	-6.6	0.0
	SLN213 ring / Begin	-12.5	-0.1	-11.6	0.0	0.0	0.3
	SLN214 ring / End	12.5	-0.1	-12.3	0.0	0.0	-0.3
	SLN434 raking strut / End	-17.8	0.0	-1.0	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.1 < 5.0%	OK
Loc. deformation	0.1 < 3%	OK
Bolts	18.1 < 100%	OK
Welds	98.2 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN212+SLN200 radial (continuous)-bfl 1	12.7	C LC_120	150.0	0.0	51.4	OK
SLN212+SLN200 radial (continuous)-tfl 1	12.7	C LC_153	71.5	0.0	0.0	OK
SLN212+SLN200 radial (continuous)-w 1	7.2	C LC_120	181.5	0.0	51.4	OK
SLN47 column below	10.0	C LC_104	275.3	0.1	0.0	OK
SLN213 ring-bfl 1	12.7	C LC_120	55.0	0.0	0.0	OK
SLN213 ring-tfl 1	12.7	C LC_104	49.9	0.0	0.0	OK
SLN213 ring-w 1	7.2	C LC_120	165.0	0.0	0.0	OK
SLN214 ring-bfl 1	12.7	C LC_146	51.4	0.0	0.0	OK
SLN214 ring-tfl 1	12.7	C LC_146	45.7	0.0	0.0	OK
SLN214 ring-w 1	7.2	C LC_146	158.1	0.0	0.0	OK
SLN434 raking strut	10.0	C LC_125	51.5	0.0	0.0	OK
End plate column below SLN47 column below	12.0	C LC_107	155.2	0.0	21.0	OK
End plate SLN213 ring	12.0	C LC_104	79.4	0.0	0.0	OK
End plate SLN214 ring	12.0	C LC_146	65.9	0.0	12.0	OK
Cap plate SLN434 raking strut	12.0	C LC_126	99.4	0.0	0.0	OK
Fin plate SLN434 raking strut	12.0	C LC_125	264.0	0.0	60.6	OK
Fin plate 2 SLN434 raking strut	12.0	C LC_126	263.1	0.0	60.5	OK
Gusset SLN434 raking strut	12.0	C LC_126	275.0	0.0	41.7	OK
Flange top a end plate SLN213 ring	12.0	C LC_104	67.6	0.0	0.0	OK
Flange top b end plate SLN213 ring	12.0	C LC_104	55.5	0.0	0.0	OK
Flange bottom a end plate SLN213 ring	12.0	C LC_104	50.6	0.0	0.0	OK
Flange bottom b end plate SLN213 ring	12.0	C LC_146	77.0	0.0	0.0	OK
Flange top a end plate SLN214 ring	12.0	C LC_146	49.8	0.0	0.0	OK
Flange top b end plate SLN214 ring	12.0	C LC_146	59.5	0.0	0.0	OK
Flange bottom a end plate SLN214 ring	12.0	C LC_146	73.3	0.0	0.0	OK
Flange bottom b end plate SLN214 ring	12.0	C LC_146	44.0	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

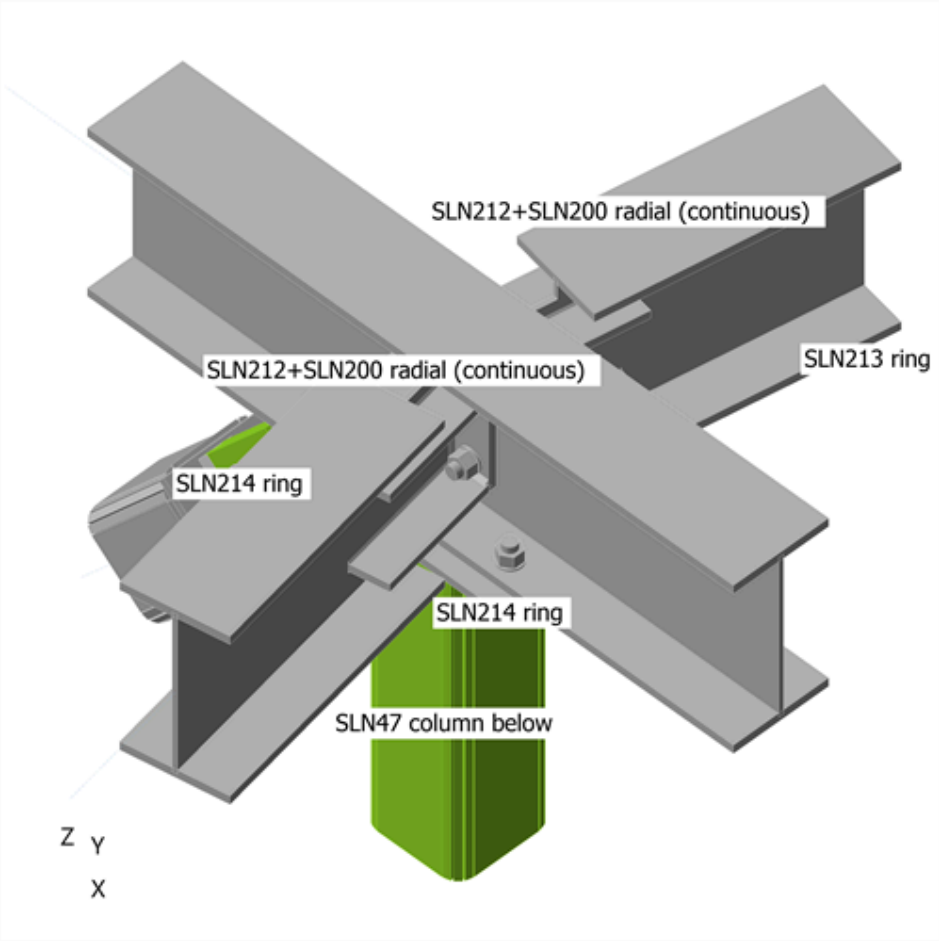
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

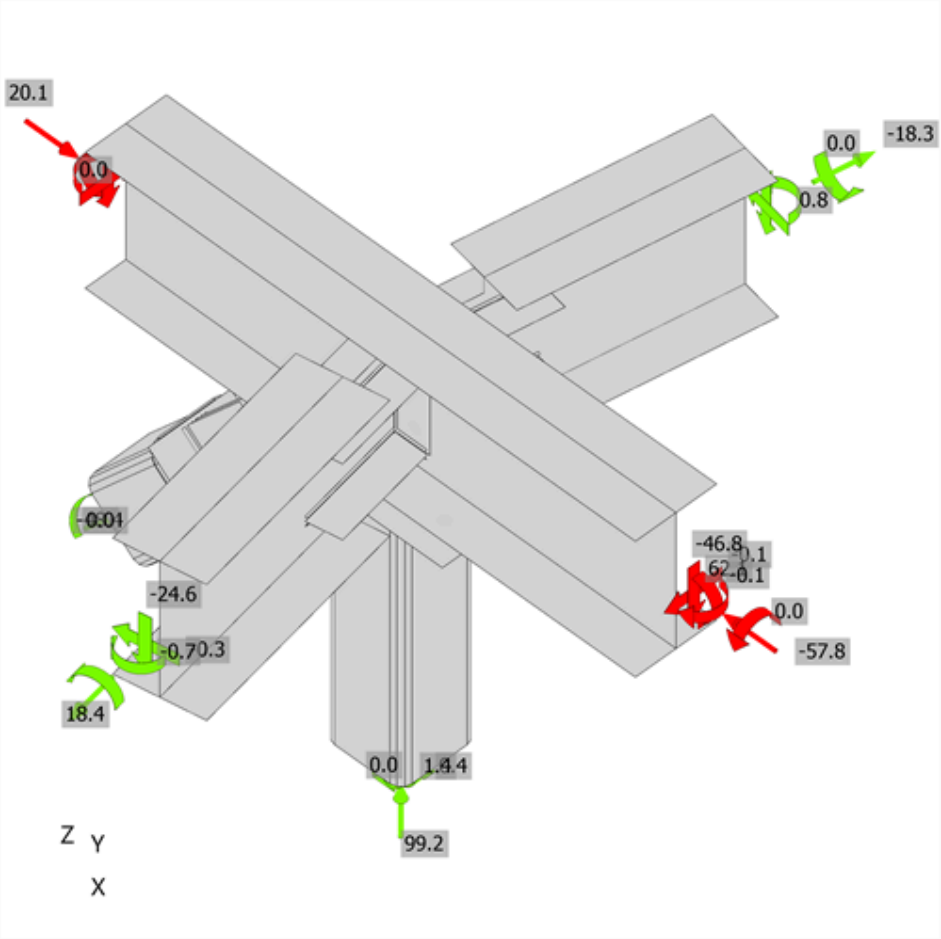
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN47 column below	150	C LC_112	0	5	0.1	OK
SLN434 raking strut	150	C LC_154	0	5	0.0	OK

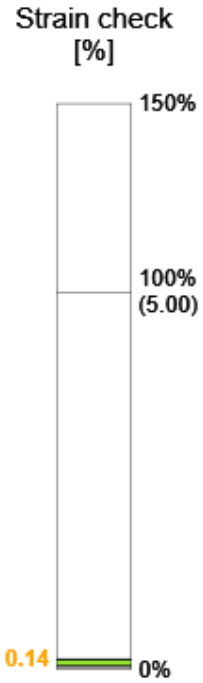
Symbol explanation

d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation

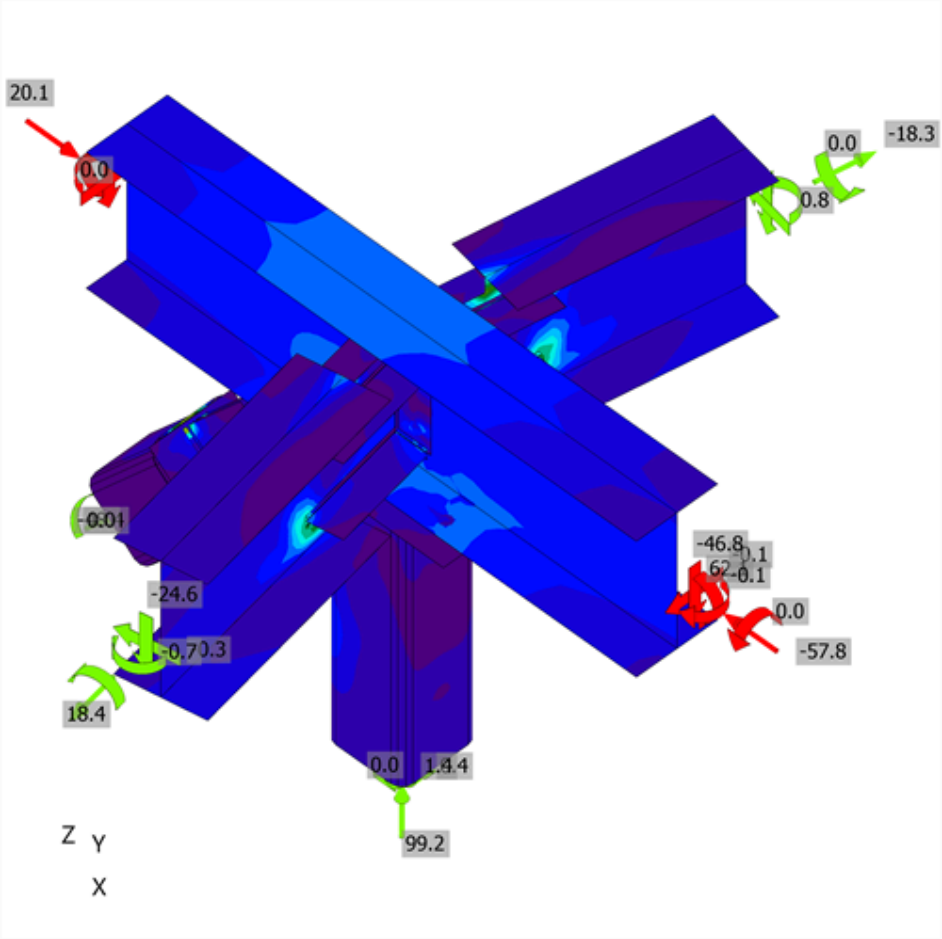


Overall check



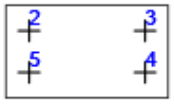
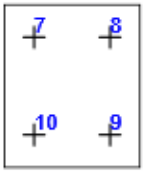
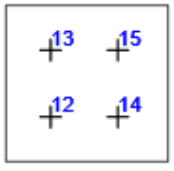


Strain check, C LC_104



Equivalent stress, C LC_104

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_152	19.2	3.5	111.7	13.6	3.8	13.5	OK	OK
	B3	M20 8.8 - 1	C LC_107	22.4	3.8	106.0	15.9	4.0	15.3	OK	OK
	B4	M20 8.8 - 1	C LC_107	21.9	3.7	102.3	15.5	3.9	15.0	OK	OK
	B5	M20 8.8 - 1	C LC_152	19.1	3.5	113.4	13.6	3.8	13.4	OK	OK
	B7	M20 8.8 - 1	C LC_104	11.5	0.9	123.8	8.1	1.0	6.7	OK	OK
	B8	M20 8.8 - 1	C LC_104	0.0	1.0	66.5	0.0	1.5	1.0	OK	OK
	B9	M20 8.8 - 1	C LC_152	0.0	1.3	123.8	0.0	2.0	1.3	OK	OK
	B10	M20 8.8 - 1	C LC_104	12.1	1.0	123.8	8.6	1.1	7.2	OK	OK
	B12	M20 8.8 - 1	C LC_125	10.8	11.9	115.3	7.6	12.6	18.1	OK	OK
	B13	M20 8.8 - 1	C LC_126	4.7	11.0	115.3	3.3	11.7	14.0	OK	OK
	B14	M20 8.8 - 1	C LC_115	0.6	5.3	106.0	0.4	9.8	5.9	OK	OK
	B15	M20 8.8 - 1	C LC_112	0.1	6.9	115.3	0.1	11.7	7.4	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN434 raking strut	SLN434 raking strut	◢ 4.2	519	C LC_126	159.4	0.0	-70.7	82.3	-5.1	39.4	21.7	OK	OK
SLN47 column below-w 3	Gusset SLN434 raking strut	◢ 5.6 ◣	288	C LC_104	397.3	0.4	-129.6	-129.1	-174.2	98.2	63.6	OK	OK
		◢ 5.6 ◣	288	C LC_104	397.3	0.4	-128.1	128.5	175.0	98.2	63.5	OK	OK
SLN213 ring-w 1	Flange top b end plate SLN213 ring	◢ 4.2 ◣	198	C LC_146	68.2	0.0	37.4	30.7	11.8	16.8	10.7	OK	OK
		◢ 4.2 ◣	197	C LC_104	33.8	0.0	-16.5	15.1	7.9	8.3	6.9	OK	OK
SLN213 ring-w 1	Flange bottom b end plate SLN213 ring	◢ 4.2 ◣	198	C LC_146	95.0	0.0	28.2	33.5	40.2	23.5	14.3	OK	OK
		◢ 4.2 ◣	198	C LC_146	140.5	0.0	57.1	-51.8	-53.1	34.7	21.3	OK	OK
SLN214 ring-w 1	Flange top a end plate SLN214 ring	◢ 4.2 ◣	198	C LC_146	63.4	0.0	34.5	28.9	-10.6	15.7	8.9	OK	OK
		◢ 4.2 ◣	197	C LC_146	30.2	0.0	-15.3	13.8	-5.8	7.5	6.6	OK	OK
SLN214 ring-w 1	Flange bottom a end plate SLN214 ring	◢ 4.2 ◣	198	C LC_146	92.4	0.0	27.6	31.6	-39.9	22.8	12.2	OK	OK
		◢ 4.2 ◣	198	C LC_146	134.2	0.0	53.2	-49.3	51.3	33.2	18.1	OK	OK
Cap plate SLN434 raking strut	Fin plate SLN434 raking strut	◢ 5.6 ◣	139	C LC_125	217.6	0.0	-96.8	-89.0	68.8	53.8	40.3	OK	OK
		◢ 5.6 ◣	139	C LC_125	145.6	0.0	-61.9	69.7	-30.5	36.0	20.1	OK	OK
Cap plate SLN434 raking strut	Fin plate 2 SLN434 raking strut	◢ 5.6 ◣	139	C LC_126	142.2	0.0	-58.9	-68.4	30.0	35.1	19.7	OK	OK
		◢ 5.6 ◣	139	C LC_126	217.3	0.0	-98.0	88.6	-68.6	53.7	40.2	OK	OK
End plate column below SLN47 column below	SLN47 column below	◢ 5.6 ◣	521	C LC_104	299.0	0.0	-193.1	-128.8	-28.1	73.9	40.4	OK	OK
		◢ 5.6 ◣	521	C LC_152	212.9	0.0	-91.8	110.0	13.9	52.6	31.6	OK	OK
SLN212+SLN200 radial (continuous)-w 1	SLN213 ring-w 1	◢ 7.2	189	C LC_119	89.6	0.0	32.2	28.1	39.2	22.1	13.6	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
SLN212+SLN200 radial (continuous)-w 1	SLN214 ring-w 1	◢ 7.2	189	C LC_121	94.3	0.0	3.3	3.1	54.3	23.3	14.7	OK	OK
End plate SLN213 ring	SLN213 ring-w 1	◢ 4.2 ◣	149	C LC_119	87.1	0.0	-34.6	-19.0	-42.1	21.5	12.3	OK	OK
		◢ 4.2 ◣	149	C LC_146	70.0	0.0	-12.1	27.5	28.8	17.3	11.2	OK	OK
End plate SLN214 ring	SLN214 ring-w 1	◢ 4.2 ◣	149	C LC_119	83.8	0.0	-23.3	-26.4	38.3	20.7	13.3	OK	OK
		◢ 4.2 ◣	149	C LC_146	72.5	0.0	-18.6	24.9	-31.9	17.9	11.5	OK	OK
SLN213 ring-w 1	Flange top a end plate SLN213 ring	◢ 4.2 ◣	199	C LC_104	117.3	0.0	-10.3	6.1	-67.2	29.0	20.4	OK	OK
		◢ 4.2 ◣	198	C LC_104	76.9	0.0	11.7	13.5	-41.8	19.0	11.1	OK	OK
SLN213 ring-w 1	Flange bottom a end plate SLN213 ring	◢ 4.2 ◣	198	C LC_120	103.5	0.0	19.0	-19.1	55.5	25.6	15.1	OK	OK
		◢ 4.2 ◣	198	C LC_104	84.5	0.0	-10.3	-2.4	48.3	20.9	12.1	OK	OK
SLN214 ring-w 1	Flange top b end plate SLN214 ring	◢ 4.2 ◣	199	C LC_146	104.0	0.0	-9.7	5.7	59.5	25.7	16.0	OK	OK
		◢ 4.2 ◣	198	C LC_146	70.3	0.0	11.2	12.8	38.0	17.4	10.2	OK	OK
SLN214 ring-w 1	Flange bottom b end plate SLN214 ring	◢ 4.2 ◣	198	C LC_146	97.3	0.0	18.2	-17.7	-52.3	24.0	14.0	OK	OK
		◢ 4.2 ◣	199	C LC_119	72.3	0.0	-12.4	13.6	-38.8	17.9	11.2	OK	OK
End plate SLN213 ring	Flange top a end plate SLN213 ring	◢ 4.2 ◣	58	C LC_104	34.7	0.0	0.9	2.1	-19.9	8.6	7.1	OK	OK
		◢ 4.2 ◣	58	C LC_104	22.2	0.0	-6.1	-10.9	-5.8	5.5	5.5	OK	OK
End plate SLN213 ring	Flange top b end plate SLN213 ring	◢ 4.2 ◣	58	C LC_120	53.9	0.0	-1.4	-0.9	-31.1	13.3	9.8	OK	OK
		◢ 4.2 ◣	58	C LC_104	66.3	0.0	37.8	26.0	-17.6	16.4	12.9	OK	OK
End plate SLN213 ring	Flange bottom a end plate SLN213 ring	◢ 4.2 ◣	58	C LC_119	53.9	0.0	12.1	-18.6	23.9	13.3	8.8	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_120	43.9	0.0	-12.0	-16.6	17.9	10.8	8.4	OK	OK
End plate SLN213 ring	Flange bottom b end plate SLN213 ring	◢ 4.2 ◣	58	C LC_120	76.7	0.0	43.4	-32.8	16.0	19.0	14.8	OK	OK
		◢ 4.2 ◣	58	C LC_120	63.8	0.0	-4.4	0.1	36.7	15.8	10.8	OK	OK
End plate SLN214 ring	Flange top a end plate SLN214 ring	◢ 4.2 ◣	58	C LC_146	43.8	0.0	0.6	-3.1	25.1	10.8	9.2	OK	OK
		◢ 4.2 ◣	58	C LC_104	51.0	0.0	6.2	10.3	27.3	12.6	10.0	OK	OK
End plate SLN214 ring	Flange top b end plate SLN214 ring	◢ 4.2 ◣	58	C LC_114	39.7	0.0	-19.8	15.9	-11.9	9.8	7.7	OK	OK
		◢ 4.2 ◣	58	C LC_114	38.5	0.0	-12.7	-16.6	12.9	9.5	8.0	OK	OK
End plate SLN214 ring	Flange bottom a end plate SLN214 ring	◢ 4.2 ◣	58	C LC_146	65.0	0.0	35.7	-28.9	-12.1	16.1	12.5	OK	OK
		◢ 4.2 ◣	58	C LC_146	52.9	0.0	-3.5	1.4	-30.5	13.1	10.0	OK	OK
End plate SLN214 ring	Flange bottom b end plate SLN214 ring	◢ 4.2 ◣	58	C LC_119	57.6	0.0	11.7	-21.8	-24.2	14.2	7.9	OK	OK
		◢ 4.2 ◣	58	C LC_121	43.8	0.0	-10.3	-19.8	-14.6	10.8	7.2	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-S_220
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN216+SLN201 radial (continuous) - SLN49 column below - SLN217 ring - SLN218 ring - SLN447 raking strut

Name	SLN216+SLN201 radial (continuous) - SLN49 column below - SLN217 ring - SLN218 ring - SLN447 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

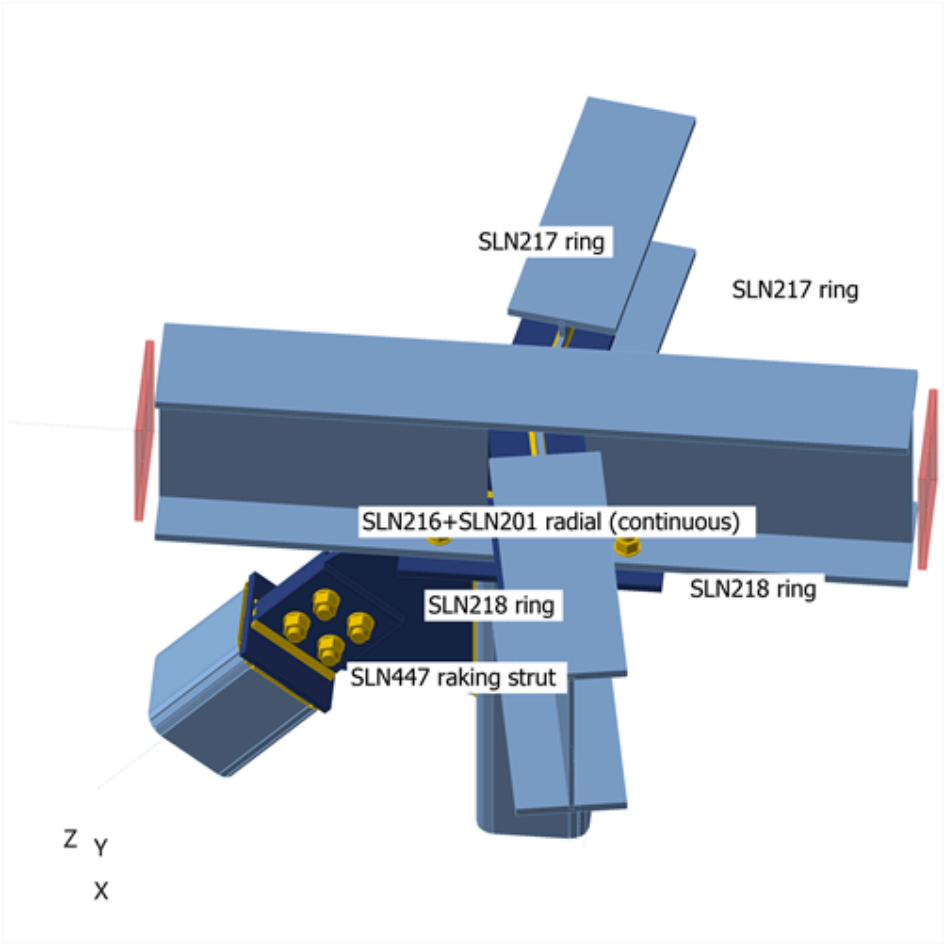
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN216+SLN201 radial (continuous)	1 - UB254x146x43	40.0	0.0	0.0	0	0	0
SLN49 column below	2 - SHS150x150x10	0.0	90.0	40.0	0	0	0
SLN217 ring	1 - UB254x146x43	-60.0	0.0	0.0	0	0	0
SLN218 ring	1 - UB254x146x43	-40.0	0.0	0.0	0	0	0
SLN447 raking strut	2 - SHS150x150x10	-139.9	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN216+SLN201 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN216+SLN201 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN49 column below / begin		Position	0
SLN217 ring / begin		Position	0
SLN218 ring / end		Position	0
SLN447 raking strut / end		Position	0



SLN216+SLN201 radial (continuous) - SLN49 column below - SLN217 ring - SLN218 ring - SLN447 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN49 column below / Begin	26.0	0.0	0.1	0.0	0.6	0.0
	SLN217 ring / Begin	-8.8	-0.1	-4.9	0.0	0.0	0.2
	SLN218 ring / End	8.8	-0.1	-5.0	0.0	0.0	-0.2
	SLN447 raking strut / End	-3.3	0.0	-1.0	0.0	0.0	0.0
C LC_102	SLN49 column below / Begin	122.8	0.0	0.3	0.0	1.2	0.0
	SLN217 ring / Begin	-27.7	-0.2	-25.6	0.0	0.0	0.6
	SLN218 ring / End	27.6	-0.2	-26.1	0.0	0.0	-0.6
	SLN447 raking strut / End	-24.5	0.0	-1.0	-0.1	0.0	0.0
C LC_103	SLN49 column below / Begin	127.4	-1.6	-1.8	0.0	-0.4	0.4
	SLN217 ring / Begin	-27.0	-0.1	-24.6	0.0	0.0	0.5
	SLN218 ring / End	25.1	0.0	-25.0	0.0	0.0	-0.2
	SLN447 raking strut / End	-10.6	0.0	-1.1	0.0	0.0	-0.4
C LC_104	SLN49 column below / Begin	107.4	0.0	0.3	0.0	1.1	0.1
	SLN217 ring / Begin	-18.3	-0.3	-24.6	0.0	0.0	0.7
	SLN218 ring / End	18.5	-0.3	-25.0	0.0	0.0	-0.6
	SLN447 raking strut / End	-37.2	-0.1	-0.9	-0.1	0.0	-0.1
C LC_105	SLN49 column below / Begin	107.9	0.0	0.3	0.0	1.1	-0.1
	SLN217 ring / Begin	-18.6	-0.2	-24.6	0.0	0.0	0.6
	SLN218 ring / End	18.0	-0.3	-25.0	0.0	0.0	-0.7
	SLN447 raking strut / End	-36.6	0.2	-0.9	0.0	0.0	0.2
C LC_106	SLN49 column below / Begin	125.4	1.9	-1.5	0.0	-0.1	-0.5
	SLN217 ring / Begin	-24.7	-0.1	-24.6	0.0	0.0	0.4
	SLN218 ring / End	26.7	-0.1	-25.0	0.0	0.0	-0.5
	SLN447 raking strut / End	-13.1	0.0	-1.1	-0.2	0.0	0.5
C LC_107	SLN49 column below / Begin	27.0	-3.1	-4.2	0.0	-2.7	0.8
	SLN217 ring / Begin	-9.4	0.0	-1.6	0.0	0.0	0.1
	SLN218 ring / End	6.1	0.1	-1.6	0.0	0.0	0.3
	SLN447 raking strut / End	23.1	0.0	-1.0	0.2	0.0	-0.9
C LC_108	SLN49 column below / Begin	-6.0	0.0	0.1	0.0	0.3	0.1
	SLN217 ring / Begin	4.3	-0.1	-1.6	0.0	0.0	0.3
	SLN218 ring / End	-4.1	0.0	-1.6	0.0	0.0	-0.1
	SLN447 raking strut / End	-21.0	-0.3	-0.5	0.0	0.0	-0.2
C LC_109	SLN49 column below / Begin	-5.0	0.0	0.1	0.0	0.3	-0.2
	SLN217 ring / Begin	4.5	0.0	-1.6	0.0	0.0	0.0
	SLN218 ring / End	-4.9	-0.1	-1.6	0.0	0.0	-0.3
	SLN447 raking strut / End	-19.6	0.4	-0.5	0.0	0.0	0.3
C LC_110	SLN49 column below / Begin	23.1	3.7	-3.5	0.0	-2.2	-1.0
	SLN217 ring / Begin	-5.3	0.1	-1.6	0.0	0.0	-0.2
	SLN218 ring / End	9.3	0.0	-1.6	0.0	0.0	0.0
	SLN447 raking strut / End	18.1	0.0	-1.0	-0.3	0.0	1.1
C LC_111	SLN49 column below / Begin	101.9	-3.1	-4.0	0.0	-2.1	0.8
	SLN217 ring / Begin	-23.4	-0.1	-17.3	0.0	0.0	0.4
	SLN218 ring / End	19.9	0.1	-17.6	0.0	0.0	0.1
	SLN447 raking strut / End	8.0	0.0	-1.3	0.2	0.0	-0.9
C LC_112	SLN49 column below / Begin	66.6	0.0	0.2	0.0	0.9	0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN217 ring / Begin	-8.5	-0.3	-17.3	0.0	0.0	0.6
	SLN218 ring / End	8.7	-0.2	-17.6	0.0	0.0	-0.4
	SLN447 raking strut / End	-39.0	-0.3	-0.8	0.0	0.0	-0.2
C LC_113	SLN49 column below / Begin	67.7	0.0	0.2	0.0	0.9	-0.2
	SLN217 ring / Begin	-8.5	-0.2	-17.3	0.0	0.0	0.4
	SLN218 ring / End	7.9	-0.3	-17.6	0.0	0.0	-0.7
	SLN447 raking strut / End	-37.6	0.4	-0.8	0.0	0.0	0.3
C LC_114	SLN49 column below / Begin	98.0	3.7	-3.3	0.0	-1.6	-1.0
	SLN217 ring / Begin	-19.1	0.0	-17.3	0.0	0.0	0.1
	SLN218 ring / End	23.1	-0.1	-17.6	0.0	0.0	-0.3
	SLN447 raking strut / End	3.1	0.0	-1.2	-0.3	0.0	1.1
C LC_115	SLN49 column below / Begin	33.8	-3.1	-4.1	0.0	-2.5	0.8
	SLN217 ring / Begin	-11.7	0.0	-2.8	0.0	0.0	0.2
	SLN218 ring / End	8.5	0.1	-2.9	0.0	0.0	0.3
	SLN447 raking strut / End	22.4	0.0	-1.3	0.2	0.0	-0.9
C LC_116	SLN49 column below / Begin	0.6	0.0	0.1	0.0	0.5	0.1
	SLN217 ring / Begin	2.1	-0.1	-2.8	0.0	0.0	0.3
	SLN218 ring / End	-1.9	-0.1	-2.9	0.0	0.0	-0.1
	SLN447 raking strut / End	-22.0	-0.3	-0.8	0.0	0.0	-0.2
C LC_117	SLN49 column below / Begin	1.6	0.0	0.1	0.0	0.5	-0.2
	SLN217 ring / Begin	2.2	0.0	-2.8	0.0	0.0	0.0
	SLN218 ring / End	-2.7	-0.1	-2.9	0.0	0.0	-0.4
	SLN447 raking strut / End	-20.7	0.4	-0.8	0.0	0.0	0.3
C LC_118	SLN49 column below / Begin	30.0	3.7	-3.4	0.0	-2.0	-1.0
	SLN217 ring / Begin	-7.7	0.1	-2.8	0.0	0.0	-0.1
	SLN218 ring / End	11.6	0.0	-2.9	0.0	0.0	-0.1
	SLN447 raking strut / End	17.4	0.0	-1.2	-0.3	0.0	1.1
C LC_119	SLN49 column below / Begin	130.9	-1.6	-1.8	0.0	-0.3	0.4
	SLN217 ring / Begin	-29.3	-0.1	-25.6	0.0	0.0	0.6
	SLN218 ring / End	27.4	-0.1	-26.1	0.0	0.0	-0.3
	SLN447 raking strut / End	-13.6	0.0	-1.1	0.0	0.0	-0.4
C LC_120	SLN49 column below / Begin	114.9	0.0	0.3	0.0	1.2	0.1
	SLN217 ring / Begin	-20.5	-0.3	-25.6	0.0	0.0	0.7
	SLN218 ring / End	20.6	-0.3	-26.1	0.0	0.0	-0.6
	SLN447 raking strut / End	-34.8	-0.1	-0.9	-0.1	0.0	-0.1
C LC_121	SLN49 column below / Begin	114.5	0.0	0.3	0.0	1.2	-0.1
	SLN217 ring / Begin	-20.8	-0.2	-25.6	0.0	0.0	0.5
	SLN218 ring / End	20.2	-0.3	-26.1	0.0	0.0	-0.7
	SLN447 raking strut / End	-35.3	0.2	-0.9	0.0	0.0	0.2
C LC_122	SLN49 column below / Begin	129.0	1.9	-1.5	0.0	0.0	-0.5
	SLN217 ring / Begin	-27.0	-0.1	-25.6	0.0	0.0	0.4
	SLN218 ring / End	29.0	-0.1	-26.1	0.0	0.0	-0.5
	SLN447 raking strut / End	-16.0	0.0	-1.1	-0.2	0.0	0.5
C LC_123	SLN49 column below / Begin	108.9	-3.1	-4.0	0.0	-1.9	0.8
	SLN217 ring / Begin	-28.0	-0.1	-19.4	0.0	0.0	0.6
	SLN218 ring / End	24.6	0.0	-19.7	0.0	0.0	-0.1
	SLN447 raking strut / End	2.1	0.0	-1.3	0.2	0.0	-0.9

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_124	SLN49 column below / Begin	81.6	0.0	0.2	0.0	1.0	0.1
	SLN217 ring / Begin	-12.8	-0.3	-19.4	0.0	0.0	0.6
	SLN218 ring / End	13.0	-0.2	-19.7	0.0	0.0	-0.4
	SLN447 raking strut / End	-34.0	-0.3	-0.8	-0.1	0.0	-0.2
C LC_125	SLN49 column below / Begin	80.7	0.0	0.2	0.0	1.0	-0.2
	SLN217 ring / Begin	-12.9	-0.2	-19.4	0.0	0.0	0.3
	SLN218 ring / End	12.2	-0.3	-19.7	0.0	0.0	-0.7
	SLN447 raking strut / End	-35.2	0.4	-0.8	0.0	0.0	0.3
C LC_126	SLN49 column below / Begin	105.1	3.7	-3.3	0.0	-1.4	-1.0
	SLN217 ring / Begin	-23.7	-0.1	-19.4	0.0	0.0	0.3
	SLN218 ring / End	27.8	-0.1	-19.7	0.0	0.0	-0.4
	SLN447 raking strut / End	-2.7	0.0	-1.2	-0.3	0.0	1.1
C LC_127	SLN49 column below / Begin	40.9	-3.1	-4.1	0.0	-2.3	0.8
	SLN217 ring / Begin	-16.4	-0.1	-4.9	0.0	0.0	0.3
	SLN218 ring / End	13.1	0.1	-5.0	0.0	0.0	0.2
	SLN447 raking strut / End	16.5	0.0	-1.3	0.2	0.0	-0.9
C LC_128	SLN49 column below / Begin	15.6	0.0	0.1	0.0	0.6	0.1
	SLN217 ring / Begin	-2.2	-0.1	-4.9	0.0	0.0	0.3
	SLN218 ring / End	2.5	-0.1	-5.0	0.0	0.0	-0.1
	SLN447 raking strut / End	-17.0	-0.3	-0.8	0.0	0.0	-0.2
C LC_129	SLN49 column below / Begin	14.7	0.0	0.1	0.0	0.6	-0.2
	SLN217 ring / Begin	-2.2	0.0	-4.9	0.0	0.0	0.0
	SLN218 ring / End	1.6	-0.2	-5.0	0.0	0.0	-0.4
	SLN447 raking strut / End	-18.2	0.4	-0.8	0.0	0.0	0.3
C LC_130	SLN49 column below / Begin	37.1	3.7	-3.4	0.0	-1.8	-1.0
	SLN217 ring / Begin	-12.2	0.0	-4.9	0.0	0.0	0.0
	SLN218 ring / End	16.3	-0.1	-5.0	0.0	0.0	-0.2
	SLN447 raking strut / End	11.7	0.0	-1.2	-0.3	0.0	1.1
C LC_135	SLN49 column below / Begin	122.4	0.0	0.3	0.0	1.2	0.0
	SLN217 ring / Begin	-28.2	-0.2	-25.6	0.0	0.0	0.6
	SLN218 ring / End	28.0	-0.2	-26.1	0.0	0.0	-0.6
	SLN447 raking strut / End	-25.0	0.0	-1.0	-0.1	0.0	0.0
C LC_141	SLN49 column below / Begin	26.5	0.0	0.1	0.0	0.6	0.0
	SLN217 ring / Begin	-9.0	0.0	-4.9	0.0	0.0	0.2
	SLN218 ring / End	9.0	0.0	-5.0	0.0	0.0	-0.1
	SLN447 raking strut / End	-2.7	0.0	-1.0	0.0	0.0	0.0
C LC_142	SLN49 column below / Begin	25.6	0.0	0.2	0.0	0.6	0.0
	SLN217 ring / Begin	-8.6	-0.1	-4.9	0.0	0.0	0.2
	SLN218 ring / End	8.6	-0.1	-5.0	0.0	0.0	-0.2
	SLN447 raking strut / End	-3.9	0.0	-1.0	0.0	0.0	0.0
C LC_143	SLN49 column below / Begin	25.6	0.0	0.2	0.0	0.6	0.0
	SLN217 ring / Begin	-8.7	-0.1	-4.9	0.0	0.0	0.2
	SLN218 ring / End	8.6	-0.1	-5.0	0.0	0.0	-0.2
	SLN447 raking strut / End	-3.9	0.0	-1.0	0.0	0.0	0.0
C LC_144	SLN49 column below / Begin	26.5	0.0	0.1	0.0	0.6	0.0
	SLN217 ring / Begin	-9.0	0.0	-4.9	0.0	0.0	0.2
	SLN218 ring / End	9.0	0.0	-5.0	0.0	0.0	-0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN447 raking strut / End	-2.7	0.0	-1.0	0.0	0.0	0.0
C LC_145	SLN49 column below / Begin	124.7	0.0	0.3	0.0	1.3	0.0
	SLN217 ring / Begin	-28.7	-0.1	-25.6	0.0	0.0	0.5
	SLN218 ring / End	28.4	-0.1	-26.1	0.0	0.0	-0.5
	SLN447 raking strut / End	-22.0	0.0	-1.0	-0.1	0.0	0.0
C LC_146	SLN49 column below / Begin	120.9	0.0	0.3	0.0	1.2	0.0
	SLN217 ring / Begin	-26.7	-0.3	-25.6	0.0	0.0	0.7
	SLN218 ring / End	26.7	-0.3	-26.1	0.0	0.0	-0.6
	SLN447 raking strut / End	-27.0	0.0	-1.0	-0.1	0.0	0.0
C LC_147	SLN49 column below / Begin	120.9	0.0	0.3	0.0	1.2	0.0
	SLN217 ring / Begin	-27.0	-0.3	-25.6	0.0	0.0	0.7
	SLN218 ring / End	26.7	-0.3	-26.1	0.0	0.0	-0.6
	SLN447 raking strut / End	-27.0	0.0	-1.0	-0.1	0.0	0.0
C LC_148	SLN49 column below / Begin	124.7	0.0	0.3	0.0	1.3	0.0
	SLN217 ring / Begin	-28.4	-0.1	-25.6	0.0	0.0	0.5
	SLN218 ring / End	28.4	-0.1	-26.1	0.0	0.0	-0.5
	SLN447 raking strut / End	-22.0	0.0	-1.0	-0.1	0.0	0.0
C LC_151	SLN49 column below / Begin	77.5	0.0	-1.6	0.0	-6.6	0.0
	SLN217 ring / Begin	-12.3	-0.1	-11.8	0.0	0.0	0.3
	SLN218 ring / End	12.3	-0.1	-11.6	0.0	0.0	-0.3
	SLN447 raking strut / End	-15.4	0.0	-1.0	0.0	0.0	0.0
C LC_152	SLN49 column below / Begin	72.2	0.0	2.1	0.0	8.7	0.0
	SLN217 ring / Begin	-28.5	-0.2	-14.1	0.0	0.0	0.5
	SLN218 ring / End	28.4	-0.2	-14.0	0.0	0.0	-0.5
	SLN447 raking strut / End	-9.5	0.0	-1.0	0.0	0.0	0.0
C LC_153	SLN49 column below / Begin	122.1	0.0	0.3	0.0	1.2	0.0
	SLN217 ring / Begin	-27.8	-0.2	-21.0	0.0	0.0	0.6
	SLN218 ring / End	27.7	-0.2	-20.6	0.0	0.0	-0.6
	SLN447 raking strut / End	-24.2	0.0	-1.0	-0.1	0.0	0.0
C LC_154	SLN49 column below / Begin	77.0	0.0	-1.6	0.0	-6.5	0.0
	SLN217 ring / Begin	-12.5	-0.1	-11.8	0.0	0.0	0.3
	SLN218 ring / End	12.4	-0.1	-11.6	0.0	0.0	-0.3
	SLN447 raking strut / End	-15.3	0.0	-1.0	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	14.4 < 100%	OK
Welds	98.1 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN216+SLN201 radial (continuous)-bfl 1	12.7	C LC_121	141.5	0.0	46.7	OK
SLN216+SLN201 radial (continuous)-tfl 1	12.7	C LC_153	71.8	0.0	0.0	OK
SLN216+SLN201 radial (continuous)-w 1	7.2	C LC_121	160.0	0.0	46.7	OK
SLN49 column below	10.0	C LC_104	275.1	0.0	0.0	OK
SLN217 ring-bfl 1	12.7	C LC_120	49.9	0.0	0.0	OK
SLN217 ring-tfl 1	12.7	C LC_120	44.6	0.0	0.0	OK
SLN217 ring-w 1	7.2	C LC_120	152.6	0.0	0.0	OK
SLN218 ring-bfl 1	12.7	C LC_121	51.3	0.0	0.0	OK
SLN218 ring-tfl 1	12.7	C LC_121	46.7	0.0	0.0	OK
SLN218 ring-w 1	7.2	C LC_121	155.8	0.0	0.0	OK
SLN447 raking strut	10.0	C LC_115	35.9	0.0	0.0	OK
End plate column below SLN49 column below	12.0	C LC_121	115.4	0.0	39.8	OK
End plate SLN217 ring	12.0	C LC_120	70.4	0.0	0.0	OK
End plate SLN218 ring	12.0	C LC_146	57.1	0.0	15.7	OK
Cap plate SLN447 raking strut	12.0	C LC_118	70.5	0.0	0.0	OK
Fin plate SLN447 raking strut	12.0	C LC_118	217.2	0.0	61.7	OK
Fin plate 2 SLN447 raking strut	12.0	C LC_115	194.9	0.0	55.4	OK
Gusset SLN447 raking strut	12.0	C LC_126	210.6	0.0	29.8	OK
Flange top a end plate SLN217 ring	12.0	C LC_120	59.5	0.0	0.0	OK
Flange top b end plate SLN217 ring	12.0	C LC_120	47.6	0.0	0.0	OK
Flange bottom a end plate SLN217 ring	12.0	C LC_120	43.7	0.0	0.0	OK
Flange bottom b end plate SLN217 ring	12.0	C LC_147	70.8	0.0	0.0	OK
Flange top a end plate SLN218 ring	12.0	C LC_121	46.3	0.0	0.0	OK
Flange top b end plate SLN218 ring	12.0	C LC_121	59.7	0.0	0.0	OK
Flange bottom a end plate SLN218 ring	12.0	C LC_121	72.1	0.0	0.0	OK
Flange bottom b end plate SLN218 ring	12.0	C LC_125	47.4	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

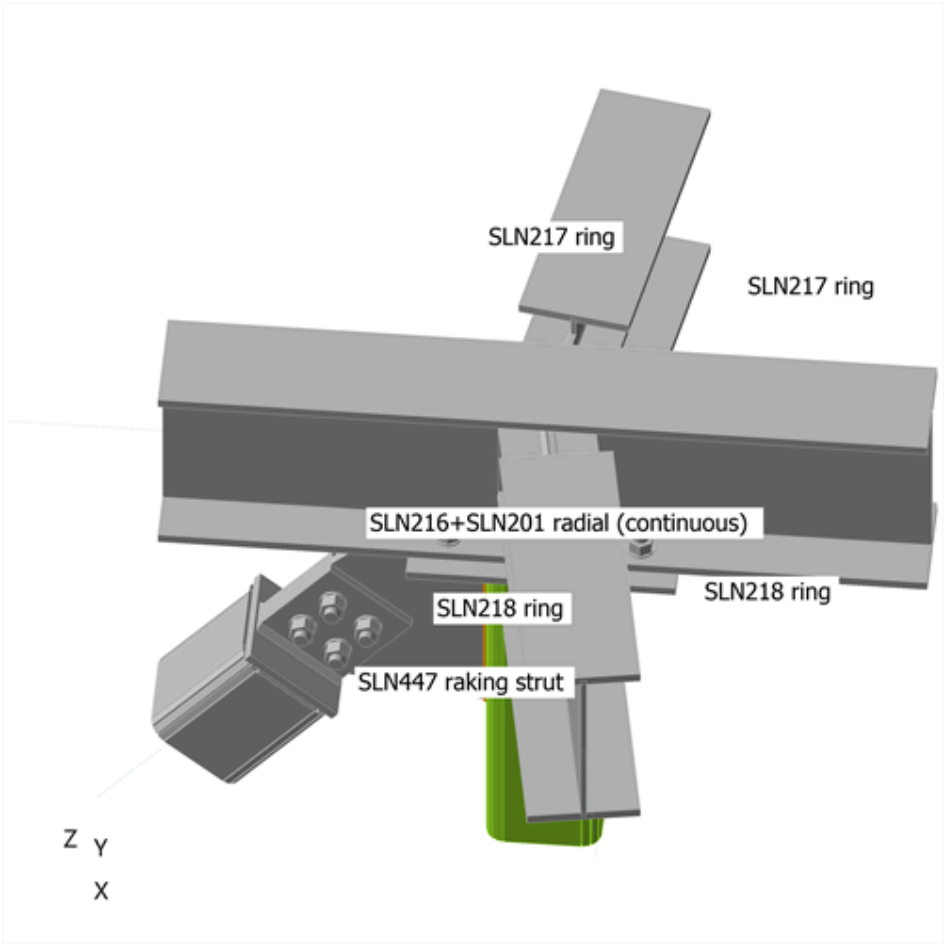
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

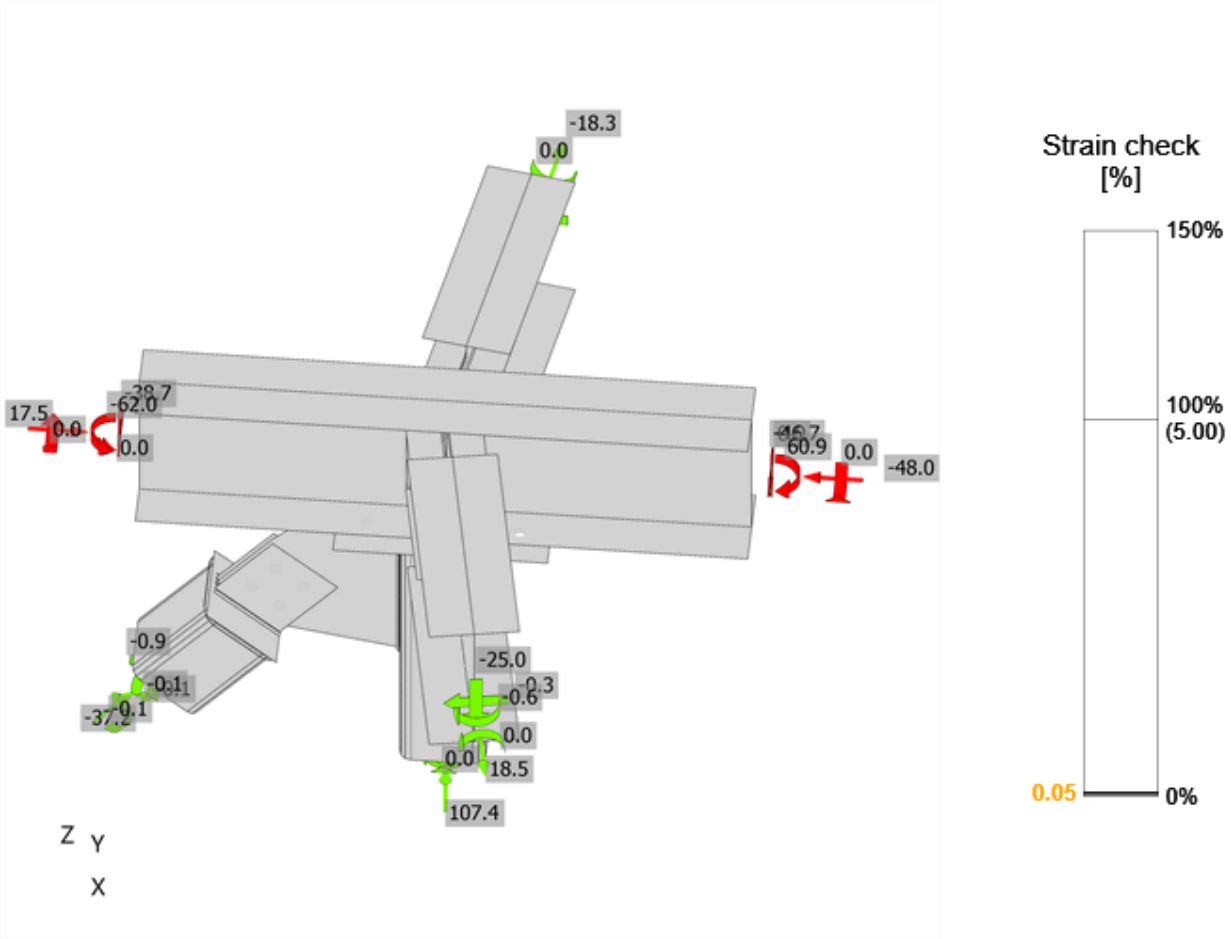
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN49 column below	150	C LC_154	0	5	0.0	OK
SLN447 raking strut	150	C LC_154	0	5	0.0	OK

Symbol explanation

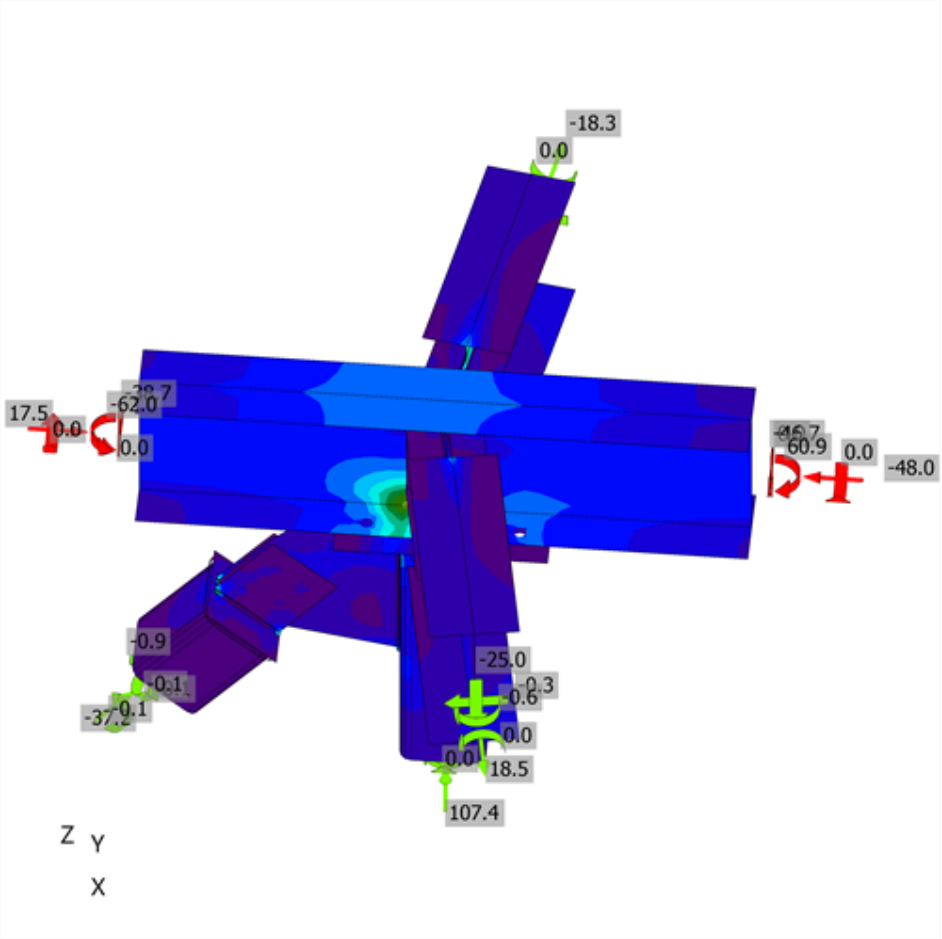
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

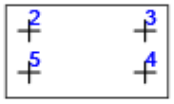
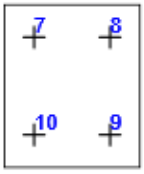
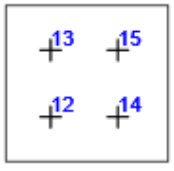


Strain check, C LC_104



Equivalent stress, C LC_104

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_152	15.4	2.7	108.5	10.9	2.9	10.7	OK	OK
	B3	M20 8.8 - 1	C LC_107	15.3	2.4	87.3	10.8	2.8	10.3	OK	OK
	B4	M20 8.8 - 1	C LC_107	14.9	2.9	168.9	10.6	3.1	10.6	OK	OK
	B5	M20 8.8 - 1	C LC_152	15.5	2.8	116.8	11.0	2.9	10.8	OK	OK
	B7	M20 8.8 - 1	C LC_120	9.8	0.6	123.8	7.0	0.8	5.6	OK	OK
	B8	M20 8.8 - 1	C LC_110	2.2	0.4	65.7	1.5	0.4	1.5	OK	OK
	B9	M20 8.8 - 1	C LC_152	0.0	1.2	123.8	0.0	1.9	1.2	OK	OK
	B10	M20 8.8 - 1	C LC_120	10.8	0.8	123.8	7.7	0.8	6.3	OK	OK
	B12	M20 8.8 - 1	C LC_118	9.2	9.1	106.0	6.5	9.7	14.4	OK	OK
	B13	M20 8.8 - 1	C LC_126	3.7	6.4	147.4	2.6	6.8	8.7	OK	OK
	B14	M20 8.8 - 1	C LC_118	1.7	6.7	106.0	1.2	7.2	8.0	OK	OK
	B15	M20 8.8 - 1	C LC_112	0.1	5.5	115.3	0.1	9.5	5.9	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN447 raking strut	SLN447 raking strut	◢ 4.2	519	C LC_112	122.5	0.0	-58.5	62.1	1.3	30.3	15.9	OK	OK
SLN49 column below-w 3	Gusset SLN447 raking strut	◢ 5.6 ◣	288	C LC_104	396.8	0.1	-122.6	-122.8	-180.0	98.1	58.5	OK	OK
		◢ 5.6 ◣	288	C LC_104	396.8	0.1	-125.0	124.4	178.3	98.1	58.8	OK	OK
SLN217 ring-w 1	Flange top b end plate SLN217 ring	◢ 4.2 ◣	198	C LC_123	59.4	0.0	31.5	26.1	12.8	14.7	9.4	OK	OK
		◢ 4.2 ◣	197	C LC_120	30.0	0.0	-14.4	13.1	7.7	7.4	6.5	OK	OK
SLN217 ring-w 1	Flange bottom b end plate SLN217 ring	◢ 4.2 ◣	198	C LC_119	93.3	0.0	27.5	31.8	40.4	23.1	14.2	OK	OK
		◢ 4.2 ◣	198	C LC_147	131.2	0.0	52.0	-47.5	-50.8	32.4	19.9	OK	OK
SLN218 ring-w 1	Flange top a end plate SLN218 ring	◢ 4.2 ◣	198	C LC_152	55.0	0.0	27.0	23.3	-14.9	13.6	7.7	OK	OK
		◢ 4.2 ◣	197	C LC_121	31.6	0.0	-16.1	14.6	-5.7	7.8	6.6	OK	OK
SLN218 ring-w 1	Flange bottom a end plate SLN218 ring	◢ 4.2 ◣	198	C LC_146	93.7	0.0	27.8	31.8	-40.7	23.1	12.4	OK	OK
		◢ 4.2 ◣	198	C LC_121	132.6	0.0	53.6	-49.3	49.7	32.8	17.9	OK	OK
Cap plate SLN447 raking strut	Fin plate SLN447 raking strut	◢ 5.6 ◣	139	C LC_118	193.1	0.0	71.0	84.8	59.6	47.7	34.0	OK	OK
		◢ 5.6 ◣	139	C LC_112	109.6	0.0	-42.8	47.1	-34.3	27.1	14.8	OK	OK
Cap plate SLN447 raking strut	Fin plate 2 SLN447 raking strut	◢ 5.6 ◣	139	C LC_113	110.6	0.0	-44.3	-48.0	33.5	27.3	14.7	OK	OK
		◢ 5.6 ◣	139	C LC_115	189.0	0.0	69.8	-82.8	-58.5	46.7	32.6	OK	OK
End plate column below SLN49 column below	SLN49 column below	◢ 5.6 ◣	521	C LC_104	270.4	0.0	-171.7	-118.5	22.5	66.8	36.8	OK	OK
		◢ 5.6 ◣	521	C LC_152	210.1	0.0	-94.3	106.0	-22.6	51.9	30.6	OK	OK
SLN216+SLN201 radial (continuous)-w 1	SLN217 ring-w 1	◢ 7.2	189	C LC_122	87.1	0.0	26.5	-28.5	-38.5	21.5	14.3	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
SLN216+SLN201 radial (continuous)-w 1	SLN218 ring-w 1	◢ 7.2	189	C LC_122	88.9	0.0	29.5	-29.9	38.0	22.0	13.6	OK	OK
End plate SLN217 ring	SLN217 ring-w 1	◢ 4.2 ◣	149	C LC_122	85.9	0.0	-25.9	-21.4	-42.2	21.2	12.0	OK	OK
		◢ 4.2 ◣	149	C LC_146	72.2	0.0	-15.9	27.4	30.0	17.8	11.4	OK	OK
End plate SLN218 ring	SLN218 ring-w 1	◢ 4.2 ◣	149	C LC_145	79.0	0.0	-24.2	-22.4	37.2	19.5	12.8	OK	OK
		◢ 4.2 ◣	149	C LC_147	68.9	0.0	-19.6	23.3	-30.2	17.0	10.9	OK	OK
SLN217 ring-w 1	Flange top a end plate SLN217 ring	◢ 4.2 ◣	199	C LC_120	102.5	0.0	-9.2	5.6	-58.7	25.3	17.9	OK	OK
		◢ 4.2 ◣	198	C LC_120	65.7	0.0	10.1	11.5	-35.7	16.2	9.6	OK	OK
SLN217 ring-w 1	Flange bottom a end plate SLN217 ring	◢ 4.2 ◣	198	C LC_120	93.2	0.0	17.3	-17.1	50.0	23.0	13.7	OK	OK
		◢ 4.2 ◣	198	C LC_104	69.9	0.0	-8.7	-1.6	40.0	17.3	10.3	OK	OK
SLN218 ring-w 1	Flange top b end plate SLN218 ring	◢ 4.2 ◣	199	C LC_121	99.5	0.0	-50.6	22.1	44.2	24.6	15.6	OK	OK
		◢ 4.2 ◣	198	C LC_125	66.1	0.0	13.6	13.2	34.9	16.3	9.3	OK	OK
SLN218 ring-w 1	Flange bottom b end plate SLN218 ring	◢ 4.2 ◣	198	C LC_121	98.5	0.0	16.7	-16.7	-53.5	24.3	13.9	OK	OK
		◢ 4.2 ◣	198	C LC_125	79.8	0.0	-7.6	-1.7	-45.9	19.7	10.9	OK	OK
End plate SLN217 ring	Flange top a end plate SLN217 ring	◢ 4.2 ◣	58	C LC_104	29.8	0.0	1.2	1.5	-17.1	7.4	6.7	OK	OK
		◢ 4.2 ◣	58	C LC_120	20.5	0.0	-5.9	-9.6	-6.0	5.1	5.1	OK	OK
End plate SLN217 ring	Flange top b end plate SLN217 ring	◢ 4.2 ◣	58	C LC_120	46.8	0.0	-1.1	-0.7	-27.0	11.6	9.0	OK	OK
		◢ 4.2 ◣	58	C LC_120	57.1	0.0	32.4	22.3	-15.5	14.1	11.6	OK	OK
End plate SLN217 ring	Flange bottom a end plate SLN217 ring	◢ 4.2 ◣	58	C LC_122	56.4	0.0	11.4	-21.4	23.7	13.9	9.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_120	44.4	0.0	-11.2	-17.7	17.4	11.0	8.4	OK	OK
End plate SLN217 ring	Flange bottom b end plate SLN217 ring	◢ 4.2 ◣	58	C LC_120	69.5	0.0	39.2	-30.0	14.1	17.2	13.8	OK	OK
		◢ 4.2 ◣	58	C LC_120	56.7	0.0	-4.3	-0.2	32.6	14.0	10.0	OK	OK
End plate SLN218 ring	Flange top a end plate SLN218 ring	◢ 4.2 ◣	58	C LC_107	43.1	0.0	-21.4	20.6	6.3	10.6	6.6	OK	OK
		◢ 4.2 ◣	58	C LC_104	44.5	0.0	5.0	8.7	24.0	11.0	9.1	OK	OK
End plate SLN218 ring	Flange top b end plate SLN218 ring	◢ 4.2 ◣	58	C LC_125	45.0	0.0	-22.1	18.1	-13.6	11.1	7.7	OK	OK
		◢ 4.2 ◣	58	C LC_125	42.8	0.0	-14.0	-17.9	15.0	10.6	8.4	OK	OK
End plate SLN218 ring	Flange bottom a end plate SLN218 ring	◢ 4.2 ◣	58	C LC_146	56.4	0.0	31.1	-25.1	-10.4	13.9	11.5	OK	OK
		◢ 4.2 ◣	58	C LC_111	53.6	0.0	-26.3	-24.3	-11.7	13.2	8.6	OK	OK
End plate SLN218 ring	Flange bottom b end plate SLN218 ring	◢ 4.2 ◣	58	C LC_119	56.7	0.0	11.3	-21.3	-23.9	14.0	7.7	OK	OK
		◢ 4.2 ◣	58	C LC_125	48.5	0.0	-25.4	-19.6	13.6	12.0	9.2	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-S_300
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN223+SLN205 radial (continuous) - SLN53 column below - SLN425 ring - SLN225 ring - SLN449 raking strut

Name	SLN223+SLN205 radial (continuous) - SLN53 column below - SLN425 ring - SLN225 ring - SLN449 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

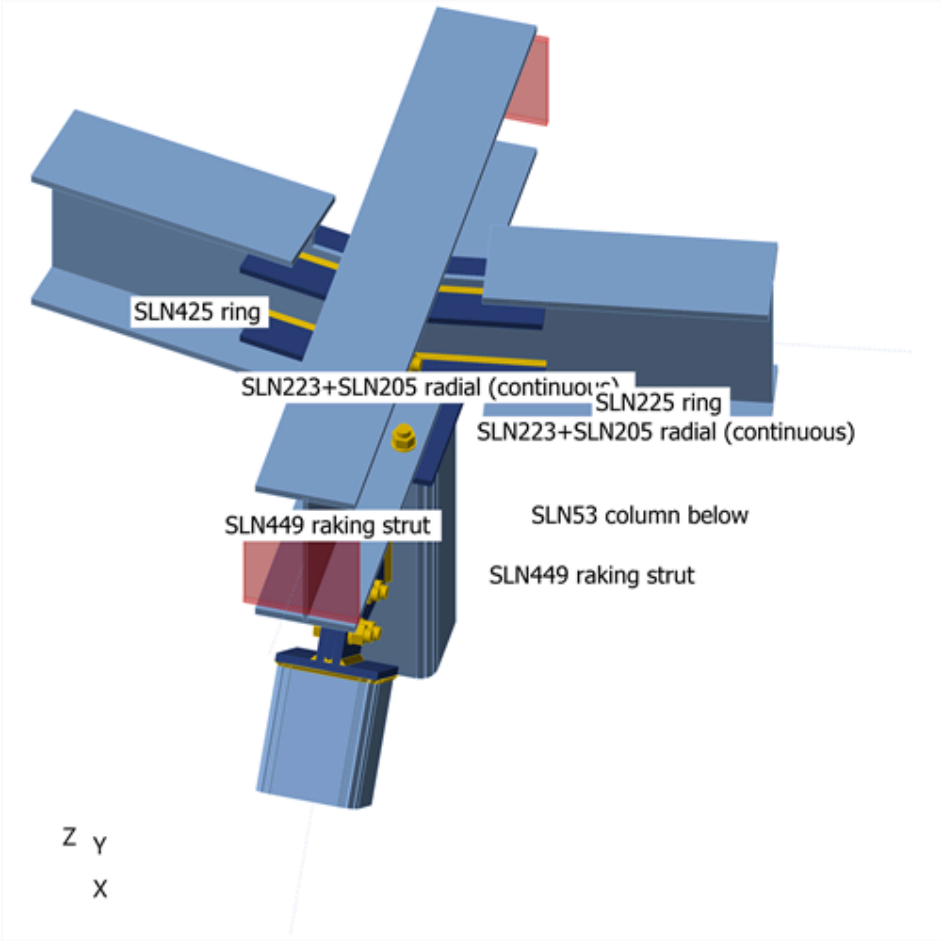
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN223+SLN205 radial (continuous)	1 - UB254x146x43	120.0	0.0	0.0	0	0	0
SLN53 column below	2 - SHS150x150x10	0.0	90.0	120.0	0	0	0
SLN425 ring	1 - UB254x146x43	20.0	0.0	0.0	0	0	0
SLN225 ring	1 - UB254x146x43	40.0	0.0	0.0	0	0	0
SLN449 raking strut	2 - SHS150x150x10	-60.0	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN223+SLN205 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN223+SLN205 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN53 column below / begin		Position	0
SLN425 ring / begin		Position	0
SLN225 ring / end		Position	0
SLN449 raking strut / end		Position	0



SLN223+SLN205 radial (continuous) - SLN53 column below - SLN425 ring - SLN225 ring - SLN449 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN53 column below / Begin	32.3	-0.1	0.3	0.0	1.3	0.2
	SLN425 ring / Begin	-8.8	0.1	-4.9	0.0	0.0	-0.2
	SLN225 ring / End	6.2	-0.2	-4.9	0.0	0.0	-0.5
	SLN449 raking strut / End	-0.5	-0.1	-1.0	-0.1	0.0	-0.3
C LC_102	SLN53 column below / Begin	150.5	-0.2	1.4	0.0	5.7	0.9
	SLN425 ring / Begin	-27.5	0.2	-25.5	0.0	0.0	-0.6
	SLN225 ring / End	18.1	-0.6	-25.4	0.0	0.0	-1.8
	SLN449 raking strut / End	-13.6	-0.3	-1.0	-0.2	0.0	-1.0
C LC_103	SLN53 column below / Begin	134.7	-0.2	1.3	0.0	5.4	0.8
	SLN425 ring / Begin	-19.2	0.1	-24.5	0.0	0.0	-0.5
	SLN225 ring / End	9.8	-0.6	-24.4	0.0	0.0	-1.7
	SLN449 raking strut / End	-25.3	-0.1	-0.9	-0.2	0.0	-0.8
C LC_104	SLN53 column below / Begin	150.7	1.9	0.0	0.0	4.7	0.3
	SLN425 ring / Begin	-22.4	0.3	-24.5	0.0	0.0	-0.7
	SLN225 ring / End	16.2	-0.5	-24.4	0.0	0.0	-1.6
	SLN449 raking strut / End	-6.2	-0.3	-1.1	-0.3	0.0	-0.3
C LC_105	SLN53 column below / Begin	132.8	-0.2	1.4	0.0	5.8	0.9
	SLN425 ring / Begin	-17.2	0.1	-24.5	0.0	0.0	-0.5
	SLN225 ring / End	9.0	-0.7	-24.4	0.0	0.0	-1.8
	SLN449 raking strut / End	-29.8	-0.4	-0.9	-0.1	0.0	-1.0
C LC_106	SLN53 column below / Begin	156.3	-1.5	-1.1	0.0	3.6	1.3
	SLN425 ring / Begin	-28.5	0.4	-24.5	0.0	0.0	-0.8
	SLN225 ring / End	17.2	-0.5	-24.4	0.0	0.0	-1.6
	SLN449 raking strut / End	2.9	-0.3	-1.2	-0.1	0.0	-1.4
C LC_107	SLN53 column below / Begin	-2.8	0.0	0.1	0.0	0.5	-0.1
	SLN425 ring / Begin	3.2	0.0	-1.6	0.0	0.0	0.0
	SLN225 ring / End	-5.0	-0.1	-1.6	0.0	0.0	-0.3
	SLN449 raking strut / End	-17.6	0.3	-0.6	-0.1	0.0	0.1
C LC_108	SLN53 column below / Begin	24.3	4.2	-2.5	0.0	-0.9	-1.1
	SLN425 ring / Begin	-1.6	0.0	-1.6	0.0	0.0	-0.1
	SLN225 ring / End	5.3	0.0	-1.6	0.0	0.0	-0.1
	SLN449 raking strut / End	14.2	-0.1	-0.9	-0.3	0.0	1.2
C LC_109	SLN53 column below / Begin	-5.7	0.0	0.3	0.0	1.4	0.1
	SLN425 ring / Begin	6.0	0.0	-1.6	0.0	0.0	0.1
	SLN225 ring / End	-6.0	-0.1	-1.6	0.0	0.0	-0.3
	SLN449 raking strut / End	-25.3	-0.2	-0.5	0.0	0.0	-0.2
C LC_110	SLN53 column below / Begin	34.6	-2.5	-4.7	0.0	-3.1	1.0
	SLN425 ring / Begin	-12.4	0.1	-1.6	0.0	0.0	-0.4
	SLN225 ring / End	6.8	-0.1	-1.6	0.0	0.0	-0.2
	SLN449 raking strut / End	31.1	-0.1	-1.0	0.0	0.0	-1.1
C LC_111	SLN53 column below / Begin	86.7	-0.1	1.0	0.0	3.9	0.4
	SLN425 ring / Begin	-9.8	0.1	-17.2	0.0	0.0	-0.3
	SLN225 ring / End	2.5	-0.5	-17.2	0.0	0.0	-1.3
	SLN449 raking strut / End	-29.0	0.2	-0.9	-0.1	0.0	-0.5
C LC_112	SLN53 column below / Begin	115.3	4.1	-1.7	0.0	2.5	-0.5

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN425 ring / Begin	-15.2	0.2	-17.2	0.0	0.0	-0.5
	SLN225 ring / End	13.6	-0.3	-17.2	0.0	0.0	-1.1
	SLN449 raking strut / End	4.9	-0.2	-1.2	-0.3	0.0	0.7
C LC_113	SLN53 column below / Begin	83.5	-0.2	1.2	0.0	4.7	0.7
	SLN425 ring / Begin	-6.6	0.0	-17.2	0.0	0.0	-0.3
	SLN225 ring / End	1.3	-0.5	-17.2	0.0	0.0	-1.4
	SLN449 raking strut / End	-37.1	-0.4	-0.7	0.0	0.0	-0.7
C LC_114	SLN53 column below / Begin	126.0	-2.7	-3.9	0.0	0.3	1.5
	SLN425 ring / Begin	-26.3	0.3	-17.2	0.0	0.0	-0.8
	SLN225 ring / End	15.3	-0.3	-17.2	0.0	0.0	-1.2
	SLN449 raking strut / End	22.2	-0.3	-1.3	-0.1	0.0	-1.7
C LC_115	SLN53 column below / Begin	5.5	0.0	0.2	0.0	0.9	-0.1
	SLN425 ring / Begin	1.0	0.0	-2.8	0.0	0.0	0.0
	SLN225 ring / End	-3.4	-0.1	-2.8	0.0	0.0	-0.4
	SLN449 raking strut / End	-17.9	0.3	-0.9	-0.1	0.0	0.0
C LC_116	SLN53 column below / Begin	32.8	4.2	-2.4	0.0	-0.5	-1.0
	SLN425 ring / Begin	-3.9	0.1	-2.8	0.0	0.0	-0.2
	SLN225 ring / End	6.9	-0.1	-2.8	0.0	0.0	-0.2
	SLN449 raking strut / End	14.2	-0.1	-1.2	-0.3	0.0	1.2
C LC_117	SLN53 column below / Begin	2.6	0.0	0.4	0.0	1.7	0.2
	SLN425 ring / Begin	3.8	0.0	-2.8	0.0	0.0	0.0
	SLN225 ring / End	-4.5	-0.2	-2.8	0.0	0.0	-0.4
	SLN449 raking strut / End	-25.6	-0.2	-0.7	0.0	0.0	-0.2
C LC_118	SLN53 column below / Begin	43.1	-2.5	-4.6	0.0	-2.7	1.0
	SLN425 ring / Begin	-14.7	0.2	-2.8	0.0	0.0	-0.4
	SLN225 ring / End	8.5	-0.1	-2.8	0.0	0.0	-0.3
	SLN449 raking strut / End	31.1	-0.1	-1.3	0.0	0.0	-1.2
C LC_119	SLN53 column below / Begin	141.5	-0.2	1.3	0.0	5.4	0.8
	SLN425 ring / Begin	-21.4	0.2	-25.5	0.0	0.0	-0.5
	SLN225 ring / End	11.7	-0.6	-25.4	0.0	0.0	-1.7
	SLN449 raking strut / End	-24.0	-0.1	-0.9	-0.2	0.0	-0.9
C LC_120	SLN53 column below / Begin	154.8	1.9	0.0	0.0	4.8	0.3
	SLN425 ring / Begin	-24.7	0.2	-25.5	0.0	0.0	-0.7
	SLN225 ring / End	18.2	-0.5	-25.4	0.0	0.0	-1.6
	SLN449 raking strut / End	-8.4	-0.3	-1.1	-0.3	0.0	-0.3
C LC_121	SLN53 column below / Begin	141.2	-0.2	1.4	0.0	5.8	0.9
	SLN425 ring / Begin	-19.3	0.1	-25.5	0.0	0.0	-0.5
	SLN225 ring / End	11.0	-0.7	-25.4	0.0	0.0	-1.8
	SLN449 raking strut / End	-26.1	-0.4	-0.9	-0.1	0.0	-1.0
C LC_122	SLN53 column below / Begin	160.0	-1.5	-1.1	0.0	3.7	1.4
	SLN425 ring / Begin	-30.8	0.4	-25.5	0.0	0.0	-0.8
	SLN225 ring / End	19.1	-0.5	-25.4	0.0	0.0	-1.7
	SLN449 raking strut / End	-0.2	-0.4	-1.2	-0.1	0.0	-1.5
C LC_123	SLN53 column below / Begin	100.2	-0.1	1.0	0.0	3.9	0.5
	SLN425 ring / Begin	-14.2	0.1	-19.3	0.0	0.0	-0.4
	SLN225 ring / End	6.4	-0.5	-19.3	0.0	0.0	-1.4
	SLN449 raking strut / End	-26.4	0.1	-0.9	-0.2	0.0	-0.5

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_124	SLN53 column below / Begin	123.7	4.1	-1.6	0.0	2.6	-0.5
	SLN425 ring / Begin	-19.7	0.2	-19.3	0.0	0.0	-0.5
	SLN225 ring / End	17.6	-0.4	-19.3	0.0	0.0	-1.2
	SLN449 raking strut / End	0.4	-0.3	-1.2	-0.4	0.0	0.6
C LC_125	SLN53 column below / Begin	100.4	-0.2	1.2	0.0	4.7	0.7
	SLN425 ring / Begin	-10.8	0.1	-19.3	0.0	0.0	-0.3
	SLN225 ring / End	5.2	-0.5	-19.3	0.0	0.0	-1.4
	SLN449 raking strut / End	-29.6	-0.4	-0.7	0.0	0.0	-0.7
C LC_126	SLN53 column below / Begin	133.3	-2.7	-3.8	0.0	0.4	1.6
	SLN425 ring / Begin	-31.1	0.3	-19.3	0.0	0.0	-0.8
	SLN225 ring / End	19.3	-0.4	-19.3	0.0	0.0	-1.4
	SLN449 raking strut / End	16.1	-0.3	-1.3	-0.1	0.0	-1.8
C LC_127	SLN53 column below / Begin	19.0	0.0	0.2	0.0	0.8	0.0
	SLN425 ring / Begin	-3.5	0.0	-4.9	0.0	0.0	-0.1
	SLN225 ring / End	0.5	-0.2	-4.9	0.0	0.0	-0.5
	SLN449 raking strut / End	-15.2	0.3	-0.9	-0.1	0.0	0.0
C LC_128	SLN53 column below / Begin	41.1	4.2	-2.4	0.0	-0.4	-1.0
	SLN425 ring / Begin	-8.5	0.0	-4.9	0.0	0.0	-0.1
	SLN225 ring / End	11.0	-0.1	-4.9	0.0	0.0	-0.4
	SLN449 raking strut / End	9.7	-0.1	-1.2	-0.3	0.0	1.1
C LC_129	SLN53 column below / Begin	19.5	0.0	0.4	0.0	1.7	0.2
	SLN425 ring / Begin	-0.4	0.0	-4.9	0.0	0.0	0.0
	SLN225 ring / End	-0.6	-0.2	-4.9	0.0	0.0	-0.5
	SLN449 raking strut / End	-18.1	-0.2	-0.7	0.0	0.0	-0.2
C LC_130	SLN53 column below / Begin	50.4	-2.6	-4.6	0.0	-2.6	1.1
	SLN425 ring / Begin	-19.5	0.1	-4.9	0.0	0.0	-0.4
	SLN225 ring / End	12.5	-0.2	-4.9	0.0	0.0	-0.6
	SLN449 raking strut / End	25.0	-0.2	-1.3	0.0	0.0	-1.3
C LC_135	SLN53 column below / Begin	154.9	-0.3	1.4	0.0	5.7	1.0
	SLN425 ring / Begin	-27.9	0.3	-25.5	0.0	0.0	-0.9
	SLN225 ring / End	15.3	-0.7	-25.4	0.0	0.0	-2.1
	SLN449 raking strut / End	-11.2	-0.4	-1.0	-0.2	0.0	-1.3
C LC_141	SLN53 column below / Begin	32.0	-0.1	0.3	0.0	1.3	0.2
	SLN425 ring / Begin	-8.7	0.0	-4.9	0.0	0.0	-0.1
	SLN225 ring / End	6.1	-0.2	-4.9	0.0	0.0	-0.5
	SLN449 raking strut / End	-0.9	-0.1	-1.0	-0.1	0.0	-0.3
C LC_142	SLN53 column below / Begin	32.7	-0.1	0.3	0.0	1.3	0.2
	SLN425 ring / Begin	-8.9	0.1	-4.9	0.0	0.0	-0.2
	SLN225 ring / End	6.4	-0.1	-4.9	0.0	0.0	-0.5
	SLN449 raking strut / End	0.0	-0.1	-1.0	-0.1	0.0	-0.3
C LC_143	SLN53 column below / Begin	31.9	-0.1	0.3	0.0	1.3	0.2
	SLN425 ring / Begin	-8.5	0.0	-4.9	0.0	0.0	-0.1
	SLN225 ring / End	6.1	-0.2	-4.9	0.0	0.0	-0.5
	SLN449 raking strut / End	-1.1	-0.1	-1.0	-0.1	0.0	-0.3
C LC_144	SLN53 column below / Begin	32.8	-0.1	0.3	0.0	1.3	0.2
	SLN425 ring / Begin	-9.0	0.1	-4.9	0.0	0.0	-0.2
	SLN225 ring / End	6.4	-0.1	-4.9	0.0	0.0	-0.5

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN449 raking strut / End	0.1	-0.1	-1.0	-0.1	0.0	-0.3
C LC_145	SLN53 column below / Begin	149.2	-0.2	1.4	0.0	5.7	0.9
	SLN425 ring / Begin	-27.0	0.2	-25.5	0.0	0.0	-0.6
	SLN225 ring / End	17.4	-0.7	-25.4	0.0	0.0	-1.8
	SLN449 raking strut / End	-15.4	-0.3	-1.0	-0.2	0.0	-1.0
C LC_146	SLN53 column below / Begin	151.9	-0.2	1.4	0.0	5.7	0.9
	SLN425 ring / Begin	-27.9	0.3	-25.5	0.0	0.0	-0.7
	SLN225 ring / End	18.8	-0.5	-25.4	0.0	0.0	-1.7
	SLN449 raking strut / End	-11.8	-0.3	-1.0	-0.2	0.0	-1.0
C LC_147	SLN53 column below / Begin	148.7	-0.2	1.4	0.0	5.7	0.9
	SLN425 ring / Begin	-26.3	0.1	-25.5	0.0	0.0	-0.5
	SLN225 ring / End	17.2	-0.7	-25.4	0.0	0.0	-1.8
	SLN449 raking strut / End	-16.1	-0.3	-1.0	-0.2	0.0	-1.0
C LC_148	SLN53 column below / Begin	152.4	-0.2	1.4	0.0	5.7	0.9
	SLN425 ring / Begin	-28.6	0.3	-25.5	0.0	0.0	-0.7
	SLN225 ring / End	19.0	-0.5	-25.4	0.0	0.0	-1.7
	SLN449 raking strut / End	-11.2	-0.3	-1.0	-0.2	0.0	-1.0
C LC_151	SLN53 column below / Begin	96.4	-0.2	-0.3	0.0	-1.1	0.7
	SLN425 ring / Begin	-12.3	0.2	-11.5	0.0	0.0	-0.4
	SLN225 ring / End	6.1	-0.4	-11.7	0.0	0.0	-1.2
	SLN449 raking strut / End	-15.1	-0.3	-1.0	-0.1	0.0	-0.8
C LC_152	SLN53 column below / Begin	101.1	-0.2	3.1	0.0	12.6	0.9
	SLN425 ring / Begin	-28.2	0.2	-14.0	0.0	0.0	-0.6
	SLN225 ring / End	19.7	-0.5	-13.8	0.0	0.0	-1.5
	SLN449 raking strut / End	3.8	-0.3	-1.0	-0.2	0.0	-0.8
C LC_153	SLN53 column below / Begin	150.2	-0.2	1.4	0.0	5.5	0.9
	SLN425 ring / Begin	-27.6	0.2	-20.6	0.0	0.0	-0.6
	SLN225 ring / End	18.2	-0.6	-20.6	0.0	0.0	-1.8
	SLN449 raking strut / End	-13.3	-0.3	-1.0	-0.2	0.0	-1.0
C LC_154	SLN53 column below / Begin	96.0	-0.2	-0.3	0.0	-1.0	0.7
	SLN425 ring / Begin	-12.4	0.1	-11.5	0.0	0.0	-0.4
	SLN225 ring / End	6.3	-0.4	-11.7	0.0	0.0	-1.2
	SLN449 raking strut / End	-14.9	-0.3	-1.0	-0.1	0.0	-0.8

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.1 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	18.5 < 100%	OK
Welds	98.1 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN223+SLN205 radial (continuous)-bfl 1	12.7	C LC_121	188.2	0.0	59.4	OK
SLN223+SLN205 radial (continuous)-tfl 1	12.7	C LC_135	146.0	0.0	0.0	OK
SLN223+SLN205 radial (continuous)-w 1	7.2	C LC_135	275.2	0.1	53.3	OK
SLN53 column below	10.0	C LC_113	275.1	0.1	0.0	OK
SLN425 ring-bfl 1	12.7	C LC_135	62.0	0.0	0.0	OK
SLN425 ring-tfl 1	12.7	C LC_135	55.4	0.0	0.0	OK
SLN425 ring-w 1	7.2	C LC_135	184.1	0.0	0.0	OK
SLN225 ring-bfl 1	12.7	C LC_135	121.6	0.0	0.0	OK
SLN225 ring-tfl 1	12.7	C LC_135	118.9	0.0	0.0	OK
SLN225 ring-w 1	7.2	C LC_135	275.1	0.0	0.0	OK
SLN449 raking strut	10.0	C LC_114	46.7	0.0	0.0	OK
End plate column below SLN53 column below	12.0	C LC_152	154.1	0.0	51.4	OK
End plate SLN425 ring	12.0	C LC_135	47.1	0.0	122.0	OK
End plate SLN225 ring	12.0	C LC_135	87.1	0.0	467.6	OK
Cap plate SLN449 raking strut	12.0	C LC_126	97.4	0.0	0.0	OK
Fin plate SLN449 raking strut	12.0	C LC_108	240.1	0.0	40.4	OK
Fin plate 2 SLN449 raking strut	12.0	C LC_114	239.0	0.0	105.0	OK
Gusset SLN449 raking strut	12.0	C LC_126	275.0	0.0	54.4	OK
Flange top a end plate SLN425 ring	12.0	C LC_135	61.9	0.0	0.0	OK
Flange top b end plate SLN425 ring	12.0	C LC_135	65.2	0.0	0.0	OK
Flange bottom a end plate SLN425 ring	12.0	C LC_135	86.4	0.0	0.0	OK
Flange bottom b end plate SLN425 ring	12.0	C LC_135	76.5	0.0	0.0	OK
Flange top a end plate SLN225 ring	12.0	C LC_135	139.4	0.0	0.0	OK
Flange top b end plate SLN225 ring	12.0	C LC_135	162.1	0.0	0.0	OK
Flange bottom a end plate SLN225 ring	12.0	C LC_135	154.3	0.0	0.0	OK
Flange bottom b end plate SLN225 ring	12.0	C LC_135	148.4	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

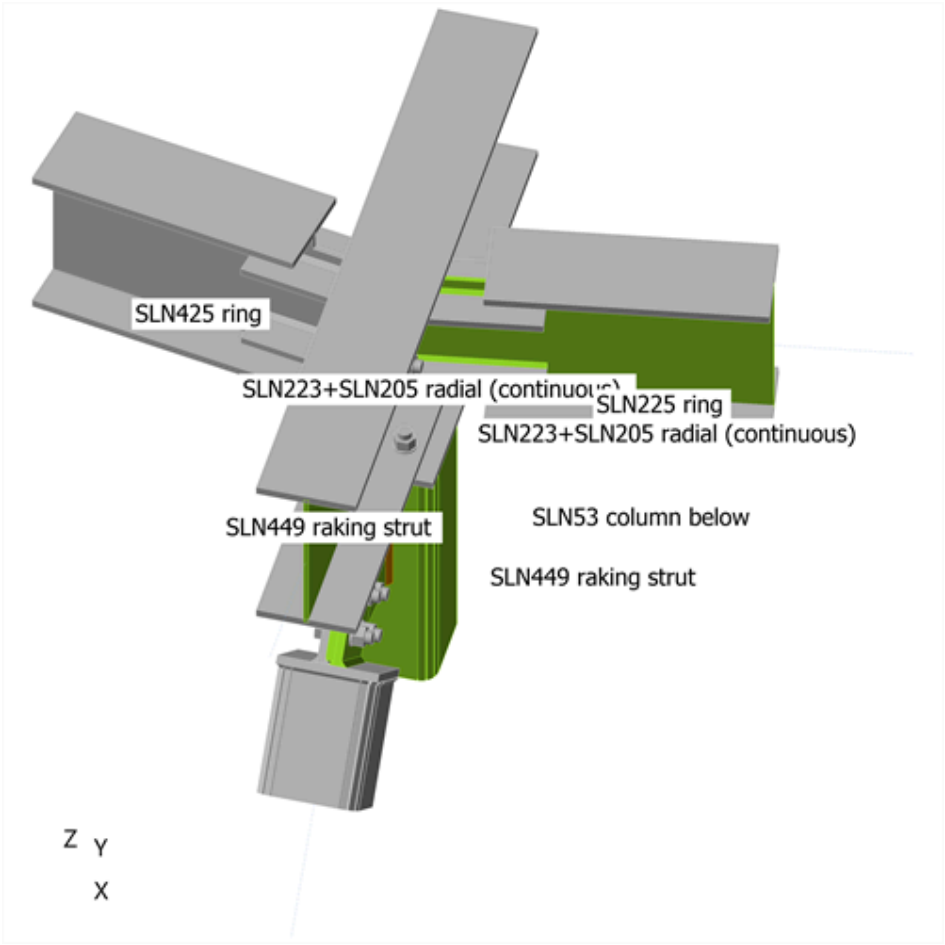
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

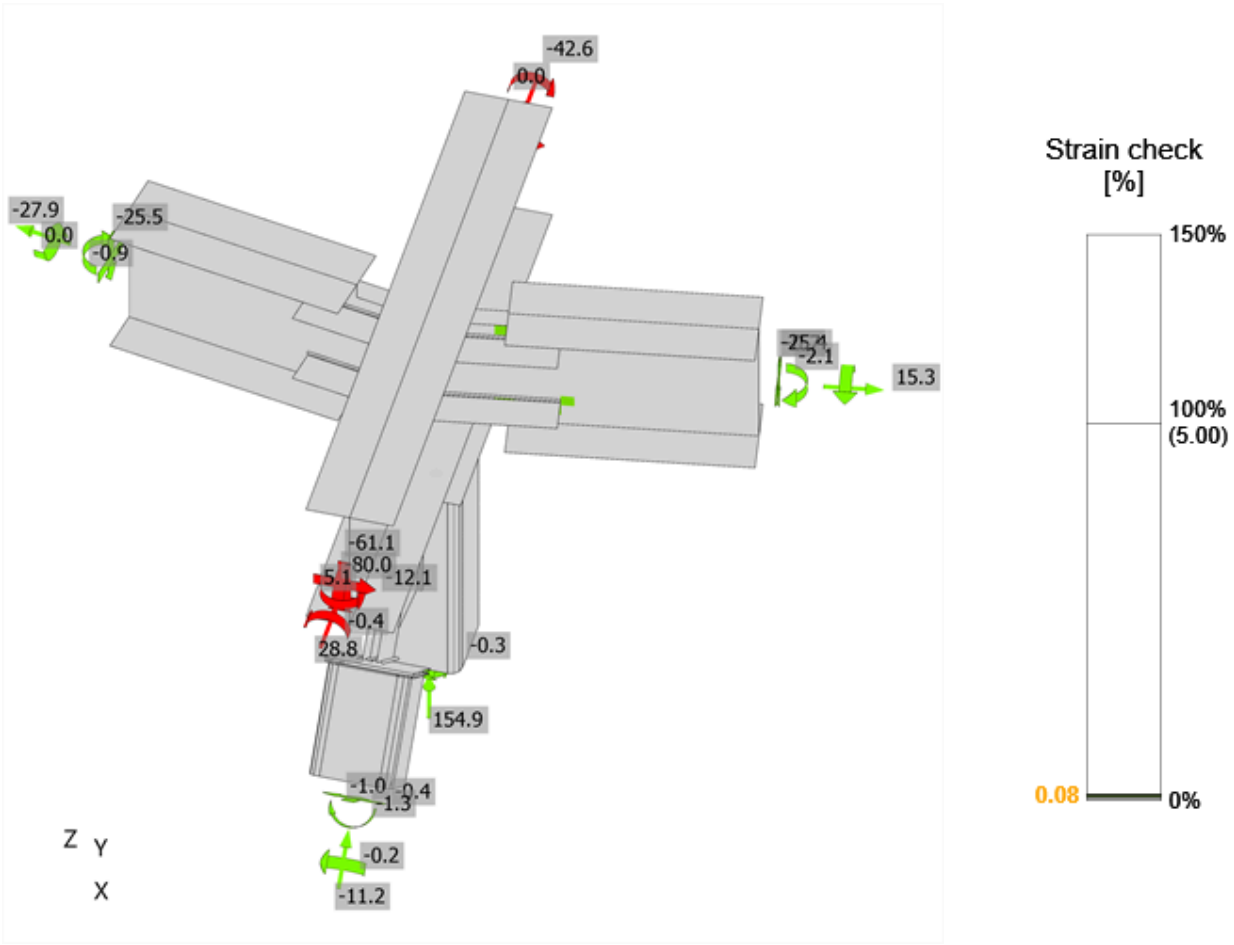
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN53 column below	150	C LC_154	0	5	0.0	OK
SLN449 raking strut	150	C LC_154	0	5	0.0	OK

Symbol explanation

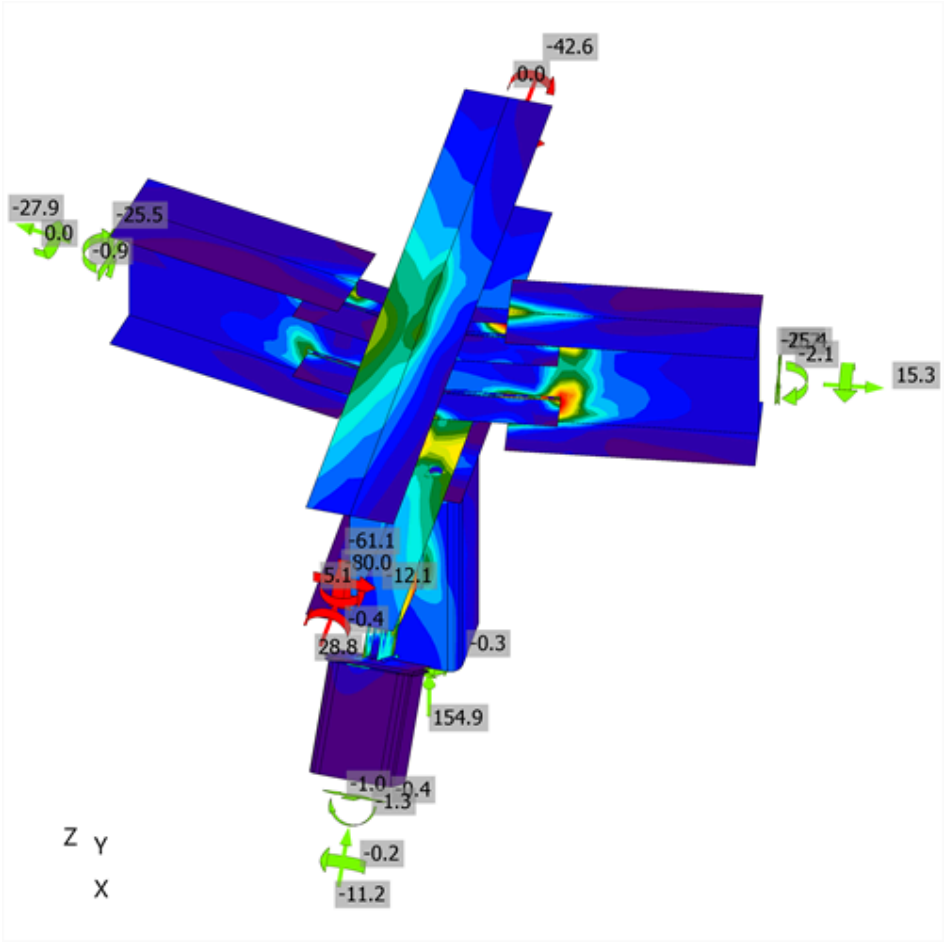
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

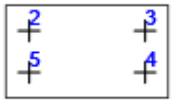
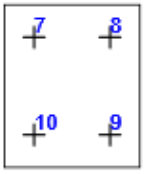
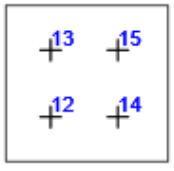


Strain check, C LC_135



Equivalent stress, C LC_135

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_152	22.5	0.4	169.7	16.0	0.5	11.9	OK	OK
	B3	M20 8.8 - 1	C LC_110	18.3	3.2	87.6	13.0	3.7	12.7	OK	OK
	B4	M20 8.8 - 1	C LC_110	20.1	4.5	169.7	14.2	4.7	14.9	OK	OK
	B5	M20 8.8 - 1	C LC_152	17.9	1.9	76.3	12.7	2.5	11.1	OK	OK
	B7	M20 8.8 - 1	C LC_135	5.6	0.6	123.8	4.0	0.9	3.4	OK	OK
	B8	M20 8.8 - 1	C LC_135	3.5	0.3	61.9	2.5	0.5	2.1	OK	OK
	B9	M20 8.8 - 1	C LC_135	5.3	1.2	123.8	3.8	2.0	4.0	OK	OK
	B10	M20 8.8 - 1	C LC_135	7.5	0.5	123.8	5.3	0.7	4.3	OK	OK
	B12	M20 8.8 - 1	C LC_114	11.4	12.0	106.0	8.1	12.7	18.5	OK	OK
	B13	M20 8.8 - 1	C LC_114	6.5	8.1	106.0	4.6	8.7	12.0	OK	OK
	B14	M20 8.8 - 1	C LC_114	2.7	8.3	106.0	1.9	8.8	10.2	OK	OK
	B15	M20 8.8 - 1	C LC_113	0.5	6.3	115.3	0.3	8.9	6.9	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN449 raking strut	SLN449 raking strut	◢ 4.2	519	C LC_118	154.4	0.0	71.0	-79.1	-2.1	38.1	21.2	OK	OK
SLN53 column below-w 3	Gusset SLN449 raking strut	◢ 5.6 ◣	288	C LC_113	396.9	0.2	-114.4	-120.4	-183.5	98.1	61.5	OK	OK
		◢ 5.6 ◣	288	C LC_113	396.9	0.2	-127.5	121.3	179.9	98.1	59.9	OK	OK
SLN425 ring-w 1	Flange top b end plate SLN425 ring	◢ 4.2 ◣	198	C LC_135	113.2	0.0	-51.5	-44.9	-37.1	28.0	16.9	OK	OK
		◢ 4.2 ◣	198	C LC_135	56.3	0.0	-15.9	22.5	21.6	13.9	9.4	OK	OK
SLN425 ring-w 1	Flange bottom b end plate SLN425 ring	◢ 4.2 ◣	198	C LC_145	48.3	0.0	13.4	8.8	25.3	11.9	9.8	OK	OK
		◢ 4.2 ◣	198	C LC_135	91.5	0.0	3.9	-15.4	50.5	22.6	12.2	OK	OK
SLN225 ring-w 1	Flange top a end plate SLN225 ring	◢ 4.2 ◣	198	C LC_135	191.3	0.0	101.8	85.1	-38.7	47.3	24.4	OK	OK
		◢ 4.2 ◣	197	C LC_135	83.7	0.0	-42.3	39.3	-13.8	20.7	11.4	OK	OK
SLN225 ring-w 1	Flange bottom a end plate SLN225 ring	◢ 4.2 ◣	198	C LC_135	126.5	0.0	38.1	53.0	-45.2	31.3	17.2	OK	OK
		◢ 4.2 ◣	198	C LC_135	258.7	0.0	120.2	-105.2	80.1	63.9	34.9	OK	OK
Cap plate SLN449 raking strut	Fin plate SLN449 raking strut	◢ 5.6 ◣	139	C LC_108	181.5	0.0	69.1	81.2	52.8	44.8	33.6	OK	OK
		◢ 5.6 ◣	139	C LC_113	116.7	0.0	-46.8	52.5	-32.4	28.8	16.1	OK	OK
Cap plate SLN449 raking strut	Fin plate 2 SLN449 raking strut	◢ 5.6 ◣	139	C LC_113	92.2	0.0	-19.7	-36.8	-36.7	22.8	15.3	OK	OK
		◢ 5.6 ◣	139	C LC_114	256.1	0.0	70.3	-111.4	-88.3	63.3	44.7	OK	OK
End plate column below SLN53 column below	SLN53 column below	◢ 5.6 ◣	521	C LC_105	321.1	0.0	-189.6	-146.1	-32.0	79.3	43.8	OK	OK
		◢ 5.6 ◣	521	C LC_152	281.2	0.0	-141.4	137.0	30.2	69.5	40.9	OK	OK
SLN223+SLN205 radial (continuous)-w 1	SLN425 ring-w 1	◢ 7.2	189	C LC_135	130.4	0.0	39.3	-59.7	-39.8	32.2	19.9	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
SLN223+SLN205 radial (continuous)-w 1	SLN225 ring-w 1	◢ 7.2	189	C LC_122	119.9	0.0	-0.3	-26.7	63.9	29.6	20.6	OK	OK
End plate SLN425 ring	SLN425 ring-w 1	◢ 4.2 ◣	149	C LC_135	88.6	0.0	-21.4	-47.0	-15.9	21.9	13.6	OK	OK
		◢ 4.2 ◣	149	C LC_135	70.9	0.0	-36.6	17.4	30.5	17.5	10.7	OK	OK
End plate SLN225 ring	SLN225 ring-w 1	◢ 4.2 ◣	149	C LC_122	123.9	0.0	-48.2	-26.5	60.3	30.6	19.5	OK	OK
		◢ 4.2 ◣	149	C LC_135	94.5	0.0	-20.9	44.0	-29.9	23.3	14.4	OK	OK
SLN425 ring-w 1	Flange top a end plate SLN425 ring	◢ 4.2 ◣	199	C LC_135	105.7	0.0	13.2	-6.1	60.2	26.1	17.8	OK	OK
		◢ 4.2 ◣	198	C LC_135	94.7	0.0	-17.1	-18.7	50.4	23.4	12.9	OK	OK
SLN425 ring-w 1	Flange bottom a end plate SLN425 ring	◢ 4.2 ◣	199	C LC_135	89.2	0.0	10.5	-44.4	25.5	22.1	14.7	OK	OK
		◢ 4.2 ◣	199	C LC_135	150.0	0.0	69.0	35.1	-68.5	37.1	26.2	OK	OK
SLN225 ring-w 1	Flange top b end plate SLN225 ring	◢ 4.2 ◣	199	C LC_135	286.6	0.0	-27.3	14.8	164.1	70.8	43.4	OK	OK
		◢ 4.2 ◣	198	C LC_135	208.6	0.0	31.8	36.1	113.4	51.5	28.7	OK	OK
SLN225 ring-w 1	Flange bottom b end plate SLN225 ring	◢ 4.2 ◣	198	C LC_135	232.0	0.0	38.6	-41.5	-125.4	57.3	31.5	OK	OK
		◢ 4.2 ◣	199	C LC_135	248.7	0.0	-27.0	-10.3	-142.4	61.5	35.3	OK	OK
End plate SLN425 ring	Flange top a end plate SLN425 ring	◢ 4.2 ◣	58	C LC_135	24.8	0.0	9.5	-7.0	11.2	6.1	6.1	OK	OK
		◢ 4.2 ◣	58	C LC_135	38.0	0.0	-3.1	-2.7	-21.7	9.4	8.6	OK	OK
End plate SLN425 ring	Flange top b end plate SLN425 ring	◢ 4.2 ◣	58	C LC_135	158.2	0.0	-75.5	77.2	-21.9	39.1	24.4	OK	OK
		◢ 4.2 ◣	58	C LC_135	140.1	0.0	-66.9	-65.4	27.9	34.6	21.7	OK	OK
End plate SLN425 ring	Flange bottom a end plate SLN425 ring	◢ 4.2 ◣	58	C LC_135	71.5	0.0	11.3	-25.7	31.6	17.7	10.4	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_114	29.7	0.0	4.3	2.7	-16.7	7.3	6.3	OK	OK
End plate SLN425 ring	Flange bottom b end plate SLN425 ring	◢ 4.2 ◣	58	C LC_135	118.5	0.0	-54.1	53.5	-29.1	29.3	21.4	OK	OK
		◢ 4.2 ◣	58	C LC_135	165.6	0.0	-79.6	-80.3	24.0	40.9	26.3	OK	OK
End plate SLN225 ring	Flange top a end plate SLN225 ring	◢ 4.2 ◣	58	C LC_135	46.0	0.0	6.0	-0.2	26.3	11.4	8.9	OK	OK
		◢ 4.2 ◣	58	C LC_135	23.7	0.0	1.1	6.9	-11.8	5.9	5.9	OK	OK
End plate SLN225 ring	Flange top b end plate SLN225 ring	◢ 4.2 ◣	58	C LC_135	264.7	0.0	-128.9	103.3	-84.6	65.4	37.5	OK	OK
		◢ 4.2 ◣	58	C LC_135	212.6	0.0	-67.4	-93.1	69.8	52.5	31.8	OK	OK
End plate SLN225 ring	Flange bottom a end plate SLN225 ring	◢ 4.2 ◣	58	C LC_122	66.4	0.0	13.3	-31.5	20.5	16.4	11.5	OK	OK
		◢ 4.2 ◣	58	C LC_135	61.0	0.0	0.4	-4.8	-34.9	15.1	10.4	OK	OK
End plate SLN225 ring	Flange bottom b end plate SLN225 ring	◢ 4.2 ◣	58	C LC_135	186.4	0.0	-54.6	82.9	-61.0	46.1	29.2	OK	OK
		◢ 4.2 ◣	58	C LC_135	263.5	0.0	-130.8	-102.5	83.3	65.1	38.9	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-S_340
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN228+SLN207 radial (continuous) - SLN37 column below - SLN226 ring - SLN227 ring - SLN450 raking strut

Name	SLN228+SLN207 radial (continuous) - SLN37 column below - SLN226 ring - SLN227 ring - SLN450 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

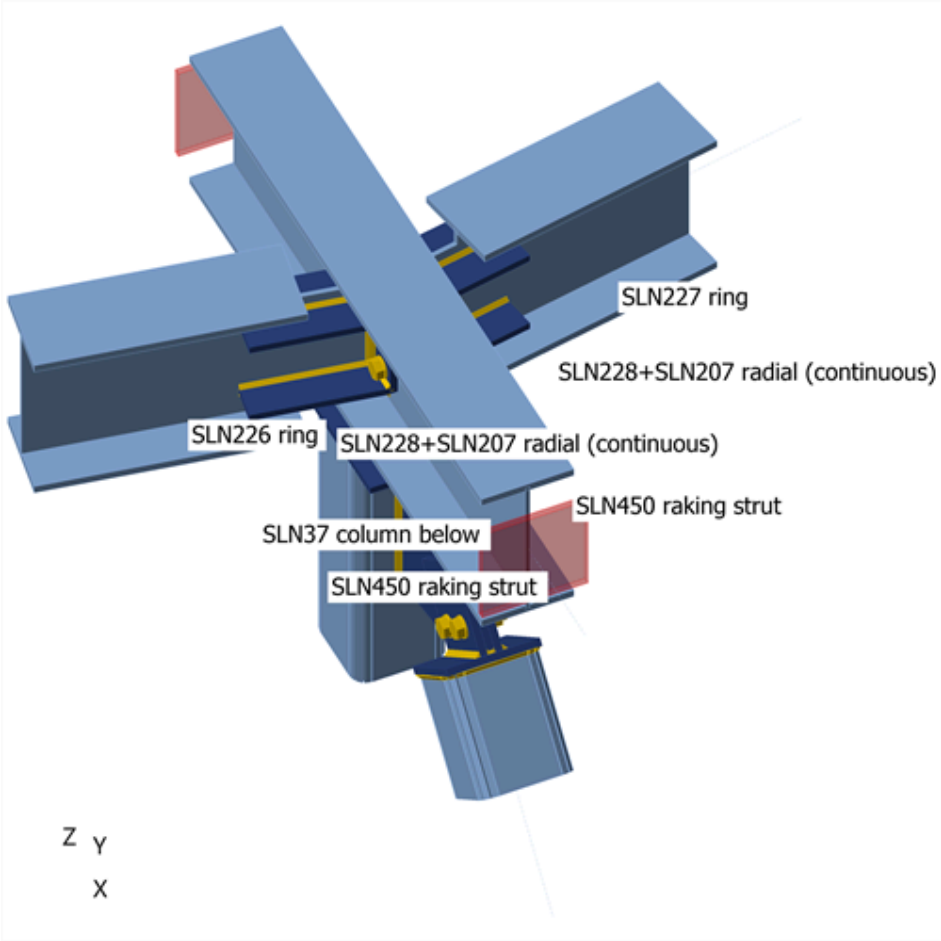
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN228+SLN207 radial (continuous)	1 - UB254x146x43	160.0	0.0	0.0	0	0	0
SLN37 column below	2 - SHS150x150x10	0.0	90.0	160.0	0	0	0
SLN226 ring	1 - UB254x146x43	60.0	0.0	0.0	0	0	0
SLN227 ring	1 - UB254x146x43	80.0	0.0	0.0	0	0	0
SLN450 raking strut	2 - SHS150x150x10	-19.9	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN228+SLN207 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN228+SLN207 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN37 column below / begin		Position	0
SLN226 ring / begin		Position	0
SLN227 ring / end		Position	0
SLN450 raking strut / end		Position	0



SLN228+SLN207 radial (continuous) - SLN37 column below - SLN226 ring - SLN227 ring - SLN450 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN37 column below / Begin	27.6	0.0	0.1	0.0	0.4	0.0
	SLN226 ring / Begin	-6.2	0.0	-4.9	0.0	0.0	0.1
	SLN227 ring / End	6.2	0.0	-4.9	0.0	0.0	-0.1
	SLN450 raking strut / End	-0.9	0.0	-1.0	0.0	0.0	0.0
C LC_102	SLN37 column below / Begin	131.8	0.0	0.0	0.0	0.0	0.0
	SLN226 ring / Begin	-17.8	-0.2	-25.6	0.0	0.0	0.5
	SLN227 ring / End	17.8	-0.2	-25.7	0.0	0.0	-0.5
	SLN450 raking strut / End	-10.3	0.0	-1.0	0.0	0.0	0.0
C LC_103	SLN37 column below / Begin	113.7	0.0	0.0	0.0	-0.1	0.0
	SLN226 ring / Begin	-9.1	-0.3	-24.6	0.0	0.0	0.7
	SLN227 ring / End	9.1	-0.3	-24.7	0.0	0.0	-0.5
	SLN450 raking strut / End	-26.6	0.1	-0.9	0.0	0.0	0.1
C LC_104	SLN37 column below / Begin	139.4	0.8	-2.6	0.0	-2.0	-0.2
	SLN226 ring / Begin	-17.6	0.0	-24.6	0.0	0.0	0.3
	SLN227 ring / End	18.5	0.0	-24.7	0.0	0.0	-0.3
	SLN450 raking strut / End	7.8	0.0	-1.2	-0.1	0.0	0.3
C LC_105	SLN37 column below / Begin	119.7	0.0	0.0	0.0	-0.2	0.2
	SLN226 ring / Begin	-9.1	-0.2	-24.6	0.0	0.0	0.4
	SLN227 ring / End	9.4	-0.2	-24.7	0.0	0.0	-0.6
	SLN450 raking strut / End	-18.6	-0.2	-1.0	0.0	0.0	-0.1
C LC_106	SLN37 column below / Begin	128.5	-2.3	-1.0	0.0	-0.8	0.6
	SLN226 ring / Begin	-14.8	-0.1	-24.6	0.0	0.0	0.5
	SLN227 ring / End	12.3	-0.1	-24.7	0.0	0.0	-0.3
	SLN450 raking strut / End	-7.0	0.0	-1.1	0.1	0.0	-0.7
C LC_107	SLN37 column below / Begin	-9.7	0.0	0.1	0.0	0.2	-0.1
	SLN226 ring / Begin	5.0	-0.1	-1.6	0.0	0.0	0.3
	SLN227 ring / End	-5.1	0.0	-1.6	0.0	0.0	-0.1
	SLN450 raking strut / End	-25.6	0.1	-0.4	0.0	0.0	0.1
C LC_108	SLN37 column below / Begin	32.9	1.7	-5.1	0.0	-3.5	-0.5
	SLN226 ring / Begin	-8.1	0.1	-1.6	0.0	0.0	-0.1
	SLN227 ring / End	9.8	0.1	-1.6	0.0	0.0	0.2
	SLN450 raking strut / End	31.5	0.0	-1.1	-0.1	0.0	0.5
C LC_109	SLN37 column below / Begin	-0.4	-0.1	0.0	0.0	0.2	0.2
	SLN226 ring / Begin	6.2	0.0	-1.6	0.0	0.0	0.0
	SLN227 ring / End	-6.0	-0.1	-1.6	0.0	0.0	-0.3
	SLN450 raking strut / End	-13.4	-0.4	-0.6	0.0	0.0	-0.3
C LC_110	SLN37 column below / Begin	14.0	-4.6	-1.8	0.0	-1.2	1.3
	SLN226 ring / Begin	-3.4	-0.1	-1.6	0.0	0.0	0.2
	SLN227 ring / End	-1.1	0.1	-1.6	0.0	0.0	0.2
	SLN450 raking strut / End	5.7	0.0	-0.9	0.3	0.0	-1.4
C LC_111	SLN37 column below / Begin	68.4	0.0	0.0	0.0	0.0	-0.1
	SLN226 ring / Begin	-2.0	-0.3	-17.3	0.0	0.0	0.6
	SLN227 ring / End	1.9	-0.2	-17.4	0.0	0.0	-0.4
	SLN450 raking strut / End	-34.9	0.1	-0.7	0.0	0.0	0.1
C LC_112	SLN37 column below / Begin	113.9	1.7	-5.2	0.0	-3.7	-0.4

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN226 ring / Begin	-16.5	0.0	-17.3	0.0	0.0	0.1
	SLN227 ring / End	18.2	0.0	-17.4	0.0	0.0	-0.1
	SLN450 raking strut / End	25.9	0.0	-1.3	-0.1	0.0	0.5
C LC_113	SLN37 column below / Begin	78.6	-0.1	0.0	0.0	-0.1	0.3
	SLN226 ring / Begin	-1.2	-0.1	-17.3	0.0	0.0	0.3
	SLN227 ring / End	1.6	-0.2	-17.4	0.0	0.0	-0.5
	SLN450 raking strut / End	-21.5	-0.4	-0.9	0.0	0.0	-0.3
C LC_114	SLN37 column below / Begin	94.0	-4.6	-1.9	0.0	-1.4	1.3
	SLN226 ring / Begin	-11.4	-0.1	-17.3	0.0	0.0	0.5
	SLN227 ring / End	6.8	0.0	-17.4	0.0	0.0	0.0
	SLN450 raking strut / End	-1.1	0.0	-1.1	0.3	0.0	-1.4
C LC_115	SLN37 column below / Begin	-2.7	0.0	0.1	0.0	0.3	-0.1
	SLN226 ring / Begin	3.5	-0.1	-2.8	0.0	0.0	0.4
	SLN227 ring / End	-3.6	-0.1	-2.9	0.0	0.0	-0.1
	SLN450 raking strut / End	-26.1	0.1	-0.7	0.0	0.0	0.1
C LC_116	SLN37 column below / Begin	40.3	1.7	-5.1	0.0	-3.4	-0.5
	SLN226 ring / Begin	-9.8	0.0	-2.8	0.0	0.0	-0.1
	SLN227 ring / End	11.4	0.1	-2.9	0.0	0.0	0.1
	SLN450 raking strut / End	31.4	0.0	-1.3	-0.1	0.0	0.5
C LC_117	SLN37 column below / Begin	6.7	-0.1	0.1	0.0	0.3	0.2
	SLN226 ring / Begin	4.6	0.0	-2.8	0.0	0.0	0.0
	SLN227 ring / End	-4.4	-0.1	-2.9	0.0	0.0	-0.3
	SLN450 raking strut / End	-13.7	-0.4	-0.9	0.0	0.0	-0.3
C LC_118	SLN37 column below / Begin	21.2	-4.6	-1.8	0.0	-1.0	1.3
	SLN226 ring / Begin	-5.0	-0.1	-2.8	0.0	0.0	0.2
	SLN227 ring / End	0.5	0.1	-2.9	0.0	0.0	0.2
	SLN450 raking strut / End	5.5	0.0	-1.1	0.3	0.0	-1.4
C LC_119	SLN37 column below / Begin	122.3	0.0	0.0	0.0	-0.1	0.0
	SLN226 ring / Begin	-11.1	-0.3	-25.6	0.0	0.0	0.6
	SLN227 ring / End	11.0	-0.3	-25.7	0.0	0.0	-0.5
	SLN450 raking strut / End	-22.7	0.1	-0.9	0.0	0.0	0.1
C LC_120	SLN37 column below / Begin	142.8	0.8	-2.6	0.0	-1.9	-0.2
	SLN226 ring / Begin	-19.6	0.0	-25.6	0.0	0.0	0.3
	SLN227 ring / End	20.6	0.0	-25.7	0.0	0.0	-0.3
	SLN450 raking strut / End	4.7	0.0	-1.2	-0.1	0.0	0.3
C LC_121	SLN37 column below / Begin	125.9	0.0	0.0	0.0	-0.1	0.2
	SLN226 ring / Begin	-11.0	-0.2	-25.6	0.0	0.0	0.4
	SLN227 ring / End	11.3	-0.2	-25.7	0.0	0.0	-0.6
	SLN450 raking strut / End	-18.0	-0.2	-1.0	0.0	0.0	-0.1
C LC_122	SLN37 column below / Begin	133.0	-2.3	-1.0	0.0	-0.7	0.7
	SLN226 ring / Begin	-16.7	-0.1	-25.6	0.0	0.0	0.6
	SLN227 ring / End	14.3	-0.1	-25.7	0.0	0.0	-0.3
	SLN450 raking strut / End	-8.6	0.0	-1.1	0.1	0.0	-0.7
C LC_123	SLN37 column below / Begin	85.5	0.0	0.0	0.0	0.1	-0.1
	SLN226 ring / Begin	-5.9	-0.3	-19.4	0.0	0.0	0.6
	SLN227 ring / End	5.8	-0.2	-19.5	0.0	0.0	-0.4
	SLN450 raking strut / End	-27.2	0.1	-0.7	0.0	0.0	0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_124	SLN37 column below / Begin	120.8	1.7	-5.2	0.0	-3.5	-0.5
	SLN226 ring / Begin	-20.5	0.0	-19.4	0.0	0.0	0.2
	SLN227 ring / End	22.2	0.0	-19.5	0.0	0.0	-0.2
	SLN450 raking strut / End	19.8	0.0	-1.3	-0.1	0.0	0.6
C LC_125	SLN37 column below / Begin	90.8	-0.1	0.0	0.0	0.0	0.2
	SLN226 ring / Begin	-5.1	-0.1	-19.4	0.0	0.0	0.3
	SLN227 ring / End	5.5	-0.2	-19.5	0.0	0.0	-0.6
	SLN450 raking strut / End	-20.2	-0.4	-0.9	0.0	0.0	-0.2
C LC_126	SLN37 column below / Begin	103.0	-4.6	-1.9	0.0	-1.3	1.3
	SLN226 ring / Begin	-15.3	-0.2	-19.4	0.0	0.0	0.6
	SLN227 ring / End	10.7	0.0	-19.5	0.0	0.0	-0.1
	SLN450 raking strut / End	-4.3	0.0	-1.1	0.3	0.0	-1.4
C LC_127	SLN37 column below / Begin	14.4	0.0	0.1	0.0	0.4	-0.1
	SLN226 ring / Begin	-0.4	-0.1	-4.9	0.0	0.0	0.4
	SLN227 ring / End	0.3	-0.1	-4.9	0.0	0.0	-0.1
	SLN450 raking strut / End	-18.3	0.1	-0.7	0.0	0.0	0.1
C LC_128	SLN37 column below / Begin	47.2	1.7	-5.1	0.0	-3.1	-0.5
	SLN226 ring / Begin	-13.8	0.0	-4.9	0.0	0.0	0.1
	SLN227 ring / End	15.5	0.0	-4.9	0.0	0.0	0.0
	SLN450 raking strut / End	25.3	0.0	-1.3	-0.1	0.0	0.5
C LC_129	SLN37 column below / Begin	19.0	-0.1	0.1	0.0	0.4	0.2
	SLN226 ring / Begin	0.7	0.0	-4.9	0.0	0.0	0.0
	SLN227 ring / End	-0.5	-0.1	-4.9	0.0	0.0	-0.3
	SLN450 raking strut / End	-12.4	-0.4	-0.9	0.0	0.0	-0.2
C LC_130	SLN37 column below / Begin	30.2	-4.6	-1.8	0.0	-0.9	1.3
	SLN226 ring / Begin	-9.0	-0.1	-4.9	0.0	0.0	0.3
	SLN227 ring / End	4.4	0.0	-4.9	0.0	0.0	0.1
	SLN450 raking strut / End	2.3	0.0	-1.1	0.3	0.0	-1.4
C LC_135	SLN37 column below / Begin	131.2	0.0	0.0	0.0	0.2	0.0
	SLN226 ring / Begin	-15.0	-0.2	-25.6	0.0	0.0	0.5
	SLN227 ring / End	15.0	-0.2	-25.7	0.0	0.0	-0.5
	SLN450 raking strut / End	-11.3	0.0	-1.0	0.0	0.0	0.0
C LC_141	SLN37 column below / Begin	27.0	0.0	0.1	0.0	0.4	0.0
	SLN226 ring / Begin	-6.0	-0.1	-4.9	0.0	0.0	0.2
	SLN227 ring / End	5.9	-0.1	-4.9	0.0	0.0	-0.2
	SLN450 raking strut / End	-1.7	0.0	-1.0	0.0	0.0	0.0
C LC_142	SLN37 column below / Begin	28.2	0.0	0.1	0.0	0.4	0.0
	SLN226 ring / Begin	-6.4	0.0	-4.9	0.0	0.0	0.1
	SLN227 ring / End	6.4	0.0	-4.9	0.0	0.0	-0.1
	SLN450 raking strut / End	-0.1	0.0	-1.0	0.0	0.0	0.0
C LC_143	SLN37 column below / Begin	27.4	0.0	0.1	0.0	0.4	0.0
	SLN226 ring / Begin	-6.1	-0.1	-4.9	0.0	0.0	0.1
	SLN227 ring / End	6.1	-0.1	-4.9	0.0	0.0	-0.1
	SLN450 raking strut / End	-1.2	0.0	-1.0	0.0	0.0	0.0
C LC_144	SLN37 column below / Begin	27.8	0.0	0.1	0.0	0.4	0.0
	SLN226 ring / Begin	-6.3	0.0	-4.9	0.0	0.0	0.1
	SLN227 ring / End	6.2	0.0	-4.9	0.0	0.0	-0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN450 raking strut / End	-0.6	0.0	-1.0	0.0	0.0	0.0
C LC_145	SLN37 column below / Begin	129.4	0.0	0.0	0.0	0.0	0.0
	SLN226 ring / Begin	-16.8	-0.3	-25.6	0.0	0.0	0.6
	SLN227 ring / End	16.7	-0.3	-25.7	0.0	0.0	-0.6
	SLN450 raking strut / End	-13.5	0.0	-1.0	0.0	0.0	0.0
C LC_146	SLN37 column below / Begin	134.2	0.0	0.0	0.0	0.0	0.0
	SLN226 ring / Begin	-18.8	-0.1	-25.6	0.0	0.0	0.4
	SLN227 ring / End	18.9	-0.1	-25.7	0.0	0.0	-0.4
	SLN450 raking strut / End	-7.1	0.0	-1.0	0.0	0.0	0.0
C LC_147	SLN37 column below / Begin	130.9	0.0	0.0	0.0	0.0	0.0
	SLN226 ring / Begin	-17.3	-0.2	-25.6	0.0	0.0	0.5
	SLN227 ring / End	17.5	-0.2	-25.7	0.0	0.0	-0.5
	SLN450 raking strut / End	-11.4	0.0	-1.0	0.0	0.0	0.0
C LC_148	SLN37 column below / Begin	132.6	0.0	0.0	0.0	0.0	0.0
	SLN226 ring / Begin	-18.3	-0.1	-25.6	0.0	0.0	0.5
	SLN227 ring / End	18.1	-0.1	-25.7	0.0	0.0	-0.4
	SLN450 raking strut / End	-9.1	0.0	-1.0	0.0	0.0	0.0
C LC_151	SLN37 column below / Begin	78.3	0.0	-2.1	0.0	-8.4	0.0
	SLN226 ring / Begin	-6.0	-0.1	-11.8	0.0	0.0	0.3
	SLN227 ring / End	6.0	-0.1	-11.5	0.0	0.0	-0.3
	SLN450 raking strut / End	-11.9	0.0	-1.0	0.0	0.0	0.0
C LC_152	SLN37 column below / Begin	79.3	0.0	1.8	0.0	7.4	0.0
	SLN226 ring / Begin	-19.4	-0.1	-14.1	0.0	0.0	0.4
	SLN227 ring / End	19.4	-0.1	-14.2	0.0	0.0	-0.4
	SLN450 raking strut / End	1.7	0.0	-1.0	0.0	0.0	0.0
C LC_153	SLN37 column below / Begin	131.4	0.0	-0.1	0.0	-0.2	0.0
	SLN226 ring / Begin	-17.8	-0.2	-21.0	0.0	0.0	0.5
	SLN227 ring / End	17.9	-0.2	-20.7	0.0	0.0	-0.5
	SLN450 raking strut / End	-10.0	0.0	-1.0	0.0	0.0	0.0
C LC_154	SLN37 column below / Begin	77.9	0.0	-2.1	0.0	-8.3	0.0
	SLN226 ring / Begin	-6.1	-0.1	-11.8	0.0	0.0	0.3
	SLN227 ring / End	6.1	-0.1	-11.5	0.0	0.0	-0.3
	SLN450 raking strut / End	-11.7	0.0	-1.0	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	15.5 < 100%	OK
Welds	98.0 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN228+SLN207 radial (continuous)-bfl 1	12.7	C LC_108	125.7	0.0	20.5	OK
SLN228+SLN207 radial (continuous)-tfl 1	12.7	C LC_153	69.1	0.0	0.0	OK
SLN228+SLN207 radial (continuous)-w 1	7.2	C LC_126	153.7	0.0	23.5	OK
SLN37 column below	10.0	C LC_111	275.0	0.0	0.0	OK
SLN226 ring-bfl 1	12.7	C LC_119	46.3	0.0	0.0	OK
SLN226 ring-tfl 1	12.7	C LC_103	43.1	0.0	0.0	OK
SLN226 ring-w 1	7.2	C LC_119	141.6	0.0	0.0	OK
SLN227 ring-bfl 1	12.7	C LC_121	44.1	0.0	0.0	OK
SLN227 ring-tfl 1	12.7	C LC_121	41.4	0.0	0.0	OK
SLN227 ring-w 1	7.2	C LC_121	134.9	0.0	0.0	OK
SLN450 raking strut	10.0	C LC_116	37.8	0.0	0.0	OK
End plate column below SLN37 column below	12.0	C LC_108	149.5	0.0	20.5	OK
End plate SLN226 ring	12.0	C LC_103	65.6	0.0	0.0	OK
End plate SLN227 ring	12.0	C LC_145	51.6	0.0	11.6	OK
Cap plate SLN450 raking strut	12.0	C LC_126	81.3	0.0	0.0	OK
Fin plate SLN450 raking strut	12.0	C LC_130	225.1	0.0	53.0	OK
Fin plate 2 SLN450 raking strut	12.0	C LC_118	241.4	0.0	86.5	OK
Gusset SLN450 raking strut	12.0	C LC_126	266.2	0.0	42.4	OK
Flange top a end plate SLN226 ring	12.0	C LC_103	59.3	0.0	0.0	OK
Flange top b end plate SLN226 ring	12.0	C LC_111	44.0	0.0	0.0	OK
Flange bottom a end plate SLN226 ring	12.0	C LC_111	41.6	0.0	0.0	OK
Flange bottom b end plate SLN226 ring	12.0	C LC_119	63.9	0.0	0.0	OK
Flange top a end plate SLN227 ring	12.0	C LC_121	37.5	0.0	0.0	OK
Flange top b end plate SLN227 ring	12.0	C LC_121	54.3	0.0	0.0	OK
Flange bottom a end plate SLN227 ring	12.0	C LC_145	61.0	0.0	0.0	OK
Flange bottom b end plate SLN227 ring	12.0	C LC_121	38.5	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

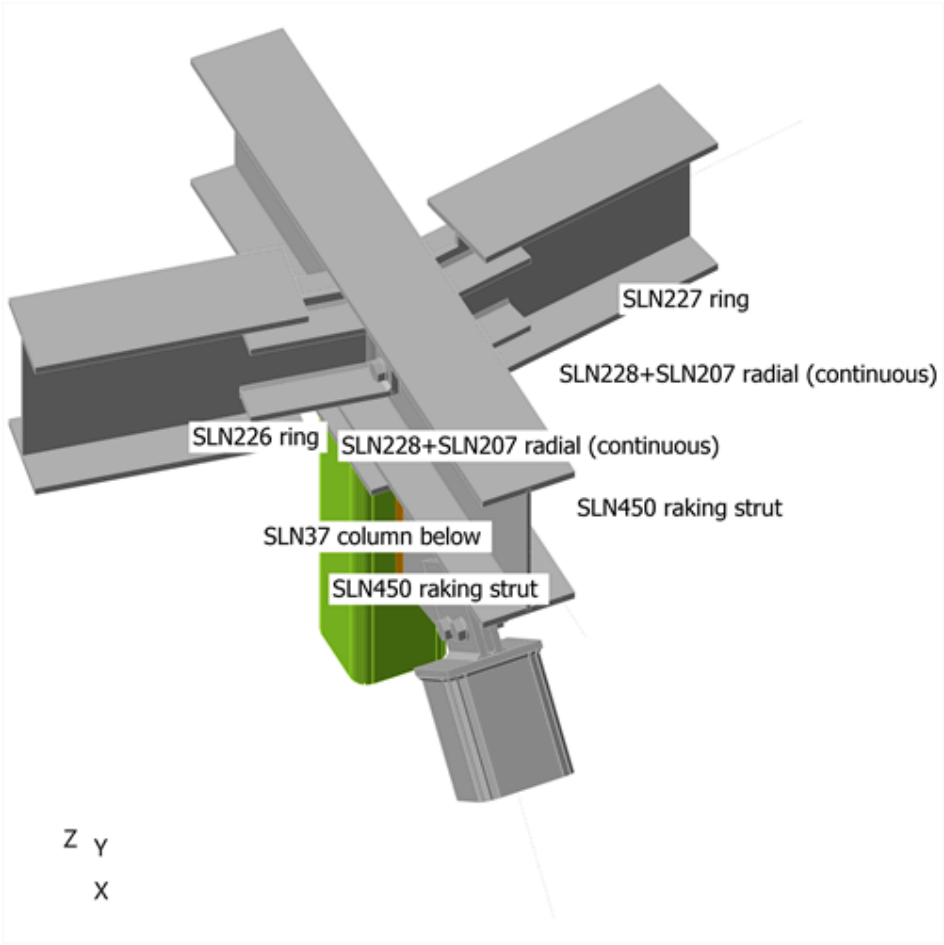
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

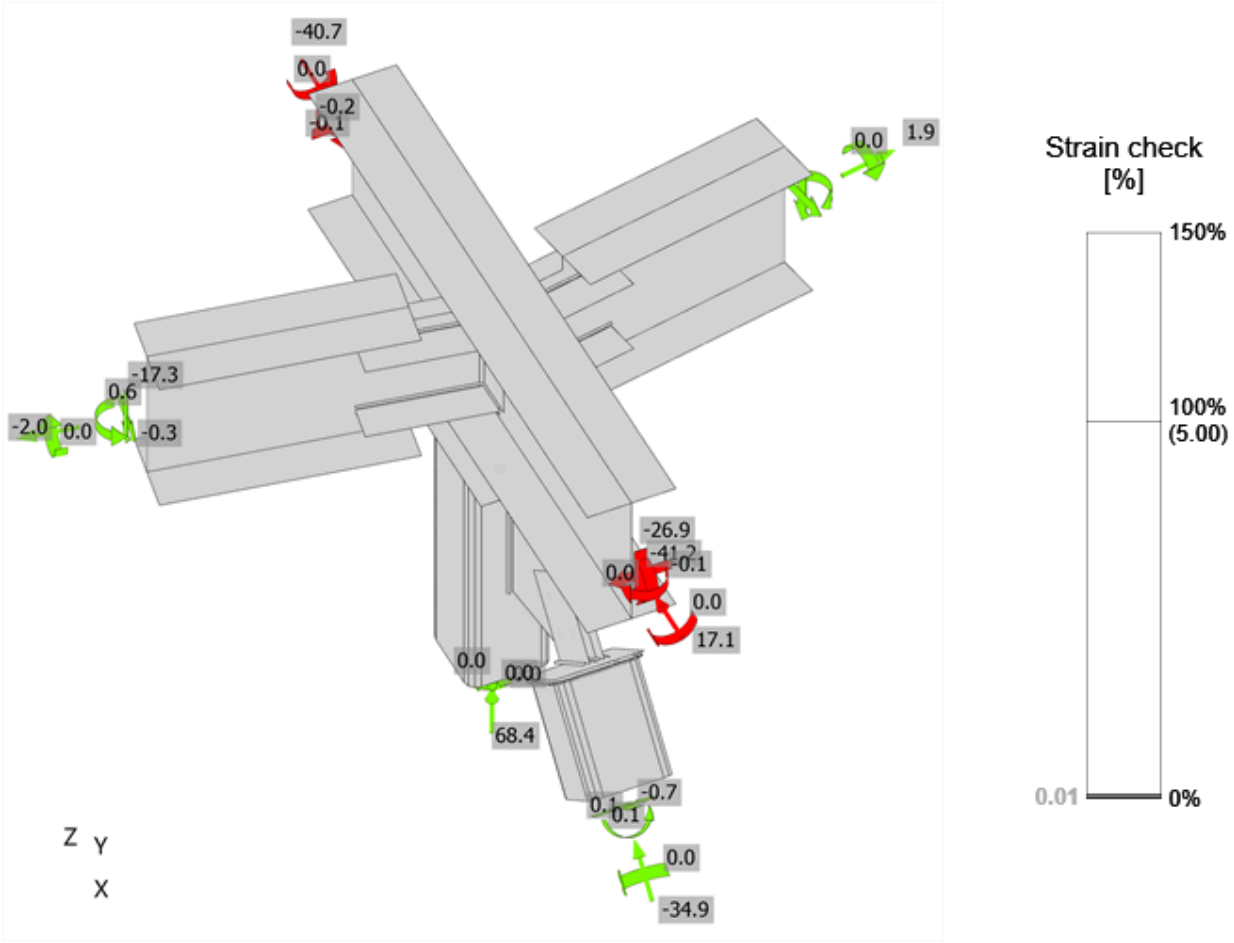
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN37 column below	150	C LC_154	0	5	0.0	OK
SLN450 raking strut	150	C LC_154	0	5	0.0	OK

Symbol explanation

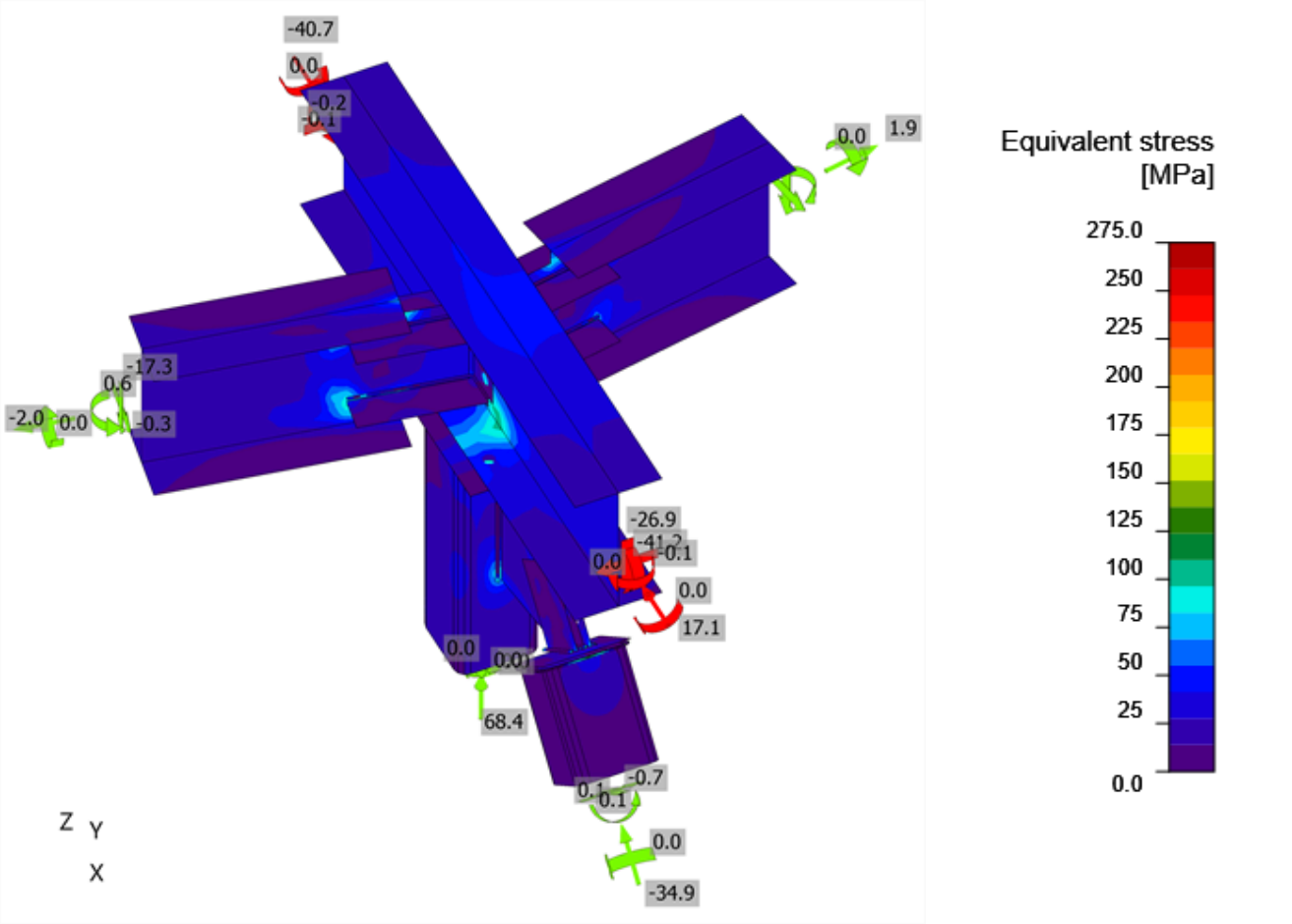
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

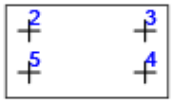
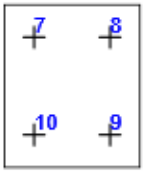
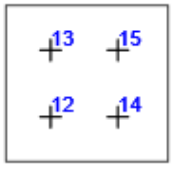


Strain check, C LC_111



Equivalent stress, C LC_111

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_107	8.7	4.2	123.5	6.2	4.4	8.9	OK	OK
	B3	M20 8.8 - 1	C LC_108	21.8	3.8	117.1	15.5	4.0	15.1	OK	OK
	B4	M20 8.8 - 1	C LC_108	21.3	3.5	93.7	15.1	3.7	14.5	OK	OK
	B5	M20 8.8 - 1	C LC_107	9.5	4.3	171.2	6.7	4.6	9.4	OK	OK
	B7	M20 8.8 - 1	C LC_103	9.2	0.6	68.9	6.5	0.7	5.3	OK	OK
	B8	M20 8.8 - 1	C LC_108	1.5	0.4	123.8	1.1	0.5	1.2	OK	OK
	B9	M20 8.8 - 1	C LC_152	0.0	1.0	123.8	0.0	1.6	1.0	OK	OK
	B10	M20 8.8 - 1	C LC_103	10.2	0.8	123.8	7.2	0.8	6.0	OK	OK
	B12	M20 8.8 - 1	C LC_118	11.0	9.3	106.0	7.8	9.9	15.4	OK	OK
	B13	M20 8.8 - 1	C LC_126	5.0	7.9	155.3	3.6	8.4	11.0	OK	OK
	B14	M20 8.8 - 1	C LC_116	1.0	6.8	106.0	0.7	9.9	7.7	OK	OK
	B15	M20 8.8 - 1	C LC_111	0.1	5.0	115.3	0.1	8.5	5.4	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN450 raking strut	SLN450 raking strut	◢ 4.2	519	C LC_116	131.9	0.0	62.5	-67.0	1.2	32.6	17.0	OK	OK
SLN37 column below-w 3	Gusset SLN450 raking strut	◢ 5.6 ◣	288	C LC_111	396.6	0.0	-130.1	-129.6	-173.2	98.0	57.4	OK	OK
		◢ 5.6 ◣	288	C LC_111	396.7	0.0	-129.7	130.4	172.7	98.0	57.9	OK	OK
SLN226 ring-w 1	Flange top b end plate SLN226 ring	◢ 4.2 ◣	198	C LC_126	48.3	0.0	27.3	22.1	6.6	11.9	8.3	OK	OK
		◢ 4.2 ◣	197	C LC_103	30.1	0.0	-13.6	12.2	9.5	7.4	6.8	OK	OK
SLN226 ring-w 1	Flange bottom b end plate SLN226 ring	◢ 4.2 ◣	198	C LC_145	79.6	0.0	23.5	27.3	34.4	19.7	12.0	OK	OK
		◢ 4.2 ◣	198	C LC_119	115.0	0.0	46.8	-42.6	-43.2	28.4	17.3	OK	OK
SLN227 ring-w 1	Flange top a end plate SLN227 ring	◢ 4.2 ◣	198	C LC_152	38.9	0.0	20.4	17.2	-8.3	9.6	6.6	OK	OK
		◢ 4.2 ◣	197	C LC_121	29.2	0.0	-13.6	12.4	-8.3	7.2	6.2	OK	OK
SLN227 ring-w 1	Flange bottom a end plate SLN227 ring	◢ 4.2 ◣	198	C LC_147	80.0	0.0	23.8	26.9	-35.0	19.8	10.8	OK	OK
		◢ 4.2 ◣	198	C LC_145	113.1	0.0	44.3	-41.1	43.8	28.0	15.2	OK	OK
Cap plate SLN450 raking strut	Fin plate SLN450 raking strut	◢ 5.6 ◣	139	C LC_116	171.8	0.0	66.5	75.5	51.7	42.5	28.4	OK	OK
		◢ 5.6 ◣	139	C LC_111	96.2	0.0	-37.2	41.1	-30.5	23.8	13.2	OK	OK
Cap plate SLN450 raking strut	Fin plate 2 SLN450 raking strut	◢ 5.6 ◣	139	C LC_116	113.7	0.0	58.0	46.6	31.8	28.1	17.5	OK	OK
		◢ 5.6 ◣	139	C LC_118	198.8	0.0	64.9	-87.5	-64.1	49.1	36.2	OK	OK
		◢ 5.6 ◣	521	C LC_151	190.5	0.0	-92.4	90.6	32.5	47.1	26.5	OK	OK
End plate column below SLN37 column below	SLN37 column below	◢ 5.6 ◣	521	C LC_103	218.6	0.0	-136.8	-96.4	-20.0	54.0	30.1	OK	OK
SLN228+SLN207 radial (continuous)-w 1	SLN226 ring-w 1	◢ 7.2	189	C LC_120	81.6	0.0	24.1	-23.3	-38.5	20.2	12.9	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
SLN228+SLN207 radial (continuous)-w 1	SLN227 ring-w 1	◢ 7.2	189	C LC_119	85.3	0.0	-0.9	7.8	48.6	21.1	12.6	OK	OK
End plate SLN226 ring	SLN226 ring-w 1	◢ 4.2 ◣	149	C LC_120	79.9	0.0	-21.6	-18.6	-40.4	19.8	11.0	OK	OK
		◢ 4.2 ◣	149	C LC_145	64.3	0.0	-12.0	22.9	28.4	15.9	10.7	OK	OK
End plate SLN227 ring	SLN227 ring-w 1	◢ 4.2 ◣	149	C LC_120	80.4	0.0	-20.3	-21.4	39.5	19.9	12.5	OK	OK
		◢ 4.2 ◣	149	C LC_145	64.2	0.0	-16.4	19.5	-30.1	15.9	10.4	OK	OK
SLN226 ring-w 1	Flange top a end plate SLN226 ring	◢ 4.2 ◣	199	C LC_103	100.6	0.0	-51.2	22.8	-44.5	24.9	17.8	OK	OK
		◢ 4.2 ◣	198	C LC_111	59.6	0.0	9.0	10.2	-32.4	14.7	9.3	OK	OK
SLN226 ring-w 1	Flange bottom a end plate SLN226 ring	◢ 4.2 ◣	198	C LC_119	87.1	0.0	15.7	-16.0	46.8	21.5	12.9	OK	OK
		◢ 4.2 ◣	198	C LC_111	73.5	0.0	-10.7	-4.2	41.8	18.2	11.1	OK	OK
SLN227 ring-w 1	Flange top b end plate SLN227 ring	◢ 4.2 ◣	199	C LC_121	92.4	0.0	-45.1	20.8	41.6	22.8	14.6	OK	OK
		◢ 4.2 ◣	199	C LC_121	52.4	0.0	-3.8	-28.1	-11.0	12.9	8.3	OK	OK
SLN227 ring-w 1	Flange bottom b end plate SLN227 ring	◢ 4.2 ◣	198	C LC_121	82.0	0.0	14.3	-14.5	-44.3	20.3	11.8	OK	OK
		◢ 4.2 ◣	199	C LC_120	63.9	0.0	-10.7	12.3	-34.2	15.8	10.1	OK	OK
End plate SLN226 ring	Flange top a end plate SLN226 ring	◢ 4.2 ◣	58	C LC_103	32.1	0.0	0.3	2.6	-18.4	7.9	6.8	OK	OK
		◢ 4.2 ◣	58	C LC_103	21.2	0.0	-7.2	-11.3	-2.3	5.2	5.2	OK	OK
End plate SLN226 ring	Flange top b end plate SLN226 ring	◢ 4.2 ◣	58	C LC_119	41.6	0.0	-1.2	0.1	-24.0	10.3	8.6	OK	OK
		◢ 4.2 ◣	58	C LC_103	54.9	0.0	4.0	8.6	-30.4	13.6	11.3	OK	OK
End plate SLN226 ring	Flange bottom a end plate SLN226 ring	◢ 4.2 ◣	58	C LC_120	49.5	0.0	10.5	-19.3	20.2	12.2	8.4	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_119	44.3	0.0	-11.9	-17.4	17.5	10.9	8.3	OK	OK
End plate SLN226 ring	Flange bottom b end plate SLN226 ring	◢ 4.2 ◣	58	C LC_103	65.1	0.0	36.9	-28.0	13.2	16.1	12.7	OK	OK
		◢ 4.2 ◣	58	C LC_119	51.9	0.0	-10.1	-28.7	6.2	12.8	9.3	OK	OK
End plate SLN227 ring	Flange top a end plate SLN227 ring	◢ 4.2 ◣	58	C LC_130	46.6	0.0	-23.7	18.3	14.2	11.5	7.4	OK	OK
		◢ 4.2 ◣	58	C LC_103	43.0	0.0	2.6	6.2	24.0	10.6	9.0	OK	OK
End plate SLN227 ring	Flange top b end plate SLN227 ring	◢ 4.2 ◣	58	C LC_125	38.6	0.0	-19.0	15.3	-11.9	9.5	7.2	OK	OK
		◢ 4.2 ◣	58	C LC_125	36.8	0.0	-11.8	-15.6	12.8	9.1	7.9	OK	OK
End plate SLN227 ring	Flange bottom a end plate SLN227 ring	◢ 4.2 ◣	58	C LC_145	52.1	0.0	28.8	-23.2	-9.5	12.9	10.6	OK	OK
		◢ 4.2 ◣	58	C LC_126	49.3	0.0	-24.0	-21.3	-12.8	12.2	8.8	OK	OK
End plate SLN227 ring	Flange bottom b end plate SLN227 ring	◢ 4.2 ◣	58	C LC_120	51.0	0.0	10.7	-20.0	-20.8	12.6	7.3	OK	OK
		◢ 4.2 ◣	58	C LC_119	42.5	0.0	-10.7	-18.9	-14.5	10.5	7.1	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-T_060
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN266+SLN188 radial (continuous) - SLN41 column below - SLN121 column above - SLN265 ring - SLN189 ring - SLN452 raking strut

Name	SLN266+SLN188 radial (continuous) - SLN41 column below - SLN121 column above - SLN265 ring - SLN189 ring - SLN452 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

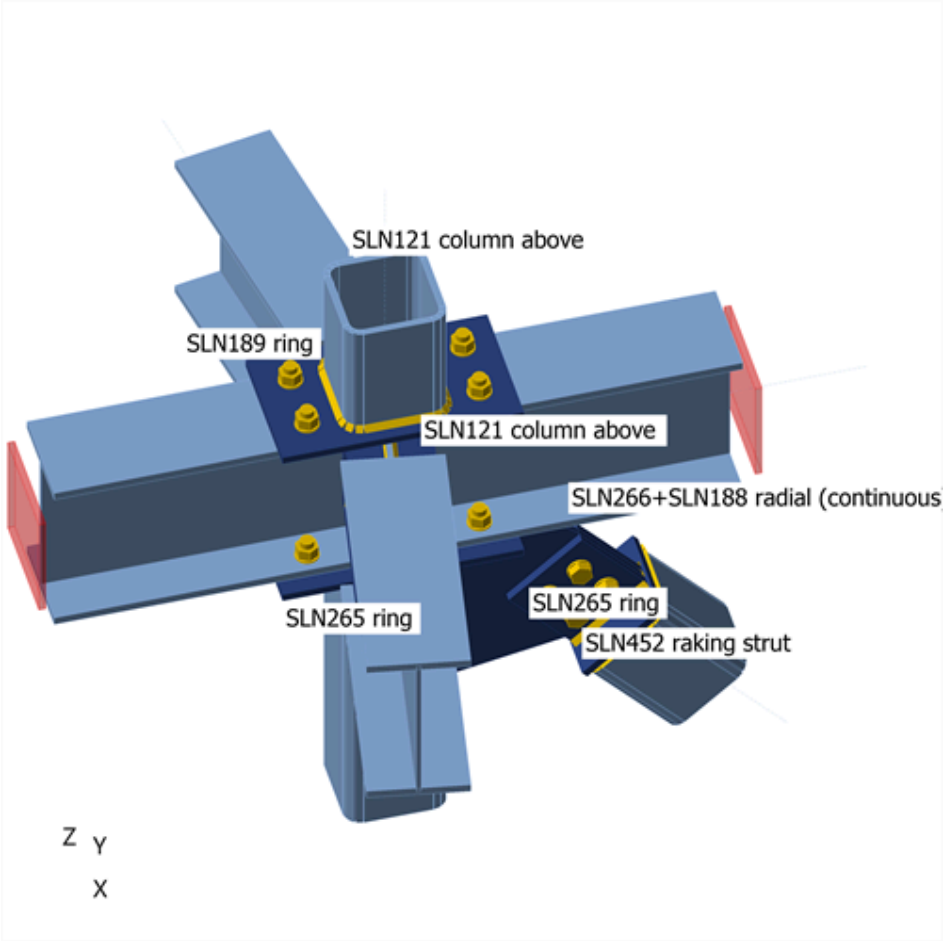
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN266+SLN188 radial (continuous)	1 - UB254x146x43	-120.0	0.0	0.0	0	0	0
SLN41 column below	2 - SHS150x150x10	0.0	90.0	-120.0	0	0	0
SLN121 column above	2 - SHS150x150x10	0.0	90.0	-120.0	0	0	0
SLN265 ring	1 - UB254x146x43	140.0	0.0	0.0	0	0	0
SLN189 ring	1 - UB254x146x43	160.0	0.0	0.0	0	0	0
SLN452 raking strut	2 - SHS150x150x10	60.0	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN266+SLN188 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN266+SLN188 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN41 column below / begin		Position	0
SLN121 column above / end		Position	0
SLN265 ring / begin		Position	0
SLN189 ring / end		Position	0
SLN452 raking strut / end		Position	0



SLN266+SLN188 radial (continuous) - SLN41 column below - SLN121 column above - SLN265 ring - SLN189 ring - SLN452 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN41 column below / Begin	60.5	0.0	0.1	0.0	0.6	0.0
	SLN121 column above / End	-40.2	0.0	-0.1	0.1	0.4	0.1
	SLN265 ring / Begin	-8.9	-0.1	-5.0	0.0	0.0	0.2
	SLN189 ring / End	8.9	0.0	-4.9	0.0	0.0	-0.1
	SLN452 raking strut / End	-10.8	0.0	-1.0	0.0	0.0	0.0
C LC_102	SLN41 column below / Begin	254.2	0.0	0.3	0.0	1.2	-0.2
	SLN121 column above / End	-156.7	0.2	0.1	0.6	0.0	0.3
	SLN265 ring / Begin	-28.0	-0.3	-26.0	0.0	0.0	0.8
	SLN189 ring / End	28.2	-0.1	-25.8	0.0	0.0	-0.4
	SLN452 raking strut / End	-56.9	-0.1	-1.0	0.2	0.0	-0.1
C LC_103	SLN41 column below / Begin	237.3	0.0	0.3	0.0	1.1	-0.2
	SLN121 column above / End	-151.3	0.2	0.2	0.6	-0.1	0.5
	SLN265 ring / Begin	-19.4	-0.4	-25.0	0.0	0.0	0.9
	SLN189 ring / End	20.0	-0.1	-24.7	0.0	0.0	-0.3
	SLN452 raking strut / End	-64.5	-0.3	-0.9	0.2	0.0	-0.3
C LC_104	SLN41 column below / Begin	246.6	-2.2	-1.1	0.0	0.1	0.9
	SLN121 column above / End	-149.5	0.0	0.1	0.5	0.3	-0.5
	SLN265 ring / Begin	-25.6	-0.2	-25.0	0.0	0.0	0.7
	SLN189 ring / End	23.0	0.0	-24.7	0.0	0.0	-0.2
	SLN452 raking strut / End	-49.7	0.0	-1.1	0.3	0.0	-0.4
C LC_105	SLN41 column below / Begin	253.3	1.3	-2.1	0.0	-0.6	-0.5
	SLN121 column above / End	-148.3	0.2	0.0	0.6	0.4	0.4
	SLN265 ring / Begin	-27.4	-0.1	-25.0	0.0	0.0	0.5
	SLN189 ring / End	28.9	0.0	-24.7	0.0	0.0	-0.3
	SLN452 raking strut / End	-39.3	-0.1	-1.2	0.2	0.0	0.3
C LC_106	SLN41 column below / Begin	233.6	0.1	0.3	0.0	1.1	-0.3
	SLN121 column above / End	-152.4	0.2	0.2	0.6	-0.2	0.5
	SLN265 ring / Begin	-18.0	-0.4	-25.0	0.0	0.0	0.9
	SLN189 ring / End	18.2	-0.2	-24.7	0.0	0.0	-0.4
	SLN452 raking strut / End	-70.8	0.0	-0.9	0.3	0.0	-0.1
C LC_107	SLN41 column below / Begin	16.9	0.0	0.1	0.0	0.3	-0.1
	SLN121 column above / End	-20.7	0.1	0.0	0.1	0.1	0.4
	SLN265 ring / Begin	3.4	-0.1	-1.6	0.0	0.0	0.4
	SLN189 ring / End	-2.8	0.0	-1.6	0.0	0.0	0.1
	SLN452 raking strut / End	-17.9	-0.4	-0.6	0.0	0.0	-0.4
C LC_108	SLN41 column below / Begin	32.2	-4.5	-2.7	0.0	-1.7	2.2
	SLN121 column above / End	-17.6	-0.4	-0.3	-0.1	0.8	-1.5
	SLN265 ring / Begin	-6.6	0.0	-1.6	0.0	0.0	0.1
	SLN189 ring / End	1.4	0.1	-1.6	0.0	0.0	0.2
	SLN452 raking strut / End	6.8	0.1	-0.9	0.2	0.0	-0.7
C LC_109	SLN41 column below / Begin	44.0	2.5	-4.7	0.0	-3.2	-0.9
	SLN121 column above / End	-15.4	0.1	-0.3	0.1	1.0	0.3
	SLN265 ring / Begin	-9.6	0.1	-1.6	0.0	0.0	-0.3
	SLN189 ring / End	12.2	0.0	-1.6	0.0	0.0	-0.1
	SLN452 raking strut / End	25.3	0.0	-1.0	-0.1	0.0	0.7

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_110	SLN41 column below / Begin	11.0	0.1	0.1	0.0	0.3	-0.3
	SLN121 column above / End	-22.6	0.1	0.0	0.1	0.1	0.4
	SLN265 ring / Begin	5.5	-0.1	-1.6	0.0	0.0	0.2
	SLN189 ring / End	-5.3	-0.1	-1.6	0.0	0.0	-0.2
	SLN452 raking strut / End	-28.2	0.2	-0.5	0.1	0.0	0.0
C LC_111	SLN41 column below / Begin	166.1	0.0	0.2	0.0	0.8	-0.1
	SLN121 column above / End	-112.1	0.2	0.1	0.5	-0.1	0.5
	SLN265 ring / Begin	-9.8	-0.3	-17.6	0.0	0.0	0.8
	SLN189 ring / End	10.6	-0.1	-17.4	0.0	0.0	-0.2
	SLN452 raking strut / End	-54.7	-0.5	-0.9	0.2	0.0	-0.4
C LC_112	SLN41 column below / Begin	182.6	-4.5	-2.6	0.0	-1.1	2.1
	SLN121 column above / End	-108.8	-0.3	-0.1	0.3	0.6	-1.3
	SLN265 ring / Begin	-20.5	-0.1	-17.6	0.0	0.0	0.6
	SLN189 ring / End	15.4	0.0	-17.4	0.0	0.0	0.0
	SLN452 raking strut / End	-28.5	0.1	-1.2	0.4	0.0	-0.7
C LC_113	SLN41 column below / Begin	194.9	2.5	-4.6	0.0	-2.6	-0.9
	SLN121 column above / End	-106.6	0.2	-0.2	0.5	0.8	0.4
	SLN265 ring / Begin	-23.8	0.0	-17.6	0.0	0.0	0.1
	SLN189 ring / End	26.5	0.0	-17.4	0.0	0.0	-0.2
	SLN452 raking strut / End	-9.2	0.0	-1.3	0.0	0.0	0.6
C LC_114	SLN41 column below / Begin	159.7	0.1	0.2	0.0	0.9	-0.5
	SLN121 column above / End	-114.0	0.2	0.2	0.4	-0.1	0.6
	SLN265 ring / Begin	-7.5	-0.3	-17.6	0.0	0.0	0.8
	SLN189 ring / End	7.7	-0.2	-17.4	0.0	0.0	-0.4
	SLN452 raking strut / End	-65.8	0.1	-0.7	0.2	0.0	-0.1
C LC_115	SLN41 column below / Begin	32.5	0.0	0.1	0.0	0.4	-0.1
	SLN121 column above / End	-31.2	0.1	-0.1	0.2	0.2	0.4
	SLN265 ring / Begin	1.1	-0.1	-2.9	0.0	0.0	0.4
	SLN189 ring / End	-0.5	0.0	-2.9	0.0	0.0	0.0
	SLN452 raking strut / End	-20.8	-0.5	-0.9	0.0	0.0	-0.4
C LC_116	SLN41 column below / Begin	48.0	-4.5	-2.7	0.0	-1.5	2.1
	SLN121 column above / End	-28.0	-0.4	-0.3	-0.1	0.9	-1.5
	SLN265 ring / Begin	-8.9	0.0	-2.9	0.0	0.0	0.2
	SLN189 ring / End	3.7	0.1	-2.9	0.0	0.0	0.1
	SLN452 raking strut / End	4.1	0.1	-1.2	0.2	0.0	-0.7
C LC_117	SLN41 column below / Begin	59.8	2.5	-4.7	0.0	-3.0	-0.9
	SLN121 column above / End	-25.8	0.1	-0.4	0.2	1.1	0.3
	SLN265 ring / Begin	-12.0	0.1	-2.9	0.0	0.0	-0.2
	SLN189 ring / End	14.5	0.0	-2.9	0.0	0.0	-0.1
	SLN452 raking strut / End	22.6	0.0	-1.3	-0.1	0.0	0.7
C LC_118	SLN41 column below / Begin	26.6	0.1	0.1	0.0	0.4	-0.4
	SLN121 column above / End	-33.0	0.1	-0.1	0.1	0.2	0.4
	SLN265 ring / Begin	3.2	-0.1	-2.9	0.0	0.0	0.3
	SLN189 ring / End	-3.1	-0.1	-2.9	0.0	0.0	-0.2
	SLN452 raking strut / End	-31.1	0.2	-0.7	0.1	0.0	-0.1
C LC_119	SLN41 column below / Begin	248.3	0.1	0.3	0.0	1.2	-0.2
	SLN121 column above / End	-156.5	0.2	0.3	0.6	-0.2	0.5

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN265 ring / Begin	-21.5	-0.4	-26.0	0.0	0.0	0.9
	SLN189 ring / End	22.2	-0.1	-25.8	0.0	0.0	-0.3
	SLN452 raking strut / End	-64.2	-0.3	-0.9	0.2	0.0	-0.3
C LC_120	SLN41 column below / Begin	255.2	-2.2	-1.1	0.0	0.2	1.0
	SLN121 column above / End	-155.0	-0.1	0.0	0.5	0.3	-0.5
	SLN265 ring / Begin	-27.9	-0.2	-26.0	0.0	0.0	0.8
	SLN189 ring / End	25.3	0.0	-25.8	0.0	0.0	-0.2
	SLN452 raking strut / End	-53.1	0.0	-1.1	0.3	0.0	-0.4
C LC_121	SLN41 column below / Begin	261.4	1.3	-2.1	0.0	-0.5	-0.6
	SLN121 column above / End	-153.9	0.2	-0.1	0.6	0.6	0.4
	SLN265 ring / Begin	-29.8	-0.1	-26.0	0.0	0.0	0.6
	SLN189 ring / End	31.3	0.0	-25.8	0.0	0.0	-0.3
	SLN452 raking strut / End	-43.6	-0.1	-1.2	0.2	0.0	0.3
C LC_122	SLN41 column below / Begin	245.8	0.1	0.3	0.0	1.2	-0.3
	SLN121 column above / End	-157.5	0.2	0.3	0.6	-0.3	0.5
	SLN265 ring / Begin	-20.1	-0.4	-26.0	0.0	0.0	0.9
	SLN189 ring / End	20.3	-0.2	-25.8	0.0	0.0	-0.4
	SLN452 raking strut / End	-68.8	0.0	-0.9	0.3	0.0	-0.1
C LC_123	SLN41 column below / Begin	188.2	0.1	0.2	0.0	1.0	-0.2
	SLN121 column above / End	-122.5	0.2	0.2	0.5	-0.2	0.7
	SLN265 ring / Begin	-14.1	-0.3	-19.7	0.0	0.0	0.9
	SLN189 ring / End	15.0	-0.1	-19.5	0.0	0.0	-0.2
	SLN452 raking strut / End	-54.1	-0.5	-0.9	0.2	0.0	-0.5
C LC_124	SLN41 column below / Begin	199.8	-4.5	-2.5	0.0	-1.0	2.1
	SLN121 column above / End	-119.7	-0.3	-0.2	0.3	0.8	-1.4
	SLN265 ring / Begin	-25.2	-0.2	-19.7	0.0	0.0	0.7
	SLN189 ring / End	20.0	0.0	-19.5	0.0	0.0	-0.1
	SLN452 raking strut / End	-35.3	0.1	-1.2	0.4	0.0	-0.7
C LC_125	SLN41 column below / Begin	211.0	2.5	-4.6	0.0	-2.4	-1.0
	SLN121 column above / End	-117.7	0.2	-0.4	0.5	1.2	0.5
	SLN265 ring / Begin	-28.6	0.0	-19.7	0.0	0.0	0.3
	SLN189 ring / End	31.3	-0.1	-19.5	0.0	0.0	-0.4
	SLN452 raking strut / End	-17.9	0.0	-1.3	0.0	0.0	0.6
C LC_126	SLN41 column below / Begin	184.3	0.1	0.2	0.0	1.0	-0.5
	SLN121 column above / End	-124.3	0.2	0.3	0.4	-0.3	0.6
	SLN265 ring / Begin	-11.7	-0.3	-19.7	0.0	0.0	0.8
	SLN189 ring / End	11.9	-0.2	-19.5	0.0	0.0	-0.4
	SLN452 raking strut / End	-61.7	0.1	-0.7	0.2	0.0	-0.1
C LC_127	SLN41 column below / Begin	54.6	0.0	0.1	0.0	0.6	-0.2
	SLN121 column above / End	-41.5	0.1	0.0	0.1	0.1	0.5
	SLN265 ring / Begin	-3.2	-0.2	-5.0	0.0	0.0	0.4
	SLN189 ring / End	3.9	0.0	-4.9	0.0	0.0	0.0
	SLN452 raking strut / End	-20.1	-0.5	-0.9	0.0	0.0	-0.4
C LC_128	SLN41 column below / Begin	65.2	-4.5	-2.6	0.0	-1.4	2.2
	SLN121 column above / End	-38.9	-0.4	-0.4	0.0	1.0	-1.6
	SLN265 ring / Begin	-13.6	-0.1	-5.0	0.0	0.0	0.3
	SLN189 ring / End	8.3	0.0	-4.9	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN452 raking strut / End	-2.7	0.1	-1.2	0.2	0.0	-0.7
C LC_129	SLN41 column below / Begin	75.9	2.5	-4.7	0.0	-2.8	-0.9
	SLN121 column above / End	-36.9	0.1	-0.5	0.2	1.4	0.3
	SLN265 ring / Begin	-16.8	0.0	-5.0	0.0	0.0	-0.1
	SLN189 ring / End	19.3	-0.1	-4.9	0.0	0.0	-0.2
	SLN452 raking strut / End	14.0	0.0	-1.3	-0.1	0.0	0.7
C LC_130	SLN41 column below / Begin	51.1	0.1	0.1	0.0	0.6	-0.4
	SLN121 column above / End	-43.3	0.1	0.1	0.1	0.0	0.5
	SLN265 ring / Begin	-1.0	-0.1	-5.0	0.0	0.0	0.3
	SLN189 ring / End	1.2	-0.1	-4.9	0.0	0.0	-0.2
	SLN452 raking strut / End	-27.1	0.1	-0.7	0.1	0.0	-0.1
C LC_135	SLN41 column below / Begin	251.0	0.0	0.3	0.0	1.3	-0.1
	SLN121 column above / End	-152.3	0.2	0.1	0.4	0.2	0.3
	SLN265 ring / Begin	-28.5	-0.3	-26.0	0.0	0.0	0.8
	SLN189 ring / End	28.6	-0.1	-25.8	0.0	0.0	-0.4
	SLN452 raking strut / End	-55.6	-0.1	-1.0	0.2	0.0	-0.1
C LC_141	SLN41 column below / Begin	60.3	0.0	0.1	0.0	0.6	0.0
	SLN121 column above / End	-40.3	0.0	-0.1	0.1	0.4	0.1
	SLN265 ring / Begin	-8.7	-0.1	-5.0	0.0	0.0	0.2
	SLN189 ring / End	8.8	-0.1	-4.9	0.0	0.0	-0.1
	SLN452 raking strut / End	-11.1	0.0	-1.0	0.0	0.0	0.0
C LC_142	SLN41 column below / Begin	60.7	0.0	0.1	0.0	0.6	0.0
	SLN121 column above / End	-40.2	0.0	-0.1	0.1	0.4	0.1
	SLN265 ring / Begin	-9.0	-0.1	-5.0	0.0	0.0	0.2
	SLN189 ring / End	9.0	0.0	-4.9	0.0	0.0	-0.1
	SLN452 raking strut / End	-10.4	0.0	-1.0	0.0	0.0	0.0
C LC_143	SLN41 column below / Begin	60.9	0.0	0.1	0.0	0.6	0.0
	SLN121 column above / End	-40.1	0.0	-0.1	0.1	0.4	0.1
	SLN265 ring / Begin	-9.1	-0.1	-5.0	0.0	0.0	0.2
	SLN189 ring / End	9.2	0.0	-4.9	0.0	0.0	-0.1
	SLN452 raking strut / End	-10.1	0.0	-1.0	0.0	0.0	0.0
C LC_144	SLN41 column below / Begin	60.1	0.0	0.1	0.0	0.6	0.0
	SLN121 column above / End	-40.3	0.0	-0.1	0.1	0.4	0.1
	SLN265 ring / Begin	-8.7	-0.1	-5.0	0.0	0.0	0.2
	SLN189 ring / End	8.7	-0.1	-4.9	0.0	0.0	-0.1
	SLN452 raking strut / End	-11.4	0.0	-1.0	0.0	0.0	0.0
C LC_145	SLN41 column below / Begin	253.3	0.0	0.3	0.0	1.2	-0.2
	SLN121 column above / End	-156.8	0.2	0.2	0.6	0.0	0.3
	SLN265 ring / Begin	-27.3	-0.3	-26.0	0.0	0.0	0.9
	SLN189 ring / End	27.7	-0.2	-25.8	0.0	0.0	-0.4
	SLN452 raking strut / End	-58.2	-0.1	-1.0	0.2	0.0	-0.1
C LC_146	SLN41 column below / Begin	255.1	0.0	0.3	0.0	1.2	-0.2
	SLN121 column above / End	-156.5	0.2	0.1	0.6	0.1	0.3
	SLN265 ring / Begin	-28.7	-0.2	-26.0	0.0	0.0	0.8
	SLN189 ring / End	28.6	-0.1	-25.8	0.0	0.0	-0.3
	SLN452 raking strut / End	-55.5	-0.1	-1.0	0.2	0.0	0.0
C LC_147	SLN41 column below / Begin	256.0	0.0	0.3	0.0	1.2	-0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN121 column above / End	-156.4	0.2	0.1	0.6	0.1	0.3
	SLN265 ring / Begin	-29.0	-0.2	-26.0	0.0	0.0	0.7
	SLN189 ring / End	29.3	0.0	-25.8	0.0	0.0	-0.3
	SLN452 raking strut / End	-54.2	-0.1	-1.0	0.2	0.0	-0.1
C LC_148	SLN41 column below / Begin	252.4	0.0	0.3	0.0	1.2	-0.2
	SLN121 column above / End	-156.9	0.2	0.2	0.6	0.0	0.3
	SLN265 ring / Begin	-27.0	-0.4	-26.0	0.0	0.0	0.9
	SLN189 ring / End	27.1	-0.2	-25.8	0.0	0.0	-0.4
	SLN452 raking strut / End	-59.5	-0.1	-1.0	0.2	0.0	-0.1
C LC_151	SLN41 column below / Begin	174.8	0.0	-1.4	0.0	-5.8	0.0
	SLN121 column above / End	-121.4	0.1	3.8	0.4	-9.3	0.2
	SLN265 ring / Begin	-12.6	-0.2	-11.2	0.0	0.0	0.5
	SLN189 ring / End	12.6	-0.1	-11.4	0.0	0.0	-0.2
	SLN452 raking strut / End	-44.2	-0.1	-1.0	0.3	0.0	-0.1
C LC_152	SLN41 column below / Begin	153.6	0.0	1.8	0.0	7.2	-0.1
	SLN121 column above / End	-94.9	0.1	-3.8	0.6	9.4	0.3
	SLN265 ring / Begin	-28.7	-0.2	-14.0	0.0	0.0	0.7
	SLN189 ring / End	28.9	-0.1	-14.1	0.0	0.0	-0.3
	SLN452 raking strut / End	-27.5	-0.1	-1.0	0.2	0.0	0.0
C LC_153	SLN41 column below / Begin	251.5	0.0	0.3	0.0	1.1	-0.2
	SLN121 column above / End	-155.1	0.2	0.1	0.6	-0.1	0.3
	SLN265 ring / Begin	-28.1	-0.3	-20.2	0.0	0.0	0.8
	SLN189 ring / End	28.3	-0.1	-20.6	0.0	0.0	-0.4
	SLN452 raking strut / End	-56.3	-0.1	-1.0	0.2	0.0	-0.1
C LC_154	SLN41 column below / Begin	176.0	0.0	-1.4	0.0	-5.7	0.0
	SLN121 column above / End	-123.3	0.1	3.8	0.4	-9.3	0.2
	SLN265 ring / Begin	-12.7	-0.2	-11.2	0.0	0.0	0.5
	SLN189 ring / End	12.8	-0.1	-11.4	0.0	0.0	-0.2
	SLN452 raking strut / End	-44.4	-0.1	-1.0	0.3	0.0	-0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.9 < 5.0%	OK
Loc. deformation	0.2 < 3%	OK
Bolts	17.1 < 100%	OK
Welds	98.8 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN266+SLN188 radial (continuous)-bfl 1	12.7	C LC_122	236.0	0.0	79.2	OK
SLN266+SLN188 radial (continuous)-tfl 1	12.7	C LC_153	136.8	0.0	33.2	OK
SLN266+SLN188 radial (continuous)-w 1	7.2	C LC_124	275.4	0.2	57.4	OK
SLN41 column below	10.0	C LC_122	277.0	0.9	0.0	OK
SLN121 column above	10.0	C LC_154	188.0	0.0	0.0	OK
SLN265 ring-bfl 1	12.7	C LC_119	63.8	0.0	0.0	OK
SLN265 ring-tfl 1	12.7	C LC_119	57.7	0.0	0.0	OK
SLN265 ring-w 1	7.2	C LC_119	188.9	0.0	0.0	OK
SLN189 ring-bfl 1	12.7	C LC_148	37.8	0.0	0.0	OK
SLN189 ring-tfl 1	12.7	C LC_106	30.9	0.0	0.0	OK
SLN189 ring-w 1	7.2	C LC_148	123.3	0.0	0.0	OK
SLN452 raking strut	10.0	C LC_106	60.7	0.0	0.0	OK
End plate column below SLN41 column below	12.0	C LC_122	238.5	0.0	66.9	OK
End plate column above SLN121 column above	12.0	C LC_154	128.7	0.0	47.8	OK
End plate SLN265 ring	12.0	C LC_119	95.1	0.0	0.0	OK
End plate SLN189 ring	12.0	C LC_120	67.2	0.0	96.0	OK
Cap plate SLN452 raking strut	12.0	C LC_106	75.7	0.0	0.0	OK
Fin plate SLN452 raking strut	12.0	C LC_124	187.2	0.0	38.3	OK
Fin plate 2 SLN452 raking strut	12.0	C LC_108	147.1	0.0	43.2	OK
Gusset SLN452 raking strut	12.0	C LC_106	237.5	0.0	7.8	OK
Flange top a end plate SLN265 ring	12.0	C LC_119	77.5	0.0	0.0	OK
Flange top b end plate SLN265 ring	12.0	C LC_119	68.0	0.0	0.0	OK
Flange bottom a end plate SLN265 ring	12.0	C LC_119	60.2	0.0	0.0	OK
Flange bottom b end plate SLN265 ring	12.0	C LC_148	87.3	0.0	0.0	OK
Flange top a end plate SLN189 ring	12.0	C LC_120	40.8	0.0	0.0	OK
Flange top b end plate SLN189 ring	12.0	C LC_106	40.2	0.0	0.0	OK
Flange bottom a end plate SLN189 ring	12.0	C LC_120	64.7	0.0	0.0	OK
Flange bottom b end plate SLN189 ring	12.0	C LC_120	48.8	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

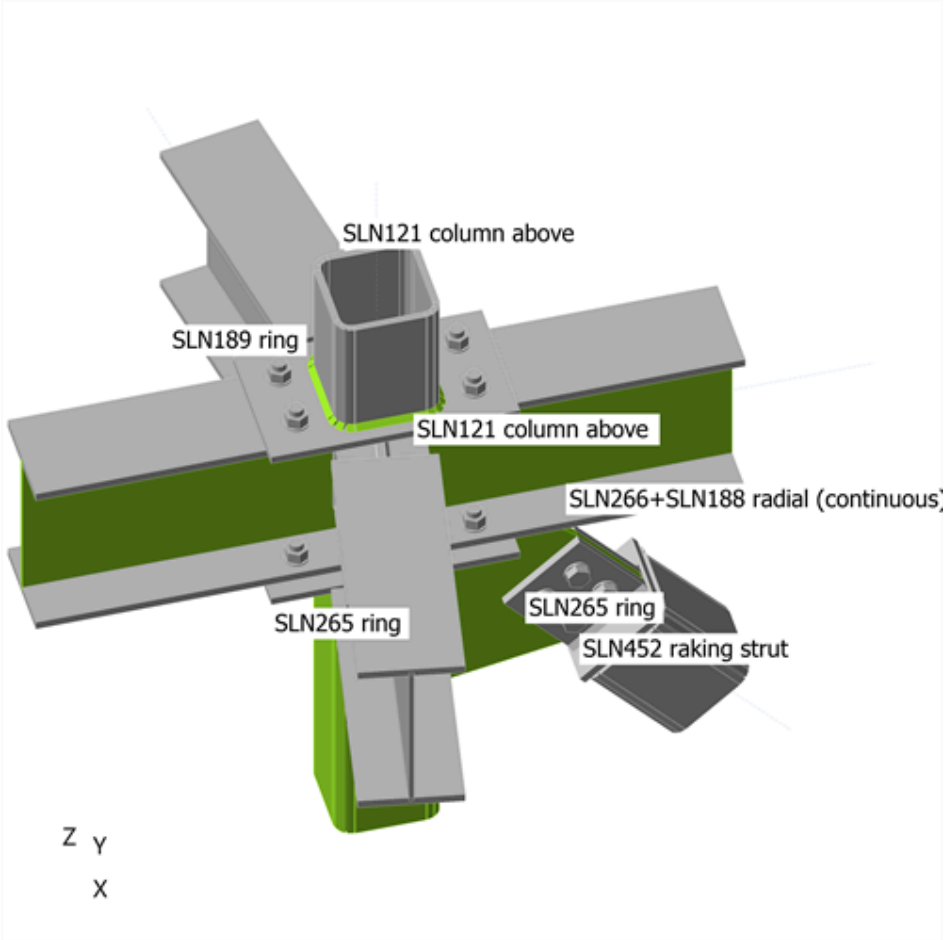
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

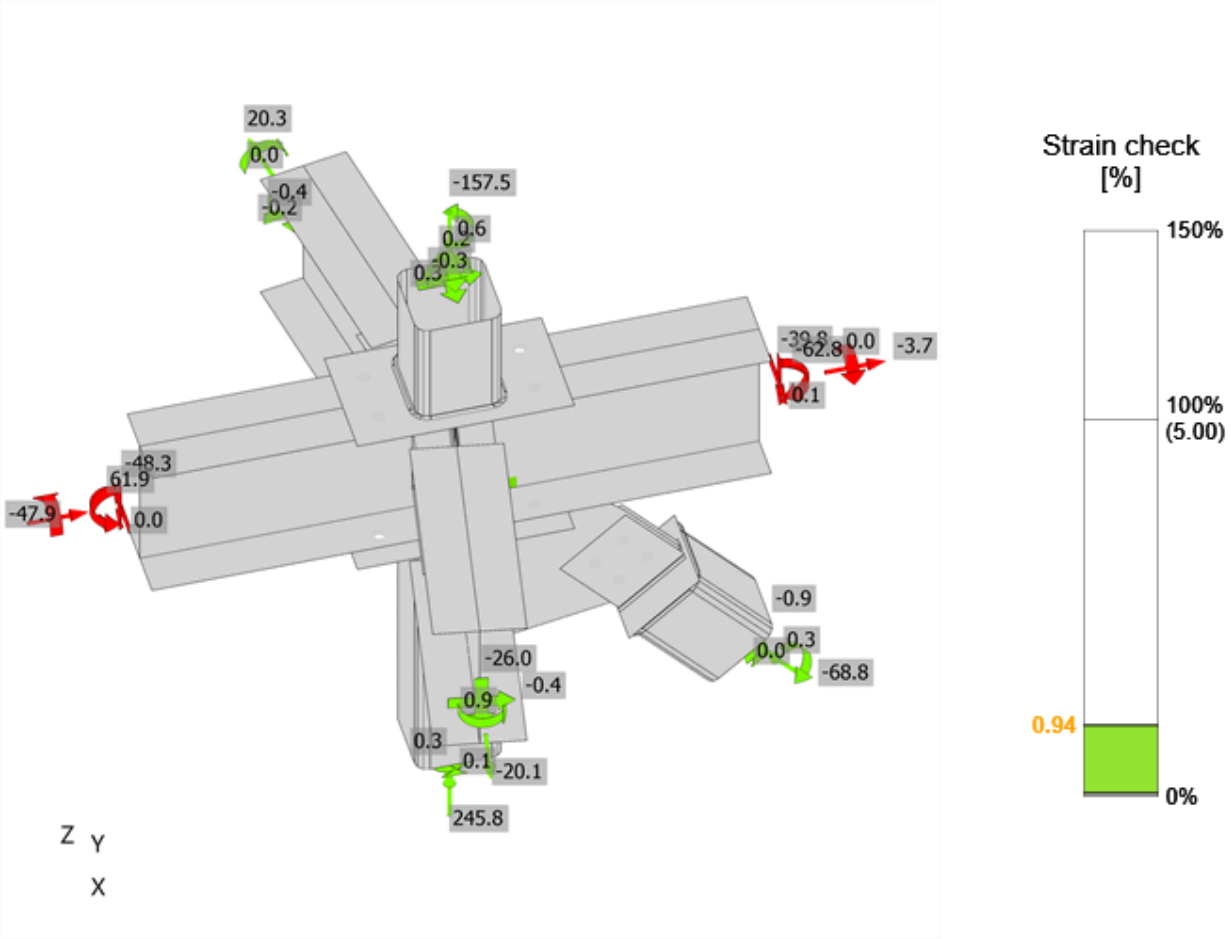
Name	d_0 [mm]	Loads	δ [mm]	$δ_{lim}$ [mm]	$δ/d_0$ [%]	Check status
SLN41 column below	150	C LC_106	0	5	0.2	OK
SLN121 column above	150	C LC_154	0	5	0.0	OK
SLN452 raking strut	150	C LC_154	0	5	0.0	OK

Symbol explanation

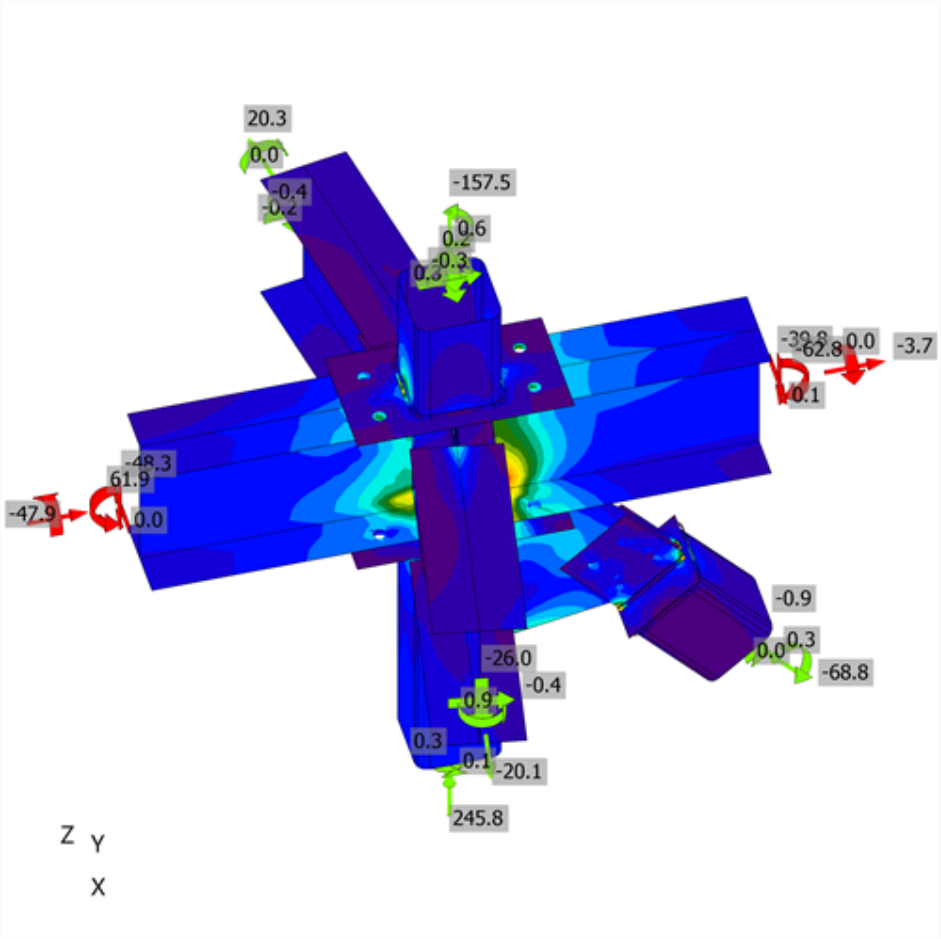
d_0	Cross-section size
δ	Local cross-section deformation
$δ_{lim}$	Allowed deformation
$δ/d_0$	Relative deformation



Overall check

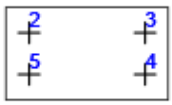
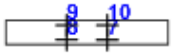
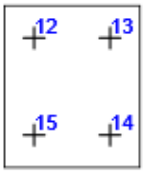
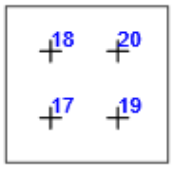


Strain check, C LC_122



Equivalent stress, C LC_122

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_106	0.5	15.9	171.5	0.3	16.9	17.1	OK	OK
	B3	M20 8.8 - 1	C LC_106	2.3	8.1	156.4	1.6	8.7	9.8	OK	OK
	B4	M20 8.8 - 1	C LC_114	2.4	8.6	156.4	1.7	9.2	10.4	OK	OK
	B5	M20 8.8 - 1	C LC_106	0.9	14.8	168.6	0.6	15.8	16.2	OK	OK
	B7	M20 8.8 - 1	C LC_151	3.5	1.7	156.4	2.5	1.8	3.6	OK	OK
	B8	M20 8.8 - 1	C LC_152	8.7	3.0	82.7	6.1	3.6	7.6	OK	OK
	B9	M20 8.8 - 1	C LC_152	7.5	1.4	156.4	5.3	1.5	5.3	OK	OK
	B10	M20 8.8 - 1	C LC_124	5.0	2.2	127.0	3.6	2.4	4.9	OK	OK
	B12	M20 8.8 - 1	C LC_119	13.4	1.4	123.8	9.5	1.9	8.2	OK	OK
	B13	M20 8.8 - 1	C LC_109	3.5	0.3	123.8	2.5	0.5	2.1	OK	OK
	B14	M20 8.8 - 1	C LC_152	0.0	2.2	123.8	0.0	3.4	2.4	OK	OK
	B15	M20 8.8 - 1	C LC_119	14.2	1.2	123.8	10.1	1.3	8.5	OK	OK
	B17	M20 8.8 - 1	C LC_124	5.9	8.4	115.3	4.2	8.9	11.9	OK	OK
	B18	M20 8.8 - 1	C LC_106	0.7	9.8	174.9	0.5	11.1	10.8	OK	OK
	B19	M20 8.8 - 1	C LC_106	0.6	8.5	115.3	0.5	14.0	9.3	OK	OK
	B20	M20 8.8 - 1	C LC_106	0.2	9.5	115.3	0.2	16.3	10.2	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN452 raking strut	SLN452 raking strut	◢ 4.2	519	C LC_106	209.8	0.0	-100.3	106.3	-3.8	51.8	28.1	OK	OK
SLN41 column below-w 3	Gusset SLN452 raking strut	◢ 5.6 ◣	287	C LC_106	399.8	1.9	-130.9	-134.1	-172.0	98.8	82.2	OK	OK
		◢ 5.6 ◣	287	C LC_106	399.5	1.8	-130.4	126.9	177.2	98.7	80.7	OK	OK
SLN265 ring-w 1	Flange top b end plate SLN265 ring	◢ 4.2 ◣	198	C LC_153	85.5	0.0	44.6	37.1	19.9	21.1	12.8	OK	OK
		◢ 4.2 ◣	197	C LC_119	39.8	0.0	-19.7	18.1	8.5	9.8	7.7	OK	OK
SLN265 ring-w 1	Flange bottom b end plate SLN265 ring	◢ 4.2 ◣	198	C LC_148	101.6	0.0	29.9	36.3	42.7	25.1	15.4	OK	OK
		◢ 4.2 ◣	198	C LC_148	157.5	0.0	65.1	-58.6	-58.6	38.9	23.9	OK	OK
SLN189 ring-w 1	Flange top a end plate SLN189 ring	◢ 4.2 ◣	198	C LC_124	71.3	0.0	-26.1	4.8	38.0	17.6	10.8	OK	OK
		◢ 4.2 ◣	198	C LC_124	28.0	0.0	20.2	10.7	3.2	6.9	6.5	OK	OK
SLN189 ring-w 1	Flange bottom a end plate SLN189 ring	◢ 4.2 ◣	198	C LC_121	88.8	0.0	25.7	28.7	-39.8	21.9	12.0	OK	OK
		◢ 4.2 ◣	198	C LC_120	128.2	0.0	-29.0	-13.3	-70.9	31.7	19.9	OK	OK
Cap plate SLN452 raking strut	Fin plate SLN452 raking strut	◢ 5.6 ◣	139	C LC_106	208.4	0.0	-91.1	-87.8	63.2	51.5	31.1	OK	OK
		◢ 5.6 ◣	139	C LC_106	202.5	0.0	-84.1	87.5	-60.4	50.0	27.0	OK	OK
Cap plate SLN452 raking strut	Fin plate 2 SLN452 raking strut	◢ 5.6 ◣	139	C LC_106	177.9	0.0	-65.2	-75.5	58.6	44.0	27.3	OK	OK
		◢ 5.6 ◣	139	C LC_106	203.2	0.0	-95.5	85.1	-58.9	50.2	28.3	OK	OK
End plate column below SLN41 column below	SLN41 column below	◢ 5.6 ◣	521	C LC_122	394.4	0.0	-251.5	-172.5	-31.6	97.5	52.7	OK	OK
		◢ 5.6 ◣	521	C LC_147	324.6	0.0	-105.5	174.6	30.3	80.2	43.7	OK	OK
End plate column above SLN121 column above	SLN121 column above	◢ 5.6 ◣	520	C LC_154	228.2	0.0	-112.0	-114.6	-6.0	56.4	30.9	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 5.6 ◣	521	C LC_154	245.7	0.0	-121.3	121.3	22.6	60.7	33.3	OK	OK
SLN266+SLN188 radial (continuous)-w 1	SLN265 ring-w 1	◢ 7.2	189	C LC_119	244.2	0.0	6.7	-13.2	140.3	60.3	33.3	OK	OK
SLN266+SLN188 radial (continuous)-w 1	SLN189 ring-w 1	◢ 7.2	189	C LC_124	289.6	0.0	0.8	20.2	-166.0	71.6	49.9	OK	OK
End plate SLN265 ring	SLN265 ring-w 1	◢ 4.2 ◣	149	C LC_119	224.1	0.0	-60.7	-26.9	-121.6	55.4	32.9	OK	OK
		◢ 4.2 ◣	149	C LC_119	161.5	0.0	-22.0	47.8	79.0	39.9	25.4	OK	OK
End plate SLN189 ring	SLN189 ring-w 1	◢ 4.2 ◣	149	C LC_124	249.6	0.0	-50.8	-61.3	127.1	61.7	36.7	OK	OK
		◢ 4.2 ◣	149	C LC_124	187.0	0.0	-51.6	44.3	-93.8	46.2	29.2	OK	OK
SLN265 ring-w 1	Flange top a end plate SLN265 ring	◢ 4.2 ◣	199	C LC_119	134.4	0.0	-12.0	7.0	-77.0	33.2	23.6	OK	OK
		◢ 4.2 ◣	198	C LC_119	91.1	0.0	14.7	16.3	-49.3	22.5	12.9	OK	OK
SLN265 ring-w 1	Flange bottom a end plate SLN265 ring	◢ 4.2 ◣	198	C LC_119	119.6	0.0	22.5	-22.3	64.1	29.6	17.6	OK	OK
		◢ 4.2 ◣	199	C LC_121	119.5	0.0	-28.4	24.6	62.3	29.5	15.6	OK	OK
SLN189 ring-w 1	Flange top b end plate SLN189 ring	◢ 4.2 ◣	199	C LC_124	77.0	0.0	-30.2	-10.1	39.6	19.0	12.0	OK	OK
		◢ 4.2 ◣	199	C LC_124	49.6	0.0	18.1	-22.2	14.7	12.3	8.2	OK	OK
SLN189 ring-w 1	Flange bottom b end plate SLN189 ring	◢ 4.2 ◣	198	C LC_148	69.2	0.0	12.6	-12.3	-37.4	17.1	11.0	OK	OK
		◢ 4.2 ◣	199	C LC_120	137.4	0.0	-34.5	22.2	-73.5	34.0	20.8	OK	OK
End plate SLN265 ring	Flange top a end plate SLN265 ring	◢ 4.2 ◣	58	C LC_119	70.1	0.0	-10.2	25.1	-31.2	17.3	10.9	OK	OK
		◢ 4.2 ◣	58	C LC_153	47.0	0.0	-3.1	11.9	-24.3	11.6	8.7	OK	OK
End plate SLN265 ring	Flange top b end plate SLN265 ring	◢ 4.2 ◣	58	C LC_119	78.6	0.0	-8.8	44.2	-8.8	19.4	11.8	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_119	79.1	0.0	44.0	32.5	-19.7	19.6	15.0	OK	OK
End plate SLN265 ring	Flange bottom a end plate SLN265 ring	◢ 4.2 ◣	58	C LC_121	94.0	0.0	11.4	-37.4	38.8	23.2	14.1	OK	OK
		◢ 4.2 ◣	58	C LC_119	103.6	0.0	-21.6	-45.6	36.6	25.6	16.2	OK	OK
End plate SLN265 ring	Flange bottom b end plate SLN265 ring	◢ 4.2 ◣	58	C LC_119	92.1	0.0	51.3	-40.3	18.2	22.8	18.8	OK	OK
		◢ 4.2 ◣	58	C LC_119	110.7	0.0	-21.7	-62.6	-2.4	27.3	15.5	OK	OK
End plate SLN189 ring	Flange top a end plate SLN189 ring	◢ 4.2 ◣	58	C LC_123	76.3	0.0	-34.5	29.9	25.6	18.9	11.3	OK	OK
		◢ 4.2 ◣	58	C LC_120	46.5	0.0	23.7	17.5	15.1	11.5	8.6	OK	OK
End plate SLN189 ring	Flange top b end plate SLN189 ring	◢ 4.2 ◣	58	C LC_124	71.7	0.0	-7.8	34.4	22.5	17.7	9.6	OK	OK
		◢ 4.2 ◣	58	C LC_124	52.2	0.0	-6.7	19.9	22.3	12.9	7.4	OK	OK
End plate SLN189 ring	Flange bottom a end plate SLN189 ring	◢ 4.2 ◣	58	C LC_120	84.5	0.0	9.0	-44.8	18.7	20.9	14.3	OK	OK
		◢ 4.2 ◣	58	C LC_124	112.7	0.0	-25.1	-60.9	17.8	27.9	20.5	OK	OK
End plate SLN189 ring	Flange bottom b end plate SLN189 ring	◢ 4.2 ◣	58	C LC_120	103.5	0.0	10.5	-42.7	-41.3	25.6	12.9	OK	OK
		◢ 4.2 ◣	58	C LC_120	97.7	0.0	-16.8	-49.0	-26.2	24.1	12.8	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-T_260
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN220+SLN203 radial (continuous) - SLN51 column below - SLN96 column above - SLN221 ring - SLN426 ring - SLN448 raking strut

Name	SLN220+SLN203 radial (continuous) - SLN51 column below - SLN96 column above - SLN221 ring - SLN426 ring - SLN448 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

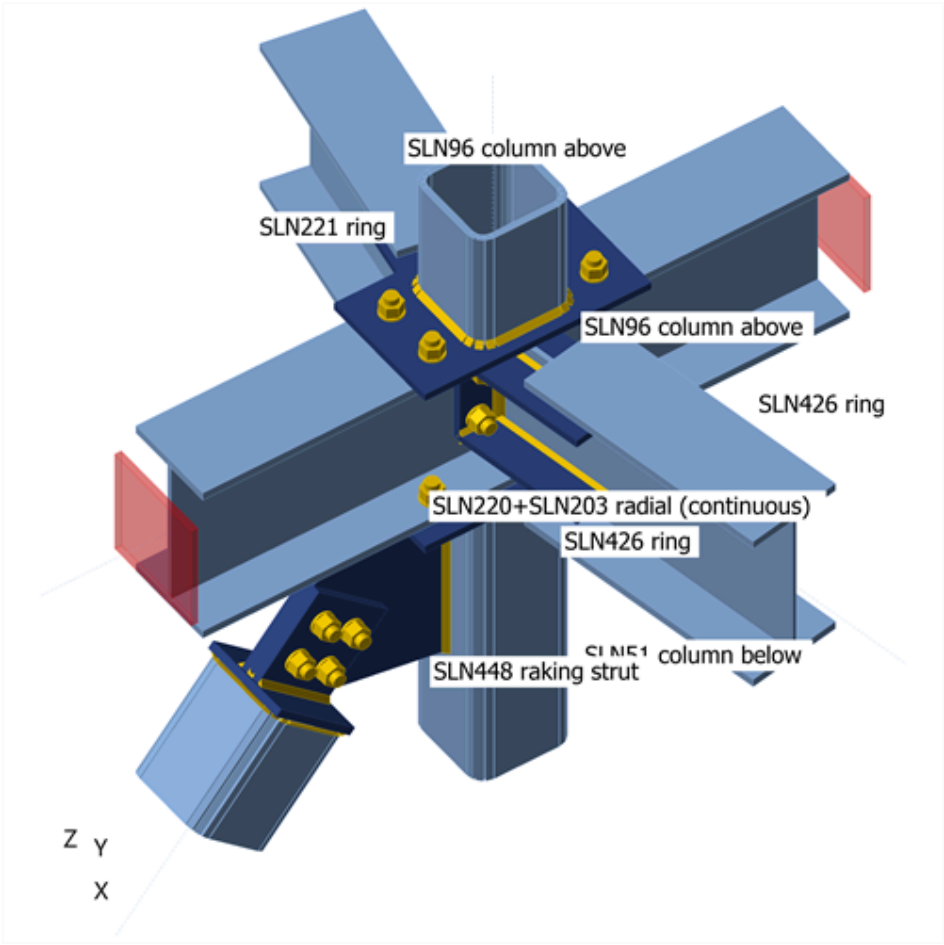
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN220+SLN203 radial (continuous)	1 - UB254x146x43	80.0	0.0	0.0	0	0	0
SLN51 column below	2 - SHS150x150x10	0.0	90.0	80.0	0	0	0
SLN96 column above	2 - SHS150x150x10	0.0	90.0	80.0	0	0	0
SLN221 ring	1 - UB254x146x43	-20.0	0.0	0.0	0	0	0
SLN426 ring	1 - UB254x146x43	0.0	0.0	0.0	0	0	0
SLN448 raking strut	2 - SHS150x150x10	-100.0	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN220+SLN203 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN220+SLN203 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN51 column below / begin		Position	0
SLN96 column above / end		Position	0
SLN221 ring / begin		Position	0
SLN426 ring / end		Position	0
SLN448 raking strut / end		Position	0



SLN220+SLN203 radial (continuous) - SLN51 column below - SLN96 column above - SLN221 ring - SLN426 ring - SLN448 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN51 column below / Begin	54.9	0.0	0.2	0.0	0.9	0.0
	SLN96 column above / End	-36.1	0.0	-0.4	-0.1	1.0	0.0
	SLN221 ring / Begin	-8.9	0.0	-4.9	0.0	0.0	0.1
	SLN426 ring / End	8.8	-0.1	-4.9	0.0	0.0	-0.2
	SLN448 raking strut / End	-10.8	0.0	-1.0	0.0	0.0	0.0
C LC_102	SLN51 column below / Begin	233.7	0.0	0.7	0.0	2.8	0.0
	SLN96 column above / End	-143.6	-0.1	-1.1	-0.6	2.7	-0.2
	SLN221 ring / Begin	-28.0	-0.2	-25.7	0.0	0.0	0.5
	SLN426 ring / End	27.7	-0.3	-25.5	0.0	0.0	-0.8
	SLN448 raking strut / End	-57.3	0.1	-1.0	-0.2	0.0	0.0
C LC_103	SLN51 column below / Begin	224.0	-2.5	0.2	0.0	2.2	1.1
	SLN96 column above / End	-138.1	-0.3	-1.0	-0.7	2.7	-0.9
	SLN221 ring / Begin	-23.1	-0.1	-24.7	0.0	0.0	0.5
	SLN426 ring / End	19.8	-0.2	-24.5	0.0	0.0	-0.6
	SLN448 raking strut / End	-55.3	0.1	-1.0	-0.1	0.0	-0.4
C LC_104	SLN51 column below / Begin	219.9	0.0	0.6	0.0	2.6	0.1
	SLN96 column above / End	-139.4	-0.1	-1.0	-0.6	2.6	-0.2
	SLN221 ring / Begin	-19.3	-0.2	-24.7	0.0	0.0	0.5
	SLN426 ring / End	19.4	-0.2	-24.5	0.0	0.0	-0.7
	SLN448 raking strut / End	-62.6	-0.2	-1.0	-0.2	0.0	-0.1
C LC_105	SLN51 column below / Begin	214.2	-0.1	0.6	0.0	2.6	0.2
	SLN96 column above / End	-141.4	-0.2	-0.9	-0.6	2.3	-0.5
	SLN221 ring / Begin	-18.1	-0.2	-24.7	0.0	0.0	0.4
	SLN426 ring / End	17.5	-0.4	-24.5	0.0	0.0	-1.0
	SLN448 raking strut / End	-72.9	0.1	-0.9	-0.2	0.0	0.2
C LC_106	SLN51 column below / Begin	236.1	0.5	-2.1	0.0	0.7	-0.3
	SLN96 column above / End	-135.9	0.0	-1.2	-0.6	3.1	0.2
	SLN221 ring / Begin	-28.6	0.0	-24.7	0.0	0.0	0.2
	SLN426 ring / End	29.1	-0.1	-24.5	0.0	0.0	-0.5
	SLN448 raking strut / End	-35.9	0.0	-1.2	-0.2	0.0	0.0
C LC_107	SLN51 column below / Begin	23.6	-5.1	-0.9	0.0	-0.3	2.3
	SLN96 column above / End	-16.4	-0.4	-0.3	-0.2	0.7	-1.5
	SLN221 ring / Begin	-2.2	-0.1	-1.6	0.0	0.0	0.2
	SLN426 ring / End	-3.4	0.1	-1.6	0.0	0.0	0.2
	SLN448 raking strut / End	-2.3	0.1	-0.8	0.2	0.0	-0.9
C LC_108	SLN51 column below / Begin	17.0	0.0	0.1	0.0	0.4	0.2
	SLN96 column above / End	-18.8	0.0	-0.2	0.0	0.5	-0.1
	SLN221 ring / Begin	4.1	-0.1	-1.6	0.0	0.0	0.2
	SLN426 ring / End	-3.9	0.0	-1.6	0.0	0.0	0.0
	SLN448 raking strut / End	-14.6	-0.4	-0.7	0.0	0.0	-0.2
C LC_109	SLN51 column below / Begin	8.2	-0.1	0.1	0.0	0.3	0.3
	SLN96 column above / End	-22.4	-0.2	-0.1	-0.1	0.1	-0.6
	SLN221 ring / Begin	5.0	0.0	-1.6	0.0	0.0	0.0
	SLN426 ring / End	-5.6	-0.2	-1.6	0.0	0.0	-0.4
	SLN448 raking strut / End	-31.0	0.2	-0.4	-0.1	0.0	0.3

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_110	SLN51 column below / Begin	45.3	1.0	-5.3	0.0	-3.4	-0.7
	SLN96 column above / End	-12.5	0.2	-0.5	0.0	1.5	0.6
	SLN221 ring / Begin	-11.8	0.0	-1.6	0.0	0.0	-0.1
	SLN426 ring / End	13.1	0.1	-1.6	0.0	0.0	0.1
	SLN448 raking strut / End	32.2	-0.1	-1.1	0.0	0.0	0.0
C LC_111	SLN51 column below / Begin	162.2	-5.1	-0.5	0.0	1.2	2.3
	SLN96 column above / End	-100.7	-0.4	-0.8	-0.6	2.1	-1.6
	SLN221 ring / Begin	-16.0	-0.1	-17.4	0.0	0.0	0.5
	SLN426 ring / End	10.1	-0.1	-17.3	0.0	0.0	-0.3
	SLN448 raking strut / End	-38.2	0.2	-1.1	0.0	0.0	-0.9
C LC_112	SLN51 column below / Begin	155.1	-0.1	0.5	0.0	1.9	0.2
	SLN96 column above / End	-103.3	-0.1	-0.8	-0.4	2.0	-0.2
	SLN221 ring / Begin	-9.2	-0.2	-17.4	0.0	0.0	0.4
	SLN426 ring / End	9.5	-0.2	-17.3	0.0	0.0	-0.5
	SLN448 raking strut / End	-51.3	-0.4	-1.0	-0.2	0.0	-0.2
C LC_113	SLN51 column below / Begin	145.4	-0.1	0.5	0.0	1.8	0.4
	SLN96 column above / End	-107.0	-0.2	-0.6	-0.5	1.5	-0.7
	SLN221 ring / Begin	-7.8	-0.2	-17.4	0.0	0.0	0.3
	SLN426 ring / End	7.1	-0.4	-17.3	0.0	0.0	-0.9
	SLN448 raking strut / End	-69.1	0.2	-0.7	-0.2	0.0	0.3
C LC_114	SLN51 column below / Begin	184.7	1.0	-5.0	0.0	-1.9	-0.7
	SLN96 column above / End	-96.7	0.1	-1.1	-0.4	2.9	0.5
	SLN221 ring / Begin	-26.0	0.0	-17.4	0.0	0.0	0.1
	SLN426 ring / End	27.3	0.0	-17.3	0.0	0.0	-0.3
	SLN448 raking strut / End	-2.4	0.0	-1.3	-0.2	0.0	0.0
C LC_115	SLN51 column below / Begin	37.9	-5.1	-0.8	0.0	-0.1	2.3
	SLN96 column above / End	-25.7	-0.4	-0.4	-0.2	0.9	-1.5
	SLN221 ring / Begin	-4.5	-0.1	-2.9	0.0	0.0	0.3
	SLN426 ring / End	-1.1	0.0	-2.8	0.0	0.0	0.1
	SLN448 raking strut / End	-5.1	0.1	-1.1	0.2	0.0	-0.9
C LC_116	SLN51 column below / Begin	31.2	0.0	0.1	0.0	0.6	0.2
	SLN96 column above / End	-28.2	0.0	-0.3	0.0	0.8	-0.1
	SLN221 ring / Begin	1.8	-0.1	-2.9	0.0	0.0	0.2
	SLN426 ring / End	-1.6	0.0	-2.8	0.0	0.0	-0.1
	SLN448 raking strut / End	-17.4	-0.4	-1.0	-0.1	0.0	-0.2
C LC_117	SLN51 column below / Begin	22.3	-0.1	0.1	0.0	0.6	0.3
	SLN96 column above / End	-31.8	-0.2	-0.2	-0.1	0.4	-0.6
	SLN221 ring / Begin	2.8	0.0	-2.9	0.0	0.0	0.1
	SLN426 ring / End	-3.4	-0.2	-2.8	0.0	0.0	-0.5
	SLN448 raking strut / End	-34.0	0.2	-0.7	-0.1	0.0	0.3
C LC_118	SLN51 column below / Begin	59.6	1.0	-5.3	0.0	-3.2	-0.7
	SLN96 column above / End	-21.8	0.2	-0.7	0.0	1.7	0.6
	SLN221 ring / Begin	-14.2	0.0	-2.9	0.0	0.0	-0.1
	SLN426 ring / End	15.5	0.1	-2.8	0.0	0.0	0.1
	SLN448 raking strut / End	29.6	-0.1	-1.3	0.0	0.0	0.0
C LC_119	SLN51 column below / Begin	232.2	-2.5	0.2	0.0	2.4	1.2
	SLN96 column above / End	-142.8	-0.3	-1.1	-0.7	2.8	-0.9

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN221 ring / Begin	-25.4	-0.1	-25.7	0.0	0.0	0.5
	SLN426 ring / End	22.1	-0.2	-25.5	0.0	0.0	-0.6
	SLN448 raking strut / End	-57.8	0.1	-1.0	-0.1	0.0	-0.4
C LC_120	SLN51 column below / Begin	228.7	0.0	0.7	0.0	2.7	0.1
	SLN96 column above / End	-143.8	-0.1	-1.0	-0.6	2.7	-0.2
	SLN221 ring / Begin	-21.5	-0.2	-25.7	0.0	0.0	0.5
	SLN426 ring / End	21.6	-0.3	-25.5	0.0	0.0	-0.8
	SLN448 raking strut / End	-63.9	-0.2	-1.0	-0.2	0.0	-0.1
C LC_121	SLN51 column below / Begin	225.5	-0.1	0.7	0.0	2.7	0.2
	SLN96 column above / End	-145.0	-0.2	-0.8	-0.6	2.3	-0.6
	SLN221 ring / Begin	-20.2	-0.2	-25.7	0.0	0.0	0.4
	SLN426 ring / End	19.6	-0.4	-25.5	0.0	0.0	-1.0
	SLN448 raking strut / End	-69.8	0.1	-0.9	-0.2	0.0	0.2
C LC_122	SLN51 column below / Begin	243.7	0.5	-2.0	0.0	0.8	-0.3
	SLN96 column above / End	-141.5	0.0	-1.3	-0.6	3.4	0.2
	SLN221 ring / Begin	-31.0	0.0	-25.7	0.0	0.0	0.3
	SLN426 ring / End	31.5	-0.1	-25.5	0.0	0.0	-0.6
	SLN448 raking strut / End	-40.8	0.0	-1.2	-0.2	0.0	0.0
C LC_123	SLN51 column below / Begin	178.6	-5.1	-0.4	0.0	1.5	2.3
	SLN96 column above / End	-110.1	-0.4	-1.0	-0.6	2.5	-1.6
	SLN221 ring / Begin	-20.6	-0.2	-19.5	0.0	0.0	0.6
	SLN426 ring / End	14.6	-0.1	-19.3	0.0	0.0	-0.3
	SLN448 raking strut / End	-43.3	0.2	-1.1	0.1	0.0	-0.9
C LC_124	SLN51 column below / Begin	172.6	0.0	0.5	0.0	2.1	0.2
	SLN96 column above / End	-112.0	0.0	-0.9	-0.5	2.2	-0.1
	SLN221 ring / Begin	-13.6	-0.2	-19.5	0.0	0.0	0.5
	SLN426 ring / End	13.9	-0.2	-19.3	0.0	0.0	-0.6
	SLN448 raking strut / End	-53.9	-0.4	-1.0	-0.2	0.0	-0.3
C LC_125	SLN51 column below / Begin	167.9	-0.1	0.5	0.0	2.1	0.4
	SLN96 column above / End	-114.2	-0.2	-0.5	-0.5	1.5	-0.8
	SLN221 ring / Begin	-12.1	-0.2	-19.5	0.0	0.0	0.3
	SLN426 ring / End	11.3	-0.4	-19.3	0.0	0.0	-0.9
	SLN448 raking strut / End	-62.9	0.2	-0.7	-0.2	0.0	0.4
C LC_126	SLN51 column below / Begin	199.7	1.0	-4.9	0.0	-1.6	-0.7
	SLN96 column above / End	-108.0	0.1	-1.3	-0.4	3.4	0.6
	SLN221 ring / Begin	-30.8	0.0	-19.5	0.0	0.0	0.3
	SLN426 ring / End	32.0	-0.1	-19.3	0.0	0.0	-0.4
	SLN448 raking strut / End	-12.1	-0.1	-1.3	-0.2	0.0	0.0
C LC_127	SLN51 column below / Begin	54.3	-5.1	-0.7	0.0	0.2	2.3
	SLN96 column above / End	-35.1	-0.4	-0.6	-0.2	1.3	-1.5
	SLN221 ring / Begin	-9.1	-0.1	-4.9	0.0	0.0	0.4
	SLN426 ring / End	3.4	0.0	-4.9	0.0	0.0	0.0
	SLN448 raking strut / End	-10.1	0.2	-1.1	0.2	0.0	-0.9
C LC_128	SLN51 column below / Begin	48.8	0.0	0.2	0.0	0.8	0.2
	SLN96 column above / End	-36.9	0.0	-0.4	-0.1	1.1	0.0
	SLN221 ring / Begin	-2.6	-0.1	-4.9	0.0	0.0	0.3
	SLN426 ring / End	2.9	0.0	-4.9	0.0	0.0	-0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN448 raking strut / End	-20.0	-0.4	-1.0	-0.1	0.0	-0.3
C LC_129	SLN51 column below / Begin	44.9	-0.1	0.2	0.0	0.8	0.4
	SLN96 column above / End	-39.0	-0.2	-0.1	-0.1	0.4	-0.7
	SLN221 ring / Begin	-1.5	0.0	-4.9	0.0	0.0	0.0
	SLN426 ring / End	0.9	-0.2	-4.9	0.0	0.0	-0.5
	SLN448 raking strut / End	-27.8	0.2	-0.7	-0.1	0.0	0.3
C LC_130	SLN51 column below / Begin	74.7	1.0	-5.2	0.0	-2.8	-0.7
	SLN96 column above / End	-33.1	0.2	-0.9	0.0	2.2	0.7
	SLN221 ring / Begin	-18.9	0.0	-4.9	0.0	0.0	0.1
	SLN426 ring / End	20.3	0.0	-4.9	0.0	0.0	-0.1
	SLN448 raking strut / End	19.9	-0.1	-1.3	0.0	0.0	0.0
C LC_135	SLN51 column below / Begin	230.6	0.0	0.7	0.0	2.8	0.0
	SLN96 column above / End	-139.4	-0.1	-1.1	-0.4	2.9	-0.2
	SLN221 ring / Begin	-28.5	-0.2	-25.7	0.0	0.0	0.5
	SLN426 ring / End	28.2	-0.3	-25.5	0.0	0.0	-0.7
	SLN448 raking strut / End	-55.9	0.1	-1.0	-0.2	0.0	0.1
C LC_141	SLN51 column below / Begin	55.0	0.0	0.2	0.0	0.9	0.0
	SLN96 column above / End	-36.1	0.0	-0.4	-0.1	1.0	0.0
	SLN221 ring / Begin	-9.0	0.0	-4.9	0.0	0.0	0.1
	SLN426 ring / End	8.9	-0.1	-4.9	0.0	0.0	-0.2
	SLN448 raking strut / End	-10.7	0.0	-1.0	0.0	0.0	0.0
C LC_142	SLN51 column below / Begin	54.8	0.0	0.2	0.0	0.9	0.0
	SLN96 column above / End	-36.1	0.0	-0.4	-0.1	1.0	0.0
	SLN221 ring / Begin	-8.8	-0.1	-4.9	0.0	0.0	0.1
	SLN426 ring / End	8.8	-0.1	-4.9	0.0	0.0	-0.2
	SLN448 raking strut / End	-11.0	0.0	-1.0	0.0	0.0	0.0
C LC_143	SLN51 column below / Begin	54.4	0.0	0.2	0.0	0.9	0.0
	SLN96 column above / End	-36.2	0.0	-0.4	-0.1	1.0	0.0
	SLN221 ring / Begin	-8.6	-0.1	-4.9	0.0	0.0	0.2
	SLN426 ring / End	8.6	-0.1	-4.9	0.0	0.0	-0.2
	SLN448 raking strut / End	-11.6	0.0	-1.0	0.0	0.0	0.0
C LC_144	SLN51 column below / Begin	55.3	0.0	0.2	0.0	0.9	0.0
	SLN96 column above / End	-36.0	0.0	-0.4	-0.1	1.0	0.0
	SLN221 ring / Begin	-9.1	0.0	-4.9	0.0	0.0	0.1
	SLN426 ring / End	9.1	0.0	-4.9	0.0	0.0	-0.2
	SLN448 raking strut / End	-10.1	0.0	-1.0	0.0	0.0	0.0
C LC_145	SLN51 column below / Begin	234.1	0.0	0.7	0.0	2.8	0.0
	SLN96 column above / End	-143.6	-0.1	-1.1	-0.6	2.8	-0.2
	SLN221 ring / Begin	-28.3	-0.1	-25.7	0.0	0.0	0.4
	SLN426 ring / End	27.9	-0.3	-25.5	0.0	0.0	-0.8
	SLN448 raking strut / End	-56.6	0.1	-1.0	-0.2	0.0	0.0
C LC_146	SLN51 column below / Begin	233.2	0.0	0.7	0.0	2.8	0.0
	SLN96 column above / End	-143.7	-0.1	-1.1	-0.6	2.7	-0.2
	SLN221 ring / Begin	-27.6	-0.2	-25.7	0.0	0.0	0.5
	SLN426 ring / End	27.6	-0.3	-25.5	0.0	0.0	-0.8
	SLN448 raking strut / End	-57.9	0.0	-1.0	-0.2	0.0	0.0
C LC_147	SLN51 column below / Begin	231.8	0.0	0.7	0.0	2.8	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN96 column above / End	-143.9	-0.1	-1.0	-0.6	2.6	-0.2
	SLN221 ring / Begin	-26.8	-0.3	-25.7	0.0	0.0	0.5
	SLN426 ring / End	26.5	-0.4	-25.5	0.0	0.0	-0.9
	SLN448 raking strut / End	-60.2	0.1	-1.0	-0.2	0.0	0.0
C LC_148	SLN51 column below / Begin	235.6	0.0	0.7	0.0	2.8	0.0
	SLN96 column above / End	-143.3	-0.1	-1.1	-0.6	2.8	-0.2
	SLN221 ring / Begin	-29.1	0.0	-25.7	0.0	0.0	0.4
	SLN426 ring / End	29.0	-0.2	-25.5	0.0	0.0	-0.7
	SLN448 raking strut / End	-54.3	0.0	-1.0	-0.2	0.0	0.0
C LC_151	SLN51 column below / Begin	153.8	0.0	-1.0	0.0	-4.1	-0.1
	SLN96 column above / End	-106.4	0.0	2.5	-0.4	-6.5	-0.1
	SLN221 ring / Begin	-12.5	-0.1	-11.7	0.0	0.0	0.3
	SLN426 ring / End	12.4	-0.2	-11.9	0.0	0.0	-0.5
	SLN448 raking strut / End	-42.4	0.1	-1.0	-0.3	0.0	0.0
C LC_152	SLN51 column below / Begin	148.6	0.0	1.8	0.0	7.4	0.1
	SLN96 column above / End	-91.7	-0.1	-4.0	-0.7	9.9	-0.3
	SLN221 ring / Begin	-28.6	-0.1	-13.9	0.0	0.0	0.3
	SLN426 ring / End	28.4	-0.2	-14.1	0.0	0.0	-0.7
	SLN448 raking strut / End	-30.0	0.1	-1.0	-0.2	0.0	0.0
C LC_153	SLN51 column below / Begin	233.1	0.0	0.6	0.0	2.6	0.0
	SLN96 column above / End	-142.7	-0.1	-1.0	-0.6	2.6	-0.2
	SLN221 ring / Begin	-28.1	-0.2	-20.6	0.0	0.0	0.5
	SLN426 ring / End	27.9	-0.3	-21.1	0.0	0.0	-0.8
	SLN448 raking strut / End	-56.9	0.1	-1.0	-0.2	0.0	0.0
C LC_154	SLN51 column below / Begin	153.4	0.0	-1.0	0.0	-4.0	-0.1
	SLN96 column above / End	-106.8	0.0	2.5	-0.4	-6.3	-0.1
	SLN221 ring / Begin	-12.7	-0.1	-11.7	0.0	0.0	0.3
	SLN426 ring / End	12.6	-0.2	-11.9	0.0	0.0	-0.5
	SLN448 raking strut / End	-42.4	0.1	-1.0	-0.3	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	1.0 < 5.0%	OK
Loc. deformation	0.2 < 3%	OK
Bolts	17.1 < 100%	OK
Welds	98.8 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN220+SLN203 radial (continuous)-bfl 1	12.7	C LC_121	239.2	0.0	79.4	OK
SLN220+SLN203 radial (continuous)-tfl 1	12.7	C LC_152	150.7	0.0	46.2	OK
SLN220+SLN203 radial (continuous)-w 1	7.2	C LC_123	275.3	0.1	62.9	OK
SLN51 column below	10.0	C LC_105	277.1	1.0	0.0	OK
SLN96 column above	10.0	C LC_152	191.4	0.0	0.0	OK
SLN221 ring-bfl 1	12.7	C LC_119	41.5	0.0	0.0	OK
SLN221 ring-tfl 1	12.7	C LC_119	37.4	0.0	0.0	OK
SLN221 ring-w 1	7.2	C LC_119	131.3	0.0	0.0	OK
SLN426 ring-bfl 1	12.7	C LC_121	64.2	0.0	0.0	OK
SLN426 ring-tfl 1	12.7	C LC_121	60.4	0.0	0.0	OK
SLN426 ring-w 1	7.2	C LC_121	189.0	0.0	0.0	OK
SLN448 raking strut	10.0	C LC_105	61.8	0.0	0.0	OK
End plate column below SLN51 column below	12.0	C LC_121	241.2	0.0	70.5	OK
End plate column above SLN96 column above	12.0	C LC_152	115.0	0.0	40.9	OK
End plate SLN221 ring	12.0	C LC_122	52.1	0.0	33.7	OK
End plate SLN426 ring	12.0	C LC_123	60.2	0.0	37.5	OK
Cap plate SLN448 raking strut	12.0	C LC_105	79.6	0.0	0.0	OK
Fin plate SLN448 raking strut	12.0	C LC_123	191.9	0.0	30.6	OK
Fin plate 2 SLN448 raking strut	12.0	C LC_107	165.6	0.0	56.3	OK
Gusset SLN448 raking strut	12.0	C LC_105	239.3	0.0	8.6	OK
Flange top a end plate SLN221 ring	12.0	C LC_123	57.7	0.0	0.0	OK
Flange top b end plate SLN221 ring	12.0	C LC_123	37.9	0.0	0.0	OK
Flange bottom a end plate SLN221 ring	12.0	C LC_122	46.9	0.0	0.0	OK
Flange bottom b end plate SLN221 ring	12.0	C LC_147	61.3	0.0	0.0	OK
Flange top a end plate SLN426 ring	12.0	C LC_121	65.3	0.0	0.0	OK
Flange top b end plate SLN426 ring	12.0	C LC_121	80.0	0.0	0.0	OK
Flange bottom a end plate SLN426 ring	12.0	C LC_121	85.4	0.0	0.0	OK
Flange bottom b end plate SLN426 ring	12.0	C LC_105	62.2	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

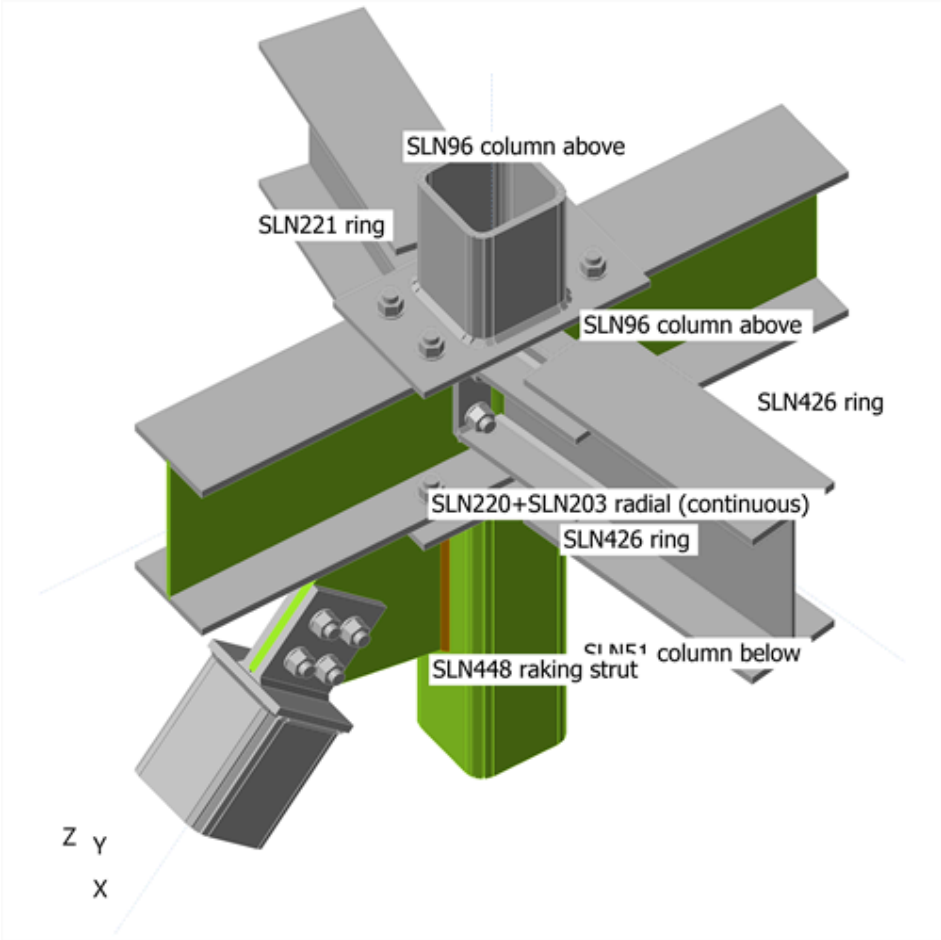
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

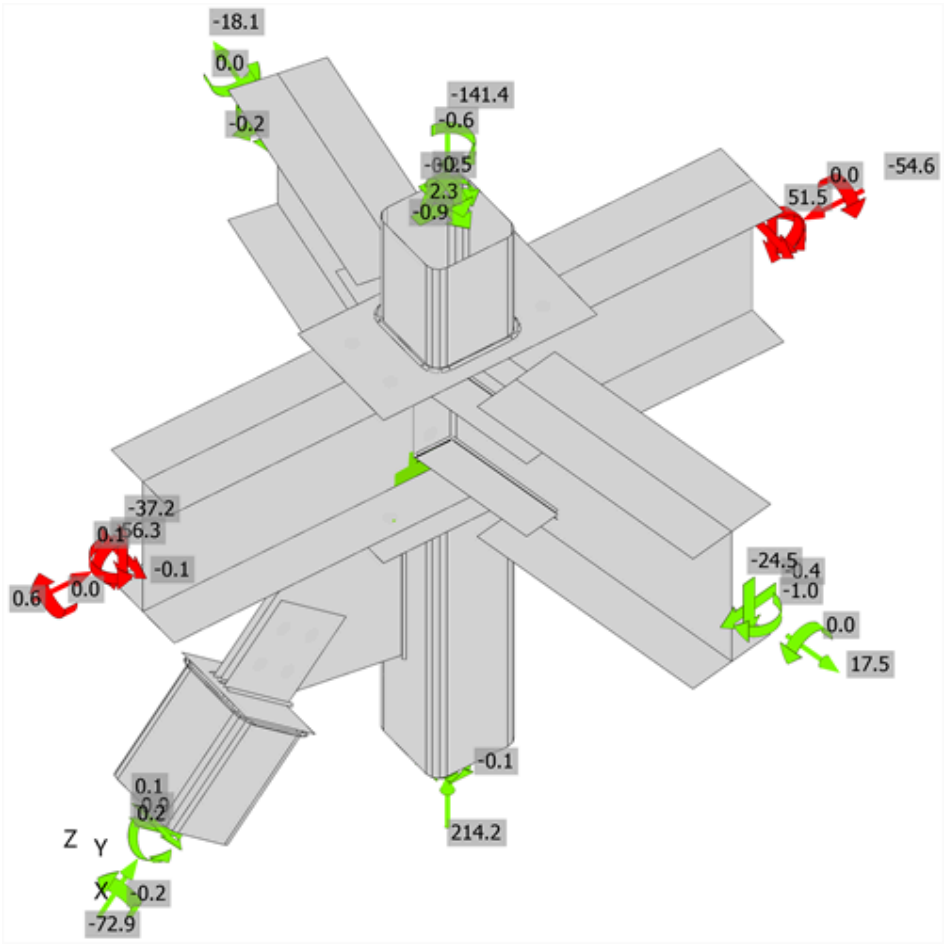
Name	d_0 [mm]	Loads	δ [mm]	$δ_{lim}$ [mm]	$δ/d_0$ [%]	Check status
SLN51 column below	150	C LC_105	0	5	0.2	OK
SLN96 column above	150	C LC_154	0	5	0.0	OK
SLN448 raking strut	150	C LC_154	0	5	0.0	OK

Symbol explanation

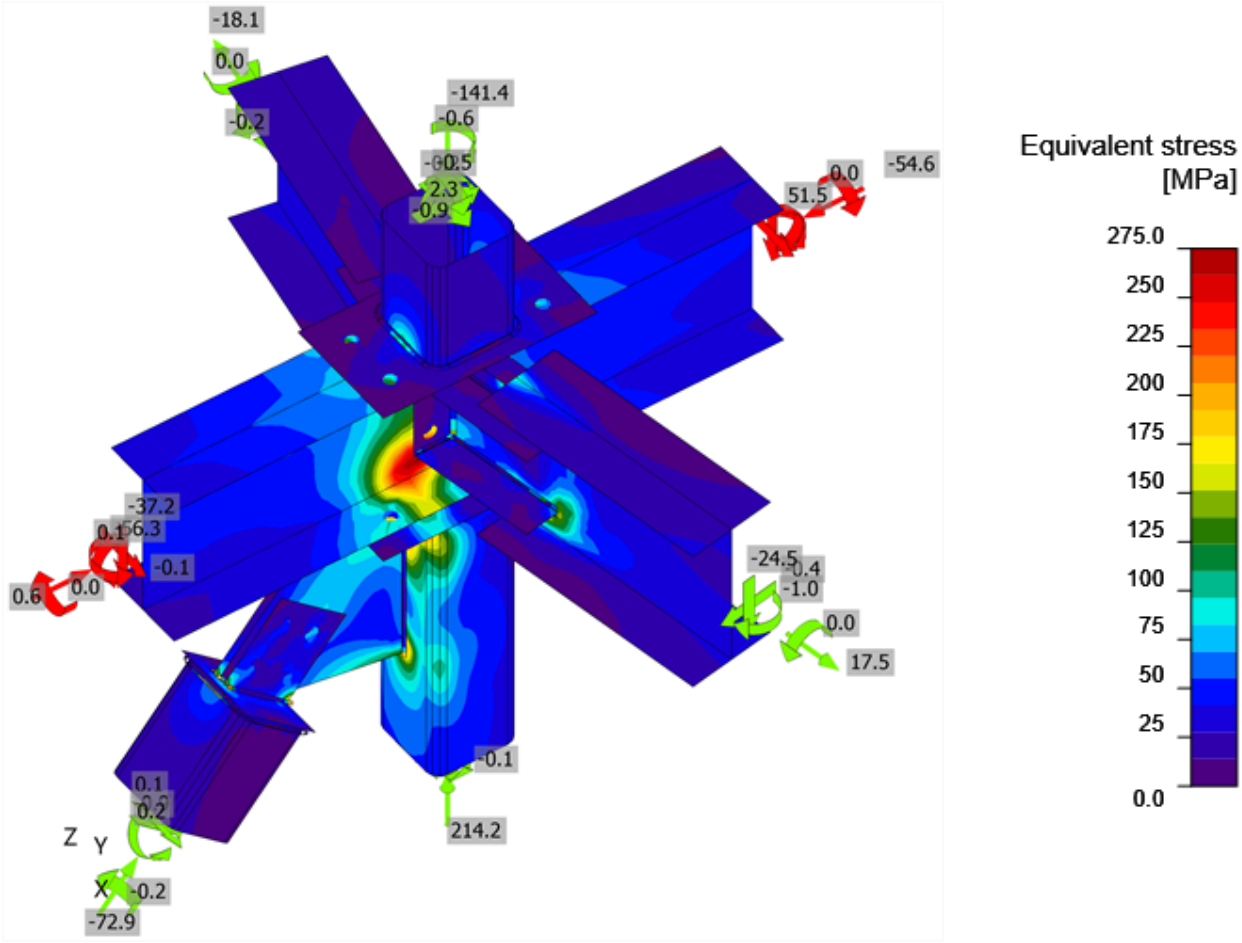
d_0	Cross-section size
δ	Local cross-section deformation
$δ_{lim}$	Allowed deformation
$δ/d_0$	Relative deformation



Overall check

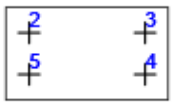
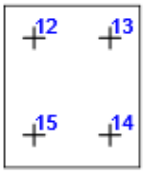
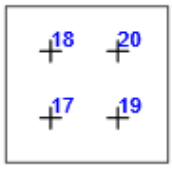


Strain check, C LC_105



Equivalent stress, C LC_105

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_105	0.8	14.8	170.0	0.6	15.7	16.1	OK	OK
	B3	M20 8.8 - 1	C LC_110	15.4	4.2	106.4	10.9	4.4	12.2	OK	OK
	B4	M20 8.8 - 1	C LC_110	19.7	3.4	115.9	14.0	3.6	13.6	OK	OK
	B5	M20 8.8 - 1	C LC_105	0.5	15.9	170.1	0.3	16.9	17.1	OK	OK
	B7	M20 8.8 - 1	C LC_151	1.3	2.1	90.4	0.9	2.3	2.9	OK	OK
	B8	M20 8.8 - 1	C LC_152	10.1	1.5	156.4	7.2	1.6	6.7	OK	OK
	B9	M20 8.8 - 1	C LC_152	11.4	3.2	81.1	8.1	3.9	9.2	OK	OK
	B10	M20 8.8 - 1	C LC_123	5.6	2.1	163.5	4.0	2.2	5.0	OK	OK
	B12	M20 8.8 - 1	C LC_152	3.4	1.8	123.8	2.4	2.8	3.6	OK	OK
	B13	M20 8.8 - 1	C LC_107	3.2	0.2	61.9	2.2	0.2	1.8	OK	OK
	B14	M20 8.8 - 1	C LC_152	0.6	2.3	123.8	0.4	3.6	2.7	OK	OK
	B15	M20 8.8 - 1	C LC_121	5.8	0.3	123.8	4.1	0.5	3.3	OK	OK
	B17	M20 8.8 - 1	C LC_123	5.6	10.1	115.3	4.0	10.7	13.6	OK	OK
	B18	M20 8.8 - 1	C LC_123	2.9	10.6	115.3	2.0	11.3	12.7	OK	OK
	B19	M20 8.8 - 1	C LC_105	0.6	8.8	115.3	0.4	14.4	9.7	OK	OK
	B20	M20 8.8 - 1	C LC_105	0.2	9.7	115.3	0.1	16.7	10.4	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN448 raking strut	SLN448 raking strut	◢ 4.2	519	C LC_105	217.2	0.0	-103.3	110.2	-5.6	53.7	29.1	OK	OK
SLN51 column below-w 3	Gusset SLN448 raking strut	◢ 5.6 ◣	287	C LC_105	399.7	1.9	-131.5	-128.2	-176.2	98.8	81.9	OK	OK
		◢ 5.6 ◣	287	C LC_105	400.0	2.1	-131.4	134.9	171.4	98.8	83.3	OK	OK
SLN221 ring-w 1	Flange top b end plate SLN221 ring	◢ 4.2 ◣	198	C LC_153	59.6	0.0	-19.6	3.7	-32.3	14.7	8.5	OK	OK
		◢ 4.2 ◣	198	C LC_127	47.0	0.0	-7.5	-7.6	25.7	11.6	7.1	OK	OK
SLN221 ring-w 1	Flange bottom b end plate SLN221 ring	◢ 4.2 ◣	198	C LC_135	87.8	0.0	25.3	28.7	39.1	21.7	13.7	OK	OK
		◢ 4.2 ◣	198	C LC_122	121.0	0.0	-27.8	-13.6	66.6	29.9	17.5	OK	OK
SLN426 ring-w 1	Flange top a end plate SLN426 ring	◢ 4.2 ◣	198	C LC_121	83.3	0.0	45.8	37.4	-14.5	20.6	11.2	OK	OK
		◢ 4.2 ◣	198	C LC_123	46.0	0.0	24.4	17.0	14.8	11.4	8.3	OK	OK
SLN426 ring-w 1	Flange bottom a end plate SLN426 ring	◢ 4.2 ◣	198	C LC_147	94.6	0.0	27.9	33.5	-40.0	23.4	13.0	OK	OK
		◢ 4.2 ◣	198	C LC_121	149.1	0.0	62.8	-56.6	53.8	36.8	21.9	OK	OK
Cap plate SLN448 raking strut	Fin plate SLN448 raking strut	◢ 5.6 ◣	139	C LC_119	215.2	0.0	-98.7	-89.7	64.4	53.2	32.5	OK	OK
		◢ 5.6 ◣	139	C LC_105	181.5	0.0	-65.8	76.2	-61.2	44.9	27.7	OK	OK
Cap plate SLN448 raking strut	Fin plate 2 SLN448 raking strut	◢ 5.6 ◣	139	C LC_105	207.8	0.0	-85.7	-89.5	62.7	51.3	27.4	OK	OK
		◢ 5.6 ◣	139	C LC_105	217.0	0.0	-94.8	90.9	-66.6	53.6	32.5	OK	OK
End plate column below SLN51 column below	SLN51 column below	◢ 5.6 ◣	521	C LC_121	395.1	0.0	-251.8	-172.5	-33.5	97.6	53.0	OK	OK
		◢ 5.6 ◣	521	C LC_148	333.7	0.0	-105.8	175.0	52.8	82.5	43.7	OK	OK
End plate column above SLN96 column above	SLN96 column above	◢ 5.6 ◣	521	C LC_152	191.3	0.0	-90.1	-97.3	-3.9	47.3	26.7	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 5.6 ◣	521	C LC_152	223.1	0.0	-114.2	110.3	9.2	55.1	31.7	OK	OK
SLN220+SLN203 radial (continuous)-w 1	SLN221 ring-w 1	◢ 7.2	189	C LC_122	215.0	0.0	3.8	2.2	-124.1	53.1	31.6	OK	OK
SLN220+SLN203 radial (continuous)-w 1	SLN426 ring-w 1	◢ 7.2	189	C LC_123	282.0	0.0	-8.6	11.0	162.4	69.7	44.9	OK	OK
End plate SLN221 ring	SLN221 ring-w 1	◢ 4.2 ◣	149	C LC_122	200.0	0.0	-42.3	-46.0	-103.0	49.4	27.4	OK	OK
		◢ 4.2 ◣	149	C LC_122	148.4	0.0	-37.7	35.3	75.0	36.7	21.9	OK	OK
End plate SLN426 ring	SLN426 ring-w 1	◢ 4.2 ◣	149	C LC_123	226.5	0.0	-39.8	-40.5	122.2	56.0	34.2	OK	OK
		◢ 4.2 ◣	149	C LC_123	163.5	0.0	-33.9	33.9	-85.9	40.4	25.3	OK	OK
SLN221 ring-w 1	Flange top a end plate SLN221 ring	◢ 4.2 ◣	199	C LC_119	81.9	0.0	-41.8	19.0	-36.0	20.2	14.0	OK	OK
		◢ 4.2 ◣	198	C LC_123	52.8	0.0	8.7	10.2	-28.3	13.1	9.0	OK	OK
SLN221 ring-w 1	Flange bottom a end plate SLN221 ring	◢ 4.2 ◣	198	C LC_119	76.7	0.0	13.8	-13.7	41.4	18.9	11.8	OK	OK
		◢ 4.2 ◣	199	C LC_122	126.4	0.0	-29.6	19.4	68.2	31.2	16.4	OK	OK
SLN426 ring-w 1	Flange top b end plate SLN426 ring	◢ 4.2 ◣	199	C LC_105	135.3	0.0	-71.7	30.9	58.6	33.4	22.7	OK	OK
		◢ 4.2 ◣	198	C LC_121	89.8	0.0	16.3	18.8	47.4	22.2	12.9	OK	OK
SLN426 ring-w 1	Flange bottom b end plate SLN426 ring	◢ 4.2 ◣	198	C LC_121	119.5	0.0	22.5	-22.4	-63.9	29.5	17.6	OK	OK
		◢ 4.2 ◣	199	C LC_119	126.8	0.0	-29.9	22.2	-67.6	31.3	19.3	OK	OK
End plate SLN221 ring	Flange top a end plate SLN221 ring	◢ 4.2 ◣	58	C LC_123	82.2	0.0	-30.9	39.3	19.8	20.3	11.9	OK	OK
		◢ 4.2 ◣	58	C LC_123	109.3	0.0	-49.7	-41.3	-38.1	27.0	14.5	OK	OK
End plate SLN221 ring	Flange top b end plate SLN221 ring	◢ 4.2 ◣	58	C LC_125	71.7	0.0	-29.4	32.4	-19.5	17.7	11.6	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_125	76.1	0.0	-36.1	-33.1	20.0	18.8	12.3	OK	OK
End plate SLN221 ring	Flange bottom a end plate SLN221 ring	◢ 4.2 ◣	58	C LC_122	98.5	0.0	11.3	-42.6	37.0	24.3	14.7	OK	OK
		◢ 4.2 ◣	58	C LC_122	87.7	0.0	-13.8	-45.2	21.4	21.7	13.6	OK	OK
End plate SLN221 ring	Flange bottom b end plate SLN221 ring	◢ 4.2 ◣	58	C LC_122	86.5	0.0	13.6	-42.6	-24.9	21.4	12.7	OK	OK
		◢ 4.2 ◣	58	C LC_122	90.6	0.0	-18.6	-47.7	-18.7	22.4	11.7	OK	OK
End plate SLN426 ring	Flange top a end plate SLN426 ring	◢ 4.2 ◣	58	C LC_123	90.2	0.0	-16.3	46.8	-20.8	22.3	13.7	OK	OK
		◢ 4.2 ◣	58	C LC_123	53.6	0.0	-4.7	25.8	-16.9	13.2	10.1	OK	OK
End plate SLN426 ring	Flange top b end plate SLN426 ring	◢ 4.2 ◣	58	C LC_123	90.8	0.0	-21.3	47.7	18.0	22.4	11.6	OK	OK
		◢ 4.2 ◣	58	C LC_105	67.2	0.0	-22.1	-29.0	22.4	16.6	12.2	OK	OK
End plate SLN426 ring	Flange bottom a end plate SLN426 ring	◢ 4.2 ◣	58	C LC_119	89.0	0.0	12.9	-44.0	25.5	22.0	16.0	OK	OK
		◢ 4.2 ◣	58	C LC_123	107.0	0.0	-23.9	-54.5	25.5	26.4	16.4	OK	OK
End plate SLN426 ring	Flange bottom b end plate SLN426 ring	◢ 4.2 ◣	58	C LC_119	99.4	0.0	10.5	-44.5	-35.7	24.6	13.2	OK	OK
		◢ 4.2 ◣	58	C LC_123	104.1	0.0	-16.7	-53.3	-26.0	25.7	13.7	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_000
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN73b+SLN73a radial (continuous) - SLN80 ring - SLN126 ring

Name	SLN73b+SLN73a radial (continuous) - SLN80 ring - SLN126 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

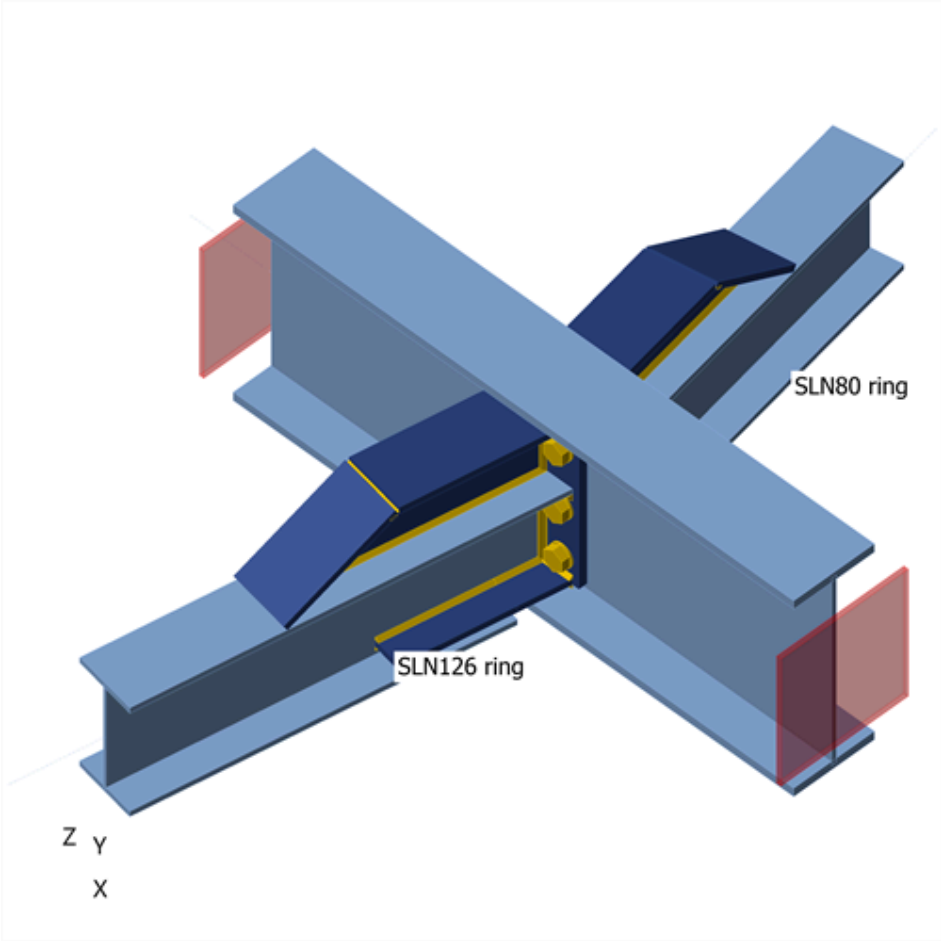
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN73b+SLN73a radial (continuous)	1 - UB457x191x98	0.0	0.0	0.0	0	0	0
SLN80 ring	2 - UB254x146x43	100.0	0.0	0.0	0	0	-102
SLN126 ring	2 - UB254x146x43	-100.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN73b+SLN73a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN73b+SLN73a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN80 ring / end		Position	0
SLN126 ring / end		Position	0



SLN73b+SLN73a radial (continuous) - SLN80 ring - SLN126 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN80 ring / End	0.6	0.0	-4.9	0.0	0.0	-0.1
	SLN126 ring / End	0.6	0.0	-4.9	0.0	0.0	0.0
C LC_102	SLN80 ring / End	-4.1	-0.1	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-4.0	0.1	-25.6	0.0	0.0	0.1
C LC_103	SLN80 ring / End	-8.9	-0.2	-24.5	0.0	0.0	-0.2
	SLN126 ring / End	-8.8	0.2	-24.6	0.0	0.0	0.1
C LC_104	SLN80 ring / End	-5.2	0.1	-24.5	0.0	0.0	0.0
	SLN126 ring / End	-5.2	0.0	-24.6	0.0	0.0	0.1
C LC_105	SLN80 ring / End	-7.1	0.0	-24.5	0.0	0.0	-0.1
	SLN126 ring / End	-7.4	0.1	-24.6	0.0	0.0	0.1
C LC_106	SLN80 ring / End	-7.0	-0.1	-24.5	0.0	0.0	-0.2
	SLN126 ring / End	-6.6	0.0	-24.6	0.0	0.0	0.0
C LC_107	SLN80 ring / End	-3.8	0.0	-1.6	0.0	0.0	0.0
	SLN126 ring / End	-3.6	0.0	-1.6	0.0	0.0	0.1
C LC_108	SLN80 ring / End	0.6	0.1	-1.6	0.0	0.0	0.2
	SLN126 ring / End	0.5	0.1	-1.6	0.0	0.0	0.1
C LC_109	SLN80 ring / End	-2.1	0.1	-1.6	0.0	0.0	0.1
	SLN126 ring / End	-2.3	0.1	-1.6	0.0	0.0	0.1
C LC_110	SLN80 ring / End	-1.1	-0.1	-1.6	0.0	0.0	-0.1
	SLN126 ring / End	-0.9	-0.2	-1.6	0.0	0.0	-0.2
C LC_111	SLN80 ring / End	-8.8	-0.1	-17.3	0.0	0.0	-0.2
	SLN126 ring / End	-8.6	0.1	-17.3	0.0	0.0	0.1
C LC_112	SLN80 ring / End	-3.4	0.1	-17.3	0.0	0.0	0.1
	SLN126 ring / End	-3.4	0.0	-17.3	0.0	0.0	0.1
C LC_113	SLN80 ring / End	-6.5	0.0	-17.3	0.0	0.0	0.0
	SLN126 ring / End	-6.8	0.1	-17.3	0.0	0.0	0.1
C LC_114	SLN80 ring / End	-5.7	-0.2	-17.3	0.0	0.0	-0.2
	SLN126 ring / End	-5.3	-0.1	-17.3	0.0	0.0	-0.1
C LC_115	SLN80 ring / End	-3.7	0.0	-2.8	0.0	0.0	-0.1
	SLN126 ring / End	-3.5	0.1	-2.8	0.0	0.0	0.1
C LC_116	SLN80 ring / End	0.8	0.1	-2.8	0.0	0.0	0.2
	SLN126 ring / End	0.8	0.1	-2.8	0.0	0.0	0.1
C LC_117	SLN80 ring / End	-2.0	0.1	-2.8	0.0	0.0	0.1
	SLN126 ring / End	-2.2	0.1	-2.8	0.0	0.0	0.1
C LC_118	SLN80 ring / End	-0.9	-0.1	-2.8	0.0	0.0	-0.2
	SLN126 ring / End	-0.7	-0.1	-2.8	0.0	0.0	-0.2
C LC_119	SLN80 ring / End	-7.5	-0.2	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-7.4	0.2	-25.6	0.0	0.0	0.2
C LC_120	SLN80 ring / End	-4.8	0.0	-25.6	0.0	0.0	0.0
	SLN126 ring / End	-4.8	0.0	-25.6	0.0	0.0	0.1
C LC_121	SLN80 ring / End	-5.9	-0.1	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-6.1	0.0	-25.6	0.0	0.0	0.1
C LC_122	SLN80 ring / End	-5.9	-0.1	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-5.5	0.0	-25.6	0.0	0.0	0.1
C LC_123	SLN80 ring / End	-5.9	-0.1	-19.4	0.0	0.0	-0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN126 ring / End	-5.8	0.2	-19.4	0.0	0.0	0.2
C LC_124	SLN80 ring / End	-2.6	0.1	-19.4	0.0	0.0	0.1
	SLN126 ring / End	-2.6	0.1	-19.4	0.0	0.0	0.2
C LC_125	SLN80 ring / End	-4.1	-0.1	-19.4	0.0	0.0	-0.1
	SLN126 ring / End	-4.3	0.0	-19.4	0.0	0.0	0.1
C LC_126	SLN80 ring / End	-3.4	-0.1	-19.4	0.0	0.0	-0.2
	SLN126 ring / End	-3.1	0.0	-19.4	0.0	0.0	0.0
C LC_127	SLN80 ring / End	-0.8	0.0	-4.9	0.0	0.0	0.0
	SLN126 ring / End	-0.7	0.1	-4.9	0.0	0.0	0.1
C LC_128	SLN80 ring / End	1.6	0.1	-4.9	0.0	0.0	0.1
	SLN126 ring / End	1.5	0.1	-4.9	0.0	0.0	0.2
C LC_129	SLN80 ring / End	0.4	0.0	-4.9	0.0	0.0	-0.1
	SLN126 ring / End	0.3	0.0	-4.9	0.0	0.0	0.0
C LC_130	SLN80 ring / End	1.3	-0.1	-4.9	0.0	0.0	-0.1
	SLN126 ring / End	1.5	0.0	-4.9	0.0	0.0	0.0
C LC_135	SLN80 ring / End	-0.4	-0.1	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-0.4	0.1	-25.6	0.0	0.0	0.1
C LC_141	SLN80 ring / End	0.4	0.0	-4.9	0.0	0.0	-0.1
	SLN126 ring / End	0.4	0.0	-4.9	0.0	0.0	0.1
C LC_142	SLN80 ring / End	0.8	0.0	-4.9	0.0	0.0	0.0
	SLN126 ring / End	0.8	0.0	-4.9	0.0	0.0	0.0
C LC_143	SLN80 ring / End	0.6	0.0	-4.9	0.0	0.0	-0.1
	SLN126 ring / End	0.6	0.0	-4.9	0.0	0.0	0.0
C LC_144	SLN80 ring / End	0.5	0.0	-4.9	0.0	0.0	0.0
	SLN126 ring / End	0.6	0.0	-4.9	0.0	0.0	0.0
C LC_145	SLN80 ring / End	-4.9	-0.2	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-4.8	0.2	-25.6	0.0	0.0	0.2
C LC_146	SLN80 ring / End	-3.2	0.0	-25.6	0.0	0.0	-0.1
	SLN126 ring / End	-3.2	0.0	-25.6	0.0	0.0	0.1
C LC_147	SLN80 ring / End	-3.9	-0.1	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-4.0	0.1	-25.6	0.0	0.0	0.1
C LC_148	SLN80 ring / End	-4.3	-0.1	-25.6	0.0	0.0	-0.1
	SLN126 ring / End	-4.0	0.1	-25.6	0.0	0.0	0.1
C LC_151	SLN80 ring / End	-1.9	-0.1	-11.7	0.0	0.0	-0.1
	SLN126 ring / End	-1.8	0.1	-11.7	0.0	0.0	0.1
C LC_152	SLN80 ring / End	-6.9	-0.1	-14.1	0.0	0.0	-0.1
	SLN126 ring / End	-6.8	0.1	-14.2	0.0	0.0	0.1
C LC_153	SLN80 ring / End	-4.2	-0.1	-20.9	0.0	0.0	-0.2
	SLN126 ring / End	-4.2	0.1	-20.9	0.0	0.0	0.1
C LC_154	SLN80 ring / End	-1.8	-0.1	-11.7	0.0	0.0	-0.1
	SLN126 ring / End	-1.7	0.1	-11.7	0.0	0.0	0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.9 < 100%	OK
Welds	18.1 < 100%	OK
Buckling	Not calculated	

Plates

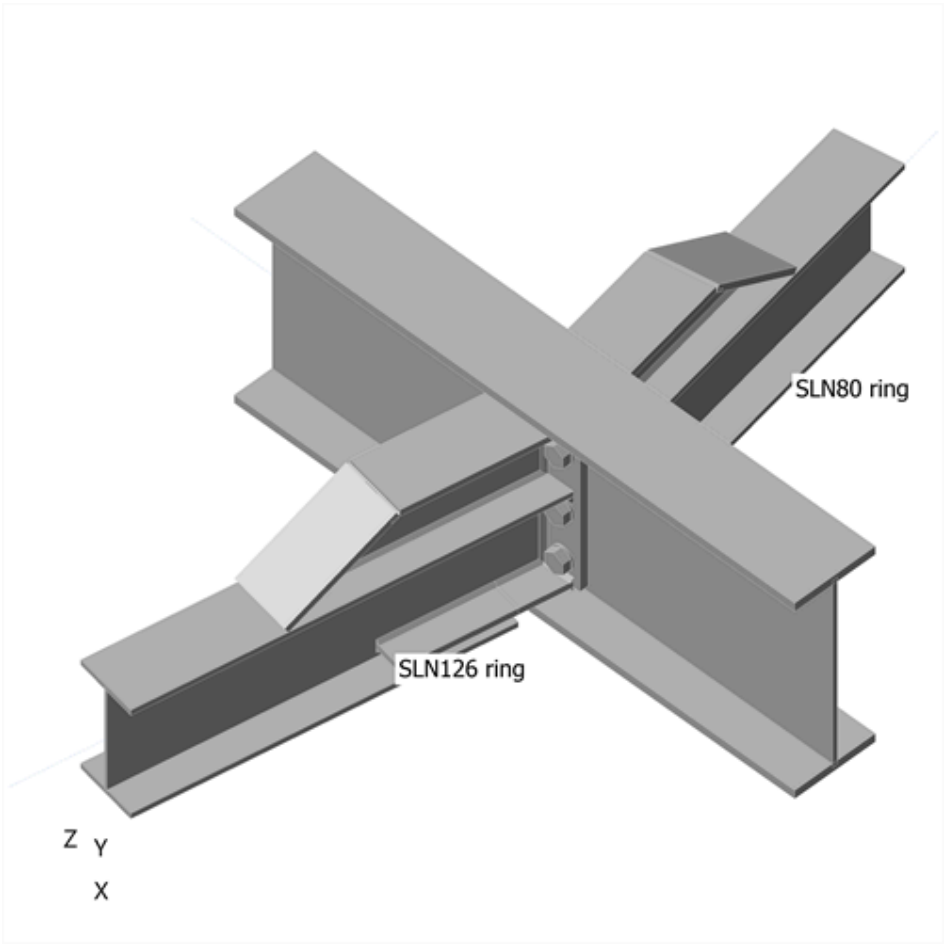
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN73b+SLN73a radial (continuous)-bfl 1	19.6	C LC_122	20.3	0.0	0.0	OK
SLN73b+SLN73a radial (continuous)-tfl 1	19.6	C LC_122	17.1	0.0	0.0	OK
SLN73b+SLN73a radial (continuous)-w 1	11.4	C LC_121	23.6	0.0	0.3	OK
SLN80 ring-bfl 1	12.7	C LC_135	46.4	0.0	0.0	OK
SLN80 ring-tfl 1	12.7	C LC_121	47.9	0.0	0.0	OK
SLN80 ring-w 1	7.2	C LC_119	47.0	0.0	0.0	OK
SLN126 ring-bfl 1	12.7	C LC_135	46.5	0.0	0.0	OK
SLN126 ring-tfl 1	12.7	C LC_119	47.8	0.0	0.0	OK
SLN126 ring-w 1	7.2	C LC_119	47.4	0.0	0.0	OK
End plate SLN80 ring	20.0	C LC_147	6.6	0.0	1.3	OK
Bottom flange plate SLN80 ring	15.0	C LC_135	13.4	0.0	0.0	OK
Bottom flange half plate L SLN80 ring	15.0	C LC_135	12.9	0.0	0.0	OK
Bottom flange half plate R SLN80 ring	15.0	C LC_135	12.6	0.0	0.0	OK
Haunch web SLN80 ring	12.0	C LC_119	31.6	0.0	0.0	OK
Haunch top plate SLN80 ring	15.0	C LC_145	5.5	0.0	0.0	OK
Haunch sloping plate SLN80 ring	15.0	C LC_119	14.3	0.0	0.0	OK
End plate SLN126 ring	20.0	C LC_148	6.6	0.0	0.7	OK
Bottom flange plate SLN126 ring	15.0	C LC_135	13.4	0.0	0.0	OK
Bottom flange half plate L SLN126 ring	15.0	C LC_135	12.7	0.0	0.0	OK
Bottom flange half plate R SLN126 ring	15.0	C LC_135	12.8	0.0	0.0	OK
Haunch web SLN126 ring	12.0	C LC_119	32.5	0.0	0.0	OK
Haunch top plate SLN126 ring	15.0	C LC_145	5.4	0.0	0.0	OK
Haunch sloping plate SLN126 ring	15.0	C LC_119	14.4	0.0	0.0	OK

Design data

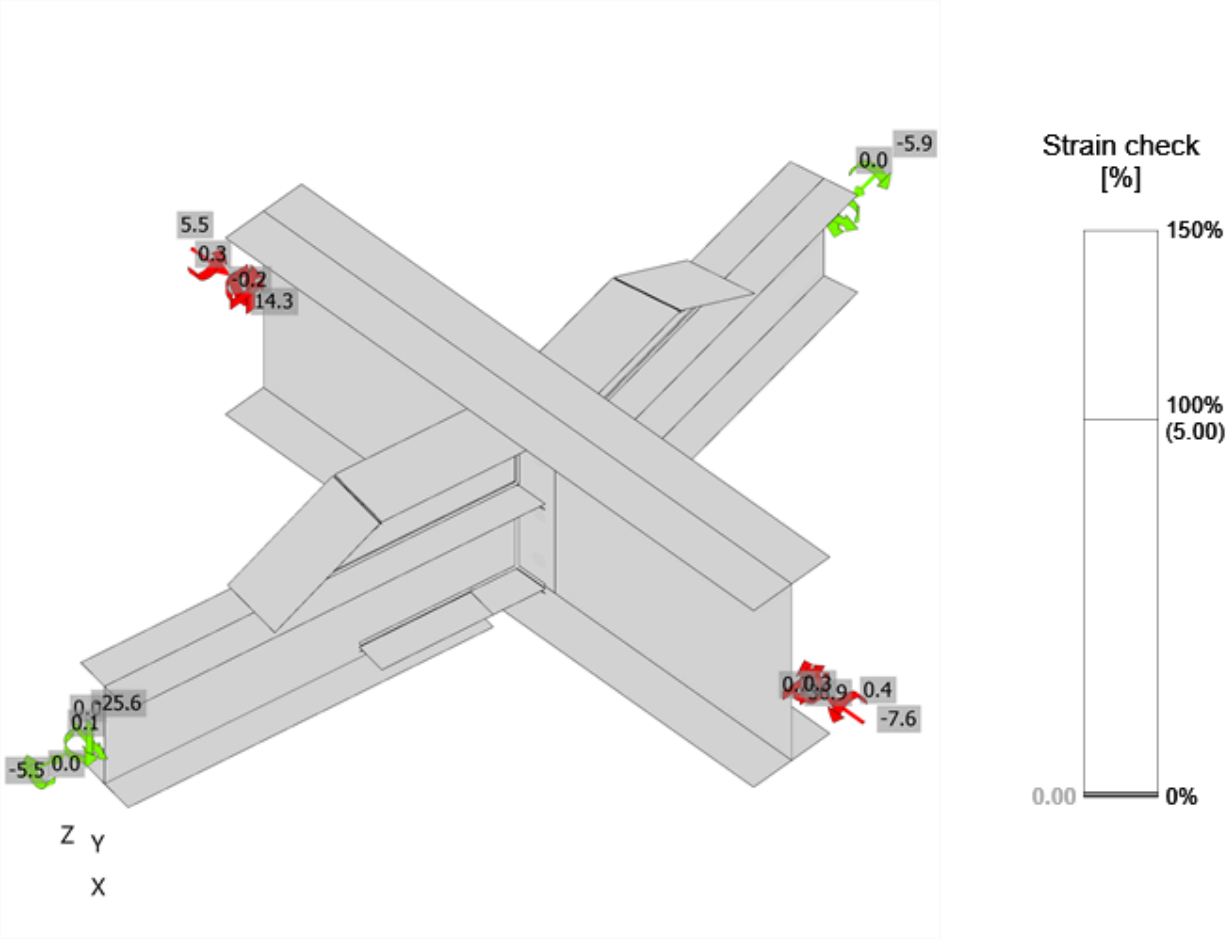
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

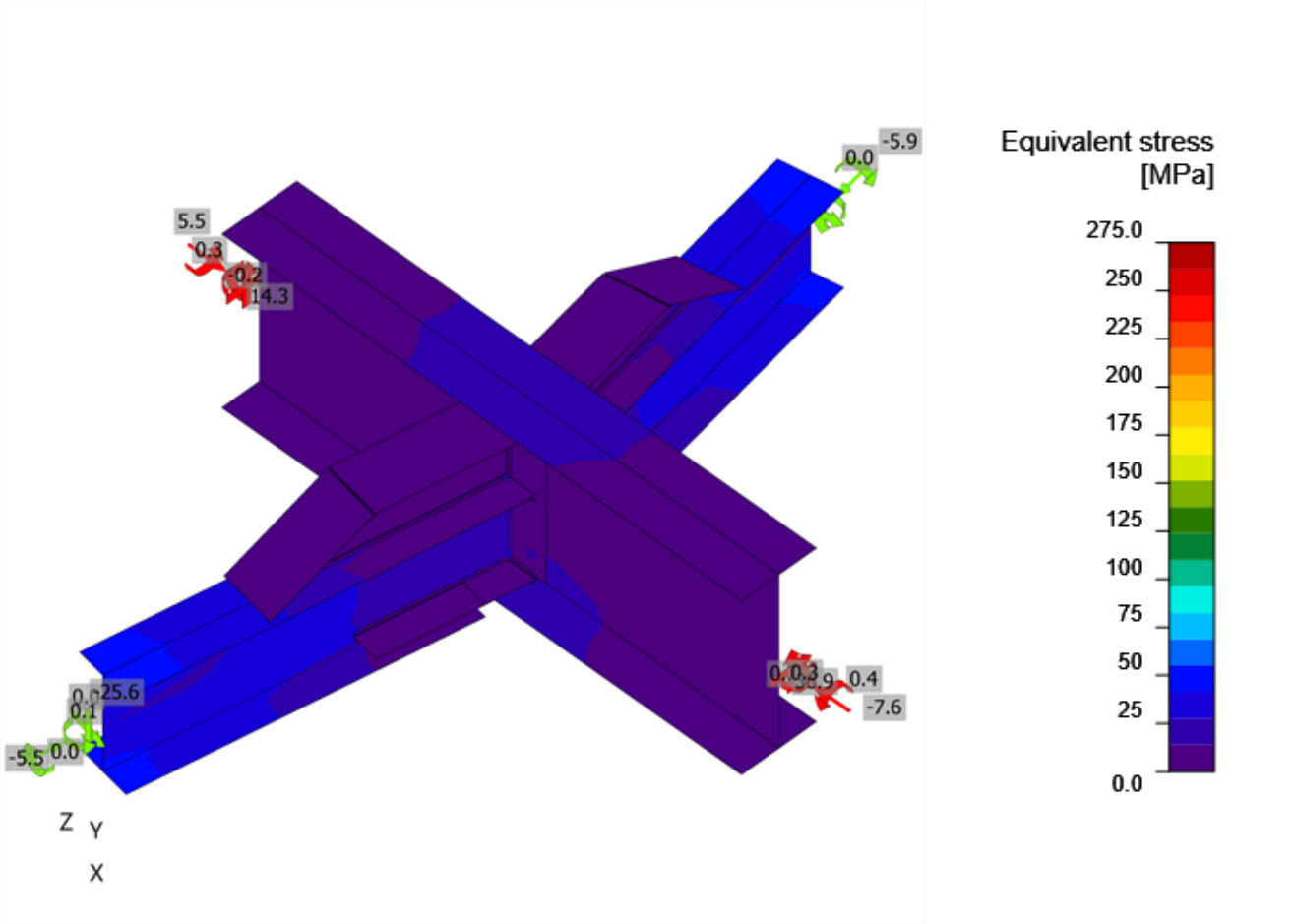
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

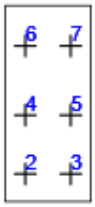


Strain check, C LC_122



Equivalent stress, C LC_122

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_145	0.7	0.4	194.1	0.2	0.4	0.4	OK	OK
	B3	M30 8.8 - 1	C LC_135	0.8	0.5	194.1	0.3	0.5	0.4	OK	OK
	B4	M30 8.8 - 1	C LC_119	0.0	0.4	232.2	0.0	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_135	0.0	0.5	232.2	0.0	0.4	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_148	2.4	0.8	249.2	0.7	0.6	0.9	OK	OK
	B7	M30 8.8 - 1	C LC_120	1.6	0.9	249.2	0.5	0.7	0.8	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN126 ring	Bottom flange plate SLN126 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN80 ring	Bottom flange plate SLN80 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN80 ring	Bottom flange plate SLN80 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN126 ring	Bottom flange plate SLN126 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN73b+SLN73a radial (continuous)-w 1	SLN80 ring-tfl 1	◢ 11.4	149	C LC_119	11.6	0.0	-10.3	-3.0	-0.2	3.3	3.3	OK	OK
SLN73b+SLN73a radial (continuous)-w 1	SLN126 ring-tfl 1	◢ 11.4	149	C LC_119	11.8	0.0	-10.3	-3.3	0.3	3.3	3.3	OK	OK
End plate SLN80 ring	SLN80 ring-w 1	◢ 6.0 ◣	208	C LC_135	10.1	0.0	0.6	-0.7	5.8	2.5	0.0	OK	OK
		◢ 6.0 ◣	208	C LC_135	8.2	0.0	-1.0	0.0	-4.7	2.0	0.0	OK	OK
Bottom flange plate SLN80 ring	SLN80 ring-w 1	◢ 6.0 ◣	100	C LC_135	20.3	0.0	1.6	1.3	11.6	5.0	5.0	OK	OK
		◢ 6.0 ◣	100	C LC_135	21.1	0.0	2.0	-2.3	-11.9	5.2	5.2	OK	OK
SLN80 ring-w 1	Bottom flange half plate L SLN80 ring	◢ 6.0 ◣	243	C LC_135	16.4	0.0	4.0	3.9	8.3	4.0	4.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	19.7	0.0	6.3	2.2	-10.5	4.9	4.9	OK	OK
SLN80 ring-w 1	Bottom flange half plate R SLN80 ring	◢ 6.0 ◣	243	C LC_135	16.9	0.0	3.9	4.0	-8.6	4.2	4.2	OK	OK
		◢ 6.0 ◣	243	C LC_135	19.5	0.0	5.5	2.5	10.5	4.8	4.8	OK	OK
SLN80 ring-tfl 1	Haunch web SLN80 ring	◢ 6.0 ◣	572	C LC_119	72.3	0.0	-17.3	-18.0	36.3	17.9	10.2	OK	OK
		◢ 6.0 ◣	572	C LC_119	69.7	0.0	-18.0	17.2	-34.9	17.2	9.9	OK	OK
End plate SLN126 ring	SLN126 ring-w 1	◢ 6.0 ◣	208	C LC_135	7.9	0.0	-0.9	0.1	4.5	2.0	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_135	10.3	0.0	0.6	0.7	-5.9	2.5	0.0	OK	OK
Bottom flange plate SLN126 ring	SLN126 ring-w 1	◢ 6.0 ◣	100	C LC_135	21.2	0.0	1.9	1.8	12.0	5.2	5.2	OK	OK
		◢ 6.0 ◣	100	C LC_135	20.3	0.0	1.7	-1.8	-11.5	5.0	5.0	OK	OK
SLN126 ring-w 1	Bottom flange half plate L SLN126 ring	◢ 6.0 ◣	243	C LC_135	17.3	0.0	4.4	4.1	8.8	4.3	4.3	OK	OK
		◢ 6.0 ◣	243	C LC_135	19.2	0.0	5.9	2.3	-10.3	4.8	4.8	OK	OK
SLN126 ring-w 1	Bottom flange half plate R SLN126 ring	◢ 6.0 ◣	243	C LC_135	16.0	0.0	3.5	3.8	-8.2	4.0	4.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	20.1	0.0	5.8	2.4	10.9	5.0	5.0	OK	OK
SLN126 ring-tfl 1	Haunch web SLN126 ring	◢ 6.0 ◣	572	C LC_119	73.1	0.0	-19.1	-19.0	36.0	18.1	10.2	OK	OK
		◢ 6.0 ◣	572	C LC_119	69.2	0.0	-16.2	16.2	-35.3	17.1	9.8	OK	OK
End plate SLN80 ring	Bottom flange plate SLN80 ring	◢ 6.0 ◣	149	C LC_118	8.2	0.0	-2.3	-3.1	3.2	2.0	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_114	11.8	0.0	-5.1	3.6	-5.0	2.9	0.0	OK	OK
End plate SLN80 ring	Haunch web SLN80 ring	◢ 6.0 ◣	145	C LC_120	6.8	0.0	0.7	0.6	3.8	1.7	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_121	10.3	0.0	1.7	-1.6	-5.7	2.6	0.0	OK	OK
End plate SLN80 ring	Haunch top plate SLN80 ring	◢ 6.0 ◣	149	C LC_110	4.9	0.0	-2.3	-1.7	1.8	1.2	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_118	3.8	0.0	-1.1	1.7	-1.2	0.9	0.0	OK	OK
Haunch top plate SLN80 ring	Haunch web SLN80 ring	◢ 6.0 ◣	342	C LC_145	3.9	0.0	1.5	1.4	1.5	1.0	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_145	3.9	0.0	-0.2	0.2	2.2	1.0	0.0	OK	OK
Haunch sloping plate SLN80 ring	Haunch web SLN80 ring	◢ 6.0 ◣	271	C LC_119	57.4	0.0	-10.0	-8.9	31.4	14.2	8.8	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_119	56.3	0.0	-9.8	10.8	-30.1	13.9	8.6	OK	OK
End plate SLN126 ring	Bottom flange plate SLN126 ring	◢ 6.0 ◣	149	C LC_110	7.8	0.0	-3.7	-3.8	1.1	1.9	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_114	11.8	0.0	-6.0	5.5	-2.1	2.9	0.0	OK	OK
End plate SLN126 ring	Haunch web SLN126 ring	◢ 6.0 ◣	145	C LC_120	10.2	0.0	1.7	1.7	5.5	2.5	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	7.4	0.0	0.7	-0.6	-4.2	1.8	0.0	OK	OK
End plate SLN126 ring	Haunch top plate SLN126 ring	◢ 6.0 ◣	149	C LC_110	4.2	0.0	-2.2	-1.8	1.1	1.0	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_148	3.8	0.0	2.3	-1.1	1.4	0.9	0.0	OK	OK
Haunch top plate SLN126 ring	Haunch web SLN126 ring	◢ 6.0 ◣	342	C LC_145	3.5	0.0	-0.2	-0.2	-2.0	0.9	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	3.7	0.0	0.0	0.0	2.1	0.9	0.0	OK	OK
Haunch sloping plate SLN126 ring	Haunch web SLN126 ring	◢ 6.0 ◣	271	C LC_119	60.0	0.0	-11.2	-11.3	32.1	14.8	9.1	OK	OK
		◢ 6.0 ◣	271	C LC_119	53.7	0.0	-8.6	8.5	-29.4	13.3	8.5	OK	OK
SLN73b+SLN73a radial (continuous)-w 1	SLN80 ring-w 1	◢ 7.2	208	C LC_135	37.8	0.0	8.2	8.9	-19.3	9.3	9.0	OK	OK
SLN73b+SLN73a radial (continuous)-w 1	SLN126 ring-w 1	◢ 7.2	208	C LC_135	36.2	0.0	8.3	6.6	-19.3	8.9	8.8	OK	OK
Haunch sloping plate SLN80 ring	Haunch top plate SLN80 ring	◢ 6.0 ◣	146	C LC_122	7.1	0.0	-4.8	3.0	-0.2	1.9	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_145	2.6	0.0	-0.6	1.3	0.6	0.6	0.0	OK	OK
Haunch sloping plate SLN126 ring	Haunch top plate SLN126 ring	◢ 6.0 ◣	146	C LC_121	7.0	0.0	-5.0	2.8	-0.3	1.9	0.0	OK	OK
		◢ 6.0 ◣	146	C LC_145	2.4	0.0	-0.5	1.3	-0.6	0.6	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_020
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN78a+SLN78b radial (continuous) - SLN80 ring - SLN83 ring

Name	SLN78a+SLN78b radial (continuous) - SLN80 ring - SLN83 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

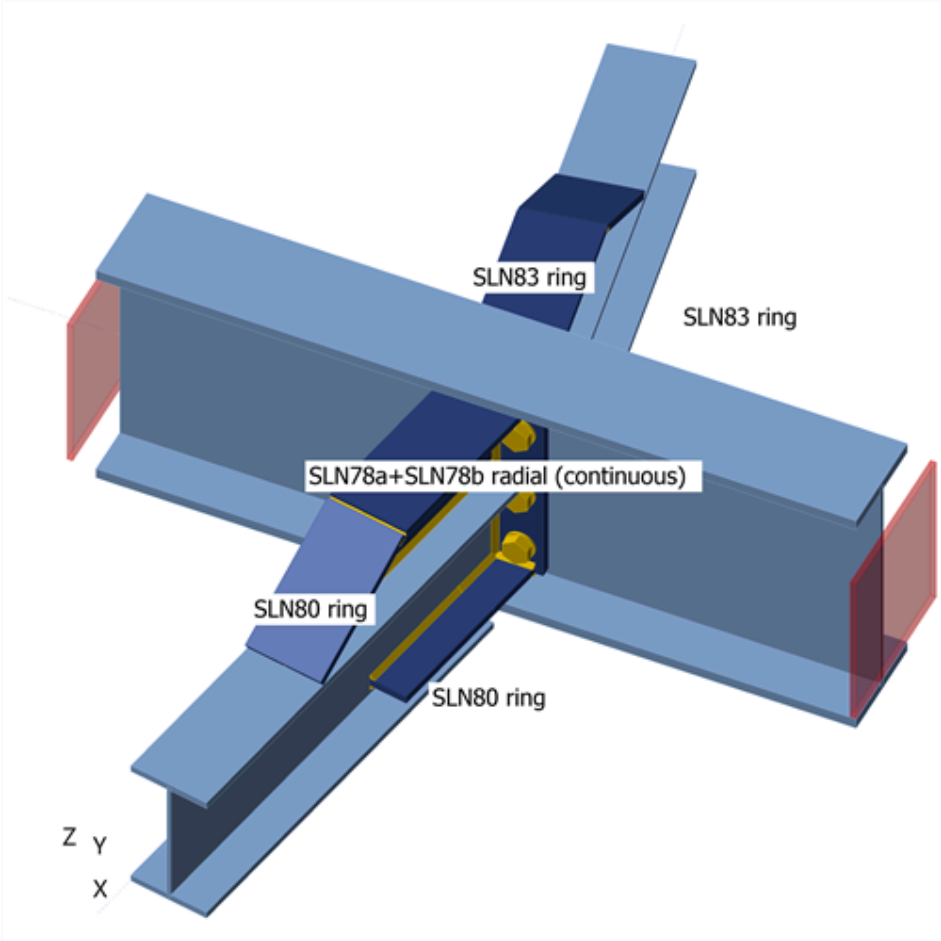
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN78a+SLN78b radial (continuous)	1 - UB457x191x98	20.0	0.0	0.0	0	0	0
SLN80 ring	2 - UB254x146x43	-80.0	0.0	0.0	0	0	-102
SLN83 ring	2 - UB254x146x43	120.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN78a+SLN78b radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN78a+SLN78b radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN80 ring / end		Position	0
SLN83 ring / end		Position	0



SLN78a+SLN78b radial (continuous) - SLN80 ring - SLN83 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN80 ring / End	0.6	0.0	-4.9	0.0	0.0	0.0
	SLN83 ring / End	0.6	0.0	-2.2	0.0	0.0	0.0
C LC_102	SLN80 ring / End	-4.1	-0.1	-25.6	0.0	0.0	-0.2
	SLN83 ring / End	-4.1	0.0	-8.8	0.0	0.0	0.0
C LC_103	SLN80 ring / End	-8.9	0.0	-24.6	0.0	0.0	-0.1
	SLN83 ring / End	-8.9	-0.1	-8.5	0.0	0.0	-0.1
C LC_104	SLN80 ring / End	-5.2	-0.1	-24.6	0.0	0.0	-0.1
	SLN83 ring / End	-5.4	0.0	-8.5	0.0	0.0	0.0
C LC_105	SLN80 ring / End	-7.0	-0.1	-24.6	0.0	0.0	-0.1
	SLN83 ring / End	-6.4	-0.1	-8.5	0.0	0.0	0.0
C LC_106	SLN80 ring / End	-7.2	-0.1	-24.6	0.0	0.0	-0.2
	SLN83 ring / End	-7.5	-0.1	-8.5	0.0	0.0	-0.1
C LC_107	SLN80 ring / End	-3.8	0.0	-1.6	0.0	0.0	0.0
	SLN83 ring / End	-3.9	0.0	-0.9	0.0	0.0	0.0
C LC_108	SLN80 ring / End	0.6	0.1	-1.6	0.0	0.0	0.1
	SLN83 ring / End	0.5	0.0	-0.9	0.0	0.0	0.0
C LC_109	SLN80 ring / End	-2.1	0.1	-1.6	0.0	0.0	0.2
	SLN83 ring / End	-1.2	0.0	-0.9	0.0	0.0	0.1
C LC_110	SLN80 ring / End	-1.1	-0.1	-1.6	0.0	0.0	-0.2
	SLN83 ring / End	-1.6	-0.1	-0.9	0.0	0.0	-0.1
C LC_111	SLN80 ring / End	-8.7	0.0	-17.3	0.0	0.0	-0.1
	SLN83 ring / End	-8.8	-0.1	-6.2	0.0	0.0	0.0
C LC_112	SLN80 ring / End	-3.4	0.0	-17.3	0.0	0.0	0.0
	SLN83 ring / End	-3.6	0.0	-6.2	0.0	0.0	0.0
C LC_113	SLN80 ring / End	-6.4	0.0	-17.3	0.0	0.0	0.1
	SLN83 ring / End	-5.4	0.0	-6.2	0.0	0.0	0.0
C LC_114	SLN80 ring / End	-5.8	-0.1	-17.3	0.0	0.0	-0.2
	SLN83 ring / End	-6.4	-0.1	-6.2	0.0	0.0	-0.1
C LC_115	SLN80 ring / End	-3.7	0.0	-2.8	0.0	0.0	0.0
	SLN83 ring / End	-3.8	0.0	-1.5	0.0	0.0	0.0
C LC_116	SLN80 ring / End	0.8	0.1	-2.8	0.0	0.0	0.1
	SLN83 ring / End	0.7	0.0	-1.5	0.0	0.0	0.0
C LC_117	SLN80 ring / End	-1.9	0.1	-2.8	0.0	0.0	0.2
	SLN83 ring / End	-1.0	0.0	-1.5	0.0	0.0	0.1
C LC_118	SLN80 ring / End	-1.0	-0.1	-2.8	0.0	0.0	-0.2
	SLN83 ring / End	-1.5	-0.1	-1.5	0.0	0.0	-0.1
C LC_119	SLN80 ring / End	-7.5	0.0	-25.6	0.0	0.0	-0.1
	SLN83 ring / End	-7.4	-0.1	-8.8	0.0	0.0	-0.1
C LC_120	SLN80 ring / End	-4.8	-0.1	-25.6	0.0	0.0	-0.1
	SLN83 ring / End	-4.9	0.0	-8.8	0.0	0.0	0.1
C LC_121	SLN80 ring / End	-5.8	-0.1	-25.6	0.0	0.0	-0.2
	SLN83 ring / End	-5.7	-0.1	-8.8	0.0	0.0	-0.1
C LC_122	SLN80 ring / End	-6.0	-0.1	-25.6	0.0	0.0	-0.1
	SLN83 ring / End	-6.2	-0.1	-8.8	0.0	0.0	0.0
C LC_123	SLN80 ring / End	-5.9	0.0	-19.4	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN83 ring / End	-5.9	-0.1	-6.8	0.0	0.0	0.0
C LC_124	SLN80 ring / End	-2.6	0.0	-19.4	0.0	0.0	0.0
	SLN83 ring / End	-2.6	0.1	-6.8	0.0	0.0	0.1
C LC_125	SLN80 ring / End	-4.0	-0.1	-19.4	0.0	0.0	-0.2
	SLN83 ring / End	-4.0	-0.1	-6.8	0.0	0.0	-0.1
C LC_126	SLN80 ring / End	-3.5	-0.1	-19.4	0.0	0.0	-0.1
	SLN83 ring / End	-3.7	-0.1	-6.8	0.0	0.0	0.0
C LC_127	SLN80 ring / End	-0.8	0.0	-4.9	0.0	0.0	0.0
	SLN83 ring / End	-0.8	0.0	-2.2	0.0	0.0	0.1
C LC_128	SLN80 ring / End	1.6	0.0	-4.9	0.0	0.0	0.1
	SLN83 ring / End	1.7	0.1	-2.2	0.0	0.0	0.2
C LC_129	SLN80 ring / End	0.5	-0.1	-4.9	0.0	0.0	-0.1
	SLN83 ring / End	0.4	0.0	-2.2	0.0	0.0	0.0
C LC_130	SLN80 ring / End	1.3	-0.1	-4.9	0.0	0.0	-0.1
	SLN83 ring / End	1.3	0.0	-2.2	0.0	0.0	0.0
C LC_135	SLN80 ring / End	-0.4	-0.1	-25.6	0.0	0.0	-0.1
	SLN83 ring / End	-0.4	0.0	-8.8	0.0	0.0	0.0
C LC_141	SLN80 ring / End	0.4	0.0	-4.9	0.0	0.0	0.0
	SLN83 ring / End	0.4	0.0	-2.2	0.0	0.0	0.0
C LC_142	SLN80 ring / End	0.8	0.0	-4.9	0.0	0.0	-0.1
	SLN83 ring / End	0.8	0.0	-2.2	0.0	0.0	0.0
C LC_143	SLN80 ring / End	0.7	0.0	-4.9	0.0	0.0	-0.1
	SLN83 ring / End	0.7	0.0	-2.2	0.0	0.0	0.0
C LC_144	SLN80 ring / End	0.5	0.0	-4.9	0.0	0.0	0.0
	SLN83 ring / End	0.5	0.0	-2.2	0.0	0.0	0.0
C LC_145	SLN80 ring / End	-4.9	0.0	-25.6	0.0	0.0	-0.1
	SLN83 ring / End	-4.8	-0.1	-8.8	0.0	0.0	0.0
C LC_146	SLN80 ring / End	-3.3	-0.2	-25.6	0.0	0.0	-0.2
	SLN83 ring / End	-3.4	0.0	-8.8	0.0	0.0	0.0
C LC_147	SLN80 ring / End	-3.7	-0.2	-25.6	0.0	0.0	-0.2
	SLN83 ring / End	-3.6	-0.1	-8.8	0.0	0.0	0.0
C LC_148	SLN80 ring / End	-4.4	-0.1	-25.6	0.0	0.0	-0.1
	SLN83 ring / End	-4.6	0.0	-8.8	0.0	0.0	0.0
C LC_151	SLN80 ring / End	-1.9	-0.1	-11.7	0.0	0.0	-0.1
	SLN83 ring / End	-1.8	0.0	-8.8	0.0	0.0	0.0
C LC_152	SLN80 ring / End	-6.9	-0.1	-14.2	0.0	0.0	-0.1
	SLN83 ring / End	-6.9	-0.1	-2.2	0.0	0.0	0.0
C LC_153	SLN80 ring / End	-4.2	-0.1	-21.0	0.0	0.0	-0.2
	SLN83 ring / End	-4.2	0.0	-8.8	0.0	0.0	0.0
C LC_154	SLN80 ring / End	-1.8	-0.1	-11.7	0.0	0.0	-0.1
	SLN83 ring / End	-1.8	0.0	-8.8	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.6 < 100%	OK
Welds	17.7 < 100%	OK
Buckling	Not calculated	

Plates

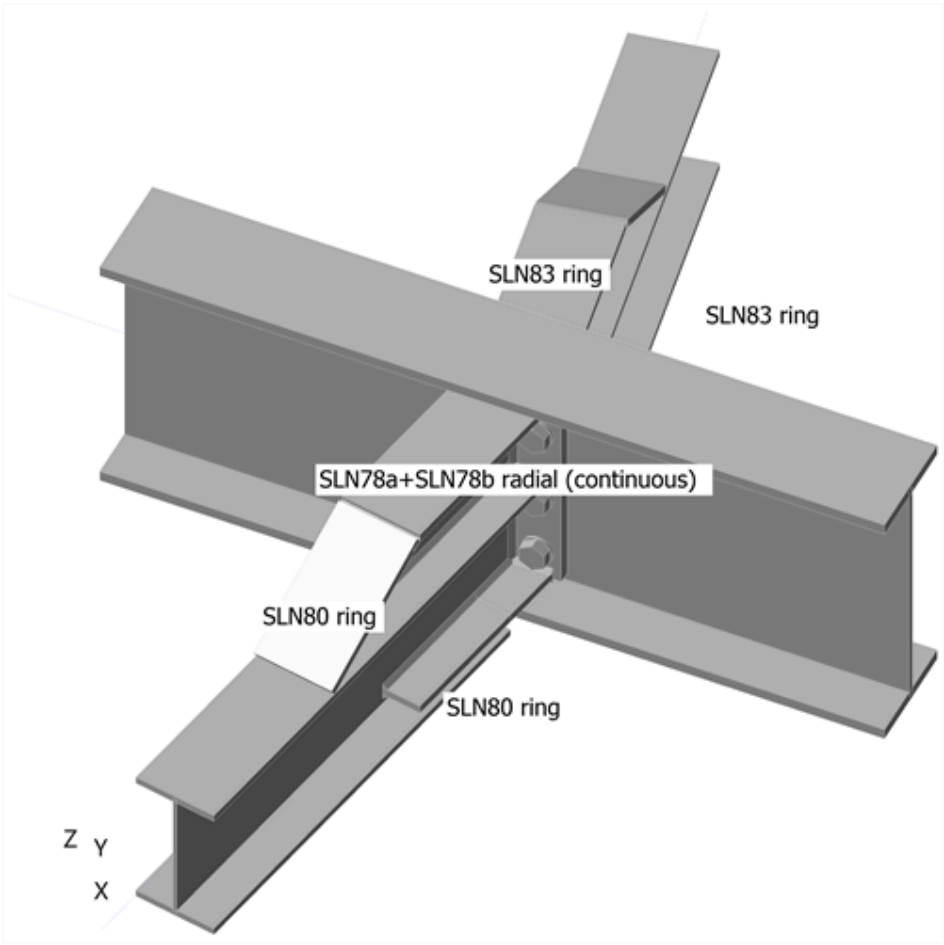
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN78a+SLN78b radial (continuous)-bfl 1	19.6	C LC_147	13.7	0.0	0.0	OK
SLN78a+SLN78b radial (continuous)-tfl 1	19.6	C LC_147	11.5	0.0	0.0	OK
SLN78a+SLN78b radial (continuous)-w 1	11.4	C LC_147	16.3	0.0	0.4	OK
SLN80 ring-bfl 1	12.7	C LC_135	46.3	0.0	0.0	OK
SLN80 ring-tfl 1	12.7	C LC_119	48.0	0.0	0.0	OK
SLN80 ring-w 1	7.2	C LC_119	47.2	0.0	0.0	OK
SLN83 ring-bfl 1	12.7	C LC_135	16.1	0.0	0.0	OK
SLN83 ring-tfl 1	12.7	C LC_119	17.6	0.0	0.0	OK
SLN83 ring-w 1	7.2	C LC_119	17.0	0.0	0.0	OK
End plate SLN83 ring	20.0	C LC_121	3.9	0.0	2.7	OK
Bottom flange plate SLN83 ring	15.0	C LC_114	8.3	0.0	0.0	OK
Bottom flange half plate L SLN83 ring	15.0	C LC_135	4.4	0.0	0.0	OK
Bottom flange half plate R SLN83 ring	15.0	C LC_135	4.5	0.0	0.0	OK
Haunch web SLN83 ring	12.0	C LC_119	11.7	0.0	0.0	OK
Haunch top plate SLN83 ring	15.0	C LC_110	2.9	0.0	0.0	OK
Haunch sloping plate SLN83 ring	15.0	C LC_119	5.3	0.0	0.0	OK
End plate SLN80 ring	20.0	C LC_147	4.3	0.0	2.8	OK
Bottom flange plate SLN80 ring	15.0	C LC_135	13.3	0.0	0.0	OK
Bottom flange half plate L SLN80 ring	15.0	C LC_135	12.8	0.0	0.0	OK
Bottom flange half plate R SLN80 ring	15.0	C LC_135	12.5	0.0	0.0	OK
Haunch web SLN80 ring	12.0	C LC_119	31.5	0.0	0.0	OK
Haunch top plate SLN80 ring	15.0	C LC_147	5.6	0.0	0.0	OK
Haunch sloping plate SLN80 ring	15.0	C LC_119	14.3	0.0	0.0	OK

Design data

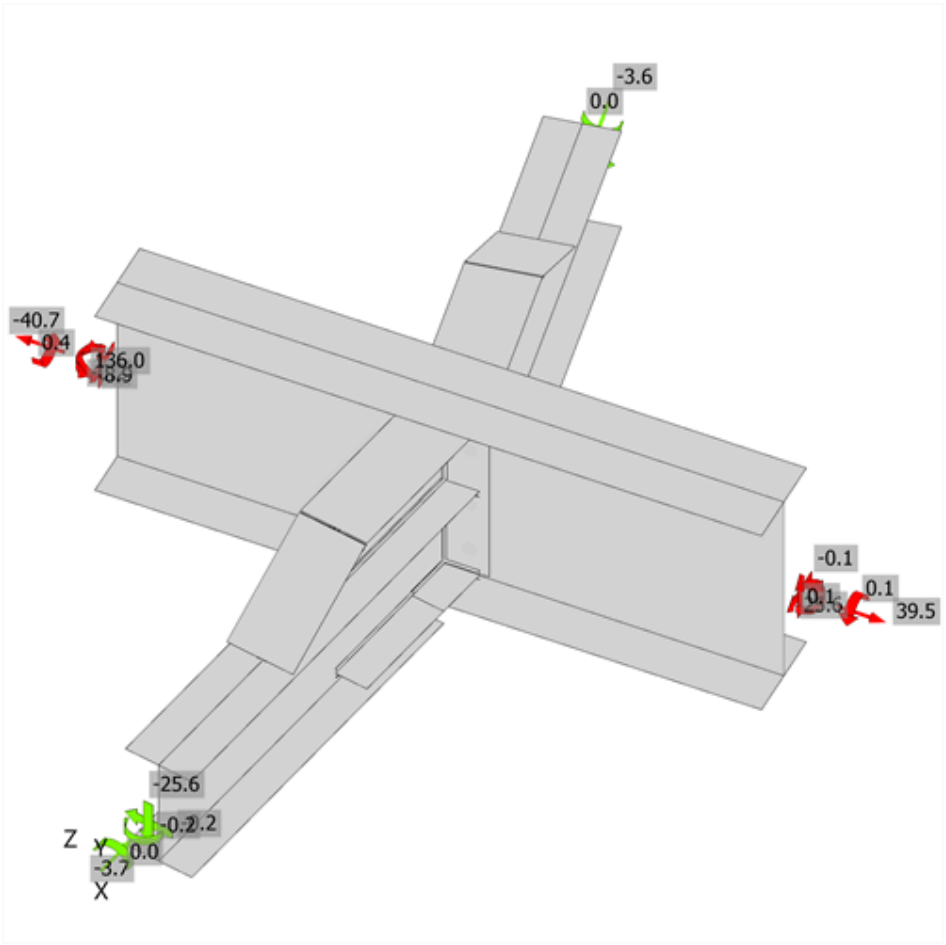
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

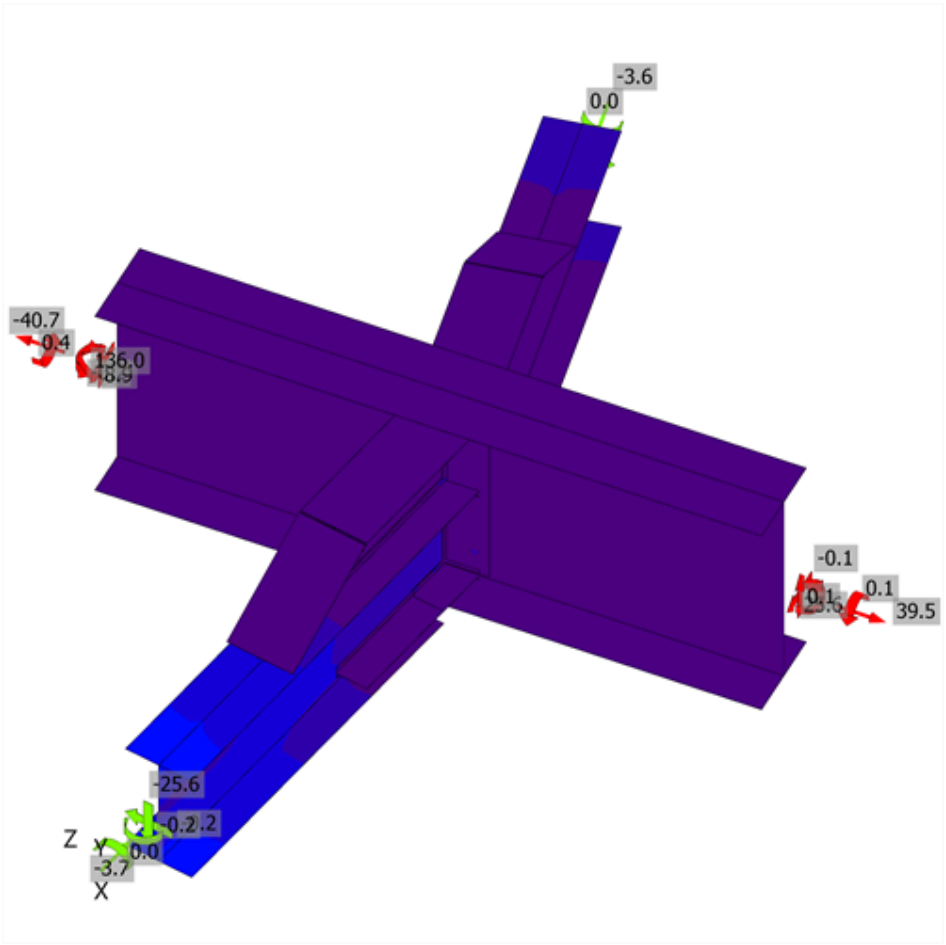
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

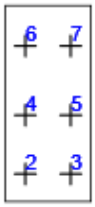


Strain check, C LC_147



Equivalent stress, C LC_147

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_147	0.4	0.3	194.1	0.1	0.3	0.2	OK	OK
	B3	M30 8.8 - 1	C LC_135	0.8	0.4	194.1	0.2	0.4	0.4	OK	OK
	B4	M30 8.8 - 1	C LC_119	0.1	0.3	232.2	0.0	0.2	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_135	0.0	0.3	232.2	0.0	0.2	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_121	1.4	0.6	249.2	0.4	0.4	0.6	OK	OK
	B7	M30 8.8 - 1	C LC_147	1.4	0.6	249.2	0.4	0.4	0.6	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate R SLN83 ring	Bottom flange plate SLN83 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN80 ring	Bottom flange plate SLN80 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN83 ring	Bottom flange plate SLN83 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN80 ring	Bottom flange plate SLN80 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN78a+SLN78b radial (continuous)-w 1	SLN80 ring-tfl 1	◢ 11.4	149	C LC_119	8.1	0.0	-7.9	-0.9	0.8	2.5	2.5	OK	OK
SLN78a+SLN78b radial (continuous)-w 1	SLN83 ring-tfl 1	◢ 11.4	149	C LC_119	7.2	0.0	-5.7	-2.5	0.0	1.8	0.0	OK	OK
End plate SLN83 ring	SLN83 ring-w 1	◢ 6.0 ◣	208	C LC_119	6.5	0.0	-0.3	-1.3	3.5	1.6	0.0	OK	OK
		◢ 6.0 ◣	208	C LC_120	4.7	0.0	-1.0	0.2	-2.6	1.2	0.0	OK	OK
Bottom flange plate SLN83 ring	SLN83 ring-w 1	◢ 6.0 ◣	100	C LC_135	7.2	0.0	0.7	0.8	4.1	1.8	1.8	OK	OK
		◢ 6.0 ◣	100	C LC_121	7.5	0.0	0.5	-1.0	-4.2	1.9	1.9	OK	OK
SLN83 ring-w 1	Bottom flange half plate L SLN83 ring	◢ 6.0 ◣	243	C LC_135	5.6	0.0	1.3	1.3	2.9	1.4	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	6.9	0.0	2.0	0.9	-3.7	1.7	0.0	OK	OK
SLN83 ring-w 1	Bottom flange half plate R SLN83 ring	◢ 6.0 ◣	243	C LC_135	5.8	0.0	1.5	1.4	-2.9	1.4	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	6.7	0.0	2.1	0.7	3.6	1.7	0.0	OK	OK
SLN83 ring-tfl 1	Haunch web SLN83 ring	◢ 6.0 ◣	572	C LC_119	25.9	0.0	-6.3	-6.2	13.1	6.4	6.0	OK	OK
		◢ 6.0 ◣	572	C LC_119	26.4	0.0	-6.5	6.6	-13.2	6.5	6.0	OK	OK
End plate SLN80 ring	SLN80 ring-w 1	◢ 6.0 ◣	208	C LC_135	6.1	0.0	-0.4	0.2	3.5	1.5	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_135	7.3	0.0	0.5	0.3	-4.2	1.8	0.0	OK	OK
Bottom flange plate SLN80 ring	SLN80 ring-w 1	◢ 6.0 ◣	100	C LC_135	20.4	0.0	1.7	1.4	11.6	5.0	5.0	OK	OK
		◢ 6.0 ◣	100	C LC_147	21.6	0.0	1.8	-2.4	-12.2	5.3	5.3	OK	OK
SLN80 ring-w 1	Bottom flange half plate L SLN80 ring	◢ 6.0 ◣	243	C LC_135	15.9	0.0	3.8	3.8	8.1	3.9	3.9	OK	OK
		◢ 6.0 ◣	243	C LC_135	19.8	0.0	6.3	2.1	-10.7	4.9	4.9	OK	OK
SLN80 ring-w 1	Bottom flange half plate R SLN80 ring	◢ 6.0 ◣	243	C LC_135	17.2	0.0	4.0	4.0	-8.8	4.2	4.2	OK	OK
		◢ 6.0 ◣	243	C LC_135	19.2	0.0	5.4	2.6	10.3	4.7	4.7	OK	OK
SLN80 ring-tfl 1	Haunch web SLN80 ring	◢ 6.0 ◣	572	C LC_119	70.9	0.0	-17.3	-17.4	35.7	17.5	10.0	OK	OK
		◢ 6.0 ◣	572	C LC_119	71.5	0.0	-18.2	18.1	-35.6	17.7	10.1	OK	OK
End plate SLN83 ring	Bottom flange plate SLN83 ring	◢ 6.0 ◣	149	C LC_114	9.5	0.0	-2.5	-3.8	3.7	2.3	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_114	12.0	0.0	-5.1	3.8	-5.0	3.0	0.0	OK	OK
End plate SLN83 ring	Haunch web SLN83 ring	◢ 6.0 ◣	145	C LC_135	3.2	0.0	-0.5	0.2	-1.8	0.8	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_121	4.8	0.0	0.7	-0.8	-2.6	1.2	0.0	OK	OK
End plate SLN83 ring	Haunch top plate SLN83 ring	◢ 6.0 ◣	149	C LC_110	5.0	0.0	-2.5	-1.7	1.9	1.2	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_114	3.8	0.0	-1.0	1.7	-1.2	0.9	0.0	OK	OK
Haunch top plate SLN83 ring	Haunch web SLN83 ring	◢ 6.0 ◣	342	C LC_120	1.4	0.0	-0.1	-0.1	-0.8	0.3	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_121	1.5	0.0	0.0	0.1	0.9	0.4	0.0	OK	OK
Haunch sloping plate SLN83 ring	Haunch web SLN83 ring	◢ 6.0 ◣	271	C LC_119	20.6	0.0	-3.5	-3.6	11.1	5.1	5.1	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_119	21.2	0.0	-3.7	3.6	-11.5	5.2	5.2	OK	OK
End plate SLN80 ring	Bottom flange plate SLN80 ring	◢ 6.0 ◣	149	C LC_109	9.5	0.0	-2.9	-3.7	-3.7	2.4	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_109	10.8	0.0	-4.5	3.7	4.4	2.7	2.7	OK	OK
End plate SLN80 ring	Haunch web SLN80 ring	◢ 6.0 ◣	145	C LC_148	9.1	0.0	1.1	1.2	5.1	2.2	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_121	6.5	0.0	0.7	-0.7	-3.7	1.6	0.0	OK	OK
End plate SLN80 ring	Haunch top plate SLN80 ring	◢ 6.0 ◣	149	C LC_109	5.1	0.0	-2.5	-1.8	-1.9	1.3	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_109	4.0	0.0	-1.2	1.8	1.2	1.0	0.0	OK	OK
Haunch top plate SLN80 ring	Haunch web SLN80 ring	◢ 6.0 ◣	342	C LC_147	4.0	0.0	1.5	1.4	1.7	1.0	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_147	3.6	0.0	-0.2	0.2	2.1	0.9	0.0	OK	OK
Haunch sloping plate SLN80 ring	Haunch web SLN80 ring	◢ 6.0 ◣	271	C LC_119	56.1	0.0	-9.5	-9.4	30.5	13.9	8.6	OK	OK
		◢ 6.0 ◣	271	C LC_119	57.6	0.0	-10.2	10.4	-31.0	14.2	8.8	OK	OK
SLN78a+SLN78b radial (continuous)-w 1	SLN83 ring-w 1	◢ 7.2	208	C LC_135	21.4	0.0	2.8	2.9	-11.9	5.3	5.3	OK	OK
SLN78a+SLN78b radial (continuous)-w 1	SLN80 ring-w 1	◢ 7.2	208	C LC_135	32.6	0.0	8.6	-9.7	15.3	8.0	8.0	OK	OK
Haunch sloping plate SLN83 ring	Haunch top plate SLN83 ring	◢ 6.0 ◣	146	C LC_154	2.4	0.0	-1.7	1.0	0.1	0.7	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_114	1.1	0.0	-0.2	0.6	-0.1	0.3	0.0	OK	OK
Haunch sloping plate SLN80 ring	Haunch top plate SLN80 ring	◢ 6.0 ◣	146	C LC_121	7.4	0.0	-4.8	3.2	-0.3	2.0	0.0	OK	OK
		◢ 6.0 ◣	146	C LC_147	2.6	0.0	-0.6	1.4	0.5	0.6	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_080
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN92b+SLN92a radial (continuous) - SLN88 ring - SLN91 ring

Name	SLN92b+SLN92a radial (continuous) - SLN88 ring - SLN91 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

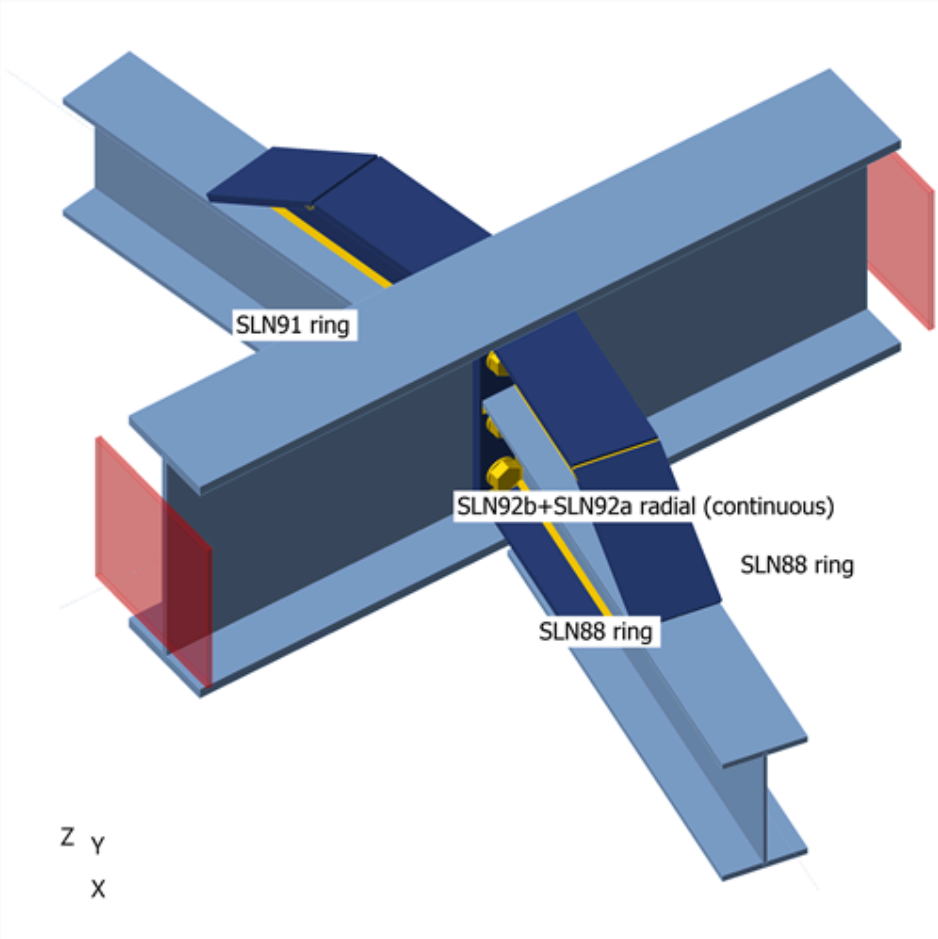
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN92b+SLN92a radial (continuous)	1 - UB457x191x98	80.0	0.0	0.0	0	0	0
SLN88 ring	2 - UB254x146x43	-20.0	0.0	0.0	0	0	-102
SLN91 ring	2 - UB254x146x43	180.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN92b+SLN92a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN92b+SLN92a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN88 ring / end		Position	0
SLN91 ring / end		Position	0



SLN92b+SLN92a radial (continuous) - SLN88 ring - SLN91 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN88 ring / End	4.8	0.2	-4.9	0.0	0.0	0.2
	SLN91 ring / End	4.8	-0.1	-4.9	0.0	0.0	-0.1
C LC_102	SLN88 ring / End	19.4	1.0	-25.8	0.0	0.0	1.0
	SLN91 ring / End	19.4	-0.2	-25.6	0.0	0.0	-0.4
C LC_103	SLN88 ring / End	16.9	1.1	-24.7	0.0	0.0	0.9
	SLN91 ring / End	17.0	-0.2	-24.6	0.0	0.0	-0.4
C LC_104	SLN88 ring / End	15.0	1.0	-24.7	0.0	0.0	0.9
	SLN91 ring / End	14.9	-0.1	-24.6	0.0	0.0	-0.2
C LC_105	SLN88 ring / End	16.6	1.0	-24.7	0.0	0.0	0.9
	SLN91 ring / End	16.8	0.0	-24.6	0.0	0.0	-0.2
C LC_106	SLN88 ring / End	15.4	1.0	-24.7	0.0	0.0	0.9
	SLN91 ring / End	15.2	-0.4	-24.6	0.0	0.0	-0.5
C LC_107	SLN88 ring / End	3.3	0.3	-1.6	0.0	0.0	0.2
	SLN91 ring / End	3.0	0.0	-1.6	0.0	0.0	-0.1
C LC_108	SLN88 ring / End	-1.3	0.0	-1.6	0.0	0.0	0.1
	SLN91 ring / End	-1.1	0.1	-1.6	0.0	0.0	0.2
C LC_109	SLN88 ring / End	0.7	0.2	-1.6	0.0	0.0	0.2
	SLN91 ring / End	1.0	0.1	-1.6	0.0	0.0	0.2
C LC_110	SLN88 ring / End	1.5	0.0	-1.6	0.0	0.0	0.0
	SLN91 ring / End	1.1	-0.2	-1.6	0.0	0.0	-0.3
C LC_111	SLN88 ring / End	13.1	0.9	-17.4	0.0	0.0	0.8
	SLN91 ring / End	12.9	-0.2	-17.3	0.0	0.0	-0.3
C LC_112	SLN88 ring / End	8.8	0.6	-17.4	0.0	0.0	0.7
	SLN91 ring / End	8.9	0.0	-17.3	0.0	0.0	0.0
C LC_113	SLN88 ring / End	11.1	0.7	-17.4	0.0	0.0	0.7
	SLN91 ring / End	11.5	0.1	-17.3	0.0	0.0	0.0
C LC_114	SLN88 ring / End	10.9	0.7	-17.4	0.0	0.0	0.6
	SLN91 ring / End	10.5	-0.4	-17.3	0.0	0.0	-0.6
C LC_115	SLN88 ring / End	4.5	0.3	-2.9	0.0	0.0	0.2
	SLN91 ring / End	4.2	0.0	-2.8	0.0	0.0	-0.1
C LC_116	SLN88 ring / End	-0.1	0.1	-2.9	0.0	0.0	0.1
	SLN91 ring / End	0.1	0.1	-2.8	0.0	0.0	0.1
C LC_117	SLN88 ring / End	1.9	0.2	-2.9	0.0	0.0	0.2
	SLN91 ring / End	2.3	0.1	-2.8	0.0	0.0	0.1
C LC_118	SLN88 ring / End	2.7	0.1	-2.9	0.0	0.0	0.0
	SLN91 ring / End	2.3	-0.2	-2.8	0.0	0.0	-0.3
C LC_119	SLN88 ring / End	18.5	1.1	-25.8	0.0	0.0	0.9
	SLN91 ring / End	18.5	-0.3	-25.6	0.0	0.0	-0.5
C LC_120	SLN88 ring / End	16.5	1.0	-25.8	0.0	0.0	1.0
	SLN91 ring / End	16.3	-0.1	-25.6	0.0	0.0	-0.2
C LC_121	SLN88 ring / End	17.6	1.0	-25.8	0.0	0.0	0.9
	SLN91 ring / End	17.8	-0.1	-25.6	0.0	0.0	-0.2
C LC_122	SLN88 ring / End	17.1	1.1	-25.8	0.0	0.0	0.9
	SLN91 ring / End	16.9	-0.4	-25.6	0.0	0.0	-0.5
C LC_123	SLN88 ring / End	16.2	0.9	-19.5	0.0	0.0	0.8

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN91 ring / End	16.1	-0.3	-19.4	0.0	0.0	-0.4
C LC_124	SLN88 ring / End	11.7	0.8	-19.5	0.0	0.0	0.8
	SLN91 ring / End	11.7	0.0	-19.4	0.0	0.0	0.0
C LC_125	SLN88 ring / End	13.1	0.8	-19.5	0.0	0.0	0.8
	SLN91 ring / End	13.6	0.0	-19.4	0.0	0.0	-0.1
C LC_126	SLN88 ring / End	14.3	0.7	-19.5	0.0	0.0	0.6
	SLN91 ring / End	13.9	-0.4	-19.4	0.0	0.0	-0.5
C LC_127	SLN88 ring / End	7.6	0.3	-4.9	0.0	0.0	0.3
	SLN91 ring / End	7.4	-0.2	-4.9	0.0	0.0	-0.2
C LC_128	SLN88 ring / End	2.8	0.2	-4.9	0.0	0.0	0.3
	SLN91 ring / End	2.9	0.1	-4.9	0.0	0.0	0.1
C LC_129	SLN88 ring / End	4.0	0.3	-4.9	0.0	0.0	0.3
	SLN91 ring / End	4.3	0.0	-4.9	0.0	0.0	0.0
C LC_130	SLN88 ring / End	6.1	0.1	-4.9	0.0	0.0	0.0
	SLN91 ring / End	5.7	-0.2	-4.9	0.0	0.0	-0.3
C LC_135	SLN88 ring / End	15.9	0.7	-25.8	0.0	0.0	0.7
	SLN91 ring / End	16.0	-0.2	-25.6	0.0	0.0	-0.3
C LC_141	SLN88 ring / End	4.8	0.2	-4.9	0.0	0.0	0.2
	SLN91 ring / End	4.8	-0.1	-4.9	0.0	0.0	-0.1
C LC_142	SLN88 ring / End	4.8	0.2	-4.9	0.0	0.0	0.2
	SLN91 ring / End	4.8	-0.1	-4.9	0.0	0.0	-0.1
C LC_143	SLN88 ring / End	5.0	0.2	-4.9	0.0	0.0	0.2
	SLN91 ring / End	5.0	0.0	-4.9	0.0	0.0	-0.1
C LC_144	SLN88 ring / End	4.6	0.2	-4.9	0.0	0.0	0.2
	SLN91 ring / End	4.6	-0.1	-4.9	0.0	0.0	-0.1
C LC_145	SLN88 ring / End	19.1	1.0	-25.8	0.0	0.0	0.9
	SLN91 ring / End	19.3	-0.3	-25.6	0.0	0.0	-0.4
C LC_146	SLN88 ring / End	19.6	1.0	-25.8	0.0	0.0	1.0
	SLN91 ring / End	19.4	-0.2	-25.6	0.0	0.0	-0.4
C LC_147	SLN88 ring / End	20.2	1.0	-25.8	0.0	0.0	0.9
	SLN91 ring / End	20.3	-0.1	-25.6	0.0	0.0	-0.3
C LC_148	SLN88 ring / End	18.5	1.1	-25.8	0.0	0.0	1.0
	SLN91 ring / End	18.4	-0.3	-25.6	0.0	0.0	-0.5
C LC_151	SLN88 ring / End	12.4	0.7	-11.6	0.0	0.0	0.6
	SLN91 ring / End	12.4	-0.2	-11.8	0.0	0.0	-0.3
C LC_152	SLN88 ring / End	19.3	1.1	-14.0	0.0	0.0	1.0
	SLN91 ring / End	19.3	-0.2	-14.2	0.0	0.0	-0.4
C LC_153	SLN88 ring / End	19.3	1.1	-20.7	0.0	0.0	1.0
	SLN91 ring / End	19.4	-0.2	-21.1	0.0	0.0	-0.4
C LC_154	SLN88 ring / End	12.5	0.7	-11.6	0.0	0.0	0.6
	SLN91 ring / End	12.5	-0.2	-11.8	0.0	0.0	-0.3

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	1.3 < 100%	OK
Welds	20.8 < 100%	OK
Buckling	Not calculated	

Plates

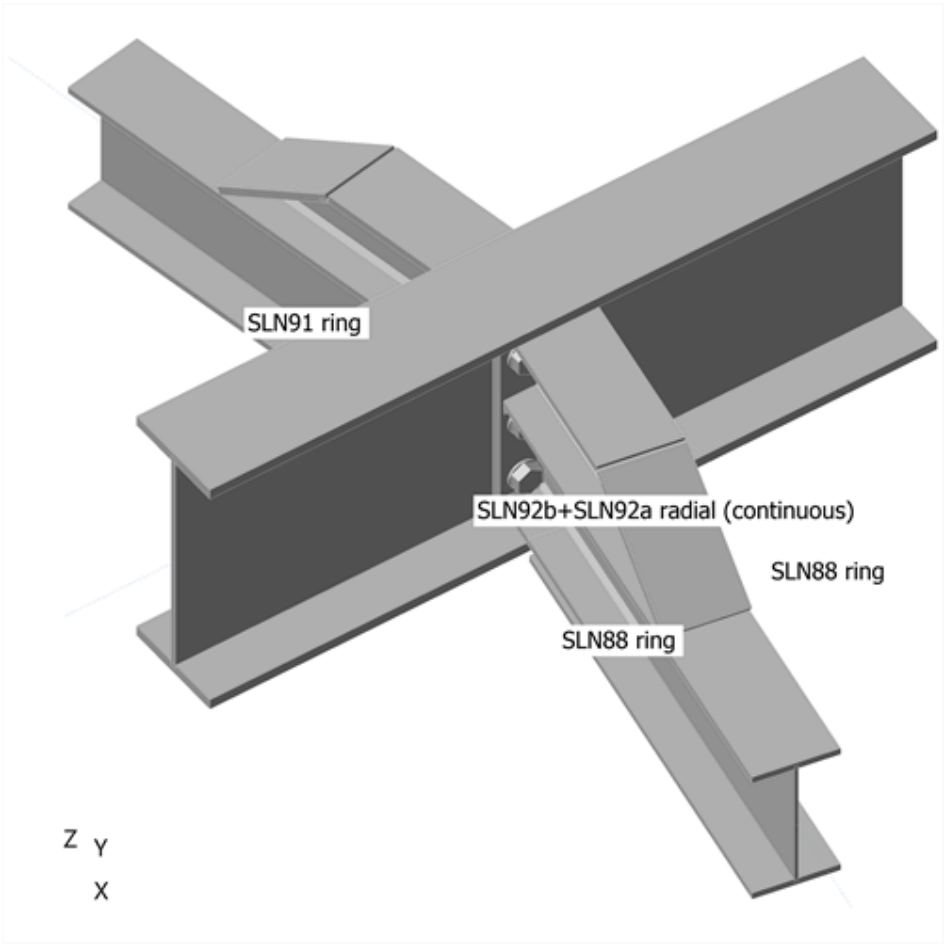
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN92b+SLN92a radial (continuous)-bfl 1	19.6	C LC_120	20.7	0.0	0.0	OK
SLN92b+SLN92a radial (continuous)-tfl 1	19.6	C LC_120	17.5	0.0	0.0	OK
SLN92b+SLN92a radial (continuous)-w 1	11.4	C LC_121	27.2	0.0	1.4	OK
SLN88 ring-bfl 1	12.7	C LC_119	51.1	0.0	0.0	OK
SLN88 ring-tfl 1	12.7	C LC_120	44.6	0.0	0.0	OK
SLN88 ring-w 1	7.2	C LC_147	49.3	0.0	0.0	OK
SLN91 ring-bfl 1	12.7	C LC_147	51.3	0.0	0.0	OK
SLN91 ring-tfl 1	12.7	C LC_122	45.2	0.0	0.0	OK
SLN91 ring-w 1	7.2	C LC_147	49.1	0.0	0.0	OK
End plate SLN91 ring	20.0	C LC_146	12.1	0.0	4.4	OK
Bottom flange plate SLN91 ring	15.0	C LC_147	20.8	0.0	0.0	OK
Bottom flange half plate L SLN91 ring	15.0	C LC_147	20.0	0.0	0.0	OK
Bottom flange half plate R SLN91 ring	15.0	C LC_147	19.8	0.0	0.0	OK
Haunch web SLN91 ring	12.0	C LC_122	28.2	0.0	0.0	OK
Haunch top plate SLN91 ring	15.0	C LC_122	5.8	0.0	0.0	OK
Haunch sloping plate SLN91 ring	15.0	C LC_120	12.6	0.0	0.0	OK
End plate SLN88 ring	20.0	C LC_148	14.5	0.0	6.9	OK
Bottom flange plate SLN88 ring	15.0	C LC_148	24.7	0.0	0.0	OK
Bottom flange half plate L SLN88 ring	15.0	C LC_147	19.7	0.0	0.0	OK
Bottom flange half plate R SLN88 ring	15.0	C LC_147	20.8	0.0	0.0	OK
Haunch web SLN88 ring	12.0	C LC_120	29.2	0.0	0.0	OK
Haunch top plate SLN88 ring	15.0	C LC_120	9.3	0.0	0.0	OK
Haunch sloping plate SLN88 ring	15.0	C LC_120	12.8	0.0	0.0	OK

Design data

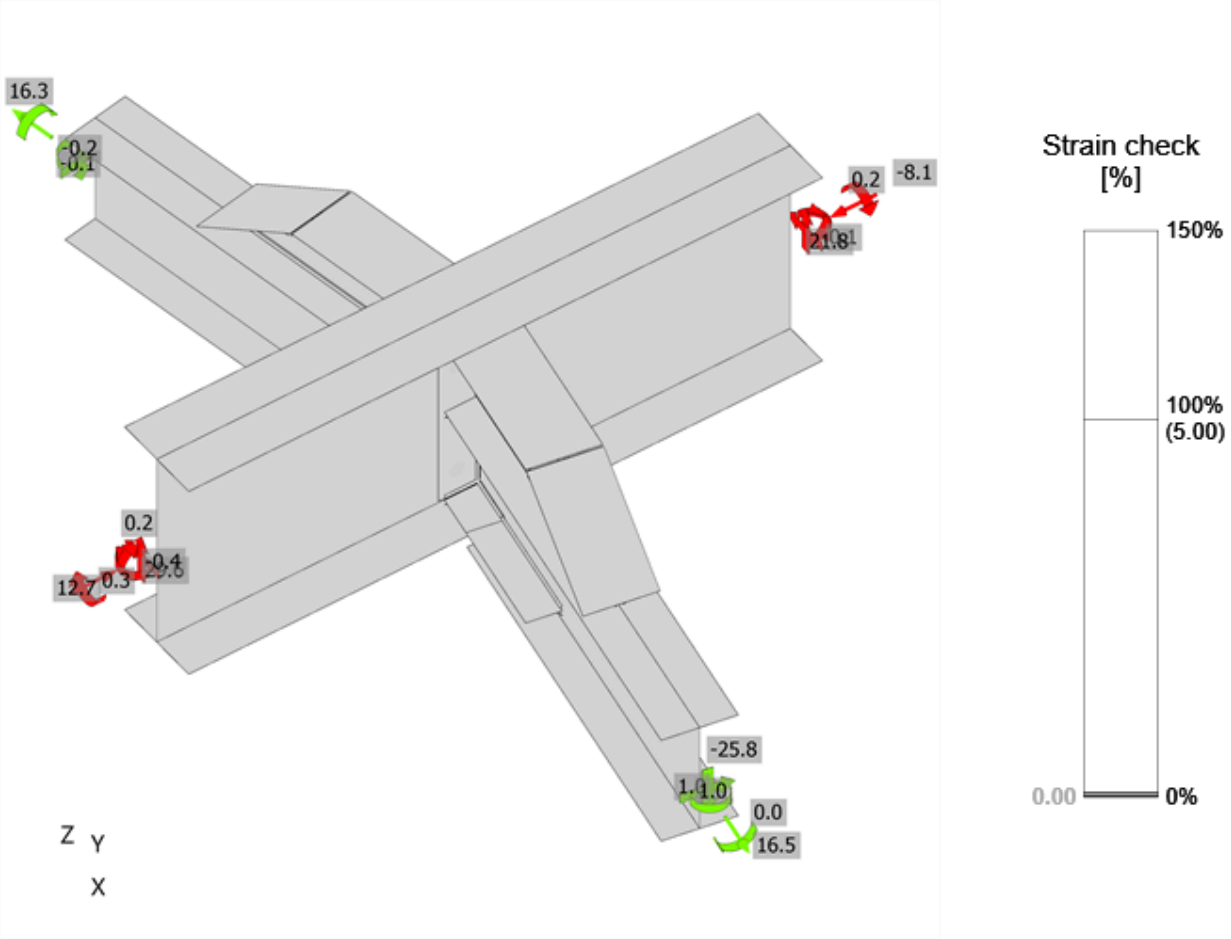
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

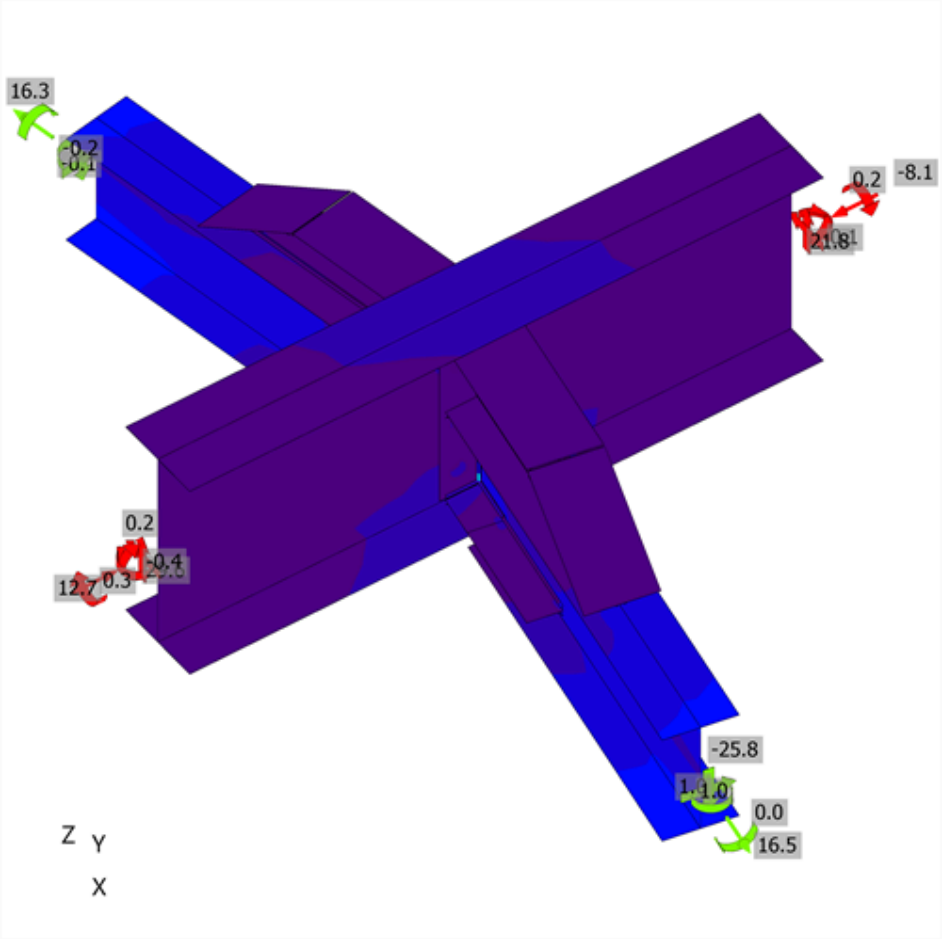
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

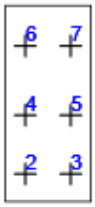


Strain check, C LC_120



Equivalent stress, C LC_120

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_148	4.3	0.5	194.1	1.3	0.5	1.2	OK	OK
	B3	M30 8.8 - 1	C LC_121	1.3	0.6	229.1	0.4	0.5	0.6	OK	OK
	B4	M30 8.8 - 1	C LC_135	0.2	0.4	232.2	0.1	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_147	0.3	0.5	232.2	0.1	0.4	0.3	OK	OK
	B6	M30 8.8 - 1	C LC_120	3.0	0.8	249.2	0.9	0.6	1.1	OK	OK
	B7	M30 8.8 - 1	C LC_120	1.4	0.9	249.2	0.4	0.7	0.7	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate R SLN91 ring	Bottom flange plate SLN91 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN88 ring	Bottom flange plate SLN88 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN91 ring	Bottom flange plate SLN91 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN88 ring	Bottom flange plate SLN88 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN92b+SLN92a radial (continuous)-w 1	SLN88 ring-tfl 1	◢ 11.4	149	C LC_152	15.7	0.0	7.5	-7.8	1.6	3.9	3.9	OK	OK
SLN92b+SLN92a radial (continuous)-w 1	SLN91 ring-tfl 1	◢ 11.4	149	C LC_106	11.9	0.0	-6.5	5.6	-1.3	2.9	2.9	OK	OK
End plate SLN91 ring	SLN91 ring-w 1	◢ 6.0 ◣	208	C LC_147	26.0	0.0	-11.2	-13.0	-4.0	6.4	6.1	OK	OK
		◢ 6.0 ◣	208	C LC_147	21.6	0.0	-11.5	10.4	-1.6	5.3	5.3	OK	OK
Bottom flange plate SLN91 ring	SLN91 ring-w 1	◢ 6.0 ◣	100	C LC_120	24.4	0.0	1.9	1.9	13.9	6.0	6.0	OK	OK
		◢ 6.0 ◣	100	C LC_145	25.9	0.0	2.9	-3.7	-14.4	6.4	6.4	OK	OK
SLN91 ring-w 1	Bottom flange half plate L SLN91 ring	◢ 6.0 ◣	243	C LC_147	24.4	0.0	6.2	6.0	12.3	6.0	6.0	OK	OK
		◢ 6.0 ◣	243	C LC_147	27.7	0.0	8.7	2.4	-15.0	6.8	6.3	OK	OK
SLN91 ring-w 1	Bottom flange half plate R SLN91 ring	◢ 6.0 ◣	243	C LC_145	25.3	0.0	5.9	6.2	-12.8	6.3	5.8	OK	OK
		◢ 6.0 ◣	243	C LC_147	27.7	0.0	7.5	3.0	15.1	6.8	6.5	OK	OK
SLN91 ring-tfl 1	Haunch web SLN91 ring	◢ 6.0 ◣	572	C LC_122	66.2	0.0	-14.7	-16.8	33.2	16.3	9.4	OK	OK
		◢ 6.0 ◣	572	C LC_135	61.8	0.0	-16.8	15.7	-30.6	15.3	9.0	OK	OK
End plate SLN88 ring	SLN88 ring-w 1	◢ 6.0 ◣	208	C LC_147	23.4	0.0	-10.9	-11.9	0.3	5.8	5.8	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_147	29.7	0.0	-14.5	13.2	7.0	7.3	6.3	OK	OK
Bottom flange plate SLN88 ring	SLN88 ring-w 1	◢ 6.0 ◣	99	C LC_146	28.5	0.0	-0.8	-3.2	16.1	7.0	7.0	OK	OK
		◢ 6.0 ◣	100	C LC_147	23.2	0.0	1.2	0.5	-13.4	5.7	5.7	OK	OK
SLN88 ring-w 1	Bottom flange half plate L SLN88 ring	◢ 6.0 ◣	243	C LC_147	28.4	0.0	6.2	7.0	14.4	7.0	6.5	OK	OK
		◢ 6.0 ◣	243	C LC_146	33.8	0.0	9.1	-8.3	-16.9	8.4	7.5	OK	OK
SLN88 ring-w 1	Bottom flange half plate R SLN88 ring	◢ 6.0 ◣	243	C LC_147	20.9	0.0	5.9	5.0	-10.4	5.2	5.2	OK	OK
		◢ 6.0 ◣	243	C LC_147	28.6	0.0	11.0	1.2	15.2	7.1	6.2	OK	OK
SLN88 ring-tfl 1	Haunch web SLN88 ring	◢ 6.0 ◣	572	C LC_135	60.2	0.0	-17.8	-15.2	29.5	14.9	8.8	OK	OK
		◢ 6.0 ◣	572	C LC_120	68.8	0.0	-13.5	17.3	-34.9	17.0	9.7	OK	OK
End plate SLN91 ring	Bottom flange plate SLN91 ring	◢ 6.0 ◣	148	C LC_147	15.7	0.0	9.3	6.7	-2.9	3.9	3.9	OK	OK
		◢ 6.0 ◣	149	C LC_124	18.5	0.0	-9.7	8.0	4.4	4.6	4.6	OK	OK
End plate SLN91 ring	Haunch web SLN91 ring	◢ 6.0 ◣	145	C LC_120	8.4	0.0	1.2	1.3	4.7	2.1	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_135	9.2	0.0	2.4	-2.2	-4.6	2.3	0.0	OK	OK
End plate SLN91 ring	Haunch top plate SLN91 ring	◢ 6.0 ◣	149	C LC_124	8.3	0.0	-4.7	-3.0	-2.6	2.1	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_118	5.3	0.0	-1.9	2.3	-1.8	1.3	0.0	OK	OK
Haunch top plate SLN91 ring	Haunch web SLN91 ring	◢ 6.0 ◣	342	C LC_122	4.5	0.0	1.8	1.6	1.8	1.1	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_120	3.5	0.0	1.1	-1.1	-1.6	0.9	0.0	OK	OK
Haunch sloping plate SLN91 ring	Haunch web SLN91 ring	◢ 6.0 ◣	271	C LC_122	52.2	0.0	-9.2	-6.3	29.0	12.9	8.4	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_135	51.2	0.0	-9.2	10.8	-27.0	12.6	8.1	OK	OK
End plate SLN88 ring	Bottom flange plate SLN88 ring	◢ 6.0 ◣	149	C LC_152	30.2	0.0	-10.5	-9.3	-13.5	7.5	7.0	OK	OK
		◢ 6.0 ◣	149	C LC_120	33.5	0.0	-11.8	11.8	13.7	8.3	6.8	OK	OK
End plate SLN88 ring	Haunch web SLN88 ring	◢ 6.0 ◣	145	C LC_120	11.2	0.0	2.7	2.8	5.6	2.8	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	6.1	0.0	0.9	-0.8	-3.4	1.5	0.0	OK	OK
End plate SLN88 ring	Haunch top plate SLN88 ring	◢ 6.0 ◣	149	C LC_152	13.1	0.0	-5.5	-5.4	-4.2	3.2	3.2	OK	OK
		◢ 6.0 ◣	149	C LC_152	13.2	0.0	-5.1	5.2	4.8	3.3	3.3	OK	OK
Haunch top plate SLN88 ring	Haunch web SLN88 ring	◢ 6.0 ◣	342	C LC_120	4.9	0.0	-0.3	-0.4	-2.8	1.2	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_120	6.7	0.0	2.4	-1.9	-3.0	1.7	0.0	OK	OK
Haunch sloping plate SLN88 ring	Haunch web SLN88 ring	◢ 6.0 ◣	271	C LC_120	51.6	0.0	-9.3	-14.4	25.5	12.7	8.2	OK	OK
		◢ 6.0 ◣	271	C LC_148	53.5	0.0	-9.3	4.0	-30.1	13.2	8.6	OK	OK
SLN92b+SLN92a radial (continuous)-w 1	SLN91 ring-w 1	◢ 7.2	208	C LC_147	84.0	0.0	35.1	39.5	-19.6	20.8	14.3	OK	OK
SLN92b+SLN92a radial (continuous)-w 1	SLN88 ring-w 1	◢ 7.2	208	C LC_147	80.4	0.0	40.6	-37.9	12.9	19.9	13.9	OK	OK
Haunch sloping plate SLN91 ring	Haunch top plate SLN91 ring	◢ 6.0 ◣	146	C LC_120	7.1	0.0	-5.2	2.8	0.3	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_122	3.3	0.0	-0.8	1.6	0.9	0.8	0.0	OK	OK
Haunch sloping plate SLN88 ring	Haunch top plate SLN88 ring	◢ 6.0 ◣	146	C LC_120	7.8	0.0	-7.0	-1.9	-0.2	2.3	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_120	5.1	0.0	-1.2	2.9	-0.2	1.3	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_100
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN92b+SLN92a radial (continuous) - SLN91 ring - SLN94 ring

Name	SLN92b+SLN92a radial (continuous) - SLN91 ring - SLN94 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

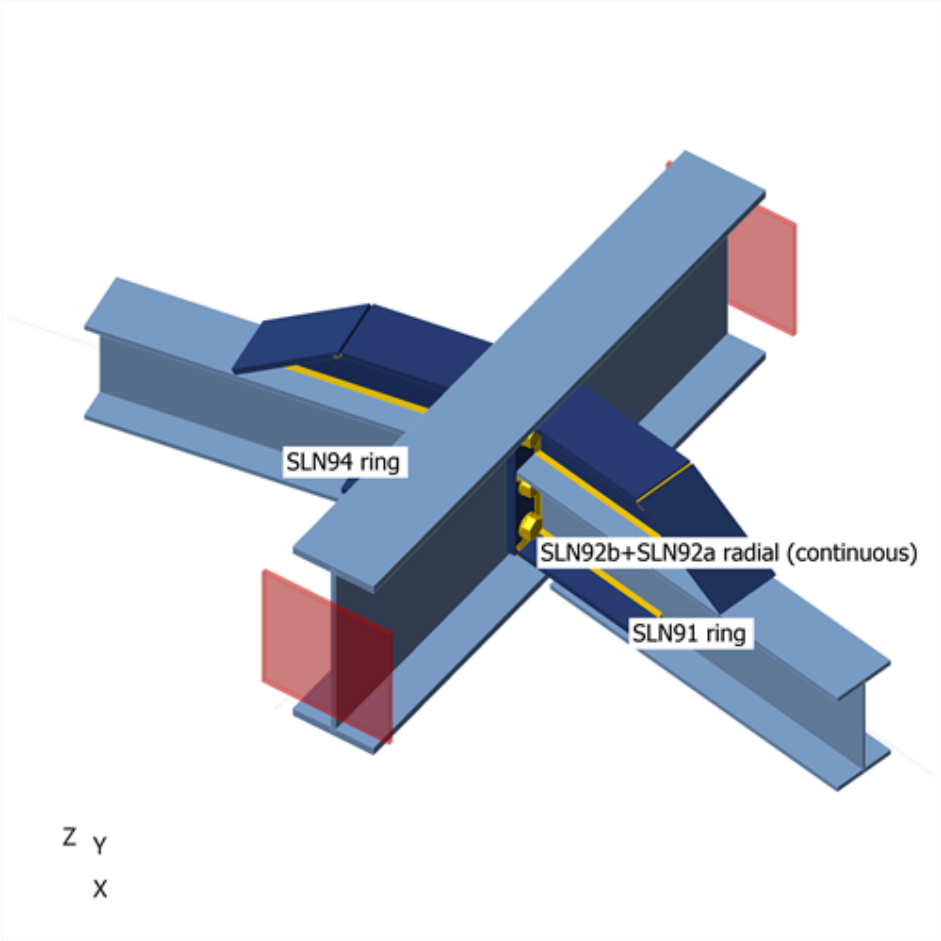
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN92b+SLN92a radial (continuous)	1 - UB457x191x98	100.0	0.0	0.0	0	0	0
SLN91 ring	2 - UB254x146x43	0.0	0.0	0.0	0	0	-102
SLN94 ring	2 - UB254x146x43	-160.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN92b+SLN92a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN92b+SLN92a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN91 ring / end		Position	0
SLN94 ring / end		Position	0



SLN92b+SLN92a radial (continuous) - SLN91 ring - SLN94 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN91 ring / End	4.8	-0.1	-4.9	0.0	0.0	-0.1
	SLN94 ring / End	4.8	0.0	-4.9	0.0	0.0	0.0
C LC_102	SLN91 ring / End	19.4	-0.2	-25.7	0.0	0.0	-0.3
	SLN94 ring / End	19.2	0.1	-25.7	0.0	0.0	0.1
C LC_103	SLN91 ring / End	17.1	-0.2	-24.6	0.0	0.0	-0.3
	SLN94 ring / End	17.4	0.1	-24.7	0.0	0.0	0.1
C LC_104	SLN91 ring / End	14.8	-0.1	-24.6	0.0	0.0	-0.1
	SLN94 ring / End	14.4	0.1	-24.7	0.0	0.0	0.2
C LC_105	SLN91 ring / End	16.8	-0.2	-24.6	0.0	0.0	-0.2
	SLN94 ring / End	16.9	0.2	-24.7	0.0	0.0	0.2
C LC_106	SLN91 ring / End	15.2	-0.2	-24.6	0.0	0.0	-0.4
	SLN94 ring / End	14.8	-0.1	-24.7	0.0	0.0	-0.1
C LC_107	SLN91 ring / End	3.0	0.0	-1.6	0.0	0.0	0.0
	SLN94 ring / End	3.4	0.0	-1.6	0.0	0.0	0.1
C LC_108	SLN91 ring / End	-1.2	0.1	-1.6	0.0	0.0	0.1
	SLN94 ring / End	-1.4	0.1	-1.6	0.0	0.0	0.1
C LC_109	SLN91 ring / End	1.0	0.1	-1.6	0.0	0.0	0.2
	SLN94 ring / End	1.3	0.1	-1.6	0.0	0.0	0.1
C LC_110	SLN91 ring / End	1.1	-0.2	-1.6	0.0	0.0	-0.3
	SLN94 ring / End	0.7	-0.1	-1.6	0.0	0.0	-0.2
C LC_111	SLN91 ring / End	13.1	-0.2	-17.4	0.0	0.0	-0.2
	SLN94 ring / End	13.5	0.0	-17.4	0.0	0.0	0.1
C LC_112	SLN91 ring / End	8.7	0.0	-17.4	0.0	0.0	0.0
	SLN94 ring / End	8.3	0.1	-17.4	0.0	0.0	0.2
C LC_113	SLN91 ring / End	11.5	-0.1	-17.4	0.0	0.0	0.0
	SLN94 ring / End	11.8	0.2	-17.4	0.0	0.0	0.2
C LC_114	SLN91 ring / End	10.5	-0.3	-17.4	0.0	0.0	-0.4
	SLN94 ring / End	10.0	-0.2	-17.4	0.0	0.0	-0.2
C LC_115	SLN91 ring / End	4.3	0.0	-2.9	0.0	0.0	0.0
	SLN94 ring / End	4.7	0.0	-2.9	0.0	0.0	0.1
C LC_116	SLN91 ring / End	0.1	0.1	-2.9	0.0	0.0	0.1
	SLN94 ring / End	-0.1	0.1	-2.9	0.0	0.0	0.1
C LC_117	SLN91 ring / End	2.3	0.1	-2.9	0.0	0.0	0.1
	SLN94 ring / End	2.6	0.1	-2.9	0.0	0.0	0.1
C LC_118	SLN91 ring / End	2.3	-0.2	-2.9	0.0	0.0	-0.3
	SLN94 ring / End	1.8	-0.1	-2.9	0.0	0.0	-0.2
C LC_119	SLN91 ring / End	18.7	-0.3	-25.7	0.0	0.0	-0.4
	SLN94 ring / End	18.5	0.0	-25.7	0.0	0.0	0.0
C LC_120	SLN91 ring / End	16.2	-0.1	-25.7	0.0	0.0	-0.1
	SLN94 ring / End	16.0	0.1	-25.7	0.0	0.0	0.2
C LC_121	SLN91 ring / End	17.8	-0.2	-25.7	0.0	0.0	-0.2
	SLN94 ring / End	17.9	0.2	-25.7	0.0	0.0	0.3
C LC_122	SLN91 ring / End	16.9	-0.2	-25.7	0.0	0.0	-0.4
	SLN94 ring / End	16.5	-0.1	-25.7	0.0	0.0	-0.1
C LC_123	SLN91 ring / End	16.2	-0.3	-19.4	0.0	0.0	-0.4

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN94 ring / End	15.8	0.0	-19.5	0.0	0.0	0.0
C LC_124	SLN91 ring / End	11.5	0.0	-19.4	0.0	0.0	0.1
	SLN94 ring / End	11.7	0.2	-19.5	0.0	0.0	0.3
C LC_125	SLN91 ring / End	13.6	-0.1	-19.4	0.0	0.0	-0.1
	SLN94 ring / End	13.9	0.2	-19.5	0.0	0.0	0.3
C LC_126	SLN91 ring / End	13.9	-0.3	-19.4	0.0	0.0	-0.4
	SLN94 ring / End	13.4	-0.2	-19.5	0.0	0.0	-0.2
C LC_127	SLN91 ring / End	7.5	-0.2	-4.9	0.0	0.0	-0.2
	SLN94 ring / End	7.0	0.0	-4.9	0.0	0.0	-0.1
C LC_128	SLN91 ring / End	2.9	0.1	-4.9	0.0	0.0	0.2
	SLN94 ring / End	3.2	0.2	-4.9	0.0	0.0	0.3
C LC_129	SLN91 ring / End	4.3	0.0	-4.9	0.0	0.0	0.1
	SLN94 ring / End	4.7	0.1	-4.9	0.0	0.0	0.2
C LC_130	SLN91 ring / End	5.7	-0.2	-4.9	0.0	0.0	-0.3
	SLN94 ring / End	5.3	-0.1	-4.9	0.0	0.0	-0.2
C LC_135	SLN91 ring / End	16.0	-0.2	-25.7	0.0	0.0	-0.2
	SLN94 ring / End	15.9	0.1	-25.7	0.0	0.0	0.2
C LC_141	SLN91 ring / End	4.8	-0.1	-4.9	0.0	0.0	-0.1
	SLN94 ring / End	4.8	0.0	-4.9	0.0	0.0	0.0
C LC_142	SLN91 ring / End	4.7	-0.1	-4.9	0.0	0.0	-0.1
	SLN94 ring / End	4.7	0.0	-4.9	0.0	0.0	0.0
C LC_143	SLN91 ring / End	5.0	-0.1	-4.9	0.0	0.0	-0.1
	SLN94 ring / End	5.0	0.0	-4.9	0.0	0.0	0.1
C LC_144	SLN91 ring / End	4.6	0.0	-4.9	0.0	0.0	-0.1
	SLN94 ring / End	4.6	0.0	-4.9	0.0	0.0	0.0
C LC_145	SLN91 ring / End	19.5	-0.3	-25.7	0.0	0.0	-0.3
	SLN94 ring / End	19.5	0.1	-25.7	0.0	0.0	0.1
C LC_146	SLN91 ring / End	19.2	-0.2	-25.7	0.0	0.0	-0.3
	SLN94 ring / End	18.9	0.1	-25.7	0.0	0.0	0.2
C LC_147	SLN91 ring / End	20.3	-0.3	-25.7	0.0	0.0	-0.3
	SLN94 ring / End	20.1	0.2	-25.7	0.0	0.0	0.2
C LC_148	SLN91 ring / End	18.4	-0.2	-25.7	0.0	0.0	-0.3
	SLN94 ring / End	18.3	0.0	-25.7	0.0	0.0	0.1
C LC_151	SLN91 ring / End	12.4	-0.2	-11.8	0.0	0.0	-0.2
	SLN94 ring / End	12.2	0.0	-11.6	0.0	0.0	0.1
C LC_152	SLN91 ring / End	19.3	-0.2	-14.3	0.0	0.0	-0.3
	SLN94 ring / End	19.1	0.1	-14.0	0.0	0.0	0.1
C LC_153	SLN91 ring / End	19.4	-0.2	-21.2	0.0	0.0	-0.3
	SLN94 ring / End	19.2	0.1	-20.7	0.0	0.0	0.1
C LC_154	SLN91 ring / End	12.5	-0.2	-11.8	0.0	0.0	-0.2
	SLN94 ring / End	12.3	0.0	-11.6	0.0	0.0	0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.9 < 100%	OK
Welds	20.0 < 100%	OK
Buckling	Not calculated	

Plates

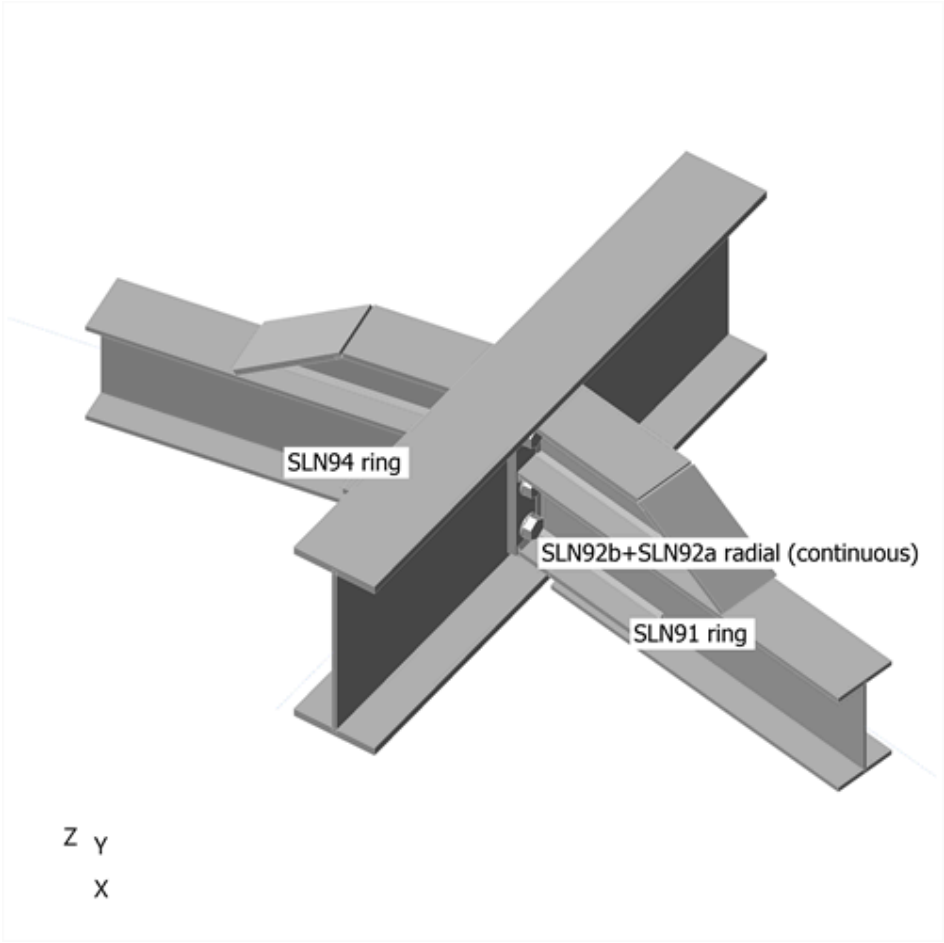
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN92b+SLN92a radial (continuous)-bfl 1	19.6	C LC_146	20.3	0.0	0.0	OK
SLN92b+SLN92a radial (continuous)-tfl 1	19.6	C LC_122	17.2	0.0	0.0	OK
SLN92b+SLN92a radial (continuous)-w 1	11.4	C LC_122	25.9	0.0	0.8	OK
SLN91 ring-bfl 1	12.7	C LC_148	50.4	0.0	0.0	OK
SLN91 ring-tfl 1	12.7	C LC_122	44.8	0.0	0.0	OK
SLN91 ring-w 1	7.2	C LC_147	49.3	0.0	0.0	OK
SLN94 ring-bfl 1	12.7	C LC_146	50.3	0.0	0.0	OK
SLN94 ring-tfl 1	12.7	C LC_120	44.6	0.0	0.0	OK
SLN94 ring-w 1	7.2	C LC_147	49.3	0.0	0.0	OK
End plate SLN94 ring	20.0	C LC_122	8.5	0.0	4.1	OK
Bottom flange plate SLN94 ring	15.0	C LC_147	20.6	0.0	0.0	OK
Bottom flange half plate L SLN94 ring	15.0	C LC_147	20.0	0.0	0.0	OK
Bottom flange half plate R SLN94 ring	15.0	C LC_147	20.1	0.0	0.0	OK
Haunch web SLN94 ring	12.0	C LC_122	28.5	0.0	0.0	OK
Haunch top plate SLN94 ring	15.0	C LC_120	5.5	0.0	0.0	OK
Haunch sloping plate SLN94 ring	15.0	C LC_135	12.7	0.0	0.0	OK
End plate SLN91 ring	20.0	C LC_147	9.1	0.0	0.7	OK
Bottom flange plate SLN91 ring	15.0	C LC_147	20.6	0.0	0.0	OK
Bottom flange half plate L SLN91 ring	15.0	C LC_147	20.3	0.0	0.0	OK
Bottom flange half plate R SLN91 ring	15.0	C LC_147	19.9	0.0	0.0	OK
Haunch web SLN91 ring	12.0	C LC_120	28.4	0.0	0.0	OK
Haunch top plate SLN91 ring	15.0	C LC_122	6.1	0.0	0.0	OK
Haunch sloping plate SLN91 ring	15.0	C LC_135	12.6	0.0	0.0	OK

Design data

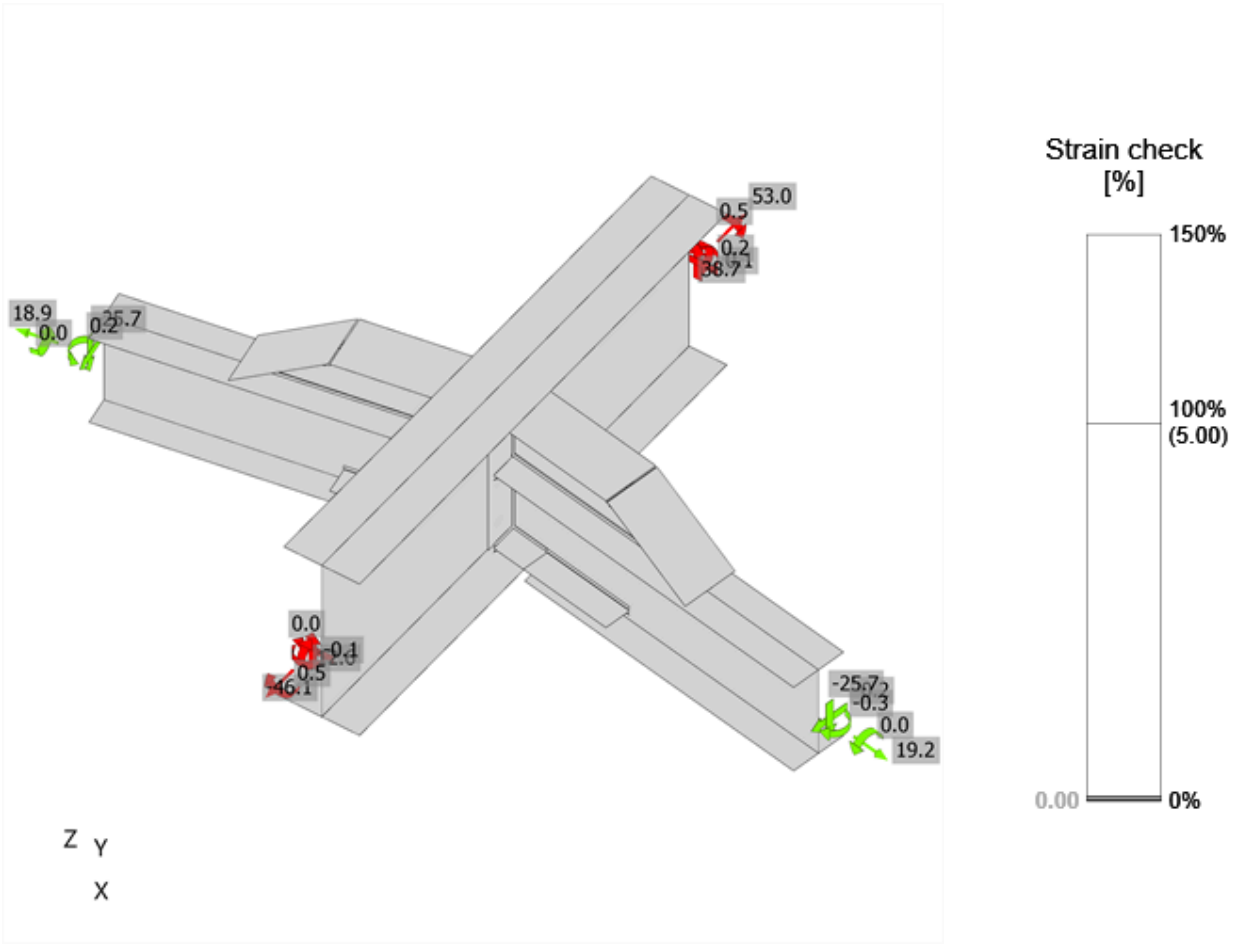
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

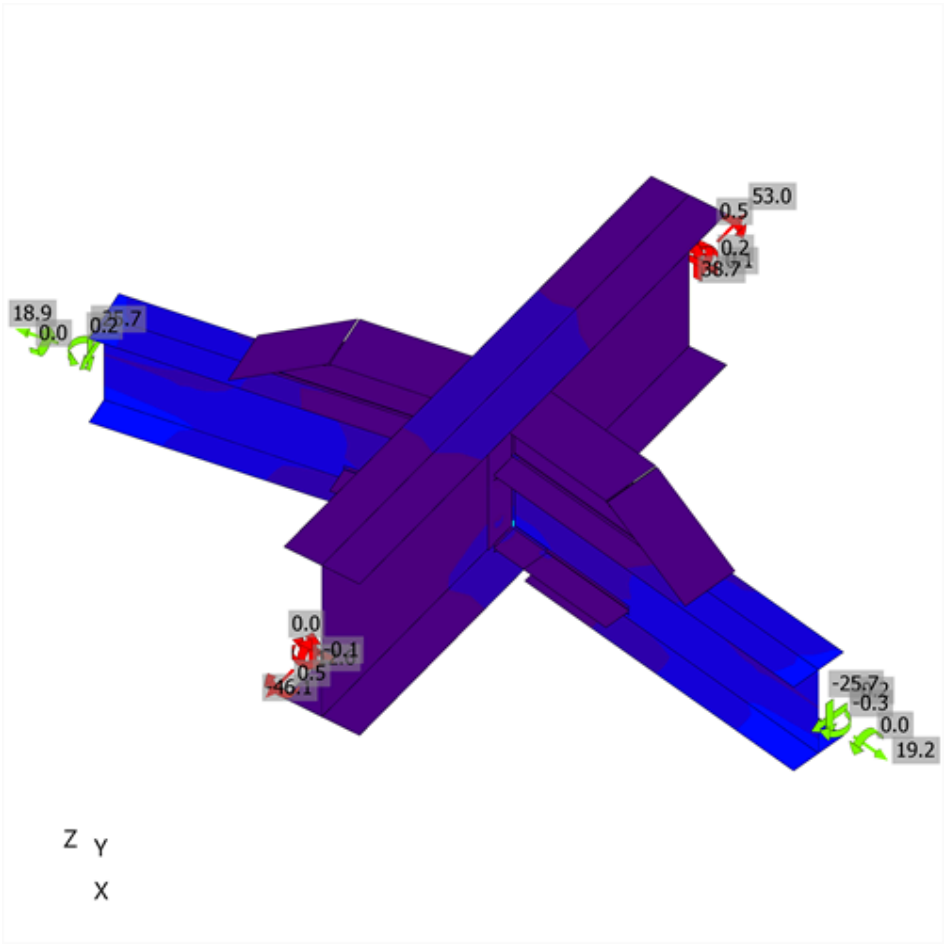
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

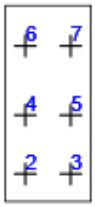


Strain check, C LC_146



Equivalent stress, C LC_146

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_122	2.5	0.5	194.1	0.8	0.5	0.8	OK	OK
	B3	M30 8.8 - 1	C LC_147	2.5	0.7	194.1	0.8	0.6	0.9	OK	OK
	B4	M30 8.8 - 1	C LC_121	0.1	0.4	232.2	0.0	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_147	0.1	0.5	232.2	0.0	0.4	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_122	1.8	0.9	249.2	0.5	0.7	0.8	OK	OK
	B7	M30 8.8 - 1	C LC_122	2.3	0.9	249.2	0.7	0.7	0.9	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate R SLN94 ring	Bottom flange plate SLN94 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN91 ring	Bottom flange plate SLN91 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN94 ring	Bottom flange plate SLN94 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN91 ring	Bottom flange plate SLN91 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN92b+SLN92a radial (continuous)-w 1	SLN91 ring-tfl 1	◢ 11.4	149	C LC_119	10.9	0.0	-5.1	4.5	-3.2	2.7	0.0	OK	OK
SLN92b+SLN92a radial (continuous)-w 1	SLN94 ring-tfl 1	◢ 11.4	149	C LC_147	10.6	0.0	-6.6	-4.8	0.1	2.6	2.6	OK	OK
End plate SLN94 ring	SLN94 ring-w 1	◢ 6.0 ◣	208	C LC_147	25.1	0.0	-10.7	-12.7	-3.3	6.2	6.0	OK	OK
		◢ 6.0 ◣	208	C LC_147	20.5	0.0	-11.2	9.9	0.1	5.1	5.1	OK	OK
Bottom flange plate SLN94 ring	SLN94 ring-w 1	◢ 6.0 ◣	100	C LC_147	24.5	0.0	2.5	2.5	13.9	6.1	6.1	OK	OK
		◢ 6.0 ◣	100	C LC_122	25.1	0.0	2.5	-3.3	-14.0	6.2	6.2	OK	OK
SLN94 ring-w 1	Bottom flange half plate L SLN94 ring	◢ 6.0 ◣	243	C LC_147	25.4	0.0	6.4	6.2	12.8	6.3	6.3	OK	OK
		◢ 6.0 ◣	243	C LC_147	27.5	0.0	8.0	2.7	-14.9	6.8	6.4	OK	OK
SLN94 ring-w 1	Bottom flange half plate R SLN94 ring	◢ 6.0 ◣	243	C LC_119	24.6	0.0	5.7	6.0	-12.4	6.1	5.8	OK	OK
		◢ 6.0 ◣	243	C LC_147	28.2	0.0	8.3	2.5	15.3	7.0	6.5	OK	OK
SLN94 ring-tfl 1	Haunch web SLN94 ring	◢ 6.0 ◣	572	C LC_122	65.5	0.0	-16.4	-17.0	32.4	16.2	9.3	OK	OK
		◢ 6.0 ◣	572	C LC_120	63.3	0.0	-14.5	15.4	-32.1	15.6	9.1	OK	OK
End plate SLN91 ring	SLN91 ring-w 1	◢ 6.0 ◣	208	C LC_147	21.0	0.0	-11.9	-10.0	-0.6	5.2	5.2	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_147	25.6	0.0	-10.3	13.2	3.2	6.3	6.1	OK	OK
Bottom flange plate SLN91 ring	SLN91 ring-w 1	◢ 6.0 ◣	100	C LC_147	23.7	0.0	2.2	1.9	13.5	5.9	5.9	OK	OK
		◢ 6.0 ◣	100	C LC_122	25.4	0.0	2.5	-3.2	-14.2	6.3	6.3	OK	OK
SLN91 ring-w 1	Bottom flange half plate L SLN91 ring	◢ 6.0 ◣	243	C LC_147	24.1	0.0	5.9	5.9	12.1	5.9	5.9	OK	OK
		◢ 6.0 ◣	243	C LC_147	28.2	0.0	8.6	2.4	-15.3	7.0	6.3	OK	OK
SLN91 ring-w 1	Bottom flange half plate R SLN91 ring	◢ 6.0 ◣	243	C LC_145	25.9	0.0	6.3	6.3	-13.1	6.4	5.9	OK	OK
		◢ 6.0 ◣	243	C LC_147	27.4	0.0	7.7	2.8	14.9	6.8	6.5	OK	OK
SLN91 ring-tfl 1	Haunch web SLN91 ring	◢ 6.0 ◣	572	C LC_122	63.5	0.0	-13.9	-15.3	32.3	15.7	9.2	OK	OK
		◢ 6.0 ◣	572	C LC_120	64.4	0.0	-16.9	16.9	-31.7	15.9	9.2	OK	OK
End plate SLN94 ring	Bottom flange plate SLN94 ring	◢ 6.0 ◣	149	C LC_126	15.7	0.0	-4.0	-5.3	7.0	3.9	3.9	OK	OK
		◢ 6.0 ◣	149	C LC_126	21.5	0.0	-8.1	6.9	-9.2	5.3	5.3	OK	OK
End plate SLN94 ring	Haunch web SLN94 ring	◢ 6.0 ◣	145	C LC_120	7.4	0.0	1.7	1.5	3.9	1.8	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	10.8	0.0	1.9	-2.4	-5.7	2.7	0.0	OK	OK
End plate SLN94 ring	Haunch top plate SLN94 ring	◢ 6.0 ◣	149	C LC_126	8.9	0.0	-4.4	-2.7	3.6	2.2	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_114	6.1	0.0	-1.3	3.0	-1.7	1.5	0.0	OK	OK
Haunch top plate SLN94 ring	Haunch web SLN94 ring	◢ 6.0 ◣	342	C LC_122	3.9	0.0	1.3	1.3	1.7	1.0	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_120	4.2	0.0	1.6	-1.4	-1.7	1.0	0.0	OK	OK
Haunch sloping plate SLN94 ring	Haunch web SLN94 ring	◢ 6.0 ◣	271	C LC_122	53.4	0.0	-10.0	-9.0	28.9	13.2	8.4	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_120	49.6	0.0	-8.3	7.0	-27.3	12.3	8.1	OK	OK
End plate SLN91 ring	Bottom flange plate SLN91 ring	◢ 6.0 ◣	148	C LC_147	15.5	0.0	7.6	4.3	6.5	3.8	3.8	OK	OK
		◢ 6.0 ◣	149	C LC_114	17.4	0.0	-8.9	7.9	-3.5	4.3	4.3	OK	OK
End plate SLN91 ring	Haunch web SLN91 ring	◢ 6.0 ◣	145	C LC_120	10.4	0.0	1.8	2.2	5.5	2.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	8.0	0.0	1.9	-1.7	-4.1	2.0	0.0	OK	OK
End plate SLN91 ring	Haunch top plate SLN91 ring	◢ 6.0 ◣	149	C LC_114	8.2	0.0	-3.8	-3.6	2.2	2.0	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_126	7.7	0.0	-2.9	3.2	-2.6	1.9	0.0	OK	OK
Haunch top plate SLN91 ring	Haunch web SLN91 ring	◢ 6.0 ◣	342	C LC_122	4.8	0.0	1.7	1.5	2.1	1.2	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	3.8	0.0	-0.2	0.2	2.2	0.9	0.0	OK	OK
Haunch sloping plate SLN91 ring	Haunch web SLN91 ring	◢ 6.0 ◣	271	C LC_135	49.5	0.0	-8.3	-7.1	27.3	12.2	8.1	OK	OK
		◢ 6.0 ◣	271	C LC_120	53.2	0.0	-10.0	10.1	-28.5	13.2	8.3	OK	OK
SLN92b+SLN92a radial (continuous)-w 1	SLN94 ring-w 1	◢ 7.2	208	C LC_147	78.6	0.0	32.6	37.6	-17.0	19.4	13.5	OK	OK
SLN92b+SLN92a radial (continuous)-w 1	SLN91 ring-w 1	◢ 7.2	208	C LC_147	80.9	0.0	32.0	-39.8	16.1	20.0	13.9	OK	OK
Haunch sloping plate SLN94 ring	Haunch top plate SLN94 ring	◢ 6.0 ◣	146	C LC_122	7.1	0.0	-5.2	2.8	0.3	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_120	2.8	0.0	-0.7	1.5	-0.6	0.7	0.0	OK	OK
Haunch sloping plate SLN91 ring	Haunch top plate SLN91 ring	◢ 6.0 ◣	146	C LC_122	7.3	0.0	-5.5	2.7	0.4	1.9	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_122	3.4	0.0	-0.7	1.9	0.2	0.8	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_120
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN95b+SLN95a radial (continuous) - SLN94 ring - SLN97 ring

Name	SLN95b+SLN95a radial (continuous) - SLN94 ring - SLN97 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

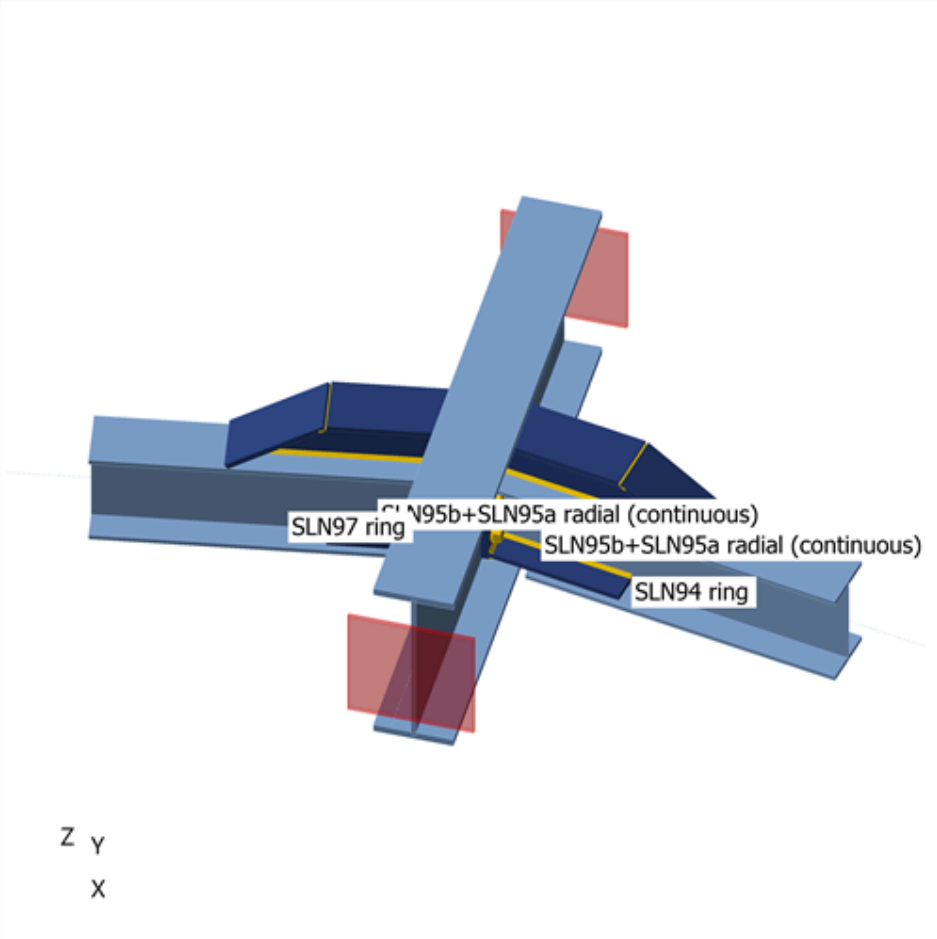
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN95b+SLN95a radial (continuous)	1 - UB457x191x98	120.0	0.0	0.0	0	0	0
SLN94 ring	2 - UB254x146x43	20.0	0.0	0.0	0	0	-102
SLN97 ring	2 - UB254x146x43	-140.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN95b+SLN95a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN95b+SLN95a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN94 ring / end		Position	0
SLN97 ring / end		Position	0



SLN95b+SLN95a radial (continuous) - SLN94 ring - SLN97 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN94 ring / End	4.8	0.0	-4.9	0.0	0.0	0.0
	SLN97 ring / End	4.8	0.0	-5.0	0.0	0.0	-0.1
C LC_102	SLN94 ring / End	19.2	0.1	-25.7	0.0	0.0	0.1
	SLN97 ring / End	19.1	-0.1	-26.0	0.0	0.0	-0.2
C LC_103	SLN94 ring / End	17.5	0.0	-24.7	0.0	0.0	0.0
	SLN97 ring / End	17.5	-0.1	-24.9	0.0	0.0	-0.1
C LC_104	SLN94 ring / End	14.2	0.2	-24.7	0.0	0.0	0.2
	SLN97 ring / End	14.1	-0.1	-24.9	0.0	0.0	-0.1
C LC_105	SLN94 ring / End	16.8	0.0	-24.7	0.0	0.0	0.1
	SLN97 ring / End	16.9	0.0	-24.9	0.0	0.0	0.0
C LC_106	SLN94 ring / End	14.9	0.1	-24.7	0.0	0.0	0.0
	SLN97 ring / End	14.7	-0.3	-24.9	0.0	0.0	-0.4
C LC_107	SLN94 ring / End	3.4	0.0	-1.6	0.0	0.0	0.0
	SLN97 ring / End	3.3	0.0	-1.6	0.0	0.0	0.0
C LC_108	SLN94 ring / End	-1.4	0.1	-1.6	0.0	0.0	0.2
	SLN97 ring / End	-1.2	0.1	-1.6	0.0	0.0	0.1
C LC_109	SLN94 ring / End	1.3	0.0	-1.6	0.0	0.0	0.1
	SLN97 ring / End	1.6	0.1	-1.6	0.0	0.0	0.2
C LC_110	SLN94 ring / End	0.7	-0.1	-1.6	0.0	0.0	-0.2
	SLN97 ring / End	0.3	-0.2	-1.6	0.0	0.0	-0.3
C LC_111	SLN94 ring / End	13.6	0.0	-17.4	0.0	0.0	0.0
	SLN97 ring / End	13.5	0.0	-17.6	0.0	0.0	-0.1
C LC_112	SLN94 ring / End	8.2	0.2	-17.4	0.0	0.0	0.3
	SLN97 ring / End	8.2	0.0	-17.6	0.0	0.0	0.0
C LC_113	SLN94 ring / End	11.7	0.0	-17.4	0.0	0.0	0.1
	SLN97 ring / End	11.9	0.1	-17.6	0.0	0.0	0.1
C LC_114	SLN94 ring / End	10.0	0.0	-17.4	0.0	0.0	-0.1
	SLN97 ring / End	9.7	-0.3	-17.6	0.0	0.0	-0.4
C LC_115	SLN94 ring / End	4.7	0.0	-2.9	0.0	0.0	0.0
	SLN97 ring / End	4.5	0.0	-2.9	0.0	0.0	0.0
C LC_116	SLN94 ring / End	-0.2	0.1	-2.9	0.0	0.0	0.2
	SLN97 ring / End	0.0	0.1	-2.9	0.0	0.0	0.1
C LC_117	SLN94 ring / End	2.6	0.1	-2.9	0.0	0.0	0.1
	SLN97 ring / End	2.9	0.1	-2.9	0.0	0.0	0.2
C LC_118	SLN94 ring / End	1.9	-0.1	-2.9	0.0	0.0	-0.2
	SLN97 ring / End	1.5	-0.2	-2.9	0.0	0.0	-0.3
C LC_119	SLN94 ring / End	18.7	0.0	-25.7	0.0	0.0	0.0
	SLN97 ring / End	18.7	-0.1	-26.0	0.0	0.0	-0.2
C LC_120	SLN94 ring / End	15.9	0.2	-25.7	0.0	0.0	0.2
	SLN97 ring / End	15.7	-0.1	-26.0	0.0	0.0	-0.1
C LC_121	SLN94 ring / End	17.9	0.1	-25.7	0.0	0.0	0.1
	SLN97 ring / End	17.9	0.0	-26.0	0.0	0.0	0.0
C LC_122	SLN94 ring / End	16.6	0.0	-25.7	0.0	0.0	0.0
	SLN97 ring / End	16.4	-0.3	-26.0	0.0	0.0	-0.4
C LC_123	SLN94 ring / End	15.9	0.0	-19.5	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN97 ring / End	15.9	-0.1	-19.7	0.0	0.0	-0.2
C LC_124	SLN94 ring / End	11.5	0.2	-19.5	0.0	0.0	0.3
	SLN97 ring / End	11.5	0.0	-19.7	0.0	0.0	0.0
C LC_125	SLN94 ring / End	13.8	0.1	-19.5	0.0	0.0	0.2
	SLN97 ring / End	14.0	0.1	-19.7	0.0	0.0	0.1
C LC_126	SLN94 ring / End	13.4	-0.1	-19.5	0.0	0.0	-0.1
	SLN97 ring / End	13.1	-0.3	-19.7	0.0	0.0	-0.4
C LC_127	SLN94 ring / End	7.0	0.0	-4.9	0.0	0.0	0.0
	SLN97 ring / End	6.9	-0.1	-5.0	0.0	0.0	-0.1
C LC_128	SLN94 ring / End	3.2	0.2	-4.9	0.0	0.0	0.2
	SLN97 ring / End	3.3	0.1	-5.0	0.0	0.0	0.1
C LC_129	SLN94 ring / End	4.7	0.1	-4.9	0.0	0.0	0.2
	SLN97 ring / End	4.9	0.1	-5.0	0.0	0.0	0.1
C LC_130	SLN94 ring / End	5.3	-0.1	-4.9	0.0	0.0	-0.2
	SLN97 ring / End	4.9	-0.2	-5.0	0.0	0.0	-0.3
C LC_135	SLN94 ring / End	15.9	0.1	-25.7	0.0	0.0	0.1
	SLN97 ring / End	15.9	-0.1	-26.0	0.0	0.0	-0.2
C LC_141	SLN94 ring / End	4.9	0.0	-4.9	0.0	0.0	0.0
	SLN97 ring / End	4.9	0.0	-5.0	0.0	0.0	-0.1
C LC_142	SLN94 ring / End	4.6	0.0	-4.9	0.0	0.0	0.1
	SLN97 ring / End	4.6	0.0	-5.0	0.0	0.0	-0.1
C LC_143	SLN94 ring / End	4.9	0.0	-4.9	0.0	0.0	0.0
	SLN97 ring / End	4.9	0.0	-5.0	0.0	0.0	0.0
C LC_144	SLN94 ring / End	4.6	0.0	-4.9	0.0	0.0	0.0
	SLN97 ring / End	4.6	-0.1	-5.0	0.0	0.0	-0.1
C LC_145	SLN94 ring / End	19.7	0.0	-25.7	0.0	0.0	0.1
	SLN97 ring / End	19.8	-0.1	-26.0	0.0	0.0	-0.2
C LC_146	SLN94 ring / End	18.7	0.1	-25.7	0.0	0.0	0.2
	SLN97 ring / End	18.5	-0.2	-26.0	0.0	0.0	-0.2
C LC_147	SLN94 ring / End	20.1	0.0	-25.7	0.0	0.0	0.1
	SLN97 ring / End	20.0	0.0	-26.0	0.0	0.0	-0.1
C LC_148	SLN94 ring / End	18.3	0.1	-25.7	0.0	0.0	0.1
	SLN97 ring / End	18.3	-0.2	-26.0	0.0	0.0	-0.3
C LC_151	SLN94 ring / End	12.2	0.0	-11.7	0.0	0.0	0.1
	SLN97 ring / End	12.2	-0.1	-11.6	0.0	0.0	-0.1
C LC_152	SLN94 ring / End	19.1	0.1	-14.1	0.0	0.0	0.1
	SLN97 ring / End	19.0	-0.1	-14.0	0.0	0.0	-0.2
C LC_153	SLN94 ring / End	19.2	0.1	-20.8	0.0	0.0	0.1
	SLN97 ring / End	19.1	-0.1	-20.6	0.0	0.0	-0.2
C LC_154	SLN94 ring / End	12.3	0.0	-11.7	0.0	0.0	0.1
	SLN97 ring / End	12.3	-0.1	-11.6	0.0	0.0	-0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.9 < 100%	OK
Welds	19.7 < 100%	OK
Buckling	Not calculated	

Plates

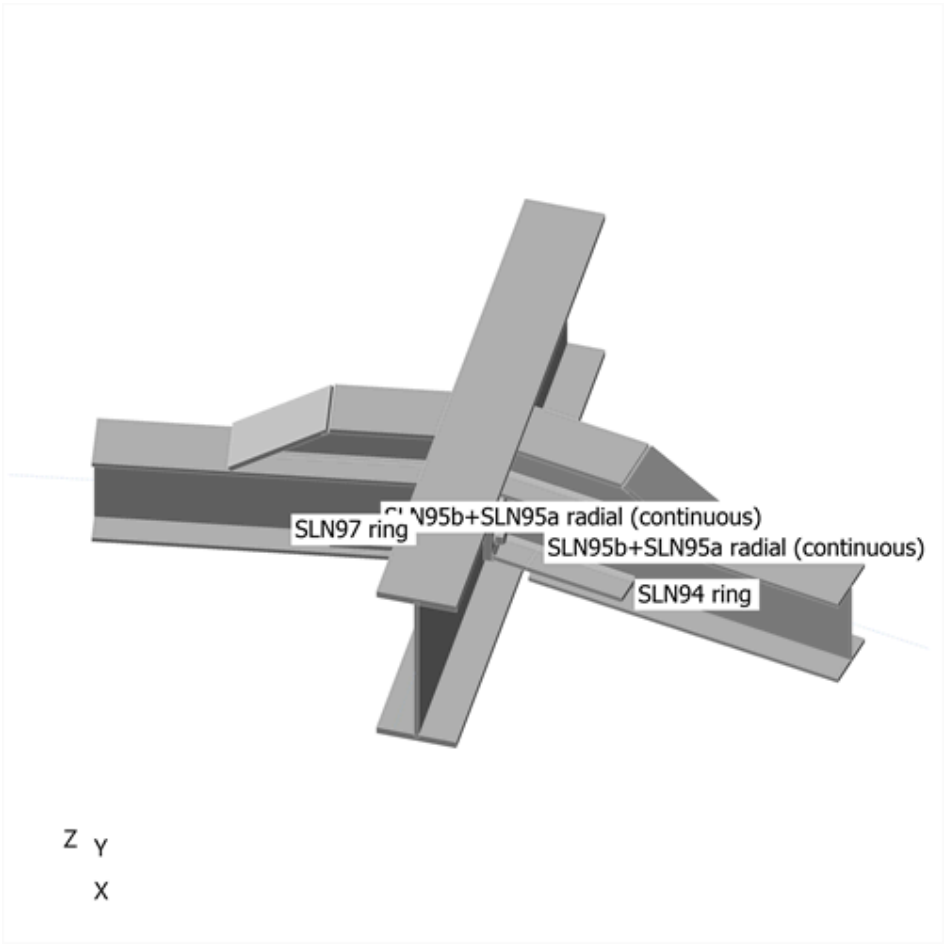
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN95b+SLN95a radial (continuous)-bfl 1	19.6	C LC_145	20.4	0.0	0.0	OK
SLN95b+SLN95a radial (continuous)-tfl 1	19.6	C LC_120	17.2	0.0	0.0	OK
SLN95b+SLN95a radial (continuous)-w 1	11.4	C LC_122	26.2	0.0	0.7	OK
SLN94 ring-bfl 1	12.7	C LC_147	50.7	0.0	0.0	OK
SLN94 ring-tfl 1	12.7	C LC_120	44.2	0.0	0.0	OK
SLN94 ring-w 1	7.2	C LC_147	49.3	0.0	0.0	OK
SLN97 ring-bfl 1	12.7	C LC_145	51.1	0.0	0.0	OK
SLN97 ring-tfl 1	12.7	C LC_122	45.0	0.0	0.0	OK
SLN97 ring-w 1	7.2	C LC_147	49.6	0.0	0.0	OK
End plate SLN97 ring	20.0	C LC_148	9.4	0.0	1.3	OK
Bottom flange plate SLN97 ring	15.0	C LC_147	20.8	0.0	0.0	OK
Bottom flange half plate L SLN97 ring	15.0	C LC_145	20.3	0.0	0.0	OK
Bottom flange half plate R SLN97 ring	15.0	C LC_147	20.1	0.0	0.0	OK
Haunch web SLN97 ring	12.0	C LC_135	28.6	0.0	0.0	OK
Haunch top plate SLN97 ring	15.0	C LC_122	5.9	0.0	0.0	OK
Haunch sloping plate SLN97 ring	15.0	C LC_120	12.8	0.0	0.0	OK
End plate SLN94 ring	20.0	C LC_119	8.3	0.0	1.9	OK
Bottom flange plate SLN94 ring	15.0	C LC_147	20.6	0.0	0.0	OK
Bottom flange half plate L SLN94 ring	15.0	C LC_147	20.0	0.0	0.0	OK
Bottom flange half plate R SLN94 ring	15.0	C LC_147	20.2	0.0	0.0	OK
Haunch web SLN94 ring	12.0	C LC_135	28.8	0.0	0.0	OK
Haunch top plate SLN94 ring	15.0	C LC_120	5.5	0.0	0.0	OK
Haunch sloping plate SLN94 ring	15.0	C LC_135	12.7	0.0	0.0	OK

Design data

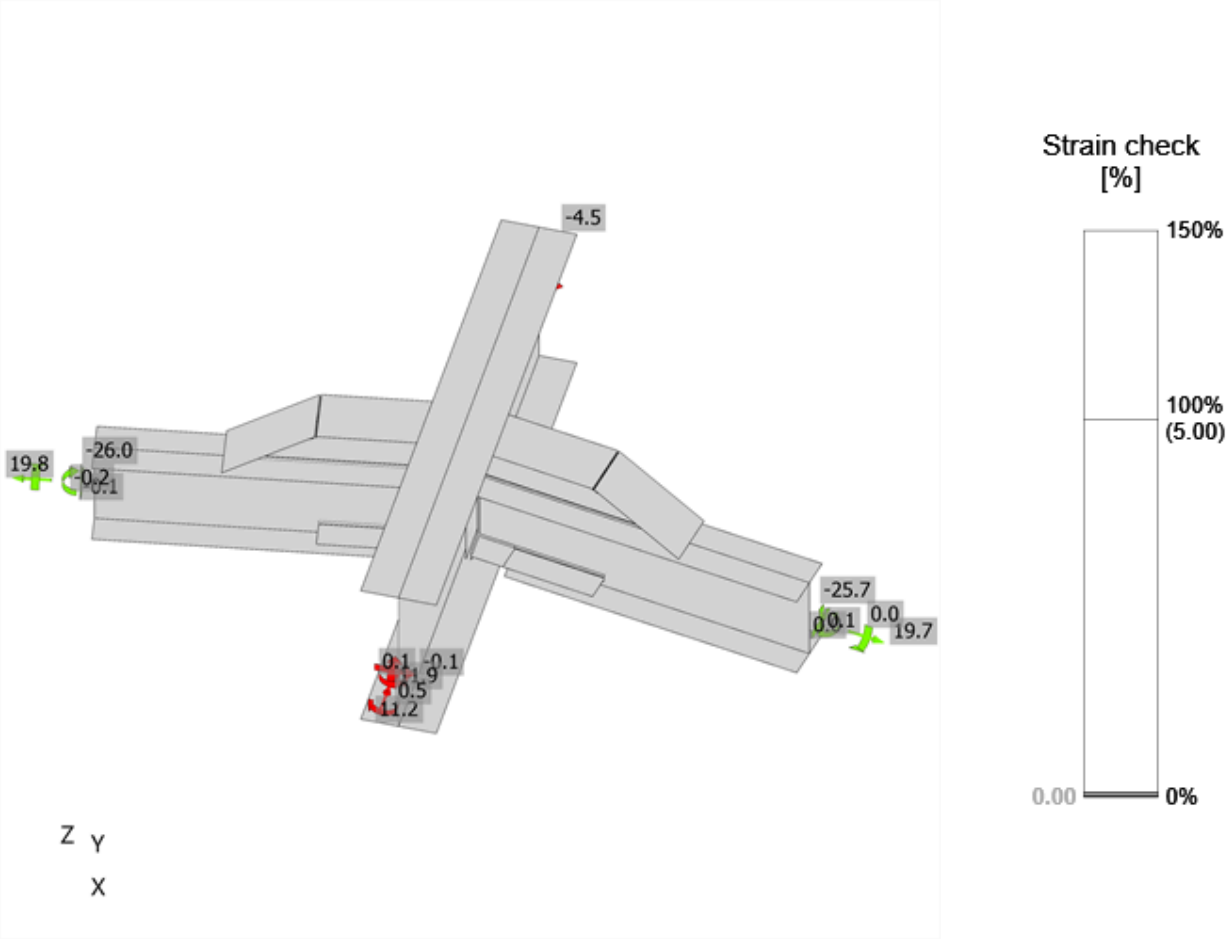
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

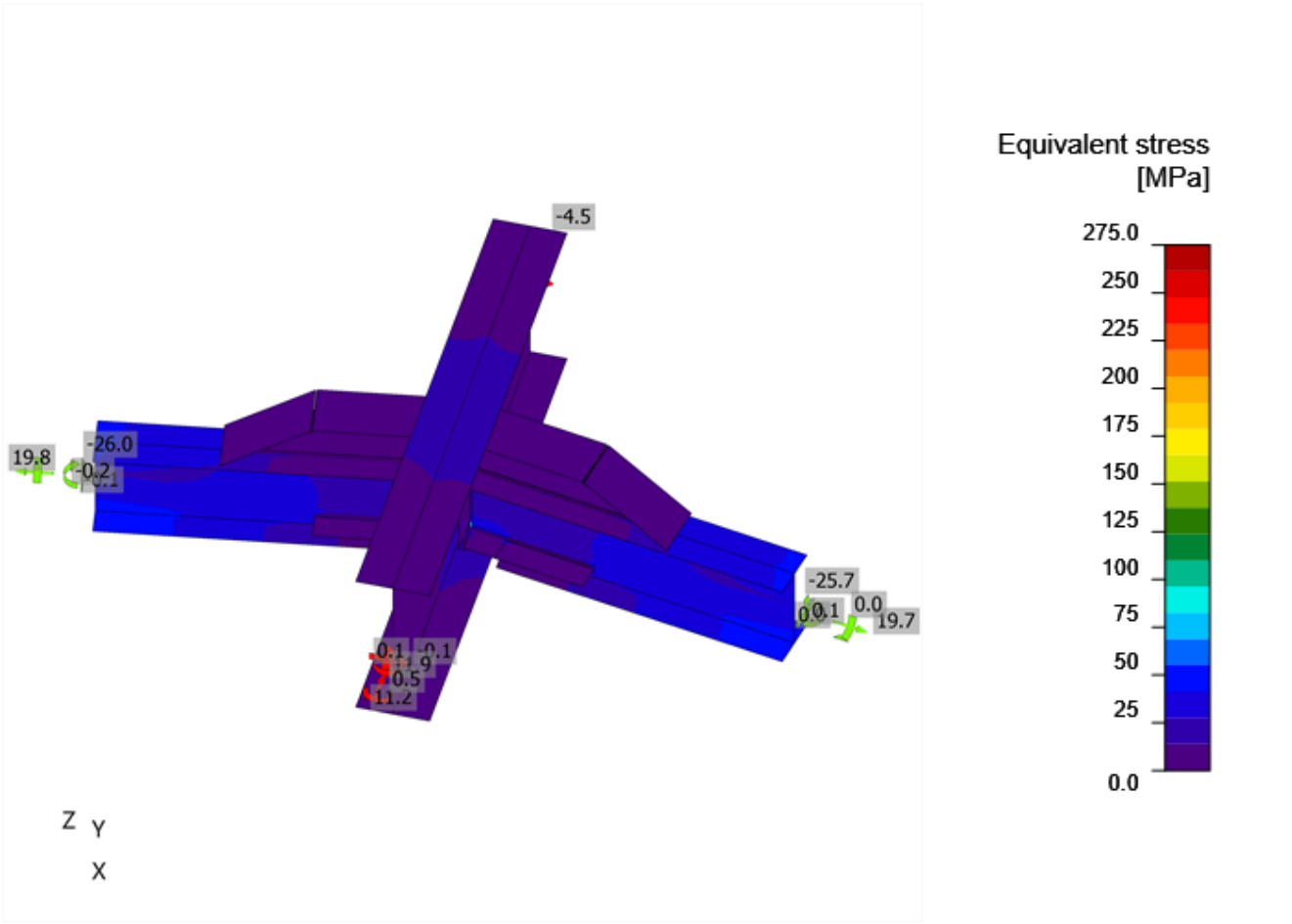
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

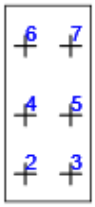


Strain check, C LC_145



Equivalent stress, C LC_145

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_148	2.6	0.5	194.1	0.8	0.5	0.8	OK	OK
	B3	M30 8.8 - 1	C LC_119	2.3	0.6	194.1	0.7	0.6	0.8	OK	OK
	B4	M30 8.8 - 1	C LC_135	0.1	0.4	232.2	0.0	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_147	0.1	0.5	232.2	0.0	0.4	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_122	2.2	0.8	249.2	0.7	0.7	0.9	OK	OK
	B7	M30 8.8 - 1	C LC_121	1.5	0.9	249.2	0.5	0.7	0.8	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN97 ring	Bottom flange plate SLN97 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN97 ring	Bottom flange plate SLN97 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN94 ring	Bottom flange plate SLN94 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN94 ring	Bottom flange plate SLN94 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN95b+SLN95a radial (continuous)-w 1	SLN94 ring-tfl 1	◢ 11.4	149	C LC_145	12.1	0.0	-7.7	-5.4	0.4	3.0	3.0	OK	OK
SLN95b+SLN95a radial (continuous)-w 1	SLN97 ring-tfl 1	◢ 11.4	149	C LC_121	12.1	0.0	-7.9	-5.3	-0.5	3.0	3.0	OK	OK
End plate SLN97 ring	SLN97 ring-w 1	◢ 6.0 ◣	208	C LC_145	25.8	0.0	-11.4	-12.8	-3.9	6.4	6.1	OK	OK
		◢ 6.0 ◣	208	C LC_145	20.8	0.0	-11.0	10.2	-0.2	5.1	5.1	OK	OK
Bottom flange plate SLN97 ring	SLN97 ring-w 1	◢ 6.0 ◣	100	C LC_147	24.1	0.0	2.2	1.9	13.7	5.9	5.9	OK	OK
		◢ 6.0 ◣	100	C LC_122	25.7	0.0	2.7	-3.8	-14.3	6.4	6.4	OK	OK
SLN97 ring-w 1	Bottom flange half plate L SLN97 ring	◢ 6.0 ◣	243	C LC_147	25.0	0.0	6.3	6.1	12.5	6.2	6.2	OK	OK
		◢ 6.0 ◣	243	C LC_145	28.1	0.0	8.9	2.2	-15.2	6.9	6.2	OK	OK
SLN97 ring-w 1	Bottom flange half plate R SLN97 ring	◢ 6.0 ◣	243	C LC_145	25.5	0.0	6.0	6.2	-12.9	6.3	5.8	OK	OK
		◢ 6.0 ◣	243	C LC_147	28.1	0.0	7.8	2.8	15.3	6.9	6.6	OK	OK
SLN97 ring-tfl 1	Haunch web SLN97 ring	◢ 6.0 ◣	572	C LC_122	67.0	0.0	-15.4	-17.1	33.5	16.6	9.5	OK	OK
		◢ 6.0 ◣	572	C LC_120	63.0	0.0	-16.0	15.6	-31.5	15.6	9.1	OK	OK
End plate SLN94 ring	SLN94 ring-w 1	◢ 6.0 ◣	208	C LC_147	20.5	0.0	-11.6	-9.8	-0.4	5.1	5.1	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_147	25.7	0.0	-10.4	13.2	3.3	6.4	6.0	OK	OK
Bottom flange plate SLN94 ring	SLN94 ring-w 1	◢ 6.0 ◣	100	C LC_147	25.1	0.0	2.6	2.6	14.2	6.2	6.2	OK	OK
		◢ 6.0 ◣	100	C LC_122	24.6	0.0	2.3	-2.9	-13.8	6.1	6.1	OK	OK
SLN94 ring-w 1	Bottom flange half plate L SLN94 ring	◢ 6.0 ◣	243	C LC_147	25.7	0.0	6.5	6.3	12.9	6.4	6.4	OK	OK
		◢ 6.0 ◣	243	C LC_145	27.3	0.0	8.2	2.5	-14.8	6.8	6.2	OK	OK
SLN94 ring-w 1	Bottom flange half plate R SLN94 ring	◢ 6.0 ◣	243	C LC_145	24.3	0.0	5.7	6.0	-12.3	6.0	5.8	OK	OK
		◢ 6.0 ◣	243	C LC_147	28.5	0.0	8.3	2.5	15.5	7.0	6.6	OK	OK
SLN94 ring-tfl 1	Haunch web SLN94 ring	◢ 6.0 ◣	572	C LC_122	65.6	0.0	-16.6	-17.2	32.4	16.2	9.3	OK	OK
		◢ 6.0 ◣	572	C LC_120	62.8	0.0	-14.4	15.1	-31.9	15.5	9.1	OK	OK
End plate SLN97 ring	Bottom flange plate SLN97 ring	◢ 6.0 ◣	149	C LC_126	16.7	0.0	-4.5	-5.5	7.5	4.1	4.1	OK	OK
		◢ 6.0 ◣	149	C LC_126	22.0	0.0	-8.2	7.3	-9.3	5.4	5.4	OK	OK
End plate SLN97 ring	Haunch web SLN97 ring	◢ 6.0 ◣	145	C LC_120	7.1	0.0	1.3	1.2	3.9	1.8	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	10.7	0.0	2.4	-2.4	-5.5	2.7	0.0	OK	OK
End plate SLN97 ring	Haunch top plate SLN97 ring	◢ 6.0 ◣	149	C LC_126	8.3	0.0	-3.7	-3.1	2.9	2.0	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_126	7.5	0.0	-2.5	3.1	-2.6	1.8	0.0	OK	OK
Haunch top plate SLN97 ring	Haunch web SLN97 ring	◢ 6.0 ◣	342	C LC_122	4.7	0.0	1.7	1.5	2.0	1.2	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	4.2	0.0	-0.1	0.2	2.4	1.0	0.0	OK	OK
Haunch sloping plate SLN97 ring	Haunch web SLN97 ring	◢ 6.0 ◣	271	C LC_135	53.3	0.0	-9.7	-8.1	29.2	13.2	8.4	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_120	50.9	0.0	-8.9	9.4	-27.3	12.6	8.1	OK	OK
End plate SLN94 ring	Bottom flange plate SLN94 ring	◢ 6.0 ◣	148	C LC_145	14.8	0.0	7.6	4.4	5.9	3.7	3.7	OK	OK
		◢ 6.0 ◣	149	C LC_126	16.7	0.0	-8.4	7.5	-3.7	4.1	4.1	OK	OK
End plate SLN94 ring	Haunch web SLN94 ring	◢ 6.0 ◣	145	C LC_120	10.2	0.0	2.3	2.4	5.3	2.5	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	7.9	0.0	1.4	-1.3	-4.3	1.9	0.0	OK	OK
End plate SLN94 ring	Haunch top plate SLN94 ring	◢ 6.0 ◣	149	C LC_130	6.3	0.0	-3.3	-2.6	1.7	1.6	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_124	5.5	0.0	-2.1	2.1	2.0	1.4	0.0	OK	OK
Haunch top plate SLN94 ring	Haunch web SLN94 ring	◢ 6.0 ◣	342	C LC_122	3.6	0.0	1.3	1.3	1.5	0.9	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_120	4.1	0.0	1.5	-1.4	-1.7	1.0	0.0	OK	OK
Haunch sloping plate SLN94 ring	Haunch web SLN94 ring	◢ 6.0 ◣	271	C LC_135	53.9	0.0	-10.2	-10.2	28.8	13.3	8.3	OK	OK
		◢ 6.0 ◣	271	C LC_120	49.1	0.0	-8.1	7.2	-27.0	12.1	8.0	OK	OK
SLN95b+SLN95a radial (continuous)-w 1	SLN97 ring-w 1	◢ 7.2	208	C LC_147	79.0	0.0	33.2	37.6	-17.3	19.5	13.6	OK	OK
SLN95b+SLN95a radial (continuous)-w 1	SLN94 ring-w 1	◢ 7.2	208	C LC_145	79.6	0.0	31.8	-38.4	17.4	19.7	13.6	OK	OK
Haunch sloping plate SLN97 ring	Haunch top plate SLN97 ring	◢ 6.0 ◣	146	C LC_122	7.2	0.0	-5.6	2.7	0.3	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_122	3.2	0.0	-0.7	1.7	0.4	0.8	0.0	OK	OK
Haunch sloping plate SLN94 ring	Haunch top plate SLN94 ring	◢ 6.0 ◣	146	C LC_122	7.0	0.0	-5.1	2.8	0.3	1.9	0.0	OK	OK
		◢ 6.0 ◣	146	C LC_120	2.7	0.0	-0.6	1.4	-0.5	0.7	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_140
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN108b+SLN108a radial (continuous) - SLN97 ring - SLN109 ring

Name	SLN108b+SLN108a radial (continuous) - SLN97 ring - SLN109 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

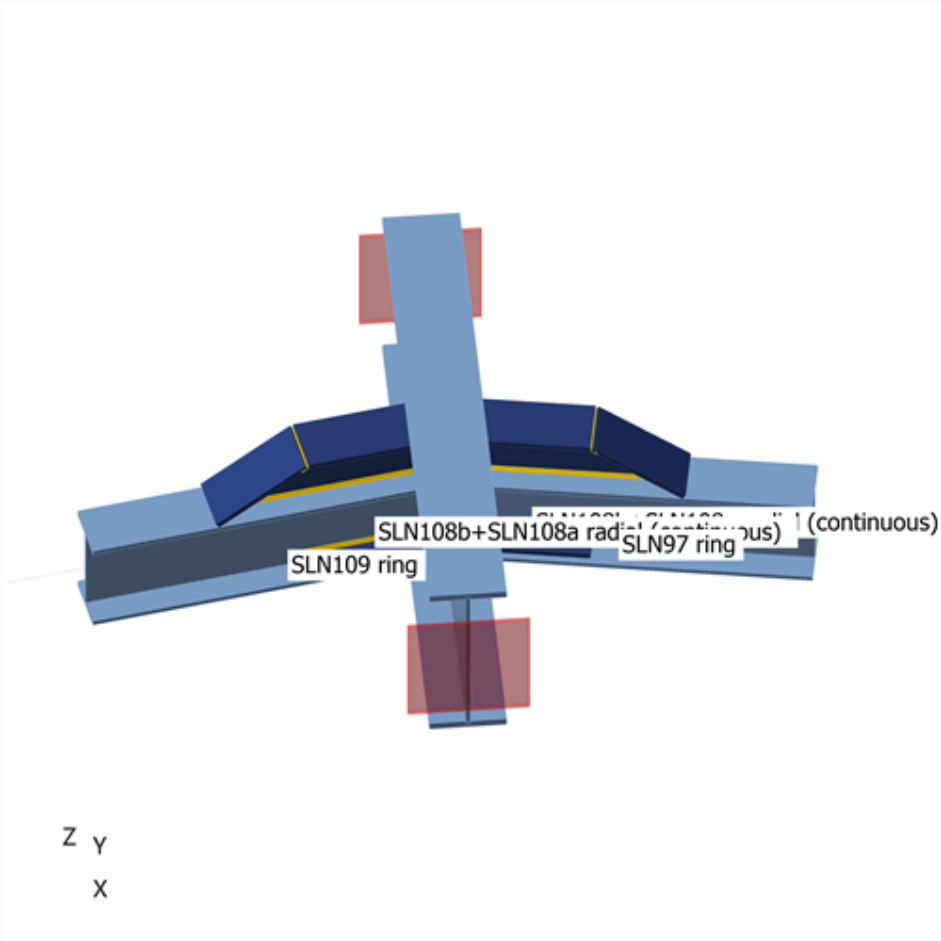
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN108b+SLN108a radial (continuous)	1 - UB457x191x98	140.0	0.0	0.0	0	0	0
SLN97 ring	2 - UB254x146x43	40.0	0.0	0.0	0	0	-102
SLN109 ring	2 - UB254x146x43	-120.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN108b+SLN108a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN108b+SLN108a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN97 ring / end		Position	0
SLN109 ring / end		Position	0



SLN108b+SLN108a radial (continuous) - SLN97 ring - SLN109 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN97 ring / End	4.8	0.0	-5.0	0.0	0.0	-0.1
	SLN109 ring / End	4.7	0.0	-4.9	0.0	0.0	0.0
C LC_102	SLN97 ring / End	19.1	-0.1	-26.0	0.0	0.0	-0.2
	SLN109 ring / End	19.1	0.1	-25.6	0.0	0.0	0.1
C LC_103	SLN97 ring / End	17.7	-0.2	-25.0	0.0	0.0	-0.2
	SLN109 ring / End	17.9	0.1	-24.5	0.0	0.0	0.1
C LC_104	SLN97 ring / End	13.9	0.0	-25.0	0.0	0.0	0.0
	SLN109 ring / End	13.8	0.1	-24.5	0.0	0.0	0.2
C LC_105	SLN97 ring / End	16.7	-0.1	-25.0	0.0	0.0	-0.1
	SLN109 ring / End	16.5	0.2	-24.5	0.0	0.0	0.2
C LC_106	SLN97 ring / End	14.8	-0.2	-25.0	0.0	0.0	-0.3
	SLN109 ring / End	14.7	-0.1	-24.5	0.0	0.0	-0.1
C LC_107	SLN97 ring / End	3.3	0.0	-1.6	0.0	0.0	0.1
	SLN109 ring / End	3.6	0.0	-1.6	0.0	0.0	0.0
C LC_108	SLN97 ring / End	-1.3	0.1	-1.6	0.0	0.0	0.1
	SLN109 ring / End	-1.1	0.1	-1.6	0.0	0.0	0.2
C LC_109	SLN97 ring / End	1.6	0.1	-1.6	0.0	0.0	0.1
	SLN109 ring / End	1.5	0.1	-1.6	0.0	0.0	0.1
C LC_110	SLN97 ring / End	0.4	-0.2	-1.6	0.0	0.0	-0.3
	SLN109 ring / End	0.0	-0.2	-1.6	0.0	0.0	-0.2
C LC_111	SLN97 ring / End	13.6	-0.1	-17.6	0.0	0.0	-0.1
	SLN109 ring / End	13.9	0.1	-17.3	0.0	0.0	0.1
C LC_112	SLN97 ring / End	8.1	0.1	-17.6	0.0	0.0	0.1
	SLN109 ring / End	8.1	0.1	-17.3	0.0	0.0	0.2
C LC_113	SLN97 ring / End	11.9	0.0	-17.6	0.0	0.0	0.0
	SLN109 ring / End	11.6	0.2	-17.3	0.0	0.0	0.2
C LC_114	SLN97 ring / End	9.8	-0.2	-17.6	0.0	0.0	-0.4
	SLN109 ring / End	9.4	-0.2	-17.3	0.0	0.0	-0.2
C LC_115	SLN97 ring / End	4.6	0.0	-2.9	0.0	0.0	0.1
	SLN109 ring / End	4.9	0.0	-2.8	0.0	0.0	0.0
C LC_116	SLN97 ring / End	-0.1	0.1	-2.9	0.0	0.0	0.1
	SLN109 ring / End	0.0	0.1	-2.8	0.0	0.0	0.2
C LC_117	SLN97 ring / End	2.9	0.1	-2.9	0.0	0.0	0.1
	SLN109 ring / End	2.7	0.1	-2.8	0.0	0.0	0.1
C LC_118	SLN97 ring / End	1.6	-0.2	-2.9	0.0	0.0	-0.3
	SLN109 ring / End	1.1	-0.2	-2.8	0.0	0.0	-0.2
C LC_119	SLN97 ring / End	18.9	-0.2	-26.0	0.0	0.0	-0.3
	SLN109 ring / End	18.8	0.1	-25.6	0.0	0.0	0.1
C LC_120	SLN97 ring / End	15.6	0.0	-26.0	0.0	0.0	0.0
	SLN109 ring / End	15.5	0.1	-25.6	0.0	0.0	0.2
C LC_121	SLN97 ring / End	17.8	-0.1	-26.0	0.0	0.0	-0.1
	SLN109 ring / End	17.9	0.2	-25.6	0.0	0.0	0.3
C LC_122	SLN97 ring / End	16.6	-0.2	-26.0	0.0	0.0	-0.3
	SLN109 ring / End	16.3	-0.1	-25.6	0.0	0.0	-0.1
C LC_123	SLN97 ring / End	16.0	-0.2	-19.7	0.0	0.0	-0.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN109 ring / End	15.7	0.1	-19.4	0.0	0.0	0.1
C LC_124	SLN97 ring / End	11.4	0.1	-19.7	0.0	0.0	0.1
	SLN109 ring / End	11.5	0.1	-19.4	0.0	0.0	0.3
C LC_125	SLN97 ring / End	13.9	0.0	-19.7	0.0	0.0	0.1
	SLN109 ring / End	14.3	0.3	-19.4	0.0	0.0	0.4
C LC_126	SLN97 ring / End	13.2	-0.2	-19.7	0.0	0.0	-0.4
	SLN109 ring / End	12.7	-0.2	-19.4	0.0	0.0	-0.2
C LC_127	SLN97 ring / End	6.9	-0.1	-5.0	0.0	0.0	-0.1
	SLN109 ring / End	6.7	0.0	-4.9	0.0	0.0	0.0
C LC_128	SLN97 ring / End	3.2	0.1	-5.0	0.0	0.0	0.1
	SLN109 ring / End	3.4	0.1	-4.9	0.0	0.0	0.2
C LC_129	SLN97 ring / End	4.9	0.1	-5.0	0.0	0.0	0.1
	SLN109 ring / End	5.4	0.2	-4.9	0.0	0.0	0.3
C LC_130	SLN97 ring / End	5.0	-0.2	-5.0	0.0	0.0	-0.3
	SLN109 ring / End	4.5	-0.2	-4.9	0.0	0.0	-0.2
C LC_135	SLN97 ring / End	15.9	-0.1	-26.0	0.0	0.0	-0.2
	SLN109 ring / End	15.8	0.1	-25.6	0.0	0.0	0.2
C LC_141	SLN97 ring / End	4.9	-0.1	-5.0	0.0	0.0	-0.1
	SLN109 ring / End	4.9	0.0	-4.9	0.0	0.0	0.0
C LC_142	SLN97 ring / End	4.6	0.0	-5.0	0.0	0.0	0.0
	SLN109 ring / End	4.5	0.0	-4.9	0.0	0.0	0.0
C LC_143	SLN97 ring / End	4.9	0.0	-5.0	0.0	0.0	-0.1
	SLN109 ring / End	4.9	0.0	-4.9	0.0	0.0	0.1
C LC_144	SLN97 ring / End	4.6	0.0	-5.0	0.0	0.0	-0.1
	SLN109 ring / End	4.6	0.0	-4.9	0.0	0.0	0.0
C LC_145	SLN97 ring / End	19.9	-0.2	-26.0	0.0	0.0	-0.3
	SLN109 ring / End	20.0	0.2	-25.6	0.0	0.0	0.2
C LC_146	SLN97 ring / End	18.4	-0.1	-26.0	0.0	0.0	-0.1
	SLN109 ring / End	18.2	0.0	-25.6	0.0	0.0	0.1
C LC_147	SLN97 ring / End	19.8	-0.2	-26.0	0.0	0.0	-0.2
	SLN109 ring / End	19.7	0.2	-25.6	0.0	0.0	0.2
C LC_148	SLN97 ring / End	18.5	-0.1	-26.0	0.0	0.0	-0.2
	SLN109 ring / End	18.5	0.0	-25.6	0.0	0.0	0.1
C LC_151	SLN97 ring / End	12.2	-0.1	-11.7	0.0	0.0	-0.1
	SLN109 ring / End	12.1	0.1	-11.5	0.0	0.0	0.1
C LC_152	SLN97 ring / End	19.0	-0.1	-14.0	0.0	0.0	-0.2
	SLN109 ring / End	19.0	0.1	-14.0	0.0	0.0	0.1
C LC_153	SLN97 ring / End	19.1	-0.1	-20.7	0.0	0.0	-0.2
	SLN109 ring / End	19.1	0.1	-20.6	0.0	0.0	0.1
C LC_154	SLN97 ring / End	12.3	-0.1	-11.7	0.0	0.0	-0.1
	SLN109 ring / End	12.2	0.1	-11.5	0.0	0.0	0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.9 < 100%	OK
Welds	20.0 < 100%	OK
Buckling	Not calculated	

Plates

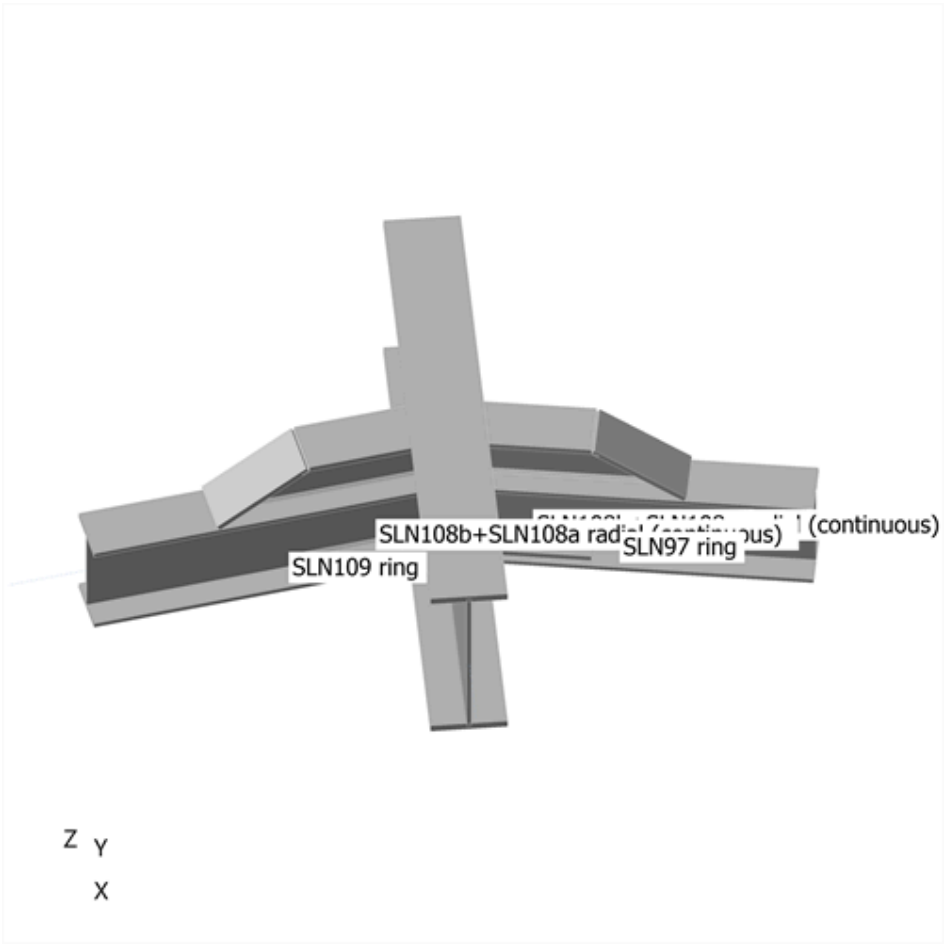
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN108b+SLN108a radial (continuous)-bfl 1	19.6	C LC_147	20.3	0.0	0.0	OK
SLN108b+SLN108a radial (continuous)-tfl 1	19.6	C LC_122	17.2	0.0	0.0	OK
SLN108b+SLN108a radial (continuous)-w 1	11.4	C LC_122	26.1	0.0	0.7	OK
SLN97 ring-bfl 1	12.7	C LC_148	51.0	0.0	0.0	OK
SLN97 ring-tfl 1	12.7	C LC_122	45.6	0.0	0.0	OK
SLN97 ring-w 1	7.2	C LC_145	49.7	0.0	0.0	OK
SLN109 ring-bfl 1	12.7	C LC_146	50.0	0.0	0.0	OK
SLN109 ring-tfl 1	12.7	C LC_120	44.4	0.0	0.0	OK
SLN109 ring-w 1	7.2	C LC_145	49.0	0.0	0.0	OK
End plate SLN109 ring	20.0	C LC_147	9.4	0.0	1.1	OK
Bottom flange plate SLN109 ring	15.0	C LC_145	20.5	0.0	0.0	OK
Bottom flange half plate L SLN109 ring	15.0	C LC_145	19.9	0.0	0.0	OK
Bottom flange half plate R SLN109 ring	15.0	C LC_145	20.1	0.0	0.0	OK
Haunch web SLN109 ring	12.0	C LC_122	28.1	0.0	0.0	OK
Haunch top plate SLN109 ring	15.0	C LC_121	5.5	0.0	0.0	OK
Haunch sloping plate SLN109 ring	15.0	C LC_120	12.6	0.0	0.0	OK
End plate SLN97 ring	20.0	C LC_145	10.0	0.0	1.3	OK
Bottom flange plate SLN97 ring	15.0	C LC_145	20.7	0.0	0.0	OK
Bottom flange half plate L SLN97 ring	15.0	C LC_145	20.4	0.0	0.0	OK
Bottom flange half plate R SLN97 ring	15.0	C LC_147	20.0	0.0	0.0	OK
Haunch web SLN97 ring	12.0	C LC_135	28.7	0.0	0.0	OK
Haunch top plate SLN97 ring	15.0	C LC_122	5.9	0.0	0.0	OK
Haunch sloping plate SLN97 ring	15.0	C LC_120	12.9	0.0	0.0	OK

Design data

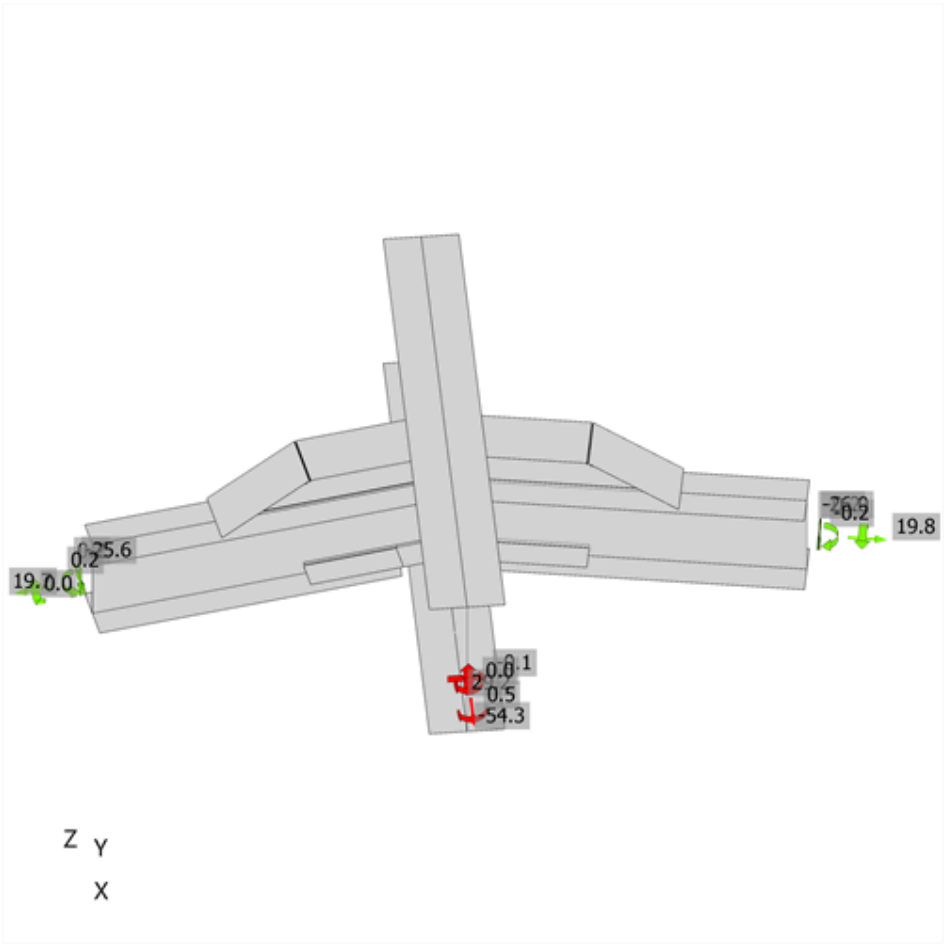
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

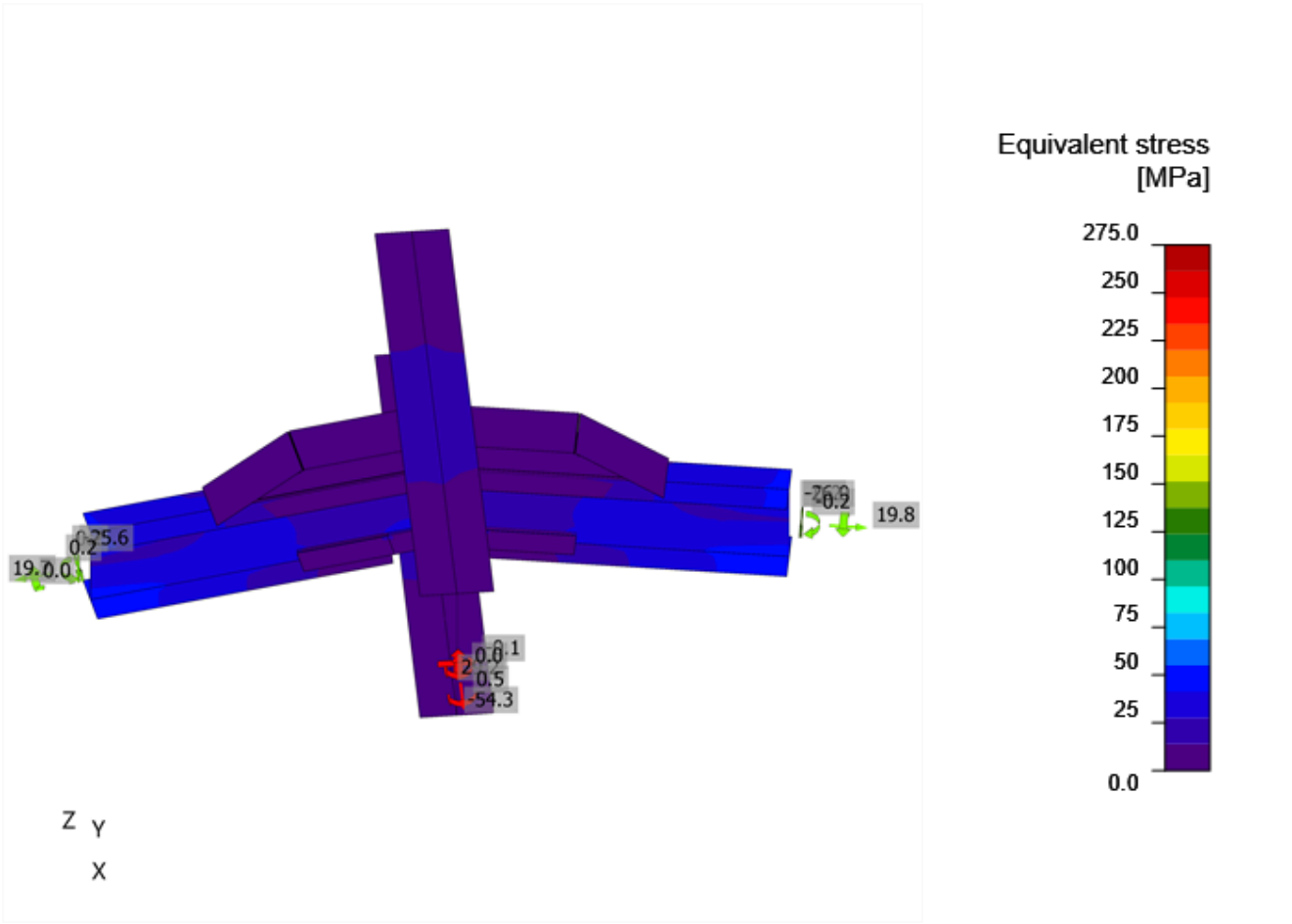
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

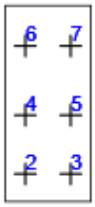


Strain check, C LC_147



Equivalent stress, C LC_147

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_122	2.1	0.5	194.1	0.7	0.5	0.7	OK	OK
	B3	M30 8.8 - 1	C LC_145	2.8	0.7	194.1	0.9	0.6	0.9	OK	OK
	B4	M30 8.8 - 1	C LC_135	0.1	0.4	232.2	0.0	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_145	0.1	0.5	232.2	0.0	0.4	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_122	1.9	0.8	249.2	0.6	0.7	0.8	OK	OK
	B7	M30 8.8 - 1	C LC_122	2.0	0.9	249.2	0.6	0.7	0.9	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN97 ring	Bottom flange plate SLN97 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN109 ring	Bottom flange plate SLN109 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN109 ring	Bottom flange plate SLN109 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN97 ring	Bottom flange plate SLN97 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN108b+SLN108a radial (continuous)-w 1	SLN97 ring-tfl 1	◢ 11.4	149	C LC_145	11.2	0.0	-7.0	-5.0	0.3	2.8	0.0	OK	OK
SLN108b+SLN108a radial (continuous)-w 1	SLN109 ring-tfl 1	◢ 11.4	149	C LC_121	11.2	0.0	-7.1	-5.0	-0.5	2.8	0.0	OK	OK
End plate SLN109 ring	SLN109 ring-w 1	◢ 6.0 ◣	208	C LC_145	25.1	0.0	-10.7	-12.8	-2.9	6.2	6.0	OK	OK
		◢ 6.0 ◣	208	C LC_145	20.3	0.0	-11.2	9.8	0.3	5.0	5.0	OK	OK
Bottom flange plate SLN109 ring	SLN109 ring-w 1	◢ 6.0 ◣	100	C LC_121	24.9	0.0	2.6	3.2	13.9	6.1	6.1	OK	OK
		◢ 6.0 ◣	100	C LC_122	24.8	0.0	2.4	-3.1	-13.9	6.1	6.1	OK	OK
SLN109 ring-w 1	Bottom flange half plate L SLN109 ring	◢ 6.0 ◣	243	C LC_147	25.5	0.0	6.2	6.2	12.8	6.3	6.3	OK	OK
		◢ 6.0 ◣	243	C LC_145	27.4	0.0	7.9	2.7	-14.9	6.8	6.4	OK	OK
SLN109 ring-w 1	Bottom flange half plate R SLN109 ring	◢ 6.0 ◣	243	C LC_145	24.1	0.0	5.9	5.9	-12.1	6.0	5.8	OK	OK
		◢ 6.0 ◣	243	C LC_145	28.0	0.0	8.3	2.4	15.2	6.9	6.5	OK	OK
SLN109 ring-tfl 1	Haunch web SLN109 ring	◢ 6.0 ◣	572	C LC_122	64.8	0.0	-16.0	-16.7	32.2	16.0	9.3	OK	OK
		◢ 6.0 ◣	572	C LC_121	63.3	0.0	-14.5	15.7	-31.9	15.7	9.1	OK	OK
End plate SLN97 ring	SLN97 ring-w 1	◢ 6.0 ◣	208	C LC_145	21.0	0.0	-12.0	-9.8	-1.1	5.2	5.2	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_145	25.8	0.0	-10.2	13.4	2.9	6.4	6.0	OK	OK
Bottom flange plate SLN97 ring	SLN97 ring-w 1	◢ 6.0 ◣	100	C LC_121	24.5	0.0	2.4	2.4	13.9	6.1	6.1	OK	OK
		◢ 6.0 ◣	100	C LC_122	25.9	0.0	2.6	-3.7	-14.4	6.4	6.4	OK	OK
SLN97 ring-w 1	Bottom flange half plate L SLN97 ring	◢ 6.0 ◣	243	C LC_147	24.7	0.0	6.3	6.0	12.4	6.1	6.1	OK	OK
		◢ 6.0 ◣	243	C LC_145	28.2	0.0	9.1	2.2	-15.2	7.0	6.2	OK	OK
SLN97 ring-w 1	Bottom flange half plate R SLN97 ring	◢ 6.0 ◣	243	C LC_145	25.8	0.0	6.0	6.3	-13.0	6.4	5.8	OK	OK
		◢ 6.0 ◣	243	C LC_147	28.1	0.0	7.6	2.9	15.3	6.9	6.6	OK	OK
SLN97 ring-tfl 1	Haunch web SLN97 ring	◢ 6.0 ◣	572	C LC_122	66.7	0.0	-15.3	-17.0	33.5	16.5	9.5	OK	OK
		◢ 6.0 ◣	572	C LC_120	63.5	0.0	-16.1	15.8	-31.8	15.7	9.1	OK	OK
End plate SLN109 ring	Bottom flange plate SLN109 ring	◢ 6.0 ◣	149	C LC_126	15.4	0.0	-3.7	-5.3	6.8	3.8	3.8	OK	OK
		◢ 6.0 ◣	149	C LC_126	21.7	0.0	-8.4	6.8	-9.3	5.4	5.4	OK	OK
End plate SLN109 ring	Haunch web SLN109 ring	◢ 6.0 ◣	145	C LC_121	7.4	0.0	1.7	1.6	3.9	1.8	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	10.7	0.0	2.0	-2.4	-5.6	2.6	0.0	OK	OK
End plate SLN109 ring	Haunch top plate SLN109 ring	◢ 6.0 ◣	149	C LC_126	9.1	0.0	-4.4	-2.9	3.6	2.2	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_126	6.6	0.0	-1.6	3.1	-2.0	1.6	0.0	OK	OK
Haunch top plate SLN109 ring	Haunch web SLN109 ring	◢ 6.0 ◣	342	C LC_122	3.9	0.0	1.4	1.3	1.7	1.0	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_121	4.4	0.0	1.6	-1.5	-1.8	1.1	0.0	OK	OK
Haunch sloping plate SLN109 ring	Haunch web SLN109 ring	◢ 6.0 ◣	271	C LC_122	52.4	0.0	-9.6	-8.6	28.5	13.0	8.3	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_120	49.9	0.0	-8.5	7.5	-27.4	12.3	8.1	OK	OK
End plate SLN97 ring	Bottom flange plate SLN97 ring	◢ 6.0 ◣	148	C LC_145	16.2	0.0	7.6	4.3	7.1	4.0	4.0	OK	OK
		◢ 6.0 ◣	149	C LC_126	20.1	0.0	-10.0	9.2	-4.0	5.0	5.0	OK	OK
End plate SLN97 ring	Haunch web SLN97 ring	◢ 6.0 ◣	145	C LC_121	10.4	0.0	1.9	2.3	5.4	2.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	8.0	0.0	1.7	-1.6	-4.2	2.0	0.0	OK	OK
End plate SLN97 ring	Haunch top plate SLN97 ring	◢ 6.0 ◣	149	C LC_126	7.9	0.0	-3.8	-3.3	2.2	1.9	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_126	6.8	0.0	-2.3	2.9	-2.3	1.7	0.0	OK	OK
Haunch top plate SLN97 ring	Haunch web SLN97 ring	◢ 6.0 ◣	342	C LC_122	4.7	0.0	1.7	1.5	2.0	1.2	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	4.0	0.0	-0.2	0.2	2.3	1.0	0.0	OK	OK
Haunch sloping plate SLN97 ring	Haunch web SLN97 ring	◢ 6.0 ◣	271	C LC_135	53.4	0.0	-9.7	-8.1	29.2	13.2	8.4	OK	OK
		◢ 6.0 ◣	271	C LC_120	51.3	0.0	-9.0	9.3	-27.6	12.7	8.2	OK	OK
SLN108b+SLN108a radial (continuous)-w 1	SLN109 ring-w 1	◢ 7.2	208	C LC_145	79.2	0.0	32.7	37.8	-17.5	19.6	13.6	OK	OK
SLN108b+SLN108a radial (continuous)-w 1	SLN97 ring-w 1	◢ 7.2	208	C LC_145	81.0	0.0	31.6	-39.7	16.6	20.0	13.9	OK	OK
Haunch sloping plate SLN109 ring	Haunch top plate SLN109 ring	◢ 6.0 ◣	146	C LC_122	7.0	0.0	-4.9	2.9	-0.3	1.9	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_121	2.9	0.0	-0.7	1.6	-0.4	0.7	0.0	OK	OK
Haunch sloping plate SLN97 ring	Haunch top plate SLN97 ring	◢ 6.0 ◣	146	C LC_122	7.3	0.0	-5.6	2.7	0.4	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_126	3.3	0.0	-0.7	1.9	0.0	0.8	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_160
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN110b+SLN110a radial (continuous) - SLN109 ring - SLN112 ring

Name	SLN110b+SLN110a radial (continuous) - SLN109 ring - SLN112 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

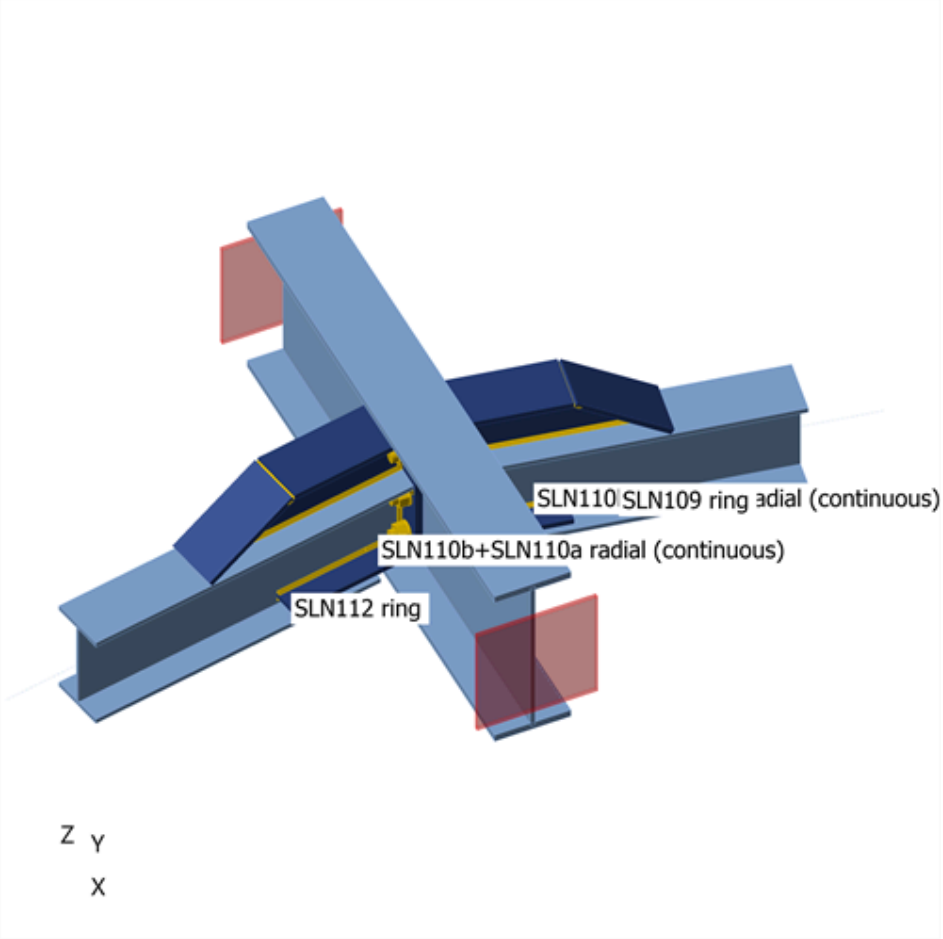
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN110b+SLN110a radial (continuous)	1 - UB457x191x98	160.0	0.0	0.0	0	0	0
SLN109 ring	2 - UB254x146x43	60.0	0.0	0.0	0	0	-102
SLN112 ring	2 - UB254x146x43	-100.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN110b+SLN110a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN110b+SLN110a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN109 ring / end		Position	0
SLN112 ring / end		Position	0



SLN110b+SLN110a radial (continuous) - SLN109 ring - SLN112 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN109 ring / End	4.7	0.0	-4.9	0.0	0.0	0.0
	SLN112 ring / End	4.7	0.0	-4.9	0.0	0.0	-0.1
C LC_102	SLN109 ring / End	19.1	0.1	-25.6	0.0	0.0	0.1
	SLN112 ring / End	19.1	-0.1	-25.5	0.0	0.0	-0.2
C LC_103	SLN109 ring / End	17.9	0.0	-24.6	0.0	0.0	0.0
	SLN112 ring / End	17.9	0.0	-24.5	0.0	0.0	-0.1
C LC_104	SLN109 ring / End	13.7	0.2	-24.6	0.0	0.0	0.3
	SLN112 ring / End	13.7	-0.1	-24.5	0.0	0.0	-0.1
C LC_105	SLN109 ring / End	16.4	0.1	-24.6	0.0	0.0	0.2
	SLN112 ring / End	16.4	0.0	-24.5	0.0	0.0	0.0
C LC_106	SLN109 ring / End	14.8	0.0	-24.6	0.0	0.0	0.0
	SLN112 ring / End	14.8	-0.2	-24.5	0.0	0.0	-0.3
C LC_107	SLN109 ring / End	3.6	0.0	-1.6	0.0	0.0	0.0
	SLN112 ring / End	3.5	0.0	-1.6	0.0	0.0	0.1
C LC_108	SLN109 ring / End	-1.2	0.1	-1.6	0.0	0.0	0.2
	SLN112 ring / End	-1.1	0.0	-1.6	0.0	0.0	0.1
C LC_109	SLN109 ring / End	1.4	0.1	-1.6	0.0	0.0	0.1
	SLN112 ring / End	1.9	0.1	-1.6	0.0	0.0	0.2
C LC_110	SLN109 ring / End	0.0	-0.2	-1.6	0.0	0.0	-0.2
	SLN112 ring / End	-0.4	-0.2	-1.6	0.0	0.0	-0.3
C LC_111	SLN109 ring / End	14.0	0.0	-17.3	0.0	0.0	0.0
	SLN112 ring / End	13.9	0.0	-17.3	0.0	0.0	0.0
C LC_112	SLN109 ring / End	8.1	0.2	-17.3	0.0	0.0	0.3
	SLN112 ring / End	8.1	-0.1	-17.3	0.0	0.0	0.0
C LC_113	SLN109 ring / End	11.5	0.1	-17.3	0.0	0.0	0.2
	SLN112 ring / End	11.8	0.1	-17.3	0.0	0.0	0.1
C LC_114	SLN109 ring / End	9.5	-0.1	-17.3	0.0	0.0	-0.2
	SLN112 ring / End	9.2	-0.2	-17.3	0.0	0.0	-0.4
C LC_115	SLN109 ring / End	4.9	0.0	-2.8	0.0	0.0	0.0
	SLN112 ring / End	4.8	0.0	-2.8	0.0	0.0	0.0
C LC_116	SLN109 ring / End	0.0	0.2	-2.8	0.0	0.0	0.2
	SLN112 ring / End	0.1	0.0	-2.8	0.0	0.0	0.1
C LC_117	SLN109 ring / End	2.7	0.1	-2.8	0.0	0.0	0.1
	SLN112 ring / End	3.1	0.1	-2.8	0.0	0.0	0.2
C LC_118	SLN109 ring / End	1.2	-0.1	-2.8	0.0	0.0	-0.2
	SLN112 ring / End	0.8	-0.2	-2.8	0.0	0.0	-0.3
C LC_119	SLN109 ring / End	18.8	0.0	-25.6	0.0	0.0	0.0
	SLN112 ring / End	18.8	0.0	-25.5	0.0	0.0	-0.1
C LC_120	SLN109 ring / End	15.4	0.2	-25.6	0.0	0.0	0.3
	SLN112 ring / End	15.4	-0.1	-25.5	0.0	0.0	-0.1
C LC_121	SLN109 ring / End	17.7	0.1	-25.6	0.0	0.0	0.2
	SLN112 ring / End	17.7	0.0	-25.5	0.0	0.0	0.0
C LC_122	SLN109 ring / End	16.5	0.0	-25.6	0.0	0.0	0.0
	SLN112 ring / End	16.4	-0.3	-25.5	0.0	0.0	-0.4
C LC_123	SLN109 ring / End	15.8	0.0	-19.4	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN112 ring / End	15.7	0.0	-19.4	0.0	0.0	-0.1
C LC_124	SLN109 ring / End	11.4	0.2	-19.4	0.0	0.0	0.3
	SLN112 ring / End	11.5	-0.1	-19.4	0.0	0.0	0.0
C LC_125	SLN109 ring / End	14.2	0.2	-19.4	0.0	0.0	0.3
	SLN112 ring / End	14.5	0.1	-19.4	0.0	0.0	0.1
C LC_126	SLN109 ring / End	12.8	-0.1	-19.4	0.0	0.0	-0.2
	SLN112 ring / End	12.6	-0.3	-19.4	0.0	0.0	-0.5
C LC_127	SLN109 ring / End	6.7	0.0	-4.9	0.0	0.0	0.0
	SLN112 ring / End	6.6	0.0	-4.9	0.0	0.0	-0.1
C LC_128	SLN109 ring / End	3.4	0.2	-4.9	0.0	0.0	0.2
	SLN112 ring / End	3.5	0.0	-4.9	0.0	0.0	0.1
C LC_129	SLN109 ring / End	5.4	0.2	-4.9	0.0	0.0	0.2
	SLN112 ring / End	5.7	0.1	-4.9	0.0	0.0	0.2
C LC_130	SLN109 ring / End	4.5	-0.1	-4.9	0.0	0.0	-0.2
	SLN112 ring / End	4.1	-0.2	-4.9	0.0	0.0	-0.3
C LC_135	SLN109 ring / End	15.8	0.1	-25.6	0.0	0.0	0.2
	SLN112 ring / End	15.8	-0.1	-25.5	0.0	0.0	-0.2
C LC_141	SLN109 ring / End	5.0	0.0	-4.9	0.0	0.0	0.0
	SLN112 ring / End	5.0	0.0	-4.9	0.0	0.0	-0.1
C LC_142	SLN109 ring / End	4.5	0.1	-4.9	0.0	0.0	0.1
	SLN112 ring / End	4.5	-0.1	-4.9	0.0	0.0	-0.1
C LC_143	SLN109 ring / End	4.8	0.0	-4.9	0.0	0.0	0.1
	SLN112 ring / End	4.8	0.0	-4.9	0.0	0.0	0.0
C LC_144	SLN109 ring / End	4.6	0.0	-4.9	0.0	0.0	0.0
	SLN112 ring / End	4.7	-0.1	-4.9	0.0	0.0	-0.1
C LC_145	SLN109 ring / End	20.1	0.0	-25.6	0.0	0.0	0.1
	SLN112 ring / End	20.1	-0.1	-25.5	0.0	0.0	-0.2
C LC_146	SLN109 ring / End	18.1	0.2	-25.6	0.0	0.0	0.2
	SLN112 ring / End	18.0	-0.2	-25.5	0.0	0.0	-0.2
C LC_147	SLN109 ring / End	19.5	0.1	-25.6	0.0	0.0	0.2
	SLN112 ring / End	19.3	-0.1	-25.5	0.0	0.0	-0.1
C LC_148	SLN109 ring / End	18.6	0.1	-25.6	0.0	0.0	0.1
	SLN112 ring / End	18.8	-0.2	-25.5	0.0	0.0	-0.3
C LC_151	SLN109 ring / End	12.1	0.1	-11.6	0.0	0.0	0.1
	SLN112 ring / End	12.1	-0.1	-11.8	0.0	0.0	-0.1
C LC_152	SLN109 ring / End	19.0	0.1	-14.0	0.0	0.0	0.1
	SLN112 ring / End	19.0	-0.1	-14.1	0.0	0.0	-0.2
C LC_153	SLN109 ring / End	19.1	0.1	-20.7	0.0	0.0	0.1
	SLN112 ring / End	19.1	-0.1	-21.0	0.0	0.0	-0.2
C LC_154	SLN109 ring / End	12.2	0.1	-11.6	0.0	0.0	0.1
	SLN112 ring / End	12.2	-0.1	-11.8	0.0	0.0	-0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.9 < 100%	OK
Welds	19.6 < 100%	OK
Buckling	Not calculated	

Plates

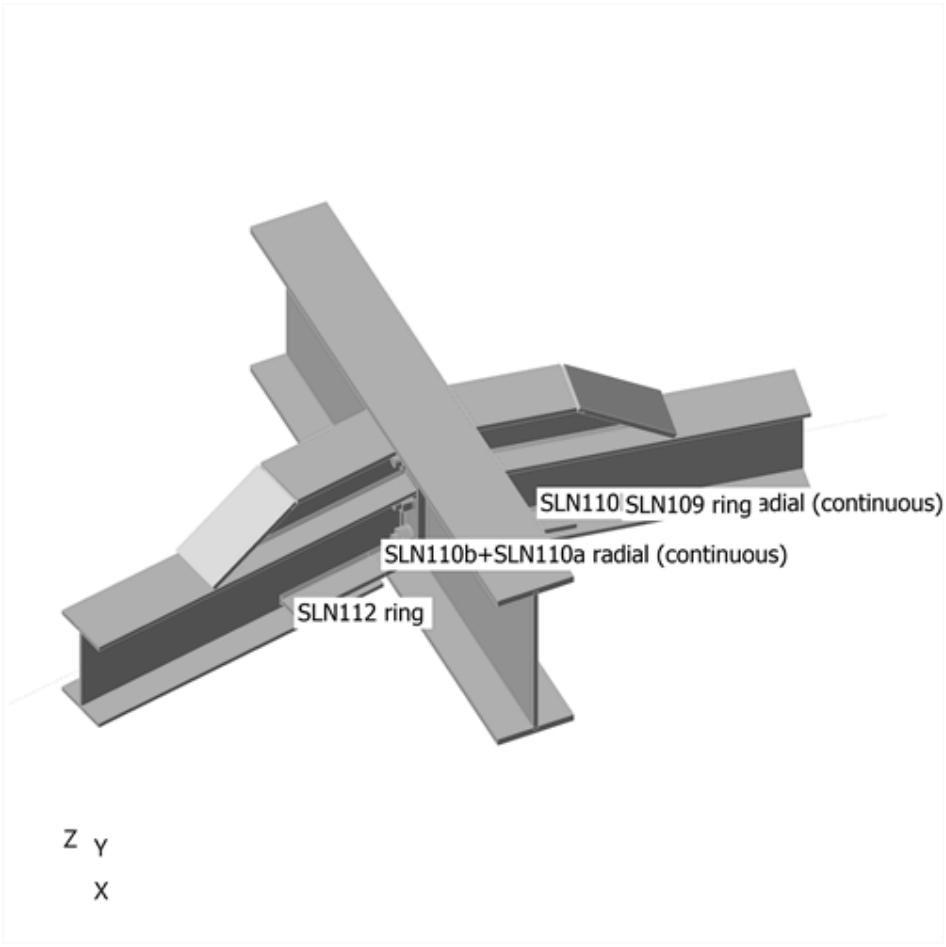
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN110b+SLN110a radial (continuous)-bfl 1	19.6	C LC_148	20.2	0.0	0.0	OK
SLN110b+SLN110a radial (continuous)-tfl 1	19.6	C LC_122	17.0	0.0	0.0	OK
SLN110b+SLN110a radial (continuous)-w 1	11.4	C LC_122	25.8	0.0	0.7	OK
SLN109 ring-bfl 1	12.7	C LC_145	50.4	0.0	0.0	OK
SLN109 ring-tfl 1	12.7	C LC_121	44.0	0.0	0.0	OK
SLN109 ring-w 1	7.2	C LC_145	49.0	0.0	0.0	OK
SLN112 ring-bfl 1	12.7	C LC_145	50.5	0.0	0.0	OK
SLN112 ring-tfl 1	12.7	C LC_122	44.4	0.0	0.0	OK
SLN112 ring-w 1	7.2	C LC_145	48.9	0.0	0.0	OK
End plate SLN112 ring	20.0	C LC_148	9.0	0.0	0.7	OK
Bottom flange plate SLN112 ring	15.0	C LC_145	20.5	0.0	0.0	OK
Bottom flange half plate L SLN112 ring	15.0	C LC_145	20.2	0.0	0.0	OK
Bottom flange half plate R SLN112 ring	15.0	C LC_145	19.9	0.0	0.0	OK
Haunch web SLN112 ring	12.0	C LC_122	28.0	0.0	0.0	OK
Haunch top plate SLN112 ring	15.0	C LC_122	5.8	0.0	0.0	OK
Haunch sloping plate SLN112 ring	15.0	C LC_120	12.6	0.0	0.0	OK
End plate SLN109 ring	20.0	C LC_147	8.5	0.0	0.6	OK
Bottom flange plate SLN109 ring	15.0	C LC_145	20.5	0.0	0.0	OK
Bottom flange half plate L SLN109 ring	15.0	C LC_145	20.0	0.0	0.0	OK
Bottom flange half plate R SLN109 ring	15.0	C LC_145	20.1	0.0	0.0	OK
Haunch web SLN109 ring	12.0	C LC_122	28.0	0.0	0.0	OK
Haunch top plate SLN109 ring	15.0	C LC_121	5.5	0.0	0.0	OK
Haunch sloping plate SLN109 ring	15.0	C LC_120	12.7	0.0	0.0	OK

Design data

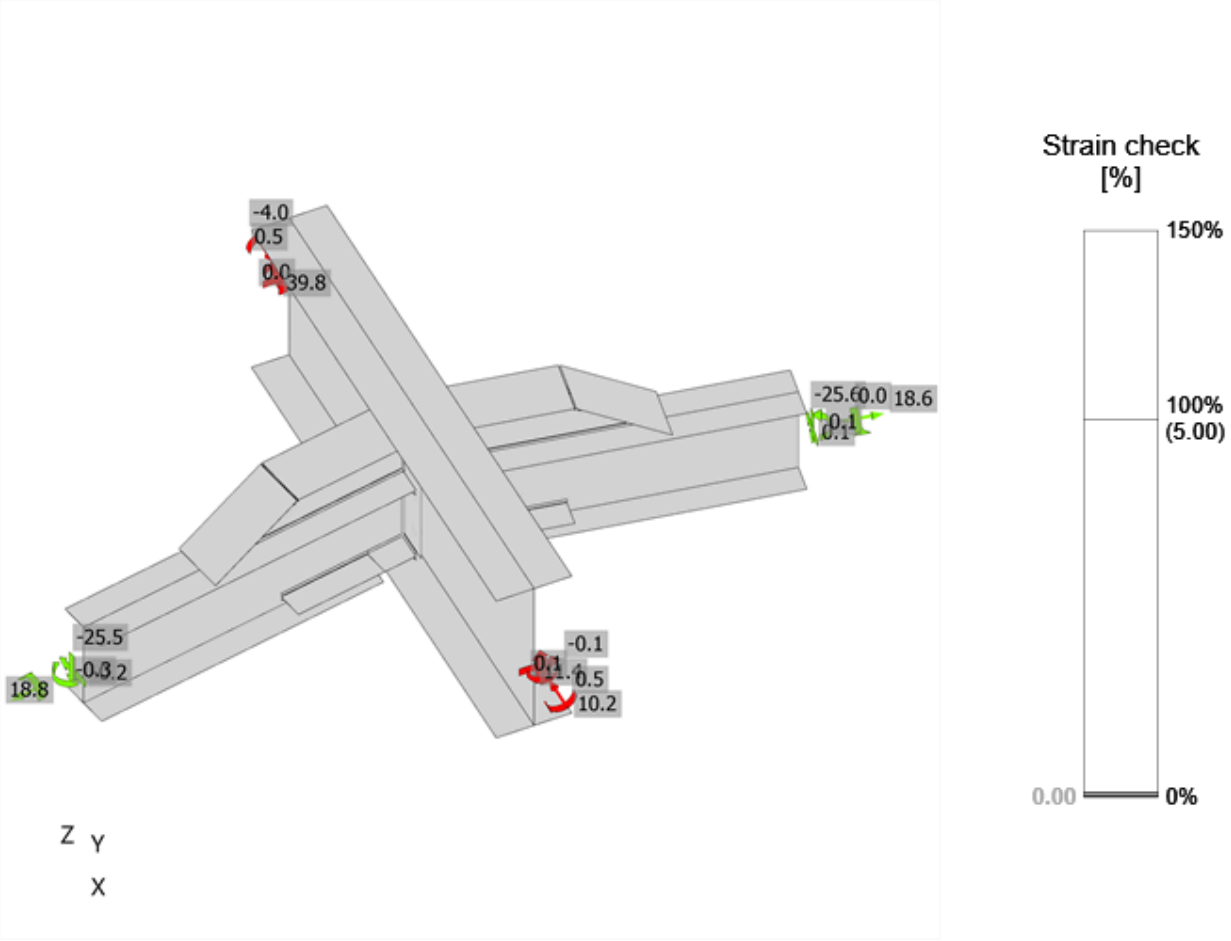
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

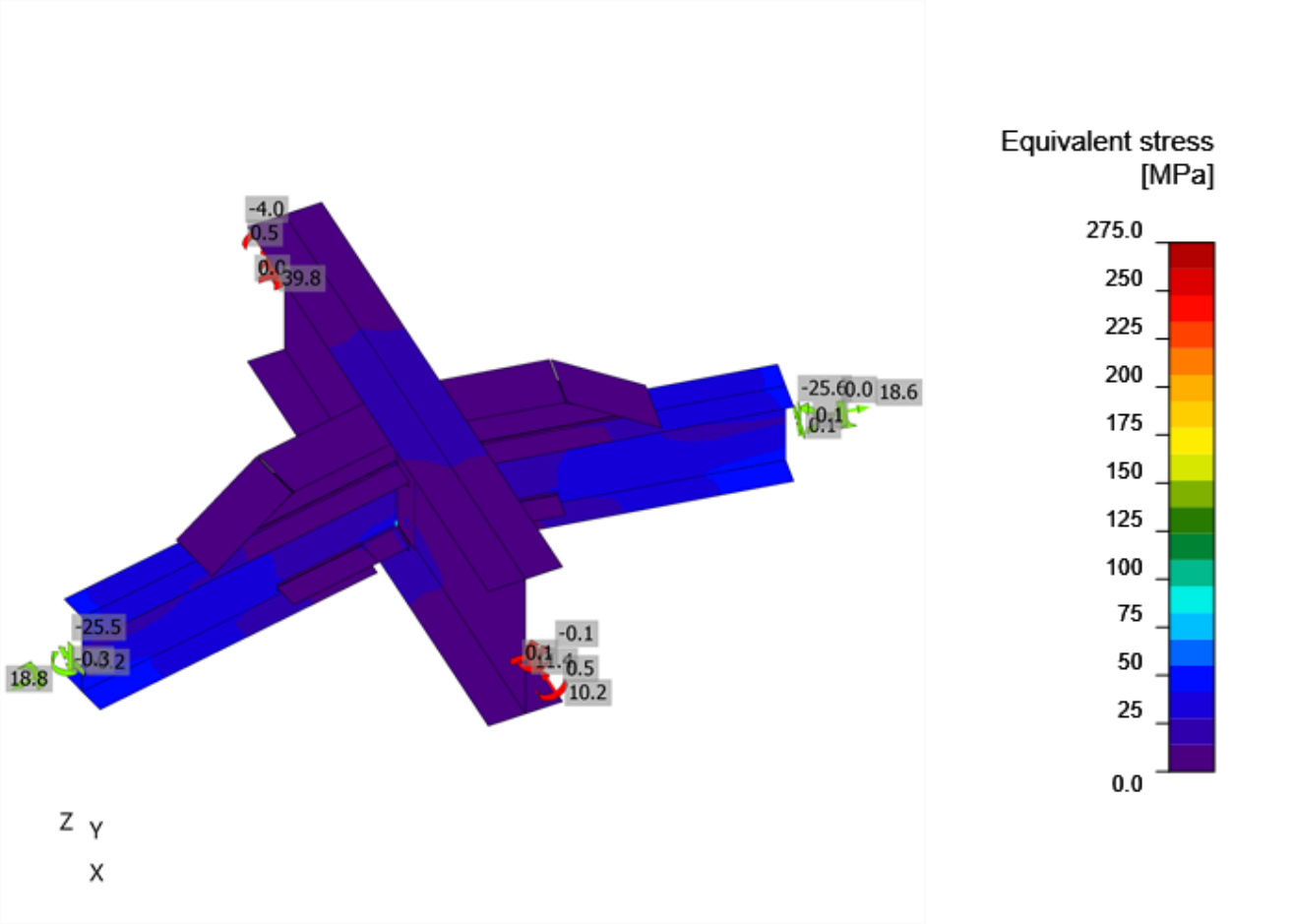
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

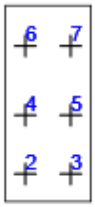


Strain check, C LC_148



Equivalent stress, C LC_148

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_146	2.5	0.5	194.1	0.8	0.5	0.8	OK	OK
	B3	M30 8.8 - 1	C LC_119	2.2	0.6	194.1	0.7	0.6	0.8	OK	OK
	B4	M30 8.8 - 1	C LC_119	0.1	0.4	232.2	0.0	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_145	0.1	0.5	232.2	0.0	0.4	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_122	2.3	0.8	249.2	0.7	0.6	0.9	OK	OK
	B7	M30 8.8 - 1	C LC_122	1.6	0.9	249.2	0.5	0.7	0.8	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN112 ring	Bottom flange plate SLN112 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN112 ring	Bottom flange plate SLN112 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN109 ring	Bottom flange plate SLN109 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN109 ring	Bottom flange plate SLN109 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN110b+SLN110a radial (continuous)-w 1	SLN109 ring-tfl 1	◢ 11.4	149	C LC_122	11.9	0.0	-7.8	-5.2	0.7	3.0	3.0	OK	OK
SLN110b+SLN110a radial (continuous)-w 1	SLN112 ring-tfl 1	◢ 11.4	149	C LC_121	11.9	0.0	-7.7	-5.2	-0.6	2.9	2.9	OK	OK
End plate SLN112 ring	SLN112 ring-w 1	◢ 6.0 ◣	208	C LC_145	25.7	0.0	-11.0	-12.9	-3.7	6.4	6.0	OK	OK
		◢ 6.0 ◣	208	C LC_145	20.6	0.0	-11.1	10.0	0.0	5.1	5.1	OK	OK
Bottom flange plate SLN112 ring	SLN112 ring-w 1	◢ 6.0 ◣	100	C LC_121	24.0	0.0	2.3	2.5	13.5	5.9	5.9	OK	OK
		◢ 6.0 ◣	100	C LC_122	25.5	0.0	2.7	-3.7	-14.2	6.3	6.3	OK	OK
SLN112 ring-w 1	Bottom flange half plate L SLN112 ring	◢ 6.0 ◣	243	C LC_145	24.3	0.0	6.1	6.0	12.2	6.0	6.0	OK	OK
		◢ 6.0 ◣	243	C LC_145	28.0	0.0	8.6	2.3	-15.2	6.9	6.3	OK	OK
SLN112 ring-w 1	Bottom flange half plate R SLN112 ring	◢ 6.0 ◣	243	C LC_145	25.2	0.0	6.1	6.2	-12.7	6.2	5.8	OK	OK
		◢ 6.0 ◣	243	C LC_145	27.5	0.0	7.6	2.9	15.0	6.8	6.5	OK	OK
SLN112 ring-tfl 1	Haunch web SLN112 ring	◢ 6.0 ◣	572	C LC_122	64.8	0.0	-14.7	-16.2	32.7	16.0	9.3	OK	OK
		◢ 6.0 ◣	572	C LC_121	62.8	0.0	-15.8	16.0	-31.2	15.5	9.0	OK	OK
End plate SLN109 ring	SLN109 ring-w 1	◢ 6.0 ◣	208	C LC_145	20.5	0.0	-11.5	-9.8	-0.3	5.1	5.1	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_145	25.4	0.0	-10.3	13.0	3.3	6.3	6.0	OK	OK
Bottom flange plate SLN109 ring	SLN109 ring-w 1	◢ 6.0 ◣	100	C LC_121	25.0	0.0	2.7	3.2	14.0	6.2	6.2	OK	OK
		◢ 6.0 ◣	100	C LC_122	24.6	0.0	2.3	-2.9	-13.8	6.1	6.1	OK	OK
SLN109 ring-w 1	Bottom flange half plate L SLN109 ring	◢ 6.0 ◣	243	C LC_147	25.3	0.0	6.2	6.2	12.7	6.3	6.3	OK	OK
		◢ 6.0 ◣	243	C LC_145	27.5	0.0	7.9	2.7	-14.9	6.8	6.3	OK	OK
SLN109 ring-w 1	Bottom flange half plate R SLN109 ring	◢ 6.0 ◣	243	C LC_145	24.3	0.0	5.9	6.0	-12.3	6.0	5.8	OK	OK
		◢ 6.0 ◣	243	C LC_145	28.0	0.0	8.2	2.5	15.3	6.9	6.5	OK	OK
SLN109 ring-tfl 1	Haunch web SLN109 ring	◢ 6.0 ◣	572	C LC_122	64.3	0.0	-16.0	-16.5	32.0	15.9	9.2	OK	OK
		◢ 6.0 ◣	572	C LC_120	63.9	0.0	-14.9	15.8	-32.2	15.8	9.2	OK	OK
End plate SLN112 ring	Bottom flange plate SLN112 ring	◢ 6.0 ◣	149	C LC_126	16.6	0.0	-4.4	-5.6	7.4	4.1	4.1	OK	OK
		◢ 6.0 ◣	149	C LC_126	22.3	0.0	-8.5	7.3	-9.4	5.5	5.5	OK	OK
End plate SLN112 ring	Haunch web SLN112 ring	◢ 6.0 ◣	145	C LC_121	7.5	0.0	1.4	1.3	4.0	1.8	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	10.6	0.0	2.3	-2.5	-5.4	2.6	0.0	OK	OK
End plate SLN112 ring	Haunch top plate SLN112 ring	◢ 6.0 ◣	149	C LC_126	10.0	0.0	-4.5	-3.7	3.6	2.5	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_126	8.9	0.0	-3.0	3.8	-3.1	2.2	0.0	OK	OK
Haunch top plate SLN112 ring	Haunch web SLN112 ring	◢ 6.0 ◣	342	C LC_122	4.7	0.0	1.7	1.5	2.0	1.2	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	4.0	0.0	-0.1	0.2	2.3	1.0	0.0	OK	OK
Haunch sloping plate SLN112 ring	Haunch web SLN112 ring	◢ 6.0 ◣	271	C LC_122	51.1	0.0	-8.8	-6.7	28.3	12.6	8.2	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_120	51.1	0.0	-9.1	9.7	-27.3	12.6	8.1	OK	OK
End plate SLN109 ring	Bottom flange plate SLN109 ring	◢ 6.0 ◣	148	C LC_145	14.7	0.0	7.6	4.4	5.8	3.6	3.6	OK	OK
		◢ 6.0 ◣	149	C LC_126	17.9	0.0	-9.1	8.0	-3.9	4.4	4.4	OK	OK
End plate SLN109 ring	Haunch web SLN109 ring	◢ 6.0 ◣	145	C LC_121	10.4	0.0	2.3	2.4	5.3	2.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	7.8	0.0	1.4	-1.4	-4.2	1.9	0.0	OK	OK
End plate SLN109 ring	Haunch top plate SLN109 ring	◢ 6.0 ◣	149	C LC_126	7.5	0.0	-4.0	-2.9	2.2	1.8	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_125	6.0	0.0	-2.1	2.5	2.1	1.5	0.0	OK	OK
Haunch top plate SLN109 ring	Haunch web SLN109 ring	◢ 6.0 ◣	342	C LC_122	3.8	0.0	1.3	1.3	1.6	0.9	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_121	4.2	0.0	1.5	-1.4	-1.8	1.0	0.0	OK	OK
Haunch sloping plate SLN109 ring	Haunch web SLN109 ring	◢ 6.0 ◣	271	C LC_122	52.1	0.0	-9.5	-8.8	28.3	12.9	8.2	OK	OK
		◢ 6.0 ◣	271	C LC_120	50.5	0.0	-8.7	7.4	-27.7	12.5	8.2	OK	OK
SLN110b+SLN110a radial (continuous)-w 1	SLN112 ring-w 1	◢ 7.2	208	C LC_145	78.6	0.0	33.5	37.5	-16.6	19.4	13.5	OK	OK
SLN110b+SLN110a radial (continuous)-w 1	SLN109 ring-w 1	◢ 7.2	208	C LC_145	79.4	0.0	31.9	-38.4	16.9	19.6	13.6	OK	OK
Haunch sloping plate SLN112 ring	Haunch top plate SLN112 ring	◢ 6.0 ◣	146	C LC_122	7.1	0.0	-5.4	2.6	0.4	1.9	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_126	3.3	0.0	-0.7	1.8	0.1	0.8	0.0	OK	OK
Haunch sloping plate SLN109 ring	Haunch top plate SLN109 ring	◢ 6.0 ◣	146	C LC_122	7.1	0.0	-5.1	2.8	0.3	1.9	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_121	2.7	0.0	-0.6	1.5	-0.5	0.7	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_180
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN111b+SLN111a radial (continuous) - SLN112 ring - SLN113 ring

Name	SLN111b+SLN111a radial (continuous) - SLN112 ring - SLN113 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

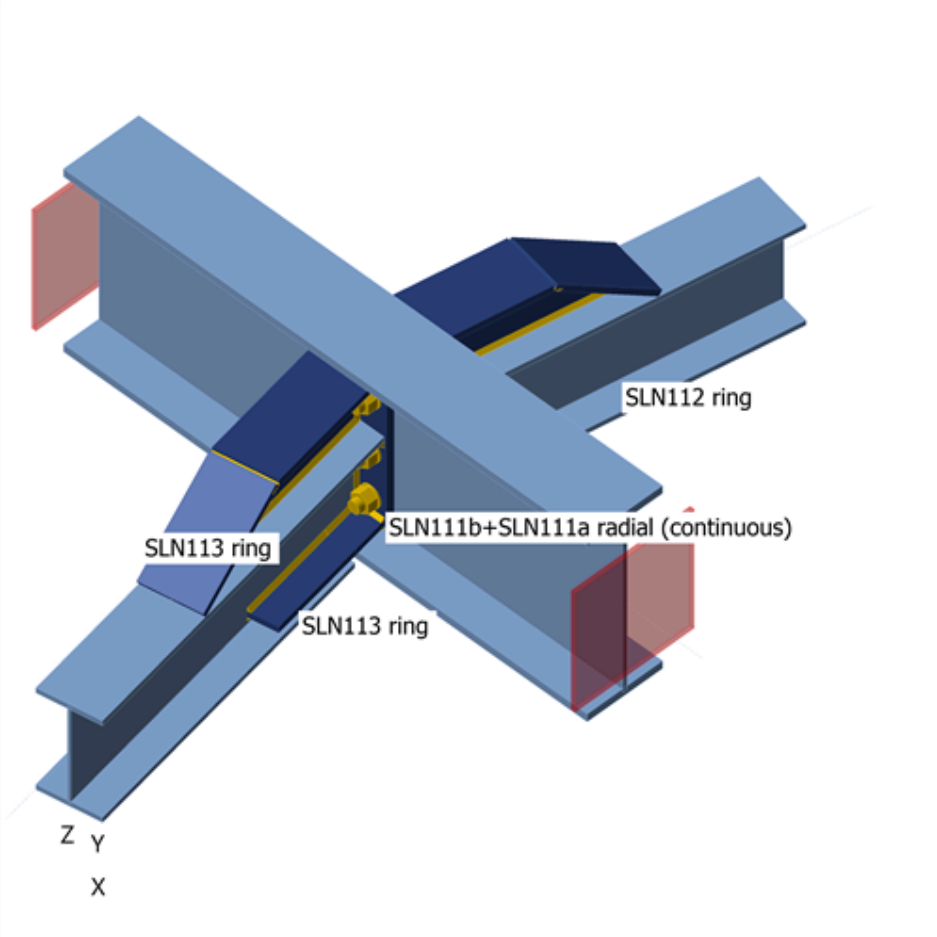
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN111b+SLN111a radial (continuous)	1 - UB457x191x98	-180.0	0.0	0.0	0	0	0
SLN112 ring	2 - UB254x146x43	80.0	0.0	0.0	0	0	-102
SLN113 ring	2 - UB254x146x43	-80.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN111b+SLN111a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN111b+SLN111a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN112 ring / end		Position	0
SLN113 ring / end		Position	0



SLN111b+SLN111a radial (continuous) - SLN112 ring - SLN113 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN112 ring / End	4.7	0.0	-4.9	0.0	0.0	-0.1
	SLN113 ring / End	4.7	0.0	-4.9	0.0	0.0	0.1
C LC_102	SLN112 ring / End	19.1	-0.1	-25.6	0.0	0.0	-0.2
	SLN113 ring / End	19.1	0.1	-25.6	0.0	0.0	0.2
C LC_103	SLN112 ring / End	18.0	-0.2	-24.5	0.0	0.0	-0.2
	SLN113 ring / End	17.9	0.2	-24.6	0.0	0.0	0.1
C LC_104	SLN112 ring / End	13.7	0.0	-24.5	0.0	0.0	0.0
	SLN113 ring / End	13.8	0.1	-24.6	0.0	0.0	0.2
C LC_105	SLN112 ring / End	16.2	0.0	-24.5	0.0	0.0	0.0
	SLN113 ring / End	16.1	0.2	-24.6	0.0	0.0	0.2
C LC_106	SLN112 ring / End	14.9	-0.2	-24.5	0.0	0.0	-0.3
	SLN113 ring / End	15.1	0.0	-24.6	0.0	0.0	0.0
C LC_107	SLN112 ring / End	3.5	0.0	-1.6	0.0	0.0	0.0
	SLN113 ring / End	3.3	0.0	-1.6	0.0	0.0	0.0
C LC_108	SLN112 ring / End	-1.1	0.1	-1.6	0.0	0.0	0.1
	SLN113 ring / End	-0.9	0.1	-1.6	0.0	0.0	0.2
C LC_109	SLN112 ring / End	1.8	0.1	-1.6	0.0	0.0	0.1
	SLN113 ring / End	1.8	0.1	-1.6	0.0	0.0	0.1
C LC_110	SLN112 ring / End	-0.4	-0.1	-1.6	0.0	0.0	-0.2
	SLN113 ring / End	-0.3	-0.1	-1.6	0.0	0.0	-0.2
C LC_111	SLN112 ring / End	13.9	-0.1	-17.3	0.0	0.0	-0.1
	SLN113 ring / End	13.8	0.1	-17.3	0.0	0.0	0.1
C LC_112	SLN112 ring / End	8.1	0.1	-17.3	0.0	0.0	0.0
	SLN113 ring / End	8.3	0.1	-17.3	0.0	0.0	0.2
C LC_113	SLN112 ring / End	11.7	0.1	-17.3	0.0	0.0	0.1
	SLN113 ring / End	11.6	0.2	-17.3	0.0	0.0	0.2
C LC_114	SLN112 ring / End	9.3	-0.2	-17.3	0.0	0.0	-0.3
	SLN113 ring / End	9.5	-0.1	-17.3	0.0	0.0	-0.1
C LC_115	SLN112 ring / End	4.8	0.0	-2.8	0.0	0.0	0.0
	SLN113 ring / End	4.6	0.0	-2.8	0.0	0.0	0.0
C LC_116	SLN112 ring / End	0.1	0.1	-2.8	0.0	0.0	0.1
	SLN113 ring / End	0.3	0.1	-2.8	0.0	0.0	0.2
C LC_117	SLN112 ring / End	3.1	0.1	-2.8	0.0	0.0	0.1
	SLN113 ring / End	3.0	0.1	-2.8	0.0	0.0	0.2
C LC_118	SLN112 ring / End	0.8	-0.2	-2.8	0.0	0.0	-0.2
	SLN113 ring / End	0.9	-0.1	-2.8	0.0	0.0	-0.2
C LC_119	SLN112 ring / End	18.9	-0.2	-25.6	0.0	0.0	-0.2
	SLN113 ring / End	18.8	0.2	-25.6	0.0	0.0	0.2
C LC_120	SLN112 ring / End	15.4	0.0	-25.6	0.0	0.0	0.0
	SLN113 ring / End	15.5	0.1	-25.6	0.0	0.0	0.2
C LC_121	SLN112 ring / End	17.6	0.0	-25.6	0.0	0.0	0.0
	SLN113 ring / End	17.8	0.2	-25.6	0.0	0.0	0.3
C LC_122	SLN112 ring / End	16.6	-0.2	-25.6	0.0	0.0	-0.4
	SLN113 ring / End	16.5	0.0	-25.6	0.0	0.0	0.0
C LC_123	SLN112 ring / End	15.8	-0.2	-19.4	0.0	0.0	-0.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN113 ring / End	15.6	0.1	-19.4	0.0	0.0	0.2
C LC_124	SLN112 ring / End	11.5	0.1	-19.4	0.0	0.0	0.0
	SLN113 ring / End	11.6	0.1	-19.4	0.0	0.0	0.2
C LC_125	SLN112 ring / End	14.3	0.1	-19.4	0.0	0.0	0.2
	SLN113 ring / End	15.0	0.3	-19.4	0.0	0.0	0.4
C LC_126	SLN112 ring / End	12.7	-0.3	-19.4	0.0	0.0	-0.5
	SLN113 ring / End	12.2	-0.1	-19.4	0.0	0.0	-0.2
C LC_127	SLN112 ring / End	6.6	-0.1	-4.9	0.0	0.0	-0.1
	SLN113 ring / End	6.4	0.1	-4.9	0.0	0.0	0.1
C LC_128	SLN112 ring / End	3.5	0.1	-4.9	0.0	0.0	0.1
	SLN113 ring / End	3.6	0.1	-4.9	0.0	0.0	0.2
C LC_129	SLN112 ring / End	5.7	0.1	-4.9	0.0	0.0	0.2
	SLN113 ring / End	6.4	0.2	-4.9	0.0	0.0	0.3
C LC_130	SLN112 ring / End	4.2	-0.2	-4.9	0.0	0.0	-0.4
	SLN113 ring / End	3.6	-0.1	-4.9	0.0	0.0	-0.2
C LC_135	SLN112 ring / End	15.8	-0.1	-25.6	0.0	0.0	-0.2
	SLN113 ring / End	15.9	0.1	-25.6	0.0	0.0	0.2
C LC_141	SLN112 ring / End	5.0	-0.1	-4.9	0.0	0.0	-0.1
	SLN113 ring / End	5.0	0.1	-4.9	0.0	0.0	0.1
C LC_142	SLN112 ring / End	4.5	0.0	-4.9	0.0	0.0	0.0
	SLN113 ring / End	4.5	0.0	-4.9	0.0	0.0	0.0
C LC_143	SLN112 ring / End	4.8	0.0	-4.9	0.0	0.0	0.0
	SLN113 ring / End	4.7	0.0	-4.9	0.0	0.0	0.1
C LC_144	SLN112 ring / End	4.7	0.0	-4.9	0.0	0.0	-0.1
	SLN113 ring / End	4.7	0.0	-4.9	0.0	0.0	0.0
C LC_145	SLN112 ring / End	20.1	-0.2	-25.6	0.0	0.0	-0.3
	SLN113 ring / End	20.2	0.2	-25.6	0.0	0.0	0.2
C LC_146	SLN112 ring / End	18.0	0.0	-25.6	0.0	0.0	-0.1
	SLN113 ring / End	18.1	0.0	-25.6	0.0	0.0	0.1
C LC_147	SLN112 ring / End	19.2	-0.1	-25.6	0.0	0.0	-0.1
	SLN113 ring / End	19.1	0.1	-25.6	0.0	0.0	0.2
C LC_148	SLN112 ring / End	18.9	-0.2	-25.6	0.0	0.0	-0.2
	SLN113 ring / End	19.1	0.1	-25.6	0.0	0.0	0.1
C LC_151	SLN112 ring / End	12.1	-0.1	-11.8	0.0	0.0	-0.1
	SLN113 ring / End	12.1	0.1	-11.5	0.0	0.0	0.1
C LC_152	SLN112 ring / End	19.0	-0.1	-14.2	0.0	0.0	-0.1
	SLN113 ring / End	19.0	0.1	-14.1	0.0	0.0	0.2
C LC_153	SLN112 ring / End	19.1	-0.1	-21.1	0.0	0.0	-0.2
	SLN113 ring / End	19.1	0.1	-20.8	0.0	0.0	0.2
C LC_154	SLN112 ring / End	12.2	-0.1	-11.8	0.0	0.0	-0.1
	SLN113 ring / End	12.2	0.1	-11.5	0.0	0.0	0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.9 < 100%	OK
Welds	19.6 < 100%	OK
Buckling	Not calculated	

Plates

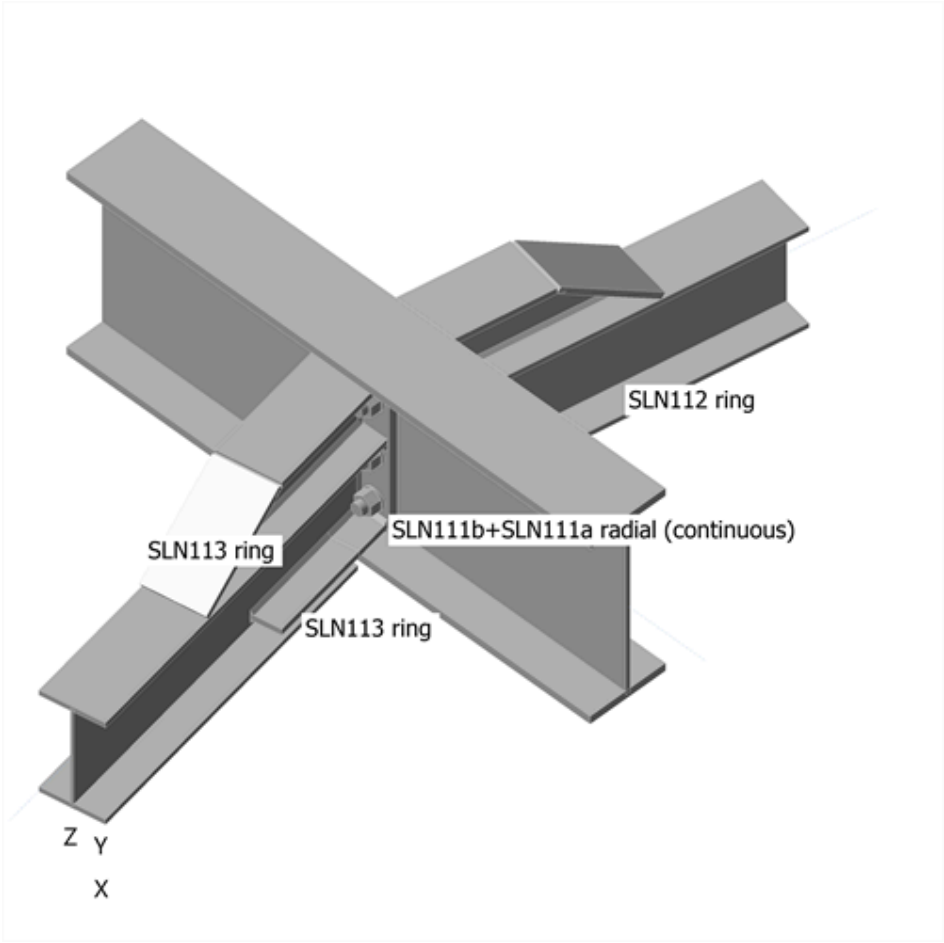
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN111b+SLN111a radial (continuous)-bfl 1	19.6	C LC_148	20.2	0.0	0.0	OK
SLN111b+SLN111a radial (continuous)-tfl 1	19.6	C LC_121	17.1	0.0	0.0	OK
SLN111b+SLN111a radial (continuous)-w 1	11.4	C LC_122	25.8	0.0	0.7	OK
SLN112 ring-bfl 1	12.7	C LC_148	50.0	0.0	0.0	OK
SLN112 ring-tfl 1	12.7	C LC_122	44.7	0.0	0.0	OK
SLN112 ring-w 1	7.2	C LC_145	48.9	0.0	0.0	OK
SLN113 ring-bfl 1	12.7	C LC_121	50.3	0.0	0.0	OK
SLN113 ring-tfl 1	12.7	C LC_120	44.4	0.0	0.0	OK
SLN113 ring-w 1	7.2	C LC_145	49.2	0.0	0.0	OK
End plate SLN113 ring	20.0	C LC_148	8.6	0.0	0.8	OK
Bottom flange plate SLN113 ring	15.0	C LC_145	20.6	0.0	0.0	OK
Bottom flange half plate L SLN113 ring	15.0	C LC_145	19.9	0.0	0.0	OK
Bottom flange half plate R SLN113 ring	15.0	C LC_145	20.1	0.0	0.0	OK
Haunch web SLN113 ring	12.0	C LC_120	29.1	0.0	0.0	OK
Haunch top plate SLN113 ring	15.0	C LC_121	5.9	0.0	0.0	OK
Haunch sloping plate SLN113 ring	15.0	C LC_120	12.7	0.0	0.0	OK
End plate SLN112 ring	20.0	C LC_145	8.9	0.0	1.1	OK
Bottom flange plate SLN112 ring	15.0	C LC_145	20.5	0.0	0.0	OK
Bottom flange half plate L SLN112 ring	15.0	C LC_145	20.1	0.0	0.0	OK
Bottom flange half plate R SLN112 ring	15.0	C LC_145	19.9	0.0	0.0	OK
Haunch web SLN112 ring	12.0	C LC_122	28.0	0.0	0.0	OK
Haunch top plate SLN112 ring	15.0	C LC_126	6.1	0.0	0.0	OK
Haunch sloping plate SLN112 ring	15.0	C LC_120	12.6	0.0	0.0	OK

Design data

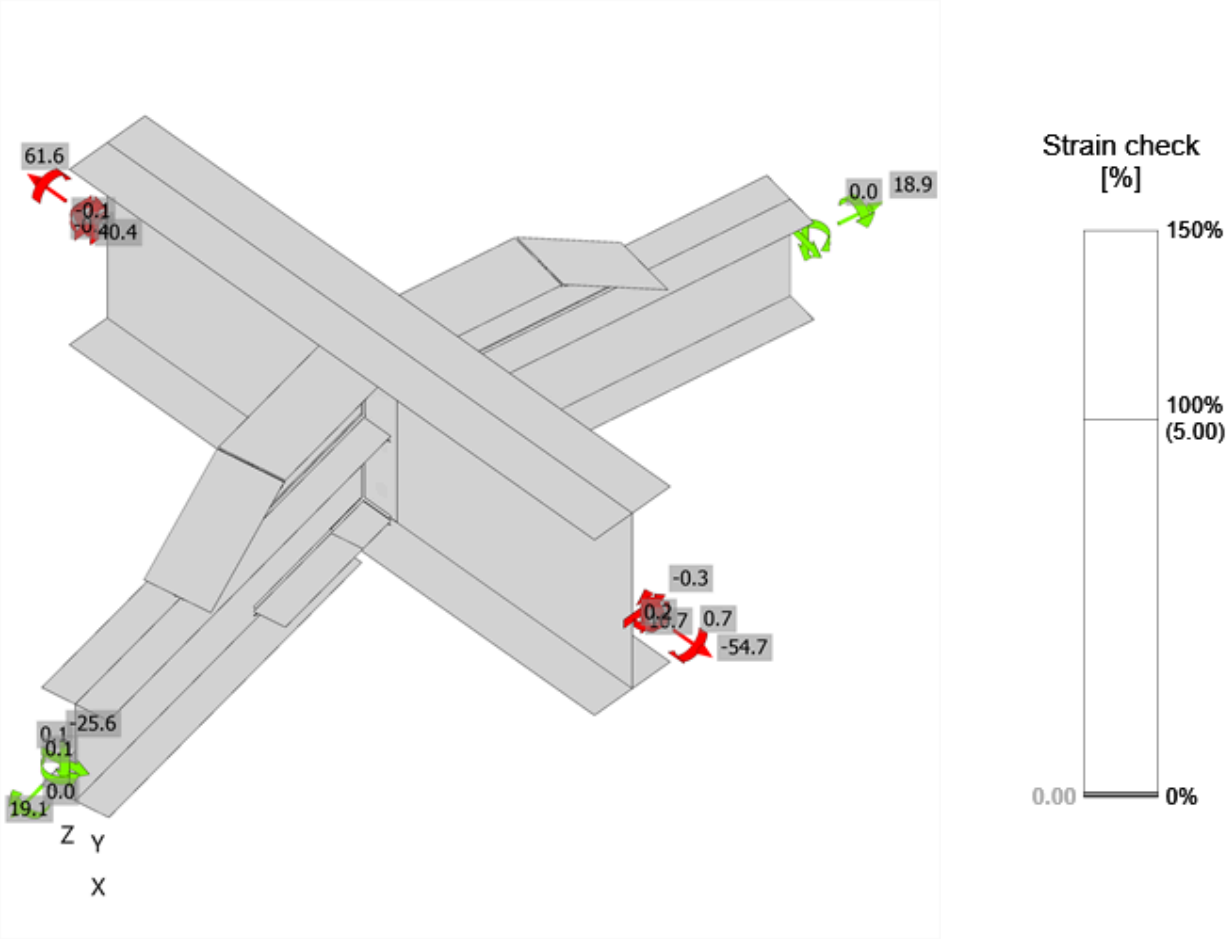
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

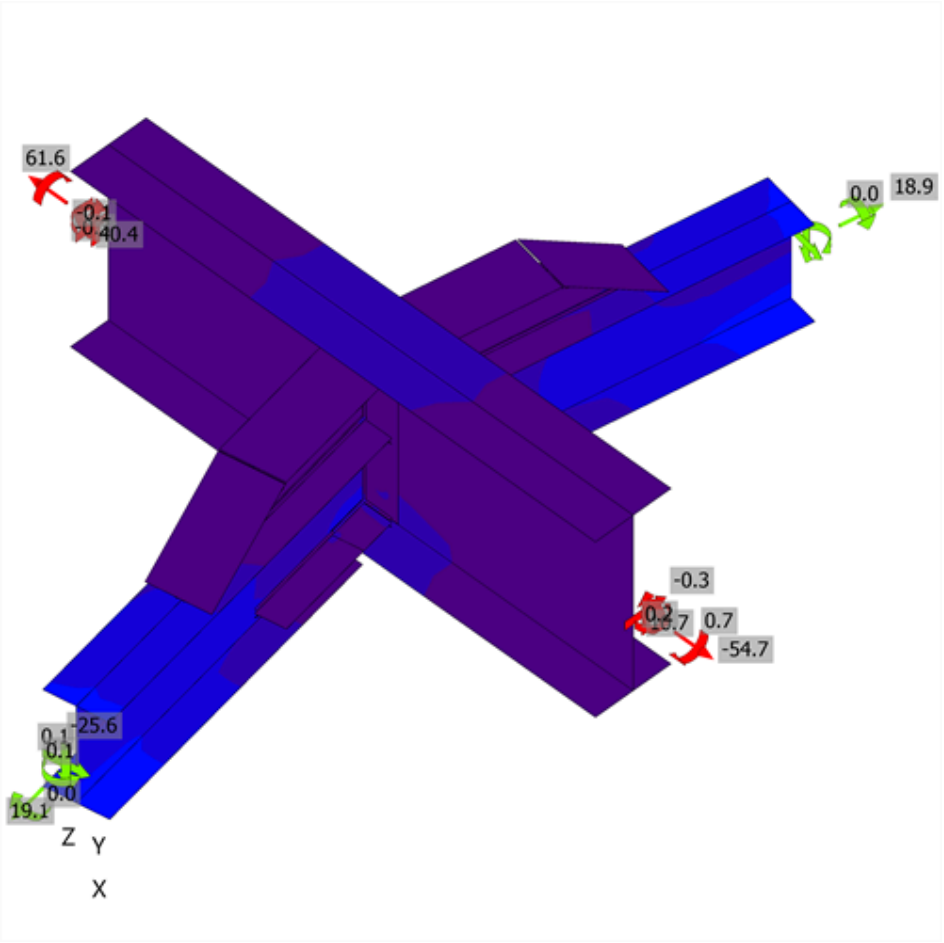
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

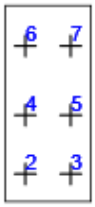


Strain check, C LC_148



Equivalent stress, C LC_148

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_121	2.3	0.5	194.1	0.7	0.5	0.8	OK	OK
	B3	M30 8.8 - 1	C LC_145	2.5	0.6	194.1	0.8	0.7	0.8	OK	OK
	B4	M30 8.8 - 1	C LC_135	0.1	0.4	232.2	0.0	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_145	0.1	0.5	232.2	0.0	0.4	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_121	1.8	0.9	249.2	0.5	0.7	0.8	OK	OK
	B7	M30 8.8 - 1	C LC_121	2.2	0.9	249.2	0.7	0.7	0.9	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN112 ring	Bottom flange plate SLN112 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN113 ring	Bottom flange plate SLN113 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN113 ring	Bottom flange plate SLN113 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN112 ring	Bottom flange plate SLN112 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN111b+SLN111a radial (continuous)-w 1	SLN112 ring-tfl 1	◢ 11.4	149	C LC_145	11.3	0.0	-6.9	-5.2	0.0	2.8	0.0	OK	OK
SLN111b+SLN111a radial (continuous)-w 1	SLN113 ring-tfl 1	◢ 11.4	149	C LC_121	11.8	0.0	-6.9	-5.5	-0.6	2.9	2.9	OK	OK
End plate SLN113 ring	SLN113 ring-w 1	◢ 6.0 ◣	208	C LC_145	24.5	0.0	-10.3	-12.3	-3.8	6.1	6.0	OK	OK
		◢ 6.0 ◣	208	C LC_145	20.2	0.0	-11.0	9.8	0.3	5.0	5.0	OK	OK
Bottom flange plate SLN113 ring	SLN113 ring-w 1	◢ 6.0 ◣	100	C LC_121	25.3	0.0	2.6	3.0	14.2	6.3	6.3	OK	OK
		◢ 6.0 ◣	100	C LC_122	24.9	0.0	2.5	-3.3	-13.9	6.1	6.1	OK	OK
SLN113 ring-w 1	Bottom flange half plate L SLN113 ring	◢ 6.0 ◣	243	C LC_145	25.9	0.0	6.6	6.3	13.0	6.4	6.4	OK	OK
		◢ 6.0 ◣	243	C LC_145	27.1	0.0	7.9	2.7	-14.7	6.7	6.3	OK	OK
SLN113 ring-w 1	Bottom flange half plate R SLN113 ring	◢ 6.0 ◣	243	C LC_122	23.9	0.0	5.3	5.9	-12.1	5.9	5.7	OK	OK
		◢ 6.0 ◣	243	C LC_145	28.4	0.0	8.3	2.5	15.5	7.0	6.6	OK	OK
SLN113 ring-tfl 1	Haunch web SLN113 ring	◢ 6.0 ◣	572	C LC_122	66.3	0.0	-16.9	-17.7	32.5	16.4	9.4	OK	OK
		◢ 6.0 ◣	572	C LC_121	61.8	0.0	-13.4	14.6	-31.6	15.3	9.0	OK	OK
End plate SLN112 ring	SLN112 ring-w 1	◢ 6.0 ◣	208	C LC_145	20.3	0.0	-11.5	-9.6	-0.6	5.0	5.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_145	25.1	0.0	-10.1	12.9	2.9	6.2	6.0	OK	OK
Bottom flange plate SLN112 ring	SLN112 ring-w 1	◢ 6.0 ◣	100	C LC_121	24.8	0.0	2.5	2.9	13.9	6.1	6.1	OK	OK
		◢ 6.0 ◣	100	C LC_122	25.4	0.0	2.5	-3.5	-14.2	6.3	6.3	OK	OK
SLN112 ring-w 1	Bottom flange half plate L SLN112 ring	◢ 6.0 ◣	243	C LC_145	24.6	0.0	6.2	6.0	12.3	6.1	6.1	OK	OK
		◢ 6.0 ◣	243	C LC_145	27.7	0.0	8.5	2.4	-15.0	6.8	6.3	OK	OK
SLN112 ring-w 1	Bottom flange half plate R SLN112 ring	◢ 6.0 ◣	243	C LC_145	25.1	0.0	6.0	6.1	-12.6	6.2	5.8	OK	OK
		◢ 6.0 ◣	243	C LC_145	27.7	0.0	7.7	2.8	15.1	6.8	6.6	OK	OK
SLN112 ring-tfl 1	Haunch web SLN112 ring	◢ 6.0 ◣	572	C LC_122	65.3	0.0	-14.7	-16.4	32.9	16.1	9.3	OK	OK
		◢ 6.0 ◣	572	C LC_120	62.8	0.0	-16.2	15.9	-31.3	15.5	9.1	OK	OK
End plate SLN113 ring	Bottom flange plate SLN113 ring	◢ 6.0 ◣	149	C LC_126	17.9	0.0	-4.7	-6.1	7.8	4.4	4.4	OK	OK
		◢ 6.0 ◣	149	C LC_126	23.8	0.0	-9.1	7.6	-10.2	5.9	5.6	OK	OK
End plate SLN113 ring	Haunch web SLN113 ring	◢ 6.0 ◣	145	C LC_121	7.7	0.0	1.9	1.7	3.9	1.9	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	10.7	0.0	1.9	-2.3	-5.6	2.7	0.0	OK	OK
End plate SLN113 ring	Haunch top plate SLN113 ring	◢ 6.0 ◣	149	C LC_130	9.2	0.0	-4.6	-3.0	3.5	2.3	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_125	8.1	0.0	-3.1	3.3	2.7	2.0	0.0	OK	OK
Haunch top plate SLN113 ring	Haunch web SLN113 ring	◢ 6.0 ◣	342	C LC_126	3.7	0.0	1.2	1.1	1.7	0.9	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_121	4.8	0.0	1.7	-1.5	-2.1	1.2	0.0	OK	OK
Haunch sloping plate SLN113 ring	Haunch web SLN113 ring	◢ 6.0 ◣	271	C LC_120	54.7	0.0	-10.6	-11.0	29.0	13.5	8.4	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_121	47.4	0.0	-7.5	6.0	-26.4	11.7	8.0	OK	OK
End plate SLN112 ring	Bottom flange plate SLN112 ring	◢ 6.0 ◣	148	C LC_145	15.1	0.0	7.4	4.2	6.3	3.7	3.7	OK	OK
		◢ 6.0 ◣	149	C LC_126	20.5	0.0	-10.2	9.5	-4.0	5.1	5.1	OK	OK
End plate SLN112 ring	Haunch web SLN112 ring	◢ 6.0 ◣	145	C LC_121	10.5	0.0	1.9	2.4	5.5	2.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	8.0	0.0	1.8	-1.6	-4.2	2.0	0.0	OK	OK
End plate SLN112 ring	Haunch top plate SLN112 ring	◢ 6.0 ◣	149	C LC_125	9.0	0.0	-4.4	-2.7	-3.6	2.2	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_126	7.7	0.0	-2.8	3.1	-2.7	1.9	0.0	OK	OK
Haunch top plate SLN112 ring	Haunch web SLN112 ring	◢ 6.0 ◣	342	C LC_122	4.8	0.0	1.7	1.5	2.1	1.2	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	4.0	0.0	-0.2	0.2	2.3	1.0	0.0	OK	OK
Haunch sloping plate SLN112 ring	Haunch web SLN112 ring	◢ 6.0 ◣	271	C LC_135	51.7	0.0	-9.2	-7.6	28.3	12.8	8.3	OK	OK
		◢ 6.0 ◣	271	C LC_120	51.1	0.0	-9.1	9.5	-27.4	12.6	8.1	OK	OK
SLN111b+SLN111a radial (continuous)-w 1	SLN113 ring-w 1	◢ 7.2	208	C LC_145	68.5	0.0	35.5	-29.9	15.8	16.9	12.2	OK	OK
SLN111b+SLN111a radial (continuous)-w 1	SLN112 ring-w 1	◢ 7.2	208	C LC_145	79.2	0.0	31.5	-38.6	16.5	19.6	13.7	OK	OK
Haunch sloping plate SLN113 ring	Haunch top plate SLN113 ring	◢ 6.0 ◣	146	C LC_121	7.2	0.0	-5.5	2.7	-0.4	1.9	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_125	3.4	0.0	-0.7	1.9	0.2	0.8	0.0	OK	OK
Haunch sloping plate SLN112 ring	Haunch top plate SLN112 ring	◢ 6.0 ◣	146	C LC_121	7.0	0.0	-5.0	2.8	-0.3	1.9	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_126	3.5	0.0	-0.7	2.0	-0.1	0.9	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_200
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN114b+SLN114a radial (continuous) - SLN113 ring - SLN116 ring

Name	SLN114b+SLN114a radial (continuous) - SLN113 ring - SLN116 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

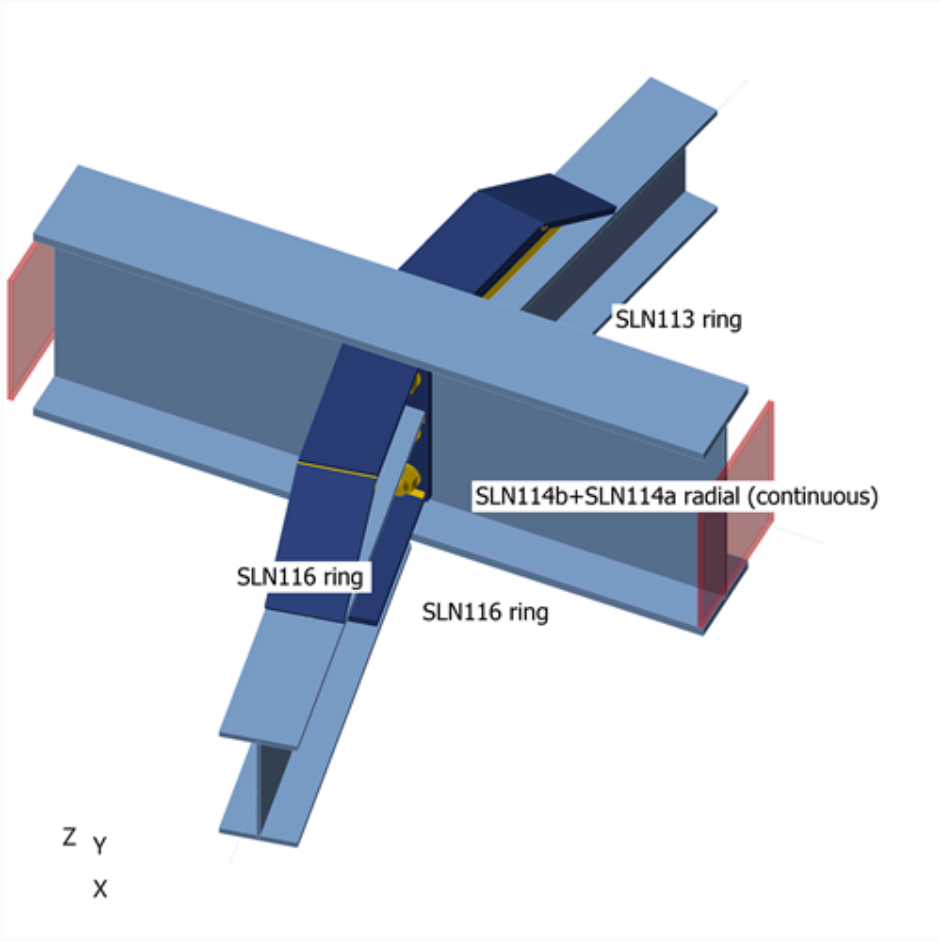
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN114b+SLN114a radial (continuous)	1 - UB457x191x98	-160.0	0.0	0.0	0	0	0
SLN113 ring	2 - UB254x146x43	100.0	0.0	0.0	0	0	-102
SLN116 ring	2 - UB254x146x43	-60.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN114b+SLN114a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN114b+SLN114a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN113 ring / end		Position	0
SLN116 ring / end		Position	0



SLN114b+SLN114a radial (continuous) - SLN113 ring - SLN116 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN113 ring / End	4.7	0.0	-4.9	0.0	0.0	0.1
	SLN116 ring / End	4.8	0.0	-4.9	0.0	0.0	-0.1
C LC_102	SLN113 ring / End	19.1	0.1	-25.6	0.0	0.0	0.2
	SLN116 ring / End	19.2	-0.1	-25.6	0.0	0.0	-0.2
C LC_103	SLN113 ring / End	17.9	0.0	-24.6	0.0	0.0	0.1
	SLN116 ring / End	17.7	0.0	-24.5	0.0	0.0	-0.1
C LC_104	SLN113 ring / End	13.9	0.2	-24.6	0.0	0.0	0.3
	SLN116 ring / End	14.0	-0.2	-24.5	0.0	0.0	-0.2
C LC_105	SLN113 ring / End	16.0	0.2	-24.6	0.0	0.0	0.3
	SLN116 ring / End	16.1	-0.1	-24.5	0.0	0.0	0.0
C LC_106	SLN113 ring / End	15.3	0.0	-24.6	0.0	0.0	-0.1
	SLN116 ring / End	15.3	-0.1	-24.5	0.0	0.0	-0.3
C LC_107	SLN113 ring / End	3.3	0.0	-1.6	0.0	0.0	0.0
	SLN116 ring / End	3.1	0.0	-1.6	0.0	0.0	0.0
C LC_108	SLN113 ring / End	-0.9	0.1	-1.6	0.0	0.0	0.2
	SLN116 ring / End	-0.8	0.0	-1.6	0.0	0.0	0.1
C LC_109	SLN113 ring / End	1.8	0.1	-1.6	0.0	0.0	0.2
	SLN116 ring / End	2.2	0.1	-1.6	0.0	0.0	0.2
C LC_110	SLN113 ring / End	-0.3	-0.1	-1.6	0.0	0.0	-0.2
	SLN116 ring / End	-0.7	-0.1	-1.6	0.0	0.0	-0.2
C LC_111	SLN113 ring / End	13.8	0.0	-17.3	0.0	0.0	0.0
	SLN116 ring / End	13.5	0.0	-17.3	0.0	0.0	0.0
C LC_112	SLN113 ring / End	8.3	0.2	-17.3	0.0	0.0	0.3
	SLN116 ring / End	8.5	-0.1	-17.3	0.0	0.0	0.0
C LC_113	SLN113 ring / End	11.5	0.2	-17.3	0.0	0.0	0.3
	SLN116 ring / End	11.9	0.0	-17.3	0.0	0.0	0.1
C LC_114	SLN113 ring / End	9.7	-0.1	-17.3	0.0	0.0	-0.2
	SLN116 ring / End	9.4	-0.2	-17.3	0.0	0.0	-0.3
C LC_115	SLN113 ring / End	4.6	0.0	-2.8	0.0	0.0	0.0
	SLN116 ring / End	4.4	0.0	-2.8	0.0	0.0	0.0
C LC_116	SLN113 ring / End	0.3	0.1	-2.8	0.0	0.0	0.2
	SLN116 ring / End	0.4	0.0	-2.8	0.0	0.0	0.1
C LC_117	SLN113 ring / End	3.0	0.1	-2.8	0.0	0.0	0.2
	SLN116 ring / End	3.4	0.1	-2.8	0.0	0.0	0.2
C LC_118	SLN113 ring / End	0.9	-0.1	-2.8	0.0	0.0	-0.2
	SLN116 ring / End	0.6	-0.1	-2.8	0.0	0.0	-0.2
C LC_119	SLN113 ring / End	18.8	0.0	-25.6	0.0	0.0	0.1
	SLN116 ring / End	18.6	0.0	-25.6	0.0	0.0	-0.1
C LC_120	SLN113 ring / End	15.5	0.2	-25.6	0.0	0.0	0.2
	SLN116 ring / End	15.7	-0.2	-25.6	0.0	0.0	-0.2
C LC_121	SLN113 ring / End	17.7	0.2	-25.6	0.0	0.0	0.3
	SLN116 ring / End	17.8	-0.1	-25.6	0.0	0.0	0.0
C LC_122	SLN113 ring / End	16.6	0.0	-25.6	0.0	0.0	0.0
	SLN116 ring / End	16.7	-0.2	-25.6	0.0	0.0	-0.3
C LC_123	SLN113 ring / End	15.6	0.0	-19.4	0.0	0.0	0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN116 ring / End	15.3	0.0	-19.4	0.0	0.0	-0.1
C LC_124	SLN113 ring / End	11.7	0.2	-19.4	0.0	0.0	0.3
	SLN116 ring / End	11.8	-0.1	-19.4	0.0	0.0	-0.1
C LC_125	SLN113 ring / End	14.9	0.3	-19.4	0.0	0.0	0.4
	SLN116 ring / End	15.2	0.0	-19.4	0.0	0.0	0.1
C LC_126	SLN113 ring / End	12.3	-0.1	-19.4	0.0	0.0	-0.2
	SLN116 ring / End	12.2	-0.3	-19.4	0.0	0.0	-0.4
C LC_127	SLN113 ring / End	6.4	0.0	-4.9	0.0	0.0	0.0
	SLN116 ring / End	6.2	0.0	-4.9	0.0	0.0	0.0
C LC_128	SLN113 ring / End	3.6	0.1	-4.9	0.0	0.0	0.2
	SLN116 ring / End	3.7	0.0	-4.9	0.0	0.0	0.0
C LC_129	SLN113 ring / End	6.4	0.2	-4.9	0.0	0.0	0.3
	SLN116 ring / End	6.8	0.1	-4.9	0.0	0.0	0.2
C LC_130	SLN113 ring / End	3.6	-0.2	-4.9	0.0	0.0	-0.2
	SLN116 ring / End	3.3	-0.2	-4.9	0.0	0.0	-0.3
C LC_135	SLN113 ring / End	15.9	0.1	-25.6	0.0	0.0	0.2
	SLN116 ring / End	15.9	-0.1	-25.6	0.0	0.0	-0.2
C LC_141	SLN113 ring / End	5.0	0.0	-4.9	0.0	0.0	0.0
	SLN116 ring / End	5.0	0.0	-4.9	0.0	0.0	0.0
C LC_142	SLN113 ring / End	4.5	0.1	-4.9	0.0	0.0	0.1
	SLN116 ring / End	4.5	-0.1	-4.9	0.0	0.0	-0.1
C LC_143	SLN113 ring / End	4.7	0.0	-4.9	0.0	0.0	0.1
	SLN116 ring / End	4.7	0.0	-4.9	0.0	0.0	0.0
C LC_144	SLN113 ring / End	4.8	0.0	-4.9	0.0	0.0	0.0
	SLN116 ring / End	4.8	0.0	-4.9	0.0	0.0	-0.1
C LC_145	SLN113 ring / End	20.1	0.0	-25.6	0.0	0.0	0.1
	SLN116 ring / End	20.1	-0.1	-25.6	0.0	0.0	-0.1
C LC_146	SLN113 ring / End	18.1	0.2	-25.6	0.0	0.0	0.2
	SLN116 ring / End	18.2	-0.2	-25.6	0.0	0.0	-0.2
C LC_147	SLN113 ring / End	18.9	0.2	-25.6	0.0	0.0	0.2
	SLN116 ring / End	18.8	-0.1	-25.6	0.0	0.0	-0.2
C LC_148	SLN113 ring / End	19.3	0.1	-25.6	0.0	0.0	0.1
	SLN116 ring / End	19.5	-0.1	-25.6	0.0	0.0	-0.2
C LC_151	SLN113 ring / End	12.1	0.1	-11.6	0.0	0.0	0.1
	SLN116 ring / End	12.2	-0.1	-11.5	0.0	0.0	-0.1
C LC_152	SLN113 ring / End	19.0	0.1	-14.2	0.0	0.0	0.2
	SLN116 ring / End	19.1	-0.1	-14.0	0.0	0.0	-0.1
C LC_153	SLN113 ring / End	19.1	0.1	-20.8	0.0	0.0	0.2
	SLN116 ring / End	19.2	-0.1	-20.6	0.0	0.0	-0.2
C LC_154	SLN113 ring / End	12.2	0.1	-11.6	0.0	0.0	0.1
	SLN116 ring / End	12.3	-0.1	-11.5	0.0	0.0	-0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.9 < 100%	OK
Welds	16.6 < 100%	OK
Buckling	Not calculated	

Plates

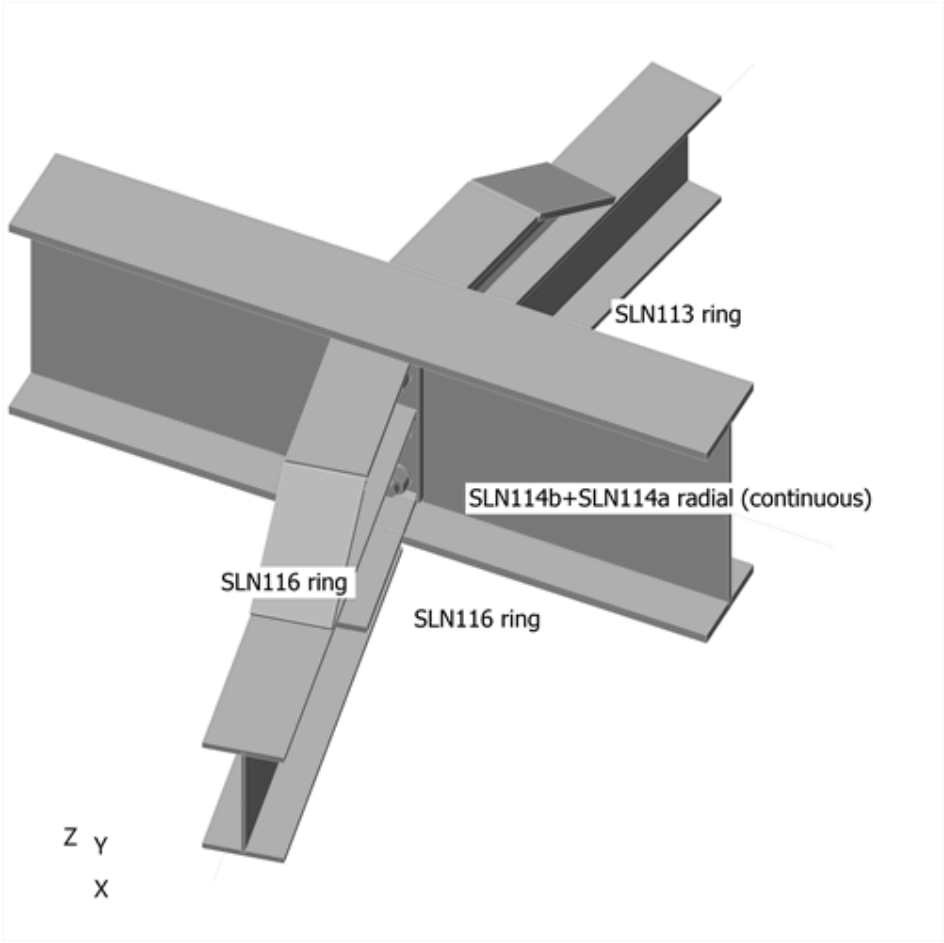
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN114b+SLN114a radial (continuous)-bfl 1	19.6	C LC_148	20.2	0.0	0.0	OK
SLN114b+SLN114a radial (continuous)-tfl 1	19.6	C LC_122	17.1	0.0	0.0	OK
SLN114b+SLN114a radial (continuous)-w 1	11.4	C LC_122	25.7	0.0	0.5	OK
SLN113 ring-bfl 1	12.7	C LC_145	50.6	0.0	0.0	OK
SLN113 ring-tfl 1	12.7	C LC_121	44.1	0.0	0.0	OK
SLN113 ring-w 1	7.2	C LC_145	49.2	0.0	0.0	OK
SLN116 ring-bfl 1	12.7	C LC_145	50.4	0.0	0.0	OK
SLN116 ring-tfl 1	12.7	C LC_122	44.4	0.0	0.0	OK
SLN116 ring-w 1	7.2	C LC_145	49.0	0.0	0.0	OK
End plate SLN116 ring	20.0	C LC_145	8.8	0.0	0.6	OK
Bottom flange plate SLN116 ring	15.0	C LC_145	20.5	0.0	0.0	OK
Bottom flange half plate L SLN116 ring	15.0	C LC_145	20.0	0.0	0.0	OK
Bottom flange half plate R SLN116 ring	15.0	C LC_145	19.9	0.0	0.0	OK
Haunch web SLN116 ring	12.0	C LC_120	28.0	0.0	0.0	OK
Haunch top plate SLN116 ring	15.0	C LC_122	5.6	0.0	0.0	OK
Haunch sloping plate SLN116 ring	15.0	C LC_120	12.6	0.0	0.0	OK
End plate SLN113 ring	20.0	C LC_103	9.4	0.0	0.7	OK
Bottom flange plate SLN113 ring	15.0	C LC_145	20.5	0.0	0.0	OK
Bottom flange half plate L SLN113 ring	15.0	C LC_145	19.9	0.0	0.0	OK
Bottom flange half plate R SLN113 ring	15.0	C LC_145	20.1	0.0	0.0	OK
Haunch web SLN113 ring	12.0	C LC_135	29.0	0.0	0.0	OK
Haunch top plate SLN113 ring	15.0	C LC_121	5.6	0.0	0.0	OK
Haunch sloping plate SLN113 ring	15.0	C LC_120	12.7	0.0	0.0	OK

Design data

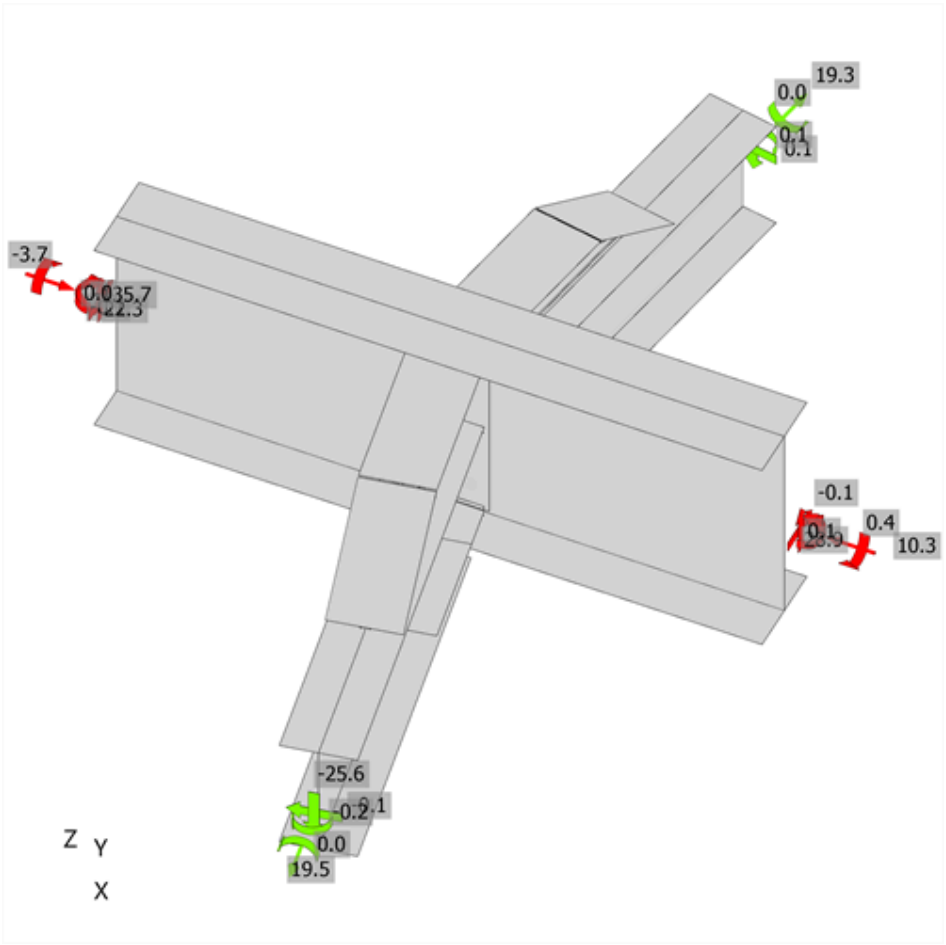
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

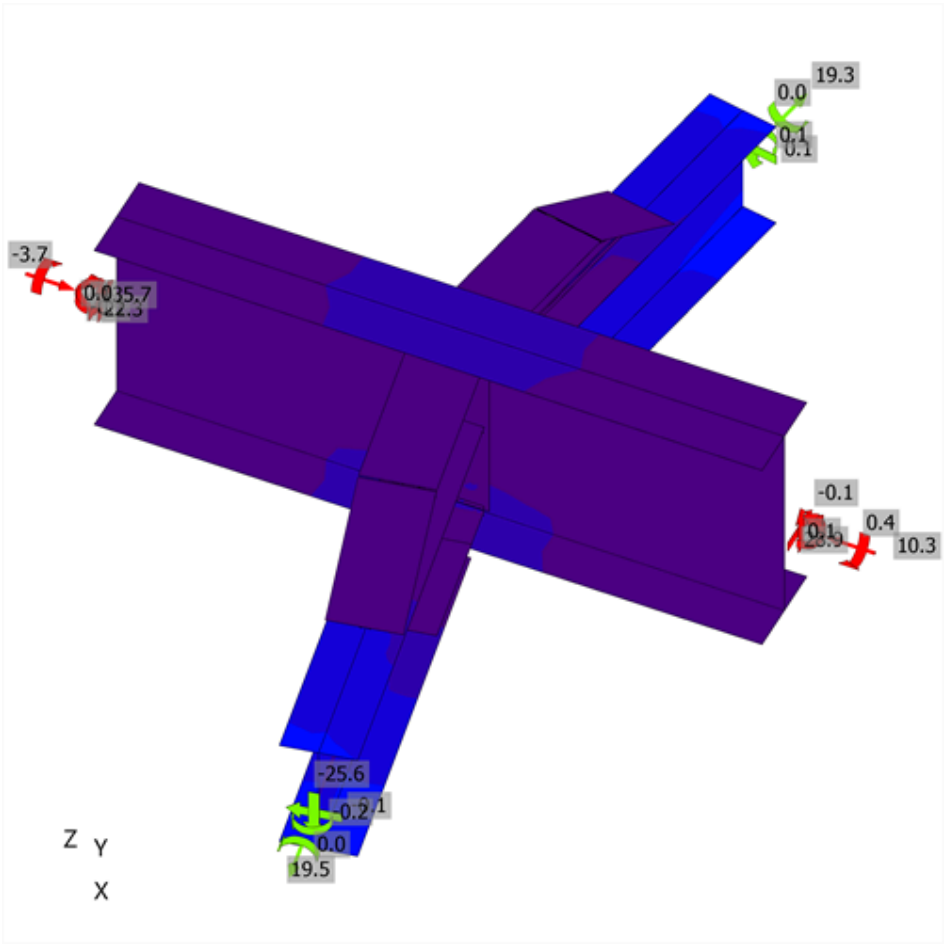
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

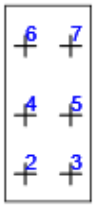


Strain check, C LC_148



Equivalent stress, C LC_148

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_146	2.3	0.5	194.1	0.7	0.5	0.7	OK	OK
	B3	M30 8.8 - 1	C LC_119	2.7	0.7	194.1	0.8	0.7	0.9	OK	OK
	B4	M30 8.8 - 1	C LC_119	0.1	0.4	232.2	0.0	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_145	0.1	0.5	232.2	0.0	0.4	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_121	2.4	0.8	249.2	0.7	0.7	0.9	OK	OK
	B7	M30 8.8 - 1	C LC_121	1.4	0.9	249.2	0.4	0.7	0.7	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN116 ring	Bottom flange plate SLN116 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN116 ring	Bottom flange plate SLN116 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN113 ring	Bottom flange plate SLN113 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN113 ring	Bottom flange plate SLN113 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN114b+SLN114a radial (continuous)-w 1	SLN113 ring-tfl 1	◢ 11.4	149	C LC_121	11.8	0.0	-7.5	-5.2	-0.5	2.9	2.9	OK	OK
SLN114b+SLN114a radial (continuous)-w 1	SLN116 ring-tfl 1	◢ 11.4	149	C LC_145	11.8	0.0	-7.4	-5.3	0.2	2.9	0.0	OK	OK
End plate SLN116 ring	SLN116 ring-w 1	◢ 6.0 ◣	208	C LC_145	24.5	0.0	-9.8	-12.5	-3.4	6.0	5.9	OK	OK
		◢ 6.0 ◣	208	C LC_145	19.5	0.0	-11.1	9.3	0.3	4.8	4.8	OK	OK
Bottom flange plate SLN116 ring	SLN116 ring-w 1	◢ 6.0 ◣	100	C LC_121	24.3	0.0	2.4	2.7	13.7	6.0	6.0	OK	OK
		◢ 6.0 ◣	100	C LC_122	25.2	0.0	2.5	-3.2	-14.1	6.2	6.2	OK	OK
SLN116 ring-w 1	Bottom flange half plate L SLN116 ring	◢ 6.0 ◣	243	C LC_145	24.6	0.0	6.0	6.1	12.4	6.1	6.1	OK	OK
		◢ 6.0 ◣	243	C LC_145	27.9	0.0	8.2	2.6	-15.2	6.9	6.5	OK	OK
SLN116 ring-w 1	Bottom flange half plate R SLN116 ring	◢ 6.0 ◣	243	C LC_148	25.0	0.0	6.1	6.1	-12.6	6.2	5.8	OK	OK
		◢ 6.0 ◣	243	C LC_145	27.4	0.0	8.0	2.6	14.9	6.8	6.4	OK	OK
SLN116 ring-tfl 1	Haunch web SLN116 ring	◢ 6.0 ◣	572	C LC_122	63.7	0.0	-14.3	-15.6	32.3	15.7	9.2	OK	OK
		◢ 6.0 ◣	572	C LC_121	63.5	0.0	-16.2	16.4	-31.4	15.7	9.1	OK	OK
End plate SLN113 ring	SLN113 ring-w 1	◢ 6.0 ◣	208	C LC_145	19.4	0.0	-11.2	-9.1	-0.6	4.8	4.8	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_145	24.8	0.0	-9.7	12.8	3.2	6.1	5.9	OK	OK
Bottom flange plate SLN113 ring	SLN113 ring-w 1	◢ 6.0 ◣	100	C LC_121	25.2	0.0	2.6	3.0	14.2	6.2	6.2	OK	OK
		◢ 6.0 ◣	100	C LC_122	24.5	0.0	2.4	-3.2	-13.7	6.1	6.1	OK	OK
SLN113 ring-w 1	Bottom flange half plate L SLN113 ring	◢ 6.0 ◣	243	C LC_145	25.6	0.0	6.6	6.2	12.9	6.3	6.3	OK	OK
		◢ 6.0 ◣	243	C LC_145	27.1	0.0	8.1	2.6	-14.7	6.7	6.2	OK	OK
SLN113 ring-w 1	Bottom flange half plate R SLN113 ring	◢ 6.0 ◣	243	C LC_145	24.1	0.0	5.6	5.9	-12.1	5.9	5.8	OK	OK
		◢ 6.0 ◣	243	C LC_145	28.4	0.0	8.1	2.6	15.5	7.0	6.7	OK	OK
SLN113 ring-tfl 1	Haunch web SLN113 ring	◢ 6.0 ◣	572	C LC_122	66.3	0.0	-17.0	-17.7	32.5	16.4	9.4	OK	OK
		◢ 6.0 ◣	572	C LC_120	61.8	0.0	-14.1	14.6	-31.6	15.3	9.0	OK	OK
End plate SLN116 ring	Bottom flange plate SLN116 ring	◢ 6.0 ◣	148	C LC_145	14.5	0.0	7.5	4.4	-5.7	3.6	3.6	OK	OK
		◢ 6.0 ◣	149	C LC_126	18.4	0.0	-7.1	5.7	-8.0	4.6	4.6	OK	OK
End plate SLN116 ring	Haunch web SLN116 ring	◢ 6.0 ◣	145	C LC_121	7.5	0.0	1.3	1.3	4.0	1.8	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	10.5	0.0	2.3	-2.4	-5.4	2.6	0.0	OK	OK
End plate SLN116 ring	Haunch top plate SLN116 ring	◢ 6.0 ◣	149	C LC_126	8.3	0.0	-3.6	-3.2	2.9	2.0	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_126	7.9	0.0	-2.8	3.2	-2.8	1.9	0.0	OK	OK
Haunch top plate SLN116 ring	Haunch web SLN116 ring	◢ 6.0 ◣	342	C LC_122	4.5	0.0	1.6	1.5	1.9	1.1	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	3.9	0.0	-0.2	0.2	2.2	1.0	0.0	OK	OK
Haunch sloping plate SLN116 ring	Haunch web SLN116 ring	◢ 6.0 ◣	271	C LC_135	50.0	0.0	-8.5	-7.4	27.5	12.4	8.1	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_120	52.0	0.0	-9.6	10.2	-27.7	12.9	8.2	OK	OK
End plate SLN113 ring	Bottom flange plate SLN113 ring	◢ 6.0 ◣	148	C LC_145	14.8	0.0	7.4	4.3	6.0	3.7	3.7	OK	OK
		◢ 6.0 ◣	149	C LC_126	17.8	0.0	-8.6	8.2	-3.7	4.4	4.4	OK	OK
End plate SLN113 ring	Haunch web SLN113 ring	◢ 6.0 ◣	145	C LC_121	10.3	0.0	2.4	2.5	5.2	2.5	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	7.7	0.0	1.3	-1.3	-4.2	1.9	0.0	OK	OK
End plate SLN113 ring	Haunch top plate SLN113 ring	◢ 6.0 ◣	149	C LC_125	8.4	0.0	-3.6	-3.3	-2.9	2.1	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_125	8.2	0.0	-3.0	3.3	2.9	2.0	0.0	OK	OK
Haunch top plate SLN113 ring	Haunch web SLN113 ring	◢ 6.0 ◣	342	C LC_121	3.7	0.0	-0.2	-0.2	-2.1	0.9	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_121	4.6	0.0	1.7	-1.5	-2.0	1.1	0.0	OK	OK
Haunch sloping plate SLN113 ring	Haunch web SLN113 ring	◢ 6.0 ◣	271	C LC_122	54.8	0.0	-10.6	-9.4	29.6	13.5	8.5	OK	OK
		◢ 6.0 ◣	271	C LC_120	47.9	0.0	-7.6	7.0	-26.4	11.8	8.0	OK	OK
SLN114b+SLN114a radial (continuous)-w 1	SLN116 ring-w 1	◢ 7.2	208	C LC_145	67.4	0.0	35.5	-28.8	16.2	16.6	11.9	OK	OK
SLN114b+SLN114a radial (continuous)-w 1	SLN113 ring-w 1	◢ 7.2	208	C LC_145	67.3	0.0	35.9	28.7	-16.0	16.6	11.9	OK	OK
Haunch sloping plate SLN116 ring	Haunch top plate SLN116 ring	◢ 6.0 ◣	146	C LC_122	7.2	0.0	-5.4	2.7	0.3	1.9	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_126	3.1	0.0	-0.7	1.7	0.1	0.8	0.0	OK	OK
Haunch sloping plate SLN113 ring	Haunch top plate SLN113 ring	◢ 6.0 ◣	146	C LC_122	7.0	0.0	-5.0	2.8	0.3	1.9	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_121	3.1	0.0	-0.7	1.7	-0.3	0.8	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_220
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN115b+SLN115a radial (continuous) - SLN116 ring - SLN117 ring

Name	SLN115b+SLN115a radial (continuous) - SLN116 ring - SLN117 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

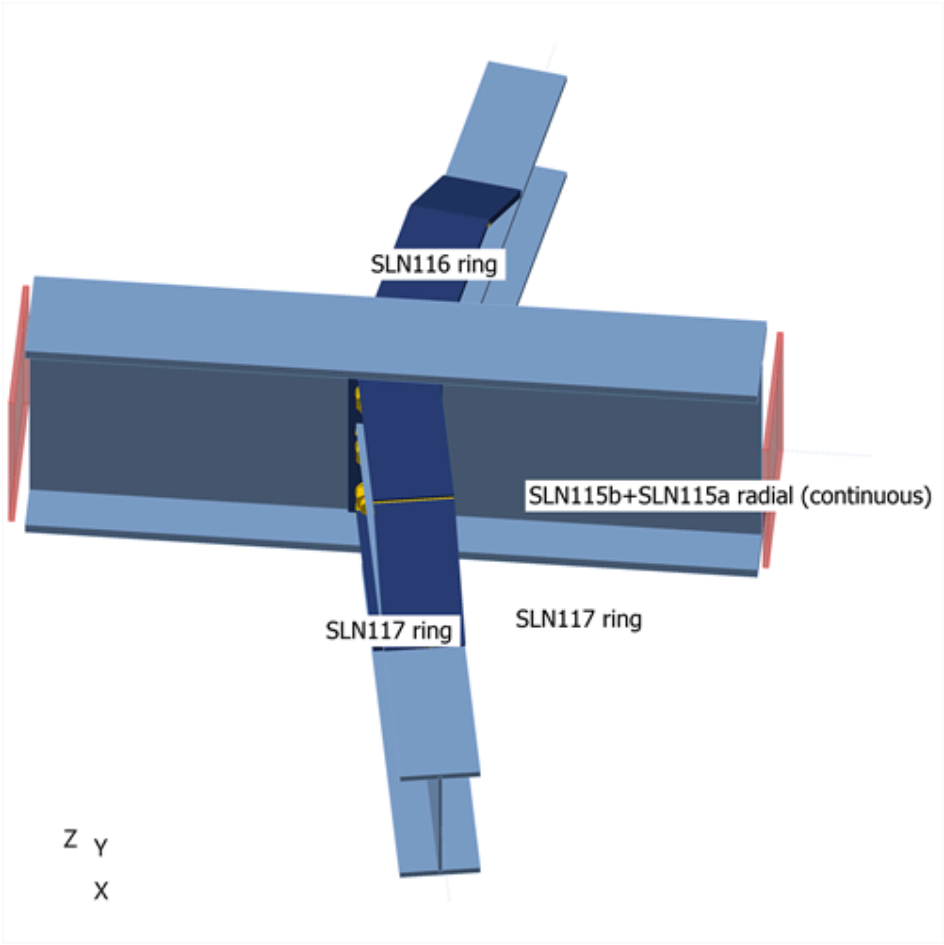
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN115b+SLN115a radial (continuous)	1 - UB457x191x98	-140.0	0.0	0.0	0	0	0
SLN116 ring	2 - UB254x146x43	120.0	0.0	0.0	0	0	-102
SLN117 ring	2 - UB254x146x43	-40.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN115b+SLN115a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN115b+SLN115a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN116 ring / end		Position	0
SLN117 ring / end		Position	0



SLN115b+SLN115a radial (continuous) - SLN116 ring - SLN117 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN116 ring / End	4.8	0.0	-4.9	0.0	0.0	-0.1
	SLN117 ring / End	4.8	0.1	-5.0	0.0	0.0	0.1
C LC_102	SLN116 ring / End	19.2	-0.1	-25.6	0.0	0.0	-0.2
	SLN117 ring / End	19.2	0.2	-26.0	0.0	0.0	0.3
C LC_103	SLN116 ring / End	17.6	-0.2	-24.6	0.0	0.0	-0.2
	SLN117 ring / End	17.2	0.2	-24.9	0.0	0.0	0.2
C LC_104	SLN116 ring / End	14.1	0.0	-24.6	0.0	0.0	-0.1
	SLN117 ring / End	14.4	0.1	-24.9	0.0	0.0	0.2
C LC_105	SLN116 ring / End	15.9	0.0	-24.6	0.0	0.0	0.0
	SLN117 ring / End	16.1	0.2	-24.9	0.0	0.0	0.3
C LC_106	SLN116 ring / End	15.5	-0.2	-24.6	0.0	0.0	-0.3
	SLN117 ring / End	15.7	0.1	-24.9	0.0	0.0	0.1
C LC_107	SLN116 ring / End	3.1	0.0	-1.6	0.0	0.0	0.0
	SLN117 ring / End	2.5	0.0	-1.6	0.0	0.0	-0.1
C LC_108	SLN116 ring / End	-0.8	0.1	-1.6	0.0	0.0	0.1
	SLN117 ring / End	-0.6	0.1	-1.6	0.0	0.0	0.2
C LC_109	SLN116 ring / End	2.2	0.1	-1.6	0.0	0.0	0.2
	SLN117 ring / End	2.6	0.2	-1.6	0.0	0.0	0.2
C LC_110	SLN116 ring / End	-0.6	-0.1	-1.6	0.0	0.0	-0.1
	SLN117 ring / End	-0.5	-0.1	-1.6	0.0	0.0	-0.2
C LC_111	SLN116 ring / End	13.5	-0.1	-17.3	0.0	0.0	-0.2
	SLN117 ring / End	12.8	0.1	-17.6	0.0	0.0	0.1
C LC_112	SLN116 ring / End	8.6	0.0	-17.3	0.0	0.0	0.0
	SLN117 ring / End	8.9	0.2	-17.6	0.0	0.0	0.2
C LC_113	SLN116 ring / End	11.7	0.1	-17.3	0.0	0.0	0.1
	SLN117 ring / End	12.1	0.2	-17.6	0.0	0.0	0.4
C LC_114	SLN116 ring / End	9.5	-0.2	-17.3	0.0	0.0	-0.3
	SLN117 ring / End	9.8	0.0	-17.6	0.0	0.0	-0.1
C LC_115	SLN116 ring / End	4.4	0.0	-2.8	0.0	0.0	-0.1
	SLN117 ring / End	3.8	0.0	-2.9	0.0	0.0	-0.1
C LC_116	SLN116 ring / End	0.4	0.1	-2.8	0.0	0.0	0.1
	SLN117 ring / End	0.6	0.1	-2.9	0.0	0.0	0.2
C LC_117	SLN116 ring / End	3.4	0.1	-2.8	0.0	0.0	0.2
	SLN117 ring / End	3.8	0.2	-2.9	0.0	0.0	0.3
C LC_118	SLN116 ring / End	0.6	-0.1	-2.8	0.0	0.0	-0.2
	SLN117 ring / End	0.8	-0.1	-2.9	0.0	0.0	-0.1
C LC_119	SLN116 ring / End	18.5	-0.2	-25.6	0.0	0.0	-0.2
	SLN117 ring / End	18.4	0.3	-26.0	0.0	0.0	0.3
C LC_120	SLN116 ring / End	15.8	0.0	-25.6	0.0	0.0	-0.1
	SLN117 ring / End	16.0	0.1	-26.0	0.0	0.0	0.2
C LC_121	SLN116 ring / End	17.6	0.0	-25.6	0.0	0.0	0.0
	SLN117 ring / End	17.9	0.2	-26.0	0.0	0.0	0.3
C LC_122	SLN116 ring / End	16.8	-0.3	-25.6	0.0	0.0	-0.4
	SLN117 ring / End	16.7	0.1	-26.0	0.0	0.0	0.1
C LC_123	SLN116 ring / End	15.2	-0.1	-19.4	0.0	0.0	-0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN117 ring / End	15.1	0.2	-19.7	0.0	0.0	0.3
C LC_124	SLN116 ring / End	11.9	0.0	-19.4	0.0	0.0	0.0
	SLN117 ring / End	12.1	0.1	-19.7	0.0	0.0	0.2
C LC_125	SLN116 ring / End	15.1	0.1	-19.4	0.0	0.0	0.1
	SLN117 ring / End	15.6	0.3	-19.7	0.0	0.0	0.4
C LC_126	SLN116 ring / End	12.3	-0.3	-19.4	0.0	0.0	-0.5
	SLN117 ring / End	11.9	0.0	-19.7	0.0	0.0	-0.1
C LC_127	SLN116 ring / End	6.2	0.0	-4.9	0.0	0.0	-0.1
	SLN117 ring / End	6.1	0.1	-5.0	0.0	0.0	0.1
C LC_128	SLN116 ring / End	3.7	0.0	-4.9	0.0	0.0	0.0
	SLN117 ring / End	3.8	0.1	-5.0	0.0	0.0	0.1
C LC_129	SLN116 ring / End	6.8	0.1	-4.9	0.0	0.0	0.2
	SLN117 ring / End	7.3	0.2	-5.0	0.0	0.0	0.3
C LC_130	SLN116 ring / End	3.4	-0.2	-4.9	0.0	0.0	-0.3
	SLN117 ring / End	2.9	-0.1	-5.0	0.0	0.0	-0.2
C LC_135	SLN116 ring / End	15.9	-0.1	-25.6	0.0	0.0	-0.2
	SLN117 ring / End	15.9	0.2	-26.0	0.0	0.0	0.2
C LC_141	SLN116 ring / End	5.0	-0.1	-4.9	0.0	0.0	-0.1
	SLN117 ring / End	4.9	0.1	-5.0	0.0	0.0	0.1
C LC_142	SLN116 ring / End	4.6	0.0	-4.9	0.0	0.0	-0.1
	SLN117 ring / End	4.6	0.0	-5.0	0.0	0.0	0.1
C LC_143	SLN116 ring / End	4.7	0.0	-4.9	0.0	0.0	0.0
	SLN117 ring / End	4.6	0.0	-5.0	0.0	0.0	0.1
C LC_144	SLN116 ring / End	4.9	-0.1	-4.9	0.0	0.0	-0.1
	SLN117 ring / End	4.9	0.1	-5.0	0.0	0.0	0.1
C LC_145	SLN116 ring / End	20.0	-0.2	-25.6	0.0	0.0	-0.2
	SLN117 ring / End	19.9	0.3	-26.0	0.0	0.0	0.3
C LC_146	SLN116 ring / End	18.3	-0.1	-25.6	0.0	0.0	-0.2
	SLN117 ring / End	18.5	0.1	-26.0	0.0	0.0	0.2
C LC_147	SLN116 ring / End	18.6	-0.1	-25.6	0.0	0.0	-0.1
	SLN117 ring / End	18.6	0.2	-26.0	0.0	0.0	0.3
C LC_148	SLN116 ring / End	19.7	-0.2	-25.6	0.0	0.0	-0.3
	SLN117 ring / End	19.8	0.2	-26.0	0.0	0.0	0.2
C LC_151	SLN116 ring / End	12.2	-0.1	-11.6	0.0	0.0	-0.1
	SLN117 ring / End	12.2	0.1	-11.9	0.0	0.0	0.2
C LC_152	SLN116 ring / End	19.1	-0.1	-14.0	0.0	0.0	-0.1
	SLN117 ring / End	19.3	0.2	-14.0	0.0	0.0	0.3
C LC_153	SLN116 ring / End	19.2	-0.1	-20.7	0.0	0.0	-0.2
	SLN117 ring / End	19.2	0.2	-20.9	0.0	0.0	0.3
C LC_154	SLN116 ring / End	12.3	-0.1	-11.6	0.0	0.0	-0.1
	SLN117 ring / End	12.3	0.1	-11.9	0.0	0.0	0.2

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.9 < 100%	OK
Welds	16.9 < 100%	OK
Buckling	Not calculated	

Plates

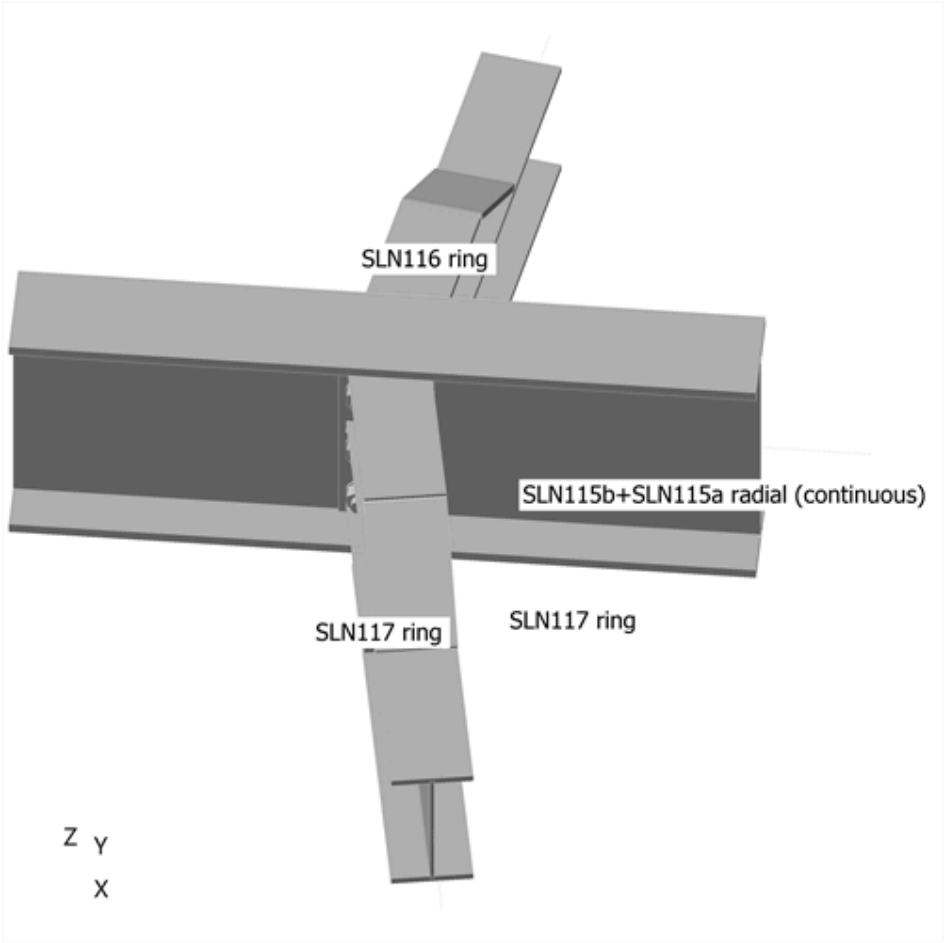
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN115b+SLN115a radial (continuous)-bfl 1	19.6	C LC_119	20.3	0.0	0.0	OK
SLN115b+SLN115a radial (continuous)-tfl 1	19.6	C LC_121	17.2	0.0	0.0	OK
SLN115b+SLN115a radial (continuous)-w 1	11.4	C LC_121	25.7	0.0	0.5	OK
SLN116 ring-bfl 1	12.7	C LC_146	50.2	0.0	0.0	OK
SLN116 ring-tfl 1	12.7	C LC_122	44.5	0.0	0.0	OK
SLN116 ring-w 1	7.2	C LC_145	49.0	0.0	0.0	OK
SLN117 ring-bfl 1	12.7	C LC_121	50.8	0.0	0.0	OK
SLN117 ring-tfl 1	12.7	C LC_121	45.0	0.0	0.0	OK
SLN117 ring-w 1	7.2	C LC_148	49.7	0.0	0.0	OK
End plate SLN117 ring	20.0	C LC_145	9.1	0.0	0.7	OK
Bottom flange plate SLN117 ring	15.0	C LC_145	20.7	0.0	0.0	OK
Bottom flange half plate L SLN117 ring	15.0	C LC_148	20.0	0.0	0.0	OK
Bottom flange half plate R SLN117 ring	15.0	C LC_145	20.2	0.0	0.0	OK
Haunch web SLN117 ring	12.0	C LC_120	29.1	0.0	0.0	OK
Haunch top plate SLN117 ring	15.0	C LC_121	6.0	0.0	0.0	OK
Haunch sloping plate SLN117 ring	15.0	C LC_135	12.8	0.0	0.0	OK
End plate SLN116 ring	20.0	C LC_148	8.9	0.0	0.6	OK
Bottom flange plate SLN116 ring	15.0	C LC_145	20.5	0.0	0.0	OK
Bottom flange half plate L SLN116 ring	15.0	C LC_145	20.1	0.0	0.0	OK
Bottom flange half plate R SLN116 ring	15.0	C LC_145	19.9	0.0	0.0	OK
Haunch web SLN116 ring	12.0	C LC_120	28.0	0.0	0.0	OK
Haunch top plate SLN116 ring	15.0	C LC_122	6.0	0.0	0.0	OK
Haunch sloping plate SLN116 ring	15.0	C LC_120	12.6	0.0	0.0	OK

Design data

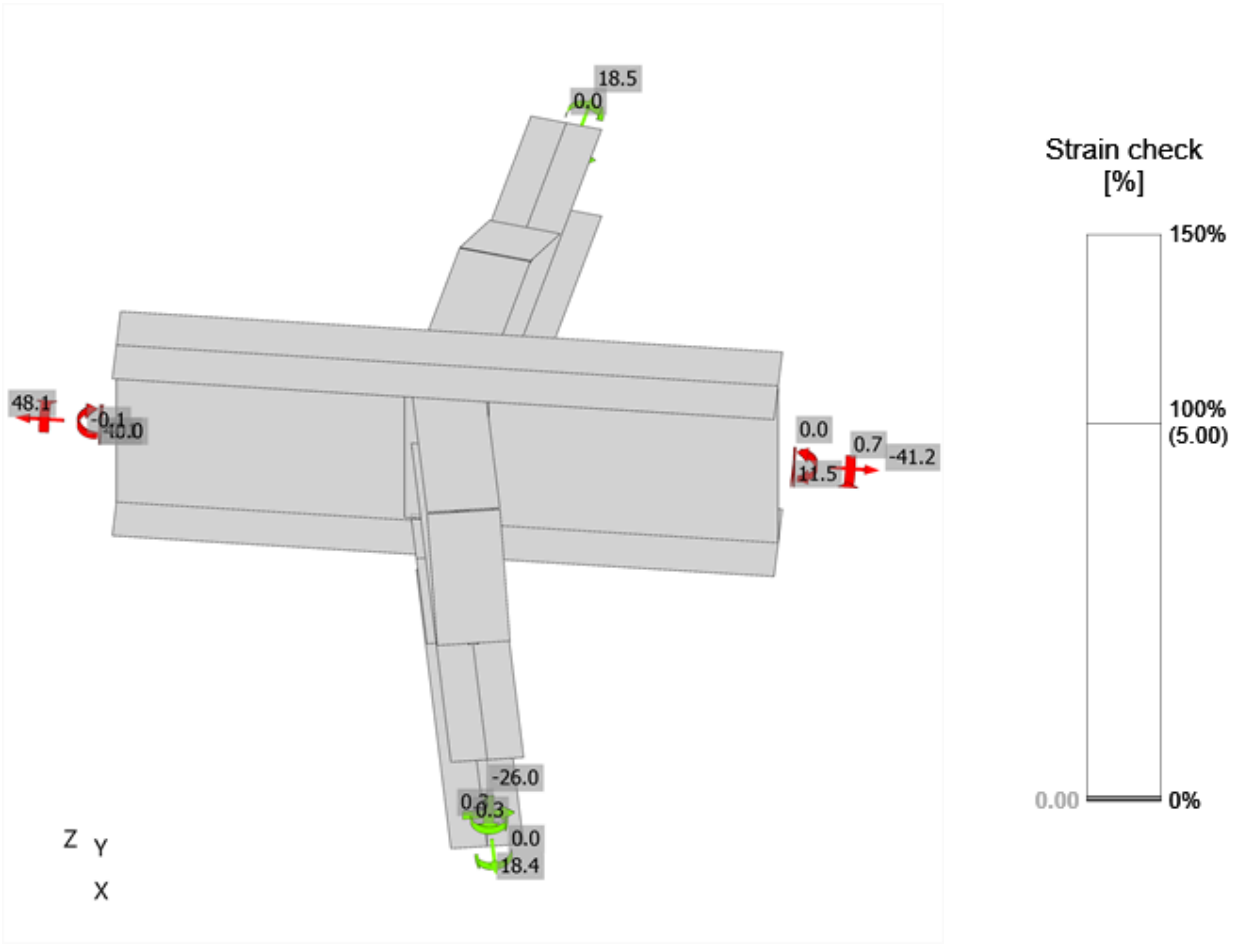
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

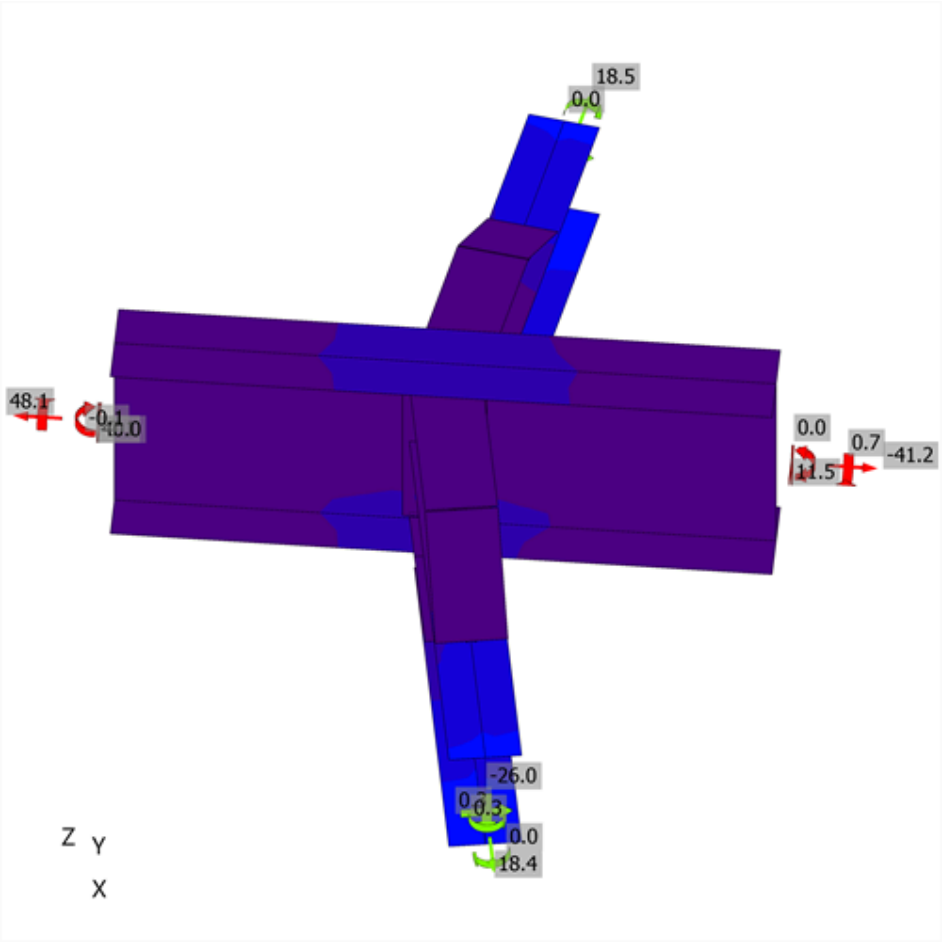
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

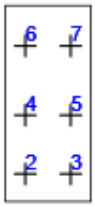


Strain check, C LC_119



Equivalent stress, C LC_119

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_121	2.6	0.5	194.1	0.8	0.5	0.8	OK	OK
	B3	M30 8.8 - 1	C LC_145	2.6	0.7	194.1	0.8	0.7	0.9	OK	OK
	B4	M30 8.8 - 1	C LC_135	0.1	0.4	232.2	0.0	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_148	0.1	0.5	232.2	0.0	0.4	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_121	1.7	0.9	249.2	0.5	0.7	0.8	OK	OK
	B7	M30 8.8 - 1	C LC_121	2.2	0.9	249.2	0.7	0.7	0.9	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate R SLN117 ring	Bottom flange plate SLN117 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN116 ring	Bottom flange plate SLN116 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN117 ring	Bottom flange plate SLN117 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN116 ring	Bottom flange plate SLN116 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN115b+SLN115a radial (continuous)-w 1	SLN116 ring-tfl 1	◢ 11.4	149	C LC_122	12.3	0.0	-7.3	-5.7	0.4	3.0	3.0	OK	OK
SLN115b+SLN115a radial (continuous)-w 1	SLN117 ring-tfl 1	◢ 11.4	149	C LC_121	12.4	0.0	-7.2	-5.8	-0.6	3.1	3.1	OK	OK
End plate SLN117 ring	SLN117 ring-w 1	◢ 6.0 ◣	208	C LC_145	24.1	0.0	-9.8	-12.4	-3.0	6.0	5.9	OK	OK
		◢ 6.0 ◣	208	C LC_148	19.7	0.0	-10.8	9.5	0.4	4.9	4.9	OK	OK
Bottom flange plate SLN117 ring	SLN117 ring-w 1	◢ 6.0 ◣	100	C LC_121	25.4	0.0	2.6	3.0	14.3	6.3	6.3	OK	OK
		◢ 6.0 ◣	100	C LC_122	24.6	0.0	2.4	-3.0	-13.8	6.1	6.1	OK	OK
SLN117 ring-w 1	Bottom flange half plate L SLN117 ring	◢ 6.0 ◣	243	C LC_145	26.0	0.0	6.5	6.3	13.1	6.4	6.4	OK	OK
		◢ 6.0 ◣	243	C LC_148	27.4	0.0	8.1	2.7	-14.9	6.8	6.3	OK	OK
SLN117 ring-w 1	Bottom flange half plate R SLN117 ring	◢ 6.0 ◣	243	C LC_148	24.0	0.0	5.7	5.9	-12.1	5.9	5.8	OK	OK
		◢ 6.0 ◣	243	C LC_145	28.4	0.0	8.6	2.4	15.4	7.0	6.5	OK	OK
SLN117 ring-tfl 1	Haunch web SLN117 ring	◢ 6.0 ◣	572	C LC_122	65.9	0.0	-17.2	-17.3	32.4	16.3	9.4	OK	OK
		◢ 6.0 ◣	572	C LC_121	63.2	0.0	-13.9	15.1	-32.3	15.6	9.1	OK	OK
End plate SLN116 ring	SLN116 ring-w 1	◢ 6.0 ◣	208	C LC_145	19.7	0.0	-10.8	-9.5	-0.4	4.9	4.9	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_145	24.2	0.0	-10.2	12.2	3.2	6.0	5.9	OK	OK
Bottom flange plate SLN116 ring	SLN116 ring-w 1	◢ 6.0 ◣	100	C LC_121	24.7	0.0	2.5	2.9	13.9	6.1	6.1	OK	OK
		◢ 6.0 ◣	100	C LC_122	25.2	0.0	2.5	-3.3	-14.1	6.2	6.2	OK	OK
SLN116 ring-w 1	Bottom flange half plate L SLN116 ring	◢ 6.0 ◣	243	C LC_121	24.4	0.0	5.7	6.0	12.3	6.0	6.0	OK	OK
		◢ 6.0 ◣	243	C LC_145	28.0	0.0	8.4	2.5	-15.2	6.9	6.4	OK	OK
SLN116 ring-w 1	Bottom flange half plate R SLN116 ring	◢ 6.0 ◣	243	C LC_148	25.2	0.0	6.1	6.1	-12.7	6.2	5.8	OK	OK
		◢ 6.0 ◣	243	C LC_145	27.4	0.0	7.9	2.7	14.9	6.8	6.5	OK	OK
SLN116 ring-tfl 1	Haunch web SLN116 ring	◢ 6.0 ◣	572	C LC_122	64.5	0.0	-14.3	-16.0	32.6	15.9	9.2	OK	OK
		◢ 6.0 ◣	572	C LC_121	63.7	0.0	-16.2	16.4	-31.6	15.7	9.1	OK	OK
End plate SLN117 ring	Bottom flange plate SLN117 ring	◢ 6.0 ◣	148	C LC_145	14.9	0.0	7.5	4.3	-6.0	3.7	3.7	OK	OK
		◢ 6.0 ◣	149	C LC_126	18.0	0.0	-6.9	5.5	-7.8	4.4	4.4	OK	OK
End plate SLN117 ring	Haunch web SLN117 ring	◢ 6.0 ◣	145	C LC_121	7.7	0.0	1.9	1.7	3.9	1.9	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	10.5	0.0	1.8	-2.3	-5.5	2.6	0.0	OK	OK
End plate SLN117 ring	Haunch top plate SLN117 ring	◢ 6.0 ◣	149	C LC_130	7.6	0.0	-3.7	-2.5	2.9	1.9	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_125	7.2	0.0	-2.8	2.9	2.4	1.8	0.0	OK	OK
Haunch top plate SLN117 ring	Haunch web SLN117 ring	◢ 6.0 ◣	342	C LC_121	3.8	0.0	-0.3	-0.2	-2.1	0.9	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_121	4.8	0.0	1.7	-1.5	-2.1	1.2	0.0	OK	OK
Haunch sloping plate SLN117 ring	Haunch web SLN117 ring	◢ 6.0 ◣	271	C LC_120	54.5	0.0	-10.4	-11.0	28.9	13.5	8.4	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_135	48.9	0.0	-7.9	7.4	-26.9	12.1	8.0	OK	OK
End plate SLN116 ring	Bottom flange plate SLN116 ring	◢ 6.0 ◣	148	C LC_148	14.7	0.0	7.3	4.3	6.0	3.6	3.6	OK	OK
		◢ 6.0 ◣	149	C LC_125	18.3	0.0	-6.9	5.7	8.0	4.5	4.5	OK	OK
End plate SLN116 ring	Haunch web SLN116 ring	◢ 6.0 ◣	145	C LC_121	10.4	0.0	1.9	2.4	5.4	2.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	7.6	0.0	1.9	-1.6	-3.9	1.9	0.0	OK	OK
End plate SLN116 ring	Haunch top plate SLN116 ring	◢ 6.0 ◣	149	C LC_125	7.8	0.0	-3.9	-2.3	-3.2	1.9	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_126	6.8	0.0	-2.7	2.7	-2.4	1.7	0.0	OK	OK
Haunch top plate SLN116 ring	Haunch web SLN116 ring	◢ 6.0 ◣	342	C LC_122	4.7	0.0	1.7	1.5	2.0	1.2	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	3.8	0.0	-0.3	0.3	2.2	0.9	0.0	OK	OK
Haunch sloping plate SLN116 ring	Haunch web SLN116 ring	◢ 6.0 ◣	271	C LC_122	50.5	0.0	-8.6	-6.4	28.0	12.5	8.2	OK	OK
		◢ 6.0 ◣	271	C LC_120	52.3	0.0	-9.6	10.1	-27.9	12.9	8.2	OK	OK
SLN115b+SLN115a radial (continuous)-w 1	SLN117 ring-w 1	◢ 7.2	208	C LC_148	68.1	0.0	34.4	-29.8	16.3	16.8	12.0	OK	OK
SLN115b+SLN115a radial (continuous)-w 1	SLN116 ring-w 1	◢ 7.2	208	C LC_145	68.3	0.0	34.4	29.9	-16.4	16.9	12.0	OK	OK
Haunch sloping plate SLN117 ring	Haunch top plate SLN117 ring	◢ 6.0 ◣	146	C LC_121	7.2	0.0	-5.5	2.7	-0.4	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_121	3.3	0.0	-0.7	1.8	-0.2	0.8	0.0	OK	OK
Haunch sloping plate SLN116 ring	Haunch top plate SLN116 ring	◢ 6.0 ◣	146	C LC_121	7.0	0.0	-5.0	2.9	0.3	1.9	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_126	3.3	0.0	-0.7	1.9	0.0	0.8	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_240
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN118a+SLN118b radial (continuous) - SLN117 ring - SLN120 ring

Name	SLN118a+SLN118b radial (continuous) - SLN117 ring - SLN120 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

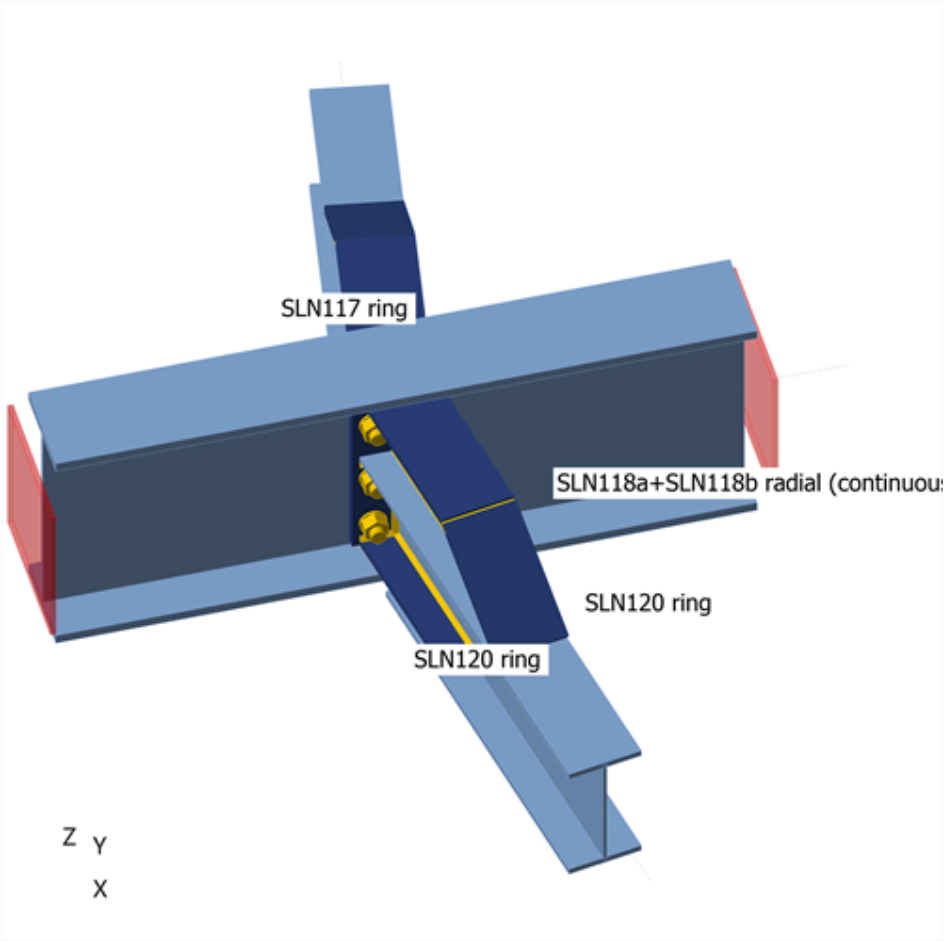
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN118a+SLN118b radial (continuous)	1 - UB457x191x98	-120.0	0.0	0.0	0	0	0
SLN117 ring	2 - UB254x146x43	140.0	0.0	0.0	0	0	-102
SLN120 ring	2 - UB254x146x43	-20.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN118a+SLN118b radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN118a+SLN118b radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN117 ring / end		Position	0
SLN120 ring / end		Position	0



SLN118a+SLN118b radial (continuous) - SLN117 ring - SLN120 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN117 ring / End	4.8	0.1	-5.0	0.0	0.0	0.1
	SLN120 ring / End	4.8	-0.2	-4.9	0.0	0.0	-0.2
C LC_102	SLN117 ring / End	19.2	0.2	-26.0	0.0	0.0	0.4
	SLN120 ring / End	19.2	-1.1	-25.7	0.0	0.0	-1.0
C LC_103	SLN117 ring / End	17.1	0.1	-24.9	0.0	0.0	0.3
	SLN120 ring / End	16.9	-1.0	-24.7	0.0	0.0	-0.9
C LC_104	SLN117 ring / End	14.5	0.3	-24.9	0.0	0.0	0.4
	SLN120 ring / End	14.8	-1.0	-24.7	0.0	0.0	-0.9
C LC_105	SLN117 ring / End	16.0	0.3	-24.9	0.0	0.0	0.5
	SLN120 ring / End	16.1	-1.1	-24.7	0.0	0.0	-1.0
C LC_106	SLN117 ring / End	15.8	0.0	-24.9	0.0	0.0	0.1
	SLN120 ring / End	15.7	-0.9	-24.7	0.0	0.0	-0.9
C LC_107	SLN117 ring / End	2.5	0.0	-1.6	0.0	0.0	0.0
	SLN120 ring / End	2.3	-0.2	-1.6	0.0	0.0	-0.1
C LC_108	SLN117 ring / End	-0.6	0.1	-1.6	0.0	0.0	0.1
	SLN120 ring / End	-0.5	0.0	-1.6	0.0	0.0	0.1
C LC_109	SLN117 ring / End	2.6	0.2	-1.6	0.0	0.0	0.3
	SLN120 ring / End	2.9	-0.2	-1.6	0.0	0.0	-0.1
C LC_110	SLN117 ring / End	-0.5	-0.1	-1.6	0.0	0.0	-0.2
	SLN120 ring / End	-0.7	0.0	-1.6	0.0	0.0	-0.1
C LC_111	SLN117 ring / End	12.7	0.0	-17.6	0.0	0.0	0.2
	SLN120 ring / End	12.4	-0.8	-17.4	0.0	0.0	-0.7
C LC_112	SLN117 ring / End	9.0	0.2	-17.6	0.0	0.0	0.3
	SLN120 ring / End	9.2	-0.6	-17.4	0.0	0.0	-0.6
C LC_113	SLN117 ring / End	12.1	0.3	-17.6	0.0	0.0	0.5
	SLN120 ring / End	12.3	-0.9	-17.4	0.0	0.0	-0.7
C LC_114	SLN117 ring / End	9.9	-0.1	-17.6	0.0	0.0	-0.1
	SLN120 ring / End	9.7	-0.6	-17.4	0.0	0.0	-0.7
C LC_115	SLN117 ring / End	3.7	0.0	-2.9	0.0	0.0	0.0
	SLN120 ring / End	3.6	-0.2	-2.9	0.0	0.0	-0.2
C LC_116	SLN117 ring / End	0.6	0.1	-2.9	0.0	0.0	0.2
	SLN120 ring / End	0.7	0.0	-2.9	0.0	0.0	0.0
C LC_117	SLN117 ring / End	3.8	0.2	-2.9	0.0	0.0	0.3
	SLN120 ring / End	4.1	-0.3	-2.9	0.0	0.0	-0.2
C LC_118	SLN117 ring / End	0.8	-0.1	-2.9	0.0	0.0	-0.2
	SLN120 ring / End	0.6	-0.1	-2.9	0.0	0.0	-0.1
C LC_119	SLN117 ring / End	18.3	0.2	-26.0	0.0	0.0	0.3
	SLN120 ring / End	18.0	-1.0	-25.7	0.0	0.0	-1.0
C LC_120	SLN117 ring / End	16.1	0.2	-26.0	0.0	0.0	0.4
	SLN120 ring / End	16.3	-1.1	-25.7	0.0	0.0	-1.0
C LC_121	SLN117 ring / End	17.7	0.3	-26.0	0.0	0.0	0.5
	SLN120 ring / End	17.8	-1.1	-25.7	0.0	0.0	-1.0
C LC_122	SLN117 ring / End	16.8	0.0	-26.0	0.0	0.0	0.1
	SLN120 ring / End	16.8	-1.0	-25.7	0.0	0.0	-1.0
C LC_123	SLN117 ring / End	15.0	0.1	-19.7	0.0	0.0	0.3

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN120 ring / End	14.7	-0.8	-19.5	0.0	0.0	-0.7
C LC_124	SLN117 ring / End	12.2	0.2	-19.7	0.0	0.0	0.3
	SLN120 ring / End	12.3	-0.7	-19.5	0.0	0.0	-0.7
C LC_125	SLN117 ring / End	15.6	0.4	-19.7	0.0	0.0	0.5
	SLN120 ring / End	15.9	-0.9	-19.5	0.0	0.0	-0.7
C LC_126	SLN117 ring / End	12.0	-0.1	-19.7	0.0	0.0	0.0
	SLN120 ring / End	11.9	-0.7	-19.5	0.0	0.0	-0.8
C LC_127	SLN117 ring / End	6.0	0.1	-5.0	0.0	0.0	0.1
	SLN120 ring / End	5.8	-0.2	-4.9	0.0	0.0	-0.2
C LC_128	SLN117 ring / End	3.8	0.1	-5.0	0.0	0.0	0.2
	SLN120 ring / End	3.9	-0.1	-4.9	0.0	0.0	-0.1
C LC_129	SLN117 ring / End	7.3	0.2	-5.0	0.0	0.0	0.3
	SLN120 ring / End	7.7	-0.3	-4.9	0.0	0.0	-0.2
C LC_130	SLN117 ring / End	2.9	-0.1	-5.0	0.0	0.0	-0.1
	SLN120 ring / End	2.8	-0.2	-4.9	0.0	0.0	-0.2
C LC_135	SLN117 ring / End	15.9	0.2	-26.0	0.0	0.0	0.3
	SLN120 ring / End	15.9	-0.8	-25.7	0.0	0.0	-0.7
C LC_141	SLN117 ring / End	4.9	0.0	-5.0	0.0	0.0	0.1
	SLN120 ring / End	4.9	-0.2	-4.9	0.0	0.0	-0.2
C LC_142	SLN117 ring / End	4.6	0.1	-5.0	0.0	0.0	0.1
	SLN120 ring / End	4.7	-0.2	-4.9	0.0	0.0	-0.2
C LC_143	SLN117 ring / End	4.6	0.1	-5.0	0.0	0.0	0.1
	SLN120 ring / End	4.6	-0.2	-4.9	0.0	0.0	-0.2
C LC_144	SLN117 ring / End	4.9	0.0	-5.0	0.0	0.0	0.1
	SLN120 ring / End	4.9	-0.2	-4.9	0.0	0.0	-0.2
C LC_145	SLN117 ring / End	19.8	0.2	-26.0	0.0	0.0	0.4
	SLN120 ring / End	19.6	-1.0	-25.7	0.0	0.0	-1.0
C LC_146	SLN117 ring / End	18.6	0.2	-26.0	0.0	0.0	0.4
	SLN120 ring / End	18.8	-1.1	-25.7	0.0	0.0	-1.1
C LC_147	SLN117 ring / End	18.5	0.3	-26.0	0.0	0.0	0.4
	SLN120 ring / End	18.4	-1.1	-25.7	0.0	0.0	-1.0
C LC_148	SLN117 ring / End	19.9	0.1	-26.0	0.0	0.0	0.3
	SLN120 ring / End	20.0	-1.0	-25.7	0.0	0.0	-1.0
C LC_151	SLN117 ring / End	12.2	0.1	-11.9	0.0	0.0	0.2
	SLN120 ring / End	12.2	-0.7	-11.8	0.0	0.0	-0.6
C LC_152	SLN117 ring / End	19.3	0.2	-14.0	0.0	0.0	0.4
	SLN120 ring / End	19.4	-1.2	-13.8	0.0	0.0	-1.1
C LC_153	SLN117 ring / End	19.2	0.2	-20.9	0.0	0.0	0.4
	SLN120 ring / End	19.2	-1.1	-20.7	0.0	0.0	-1.0
C LC_154	SLN117 ring / End	12.3	0.1	-11.9	0.0	0.0	0.2
	SLN120 ring / End	12.3	-0.7	-11.8	0.0	0.0	-0.6

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	1.5 < 100%	OK
Welds	19.4 < 100%	OK
Buckling	Not calculated	

Plates

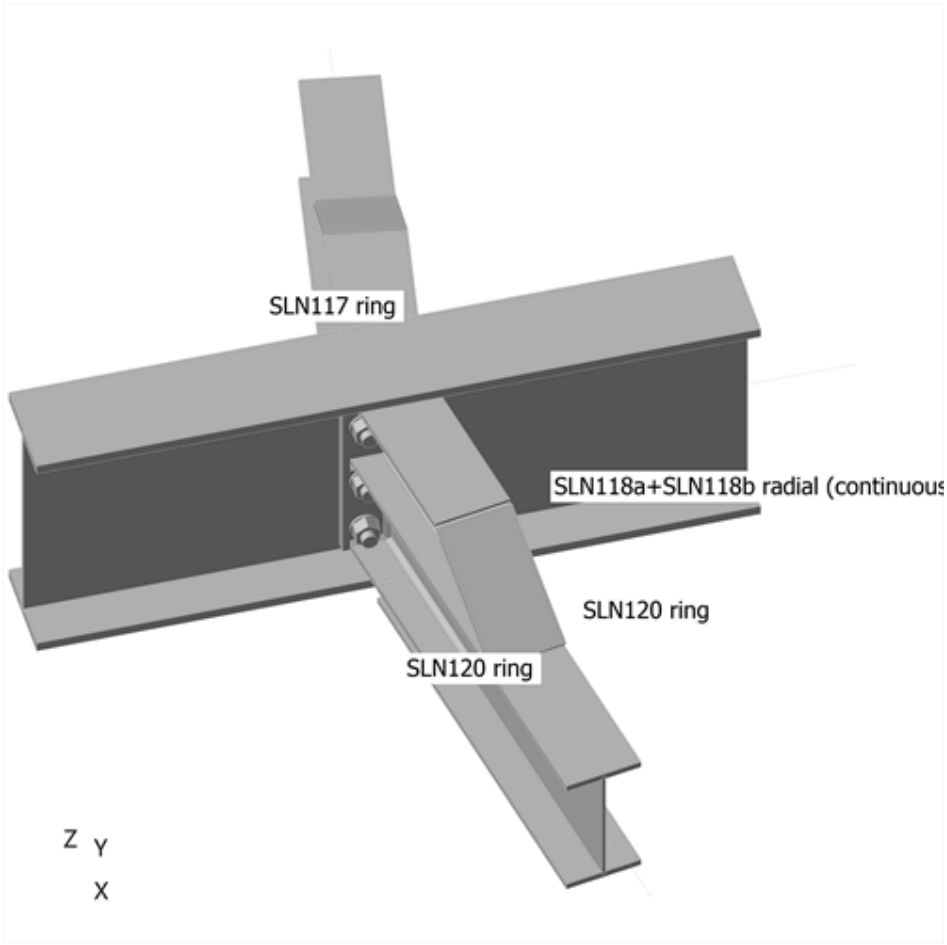
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN118a+SLN118b radial (continuous)-bfl 1	19.6	C LC_146	20.9	0.0	0.0	OK
SLN118a+SLN118b radial (continuous)-tfl 1	19.6	C LC_122	17.5	0.0	0.0	OK
SLN118a+SLN118b radial (continuous)-w 1	11.4	C LC_122	28.0	0.0	1.7	OK
SLN117 ring-bfl 1	12.7	C LC_145	51.8	0.0	0.0	OK
SLN117 ring-tfl 1	12.7	C LC_121	45.6	0.0	0.0	OK
SLN117 ring-w 1	7.2	C LC_148	49.7	0.0	0.0	OK
SLN120 ring-bfl 1	12.7	C LC_121	51.0	0.0	0.0	OK
SLN120 ring-tfl 1	12.7	C LC_122	44.7	0.0	0.0	OK
SLN120 ring-w 1	7.2	C LC_148	49.1	0.0	0.0	OK
End plate SLN120 ring	20.0	C LC_146	15.7	0.0	8.7	OK
Bottom flange plate SLN120 ring	15.0	C LC_146	26.9	0.0	0.0	OK
Bottom flange half plate L SLN120 ring	15.0	C LC_148	21.0	0.0	0.0	OK
Bottom flange half plate R SLN120 ring	15.0	C LC_146	19.9	0.0	0.0	OK
Haunch web SLN120 ring	12.0	C LC_120	29.4	0.0	0.0	OK
Haunch top plate SLN120 ring	15.0	C LC_146	9.4	0.0	0.0	OK
Haunch sloping plate SLN120 ring	15.0	C LC_120	12.8	0.0	0.0	OK
End plate SLN117 ring	20.0	C LC_146	12.8	0.0	4.9	OK
Bottom flange plate SLN117 ring	15.0	C LC_148	20.9	0.0	0.0	OK
Bottom flange half plate L SLN117 ring	15.0	C LC_148	19.8	0.0	0.0	OK
Bottom flange half plate R SLN117 ring	15.0	C LC_145	20.0	0.0	0.0	OK
Haunch web SLN117 ring	12.0	C LC_135	28.4	0.0	0.0	OK
Haunch top plate SLN117 ring	15.0	C LC_122	5.8	0.0	0.0	OK
Haunch sloping plate SLN117 ring	15.0	C LC_135	12.8	0.0	0.0	OK

Design data

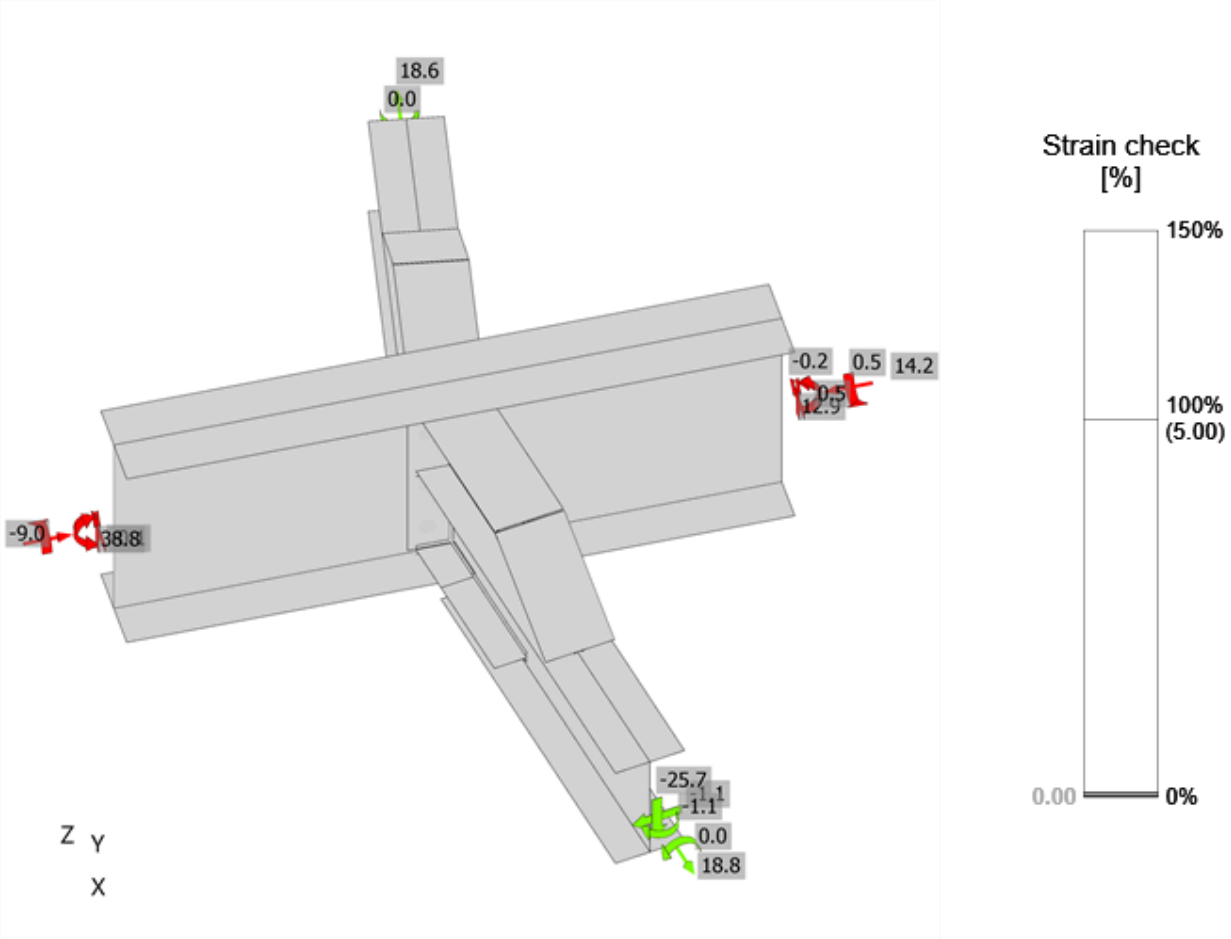
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

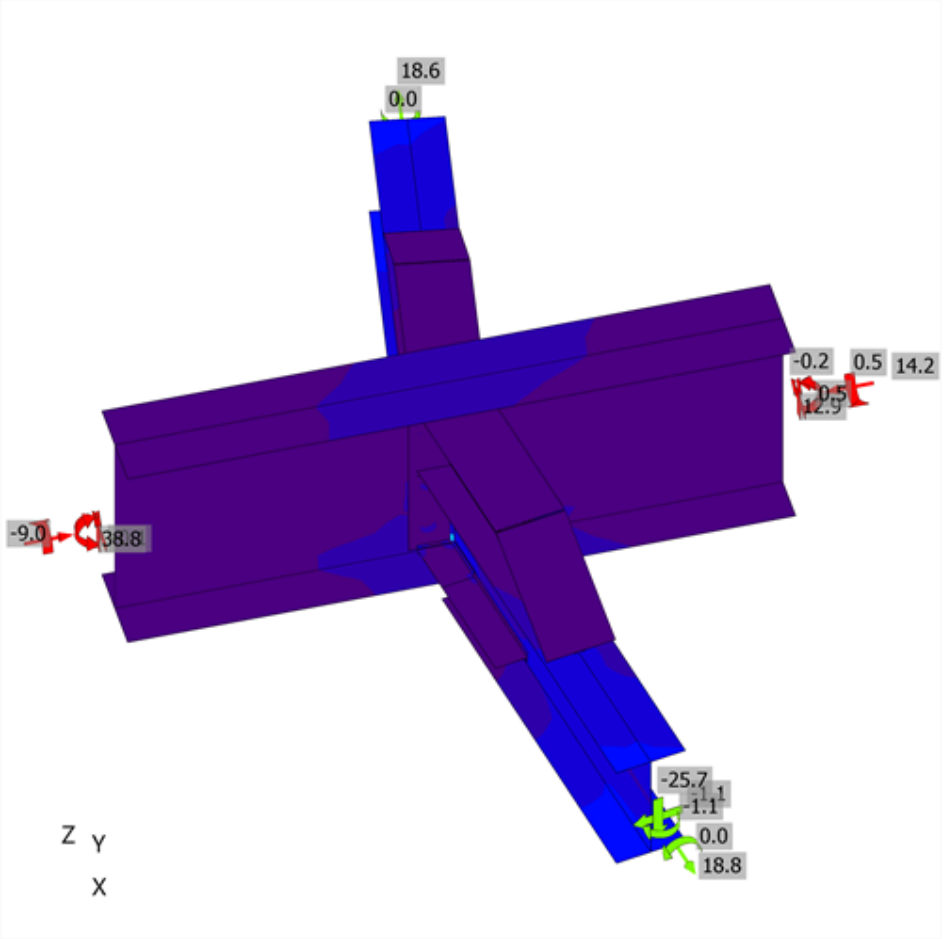
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

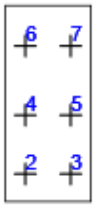


Strain check, C LC_146



Equivalent stress, C LC_146

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_146	4.8	0.6	194.1	1.5	0.5	1.3	OK	OK
	B3	M30 8.8 - 1	C LC_122	1.5	0.6	194.1	0.5	0.6	0.6	OK	OK
	B4	M30 8.8 - 1	C LC_135	0.2	0.4	232.2	0.1	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_148	0.4	0.5	232.2	0.1	0.4	0.3	OK	OK
	B6	M30 8.8 - 1	C LC_146	2.9	0.9	249.2	0.9	0.6	1.0	OK	OK
	B7	M30 8.8 - 1	C LC_122	1.4	0.9	249.2	0.4	0.7	0.7	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN117 ring	Bottom flange plate SLN117 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN120 ring	Bottom flange plate SLN120 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN120 ring	Bottom flange plate SLN120 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN117 ring	Bottom flange plate SLN117 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN118a+SLN118b radial (continuous)-w 1	SLN117 ring-tfl 1	◢ 11.4	149	C LC_148	12.2	0.0	-7.2	-5.7	0.5	3.0	3.0	OK	OK
SLN118a+SLN118b radial (continuous)-w 1	SLN120 ring-tfl 1	◢ 11.4	149	C LC_152	15.7	0.0	7.6	-7.9	-1.1	3.9	3.9	OK	OK
End plate SLN120 ring	SLN120 ring-w 1	◢ 6.0 ◣	208	C LC_148	27.8	0.0	-13.4	-11.8	-7.6	6.9	6.2	OK	OK
		◢ 6.0 ◣	208	C LC_148	24.4	0.0	-11.0	12.5	-0.4	6.0	6.0	OK	OK
Bottom flange plate SLN120 ring	SLN120 ring-w 1	◢ 6.0 ◣	100	C LC_145	23.4	0.0	1.1	-1.1	13.5	5.8	5.8	OK	OK
		◢ 6.0 ◣	100	C LC_148	28.6	0.0	-0.9	3.3	-16.2	7.1	7.1	OK	OK
SLN120 ring-w 1	Bottom flange half plate L SLN120 ring	◢ 6.0 ◣	243	C LC_145	20.9	0.0	6.1	5.0	10.4	5.2	5.2	OK	OK
		◢ 6.0 ◣	243	C LC_148	28.5	0.0	11.5	0.9	-15.0	7.0	6.1	OK	OK
SLN120 ring-w 1	Bottom flange half plate R SLN120 ring	◢ 6.0 ◣	243	C LC_148	28.6	0.0	5.9	7.1	-14.6	7.1	6.0	OK	OK
		◢ 6.0 ◣	243	C LC_146	35.4	0.0	9.8	-8.5	17.7	8.7	7.2	OK	OK
SLN120 ring-tfl 1	Haunch web SLN120 ring	◢ 6.0 ◣	572	C LC_120	70.7	0.0	-14.6	-18.6	35.4	17.5	9.9	OK	OK
		◢ 6.0 ◣	572	C LC_135	57.6	0.0	-17.0	13.8	-28.6	14.2	8.6	OK	OK
End plate SLN117 ring	SLN117 ring-w 1	◢ 6.0 ◣	208	C LC_148	21.1	0.0	-11.4	-10.1	1.8	5.2	5.2	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_148	25.0	0.0	-10.4	12.5	4.2	6.2	6.0	OK	OK
Bottom flange plate SLN117 ring	SLN117 ring-w 1	◢ 6.0 ◣	100	C LC_145	26.1	0.0	2.8	3.3	14.6	6.5	6.5	OK	OK
		◢ 6.0 ◣	100	C LC_122	25.0	0.0	1.9	-2.2	-14.2	6.2	6.2	OK	OK
SLN117 ring-w 1	Bottom flange half plate L SLN117 ring	◢ 6.0 ◣	243	C LC_145	25.5	0.0	6.2	6.2	12.9	6.3	6.3	OK	OK
		◢ 6.0 ◣	243	C LC_148	27.7	0.0	7.9	2.9	-15.1	6.9	6.4	OK	OK
SLN117 ring-w 1	Bottom flange half plate R SLN117 ring	◢ 6.0 ◣	243	C LC_148	24.5	0.0	6.1	6.0	-12.4	6.1	5.9	OK	OK
		◢ 6.0 ◣	243	C LC_145	27.8	0.0	8.8	2.3	15.0	6.9	6.4	OK	OK
SLN117 ring-tfl 1	Haunch web SLN117 ring	◢ 6.0 ◣	572	C LC_122	64.1	0.0	-16.6	-16.3	31.8	15.8	9.2	OK	OK
		◢ 6.0 ◣	572	C LC_121	65.9	0.0	-14.7	16.5	-33.2	16.3	9.4	OK	OK
End plate SLN120 ring	Bottom flange plate SLN120 ring	◢ 6.0 ◣	149	C LC_122	32.7	0.0	-10.4	-10.6	14.5	8.1	6.9	OK	OK
		◢ 6.0 ◣	149	C LC_122	37.3	0.0	-13.3	13.1	-15.3	9.2	6.4	OK	OK
End plate SLN120 ring	Haunch web SLN120 ring	◢ 6.0 ◣	145	C LC_135	5.7	0.0	0.9	0.8	3.1	1.4	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	11.2	0.0	2.7	-2.7	-5.6	2.8	0.0	OK	OK
End plate SLN120 ring	Haunch top plate SLN120 ring	◢ 6.0 ◣	149	C LC_152	13.2	0.0	-5.6	-5.4	4.2	3.2	3.2	OK	OK
		◢ 6.0 ◣	149	C LC_152	13.1	0.0	-5.0	5.1	-4.7	3.2	3.2	OK	OK
Haunch top plate SLN120 ring	Haunch web SLN120 ring	◢ 6.0 ◣	342	C LC_122	6.8	0.0	2.4	2.0	3.1	1.7	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_146	5.1	0.0	-0.4	0.5	2.9	1.3	0.0	OK	OK
Haunch sloping plate SLN120 ring	Haunch web SLN120 ring	◢ 6.0 ◣	271	C LC_121	55.9	0.0	-10.3	-4.6	31.4	13.8	8.8	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_122	49.0	0.0	-8.3	14.1	-24.1	12.1	8.0	OK	OK
End plate SLN117 ring	Bottom flange plate SLN117 ring	◢ 6.0 ◣	148	C LC_148	15.8	0.0	9.5	6.8	2.5	3.9	3.9	OK	OK
		◢ 6.0 ◣	149	C LC_122	20.7	0.0	-10.8	8.6	-5.5	5.1	5.1	OK	OK
End plate SLN117 ring	Haunch web SLN117 ring	◢ 6.0 ◣	145	C LC_135	9.0	0.0	2.3	2.2	4.5	2.2	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	8.8	0.0	1.3	-1.4	-4.8	2.2	2.2	OK	OK
End plate SLN117 ring	Haunch top plate SLN117 ring	◢ 6.0 ◣	149	C LC_126	8.5	0.0	-4.8	-3.1	2.6	2.1	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_114	4.7	0.0	-0.8	2.3	-1.4	1.2	0.0	OK	OK
Haunch top plate SLN117 ring	Haunch web SLN117 ring	◢ 6.0 ◣	342	C LC_122	3.8	0.0	1.2	1.2	1.7	0.9	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_121	4.4	0.0	1.8	-1.6	-1.7	1.1	0.0	OK	OK
Haunch sloping plate SLN117 ring	Haunch web SLN117 ring	◢ 6.0 ◣	271	C LC_135	53.1	0.0	-9.8	-10.9	28.1	13.1	8.2	OK	OK
		◢ 6.0 ◣	271	C LC_121	51.9	0.0	-9.0	6.5	-28.8	12.8	8.3	OK	OK
SLN118a+SLN118b radial (continuous)-w 1	SLN120 ring-w 1	◢ 7.2	208	C LC_148	78.5	0.0	35.0	-38.8	12.0	19.4	13.5	OK	OK
SLN118a+SLN118b radial (continuous)-w 1	SLN117 ring-w 1	◢ 7.2	208	C LC_148	74.6	0.0	36.4	31.8	-20.0	18.4	13.0	OK	OK
Haunch sloping plate SLN120 ring	Haunch top plate SLN120 ring	◢ 6.0 ◣	146	C LC_122	7.8	0.0	-7.1	-1.9	0.2	2.3	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_122	5.2	0.0	-1.2	2.9	0.2	1.3	0.0	OK	OK
Haunch sloping plate SLN117 ring	Haunch top plate SLN117 ring	◢ 6.0 ◣	146	C LC_122	7.2	0.0	-5.2	2.8	-0.3	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_121	3.2	0.0	-0.8	1.5	-1.0	0.8	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_300
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN122a+SLN122b radial (continuous) - SLN124 ring - SLN395 ring

Name	SLN122a+SLN122b radial (continuous) - SLN124 ring - SLN395 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

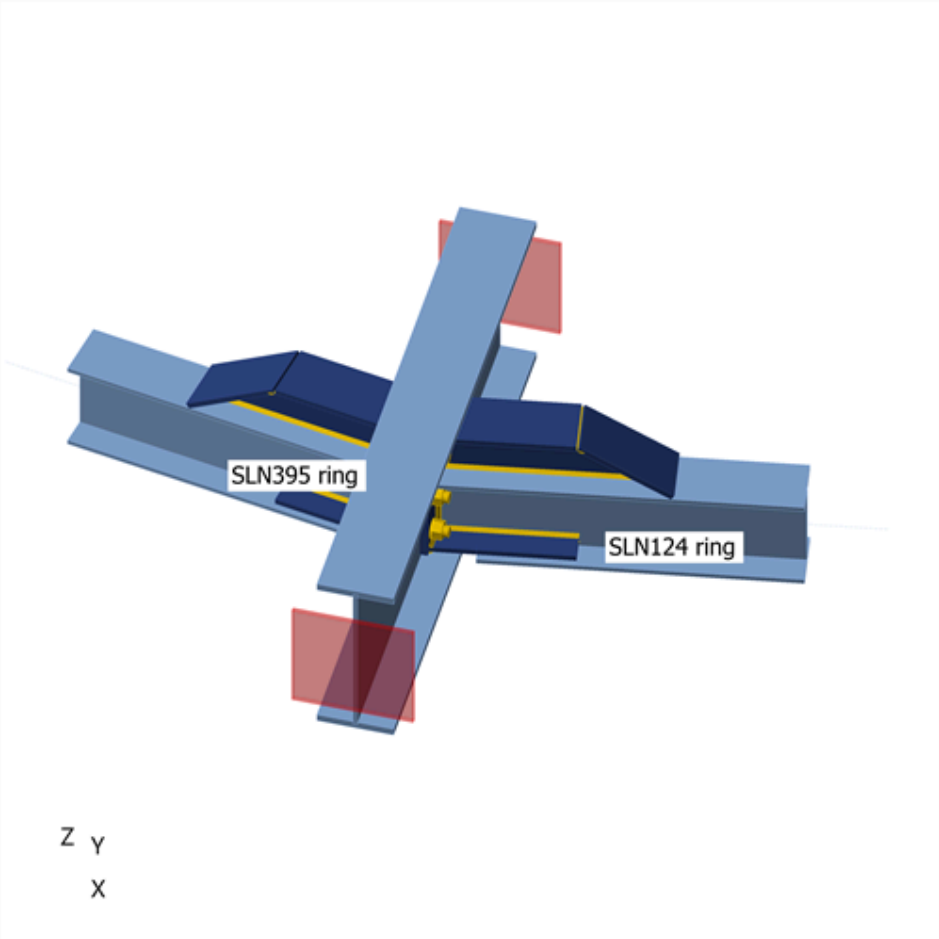
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN122a+SLN122b radial (continuous)	1 - UB457x191x98	-60.0	0.0	0.0	0	0	0
SLN124 ring	2 - UB254x146x43	40.0	0.0	0.0	0	0	-102
SLN395 ring	2 - UB254x146x43	-160.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN122a+SLN122b radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN122a+SLN122b radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN124 ring / end		Position	0
SLN395 ring / end		Position	0



SLN122a+SLN122b radial (continuous) - SLN124 ring - SLN395 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN124 ring / End	0.6	0.0	-4.9	0.0	0.0	0.0
	SLN395 ring / End	0.6	0.0	-2.2	0.0	0.0	0.0
C LC_102	SLN124 ring / End	-4.1	0.1	-25.6	0.0	0.0	0.2
	SLN395 ring / End	-4.2	0.0	-8.7	0.0	0.0	0.0
C LC_103	SLN124 ring / End	-8.3	0.1	-24.5	0.0	0.0	0.1
	SLN395 ring / End	-8.1	0.1	-8.4	0.0	0.0	0.1
C LC_104	SLN124 ring / End	-6.0	0.2	-24.5	0.0	0.0	0.2
	SLN395 ring / End	-6.7	0.1	-8.4	0.0	0.0	0.1
C LC_105	SLN124 ring / End	-8.6	0.0	-24.5	0.0	0.0	0.1
	SLN395 ring / End	-8.9	0.1	-8.4	0.0	0.0	0.0
C LC_106	SLN124 ring / End	-5.3	0.1	-24.5	0.0	0.0	0.0
	SLN395 ring / End	-5.0	0.0	-8.4	0.0	0.0	0.0
C LC_107	SLN124 ring / End	-3.3	0.0	-1.6	0.0	0.0	0.0
	SLN395 ring / End	-3.0	0.1	-0.9	0.0	0.0	0.0
C LC_108	SLN124 ring / End	-0.1	0.1	-1.6	0.0	0.0	0.2
	SLN395 ring / End	-0.9	0.1	-0.9	0.0	0.0	0.2
C LC_109	SLN124 ring / End	-3.2	0.1	-1.6	0.0	0.0	0.1
	SLN395 ring / End	-3.5	0.0	-0.9	0.0	0.0	0.0
C LC_110	SLN124 ring / End	0.5	-0.2	-1.6	0.0	0.0	-0.3
	SLN395 ring / End	1.2	-0.1	-0.9	0.0	0.0	-0.1
C LC_111	SLN124 ring / End	-8.0	0.1	-17.3	0.0	0.0	0.1
	SLN395 ring / End	-7.8	0.1	-6.1	0.0	0.0	0.1
C LC_112	SLN124 ring / End	-4.4	0.2	-17.3	0.0	0.0	0.3
	SLN395 ring / End	-5.4	0.1	-6.1	0.0	0.0	0.2
C LC_113	SLN124 ring / End	-8.2	0.1	-17.3	0.0	0.0	0.2
	SLN395 ring / End	-8.6	0.1	-6.1	0.0	0.0	0.1
C LC_114	SLN124 ring / End	-3.5	-0.1	-17.3	0.0	0.0	-0.2
	SLN395 ring / End	-2.9	0.0	-6.1	0.0	0.0	-0.1
C LC_115	SLN124 ring / End	-3.2	0.0	-2.8	0.0	0.0	0.1
	SLN395 ring / End	-2.9	0.1	-1.5	0.0	0.0	0.0
C LC_116	SLN124 ring / End	0.1	0.1	-2.8	0.0	0.0	0.2
	SLN395 ring / End	-0.7	0.1	-1.5	0.0	0.0	0.2
C LC_117	SLN124 ring / End	-3.1	0.1	-2.8	0.0	0.0	0.1
	SLN395 ring / End	-3.4	0.0	-1.5	0.0	0.0	0.0
C LC_118	SLN124 ring / End	0.7	-0.1	-2.8	0.0	0.0	-0.3
	SLN395 ring / End	1.4	-0.1	-1.5	0.0	0.0	-0.2
C LC_119	SLN124 ring / End	-7.0	0.1	-25.6	0.0	0.0	0.2
	SLN395 ring / End	-7.0	0.1	-8.7	0.0	0.0	0.1
C LC_120	SLN124 ring / End	-5.5	0.2	-25.6	0.0	0.0	0.2
	SLN395 ring / End	-5.8	0.0	-8.7	0.0	0.0	0.0
C LC_121	SLN124 ring / End	-7.1	0.0	-25.6	0.0	0.0	0.1
	SLN395 ring / End	-7.3	0.1	-8.7	0.0	0.0	0.0
C LC_122	SLN124 ring / End	-4.6	0.1	-25.6	0.0	0.0	0.1
	SLN395 ring / End	-4.6	0.0	-8.7	0.0	0.0	0.0
C LC_123	SLN124 ring / End	-5.4	0.1	-19.4	0.0	0.0	0.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN395 ring / End	-5.5	0.1	-6.8	0.0	0.0	0.2
C LC_124	SLN124 ring / End	-3.4	0.2	-19.4	0.0	0.0	0.2
	SLN395 ring / End	-3.6	0.0	-6.8	0.0	0.0	0.0
C LC_125	SLN124 ring / End	-5.2	0.0	-19.4	0.0	0.0	0.1
	SLN395 ring / End	-5.4	0.0	-6.8	0.0	0.0	0.0
C LC_126	SLN124 ring / End	-2.1	0.0	-19.4	0.0	0.0	0.0
	SLN395 ring / End	-2.0	0.0	-6.8	0.0	0.0	-0.1
C LC_127	SLN124 ring / End	-0.6	0.1	-4.9	0.0	0.0	0.1
	SLN395 ring / End	-0.6	0.1	-2.2	0.0	0.0	0.1
C LC_128	SLN124 ring / End	1.1	0.1	-4.9	0.0	0.0	0.2
	SLN395 ring / End	1.0	0.0	-2.2	0.0	0.0	0.0
C LC_129	SLN124 ring / End	-0.1	0.0	-4.9	0.0	0.0	0.0
	SLN395 ring / End	-0.2	0.0	-2.2	0.0	0.0	0.0
C LC_130	SLN124 ring / End	2.1	0.0	-4.9	0.0	0.0	-0.1
	SLN395 ring / End	2.3	-0.1	-2.2	0.0	0.0	-0.1
C LC_135	SLN124 ring / End	-0.5	0.1	-25.6	0.0	0.0	0.1
	SLN395 ring / End	-0.5	0.0	-8.7	0.0	0.0	-0.1
C LC_141	SLN124 ring / End	0.5	0.0	-4.9	0.0	0.0	0.0
	SLN395 ring / End	0.5	0.0	-2.2	0.0	0.0	0.0
C LC_142	SLN124 ring / End	0.7	0.0	-4.9	0.0	0.0	0.0
	SLN395 ring / End	0.6	0.0	-2.2	0.0	0.0	0.0
C LC_143	SLN124 ring / End	0.4	0.0	-4.9	0.0	0.0	0.0
	SLN395 ring / End	0.4	0.0	-2.2	0.0	0.0	0.0
C LC_144	SLN124 ring / End	0.7	0.1	-4.9	0.0	0.0	0.1
	SLN395 ring / End	0.7	0.0	-2.2	0.0	0.0	0.0
C LC_145	SLN124 ring / End	-4.5	0.1	-25.6	0.0	0.0	0.2
	SLN395 ring / End	-4.4	0.1	-8.7	0.0	0.0	0.0
C LC_146	SLN124 ring / End	-3.7	0.1	-25.6	0.0	0.0	0.1
	SLN395 ring / End	-4.0	0.0	-8.7	0.0	0.0	-0.1
C LC_147	SLN124 ring / End	-4.9	0.0	-25.6	0.0	0.0	0.1
	SLN395 ring / End	-5.0	0.0	-8.7	0.0	0.0	0.0
C LC_148	SLN124 ring / End	-3.3	0.2	-25.6	0.0	0.0	0.2
	SLN395 ring / End	-3.4	0.0	-8.7	0.0	0.0	0.0
C LC_151	SLN124 ring / End	-1.8	0.1	-11.6	0.0	0.0	0.1
	SLN395 ring / End	-1.8	0.0	-8.7	0.0	0.0	0.0
C LC_152	SLN124 ring / End	-7.0	0.1	-13.9	0.0	0.0	0.2
	SLN395 ring / End	-7.1	0.1	-2.2	0.0	0.0	0.1
C LC_153	SLN124 ring / End	-4.3	0.1	-20.6	0.0	0.0	0.1
	SLN395 ring / End	-4.4	0.0	-8.7	0.0	0.0	0.0
C LC_154	SLN124 ring / End	-1.8	0.1	-11.6	0.0	0.0	0.1
	SLN395 ring / End	-1.8	0.0	-8.7	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.6 < 100%	OK
Welds	18.2 < 100%	OK
Buckling	Not calculated	

Plates

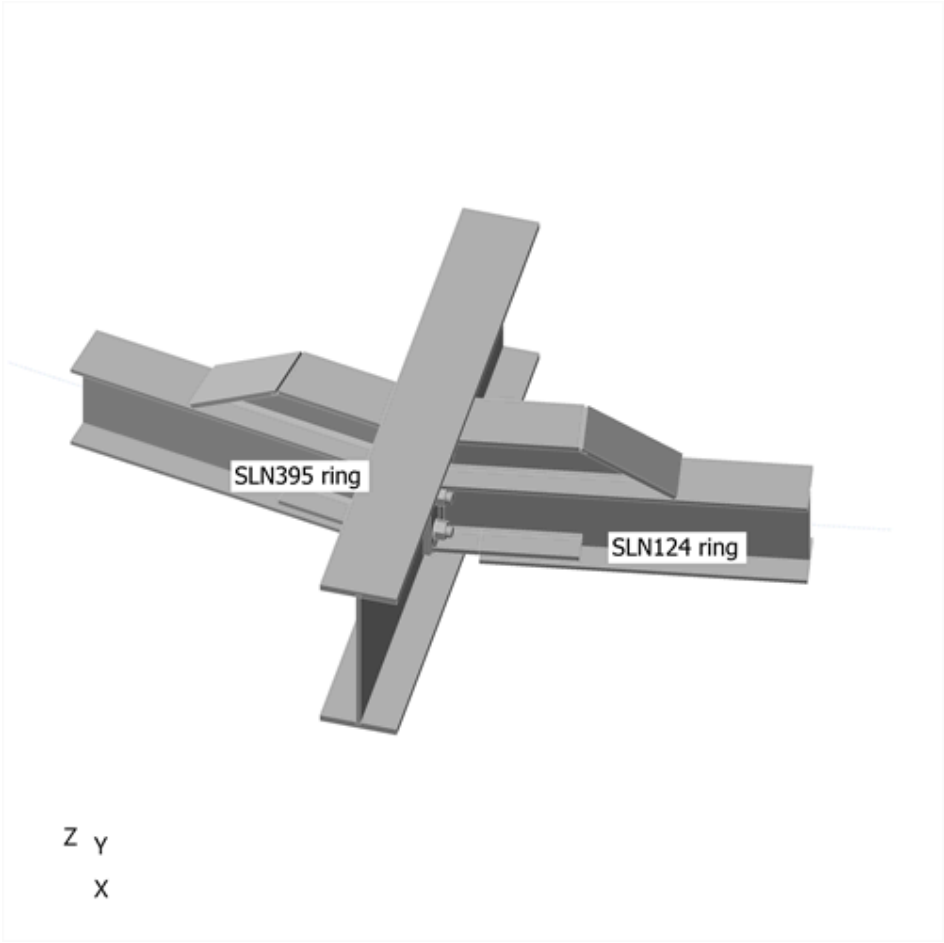
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN122a+SLN122b radial (continuous)-bfl 1	19.6	C LC_146	13.6	0.0	0.0	OK
SLN122a+SLN122b radial (continuous)-tfl 1	19.6	C LC_119	11.5	0.0	0.0	OK
SLN122a+SLN122b radial (continuous)-w 1	11.4	C LC_119	16.7	0.0	0.8	OK
SLN124 ring-bfl 1	12.7	C LC_135	46.4	0.0	0.0	OK
SLN124 ring-tfl 1	12.7	C LC_119	48.2	0.0	0.0	OK
SLN124 ring-w 1	7.2	C LC_119	47.4	0.0	0.0	OK
SLN395 ring-bfl 1	12.7	C LC_135	15.9	0.0	0.0	OK
SLN395 ring-tfl 1	12.7	C LC_121	17.6	0.0	0.0	OK
SLN395 ring-w 1	7.2	C LC_121	16.8	0.0	0.0	OK
End plate SLN124 ring	20.0	C LC_148	4.9	0.0	1.7	OK
Bottom flange plate SLN124 ring	15.0	C LC_135	13.2	0.0	0.0	OK
Bottom flange half plate L SLN124 ring	15.0	C LC_135	12.5	0.0	0.0	OK
Bottom flange half plate R SLN124 ring	15.0	C LC_135	12.7	0.0	0.0	OK
Haunch web SLN124 ring	12.0	C LC_119	32.8	0.0	0.0	OK
Haunch top plate SLN124 ring	15.0	C LC_119	5.6	0.0	0.0	OK
Haunch sloping plate SLN124 ring	15.0	C LC_119	14.3	0.0	0.0	OK
End plate SLN395 ring	20.0	C LC_112	4.8	0.0	5.0	OK
Bottom flange plate SLN395 ring	15.0	C LC_112	11.5	0.0	0.0	OK
Bottom flange half plate L SLN395 ring	15.0	C LC_135	4.3	0.0	0.0	OK
Bottom flange half plate R SLN395 ring	15.0	C LC_135	4.4	0.0	0.0	OK
Haunch web SLN395 ring	12.0	C LC_121	11.8	0.0	0.0	OK
Haunch top plate SLN395 ring	15.0	C LC_112	4.5	0.0	0.0	OK
Haunch sloping plate SLN395 ring	15.0	C LC_121	5.3	0.0	0.0	OK

Design data

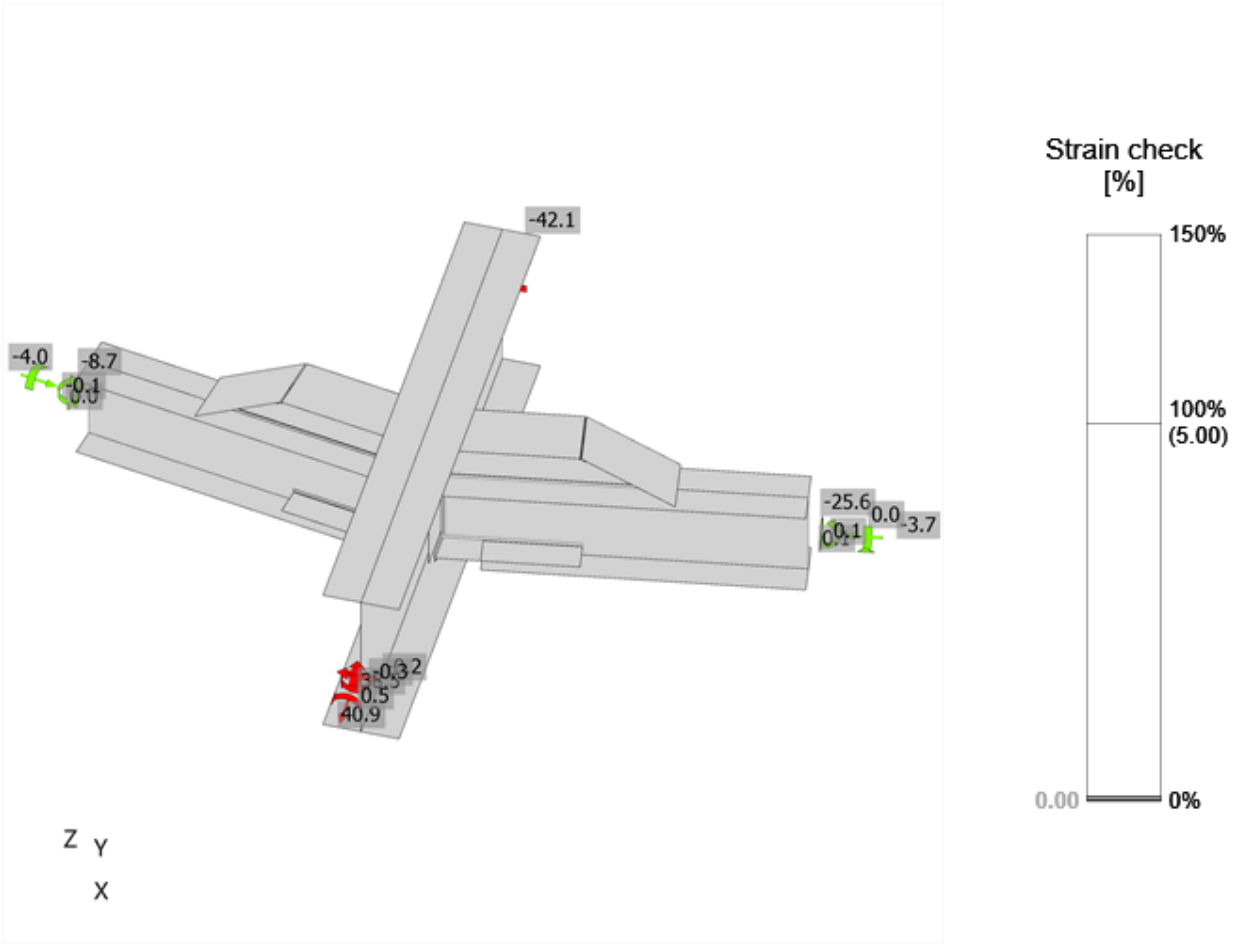
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

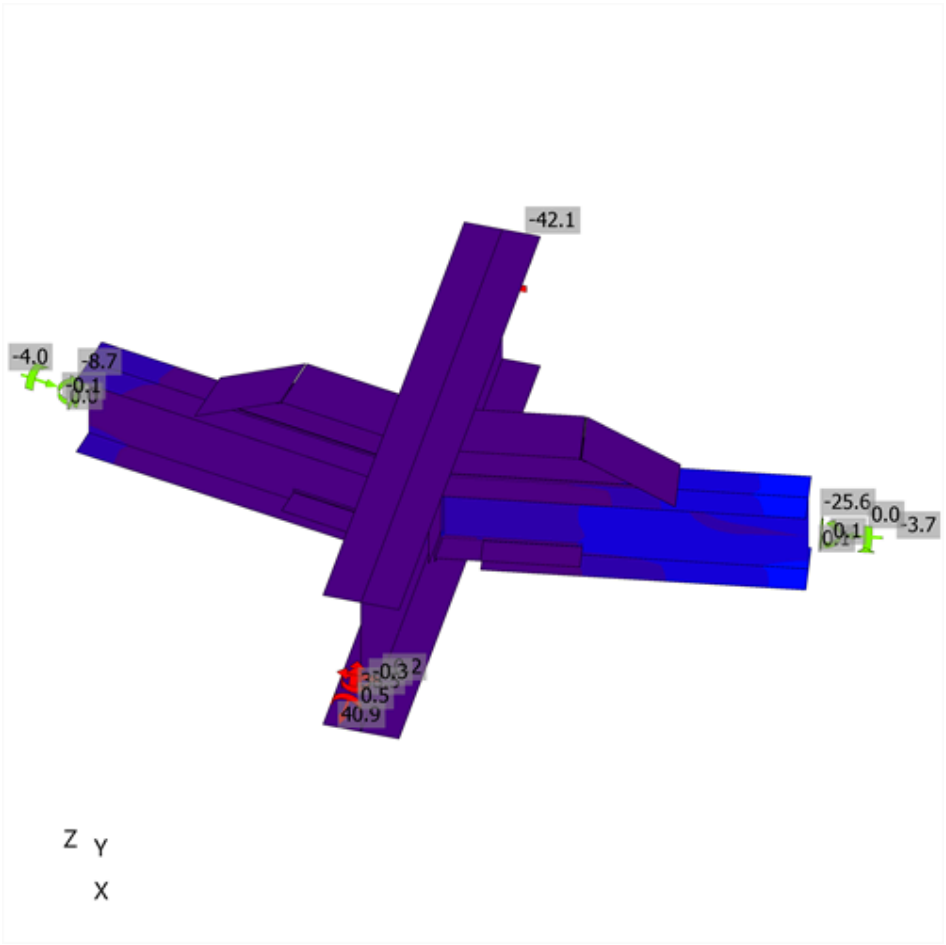
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

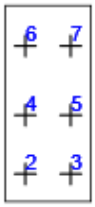


Strain check, C LC_146



Equivalent stress, C LC_146

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_135	0.7	0.4	194.1	0.2	0.3	0.3	OK	OK
	B3	M30 8.8 - 1	C LC_135	0.5	0.4	194.1	0.2	0.4	0.3	OK	OK
	B4	M30 8.8 - 1	C LC_121	0.1	0.3	232.2	0.0	0.2	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_135	0.0	0.3	232.2	0.0	0.3	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_119	1.4	0.6	249.2	0.4	0.4	0.6	OK	OK
	B7	M30 8.8 - 1	C LC_148	1.6	0.6	249.2	0.5	0.4	0.6	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN124 ring	Bottom flange plate SLN124 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN124 ring	Bottom flange plate SLN124 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN395 ring	Bottom flange plate SLN395 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN395 ring	Bottom flange plate SLN395 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN122a+SLN122b radial (continuous)-w 1	SLN124 ring-tfl 1	◢ 11.4	149	C LC_121	9.3	0.0	-8.6	-2.1	-0.4	2.8	2.8	OK	OK
SLN122a+SLN122b radial (continuous)-w 1	SLN395 ring-tfl 1	◢ 11.4	149	C LC_135	8.7	0.0	-5.6	-3.8	-0.1	2.1	0.0	OK	OK
End plate SLN124 ring	SLN124 ring-w 1	◢ 6.0 ◣	208	C LC_135	7.9	0.0	0.3	-0.6	4.5	2.0	0.0	OK	OK
		◢ 6.0 ◣	208	C LC_135	6.3	0.0	-0.3	-0.4	-3.6	1.6	0.0	OK	OK
Bottom flange plate SLN124 ring	SLN124 ring-w 1	◢ 6.0 ◣	100	C LC_148	21.1	0.0	1.8	1.9	12.0	5.2	5.2	OK	OK
		◢ 6.0 ◣	100	C LC_135	20.1	0.0	1.8	-1.9	-11.4	5.0	5.0	OK	OK
SLN124 ring-w 1	Bottom flange half plate L SLN124 ring	◢ 6.0 ◣	243	C LC_135	17.3	0.0	4.4	4.1	8.8	4.3	4.3	OK	OK
		◢ 6.0 ◣	243	C LC_135	18.8	0.0	5.8	2.4	-10.0	4.6	4.6	OK	OK
SLN124 ring-w 1	Bottom flange half plate R SLN124 ring	◢ 6.0 ◣	243	C LC_135	15.7	0.0	3.4	3.7	-8.0	3.9	3.9	OK	OK
		◢ 6.0 ◣	243	C LC_135	20.1	0.0	5.9	2.3	10.9	5.0	5.0	OK	OK
SLN124 ring-tfl 1	Haunch web SLN124 ring	◢ 6.0 ◣	572	C LC_121	73.7	0.0	-19.2	-19.4	36.3	18.2	10.3	OK	OK
		◢ 6.0 ◣	572	C LC_119	68.6	0.0	-15.7	15.8	-35.2	17.0	9.8	OK	OK
End plate SLN395 ring	SLN395 ring-w 1	◢ 6.0 ◣	208	C LC_135	4.9	0.0	-0.8	0.0	2.8	1.2	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_135	7.4	0.0	-0.1	1.2	-4.1	1.8	0.0	OK	OK
Bottom flange plate SLN395 ring	SLN395 ring-w 1	◢ 6.0 ◣	100	C LC_119	7.6	0.0	0.5	1.2	4.2	1.9	0.0	OK	OK
		◢ 6.0 ◣	100	C LC_135	6.9	0.0	0.6	-0.6	-3.9	1.7	0.0	OK	OK
SLN395 ring-w 1	Bottom flange half plate L SLN395 ring	◢ 6.0 ◣	243	C LC_135	5.9	0.0	1.5	1.4	3.0	1.5	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	6.5	0.0	2.0	0.8	-3.5	1.6	0.0	OK	OK
SLN395 ring-w 1	Bottom flange half plate R SLN395 ring	◢ 6.0 ◣	243	C LC_135	5.3	0.0	1.2	1.2	-2.7	1.3	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	6.8	0.0	2.0	0.8	3.7	1.7	0.0	OK	OK
SLN395 ring-tfl 1	Haunch web SLN395 ring	◢ 6.0 ◣	572	C LC_121	26.8	0.0	-6.4	-6.8	13.4	6.6	6.1	OK	OK
		◢ 6.0 ◣	572	C LC_119	25.6	0.0	-5.9	6.0	-13.1	6.3	6.0	OK	OK
End plate SLN124 ring	Bottom flange plate SLN124 ring	◢ 6.0 ◣	149	C LC_110	13.9	0.0	-4.4	-5.2	5.6	3.4	3.4	OK	OK
		◢ 6.0 ◣	149	C LC_110	15.3	0.0	-6.1	5.3	-6.2	3.8	3.8	OK	OK
End plate SLN124 ring	Haunch web SLN124 ring	◢ 6.0 ◣	145	C LC_119	6.3	0.0	0.8	0.7	3.5	1.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_146	9.3	0.0	1.1	-1.3	-5.2	2.3	0.0	OK	OK
End plate SLN124 ring	Haunch top plate SLN124 ring	◢ 6.0 ◣	149	C LC_110	7.2	0.0	-3.4	-2.6	2.6	1.8	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_112	5.8	0.0	-2.3	2.4	1.9	1.4	0.0	OK	OK
Haunch top plate SLN124 ring	Haunch web SLN124 ring	◢ 6.0 ◣	342	C LC_119	3.4	0.0	-0.2	-0.2	-2.0	0.8	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_119	4.1	0.0	1.5	-1.4	-1.7	1.0	0.0	OK	OK
Haunch sloping plate SLN124 ring	Haunch web SLN124 ring	◢ 6.0 ◣	271	C LC_119	60.9	0.0	-11.6	-11.8	32.5	15.1	9.2	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_121	52.9	0.0	-8.3	8.7	-28.9	13.1	8.4	OK	OK
End plate SLN395 ring	Bottom flange plate SLN395 ring	◢ 6.0 ◣	149	C LC_112	14.0	0.0	-3.9	-5.5	-5.4	3.5	3.5	OK	OK
		◢ 6.0 ◣	149	C LC_112	17.0	0.0	-7.2	5.5	7.0	4.2	4.2	OK	OK
End plate SLN395 ring	Haunch web SLN395 ring	◢ 6.0 ◣	145	C LC_119	4.9	0.0	0.7	0.8	2.7	1.2	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_135	3.4	0.0	-0.4	-0.3	1.9	0.8	0.0	OK	OK
End plate SLN395 ring	Haunch top plate SLN395 ring	◢ 6.0 ◣	149	C LC_112	8.1	0.0	-3.9	-2.7	-3.1	2.0	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_112	6.0	0.0	-1.6	2.8	1.8	1.5	0.0	OK	OK
Haunch top plate SLN395 ring	Haunch web SLN395 ring	◢ 6.0 ◣	342	C LC_112	1.5	0.0	0.0	0.0	-0.9	0.4	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_112	1.8	0.0	0.6	-0.5	-0.9	0.4	0.0	OK	OK
Haunch sloping plate SLN395 ring	Haunch web SLN395 ring	◢ 6.0 ◣	271	C LC_121	21.5	0.0	-3.9	-3.3	11.7	5.3	5.3	OK	OK
		◢ 6.0 ◣	271	C LC_119	19.9	0.0	-3.2	3.1	-10.9	4.9	4.9	OK	OK
SLN122a+SLN122b radial (continuous)-w 1	SLN124 ring-w 1	◢ 7.2	208	C LC_135	30.1	0.0	7.6	-7.2	15.2	7.4	7.4	OK	OK
SLN122a+SLN122b radial (continuous)-w 1	SLN395 ring-w 1	◢ 7.2	208	C LC_135	18.5	0.0	1.6	1.8	-10.5	4.6	4.6	OK	OK
Haunch sloping plate SLN124 ring	Haunch top plate SLN124 ring	◢ 6.0 ◣	146	C LC_119	7.4	0.0	-5.3	3.0	-0.3	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_119	2.7	0.0	-0.5	1.5	-0.3	0.7	0.0	OK	OK
Haunch sloping plate SLN395 ring	Haunch top plate SLN395 ring	◢ 6.0 ◣	146	C LC_119	2.3	0.0	-1.8	0.9	-0.1	0.7	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_112	1.6	0.0	-0.3	0.9	0.2	0.4	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_320
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN123b+SLN123a radial (continuous) - SLN124 ring - SLN125 ring

Name	SLN123b+SLN123a radial (continuous) - SLN124 ring - SLN125 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

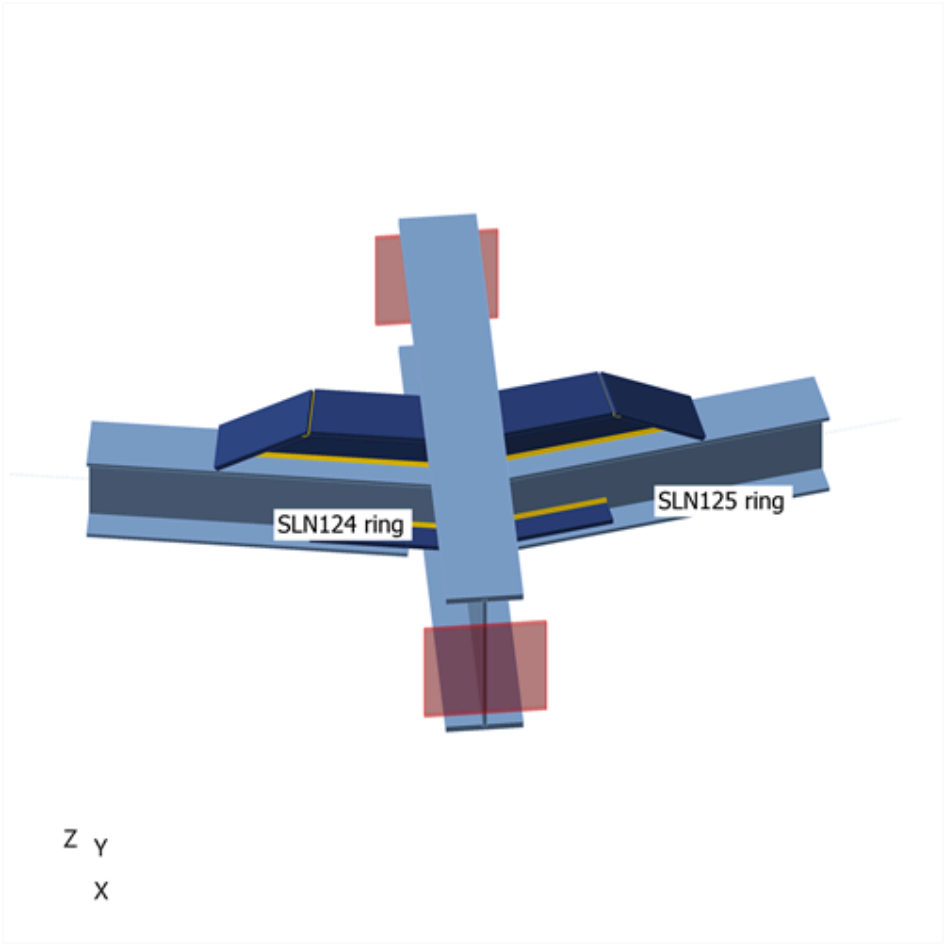
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN123b+SLN123a radial (continuous)	1 - UB457x191x98	-40.0	0.0	0.0	0	0	0
SLN124 ring	2 - UB254x146x43	-140.0	0.0	0.0	0	0	-102
SLN125 ring	2 - UB254x146x43	60.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN123b+SLN123a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN123b+SLN123a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN124 ring / end		Position	0
SLN125 ring / end		Position	0



SLN123b+SLN123a radial (continuous) - SLN124 ring - SLN125 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN124 ring / End	0.6	0.0	-4.9	0.0	0.0	0.1
	SLN125 ring / End	0.6	0.0	-4.9	0.0	0.0	0.0
C LC_102	SLN124 ring / End	-4.1	0.1	-25.6	0.0	0.0	0.2
	SLN125 ring / End	-4.0	-0.1	-25.6	0.0	0.0	-0.1
C LC_103	SLN124 ring / End	-8.4	0.2	-24.6	0.0	0.0	0.2
	SLN125 ring / End	-8.5	-0.1	-24.6	0.0	0.0	-0.1
C LC_104	SLN124 ring / End	-5.9	0.1	-24.6	0.0	0.0	0.1
	SLN125 ring / End	-5.5	0.1	-24.6	0.0	0.0	0.0
C LC_105	SLN124 ring / End	-8.5	0.2	-24.6	0.0	0.0	0.2
	SLN125 ring / End	-8.1	-0.1	-24.6	0.0	0.0	-0.1
C LC_106	SLN124 ring / End	-5.4	-0.1	-24.6	0.0	0.0	0.0
	SLN125 ring / End	-5.6	-0.1	-24.6	0.0	0.0	-0.1
C LC_107	SLN124 ring / End	-3.3	0.0	-1.6	0.0	0.0	0.1
	SLN125 ring / End	-3.5	0.0	-1.6	0.0	0.0	0.0
C LC_108	SLN124 ring / End	-0.1	0.1	-1.6	0.0	0.0	0.1
	SLN125 ring / End	0.1	0.1	-1.6	0.0	0.0	0.2
C LC_109	SLN124 ring / End	-3.2	0.1	-1.6	0.0	0.0	0.1
	SLN125 ring / End	-2.9	0.0	-1.6	0.0	0.0	0.1
C LC_110	SLN124 ring / End	0.5	-0.2	-1.6	0.0	0.0	-0.2
	SLN125 ring / End	0.2	-0.1	-1.6	0.0	0.0	-0.2
C LC_111	SLN124 ring / End	-8.1	0.1	-17.3	0.0	0.0	0.2
	SLN125 ring / End	-8.3	-0.1	-17.3	0.0	0.0	-0.1
C LC_112	SLN124 ring / End	-4.3	0.1	-17.3	0.0	0.0	0.2
	SLN125 ring / End	-3.9	0.1	-17.3	0.0	0.0	0.1
C LC_113	SLN124 ring / End	-8.2	0.2	-17.3	0.0	0.0	0.2
	SLN125 ring / End	-7.7	0.0	-17.3	0.0	0.0	0.0
C LC_114	SLN124 ring / End	-3.6	-0.2	-17.3	0.0	0.0	-0.2
	SLN125 ring / End	-4.0	-0.1	-17.3	0.0	0.0	-0.2
C LC_115	SLN124 ring / End	-3.2	0.1	-2.8	0.0	0.0	0.1
	SLN125 ring / End	-3.3	0.0	-2.8	0.0	0.0	0.0
C LC_116	SLN124 ring / End	0.1	0.1	-2.8	0.0	0.0	0.1
	SLN125 ring / End	0.3	0.1	-2.8	0.0	0.0	0.2
C LC_117	SLN124 ring / End	-3.1	0.1	-2.8	0.0	0.0	0.1
	SLN125 ring / End	-2.8	0.0	-2.8	0.0	0.0	0.0
C LC_118	SLN124 ring / End	0.7	-0.2	-2.8	0.0	0.0	-0.2
	SLN125 ring / End	0.4	-0.2	-2.8	0.0	0.0	-0.2
C LC_119	SLN124 ring / End	-7.1	0.2	-25.6	0.0	0.0	0.2
	SLN125 ring / End	-7.2	-0.1	-25.6	0.0	0.0	-0.1
C LC_120	SLN124 ring / End	-5.4	0.1	-25.6	0.0	0.0	0.1
	SLN125 ring / End	-5.0	0.0	-25.6	0.0	0.0	0.0
C LC_121	SLN124 ring / End	-7.0	0.1	-25.6	0.0	0.0	0.2
	SLN125 ring / End	-6.7	-0.1	-25.6	0.0	0.0	-0.2
C LC_122	SLN124 ring / End	-4.7	0.0	-25.6	0.0	0.0	0.0
	SLN125 ring / End	-4.9	-0.1	-25.6	0.0	0.0	-0.1
C LC_123	SLN124 ring / End	-5.5	0.2	-19.4	0.0	0.0	0.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN125 ring / End	-5.6	-0.1	-19.4	0.0	0.0	-0.1
C LC_124	SLN124 ring / End	-3.3	0.1	-19.4	0.0	0.0	0.2
	SLN125 ring / End	-2.9	0.1	-19.4	0.0	0.0	0.1
C LC_125	SLN124 ring / End	-5.1	0.1	-19.4	0.0	0.0	0.1
	SLN125 ring / End	-4.8	-0.1	-19.4	0.0	0.0	-0.1
C LC_126	SLN124 ring / End	-2.2	-0.1	-19.4	0.0	0.0	0.0
	SLN125 ring / End	-2.6	-0.1	-19.4	0.0	0.0	-0.2
C LC_127	SLN124 ring / End	-0.6	0.1	-4.9	0.0	0.0	0.1
	SLN125 ring / End	-0.7	0.0	-4.9	0.0	0.0	0.0
C LC_128	SLN124 ring / End	1.1	0.1	-4.9	0.0	0.0	0.2
	SLN125 ring / End	1.3	0.1	-4.9	0.0	0.0	0.1
C LC_129	SLN124 ring / End	-0.1	0.1	-4.9	0.0	0.0	0.1
	SLN125 ring / End	0.1	0.0	-4.9	0.0	0.0	-0.1
C LC_130	SLN124 ring / End	2.1	-0.1	-4.9	0.0	0.0	-0.1
	SLN125 ring / End	1.7	-0.1	-4.9	0.0	0.0	-0.2
C LC_135	SLN124 ring / End	-0.5	0.1	-25.6	0.0	0.0	0.2
	SLN125 ring / End	-0.4	-0.1	-25.6	0.0	0.0	-0.1
C LC_141	SLN124 ring / End	0.5	0.0	-4.9	0.0	0.0	0.1
	SLN125 ring / End	0.4	0.0	-4.9	0.0	0.0	-0.1
C LC_142	SLN124 ring / End	0.7	0.0	-4.9	0.0	0.0	0.0
	SLN125 ring / End	0.7	0.0	-4.9	0.0	0.0	0.0
C LC_143	SLN124 ring / End	0.4	0.0	-4.9	0.0	0.0	0.1
	SLN125 ring / End	0.5	0.0	-4.9	0.0	0.0	-0.1
C LC_144	SLN124 ring / End	0.7	0.0	-4.9	0.0	0.0	0.1
	SLN125 ring / End	0.7	0.0	-4.9	0.0	0.0	0.0
C LC_145	SLN124 ring / End	-4.6	0.2	-25.6	0.0	0.0	0.2
	SLN125 ring / End	-4.6	-0.2	-25.6	0.0	0.0	-0.2
C LC_146	SLN124 ring / End	-3.6	0.0	-25.6	0.0	0.0	0.1
	SLN125 ring / End	-3.3	0.0	-25.6	0.0	0.0	-0.1
C LC_147	SLN124 ring / End	-4.8	0.2	-25.6	0.0	0.0	0.2
	SLN125 ring / End	-4.5	-0.2	-25.6	0.0	0.0	-0.2
C LC_148	SLN124 ring / End	-3.4	0.1	-25.6	0.0	0.0	0.2
	SLN125 ring / End	-3.5	0.0	-25.6	0.0	0.0	-0.1
C LC_151	SLN124 ring / End	-1.8	0.1	-11.6	0.0	0.0	0.1
	SLN125 ring / End	-1.8	-0.1	-11.5	0.0	0.0	-0.1
C LC_152	SLN124 ring / End	-7.0	0.1	-13.9	0.0	0.0	0.2
	SLN125 ring / End	-6.8	-0.1	-14.0	0.0	0.0	-0.1
C LC_153	SLN124 ring / End	-4.3	0.1	-20.6	0.0	0.0	0.2
	SLN125 ring / End	-4.1	-0.1	-20.6	0.0	0.0	-0.1
C LC_154	SLN124 ring / End	-1.8	0.1	-11.6	0.0	0.0	0.1
	SLN125 ring / End	-1.7	-0.1	-11.5	0.0	0.0	-0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.9 < 100%	OK
Welds	17.9 < 100%	OK
Buckling	Not calculated	

Plates

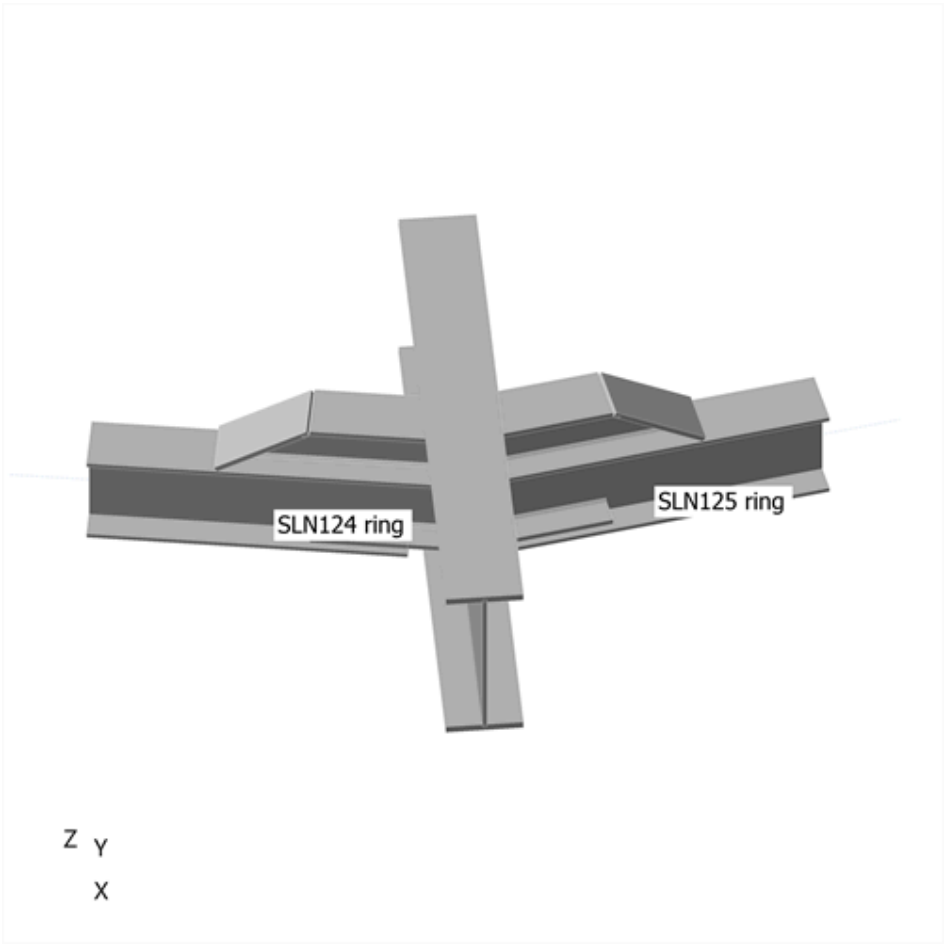
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN123b+SLN123a radial (continuous)-bfl 1	19.6	C LC_121	20.3	0.0	0.0	OK
SLN123b+SLN123a radial (continuous)-tfl 1	19.6	C LC_121	17.1	0.0	0.0	OK
SLN123b+SLN123a radial (continuous)-w 1	11.4	C LC_119	23.4	0.0	0.4	OK
SLN124 ring-bfl 1	12.7	C LC_135	46.5	0.0	0.0	OK
SLN124 ring-tfl 1	12.7	C LC_119	48.0	0.0	0.0	OK
SLN124 ring-w 1	7.2	C LC_119	47.3	0.0	0.0	OK
SLN125 ring-bfl 1	12.7	C LC_135	46.4	0.0	0.0	OK
SLN125 ring-tfl 1	12.7	C LC_119	47.7	0.0	0.0	OK
SLN125 ring-w 1	7.2	C LC_119	47.3	0.0	0.0	OK
End plate SLN125 ring	20.0	C LC_147	7.1	0.0	0.8	OK
Bottom flange plate SLN125 ring	15.0	C LC_135	13.3	0.0	0.0	OK
Bottom flange half plate L SLN125 ring	15.0	C LC_135	12.8	0.0	0.0	OK
Bottom flange half plate R SLN125 ring	15.0	C LC_135	12.7	0.0	0.0	OK
Haunch web SLN125 ring	12.0	C LC_119	31.9	0.0	0.0	OK
Haunch top plate SLN125 ring	15.0	C LC_147	5.5	0.0	0.0	OK
Haunch sloping plate SLN125 ring	15.0	C LC_119	14.3	0.0	0.0	OK
End plate SLN124 ring	20.0	C LC_119	7.1	0.0	1.2	OK
Bottom flange plate SLN124 ring	15.0	C LC_135	13.3	0.0	0.0	OK
Bottom flange half plate L SLN124 ring	15.0	C LC_135	12.7	0.0	0.0	OK
Bottom flange half plate R SLN124 ring	15.0	C LC_135	12.8	0.0	0.0	OK
Haunch web SLN124 ring	12.0	C LC_119	32.4	0.0	0.0	OK
Haunch top plate SLN124 ring	15.0	C LC_119	5.5	0.0	0.0	OK
Haunch sloping plate SLN124 ring	15.0	C LC_119	14.3	0.0	0.0	OK

Design data

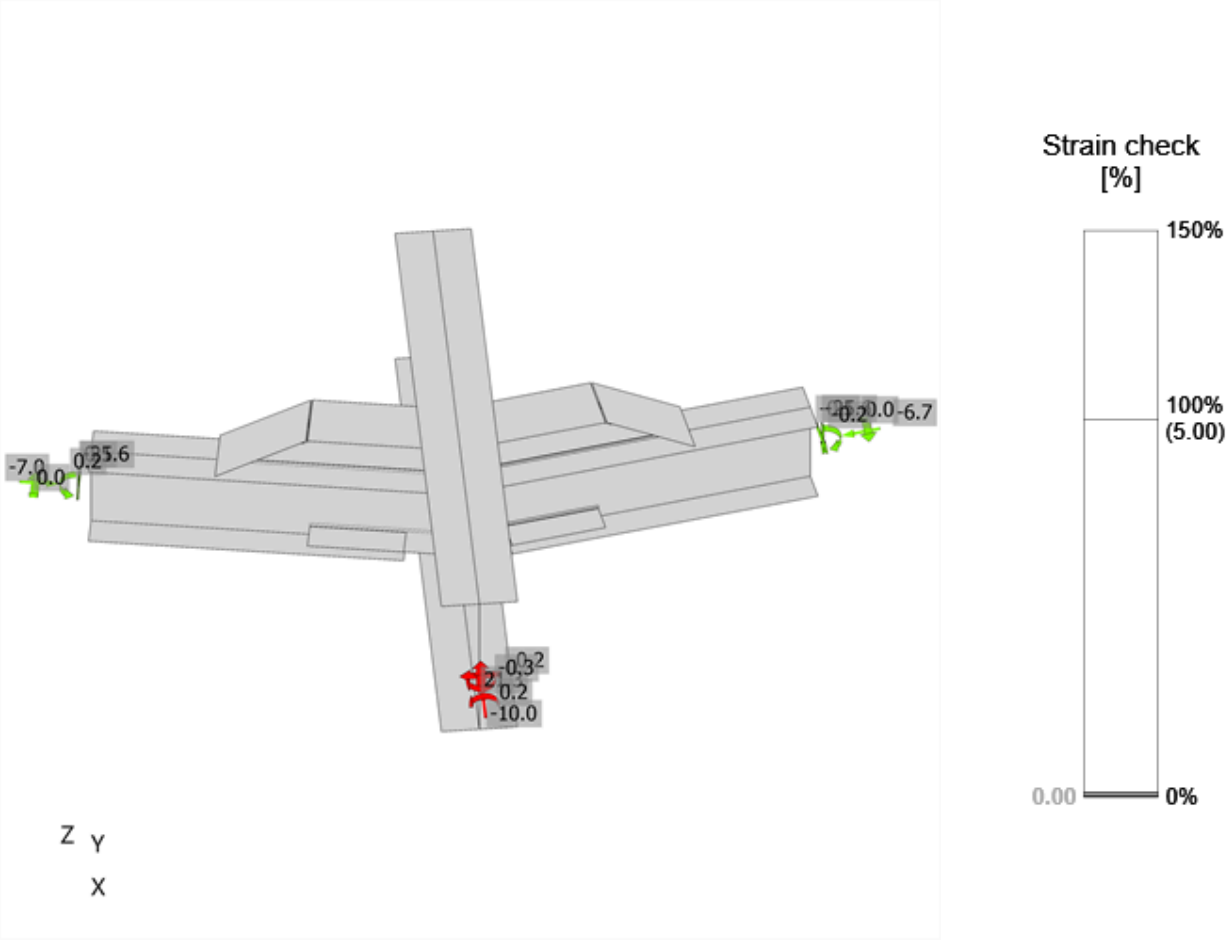
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

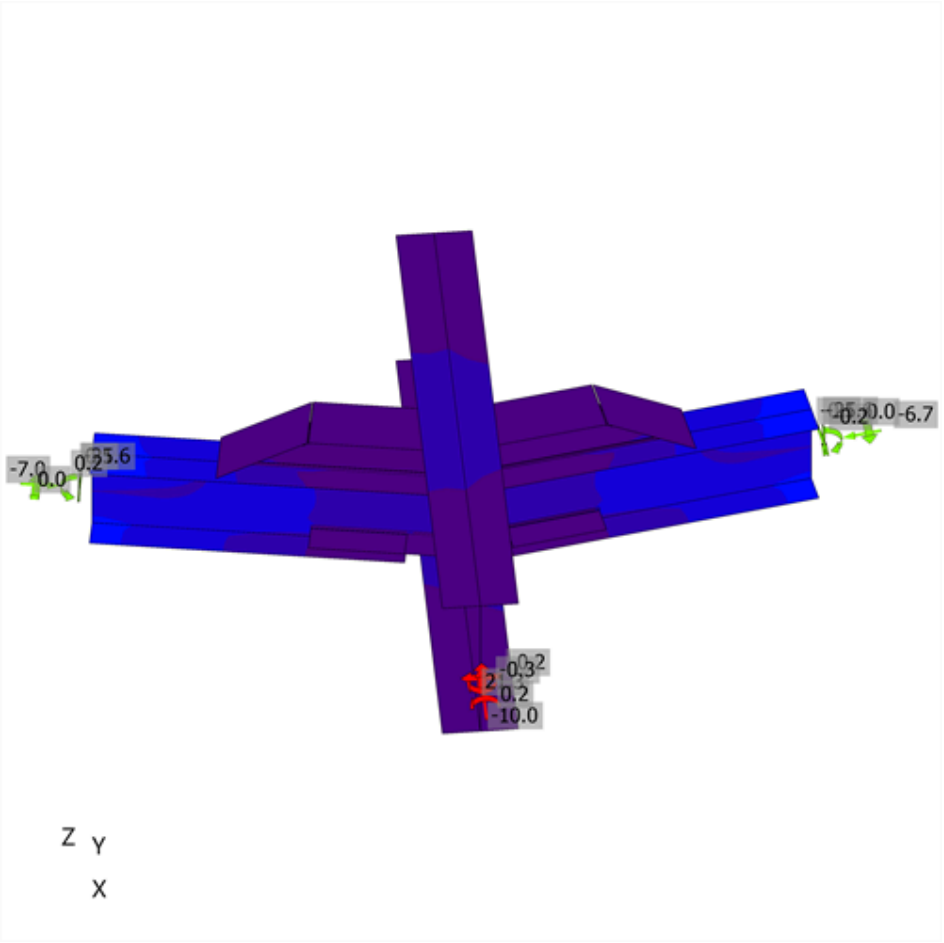
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

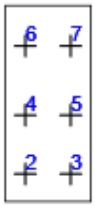


Strain check, C LC_121



Equivalent stress, C LC_121

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_119	0.4	0.4	194.1	0.1	0.4	0.3	OK	OK
	B3	M30 8.8 - 1	C LC_135	1.1	0.5	194.1	0.3	0.5	0.5	OK	OK
	B4	M30 8.8 - 1	C LC_119	0.1	0.4	232.2	0.0	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_135	0.0	0.5	232.2	0.0	0.4	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_147	2.5	0.8	249.2	0.8	0.6	0.9	OK	OK
	B7	M30 8.8 - 1	C LC_120	1.5	0.9	249.2	0.5	0.7	0.7	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN124 ring	Bottom flange plate SLN124 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN125 ring	Bottom flange plate SLN125 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN125 ring	Bottom flange plate SLN125 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN124 ring	Bottom flange plate SLN124 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN123b+SLN123a radial (continuous)-w 1	SLN124 ring-tfl 1	◢ 11.4	149	C LC_119	12.4	0.0	-10.7	-3.7	0.3	3.4	3.4	OK	OK
SLN123b+SLN123a radial (continuous)-w 1	SLN125 ring-tfl 1	◢ 11.4	149	C LC_119	12.5	0.0	-10.7	-3.6	-0.6	3.4	3.4	OK	OK
End plate SLN125 ring	SLN125 ring-w 1	◢ 6.0 ◣	208	C LC_135	9.9	0.0	0.5	-0.7	5.7	2.5	0.0	OK	OK
		◢ 6.0 ◣	208	C LC_135	8.1	0.0	-0.9	0.1	-4.7	2.0	0.0	OK	OK
Bottom flange plate SLN125 ring	SLN125 ring-w 1	◢ 6.0 ◣	100	C LC_120	20.3	0.0	1.7	2.1	11.5	5.0	5.0	OK	OK
		◢ 6.0 ◣	100	C LC_135	21.0	0.0	1.9	-1.9	-11.9	5.2	5.2	OK	OK
SLN125 ring-w 1	Bottom flange half plate L SLN125 ring	◢ 6.0 ◣	243	C LC_135	16.3	0.0	3.7	3.9	8.3	4.0	4.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	20.0	0.0	5.8	2.4	-10.8	4.9	4.9	OK	OK
SLN125 ring-w 1	Bottom flange half plate R SLN125 ring	◢ 6.0 ◣	243	C LC_135	17.0	0.0	4.2	4.0	-8.6	4.2	4.2	OK	OK
		◢ 6.0 ◣	243	C LC_135	19.3	0.0	5.9	2.3	10.3	4.8	4.8	OK	OK
SLN125 ring-tfl 1	Haunch web SLN125 ring	◢ 6.0 ◣	572	C LC_119	70.2	0.0	-16.6	-16.9	35.6	17.4	10.0	OK	OK
		◢ 6.0 ◣	572	C LC_119	71.8	0.0	-18.6	18.3	-35.6	17.7	10.1	OK	OK
End plate SLN124 ring	SLN124 ring-w 1	◢ 6.0 ◣	208	C LC_135	8.3	0.0	-0.8	0.0	4.7	2.0	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_135	10.2	0.0	0.5	0.6	-5.8	2.5	0.0	OK	OK
Bottom flange plate SLN124 ring	SLN124 ring-w 1	◢ 6.0 ◣	100	C LC_135	21.0	0.0	1.9	1.9	11.9	5.2	5.2	OK	OK
		◢ 6.0 ◣	100	C LC_135	20.1	0.0	1.7	-1.8	-11.4	5.0	5.0	OK	OK
SLN124 ring-w 1	Bottom flange half plate L SLN124 ring	◢ 6.0 ◣	243	C LC_135	17.2	0.0	4.3	4.0	8.7	4.2	4.2	OK	OK
		◢ 6.0 ◣	243	C LC_135	19.1	0.0	5.9	2.3	-10.3	4.7	4.7	OK	OK
SLN124 ring-w 1	Bottom flange half plate R SLN124 ring	◢ 6.0 ◣	243	C LC_135	16.0	0.0	3.6	3.8	-8.2	4.0	4.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	20.0	0.0	5.9	2.3	10.8	5.0	5.0	OK	OK
SLN124 ring-tfl 1	Haunch web SLN124 ring	◢ 6.0 ◣	572	C LC_119	72.4	0.0	-19.2	-18.8	35.6	17.9	10.2	OK	OK
		◢ 6.0 ◣	572	C LC_119	69.6	0.0	-16.0	16.4	-35.5	17.2	9.9	OK	OK
End plate SLN125 ring	Bottom flange plate SLN125 ring	◢ 6.0 ◣	149	C LC_110	9.5	0.0	-2.7	-3.7	3.7	2.3	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_112	12.0	0.0	-6.0	5.6	2.1	3.0	3.0	OK	OK
End plate SLN125 ring	Haunch web SLN125 ring	◢ 6.0 ◣	145	C LC_120	7.0	0.0	0.6	0.6	4.0	1.7	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_147	10.1	0.0	1.9	-1.7	-5.5	2.5	0.0	OK	OK
End plate SLN125 ring	Haunch top plate SLN125 ring	◢ 6.0 ◣	149	C LC_110	6.3	0.0	-3.0	-2.2	2.3	1.6	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_118	4.9	0.0	-1.5	2.3	-1.5	1.2	0.0	OK	OK
Haunch top plate SLN125 ring	Haunch web SLN125 ring	◢ 6.0 ◣	342	C LC_147	3.8	0.0	1.5	1.4	1.5	0.9	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_147	3.6	0.0	-0.2	0.2	2.1	0.9	0.0	OK	OK
Haunch sloping plate SLN125 ring	Haunch web SLN125 ring	◢ 6.0 ◣	271	C LC_119	55.0	0.0	-9.1	-8.7	30.1	13.6	8.6	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_121	58.7	0.0	-10.8	11.2	-31.4	14.5	8.9	OK	OK
End plate SLN124 ring	Bottom flange plate SLN124 ring	◢ 6.0 ◣	149	C LC_110	9.7	0.0	-4.6	-4.7	1.5	2.4	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_114	13.4	0.0	-6.7	6.3	-2.3	3.3	0.0	OK	OK
End plate SLN124 ring	Haunch web SLN124 ring	◢ 6.0 ◣	145	C LC_148	10.3	0.0	1.9	1.8	5.6	2.5	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	7.2	0.0	0.6	-0.6	-4.1	1.8	0.0	OK	OK
End plate SLN124 ring	Haunch top plate SLN124 ring	◢ 6.0 ◣	149	C LC_110	5.4	0.0	-2.8	-2.3	1.4	1.3	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_119	4.2	0.0	2.5	-1.2	1.5	1.0	0.0	OK	OK
Haunch top plate SLN124 ring	Haunch web SLN124 ring	◢ 6.0 ◣	342	C LC_145	3.6	0.0	-0.2	-0.2	-2.1	0.9	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_145	3.9	0.0	1.5	-1.4	-1.5	1.0	0.0	OK	OK
Haunch sloping plate SLN124 ring	Haunch web SLN124 ring	◢ 6.0 ◣	271	C LC_119	59.9	0.0	-11.2	-11.7	31.9	14.8	9.0	OK	OK
		◢ 6.0 ◣	271	C LC_121	54.0	0.0	-8.7	8.4	-29.6	13.4	8.5	OK	OK
SLN123b+SLN123a radial (continuous)-w 1	SLN125 ring-w 1	◢ 7.2	208	C LC_122	36.1	0.0	3.4	-3.5	20.5	8.9	8.8	OK	OK
SLN123b+SLN123a radial (continuous)-w 1	SLN124 ring-w 1	◢ 7.2	208	C LC_135	36.8	0.0	7.5	6.2	-19.9	9.1	8.9	OK	OK
Haunch sloping plate SLN125 ring	Haunch top plate SLN125 ring	◢ 6.0 ◣	146	C LC_119	7.0	0.0	-5.1	2.8	0.3	1.9	0.0	OK	OK
		◢ 6.0 ◣	146	C LC_147	2.6	0.0	-0.6	1.3	0.5	0.6	0.0	OK	OK
Haunch sloping plate SLN124 ring	Haunch top plate SLN124 ring	◢ 6.0 ◣	146	C LC_120	7.1	0.0	-5.2	2.8	-0.3	1.9	0.0	OK	OK
		◢ 6.0 ◣	146	C LC_145	2.6	0.0	-0.6	1.4	-0.5	0.7	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_340
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN127a+SLN127b radial (continuous) - SLN125 ring - SLN126 ring

Name	SLN127a+SLN127b radial (continuous) - SLN125 ring - SLN126 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

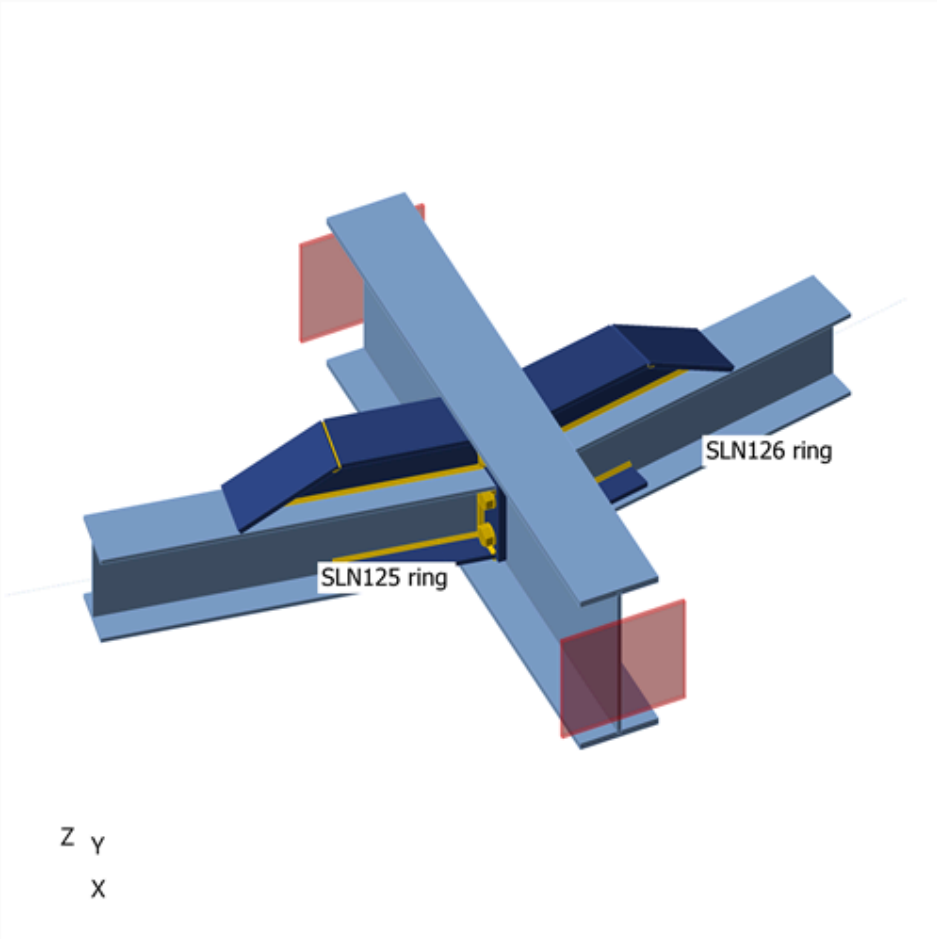
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN127a+SLN127b radial (continuous)	1 - UB457x191x98	-20.0	0.0	0.0	0	0	0
SLN125 ring	2 - UB254x146x43	-120.0	0.0	0.0	0	0	-102
SLN126 ring	2 - UB254x146x43	80.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN127a+SLN127b radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN127a+SLN127b radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN125 ring / end		Position	0
SLN126 ring / end		Position	0



SLN127a+SLN127b radial (continuous) - SLN125 ring - SLN126 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN125 ring / End	0.6	0.0	-4.9	0.0	0.0	0.0
	SLN126 ring / End	0.6	0.0	-4.9	0.0	0.0	0.0
C LC_102	SLN125 ring / End	-4.0	-0.1	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-4.0	0.1	-25.6	0.0	0.0	0.1
C LC_103	SLN125 ring / End	-8.6	0.0	-24.6	0.0	0.0	-0.1
	SLN126 ring / End	-8.7	0.0	-24.5	0.0	0.0	0.1
C LC_104	SLN125 ring / End	-5.5	-0.1	-24.6	0.0	0.0	-0.1
	SLN126 ring / End	-5.2	0.2	-24.5	0.0	0.0	0.2
C LC_105	SLN125 ring / End	-8.0	0.0	-24.6	0.0	0.0	-0.1
	SLN126 ring / End	-7.5	0.0	-24.5	0.0	0.0	0.1
C LC_106	SLN125 ring / End	-5.8	-0.2	-24.6	0.0	0.0	-0.2
	SLN126 ring / End	-6.5	0.0	-24.5	0.0	0.0	0.0
C LC_107	SLN125 ring / End	-3.5	0.0	-1.6	0.0	0.0	0.0
	SLN126 ring / End	-3.6	0.0	-1.6	0.0	0.0	0.0
C LC_108	SLN125 ring / End	0.1	0.1	-1.6	0.0	0.0	0.2
	SLN126 ring / End	0.5	0.1	-1.6	0.0	0.0	0.1
C LC_109	SLN125 ring / End	-2.9	0.1	-1.6	0.0	0.0	0.1
	SLN126 ring / End	-2.3	0.1	-1.6	0.0	0.0	0.1
C LC_110	SLN125 ring / End	0.1	-0.2	-1.6	0.0	0.0	-0.3
	SLN126 ring / End	-0.9	-0.2	-1.6	0.0	0.0	-0.3
C LC_111	SLN125 ring / End	-8.4	0.0	-17.3	0.0	0.0	0.0
	SLN126 ring / End	-8.6	0.0	-17.3	0.0	0.0	0.1
C LC_112	SLN125 ring / End	-3.9	0.0	-17.3	0.0	0.0	0.1
	SLN126 ring / End	-3.4	0.2	-17.3	0.0	0.0	0.2
C LC_113	SLN125 ring / End	-7.6	0.0	-17.3	0.0	0.0	0.0
	SLN126 ring / End	-6.9	0.1	-17.3	0.0	0.0	0.1
C LC_114	SLN125 ring / End	-4.1	-0.2	-17.3	0.0	0.0	-0.3
	SLN126 ring / End	-5.2	-0.1	-17.3	0.0	0.0	-0.2
C LC_115	SLN125 ring / End	-3.4	0.0	-2.8	0.0	0.0	0.0
	SLN126 ring / End	-3.5	0.0	-2.8	0.0	0.0	0.1
C LC_116	SLN125 ring / End	0.3	0.1	-2.8	0.0	0.0	0.2
	SLN126 ring / End	0.7	0.1	-2.8	0.0	0.0	0.1
C LC_117	SLN125 ring / End	-2.8	0.1	-2.8	0.0	0.0	0.1
	SLN126 ring / End	-2.2	0.1	-2.8	0.0	0.0	0.1
C LC_118	SLN125 ring / End	0.3	-0.2	-2.8	0.0	0.0	-0.3
	SLN126 ring / End	-0.7	-0.1	-2.8	0.0	0.0	-0.3
C LC_119	SLN125 ring / End	-7.3	0.0	-25.6	0.0	0.0	0.0
	SLN126 ring / End	-7.3	0.0	-25.6	0.0	0.0	0.1
C LC_120	SLN125 ring / End	-4.9	-0.1	-25.6	0.0	0.0	-0.1
	SLN126 ring / End	-4.8	0.2	-25.6	0.0	0.0	0.2
C LC_121	SLN125 ring / End	-6.5	-0.1	-25.6	0.0	0.0	-0.1
	SLN126 ring / End	-6.3	0.0	-25.6	0.0	0.0	0.0
C LC_122	SLN125 ring / End	-5.1	-0.1	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-5.3	0.1	-25.6	0.0	0.0	0.1
C LC_123	SLN125 ring / End	-5.7	0.0	-19.4	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN126 ring / End	-5.8	0.0	-19.4	0.0	0.0	0.1
C LC_124	SLN125 ring / End	-2.8	0.0	-19.4	0.0	0.0	0.0
	SLN126 ring / End	-2.7	0.2	-19.4	0.0	0.0	0.2
C LC_125	SLN125 ring / End	-4.6	0.0	-19.4	0.0	0.0	-0.1
	SLN126 ring / End	-4.5	0.0	-19.4	0.0	0.0	0.0
C LC_126	SLN125 ring / End	-2.7	-0.2	-19.4	0.0	0.0	-0.2
	SLN126 ring / End	-2.9	0.0	-19.4	0.0	0.0	0.0
C LC_127	SLN125 ring / End	-0.7	0.0	-4.9	0.0	0.0	0.0
	SLN126 ring / End	-0.7	0.0	-4.9	0.0	0.0	0.1
C LC_128	SLN125 ring / End	1.4	0.0	-4.9	0.0	0.0	0.1
	SLN126 ring / End	1.5	0.1	-4.9	0.0	0.0	0.2
C LC_129	SLN125 ring / End	0.1	0.0	-4.9	0.0	0.0	-0.1
	SLN126 ring / End	0.2	0.0	-4.9	0.0	0.0	0.0
C LC_130	SLN125 ring / End	1.7	-0.1	-4.9	0.0	0.0	-0.2
	SLN126 ring / End	1.6	0.0	-4.9	0.0	0.0	0.0
C LC_135	SLN125 ring / End	-0.4	-0.1	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-0.4	0.1	-25.6	0.0	0.0	0.1
C LC_141	SLN125 ring / End	0.4	0.0	-4.9	0.0	0.0	0.0
	SLN126 ring / End	0.4	0.0	-4.9	0.0	0.0	0.0
C LC_142	SLN125 ring / End	0.8	0.0	-4.9	0.0	0.0	-0.1
	SLN126 ring / End	0.8	0.0	-4.9	0.0	0.0	0.1
C LC_143	SLN125 ring / End	0.5	0.0	-4.9	0.0	0.0	-0.1
	SLN126 ring / End	0.6	0.0	-4.9	0.0	0.0	0.0
C LC_144	SLN125 ring / End	0.7	0.0	-4.9	0.0	0.0	0.0
	SLN126 ring / End	0.6	0.0	-4.9	0.0	0.0	0.1
C LC_145	SLN125 ring / End	-4.7	0.0	-25.6	0.0	0.0	-0.1
	SLN126 ring / End	-4.8	0.0	-25.6	0.0	0.0	0.1
C LC_146	SLN125 ring / End	-3.3	-0.2	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-3.2	0.2	-25.6	0.0	0.0	0.2
C LC_147	SLN125 ring / End	-4.4	-0.1	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-4.2	0.0	-25.6	0.0	0.0	0.1
C LC_148	SLN125 ring / End	-3.6	-0.1	-25.6	0.0	0.0	-0.1
	SLN126 ring / End	-3.8	0.1	-25.6	0.0	0.0	0.2
C LC_151	SLN125 ring / End	-1.8	-0.1	-11.6	0.0	0.0	-0.1
	SLN126 ring / End	-1.8	0.1	-11.6	0.0	0.0	0.1
C LC_152	SLN125 ring / End	-6.8	-0.1	-14.0	0.0	0.0	-0.1
	SLN126 ring / End	-6.8	0.1	-14.1	0.0	0.0	0.1
C LC_153	SLN125 ring / End	-4.1	-0.1	-20.7	0.0	0.0	-0.2
	SLN126 ring / End	-4.2	0.1	-20.8	0.0	0.0	0.1
C LC_154	SLN125 ring / End	-1.7	-0.1	-11.6	0.0	0.0	-0.1
	SLN126 ring / End	-1.7	0.1	-11.6	0.0	0.0	0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.9 < 100%	OK
Welds	18.0 < 100%	OK
Buckling	Not calculated	

Plates

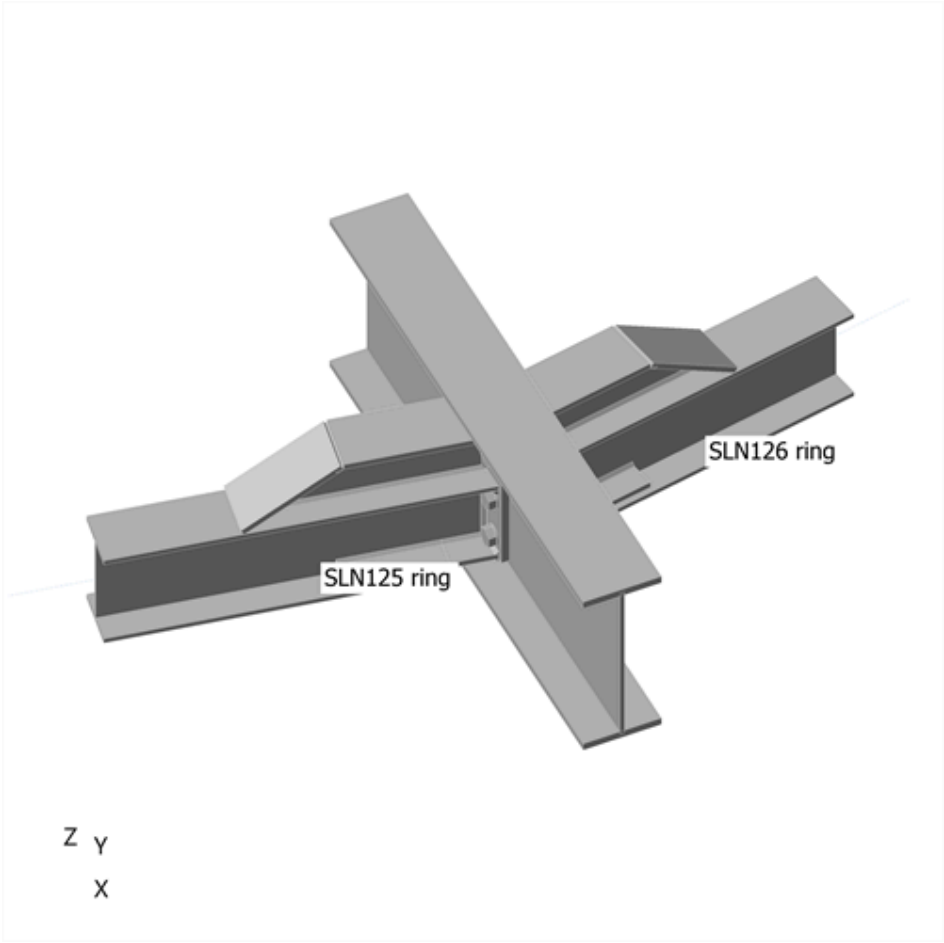
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN127a+SLN127b radial (continuous)-bfl 1	19.6	C LC_121	20.3	0.0	0.0	OK
SLN127a+SLN127b radial (continuous)-tfl 1	19.6	C LC_121	17.0	0.0	0.0	OK
SLN127a+SLN127b radial (continuous)-w 1	11.4	C LC_121	23.4	0.0	0.5	OK
SLN125 ring-bfl 1	12.7	C LC_135	46.5	0.0	0.0	OK
SLN125 ring-tfl 1	12.7	C LC_121	48.2	0.0	0.0	OK
SLN125 ring-w 1	7.2	C LC_119	47.4	0.0	0.0	OK
SLN126 ring-bfl 1	12.7	C LC_135	46.4	0.0	0.0	OK
SLN126 ring-tfl 1	12.7	C LC_119	48.2	0.0	0.0	OK
SLN126 ring-w 1	7.2	C LC_119	47.3	0.0	0.0	OK
End plate SLN126 ring	20.0	C LC_120	6.7	0.0	0.9	OK
Bottom flange plate SLN126 ring	15.0	C LC_114	14.0	0.0	0.0	OK
Bottom flange half plate L SLN126 ring	15.0	C LC_135	12.7	0.0	0.0	OK
Bottom flange half plate R SLN126 ring	15.0	C LC_135	12.8	0.0	0.0	OK
Haunch web SLN126 ring	12.0	C LC_119	32.4	0.0	0.0	OK
Haunch top plate SLN126 ring	15.0	C LC_120	5.5	0.0	0.0	OK
Haunch sloping plate SLN126 ring	15.0	C LC_119	14.3	0.0	0.0	OK
End plate SLN125 ring	20.0	C LC_106	6.8	0.0	2.1	OK
Bottom flange plate SLN125 ring	15.0	C LC_135	13.4	0.0	0.0	OK
Bottom flange half plate L SLN125 ring	15.0	C LC_135	12.9	0.0	0.0	OK
Bottom flange half plate R SLN125 ring	15.0	C LC_135	12.7	0.0	0.0	OK
Haunch web SLN125 ring	12.0	C LC_119	32.1	0.0	0.0	OK
Haunch top plate SLN125 ring	15.0	C LC_146	5.5	0.0	0.0	OK
Haunch sloping plate SLN125 ring	15.0	C LC_119	14.4	0.0	0.0	OK

Design data

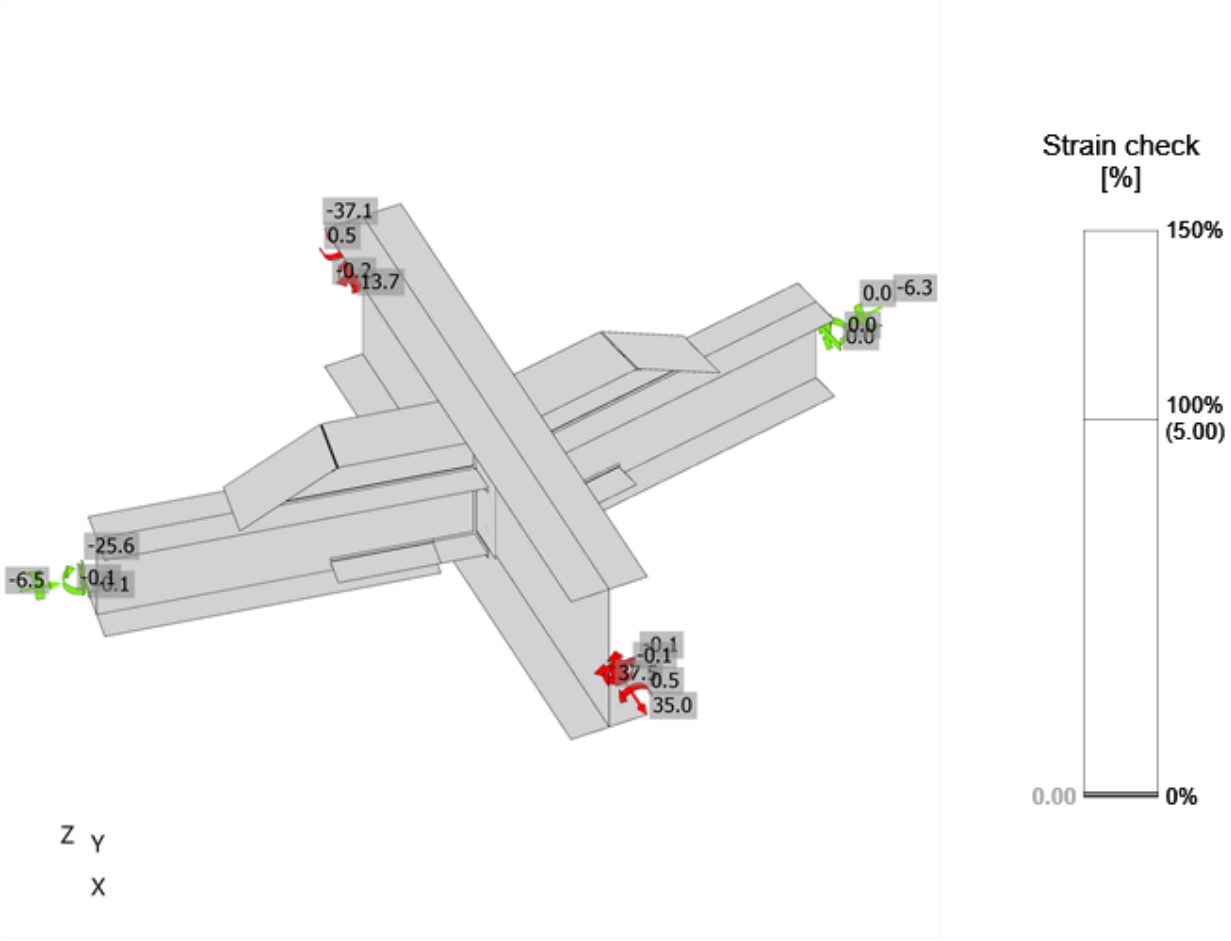
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

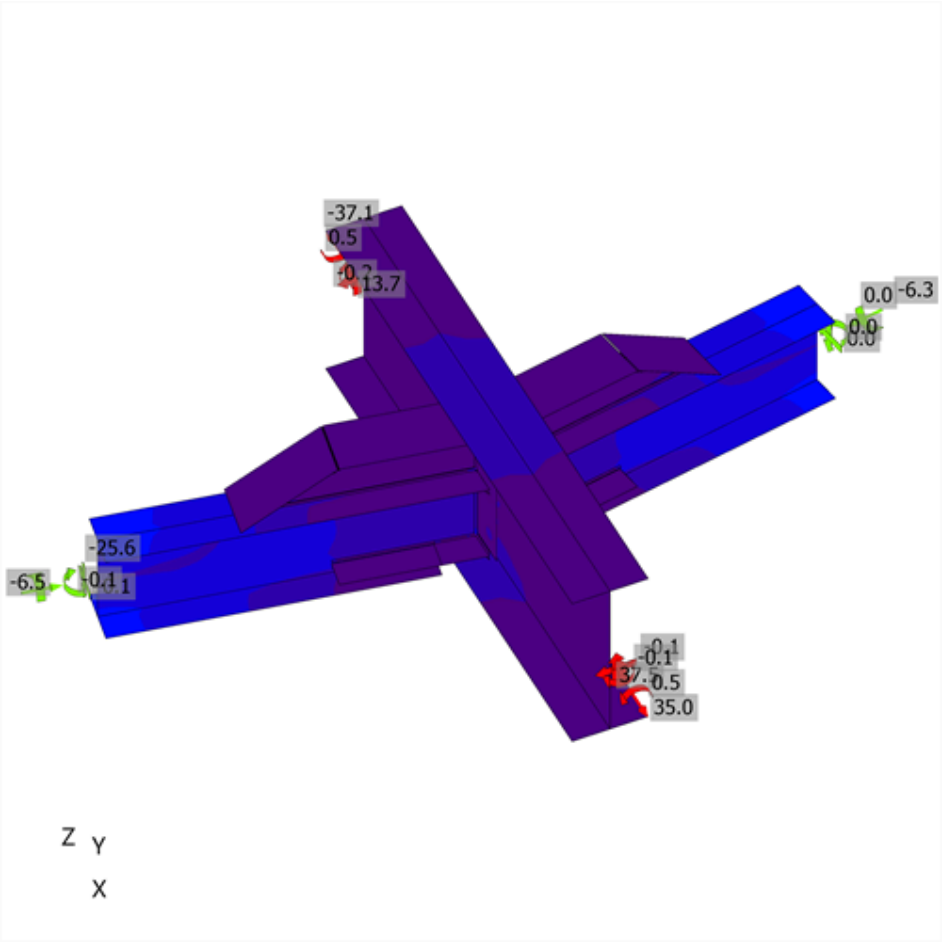
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

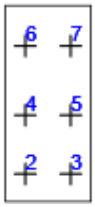


Strain check, C LC_121



Equivalent stress, C LC_121

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_135	1.1	0.4	194.1	0.4	0.5	0.5	OK	OK
	B3	M30 8.8 - 1	C LC_135	0.5	0.5	194.1	0.2	0.5	0.4	OK	OK
	B4	M30 8.8 - 1	C LC_119	0.0	0.4	232.2	0.0	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_135	0.0	0.5	232.2	0.0	0.4	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_121	1.7	0.9	249.2	0.5	0.7	0.8	OK	OK
	B7	M30 8.8 - 1	C LC_120	2.2	0.8	249.2	0.7	0.7	0.9	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate R SLN126 ring	Bottom flange plate SLN126 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN125 ring	Bottom flange plate SLN125 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN126 ring	Bottom flange plate SLN126 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN125 ring	Bottom flange plate SLN125 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN127a+SLN127b radial (continuous)-w 1	SLN125 ring-tfl 1	◢ 11.4	149	C LC_119	11.8	0.0	-10.1	-3.4	0.1	3.3	3.3	OK	OK
SLN127a+SLN127b radial (continuous)-w 1	SLN126 ring-tfl 1	◢ 11.4	149	C LC_119	11.8	0.0	-10.1	-3.5	-0.2	3.3	3.3	OK	OK
End plate SLN126 ring	SLN126 ring-w 1	◢ 6.0 ◣	208	C LC_135	10.8	0.0	0.4	-1.3	6.1	2.7	0.0	OK	OK
		◢ 6.0 ◣	208	C LC_135	8.0	0.0	-0.8	-0.5	-4.6	2.0	0.0	OK	OK
Bottom flange plate SLN126 ring	SLN126 ring-w 1	◢ 6.0 ◣	100	C LC_135	20.8	0.0	1.8	1.6	11.9	5.1	5.1	OK	OK
		◢ 6.0 ◣	100	C LC_135	20.4	0.0	1.8	-2.0	-11.6	5.0	5.0	OK	OK
SLN126 ring-w 1	Bottom flange half plate L SLN126 ring	◢ 6.0 ◣	243	C LC_135	17.0	0.0	4.3	4.0	8.6	4.2	4.2	OK	OK
		◢ 6.0 ◣	243	C LC_135	19.1	0.0	6.0	2.3	-10.2	4.7	4.7	OK	OK
SLN126 ring-w 1	Bottom flange half plate R SLN126 ring	◢ 6.0 ◣	243	C LC_135	16.2	0.0	3.6	3.9	-8.3	4.0	4.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	20.1	0.0	5.8	2.4	10.9	5.0	5.0	OK	OK
SLN126 ring-tfl 1	Haunch web SLN126 ring	◢ 6.0 ◣	572	C LC_119	72.7	0.0	-19.0	-18.8	35.9	18.0	10.2	OK	OK
		◢ 6.0 ◣	572	C LC_119	69.4	0.0	-16.2	16.4	-35.3	17.1	9.9	OK	OK
End plate SLN125 ring	SLN125 ring-w 1	◢ 6.0 ◣	208	C LC_135	7.9	0.0	-0.6	0.6	4.5	2.0	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_135	10.5	0.0	0.6	1.0	-6.0	2.6	0.0	OK	OK
Bottom flange plate SLN125 ring	SLN125 ring-w 1	◢ 6.0 ◣	100	C LC_135	20.3	0.0	1.8	1.8	11.5	5.0	5.0	OK	OK
		◢ 6.0 ◣	100	C LC_135	21.0	0.0	1.8	-1.8	-11.9	5.2	5.2	OK	OK
SLN125 ring-w 1	Bottom flange half plate L SLN125 ring	◢ 6.0 ◣	243	C LC_135	16.2	0.0	3.7	3.9	8.3	4.0	4.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	20.1	0.0	5.9	2.3	-10.9	5.0	5.0	OK	OK
SLN125 ring-w 1	Bottom flange half plate R SLN125 ring	◢ 6.0 ◣	243	C LC_135	17.0	0.0	4.2	4.0	-8.7	4.2	4.2	OK	OK
		◢ 6.0 ◣	243	C LC_135	19.2	0.0	5.9	2.3	10.3	4.7	4.7	OK	OK
SLN125 ring-tfl 1	Haunch web SLN125 ring	◢ 6.0 ◣	572	C LC_119	69.9	0.0	-16.6	-16.8	35.5	17.3	9.9	OK	OK
		◢ 6.0 ◣	572	C LC_119	72.3	0.0	-18.7	18.5	-35.8	17.9	10.2	OK	OK
End plate SLN126 ring	Bottom flange plate SLN126 ring	◢ 6.0 ◣	149	C LC_114	15.7	0.0	-4.3	-6.0	6.2	3.9	3.9	OK	OK
		◢ 6.0 ◣	149	C LC_114	20.2	0.0	-8.3	6.5	-8.4	5.0	5.0	OK	OK
End plate SLN126 ring	Haunch web SLN126 ring	◢ 6.0 ◣	145	C LC_120	6.9	0.0	1.1	0.9	3.9	1.7	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_146	10.4	0.0	1.4	-1.7	-5.7	2.6	0.0	OK	OK
End plate SLN126 ring	Haunch top plate SLN126 ring	◢ 6.0 ◣	149	C LC_110	8.9	0.0	-4.3	-3.1	3.3	2.2	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_110	6.7	0.0	-1.9	3.2	-1.9	1.6	0.0	OK	OK
Haunch top plate SLN126 ring	Haunch web SLN126 ring	◢ 6.0 ◣	342	C LC_120	3.5	0.0	-0.3	-0.2	-2.0	0.9	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_120	3.8	0.0	1.5	-1.4	-1.5	0.9	0.0	OK	OK
Haunch sloping plate SLN126 ring	Haunch web SLN126 ring	◢ 6.0 ◣	271	C LC_119	59.7	0.0	-11.1	-11.3	31.9	14.7	9.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_119	53.9	0.0	-8.7	8.5	-29.5	13.3	8.5	OK	OK
End plate SLN125 ring	Bottom flange plate SLN125 ring	◢ 6.0 ◣	149	C LC_110	10.8	0.0	-4.9	-5.3	1.8	2.7	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_114	13.6	0.0	-7.4	6.1	-2.6	3.4	0.0	OK	OK
End plate SLN125 ring	Haunch web SLN125 ring	◢ 6.0 ◣	145	C LC_148	10.4	0.0	1.4	1.6	5.7	2.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	6.9	0.0	1.1	-0.9	-3.8	1.7	0.0	OK	OK
End plate SLN125 ring	Haunch top plate SLN125 ring	◢ 6.0 ◣	149	C LC_110	7.1	0.0	-3.6	-3.0	1.8	1.8	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_118	5.6	0.0	-1.9	2.5	-1.8	1.4	0.0	OK	OK
Haunch top plate SLN125 ring	Haunch web SLN125 ring	◢ 6.0 ◣	342	C LC_146	3.9	0.0	1.5	1.4	1.6	1.0	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_146	3.5	0.0	-0.3	0.2	2.0	0.9	0.0	OK	OK
Haunch sloping plate SLN125 ring	Haunch web SLN125 ring	◢ 6.0 ◣	271	C LC_119	54.8	0.0	-9.0	-8.9	29.9	13.5	8.5	OK	OK
		◢ 6.0 ◣	271	C LC_119	59.0	0.0	-10.8	11.0	-31.6	14.6	8.9	OK	OK
SLN127a+SLN127b radial (continuous)-w 1	SLN126 ring-w 1	◢ 7.2	208	C LC_135	36.0	0.0	6.9	-7.4	19.0	8.9	8.8	OK	OK
SLN127a+SLN127b radial (continuous)-w 1	SLN125 ring-w 1	◢ 7.2	208	C LC_135	36.4	0.0	7.1	7.0	-19.4	9.0	8.8	OK	OK
Haunch sloping plate SLN126 ring	Haunch top plate SLN126 ring	◢ 6.0 ◣	146	C LC_119	7.1	0.0	-5.1	2.9	-0.3	1.9	0.0	OK	OK
		◢ 6.0 ◣	146	C LC_120	2.6	0.0	-0.6	1.3	-0.6	0.6	0.0	OK	OK
Haunch sloping plate SLN125 ring	Haunch top plate SLN125 ring	◢ 6.0 ◣	146	C LC_121	7.0	0.0	-5.1	2.8	0.3	1.9	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_146	2.6	0.0	-0.6	1.4	0.6	0.7	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-B_040
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN81+stub radial (ends 200 past the node) - SLN121 column below - SLN83 ring - SLN84 ring

Name	SLN81+stub radial (ends 200 past the node) - SLN121 column below - SLN83 ring - SLN84 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

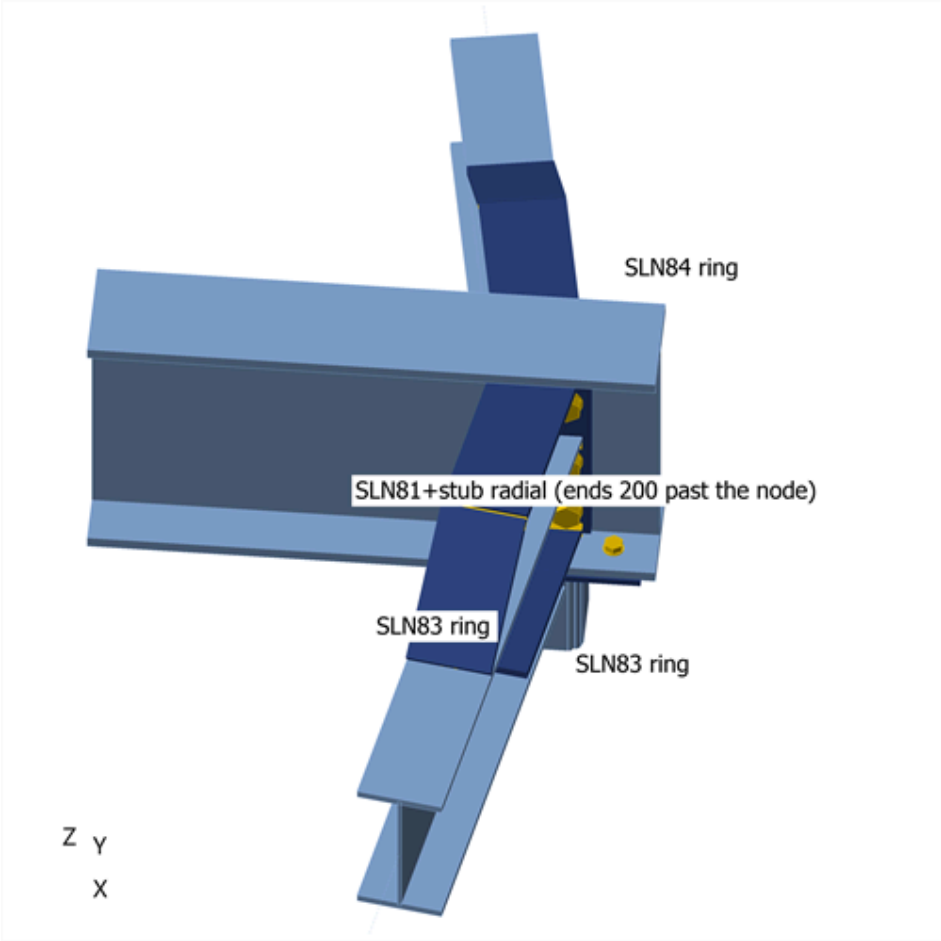
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN81+stub radial (ends 200 past the node)	1 - UB457x191x98	40.0	0.0	0.0	200	0	0
SLN121 column below	2 - SHS150x150x10	0.0	90.0	-140.0	0	0	0
SLN83 ring	3 - UB254x146x43	-60.0	0.0	0.0	0	0	-102
SLN84 ring	3 - UB254x146x43	140.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN81+stub radial (ends 200 past the node) / begin		Position	0
SLN121 column below / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN83 ring / end		Position	0
SLN84 ring / end		Position	0



SLN81+stub radial (ends 200 past the node) - SLN121 column below - SLN83 ring - SLN84 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - SHS150x150x10	S 275
3 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN81+stub radial (ends 200 past the node) / Begin	0.6	0.0	-11.1	0.0	-1.0	-0.1
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	0.6	0.0	-2.2	0.0	0.0	0.0
	SLN84 ring / End	0.6	0.1	-2.2	0.0	0.0	0.1
C LC_102	SLN81+stub radial (ends 200 past the node) / Begin	0.8	0.1	-45.9	0.0	-4.4	-0.4
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-4.1	0.0	-8.9	0.0	0.0	-0.1
	SLN84 ring / End	-4.0	0.9	-8.8	0.0	0.0	0.7
C LC_103	SLN81+stub radial (ends 200 past the node) / Begin	-0.7	0.1	-44.0	0.0	-4.4	-0.5
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-8.8	-0.1	-8.5	0.0	0.0	-0.2
	SLN84 ring / End	-8.8	0.9	-8.5	0.0	0.0	0.7
C LC_104	SLN81+stub radial (ends 200 past the node) / Begin	0.2	0.2	-44.1	0.0	-4.0	-0.5
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-5.4	-0.1	-8.5	0.0	0.0	-0.1
	SLN84 ring / End	-5.3	0.9	-8.5	0.0	0.0	0.8
C LC_105	SLN81+stub radial (ends 200 past the node) / Begin	0.1	0.0	-44.2	0.0	-4.2	-0.4
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-6.4	-0.1	-8.5	0.0	0.0	-0.2
	SLN84 ring / End	-6.4	0.9	-8.5	0.0	0.0	0.7
C LC_106	SLN81+stub radial (ends 200 past the node) / Begin	-0.6	0.2	-44.0	0.0	-4.1	-0.4
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-7.6	-0.1	-8.5	0.0	0.0	-0.1
	SLN84 ring / End	-7.7	0.8	-8.5	0.0	0.0	0.7
C LC_107	SLN81+stub radial (ends 200 past the node) / Begin	-1.0	0.1	-4.5	0.0	-0.5	-0.2
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-3.9	0.0	-0.9	0.0	0.0	0.0
	SLN84 ring / End	-3.9	0.3	-0.9	0.0	0.0	0.2
C LC_108	SLN81+stub radial (ends 200 past the node) / Begin	0.0	0.1	-4.8	0.0	0.0	0.0
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	0.5	0.0	-0.9	0.0	0.0	0.0
	SLN84 ring / End	0.8	0.0	-0.9	0.0	0.0	0.1
C LC_109	SLN81+stub radial (ends 200 past the node) / Begin	-0.2	0.0	-4.9	0.0	-0.4	-0.1
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-1.2	0.0	-0.9	0.0	0.0	-0.1
	SLN84 ring / End	-1.4	0.2	-0.9	0.0	0.0	0.1
C LC_110	SLN81+stub radial (ends 200 past the node) / Begin	-0.7	0.0	-4.4	0.0	0.0	0.1
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-1.6	-0.1	-0.9	0.0	0.0	-0.1
	SLN84 ring / End	-1.7	0.0	-0.9	0.0	0.0	0.0
C LC_111	SLN81+stub radial (ends 200 past the node) / Begin	-1.3	0.1	-31.7	0.0	-3.2	-0.4
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-8.8	0.0	-6.2	0.0	0.0	-0.1
	SLN84 ring / End	-8.8	0.8	-6.2	0.0	0.0	0.7
C LC_112	SLN81+stub radial (ends 200 past the node) / Begin	0.0	0.2	-32.0	0.0	-2.7	-0.4

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-3.6	0.0	-6.2	0.0	0.0	-0.1
	SLN84 ring / End	-3.4	0.6	-6.2	0.0	0.0	0.6
C LC_113	SLN81+stub radial (ends 200 past the node) / Begin	-0.2	0.0	-32.1	0.0	-3.0	-0.4
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-5.3	-0.1	-6.2	0.0	0.0	-0.2
	SLN84 ring / End	-5.5	0.7	-6.2	0.0	0.0	0.6
C LC_114	SLN81+stub radial (ends 200 past the node) / Begin	-1.0	0.1	-31.6	0.0	-2.7	-0.2
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-6.4	-0.1	-6.2	0.0	0.0	-0.1
	SLN84 ring / End	-6.5	0.6	-6.2	0.0	0.0	0.5
C LC_115	SLN81+stub radial (ends 200 past the node) / Begin	-0.9	0.1	-7.4	0.0	-0.7	-0.2
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-3.8	0.0	-1.5	0.0	0.0	0.0
	SLN84 ring / End	-3.8	0.3	-1.5	0.0	0.0	0.2
C LC_116	SLN81+stub radial (ends 200 past the node) / Begin	0.1	0.1	-7.7	0.0	-0.3	-0.1
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	0.7	0.0	-1.5	0.0	0.0	0.0
	SLN84 ring / End	1.0	0.1	-1.5	0.0	0.0	0.1
C LC_117	SLN81+stub radial (ends 200 past the node) / Begin	-0.1	0.0	-7.8	0.0	-0.7	-0.1
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-1.0	0.0	-1.5	0.0	0.0	-0.1
	SLN84 ring / End	-1.2	0.2	-1.5	0.0	0.0	0.1
C LC_118	SLN81+stub radial (ends 200 past the node) / Begin	-0.6	0.0	-7.3	0.0	-0.3	0.1
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-1.5	-0.1	-1.5	0.0	0.0	-0.1
	SLN84 ring / End	-1.5	0.1	-1.5	0.0	0.0	0.0
C LC_119	SLN81+stub radial (ends 200 past the node) / Begin	0.0	0.1	-46.0	0.0	-4.8	-0.5
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-7.4	0.0	-8.9	0.0	0.0	-0.2
	SLN84 ring / End	-7.3	0.9	-8.8	0.0	0.0	0.7
C LC_120	SLN81+stub radial (ends 200 past the node) / Begin	0.4	0.2	-45.7	0.0	-4.1	-0.5
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-4.9	0.0	-8.9	0.0	0.0	-0.1
	SLN84 ring / End	-4.8	0.9	-8.8	0.0	0.0	0.8
C LC_121	SLN81+stub radial (ends 200 past the node) / Begin	0.3	0.0	-45.8	0.0	-4.3	-0.4
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-5.7	-0.1	-8.9	0.0	0.0	-0.2
	SLN84 ring / End	-5.7	0.9	-8.8	0.0	0.0	0.7
C LC_122	SLN81+stub radial (ends 200 past the node) / Begin	0.1	0.2	-45.9	0.0	-4.6	-0.5
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-6.2	0.0	-8.9	0.0	0.0	-0.1
	SLN84 ring / End	-6.3	0.9	-8.8	0.0	0.0	0.7
C LC_123	SLN81+stub radial (ends 200 past the node) / Begin	0.2	0.1	-35.6	0.0	-4.1	-0.4
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-5.9	0.0	-6.9	0.0	0.0	-0.1
	SLN84 ring / End	-5.9	0.8	-6.8	0.0	0.0	0.7

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_124	SLN81+stub radial (ends 200 past the node) / Begin	0.4	0.2	-35.2	0.0	-2.8	-0.4
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-2.6	0.0	-6.9	0.0	0.0	0.0
	SLN84 ring / End	-2.3	0.6	-6.8	0.0	0.0	0.6
C LC_125	SLN81+stub radial (ends 200 past the node) / Begin	0.3	0.0	-35.3	0.0	-3.2	-0.3
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-3.9	-0.1	-6.9	0.0	0.0	-0.2
	SLN84 ring / End	-4.1	0.7	-6.8	0.0	0.0	0.5
C LC_126	SLN81+stub radial (ends 200 past the node) / Begin	0.3	0.1	-35.5	0.0	-3.5	-0.3
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-3.7	0.0	-6.9	0.0	0.0	-0.1
	SLN84 ring / End	-3.8	0.6	-6.8	0.0	0.0	0.5
C LC_127	SLN81+stub radial (ends 200 past the node) / Begin	0.6	0.1	-11.3	0.0	-1.7	-0.2
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-0.8	0.0	-2.2	0.0	0.0	0.0
	SLN84 ring / End	-0.9	0.3	-2.2	0.0	0.0	0.2
C LC_128	SLN81+stub radial (ends 200 past the node) / Begin	0.5	0.1	-10.9	0.0	-0.4	-0.1
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	1.7	0.1	-2.2	0.0	0.0	0.1
	SLN84 ring / End	2.0	0.1	-2.2	0.0	0.0	0.1
C LC_129	SLN81+stub radial (ends 200 past the node) / Begin	0.5	0.0	-11.0	0.0	-0.8	-0.1
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	0.4	0.0	-2.2	0.0	0.0	-0.1
	SLN84 ring / End	0.2	0.1	-2.2	0.0	0.0	0.1
C LC_130	SLN81+stub radial (ends 200 past the node) / Begin	0.7	0.0	-11.1	0.0	-1.1	0.0
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	1.2	0.0	-2.2	0.0	0.0	0.0
	SLN84 ring / End	1.2	0.1	-2.2	0.0	0.0	0.0
C LC_135	SLN81+stub radial (ends 200 past the node) / Begin	1.6	0.1	-45.8	0.0	-4.2	-0.3
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-0.4	0.0	-8.9	0.0	0.0	0.0
	SLN84 ring / End	-0.4	0.6	-8.8	0.0	0.0	0.4
C LC_141	SLN81+stub radial (ends 200 past the node) / Begin	0.5	0.0	-11.1	0.0	-1.0	-0.1
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	0.4	0.0	-2.2	0.0	0.0	0.0
	SLN84 ring / End	0.5	0.1	-2.2	0.0	0.0	0.1
C LC_142	SLN81+stub radial (ends 200 past the node) / Begin	0.6	0.0	-11.1	0.0	-1.0	-0.1
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	0.7	0.0	-2.2	0.0	0.0	0.0
	SLN84 ring / End	0.7	0.1	-2.2	0.0	0.0	0.1
C LC_143	SLN81+stub radial (ends 200 past the node) / Begin	0.6	0.0	-11.1	0.0	-1.0	-0.1
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	0.7	0.0	-2.2	0.0	0.0	0.0
	SLN84 ring / End	0.7	0.1	-2.2	0.0	0.0	0.1
C LC_144	SLN81+stub radial (ends 200 past the node) / Begin	0.5	0.0	-11.1	0.0	-1.0	-0.1
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	0.5	0.0	-2.2	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN84 ring / End	0.5	0.1	-2.2	0.0	0.0	0.1
C LC_145	SLN81+stub radial (ends 200 past the node) / Begin	0.6	0.0	-45.9	0.0	-4.5	-0.4
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-4.7	0.0	-8.9	0.0	0.0	-0.1
	SLN84 ring / End	-4.6	0.8	-8.8	0.0	0.0	0.6
C LC_146	SLN81+stub radial (ends 200 past the node) / Begin	1.1	0.2	-45.9	0.0	-4.4	-0.5
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-3.4	-0.1	-8.9	0.0	0.0	-0.1
	SLN84 ring / End	-3.4	0.9	-8.8	0.0	0.0	0.7
C LC_147	SLN81+stub radial (ends 200 past the node) / Begin	1.0	0.0	-45.9	0.0	-4.4	-0.4
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-3.5	-0.1	-8.9	0.0	0.0	-0.2
	SLN84 ring / End	-3.4	0.9	-8.8	0.0	0.0	0.7
C LC_148	SLN81+stub radial (ends 200 past the node) / Begin	0.6	0.2	-45.9	0.0	-4.5	-0.5
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-4.6	0.0	-8.9	0.0	0.0	-0.1
	SLN84 ring / End	-4.7	0.9	-8.8	0.0	0.0	0.7
C LC_151	SLN81+stub radial (ends 200 past the node) / Begin	3.0	0.1	-46.4	0.0	-6.8	-0.3
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-1.8	0.0	-8.9	0.0	0.0	-0.1
	SLN84 ring / End	-1.8	0.5	-8.8	0.0	0.0	0.4
C LC_152	SLN81+stub radial (ends 200 past the node) / Begin	-4.1	0.1	-10.2	0.0	3.1	-0.5
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-6.9	-0.1	-2.2	0.0	0.0	-0.2
	SLN84 ring / End	-6.9	1.0	-2.2	0.0	0.0	0.8
C LC_153	SLN81+stub radial (ends 200 past the node) / Begin	0.8	0.1	-45.2	0.0	-4.3	-0.4
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-4.2	0.0	-8.9	0.0	0.0	-0.1
	SLN84 ring / End	-4.2	0.9	-8.8	0.0	0.0	0.7
C LC_154	SLN81+stub radial (ends 200 past the node) / Begin	3.0	0.1	-47.2	0.0	-7.0	-0.3
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-1.8	0.0	-8.9	0.0	0.0	-0.1
	SLN84 ring / End	-1.8	0.5	-8.8	0.0	0.0	0.4

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	9.4 < 100%	OK
Welds	36.9 < 100%	OK
Buckling	Not calculated	
GMNA	Calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN81+stub radial (ends 200 past the node)-bfl 1	19.6	C LC_120	51.7	0.0	19.4	OK
SLN81+stub radial (ends 200 past the node)-tfl 1	19.6	C LC_147	20.7	0.0	0.0	OK
SLN81+stub radial (ends 200 past the node)-w 1	11.4	C LC_120	61.1	0.0	19.4	OK
SLN121 column below	10.0	C LC_152	95.8	0.0	0.0	OK
SLN83 ring-bfl 1	12.7	C LC_135	16.1	0.0	0.0	OK
SLN83 ring-tfl 1	12.7	C LC_119	18.6	0.0	0.0	OK
SLN83 ring-w 1	7.2	C LC_119	17.2	0.0	0.0	OK
SLN84 ring-bfl 1	12.7	C LC_147	16.8	0.0	0.0	OK
SLN84 ring-tfl 1	12.7	C LC_121	17.8	0.0	0.0	OK
SLN84 ring-w 1	7.2	C LC_119	17.0	0.0	0.0	OK
End plate column below SLN121 column below	12.0	C LC_152	93.8	0.0	12.4	OK
End plate SLN84 ring	20.0	C LC_146	4.8	0.0	8.3	OK
Bottom flange plate SLN84 ring	15.0	C LC_120	16.0	0.0	0.0	OK
Bottom flange half plate L SLN84 ring	15.0	C LC_120	6.6	0.0	0.0	OK
Bottom flange half plate R SLN84 ring	15.0	C LC_152	7.4	0.0	0.0	OK
Haunch web SLN84 ring	12.0	C LC_119	12.5	0.0	0.0	OK
Haunch top plate SLN84 ring	15.0	C LC_120	5.3	0.0	0.0	OK
Haunch sloping plate SLN84 ring	15.0	C LC_119	5.3	0.0	0.0	OK
End plate SLN83 ring	20.0	C LC_120	7.1	0.0	3.7	OK
Bottom flange plate SLN83 ring	15.0	C LC_120	11.1	0.0	0.0	OK
Bottom flange half plate L SLN83 ring	15.0	C LC_135	4.4	0.0	0.0	OK
Bottom flange half plate R SLN83 ring	15.0	C LC_135	4.6	0.0	0.0	OK
Haunch web SLN83 ring	12.0	C LC_119	12.8	0.0	0.0	OK
Haunch top plate SLN83 ring	15.0	C LC_120	5.3	0.0	0.0	OK
Haunch sloping plate SLN83 ring	15.0	C LC_119	5.4	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

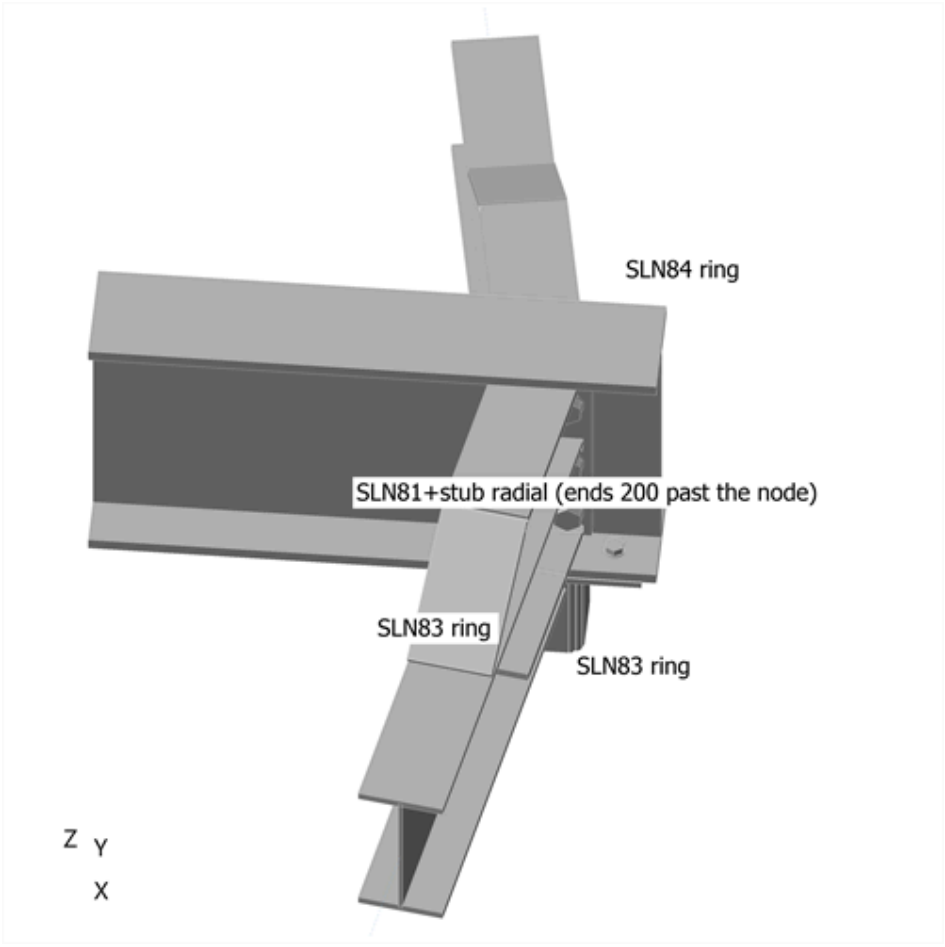
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

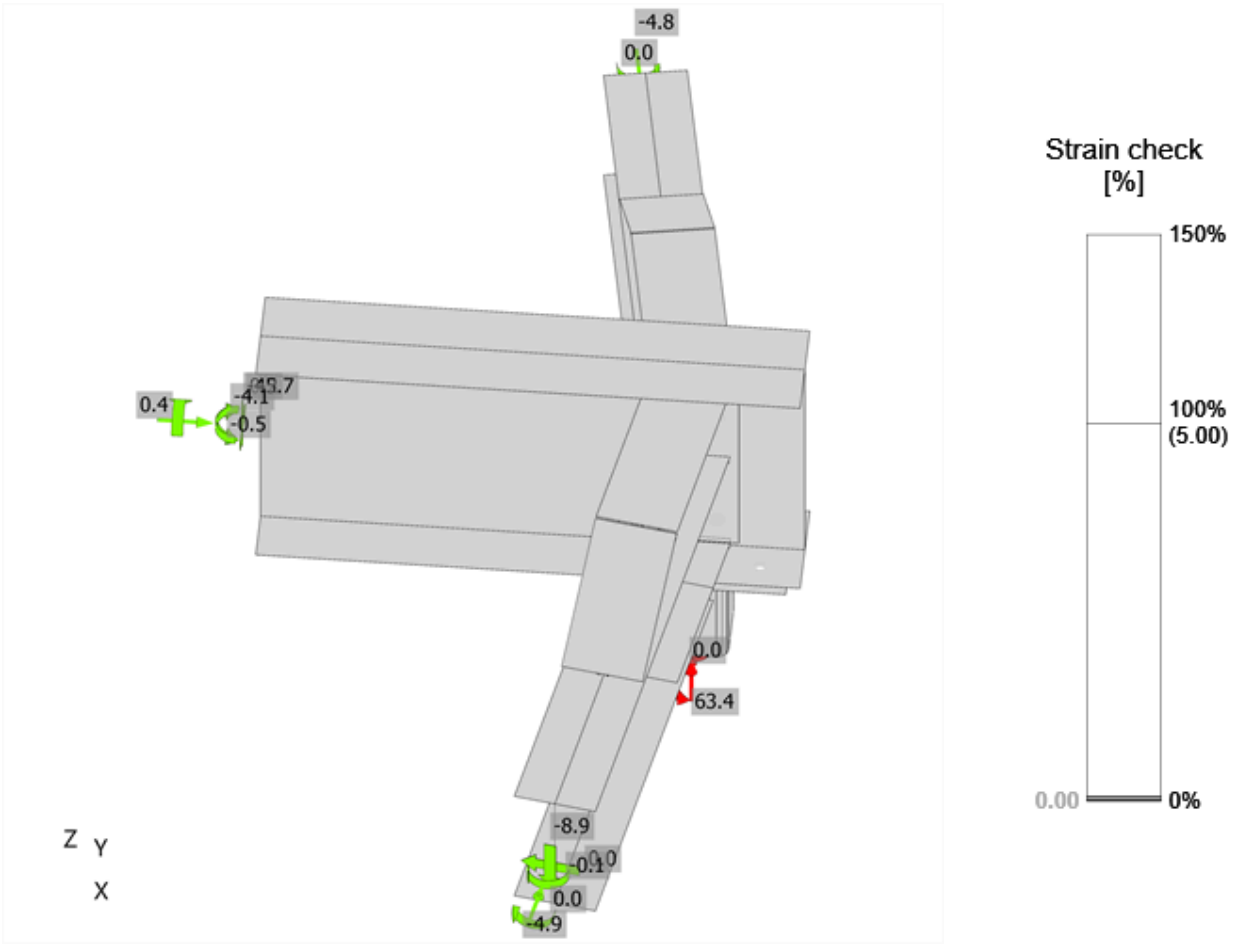
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN121 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

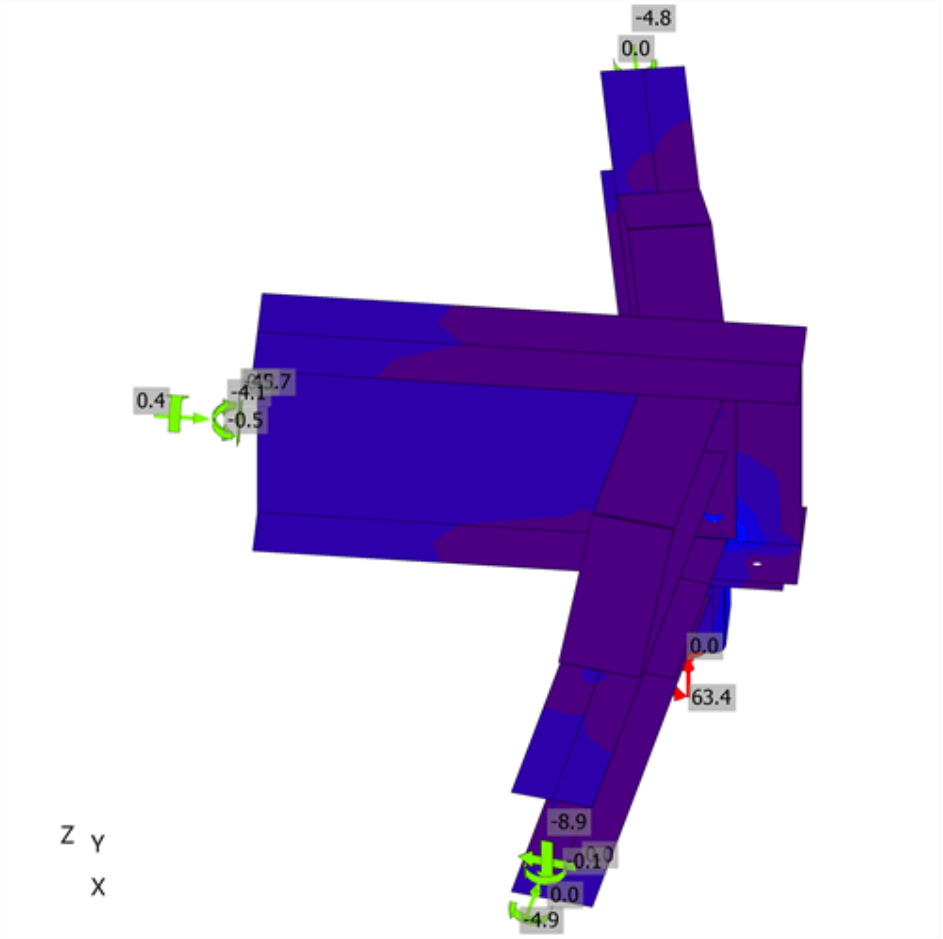
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

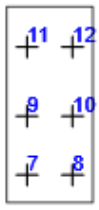


Strain check, C LC_120



Equivalent stress, C LC_120

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_152	12.8	0.9	156.4	9.0	0.9	7.4	OK	OK
	B3	M20 8.8 - 1	C LC_154	0.6	0.7	156.4	0.4	0.7	1.0	OK	OK
	B4	M20 8.8 - 1	C LC_120	0.6	0.9	172.0	0.4	0.9	1.2	OK	OK
	B5	M20 8.8 - 1	C LC_152	13.3	1.1	156.4	9.4	1.2	7.9	OK	OK
	B7	M30 8.8 - 2	C LC_135	0.1	0.7	232.2	0.0	0.5	0.3	OK	OK
	B8	M30 8.8 - 2	C LC_135	1.2	1.6	194.1	0.4	1.6	1.0	OK	OK
	B9	M30 8.8 - 2	C LC_135	0.2	0.7	249.2	0.0	0.5	0.3	OK	OK
	B10	M30 8.8 - 2	C LC_135	0.1	0.6	232.2	0.0	0.5	0.3	OK	OK
	B11	M30 8.8 - 2	C LC_120	0.8	0.8	249.2	0.2	0.6	0.5	OK	OK
	B12	M30 8.8 - 2	C LC_120	1.3	0.3	142.8	0.4	0.2	0.4	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1
M30 8.8 - 2	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN84 ring	Bottom flange plate SLN84 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN84 ring	Bottom flange plate SLN84 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN83 ring	Bottom flange plate SLN83 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN83 ring	Bottom flange plate SLN83 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN81+stub radial (ends 200 past the node)-w 1	SLN83 ring-tfl 1	◢ 11.4	149	C LC_105	7.7	0.0	-3.4	3.9	0.6	1.9	0.0	OK	OK
SLN81+stub radial (ends 200 past the node)-w 1	SLN84 ring-tfl 1	◢ 11.4	149	C LC_103	10.7	0.0	-5.0	5.5	0.5	2.6	2.6	OK	OK
End plate SLN84 ring	SLN84 ring-w 1	◢ 6.0 ◣	207	C LC_154	19.9	0.0	-1.8	0.4	11.5	4.9	4.9	OK	OK
		◢ 6.0 ◣	207	C LC_154	14.9	0.0	0.6	1.0	-8.5	3.7	3.7	OK	OK
Bottom flange plate SLN84 ring	SLN84 ring-w 1	◢ 6.0 ◣	100	C LC_120	11.2	0.0	0.9	2.8	5.8	2.8	2.8	OK	OK
		◢ 6.0 ◣	100	C LC_154	12.9	0.0	5.7	-6.4	-1.6	3.2	3.2	OK	OK
SLN84 ring-w 1	Bottom flange half plate L SLN84 ring	◢ 6.0 ◣	243	C LC_146	8.3	0.0	1.4	1.9	4.3	2.1	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_120	12.0	0.0	3.5	-2.9	-6.0	3.0	3.0	OK	OK
SLN84 ring-w 1	Bottom flange half plate R SLN84 ring	◢ 6.0 ◣	243	C LC_152	6.2	0.0	-0.9	-1.6	3.2	1.5	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_152	10.6	0.0	-3.4	2.7	-5.1	2.6	0.0	OK	OK
SLN84 ring-tfl 1	Haunch web SLN84 ring	◢ 6.0 ◣	572	C LC_119	24.1	0.0	-8.3	-6.3	11.5	6.0	5.8	OK	OK
		◢ 6.0 ◣	572	C LC_119	28.7	0.0	-4.8	6.9	-14.9	7.1	6.2	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate SLN83 ring	SLN83 ring-w 1	◢ 6.0 ◣	207	C LC_154	14.3	0.0	1.0	-0.9	8.2	3.5	3.5	OK	OK
		◢ 6.0 ◣	207	C LC_119	21.9	0.0	-2.8	-1.3	-12.4	5.4	5.4	OK	OK
Bottom flange plate SLN83 ring	SLN83 ring-w 1	◢ 6.0 ◣	100	C LC_146	14.2	0.0	5.6	7.4	1.4	3.5	3.5	OK	OK
		◢ 6.0 ◣	100	C LC_154	10.8	0.0	5.9	-4.9	-1.7	2.7	2.7	OK	OK
SLN83 ring-w 1	Bottom flange half plate L SLN83 ring	◢ 6.0 ◣	243	C LC_135	6.3	0.0	1.6	1.5	3.2	1.6	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	6.5	0.0	1.9	1.0	-3.4	1.6	0.0	OK	OK
SLN83 ring-w 1	Bottom flange half plate R SLN83 ring	◢ 6.0 ◣	243	C LC_135	4.8	0.0	1.0	1.1	-2.5	1.2	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	7.5	0.0	2.3	0.9	4.0	1.9	0.0	OK	OK
SLN83 ring-tfl 1	Haunch web SLN83 ring	◢ 6.0 ◣	572	C LC_119	29.3	0.0	-7.3	-8.1	14.2	7.2	6.2	OK	OK
		◢ 6.0 ◣	572	C LC_119	23.6	0.0	-5.9	5.1	-12.2	5.8	5.8	OK	OK
End plate SLN84 ring	Bottom flange plate SLN84 ring	◢ 6.0 ◣	149	C LC_152	25.0	0.0	-12.2	-12.3	-2.8	6.2	6.1	OK	OK
		◢ 6.0 ◣	149	C LC_120	26.3	0.0	-13.3	12.7	3.3	6.5	6.2	OK	OK
End plate SLN84 ring	Haunch web SLN84 ring	◢ 6.0 ◣	145	C LC_154	3.8	0.0	-0.3	-0.3	2.2	0.9	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_121	7.8	0.0	-0.8	0.9	-4.4	1.9	0.0	OK	OK
End plate SLN84 ring	Haunch top plate SLN84 ring	◢ 6.0 ◣	149	C LC_120	10.7	0.0	-5.0	-4.8	-2.6	2.7	2.7	OK	OK
		◢ 6.0 ◣	149	C LC_120	9.9	0.0	-3.9	4.1	3.3	2.5	0.0	OK	OK
Haunch top plate SLN84 ring	Haunch web SLN84 ring	◢ 6.0 ◣	342	C LC_120	2.7	0.0	-0.7	-1.3	-0.8	0.7	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_120	3.6	0.0	1.2	-0.9	-1.7	0.9	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Haunch sloping plate SLN84 ring	Haunch web SLN84 ring	◢ 6.0 ◣	271	C LC_119	21.9	0.0	-4.0	-6.9	10.4	5.4	5.4	OK	OK
		◢ 6.0 ◣	271	C LC_119	21.3	0.0	-3.1	0.2	-12.2	5.3	5.3	OK	OK
End plate SLN83 ring	Bottom flange plate SLN83 ring	◢ 6.0 ◣	149	C LC_120	10.5	0.0	-2.4	-4.2	-4.2	2.6	2.6	OK	OK
		◢ 6.0 ◣	149	C LC_120	15.1	0.0	-6.3	4.5	6.5	3.7	3.7	OK	OK
End plate SLN83 ring	Haunch web SLN83 ring	◢ 6.0 ◣	145	C LC_119	8.8	0.0	-0.7	-0.6	5.0	2.2	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_154	3.1	0.0	-0.2	0.3	-1.8	0.8	0.0	OK	OK
End plate SLN83 ring	Haunch top plate SLN83 ring	◢ 6.0 ◣	149	C LC_120	8.5	0.0	-4.3	-2.4	-3.5	2.1	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_120	5.5	0.0	-1.0	2.8	1.2	1.3	0.0	OK	OK
Haunch top plate SLN83 ring	Haunch web SLN83 ring	◢ 6.0 ◣	342	C LC_121	1.5	0.0	0.6	0.6	0.5	0.4	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_120	1.8	0.0	0.5	-0.4	-0.9	0.5	0.0	OK	OK
Haunch sloping plate SLN83 ring	Haunch web SLN83 ring	◢ 6.0 ◣	271	C LC_119	23.8	0.0	-4.7	-3.6	13.0	5.9	5.8	OK	OK
		◢ 6.0 ◣	271	C LC_119	17.9	0.0	-2.4	3.6	-9.6	4.4	4.4	OK	OK
End plate column below SLN121 column below	SLN121 column below	◢ 4.2 ◣	521	C LC_135	133.8	0.0	-65.7	-67.1	-4.6	33.0	17.7	OK	OK
		◢ 4.2 ◣	521	C LC_135	149.5	0.0	-75.6	74.3	3.9	36.9	20.5	OK	OK
SLN81+stub radial (ends 200 past the node)-w 1	SLN84 ring-w 1	◢ 7.2	208	C LC_154	58.6	0.0	8.1	7.8	-32.6	14.5	9.0	OK	OK
SLN81+stub radial (ends 200 past the node)-w 1	SLN83 ring-w 1	◢ 7.2	208	C LC_154	61.4	0.0	7.6	-5.2	34.8	15.2	9.4	OK	OK
Haunch sloping plate SLN84 ring	Haunch top plate SLN84 ring	◢ 6.0 ◣	146	C LC_120	4.5	0.0	-4.2	-1.0	0.1	1.3	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_120	3.3	0.0	-0.6	1.8	0.5	0.8	0.0	OK	OK
Haunch sloping plate SLN83 ring	Haunch top plate SLN83 ring	◢ 6.0 ◣	146	C LC_120	3.0	0.0	-1.9	1.3	-0.1	0.7	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	147	C LC_121	1.2	0.0	-0.3	0.4	0.5	0.3	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		

Item	Value	Unit	Reference
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-B_060
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN89b+SLN89a radial (continuous) - SLN131 column below - SLN84 ring - SLN88 ring

Name	SLN89b+SLN89a radial (continuous) - SLN131 column below - SLN84 ring - SLN88 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

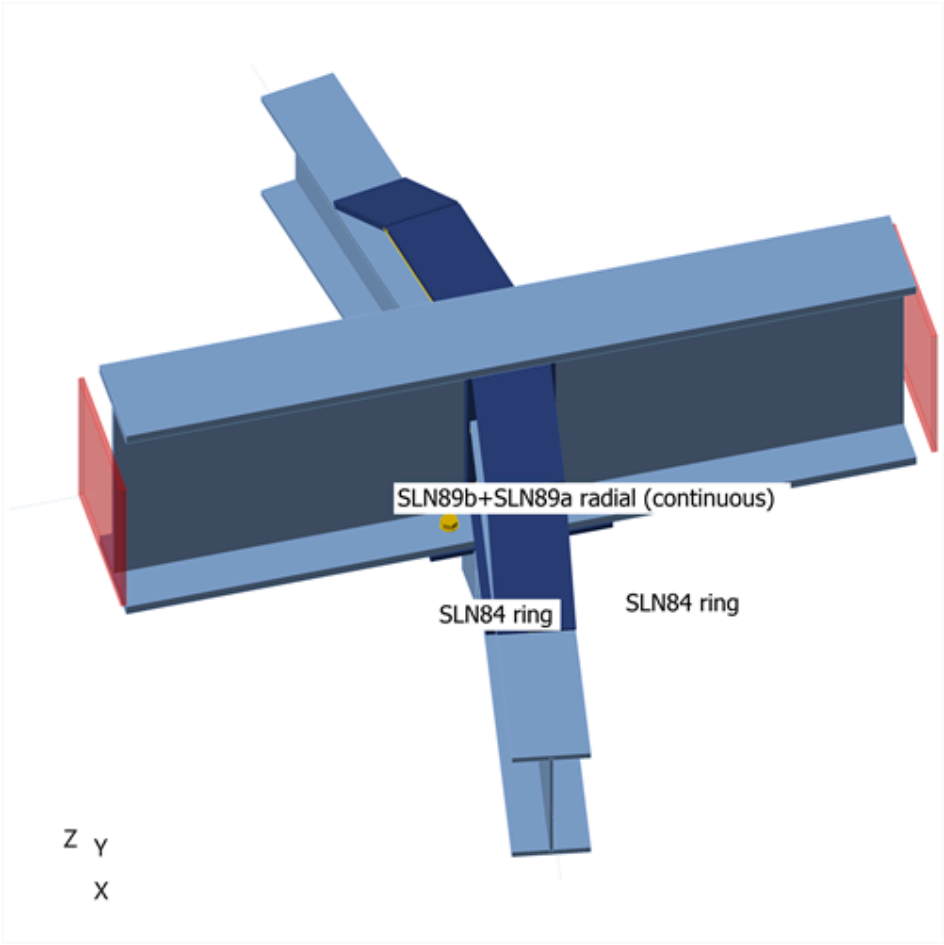
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN89b+SLN89a radial (continuous)	1 - UB457x191x98	60.0	0.0	0.0	0	0	0
SLN131 column below	2 - SHS150x150x10	0.0	90.0	-120.0	0	0	0
SLN84 ring	3 - UB254x146x43	-40.0	0.0	0.0	0	0	-102
SLN88 ring	3 - UB254x146x43	160.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN89b+SLN89a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN89b+SLN89a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN131 column below / begin		Position	0
SLN84 ring / end		Position	0
SLN88 ring / end		Position	0



SLN89b+SLN89a radial (continuous) - SLN131 column below - SLN84 ring - SLN88 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - SHS150x150x10	S 275
3 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN131 column below / Begin	37.9	0.0	0.1	-0.1	0.1	0.1
	SLN84 ring / End	0.6	0.1	-2.2	0.0	0.0	0.3
	SLN88 ring / End	4.8	0.2	-4.9	0.0	0.0	0.4
C LC_102	SLN131 column below / Begin	154.3	-0.2	-0.1	-0.6	-0.6	0.3
	SLN84 ring / End	-4.0	0.9	-8.9	0.0	0.0	1.9
	SLN88 ring / End	19.4	1.0	-25.8	0.0	0.0	2.1
C LC_103	SLN131 column below / Begin	149.0	-0.2	-0.2	-0.6	-0.7	0.3
	SLN84 ring / End	-8.7	1.0	-8.5	0.0	0.0	2.0
	SLN88 ring / End	16.7	1.0	-24.7	0.0	0.0	2.2
C LC_104	SLN131 column below / Begin	147.2	0.0	-0.1	-0.5	-0.5	0.3
	SLN84 ring / End	-5.4	0.9	-8.5	0.0	0.0	1.8
	SLN88 ring / End	15.2	1.0	-24.7	0.0	0.0	2.0
C LC_105	SLN131 column below / Begin	146.0	-0.2	0.0	-0.6	-0.5	0.3
	SLN84 ring / End	-6.3	0.9	-8.5	0.0	0.0	1.9
	SLN88 ring / End	16.5	1.1	-24.7	0.0	0.0	2.1
C LC_106	SLN131 column below / Begin	150.0	-0.2	-0.2	-0.6	-0.7	0.3
	SLN84 ring / End	-7.7	0.9	-8.5	0.0	0.0	1.8
	SLN88 ring / End	15.5	0.9	-24.7	0.0	0.0	1.9
C LC_107	SLN131 column below / Begin	19.0	-0.1	0.0	-0.1	0.1	0.0
	SLN84 ring / End	-3.9	0.3	-0.9	0.0	0.0	0.5
	SLN88 ring / End	3.2	0.2	-1.6	0.0	0.0	0.5
C LC_108	SLN131 column below / Begin	15.9	0.4	0.3	0.1	0.3	0.0
	SLN84 ring / End	0.8	0.0	-0.9	0.0	0.0	0.0
	SLN88 ring / End	-1.3	0.0	-1.6	0.0	0.0	0.0
C LC_109	SLN131 column below / Begin	13.7	-0.1	0.3	-0.1	0.3	0.0
	SLN84 ring / End	-1.4	0.2	-0.9	0.0	0.0	0.3
	SLN88 ring / End	0.6	0.2	-1.6	0.0	0.0	0.3
C LC_110	SLN131 column below / Begin	20.8	-0.1	0.0	-0.1	0.1	0.0
	SLN84 ring / End	-1.7	0.0	-0.9	0.0	0.0	0.1
	SLN88 ring / End	1.5	0.0	-1.6	0.0	0.0	0.1
C LC_111	SLN131 column below / Begin	109.8	-0.2	-0.1	-0.5	-0.5	0.2
	SLN84 ring / End	-8.7	0.8	-6.2	0.0	0.0	1.7
	SLN88 ring / End	13.0	0.8	-17.4	0.0	0.0	1.8
C LC_112	SLN131 column below / Begin	106.5	0.3	0.1	-0.3	-0.1	0.2
	SLN84 ring / End	-3.4	0.6	-6.2	0.0	0.0	1.2
	SLN88 ring / End	8.9	0.7	-17.4	0.0	0.0	1.3
C LC_113	SLN131 column below / Begin	104.3	-0.2	0.2	-0.5	-0.1	0.2
	SLN84 ring / End	-5.5	0.7	-6.2	0.0	0.0	1.5
	SLN88 ring / End	11.1	0.9	-17.4	0.0	0.0	1.6
C LC_114	SLN131 column below / Begin	111.7	-0.2	-0.1	-0.4	-0.5	0.2
	SLN84 ring / End	-6.6	0.6	-6.2	0.0	0.0	1.3
	SLN88 ring / End	11.0	0.6	-17.4	0.0	0.0	1.4
C LC_115	SLN131 column below / Begin	28.8	-0.1	0.1	-0.2	0.1	0.1
	SLN84 ring / End	-3.8	0.3	-1.5	0.0	0.0	0.6
	SLN88 ring / End	4.5	0.3	-2.9	0.0	0.0	0.6

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN131 column below / Begin	25.7	0.4	0.3	0.1	0.4	0.0
	SLN84 ring / End	1.0	0.1	-1.5	0.0	0.0	0.1
	SLN88 ring / End	0.0	0.1	-2.9	0.0	0.0	0.1
C LC_117	SLN131 column below / Begin	23.5	-0.1	0.4	-0.2	0.4	0.0
	SLN84 ring / End	-1.2	0.2	-1.5	0.0	0.0	0.4
	SLN88 ring / End	1.9	0.2	-2.9	0.0	0.0	0.4
C LC_118	SLN131 column below / Begin	30.7	-0.1	0.1	-0.1	0.1	0.1
	SLN84 ring / End	-1.6	0.1	-1.5	0.0	0.0	0.1
	SLN88 ring / End	2.7	0.1	-2.9	0.0	0.0	0.2
C LC_119	SLN131 column below / Begin	154.2	-0.2	-0.3	-0.6	-0.8	0.3
	SLN84 ring / End	-7.3	1.0	-8.9	0.0	0.0	2.0
	SLN88 ring / End	18.3	1.0	-25.8	0.0	0.0	2.2
C LC_120	SLN131 column below / Begin	152.7	0.0	0.0	-0.5	-0.4	0.3
	SLN84 ring / End	-4.8	0.9	-8.9	0.0	0.0	1.9
	SLN88 ring / End	16.6	1.1	-25.8	0.0	0.0	2.1
C LC_121	SLN131 column below / Begin	151.6	-0.2	0.1	-0.6	-0.3	0.3
	SLN84 ring / End	-5.6	0.8	-8.9	0.0	0.0	1.9
	SLN88 ring / End	17.5	1.1	-25.8	0.0	0.0	2.2
C LC_122	SLN131 column below / Begin	155.2	-0.2	-0.3	-0.6	-0.9	0.3
	SLN84 ring / End	-6.4	0.9	-8.9	0.0	0.0	1.9
	SLN88 ring / End	17.2	0.9	-25.8	0.0	0.0	2.0
C LC_123	SLN131 column below / Begin	120.1	-0.2	-0.2	-0.5	-0.7	0.3
	SLN84 ring / End	-5.8	0.8	-6.9	0.0	0.0	1.7
	SLN88 ring / End	16.1	0.9	-19.5	0.0	0.0	1.8
C LC_124	SLN131 column below / Begin	117.4	0.3	0.2	-0.3	0.0	0.2
	SLN84 ring / End	-2.3	0.6	-6.9	0.0	0.0	1.3
	SLN88 ring / End	11.8	0.8	-19.5	0.0	0.0	1.5
C LC_125	SLN131 column below / Begin	115.3	-0.2	0.4	-0.5	0.3	0.2
	SLN84 ring / End	-4.1	0.6	-6.9	0.0	0.0	1.4
	SLN88 ring / End	13.1	0.9	-19.5	0.0	0.0	1.7
C LC_126	SLN131 column below / Begin	122.0	-0.2	-0.3	-0.4	-0.8	0.3
	SLN84 ring / End	-3.9	0.6	-6.9	0.0	0.0	1.4
	SLN88 ring / End	14.3	0.6	-19.5	0.0	0.0	1.4
C LC_127	SLN131 column below / Begin	39.2	-0.1	0.0	-0.1	-0.2	0.1
	SLN84 ring / End	-0.9	0.3	-2.2	0.0	0.0	0.6
	SLN88 ring / End	7.6	0.3	-4.9	0.0	0.0	0.7
C LC_128	SLN131 column below / Begin	36.6	0.4	0.4	0.0	0.5	0.1
	SLN84 ring / End	2.0	0.1	-2.2	0.0	0.0	0.2
	SLN88 ring / End	2.9	0.2	-4.9	0.0	0.0	0.3
C LC_129	SLN131 column below / Begin	34.6	-0.1	0.5	-0.2	0.8	0.1
	SLN84 ring / End	0.2	0.1	-2.2	0.0	0.0	0.3
	SLN88 ring / End	3.9	0.3	-4.9	0.0	0.0	0.5
C LC_130	SLN131 column below / Begin	41.0	-0.1	0.0	-0.1	-0.2	0.1
	SLN84 ring / End	1.2	0.1	-2.2	0.0	0.0	0.2
	SLN88 ring / End	6.1	0.1	-4.9	0.0	0.0	0.3
C LC_135	SLN131 column below / Begin	150.0	-0.2	-0.1	-0.4	-0.5	0.3
	SLN84 ring / End	-0.4	0.6	-8.9	0.0	0.0	1.3

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN88 ring / End	15.9	0.7	-25.8	0.0	0.0	1.5
C LC_141	SLN131 column below / Begin	37.9	0.0	0.1	-0.1	0.1	0.1
	SLN84 ring / End	0.5	0.1	-2.2	0.0	0.0	0.3
	SLN88 ring / End	4.7	0.2	-4.9	0.0	0.0	0.4
C LC_142	SLN131 column below / Begin	37.8	0.0	0.1	-0.1	0.1	0.1
	SLN84 ring / End	0.7	0.1	-2.2	0.0	0.0	0.3
	SLN88 ring / End	4.9	0.2	-4.9	0.0	0.0	0.4
C LC_143	SLN131 column below / Begin	37.8	0.0	0.1	-0.1	0.1	0.1
	SLN84 ring / End	0.8	0.1	-2.2	0.0	0.0	0.3
	SLN88 ring / End	5.0	0.2	-4.9	0.0	0.0	0.4
C LC_144	SLN131 column below / Begin	37.9	0.0	0.1	-0.1	0.1	0.1
	SLN84 ring / End	0.5	0.1	-2.2	0.0	0.0	0.3
	SLN88 ring / End	4.6	0.2	-4.9	0.0	0.0	0.4
C LC_145	SLN131 column below / Begin	154.5	-0.2	-0.1	-0.6	-0.6	0.3
	SLN84 ring / End	-4.6	0.9	-8.9	0.0	0.0	1.8
	SLN88 ring / End	19.0	1.0	-25.8	0.0	0.0	2.1
C LC_146	SLN131 column below / Begin	154.2	-0.2	-0.1	-0.6	-0.5	0.3
	SLN84 ring / End	-3.5	0.9	-8.9	0.0	0.0	1.9
	SLN88 ring / End	19.7	1.1	-25.8	0.0	0.0	2.2
C LC_147	SLN131 column below / Begin	154.1	-0.1	-0.1	-0.6	-0.5	0.3
	SLN84 ring / End	-3.4	0.8	-8.9	0.0	0.0	1.8
	SLN88 ring / End	20.1	1.1	-25.8	0.0	0.0	2.2
C LC_148	SLN131 column below / Begin	154.6	-0.2	-0.2	-0.6	-0.6	0.3
	SLN84 ring / End	-4.7	0.9	-8.9	0.0	0.0	1.9
	SLN88 ring / End	18.6	1.0	-25.8	0.0	0.0	2.1
C LC_151	SLN131 column below / Begin	119.1	-0.1	-3.8	-0.4	-5.9	0.2
	SLN84 ring / End	-1.8	0.5	-8.9	0.0	0.0	1.2
	SLN88 ring / End	12.4	0.7	-11.6	0.0	0.0	1.3
C LC_152	SLN131 column below / Begin	92.6	-0.1	3.8	-0.6	5.6	0.3
	SLN84 ring / End	-6.9	1.0	-2.2	0.0	0.0	2.1
	SLN88 ring / End	19.3	1.1	-14.1	0.0	0.0	2.3
C LC_153	SLN131 column below / Begin	152.8	-0.2	-0.1	-0.6	-0.5	0.3
	SLN84 ring / End	-4.2	0.9	-8.9	0.0	0.0	1.9
	SLN88 ring / End	19.3	1.1	-20.7	0.0	0.0	2.1
C LC_154	SLN131 column below / Begin	121.0	-0.1	-3.8	-0.4	-5.9	0.2
	SLN84 ring / End	-1.8	0.5	-8.9	0.0	0.0	1.2
	SLN88 ring / End	12.5	0.7	-11.6	0.0	0.0	1.3

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	5.1 < 100%	OK
Welds	50.1 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN89b+SLN89a radial (continuous)-bfl 1	19.6	C LC_152	131.3	0.0	16.0	OK
SLN89b+SLN89a radial (continuous)-tfl 1	19.6	C LC_119	73.7	0.0	0.0	OK
SLN89b+SLN89a radial (continuous)-w 1	11.4	C LC_152	106.6	0.0	16.0	OK
SLN131 column below	10.0	C LC_154	112.8	0.0	0.0	OK
SLN84 ring-bfl 1	12.7	C LC_146	24.6	0.0	0.0	OK
SLN84 ring-tfl 1	12.7	C LC_119	28.4	0.0	0.0	OK
SLN84 ring-w 1	7.2	C LC_119	57.4	0.0	0.0	OK
SLN88 ring-bfl 1	12.7	C LC_119	59.0	0.0	0.0	OK
SLN88 ring-tfl 1	12.7	C LC_119	55.9	0.0	0.0	OK
SLN88 ring-w 1	7.2	C LC_147	55.0	0.0	0.0	OK
End plate column below SLN131 column below	12.0	C LC_154	70.1	0.0	32.8	OK
End plate SLN88 ring	20.0	C LC_147	26.4	0.0	24.2	OK
Bottom flange plate SLN88 ring	15.0	C LC_120	47.1	0.0	0.0	OK
Bottom flange half plate L SLN88 ring	15.0	C LC_146	29.7	0.0	0.0	OK
Bottom flange half plate R SLN88 ring	15.0	C LC_146	28.0	0.0	0.0	OK
Haunch web SLN88 ring	12.0	C LC_121	33.3	0.0	0.0	OK
Haunch top plate SLN88 ring	15.0	C LC_152	25.4	0.0	0.0	OK
Haunch sloping plate SLN88 ring	15.0	C LC_119	18.0	0.0	0.0	OK
End plate SLN84 ring	20.0	C LC_152	54.4	0.0	44.3	OK
Bottom flange plate SLN84 ring	15.0	C LC_152	107.3	0.0	0.0	OK
Bottom flange half plate L SLN84 ring	15.0	C LC_152	21.6	0.0	0.0	OK
Bottom flange half plate R SLN84 ring	15.0	C LC_152	31.7	0.0	0.0	OK
Haunch web SLN84 ring	12.0	C LC_119	24.7	0.0	0.0	OK
Haunch top plate SLN84 ring	15.0	C LC_152	49.0	0.0	0.0	OK
Haunch sloping plate SLN84 ring	15.0	C LC_152	17.5	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

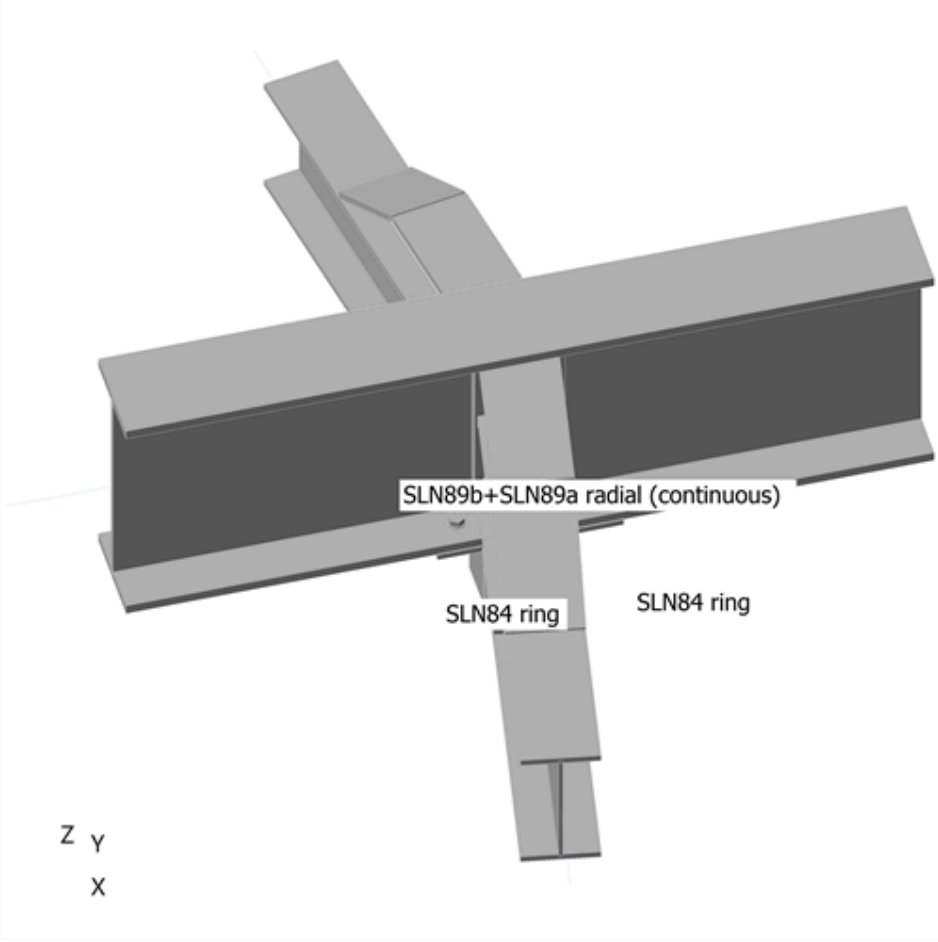
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

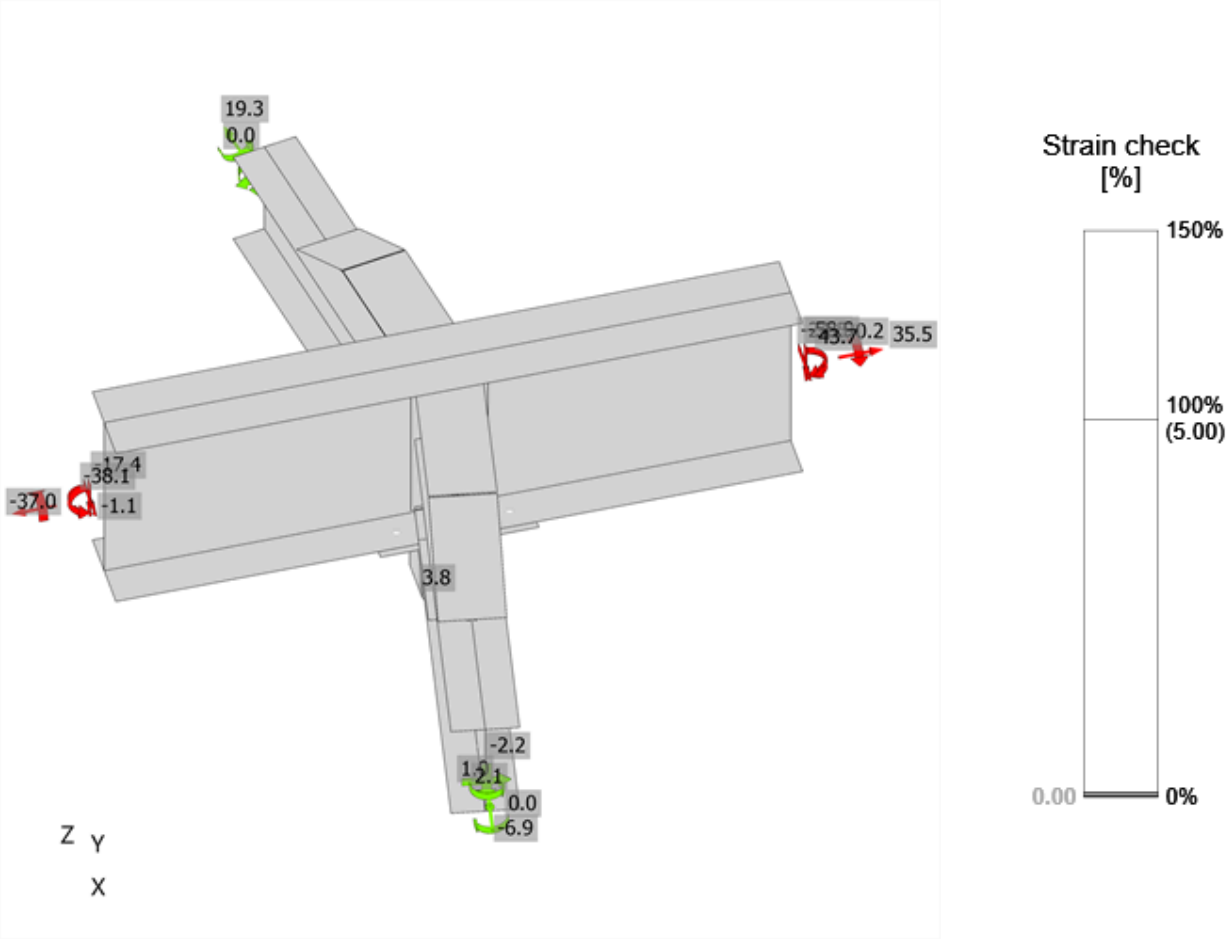
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN131 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

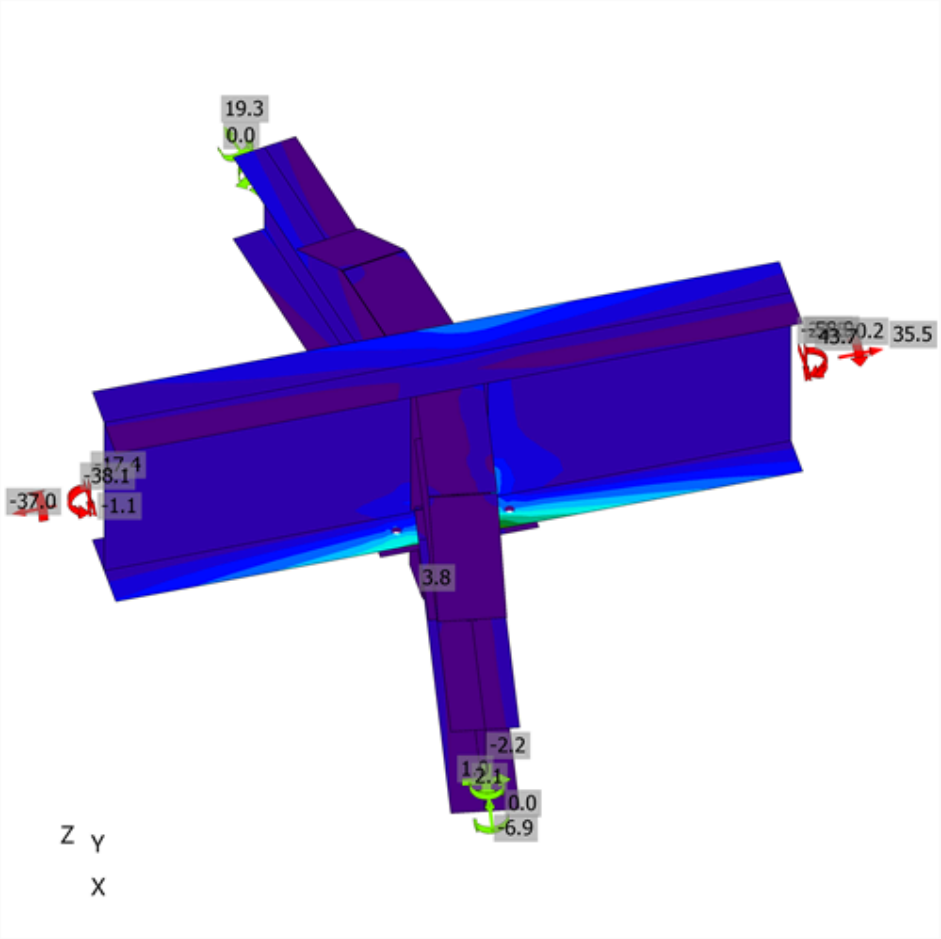
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

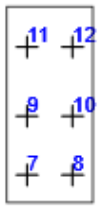


Strain check, C LC_152



Equivalent stress, C LC_152

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_119	0.3	4.6	206.4	0.2	4.9	5.1	OK	OK
	B3	M20 8.8 - 1	C LC_152	0.5	3.7	206.4	0.4	4.0	4.2	OK	OK
	B4	M20 8.8 - 1	C LC_152	0.6	2.0	172.0	0.4	2.1	2.4	OK	OK
	B5	M20 8.8 - 1	C LC_152	0.9	1.9	156.4	0.6	2.0	2.5	OK	OK
	B7	M30 8.8 - 2	C LC_154	0.5	1.3	194.1	0.2	1.2	0.7	OK	OK
	B8	M30 8.8 - 2	C LC_152	9.8	2.0	229.1	3.0	1.3	3.1	OK	OK
	B9	M30 8.8 - 2	C LC_152	1.2	0.7	249.2	0.4	0.4	0.6	OK	OK
	B10	M30 8.8 - 2	C LC_153	1.9	0.5	249.2	0.6	0.3	0.6	OK	OK
	B11	M30 8.8 - 2	C LC_119	3.3	1.4	148.6	1.0	1.0	1.4	OK	OK
	B12	M30 8.8 - 2	C LC_119	6.7	1.4	249.2	2.1	0.9	2.1	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1
M30 8.8 - 2	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate R SLN88 ring	Bottom flange plate SLN88 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN88 ring	Bottom flange plate SLN88 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN84 ring	Bottom flange plate SLN84 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN84 ring	Bottom flange plate SLN84 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN89b+SLN89a radial (continuous)-w 1	SLN84 ring-tfl 1	◢ 11.4	149	C LC_119	19.0	0.0	3.7	-3.4	10.2	4.7	4.7	OK	OK
SLN89b+SLN89a radial (continuous)-w 1	SLN88 ring-tfl 1	◢ 11.4	149	C LC_152	34.2	0.0	9.3	-9.7	16.3	8.5	7.2	OK	OK
End plate SLN88 ring	SLN88 ring-w 1	◢ 6.0 ◣	208	C LC_147	45.8	0.0	-17.9	-22.5	9.2	11.3	8.2	OK	OK
		◢ 6.0 ◣	208	C LC_146	38.7	0.0	-20.7	17.9	-6.0	9.6	7.4	OK	OK
Bottom flange plate SLN88 ring	SLN88 ring-w 1	◢ 6.0 ◣	100	C LC_146	43.3	0.0	5.1	12.9	21.2	10.7	9.8	OK	OK
		◢ 6.0 ◣	100	C LC_147	37.8	0.0	13.0	-16.8	-11.7	9.3	8.1	OK	OK
SLN88 ring-w 1	Bottom flange half plate L SLN88 ring	◢ 6.0 ◣	243	C LC_146	38.1	0.0	7.0	9.4	19.5	9.4	7.2	OK	OK
		◢ 6.0 ◣	243	C LC_146	53.1	0.0	15.5	-13.1	-26.2	13.1	9.8	OK	OK
SLN88 ring-w 1	Bottom flange half plate R SLN88 ring	◢ 6.0 ◣	243	C LC_147	19.8	0.0	3.0	2.0	-11.1	4.9	4.9	OK	OK
		◢ 6.0 ◣	243	C LC_146	36.4	0.0	20.3	-3.1	17.1	9.0	7.3	OK	OK
SLN88 ring-tfl 1	Haunch web SLN88 ring	◢ 6.0 ◣	572	C LC_135	56.1	0.0	-21.7	-14.8	25.9	13.9	8.2	OK	OK
		◢ 6.0 ◣	572	C LC_119	77.7	0.0	-8.7	18.8	-40.4	19.2	10.4	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate SLN84 ring	SLN84 ring-w 1	◢ 6.0 ◣	208	C LC_119	32.7	0.0	-12.5	-15.7	7.6	8.1	7.4	OK	OK
		◢ 6.0 ◣	208	C LC_119	51.3	0.0	-8.0	3.0	-29.1	12.7	8.9	OK	OK
Bottom flange plate SLN84 ring	SLN84 ring-w 1	◢ 6.0 ◣	100	C LC_147	42.9	0.0	13.9	20.6	11.2	10.6	8.8	OK	OK
		◢ 6.0 ◣	100	C LC_119	36.7	0.0	1.9	1.4	-21.1	9.1	8.0	OK	OK
SLN84 ring-w 1	Bottom flange half plate L SLN84 ring	◢ 6.0 ◣	243	C LC_146	21.5	0.0	3.2	4.9	11.2	5.3	5.3	OK	OK
		◢ 6.0 ◣	243	C LC_119	33.5	0.0	10.0	-8.1	-16.6	8.3	7.5	OK	OK
SLN84 ring-w 1	Bottom flange half plate R SLN84 ring	◢ 6.0 ◣	243	C LC_152	25.9	0.0	-4.3	-6.5	13.2	6.4	5.7	OK	OK
		◢ 6.0 ◣	243	C LC_152	39.3	0.0	-12.1	9.9	-19.2	9.7	7.4	OK	OK
SLN84 ring-tfl 1	Haunch web SLN84 ring	◢ 6.0 ◣	572	C LC_119	28.2	0.0	-17.5	-9.8	8.1	7.0	6.0	OK	OK
		◢ 6.0 ◣	572	C LC_119	36.6	0.0	2.4	5.4	-20.4	9.1	6.6	OK	OK
End plate SLN88 ring	Bottom flange plate SLN88 ring	◢ 6.0 ◣	149	C LC_152	77.7	0.0	-37.8	-35.8	-16.0	19.2	12.6	OK	OK
		◢ 6.0 ◣	149	C LC_152	85.3	0.0	-40.0	42.1	10.6	21.1	13.0	OK	OK
End plate SLN88 ring	Haunch web SLN88 ring	◢ 6.0 ◣	145	C LC_119	12.8	0.0	-0.1	-0.1	7.4	3.2	3.2	OK	OK
		◢ 6.0 ◣	145	C LC_135	8.9	0.0	-1.5	1.4	-4.9	2.2	0.0	OK	OK
End plate SLN88 ring	Haunch top plate SLN88 ring	◢ 6.0 ◣	149	C LC_152	48.9	0.0	-24.7	-21.5	-11.5	12.1	8.5	OK	OK
		◢ 6.0 ◣	149	C LC_152	39.9	0.0	-14.5	17.8	12.1	9.9	7.3	OK	OK
Haunch top plate SLN88 ring	Haunch web SLN88 ring	◢ 6.0 ◣	342	C LC_152	11.1	0.0	-1.6	-2.4	-5.9	2.7	2.7	OK	OK
		◢ 6.0 ◣	342	C LC_119	13.5	0.0	4.4	-3.2	-6.6	3.3	3.3	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Haunch sloping plate SLN88 ring	Haunch web SLN88 ring	◢ 6.0 ◣	271	C LC_120	58.0	0.0	-10.4	-23.8	22.8	14.3	9.4	OK	OK
		◢ 6.0 ◣	271	C LC_119	59.2	0.0	-8.0	-6.3	-33.3	14.6	9.5	OK	OK
End plate SLN84 ring	Bottom flange plate SLN84 ring	◢ 6.0 ◣	149	C LC_152	125.7	0.0	-36.7	-52.2	-45.7	31.0	19.0	OK	OK
		◢ 6.0 ◣	149	C LC_152	153.4	0.0	-66.8	51.3	61.1	37.9	22.1	OK	OK
End plate SLN84 ring	Haunch web SLN84 ring	◢ 6.0 ◣	145	C LC_119	15.7	0.0	3.4	5.9	6.6	3.9	3.9	OK	OK
		◢ 6.0 ◣	145	C LC_152	15.9	0.0	4.5	-2.9	8.3	3.9	3.9	OK	OK
End plate SLN84 ring	Haunch top plate SLN84 ring	◢ 6.0 ◣	149	C LC_152	84.2	0.0	-40.6	-28.6	-31.5	20.8	12.4	OK	OK
		◢ 6.0 ◣	149	C LC_152	62.0	0.0	-17.3	29.2	18.1	15.3	9.8	OK	OK
Haunch top plate SLN84 ring	Haunch web SLN84 ring	◢ 6.0 ◣	342	C LC_119	10.7	0.0	-1.5	-5.1	-3.3	2.7	2.7	OK	OK
		◢ 6.0 ◣	342	C LC_152	17.5	0.0	-1.7	0.5	-10.0	4.3	4.3	OK	OK
Haunch sloping plate SLN84 ring	Haunch web SLN84 ring	◢ 6.0 ◣	271	C LC_119	41.3	0.0	-8.7	-19.9	12.0	10.2	7.8	OK	OK
		◢ 6.0 ◣	271	C LC_119	30.5	0.0	0.6	-11.9	-13.0	7.5	6.9	OK	OK
End plate column below SLN131 column below	SLN131 column below	◢ 4.2 ◣	520	C LC_154	190.2	0.0	-95.0	-94.8	-8.4	47.0	26.3	OK	OK
		◢ 4.2 ◣	521	C LC_154	202.8	0.0	-100.7	100.9	12.3	50.1	28.1	OK	OK
SLN89b+SLN89a radial (continuous)-w 1	SLN88 ring-w 1	◢ 7.2	208	C LC_146	156.5	0.0	52.7	67.3	-52.1	38.7	23.6	OK	OK
SLN89b+SLN89a radial (continuous)-w 1	SLN84 ring-w 1	◢ 7.2	208	C LC_119	157.6	0.0	52.9	-46.5	72.0	39.0	23.7	OK	OK
Haunch sloping plate SLN88 ring	Haunch top plate SLN88 ring	◢ 6.0 ◣	146	C LC_119	18.0	0.0	-16.7	-3.8	0.7	5.4	5.4	OK	OK
		◢ 6.0 ◣	147	C LC_152	13.0	0.0	-2.4	7.0	2.3	3.2	3.2	OK	OK
Haunch sloping plate SLN84 ring	Haunch top plate SLN84 ring	◢ 6.0 ◣	146	C LC_119	16.8	0.0	-15.5	-3.7	0.7	5.0	5.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	147	C LC_152	12.7	0.0	-2.3	6.9	2.3	3.1	3.1	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		

Item	Value	Unit	Reference
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-B_260
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN119a+SLN119b radial (continuous) - SLN96 column below - SLN120 ring - SLN394 ring

Name	SLN119a+SLN119b radial (continuous) - SLN96 column below - SLN120 ring - SLN394 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

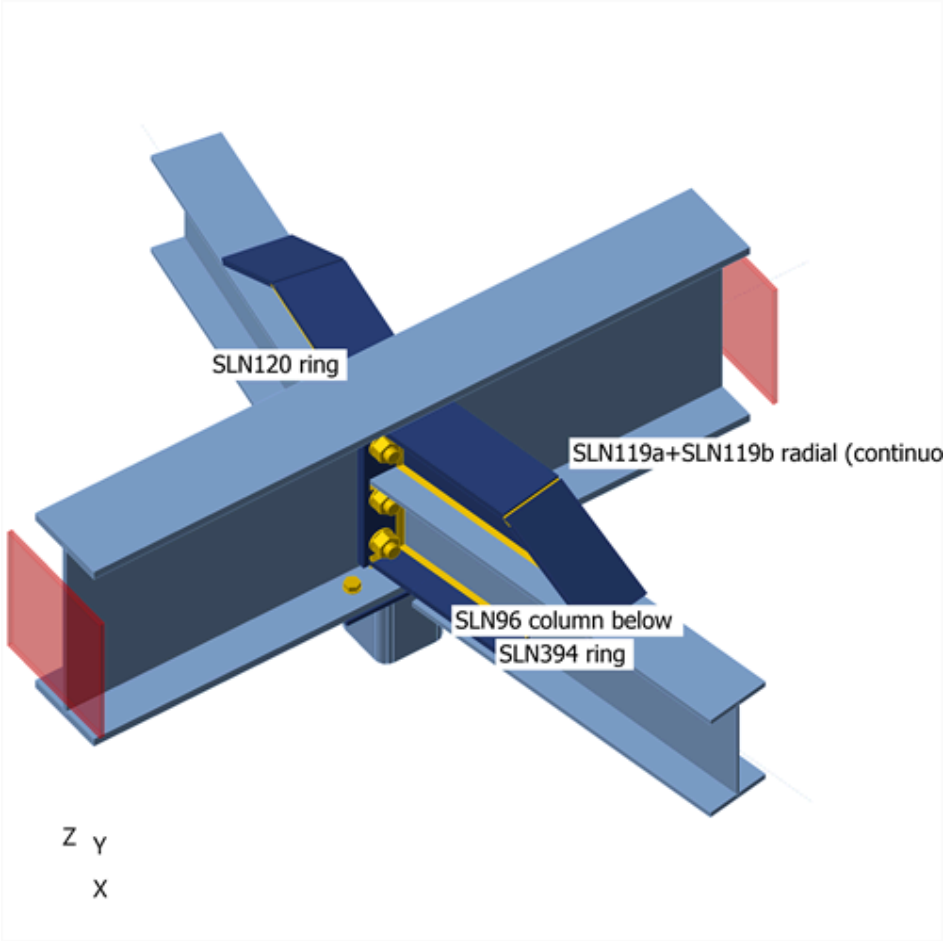
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN119a+SLN119b radial (continuous)	1 - UB457x191x98	-100.0	0.0	0.0	0	0	0
SLN96 column below	2 - SHS150x150x10	0.0	90.0	80.0	0	0	0
SLN120 ring	3 - UB254x146x43	160.0	0.0	0.0	0	0	-102
SLN394 ring	3 - UB254x146x43	0.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN119a+SLN119b radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN119a+SLN119b radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN96 column below / begin		Position	0
SLN120 ring / end		Position	0
SLN394 ring / end		Position	0



SLN119a+SLN119b radial (continuous) - SLN96 column below - SLN120 ring - SLN394 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - SHS150x150x10	S 275
3 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN96 column below / Begin	33.7	0.0	0.4	0.1	0.7	0.0
	SLN120 ring / End	4.8	-0.2	-4.9	0.0	0.0	-0.4
	SLN394 ring / End	0.6	-0.1	-2.2	0.0	0.0	-0.3
C LC_102	SLN96 column below / Begin	141.3	0.1	1.1	0.6	1.6	-0.1
	SLN120 ring / End	19.2	-1.1	-25.8	0.0	0.0	-2.2
	SLN394 ring / End	-4.2	-0.9	-9.0	0.0	0.0	-1.9
C LC_103	SLN96 column below / Begin	135.7	0.2	1.0	0.7	1.4	-0.1
	SLN120 ring / End	16.7	-1.1	-24.8	0.0	0.0	-2.2
	SLN394 ring / End	-7.8	-0.9	-8.7	0.0	0.0	-2.0
C LC_104	SLN96 column below / Begin	137.1	0.1	1.0	0.6	1.4	-0.1
	SLN120 ring / End	14.9	-1.0	-24.8	0.0	0.0	-2.0
	SLN394 ring / End	-6.6	-0.9	-8.7	0.0	0.0	-1.8
C LC_105	SLN96 column below / Begin	139.1	0.2	0.9	0.6	1.1	-0.1
	SLN120 ring / End	16.0	-1.0	-24.8	0.0	0.0	-2.1
	SLN394 ring / End	-9.0	-1.0	-8.7	0.0	0.0	-2.0
C LC_106	SLN96 column below / Begin	133.5	0.0	1.2	0.6	1.6	-0.1
	SLN120 ring / End	15.8	-1.1	-24.8	0.0	0.0	-2.1
	SLN394 ring / End	-5.0	-0.9	-8.7	0.0	0.0	-1.9
C LC_107	SLN96 column below / Begin	14.6	0.4	0.3	0.2	0.4	0.0
	SLN120 ring / End	2.3	-0.2	-1.6	0.0	0.0	-0.4
	SLN394 ring / End	-2.9	-0.2	-0.9	0.0	0.0	-0.4
C LC_108	SLN96 column below / Begin	17.1	0.0	0.2	0.0	0.4	0.0
	SLN120 ring / End	-0.5	0.1	-1.6	0.0	0.0	0.1
	SLN394 ring / End	-0.4	0.0	-0.9	0.0	0.0	0.1
C LC_109	SLN96 column below / Begin	20.7	0.2	0.1	0.1	0.1	0.0
	SLN120 ring / End	2.9	-0.2	-1.6	0.0	0.0	-0.4
	SLN394 ring / End	-3.7	-0.2	-0.9	0.0	0.0	-0.4
C LC_110	SLN96 column below / Begin	10.8	-0.2	0.5	0.0	0.7	0.0
	SLN120 ring / End	-0.7	-0.1	-1.6	0.0	0.0	0.0
	SLN394 ring / End	1.2	-0.1	-0.9	0.0	0.0	-0.1
C LC_111	SLN96 column below / Begin	98.4	0.4	0.8	0.6	1.1	-0.1
	SLN120 ring / End	12.3	-0.9	-17.5	0.0	0.0	-1.7
	SLN394 ring / End	-7.4	-0.7	-6.3	0.0	0.0	-1.6
C LC_112	SLN96 column below / Begin	100.9	0.1	0.8	0.4	1.1	-0.1
	SLN120 ring / End	9.3	-0.6	-17.5	0.0	0.0	-1.3
	SLN394 ring / End	-5.0	-0.6	-6.3	0.0	0.0	-1.2
C LC_113	SLN96 column below / Begin	104.6	0.2	0.6	0.5	0.7	-0.1
	SLN120 ring / End	12.3	-0.8	-17.5	0.0	0.0	-1.7
	SLN394 ring / End	-8.8	-0.8	-6.3	0.0	0.0	-1.6
C LC_114	SLN96 column below / Begin	94.4	-0.1	1.1	0.4	1.5	-0.1
	SLN120 ring / End	9.7	-0.8	-17.5	0.0	0.0	-1.4
	SLN394 ring / End	-2.9	-0.6	-6.3	0.0	0.0	-1.3
C LC_115	SLN96 column below / Begin	23.4	0.4	0.4	0.2	0.6	0.0
	SLN120 ring / End	3.6	-0.2	-2.9	0.0	0.0	-0.5
	SLN394 ring / End	-2.7	-0.2	-1.5	0.0	0.0	-0.5

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN96 column below / Begin	25.8	0.0	0.3	0.0	0.6	0.0
	SLN120 ring / End	0.8	0.0	-2.9	0.0	0.0	0.0
	SLN394 ring / End	-0.2	0.0	-1.5	0.0	0.0	0.0
C LC_117	SLN96 column below / Begin	29.4	0.2	0.2	0.1	0.2	0.0
	SLN120 ring / End	4.1	-0.2	-2.9	0.0	0.0	-0.5
	SLN394 ring / End	-3.6	-0.2	-1.5	0.0	0.0	-0.5
C LC_118	SLN96 column below / Begin	19.5	-0.2	0.6	0.0	0.9	0.0
	SLN120 ring / End	0.6	-0.1	-2.9	0.0	0.0	-0.2
	SLN394 ring / End	1.4	-0.1	-1.5	0.0	0.0	-0.2
C LC_119	SLN96 column below / Begin	140.4	0.3	1.1	0.7	1.6	-0.1
	SLN120 ring / End	17.8	-1.1	-25.8	0.0	0.0	-2.2
	SLN394 ring / End	-6.7	-0.9	-9.0	0.0	0.0	-1.9
C LC_120	SLN96 column below / Begin	141.4	0.1	1.0	0.6	1.5	-0.1
	SLN120 ring / End	16.5	-1.0	-25.8	0.0	0.0	-2.1
	SLN394 ring / End	-5.7	-0.9	-9.0	0.0	0.0	-1.9
C LC_121	SLN96 column below / Begin	142.7	0.2	0.8	0.6	1.1	-0.1
	SLN120 ring / End	17.8	-1.0	-25.8	0.0	0.0	-2.2
	SLN394 ring / End	-7.4	-1.0	-9.0	0.0	0.0	-2.0
C LC_122	SLN96 column below / Begin	139.2	0.0	1.3	0.6	1.9	-0.1
	SLN120 ring / End	16.9	-1.2	-25.8	0.0	0.0	-2.2
	SLN394 ring / End	-4.6	-0.9	-9.0	0.0	0.0	-1.9
C LC_123	SLN96 column below / Begin	107.8	0.4	1.0	0.6	1.4	-0.1
	SLN120 ring / End	14.5	-0.9	-19.5	0.0	0.0	-1.7
	SLN394 ring / End	-5.2	-0.7	-7.0	0.0	0.0	-1.5
C LC_124	SLN96 column below / Begin	109.6	0.0	0.9	0.5	1.3	-0.1
	SLN120 ring / End	12.5	-0.7	-19.5	0.0	0.0	-1.4
	SLN394 ring / End	-3.2	-0.6	-7.0	0.0	0.0	-1.3
C LC_125	SLN96 column below / Begin	111.9	0.2	0.5	0.5	0.6	-0.1
	SLN120 ring / End	15.8	-0.8	-19.5	0.0	0.0	-1.7
	SLN394 ring / End	-5.6	-0.8	-7.0	0.0	0.0	-1.6
C LC_126	SLN96 column below / Begin	105.7	-0.1	1.3	0.4	2.0	-0.1
	SLN120 ring / End	11.9	-0.9	-19.5	0.0	0.0	-1.6
	SLN394 ring / End	-2.0	-0.6	-7.0	0.0	0.0	-1.3
C LC_127	SLN96 column below / Begin	32.8	0.4	0.5	0.2	0.9	0.0
	SLN120 ring / End	5.8	-0.3	-4.9	0.0	0.0	-0.5
	SLN394 ring / End	-0.4	-0.2	-2.2	0.0	0.0	-0.4
C LC_128	SLN96 column below / Begin	34.6	0.0	0.4	0.1	0.7	0.0
	SLN120 ring / End	3.9	-0.1	-4.9	0.0	0.0	-0.2
	SLN394 ring / End	1.5	0.0	-2.2	0.0	0.0	-0.1
C LC_129	SLN96 column below / Begin	36.7	0.2	0.1	0.1	0.1	0.0
	SLN120 ring / End	7.7	-0.2	-4.9	0.0	0.0	-0.5
	SLN394 ring / End	-0.3	-0.2	-2.2	0.0	0.0	-0.5
C LC_130	SLN96 column below / Begin	30.8	-0.2	0.9	0.0	1.4	0.0
	SLN120 ring / End	2.8	-0.2	-4.9	0.0	0.0	-0.3
	SLN394 ring / End	2.3	-0.1	-2.2	0.0	0.0	-0.2
C LC_135	SLN96 column below / Begin	137.0	0.1	1.1	0.4	1.6	-0.1
	SLN120 ring / End	15.9	-0.8	-25.8	0.0	0.0	-1.6

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN394 ring / End	-0.5	-0.6	-9.0	0.0	0.0	-1.3
C LC_141	SLN96 column below / Begin	33.7	0.0	0.4	0.1	0.7	0.0
	SLN120 ring / End	4.8	-0.2	-4.9	0.0	0.0	-0.4
	SLN394 ring / End	0.6	-0.1	-2.2	0.0	0.0	-0.3
C LC_142	SLN96 column below / Begin	33.8	0.0	0.4	0.1	0.6	0.0
	SLN120 ring / End	4.7	-0.2	-4.9	0.0	0.0	-0.4
	SLN394 ring / End	0.6	-0.1	-2.2	0.0	0.0	-0.3
C LC_143	SLN96 column below / Begin	33.8	0.0	0.4	0.1	0.6	0.0
	SLN120 ring / End	4.6	-0.2	-4.9	0.0	0.0	-0.4
	SLN394 ring / End	0.4	-0.1	-2.2	0.0	0.0	-0.3
C LC_144	SLN96 column below / Begin	33.6	0.0	0.4	0.1	0.7	0.0
	SLN120 ring / End	4.9	-0.2	-4.9	0.0	0.0	-0.4
	SLN394 ring / End	0.8	-0.1	-2.2	0.0	0.0	-0.3
C LC_145	SLN96 column below / Begin	141.2	0.1	1.1	0.6	1.6	-0.1
	SLN120 ring / End	19.4	-1.1	-25.8	0.0	0.0	-2.2
	SLN394 ring / End	-4.2	-0.9	-9.0	0.0	0.0	-1.9
C LC_146	SLN96 column below / Begin	141.3	0.1	1.1	0.6	1.5	-0.1
	SLN120 ring / End	19.0	-1.1	-25.8	0.0	0.0	-2.2
	SLN394 ring / End	-4.2	-0.9	-9.0	0.0	0.0	-1.9
C LC_147	SLN96 column below / Begin	141.6	0.1	1.0	0.6	1.5	-0.1
	SLN120 ring / End	18.4	-1.0	-25.8	0.0	0.0	-2.1
	SLN394 ring / End	-5.1	-0.9	-9.0	0.0	0.0	-1.9
C LC_148	SLN96 column below / Begin	140.9	0.1	1.1	0.6	1.7	-0.1
	SLN120 ring / End	20.1	-1.2	-25.8	0.0	0.0	-2.3
	SLN394 ring / End	-3.3	-0.9	-9.0	0.0	0.0	-1.9
C LC_151	SLN96 column below / Begin	104.0	0.0	-2.5	0.4	-3.6	0.0
	SLN120 ring / End	12.2	-0.7	-11.9	0.0	0.0	-1.4
	SLN394 ring / End	-1.8	-0.5	-9.0	0.0	0.0	-1.2
C LC_152	SLN96 column below / Begin	89.4	0.1	4.0	0.7	6.1	-0.1
	SLN120 ring / End	19.4	-1.2	-13.9	0.0	0.0	-2.4
	SLN394 ring / End	-7.2	-1.0	-2.2	0.0	0.0	-2.1
C LC_153	SLN96 column below / Begin	140.4	0.1	1.0	0.6	1.5	-0.1
	SLN120 ring / End	19.2	-1.1	-20.9	0.0	0.0	-2.2
	SLN394 ring / End	-4.4	-0.9	-9.0	0.0	0.0	-1.9
C LC_154	SLN96 column below / Begin	104.5	0.0	-2.5	0.4	-3.6	0.0
	SLN120 ring / End	12.3	-0.7	-11.9	0.0	0.0	-1.4
	SLN394 ring / End	-1.7	-0.5	-9.0	0.0	0.0	-1.2

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	4.5 < 100%	OK
Welds	43.0 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN119a+SLN119b radial (continuous)-bfl 1	19.6	C LC_152	125.1	0.0	16.4	OK
SLN119a+SLN119b radial (continuous)-tfl 1	19.6	C LC_121	71.1	0.0	0.0	OK
SLN119a+SLN119b radial (continuous)-w 1	11.4	C LC_152	114.1	0.0	16.4	OK
SLN96 column below	10.0	C LC_152	98.2	0.0	0.0	OK
SLN120 ring-bfl 1	12.7	C LC_121	59.1	0.0	0.0	OK
SLN120 ring-tfl 1	12.7	C LC_121	56.3	0.0	0.0	OK
SLN120 ring-w 1	7.2	C LC_148	55.5	0.0	0.0	OK
SLN394 ring-bfl 1	12.7	C LC_148	25.0	0.0	0.0	OK
SLN394 ring-tfl 1	12.7	C LC_121	28.8	0.0	0.0	OK
SLN394 ring-w 1	7.2	C LC_148	57.1	0.0	0.0	OK
End plate column below SLN96 column below	12.0	C LC_122	63.5	0.0	32.3	OK
End plate SLN394 ring	20.0	C LC_152	56.4	0.0	46.9	OK
Bottom flange plate SLN394 ring	15.0	C LC_152	110.7	0.0	0.0	OK
Bottom flange half plate L SLN394 ring	15.0	C LC_152	32.6	0.0	0.0	OK
Bottom flange half plate R SLN394 ring	15.0	C LC_152	22.0	0.0	0.0	OK
Haunch web SLN394 ring	12.0	C LC_121	24.7	0.0	0.0	OK
Haunch top plate SLN394 ring	15.0	C LC_152	50.4	0.0	0.0	OK
Haunch sloping plate SLN394 ring	15.0	C LC_152	18.1	0.0	0.0	OK
End plate SLN120 ring	20.0	C LC_148	27.2	0.0	25.3	OK
Bottom flange plate SLN120 ring	15.0	C LC_152	46.2	0.0	0.0	OK
Bottom flange half plate L SLN120 ring	15.0	C LC_148	28.1	0.0	0.0	OK
Bottom flange half plate R SLN120 ring	15.0	C LC_148	31.2	0.0	0.0	OK
Haunch web SLN120 ring	12.0	C LC_122	33.1	0.0	0.0	OK
Haunch top plate SLN120 ring	15.0	C LC_152	25.7	0.0	0.0	OK
Haunch sloping plate SLN120 ring	15.0	C LC_152	18.3	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

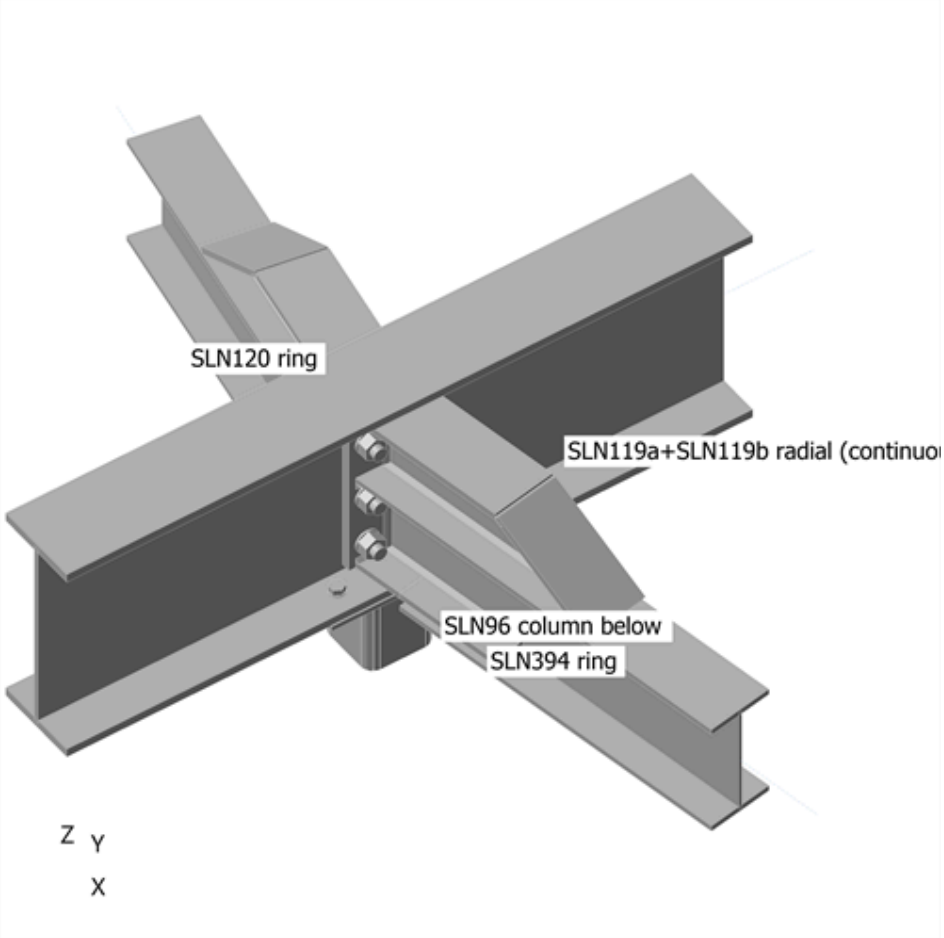
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

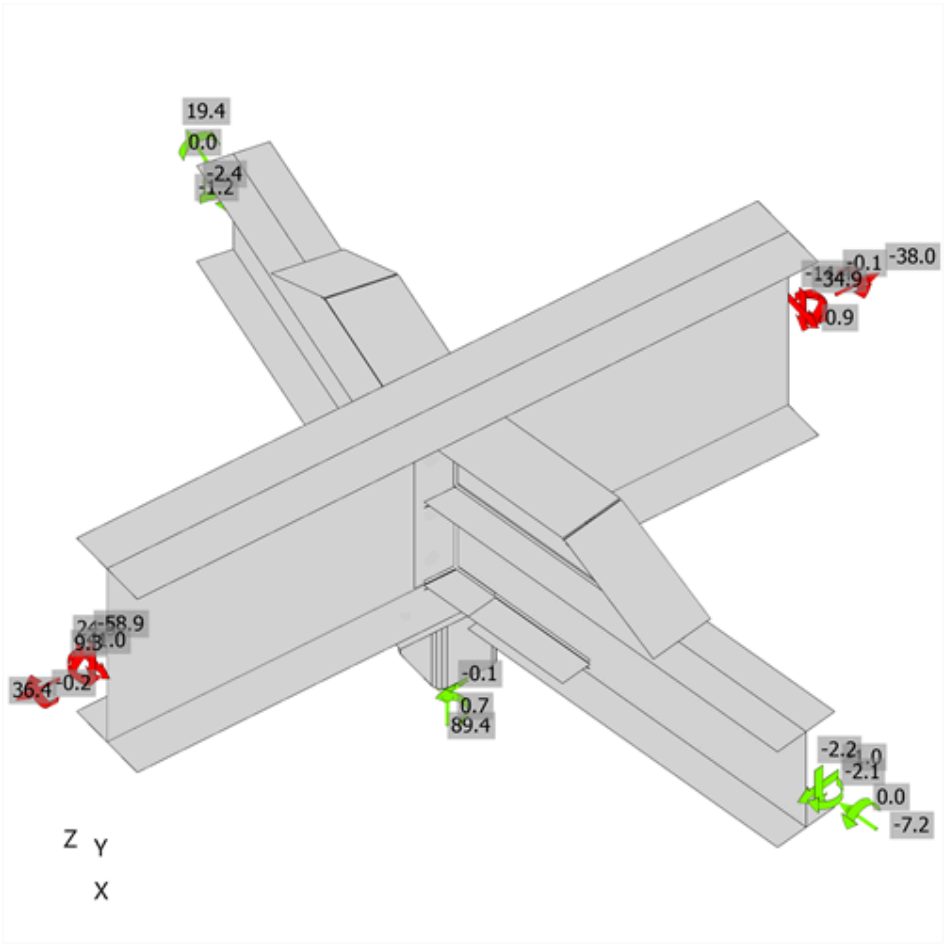
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN96 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

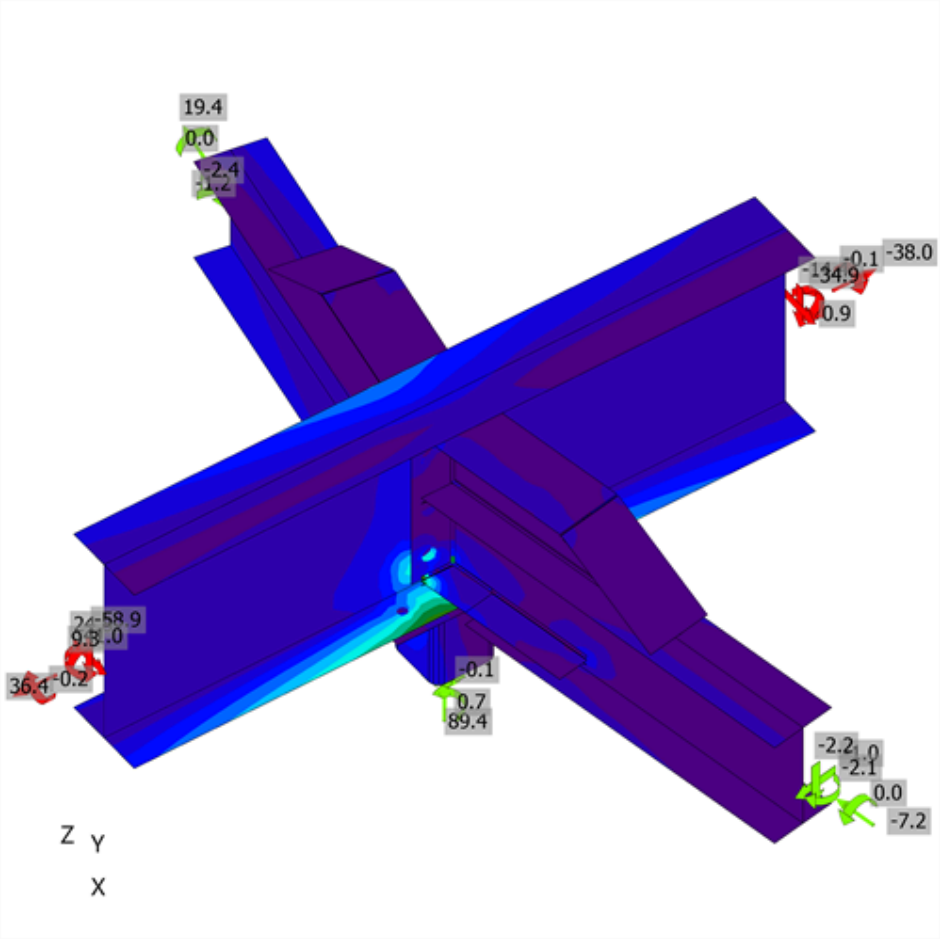
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

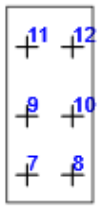


Strain check, C LC_152



Equivalent stress, C LC_152

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_152	1.4	2.1	156.4	1.0	2.2	3.0	OK	OK
	B3	M20 8.8 - 1	C LC_122	0.5	2.0	172.4	0.3	2.2	2.4	OK	OK
	B4	M20 8.8 - 1	C LC_152	0.6	3.7	206.4	0.4	4.0	4.3	OK	OK
	B5	M20 8.8 - 1	C LC_121	0.2	4.2	206.4	0.2	4.4	4.5	OK	OK
	B7	M30 8.8 - 2	C LC_151	0.8	1.0	194.1	0.2	0.9	0.7	OK	OK
	B8	M30 8.8 - 2	C LC_152	10.2	2.1	194.1	3.2	1.6	3.2	OK	OK
	B9	M30 8.8 - 2	C LC_152	1.4	0.7	249.2	0.4	0.4	0.6	OK	OK
	B10	M30 8.8 - 2	C LC_152	1.6	0.3	249.2	0.5	0.1	0.5	OK	OK
	B11	M30 8.8 - 2	C LC_121	3.1	1.3	148.1	1.0	0.9	1.3	OK	OK
	B12	M30 8.8 - 2	C LC_152	6.5	1.0	149.3	2.0	0.6	1.9	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1
M30 8.8 - 2	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN394 ring	Bottom flange plate SLN394 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN394 ring	Bottom flange plate SLN394 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN120 ring	Bottom flange plate SLN120 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN120 ring	Bottom flange plate SLN120 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN119a+SLN119b radial (continuous)-w 1	SLN120 ring-tfl 1	◢ 11.4	149	C LC_152	34.4	0.0	10.2	-10.7	-15.7	8.5	7.9	OK	OK
SLN119a+SLN119b radial (continuous)-w 1	SLN394 ring-tfl 1	◢ 11.4	149	C LC_121	21.0	0.0	3.5	-3.6	-11.4	5.2	5.2	OK	OK
End plate SLN394 ring	SLN394 ring-w 1	◢ 6.0 ◣	208	C LC_121	48.0	0.0	-7.1	-1.0	27.4	11.8	8.6	OK	OK
		◢ 6.0 ◣	208	C LC_121	34.5	0.0	-13.6	17.2	-6.2	8.5	7.4	OK	OK
Bottom flange plate SLN394 ring	SLN394 ring-w 1	◢ 6.0 ◣	99	C LC_148	37.2	0.0	0.7	-2.5	21.3	9.2	8.0	OK	OK
		◢ 6.0 ◣	100	C LC_148	41.8	0.0	12.8	-19.5	-12.2	10.3	8.9	OK	OK
SLN394 ring-w 1	Bottom flange half plate L SLN394 ring	◢ 6.0 ◣	243	C LC_152	25.1	0.0	-4.5	-6.5	-12.7	6.2	6.1	OK	OK
		◢ 6.0 ◣	243	C LC_152	38.8	0.0	-12.3	10.3	18.6	9.6	8.4	OK	OK
SLN394 ring-w 1	Bottom flange half plate R SLN394 ring	◢ 6.0 ◣	243	C LC_148	21.7	0.0	3.2	5.0	-11.3	5.4	5.4	OK	OK
		◢ 6.0 ◣	243	C LC_146	33.7	0.0	9.9	-8.0	16.8	8.3	7.1	OK	OK
SLN394 ring-tfl 1	Haunch web SLN394 ring	◢ 6.0 ◣	572	C LC_121	37.6	0.0	2.2	-5.8	20.9	9.3	6.7	OK	OK
		◢ 6.0 ◣	572	C LC_119	28.2	0.0	-17.4	9.7	-8.3	7.0	6.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate SLN120 ring	SLN120 ring-w 1	◢ 6.0 ◣	208	C LC_148	35.5	0.0	-19.8	-16.7	3.2	8.8	7.1	OK	OK
		◢ 6.0 ◣	208	C LC_148	43.6	0.0	-17.4	22.2	-6.5	10.8	8.0	OK	OK
Bottom flange plate SLN120 ring	SLN120 ring-w 1	◢ 6.0 ◣	100	C LC_148	35.5	0.0	11.6	16.6	10.0	8.8	8.0	OK	OK
		◢ 6.0 ◣	100	C LC_148	43.2	0.0	5.2	-13.2	-21.0	10.7	9.8	OK	OK
SLN120 ring-w 1	Bottom flange half plate L SLN120 ring	◢ 6.0 ◣	243	C LC_148	20.2	0.0	3.0	2.1	11.4	5.0	5.0	OK	OK
		◢ 6.0 ◣	243	C LC_148	36.3	0.0	20.8	-3.4	-16.8	9.0	6.7	OK	OK
SLN120 ring-w 1	Bottom flange half plate R SLN120 ring	◢ 6.0 ◣	243	C LC_148	39.4	0.0	6.5	9.6	-20.3	9.7	6.8	OK	OK
		◢ 6.0 ◣	243	C LC_148	56.3	0.0	16.5	-13.4	28.0	13.9	9.0	OK	OK
SLN120 ring-tfl 1	Haunch web SLN120 ring	◢ 6.0 ◣	572	C LC_119	80.1	0.0	-9.7	-20.2	41.2	19.8	10.7	OK	OK
		◢ 6.0 ◣	572	C LC_135	53.3	0.0	-20.8	13.4	-25.0	13.2	7.9	OK	OK
End plate SLN394 ring	Bottom flange plate SLN394 ring	◢ 6.0 ◣	149	C LC_152	126.0	0.0	-35.7	-52.9	45.5	31.1	17.7	OK	OK
		◢ 6.0 ◣	149	C LC_152	156.3	0.0	-68.7	51.5	-62.6	38.6	21.4	OK	OK
End plate SLN394 ring	Haunch web SLN394 ring	◢ 6.0 ◣	145	C LC_152	16.2	0.0	4.6	2.9	-8.5	4.0	4.0	OK	OK
		◢ 6.0 ◣	145	C LC_121	15.8	0.0	3.6	-5.9	-6.6	3.9	3.9	OK	OK
End plate SLN394 ring	Haunch top plate SLN394 ring	◢ 6.0 ◣	149	C LC_152	85.6	0.0	-41.5	-29.1	32.0	21.2	10.9	OK	OK
		◢ 6.0 ◣	149	C LC_152	62.4	0.0	-17.2	29.6	-17.9	15.4	8.5	OK	OK
Haunch top plate SLN394 ring	Haunch web SLN394 ring	◢ 6.0 ◣	342	C LC_152	18.1	0.0	-1.7	-0.4	10.4	4.5	4.5	OK	OK
		◢ 6.0 ◣	342	C LC_152	10.9	0.0	-1.5	5.2	3.5	2.7	2.7	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Haunch sloping plate SLN394 ring	Haunch web SLN394 ring	◢ 6.0 ◣	271	C LC_121	31.3	0.0	0.4	12.0	13.6	7.7	7.0	OK	OK
		◢ 6.0 ◣	271	C LC_121	41.6	0.0	-8.7	20.2	-11.9	10.3	7.8	OK	OK
End plate SLN120 ring	Bottom flange plate SLN120 ring	◢ 6.0 ◣	149	C LC_152	83.2	0.0	-40.8	-37.8	18.0	20.6	12.5	OK	OK
		◢ 6.0 ◣	149	C LC_152	87.9	0.0	-40.6	43.6	-11.3	21.7	12.0	OK	OK
End plate SLN120 ring	Haunch web SLN120 ring	◢ 6.0 ◣	145	C LC_135	8.2	0.0	-1.3	-1.4	4.5	2.0	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_148	12.3	0.0	4.9	-4.1	-5.1	3.0	3.0	OK	OK
End plate SLN120 ring	Haunch top plate SLN120 ring	◢ 6.0 ◣	149	C LC_152	49.8	0.0	-25.0	-21.9	11.7	12.3	7.8	OK	OK
		◢ 6.0 ◣	149	C LC_152	40.7	0.0	-14.9	18.0	-12.4	10.1	6.9	OK	OK
Haunch top plate SLN120 ring	Haunch web SLN120 ring	◢ 6.0 ◣	342	C LC_148	13.7	0.0	4.5	3.2	6.7	3.4	3.4	OK	OK
		◢ 6.0 ◣	342	C LC_152	11.3	0.0	-1.6	2.4	6.0	2.8	2.8	OK	OK
Haunch sloping plate SLN120 ring	Haunch web SLN120 ring	◢ 6.0 ◣	271	C LC_119	61.9	0.0	-9.2	5.7	34.8	15.3	9.8	OK	OK
		◢ 6.0 ◣	271	C LC_122	55.5	0.0	-9.1	23.7	-20.9	13.7	9.1	OK	OK
End plate column below SLN96 column below	SLN96 column below	◢ 4.2 ◣	520	C LC_122	166.3	0.0	-85.1	-82.2	-6.0	41.1	23.4	OK	OK
		◢ 4.2 ◣	520	C LC_152	174.1	0.0	-85.8	86.0	15.9	43.0	23.4	OK	OK
SLN119a+SLN119b radial (continuous)-w 1	SLN394 ring-w 1	◢ 7.2	208	C LC_121	147.4	0.0	41.6	-49.4	65.0	36.4	22.6	OK	OK
SLN119a+SLN119b radial (continuous)-w 1	SLN120 ring-w 1	◢ 7.2	208	C LC_148	125.1	0.0	60.6	45.6	-43.7	30.9	18.9	OK	OK
Haunch sloping plate SLN394 ring	Haunch top plate SLN394 ring	◢ 6.0 ◣	146	C LC_152	17.4	0.0	-16.1	-3.8	-0.7	5.2	5.2	OK	OK
		◢ 6.0 ◣	147	C LC_152	12.9	0.0	-2.4	7.0	-2.3	3.2	3.2	OK	OK
Haunch sloping plate SLN120 ring	Haunch top plate SLN120 ring	◢ 6.0 ◣	146	C LC_152	18.0	0.0	-16.6	-3.8	-0.9	5.4	5.4	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	147	C LC_152	13.2	0.0	-2.4	7.1	-2.3	3.3	3.3	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		

Item	Value	Unit	Reference
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-B_280
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN105+stub radial (ends 200 past the node) - SLN85 column below - SLN394 ring - SLN395 ring

Name	SLN105+stub radial (ends 200 past the node) - SLN85 column below - SLN394 ring - SLN395 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

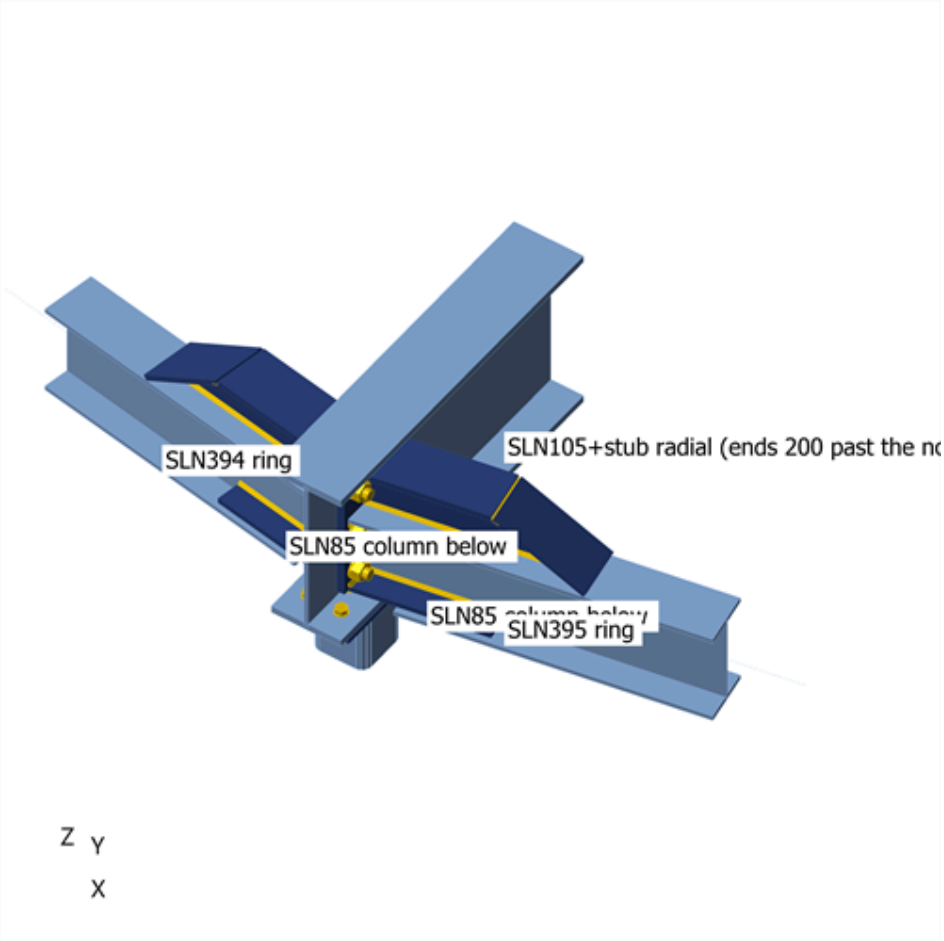
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN105+stub radial (ends 200 past the node)	1 - UB457x191x98	-80.0	0.0	0.0	200	0	0
SLN85 column below	2 - SHS150x150x10	0.0	90.0	100.0	0	0	0
SLN394 ring	3 - UB254x146x43	180.0	0.0	0.0	0	0	-102
SLN395 ring	3 - UB254x146x43	20.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN105+stub radial (ends 200 past the node) / begin		Position	0
SLN85 column below / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN394 ring / end		Position	0
SLN395 ring / end		Position	0



SLN105+stub radial (ends 200 past the node) - SLN85 column below - SLN394 ring - SLN395 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - SHS150x150x10	S 275
3 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN105+stub radial (ends 200 past the node) / Begin	0.6	0.0	-11.7	0.0	-1.2	0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	0.6	-0.1	-2.2	0.0	0.0	-0.1
	SLN395 ring / End	0.6	0.0	-2.1	0.0	0.0	0.0
C LC_102	SLN105+stub radial (ends 200 past the node) / Begin	1.1	-0.1	-48.6	0.0	-5.3	0.5
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-4.2	-0.9	-9.1	0.0	0.0	-0.7
	SLN395 ring / End	-4.2	0.0	-8.7	0.0	0.0	0.1
C LC_103	SLN105+stub radial (ends 200 past the node) / Begin	-0.1	0.0	-46.6	0.0	-5.2	0.4
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-7.9	-0.9	-8.7	0.0	0.0	-0.7
	SLN395 ring / End	-8.1	0.1	-8.4	0.0	0.0	0.2
C LC_104	SLN105+stub radial (ends 200 past the node) / Begin	0.0	-0.2	-46.6	0.0	-4.8	0.5
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-6.5	-0.9	-8.7	0.0	0.0	-0.7
	SLN395 ring / End	-6.7	0.1	-8.4	0.0	0.0	0.2
C LC_105	SLN105+stub radial (ends 200 past the node) / Begin	-0.7	-0.2	-46.5	0.0	-5.1	0.5
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-9.0	-0.9	-8.7	0.0	0.0	-0.8
	SLN395 ring / End	-8.9	0.0	-8.4	0.0	0.0	0.1
C LC_106	SLN105+stub radial (ends 200 past the node) / Begin	0.7	-0.1	-46.7	0.0	-5.0	0.5
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-5.0	-0.9	-8.7	0.0	0.0	-0.8
	SLN395 ring / End	-5.0	0.1	-8.4	0.0	0.0	0.2
C LC_107	SLN105+stub radial (ends 200 past the node) / Begin	-0.6	0.0	-5.0	0.0	-0.6	0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-2.9	-0.2	-0.9	0.0	0.0	-0.1
	SLN395 ring / End	-3.0	0.0	-0.9	0.0	0.0	0.1
C LC_108	SLN105+stub radial (ends 200 past the node) / Begin	-0.4	0.1	-4.9	0.0	0.1	-0.2
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-0.4	0.0	-0.9	0.0	0.0	0.0
	SLN395 ring / End	-0.9	0.1	-0.9	0.0	0.0	0.1
C LC_109	SLN105+stub radial (ends 200 past the node) / Begin	-1.1	0.0	-4.7	0.0	-0.3	0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-3.7	-0.2	-0.9	0.0	0.0	-0.2
	SLN395 ring / End	-3.5	0.0	-0.9	0.0	0.0	0.0
C LC_110	SLN105+stub radial (ends 200 past the node) / Begin	0.4	0.0	-5.2	0.0	-0.4	0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	1.2	-0.1	-0.9	0.0	0.0	-0.1
	SLN395 ring / End	1.2	-0.1	-0.9	0.0	0.0	0.0
C LC_111	SLN105+stub radial (ends 200 past the node) / Begin	-0.6	0.0	-33.7	0.0	-3.8	0.3
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-7.5	-0.7	-6.3	0.0	0.0	-0.5
	SLN395 ring / End	-7.7	0.1	-6.1	0.0	0.0	0.2
C LC_112	SLN105+stub radial (ends 200 past the node) / Begin	-0.4	-0.1	-33.7	0.0	-3.1	0.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-4.9	-0.6	-6.3	0.0	0.0	-0.5
	SLN395 ring / End	-5.4	0.2	-6.1	0.0	0.0	0.2
C LC_113	SLN105+stub radial (ends 200 past the node) / Begin	-1.3	-0.1	-33.4	0.0	-3.5	0.4
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-8.8	-0.7	-6.3	0.0	0.0	-0.6
	SLN395 ring / End	-8.6	0.0	-6.1	0.0	0.0	0.1
C LC_114	SLN105+stub radial (ends 200 past the node) / Begin	0.6	-0.1	-34.0	0.0	-3.5	0.4
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-2.9	-0.7	-6.3	0.0	0.0	-0.6
	SLN395 ring / End	-2.9	0.0	-6.1	0.0	0.0	0.1
C LC_115	SLN105+stub radial (ends 200 past the node) / Begin	-0.4	0.0	-8.0	0.0	-0.9	0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-2.7	-0.2	-1.5	0.0	0.0	-0.1
	SLN395 ring / End	-2.9	0.0	-1.5	0.0	0.0	0.1
C LC_116	SLN105+stub radial (ends 200 past the node) / Begin	-0.3	0.0	-8.0	0.0	-0.3	-0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-0.2	0.0	-1.5	0.0	0.0	0.0
	SLN395 ring / End	-0.8	0.1	-1.5	0.0	0.0	0.1
C LC_117	SLN105+stub radial (ends 200 past the node) / Begin	-1.0	-0.1	-7.7	0.0	-0.6	0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-3.6	-0.2	-1.5	0.0	0.0	-0.2
	SLN395 ring / End	-3.4	0.0	-1.5	0.0	0.0	0.0
C LC_118	SLN105+stub radial (ends 200 past the node) / Begin	0.6	-0.1	-8.3	0.0	-0.7	0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	1.4	-0.1	-1.5	0.0	0.0	-0.1
	SLN395 ring / End	1.4	-0.1	-1.5	0.0	0.0	0.0
C LC_119	SLN105+stub radial (ends 200 past the node) / Begin	0.4	0.0	-48.6	0.0	-5.4	0.4
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-6.7	-0.9	-9.1	0.0	0.0	-0.7
	SLN395 ring / End	-6.9	0.1	-8.7	0.0	0.0	0.2
C LC_120	SLN105+stub radial (ends 200 past the node) / Begin	0.4	-0.2	-48.4	0.0	-5.1	0.5
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-5.7	-0.9	-9.1	0.0	0.0	-0.7
	SLN395 ring / End	-5.8	0.1	-8.7	0.0	0.0	0.1
C LC_121	SLN105+stub radial (ends 200 past the node) / Begin	0.2	-0.2	-48.6	0.0	-5.7	0.5
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-7.4	-0.9	-9.1	0.0	0.0	-0.8
	SLN395 ring / End	-7.3	0.0	-8.7	0.0	0.0	0.1
C LC_122	SLN105+stub radial (ends 200 past the node) / Begin	0.9	-0.1	-48.4	0.0	-5.0	0.5
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-4.6	-0.9	-9.1	0.0	0.0	-0.7
	SLN395 ring / End	-4.6	0.1	-8.7	0.0	0.0	0.1
C LC_123	SLN105+stub radial (ends 200 past the node) / Begin	0.4	0.0	-37.6	0.0	-4.3	0.3
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-5.2	-0.7	-7.0	0.0	0.0	-0.5
	SLN395 ring / End	-5.4	0.1	-6.7	0.0	0.0	0.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_124	SLN105+stub radial (ends 200 past the node) / Begin	0.4	-0.1	-37.3	0.0	-3.6	0.3
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-3.2	-0.6	-7.0	0.0	0.0	-0.5
	SLN395 ring / End	-3.7	0.1	-6.7	0.0	0.0	0.1
C LC_125	SLN105+stub radial (ends 200 past the node) / Begin	0.3	-0.2	-37.7	0.0	-4.7	0.4
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-5.6	-0.7	-7.0	0.0	0.0	-0.6
	SLN395 ring / End	-5.4	0.0	-6.7	0.0	0.0	0.0
C LC_126	SLN105+stub radial (ends 200 past the node) / Begin	0.9	-0.1	-37.3	0.0	-3.6	0.4
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-2.0	-0.7	-7.0	0.0	0.0	-0.5
	SLN395 ring / End	-2.0	0.0	-6.7	0.0	0.0	0.0
C LC_127	SLN105+stub radial (ends 200 past the node) / Begin	0.6	0.0	-11.9	0.0	-1.5	0.0
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-0.5	-0.2	-2.2	0.0	0.0	-0.1
	SLN395 ring / End	-0.6	0.1	-2.1	0.0	0.0	0.1
C LC_128	SLN105+stub radial (ends 200 past the node) / Begin	0.5	0.0	-11.6	0.0	-0.7	0.0
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	1.6	0.0	-2.2	0.0	0.0	0.0
	SLN395 ring / End	1.0	0.0	-2.1	0.0	0.0	0.0
C LC_129	SLN105+stub radial (ends 200 past the node) / Begin	0.7	-0.1	-12.0	0.0	-1.8	0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-0.3	-0.2	-2.2	0.0	0.0	-0.2
	SLN395 ring / End	-0.2	0.0	-2.1	0.0	0.0	0.0
C LC_130	SLN105+stub radial (ends 200 past the node) / Begin	0.9	0.0	-11.6	0.0	-0.8	0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	2.3	-0.1	-2.2	0.0	0.0	-0.1
	SLN395 ring / End	2.3	-0.1	-2.1	0.0	0.0	-0.1
C LC_135	SLN105+stub radial (ends 200 past the node) / Begin	1.8	-0.1	-48.5	0.0	-5.1	0.4
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-0.5	-0.6	-9.1	0.0	0.0	-0.5
	SLN395 ring / End	-0.5	0.0	-8.7	0.0	0.0	0.0
C LC_141	SLN105+stub radial (ends 200 past the node) / Begin	0.6	0.0	-11.7	0.0	-1.2	0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	0.6	-0.1	-2.2	0.0	0.0	-0.1
	SLN395 ring / End	0.5	0.0	-2.1	0.0	0.0	0.0
C LC_142	SLN105+stub radial (ends 200 past the node) / Begin	0.7	-0.1	-11.7	0.0	-1.2	0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	0.6	-0.1	-2.2	0.0	0.0	-0.1
	SLN395 ring / End	0.6	0.0	-2.1	0.0	0.0	0.0
C LC_143	SLN105+stub radial (ends 200 past the node) / Begin	0.6	0.0	-11.8	0.0	-1.3	0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	0.4	-0.1	-2.2	0.0	0.0	-0.1
	SLN395 ring / End	0.4	0.0	-2.1	0.0	0.0	0.0
C LC_144	SLN105+stub radial (ends 200 past the node) / Begin	0.7	0.0	-11.7	0.0	-1.2	0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	0.8	-0.2	-2.2	0.0	0.0	-0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN395 ring / End	0.7	0.0	-2.1	0.0	0.0	0.0
C LC_145	SLN105+stub radial (ends 200 past the node) / Begin	1.0	0.0	-48.6	0.0	-5.3	0.4
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-4.2	-0.9	-9.1	0.0	0.0	-0.7
	SLN395 ring / End	-4.4	0.1	-8.7	0.0	0.0	0.2
C LC_146	SLN105+stub radial (ends 200 past the node) / Begin	1.1	-0.3	-48.5	0.0	-5.3	0.6
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-4.2	-0.9	-9.1	0.0	0.0	-0.8
	SLN395 ring / End	-4.0	0.0	-8.7	0.0	0.0	0.1
C LC_147	SLN105+stub radial (ends 200 past the node) / Begin	0.7	-0.2	-48.6	0.0	-5.4	0.5
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-5.1	-0.9	-9.1	0.0	0.0	-0.7
	SLN395 ring / End	-5.0	0.0	-8.7	0.0	0.0	0.1
C LC_148	SLN105+stub radial (ends 200 past the node) / Begin	1.4	-0.1	-48.5	0.0	-5.2	0.5
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-3.3	-0.9	-9.1	0.0	0.0	-0.7
	SLN395 ring / End	-3.4	0.1	-8.7	0.0	0.0	0.1
C LC_151	SLN105+stub radial (ends 200 past the node) / Begin	3.1	-0.1	-48.4	0.0	-7.4	0.3
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-1.8	-0.5	-9.1	0.0	0.0	-0.4
	SLN395 ring / End	-1.8	0.0	-8.7	0.0	0.0	0.0
C LC_152	SLN105+stub radial (ends 200 past the node) / Begin	-4.1	-0.2	-11.6	0.0	2.7	0.5
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-7.2	-1.0	-2.2	0.0	0.0	-0.8
	SLN395 ring / End	-7.1	0.1	-2.1	0.0	0.0	0.2
C LC_153	SLN105+stub radial (ends 200 past the node) / Begin	1.0	-0.1	-47.8	0.0	-5.1	0.5
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-4.4	-0.9	-9.1	0.0	0.0	-0.7
	SLN395 ring / End	-4.4	0.0	-8.7	0.0	0.0	0.1
C LC_154	SLN105+stub radial (ends 200 past the node) / Begin	3.2	-0.1	-49.2	0.0	-7.7	0.3
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-1.7	-0.5	-9.1	0.0	0.0	-0.4
	SLN395 ring / End	-1.8	0.0	-8.7	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	8.9 < 100%	OK
Welds	36.4 < 100%	OK
Buckling	Not calculated	
GMNA	Calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN105+stub radial (ends 200 past the node)-bfl 1	19.6	C LC_135	47.5	0.0	19.5	OK
SLN105+stub radial (ends 200 past the node)-tfl 1	19.6	C LC_148	21.6	0.0	0.0	OK
SLN105+stub radial (ends 200 past the node)-w 1	11.4	C LC_120	61.9	0.0	19.2	OK
SLN85 column below	10.0	C LC_152	91.2	0.0	0.0	OK
SLN394 ring-bfl 1	12.7	C LC_135	17.1	0.0	0.0	OK
SLN394 ring-tfl 1	12.7	C LC_119	18.4	0.0	0.0	OK
SLN394 ring-w 1	7.2	C LC_121	17.5	0.0	0.0	OK
SLN395 ring-bfl 1	12.7	C LC_135	15.6	0.0	0.0	OK
SLN395 ring-tfl 1	12.7	C LC_119	18.4	0.0	0.0	OK
SLN395 ring-w 1	7.2	C LC_121	17.0	0.0	0.0	OK
End plate column below SLN85 column below	12.0	C LC_152	85.6	0.0	12.2	OK
End plate SLN395 ring	20.0	C LC_146	7.5	0.0	4.1	OK
Bottom flange plate SLN395 ring	15.0	C LC_121	12.1	0.0	0.0	OK
Bottom flange half plate L SLN395 ring	15.0	C LC_135	4.6	0.0	0.0	OK
Bottom flange half plate R SLN395 ring	15.0	C LC_135	4.3	0.0	0.0	OK
Haunch web SLN395 ring	12.0	C LC_121	13.0	0.0	0.0	OK
Haunch top plate SLN395 ring	15.0	C LC_146	5.6	0.0	0.0	OK
Haunch sloping plate SLN395 ring	15.0	C LC_121	5.3	0.0	0.0	OK
End plate SLN394 ring	20.0	C LC_146	4.9	0.0	9.1	OK
Bottom flange plate SLN394 ring	15.0	C LC_121	16.3	0.0	0.0	OK
Bottom flange half plate L SLN394 ring	15.0	C LC_152	7.8	0.0	0.0	OK
Bottom flange half plate R SLN394 ring	15.0	C LC_146	6.6	0.0	0.0	OK
Haunch web SLN394 ring	12.0	C LC_121	13.0	0.0	0.0	OK
Haunch top plate SLN394 ring	15.0	C LC_152	5.5	0.0	0.0	OK
Haunch sloping plate SLN394 ring	15.0	C LC_121	5.4	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

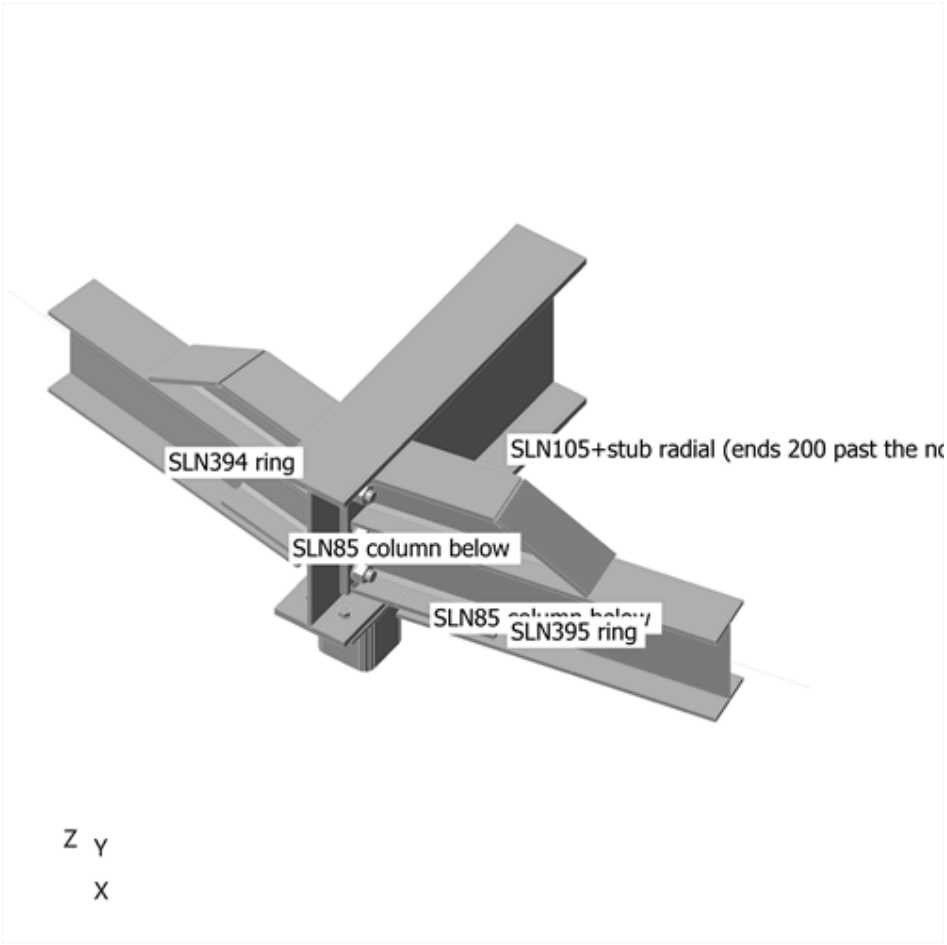
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

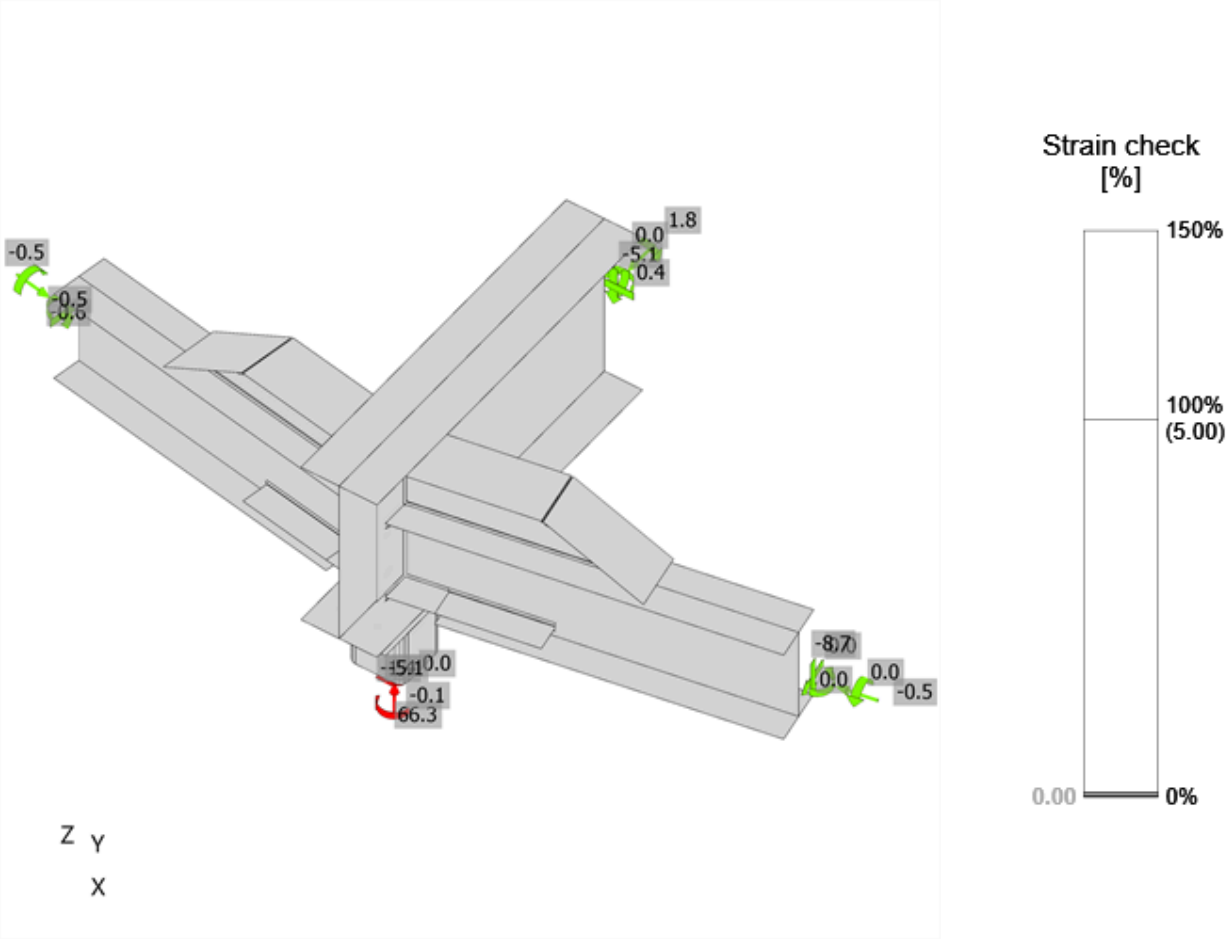
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN85 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

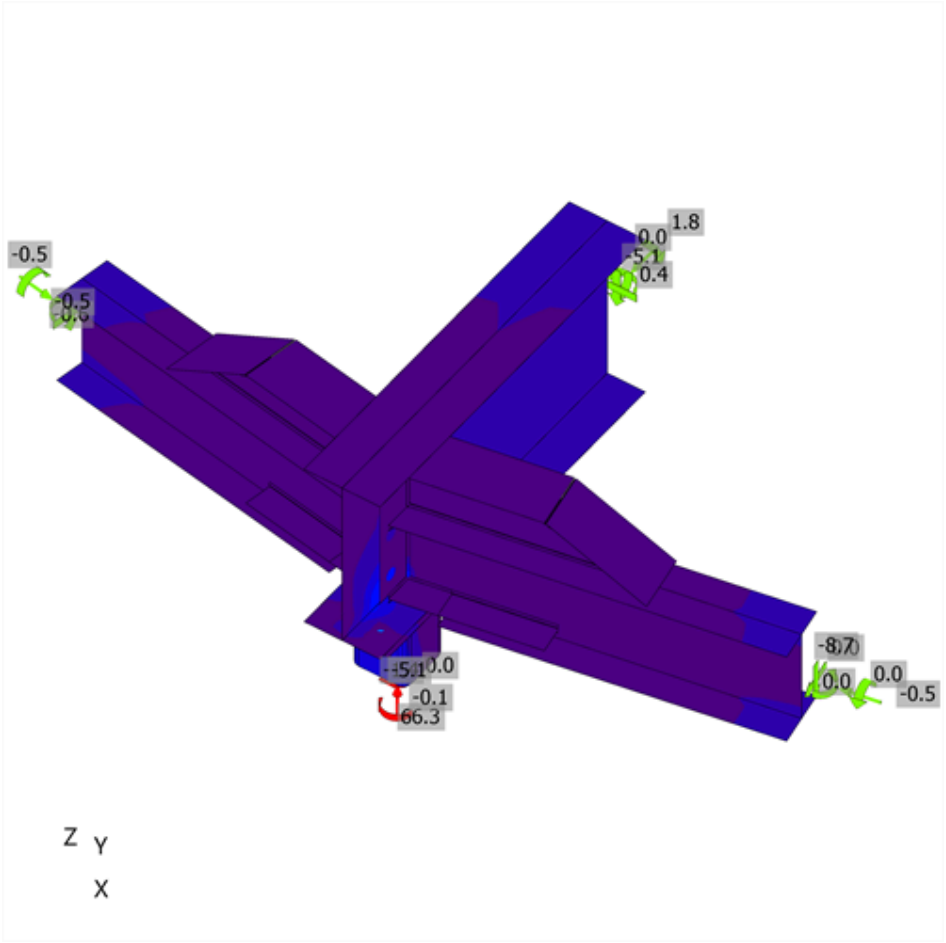
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

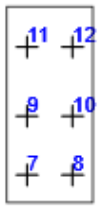


Strain check, C LC_135



Equivalent stress, C LC_135

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_152	12.5	1.1	156.4	8.9	1.1	7.5	OK	OK
	B3	M20 8.8 - 1	C LC_120	0.5	0.8	172.0	0.4	0.9	1.1	OK	OK
	B4	M20 8.8 - 1	C LC_154	0.6	0.8	156.4	0.5	0.8	1.1	OK	OK
	B5	M20 8.8 - 1	C LC_152	11.9	0.8	156.4	8.4	0.9	6.9	OK	OK
	B7	M30 8.8 - 2	C LC_135	0.2	0.6	232.2	0.0	0.5	0.3	OK	OK
	B8	M30 8.8 - 2	C LC_135	1.3	1.7	194.1	0.4	1.6	1.0	OK	OK
	B9	M30 8.8 - 2	C LC_135	0.2	0.7	249.2	0.1	0.5	0.4	OK	OK
	B10	M30 8.8 - 2	C LC_135	0.1	0.6	232.2	0.0	0.5	0.3	OK	OK
	B11	M30 8.8 - 2	C LC_135	0.6	0.8	249.2	0.2	0.6	0.5	OK	OK
	B12	M30 8.8 - 2	C LC_146	1.3	0.3	141.8	0.4	0.2	0.4	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1
M30 8.8 - 2	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate R SLN395 ring	Bottom flange plate SLN395 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN394 ring	Bottom flange plate SLN394 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN395 ring	Bottom flange plate SLN395 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN394 ring	Bottom flange plate SLN394 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN105+stub radial (ends 200 past the node)-w 1	SLN394 ring-tfl 1	◢ 11.4	149	C LC_105	10.7	0.0	-5.0	5.5	-0.3	2.7	2.7	OK	OK
SLN105+stub radial (ends 200 past the node)-w 1	SLN395 ring-tfl 1	◢ 11.4	149	C LC_103	7.8	0.0	-3.4	4.0	-0.8	1.9	0.0	OK	OK
End plate SLN395 ring	SLN395 ring-w 1	◢ 6.0 ◣	207	C LC_121	21.0	0.0	-1.9	1.9	11.9	5.2	5.2	OK	OK
		◢ 6.0 ◣	208	C LC_154	13.3	0.0	1.0	0.8	-7.6	3.3	3.3	OK	OK
Bottom flange plate SLN395 ring	SLN395 ring-w 1	◢ 6.0 ◣	100	C LC_154	10.4	0.0	5.7	4.8	1.4	2.6	2.6	OK	OK
		◢ 6.0 ◣	100	C LC_148	13.5	0.0	5.4	-7.0	-1.5	3.3	3.3	OK	OK
SLN395 ring-w 1	Bottom flange half plate L SLN395 ring	◢ 6.0 ◣	243	C LC_135	4.3	0.0	0.8	1.0	2.2	1.1	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	7.5	0.0	2.4	0.8	-4.0	1.8	0.0	OK	OK
SLN395 ring-w 1	Bottom flange half plate R SLN395 ring	◢ 6.0 ◣	243	C LC_135	6.5	0.0	1.7	1.5	-3.3	1.6	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	6.1	0.0	1.7	1.1	3.2	1.5	0.0	OK	OK
SLN395 ring-tfl 1	Haunch web SLN395 ring	◢ 6.0 ◣	572	C LC_121	22.9	0.0	-5.4	-4.7	12.0	5.7	5.7	OK	OK
		◢ 6.0 ◣	572	C LC_121	29.3	0.0	-7.6	8.3	-14.1	7.2	6.1	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate SLN394 ring	SLN394 ring-w 1	◢ 6.0 ◣	208	C LC_154	14.0	0.0	0.7	-0.8	8.0	3.5	3.5	OK	OK
		◢ 6.0 ◣	207	C LC_154	18.1	0.0	-0.9	-1.0	-10.4	4.5	4.5	OK	OK
Bottom flange plate SLN394 ring	SLN394 ring-w 1	◢ 6.0 ◣	100	C LC_148	12.7	0.0	5.6	6.4	1.7	3.1	3.1	OK	OK
		◢ 6.0 ◣	100	C LC_146	11.5	0.0	5.8	-5.3	-2.1	2.8	2.8	OK	OK
SLN394 ring-w 1	Bottom flange half plate L SLN394 ring	◢ 6.0 ◣	243	C LC_152	6.6	0.0	-1.0	-1.7	-3.4	1.6	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_152	11.1	0.0	-3.6	2.9	5.4	2.8	2.8	OK	OK
SLN394 ring-w 1	Bottom flange half plate R SLN394 ring	◢ 6.0 ◣	243	C LC_146	8.7	0.0	1.5	2.0	-4.5	2.1	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_146	12.3	0.0	3.5	-2.9	6.2	3.0	3.0	OK	OK
SLN394 ring-tfl 1	Haunch web SLN394 ring	◢ 6.0 ◣	572	C LC_121	29.0	0.0	-4.7	-6.7	15.1	7.2	6.2	OK	OK
		◢ 6.0 ◣	572	C LC_121	25.3	0.0	-8.8	6.8	-11.9	6.2	5.9	OK	OK
End plate SLN395 ring	Bottom flange plate SLN395 ring	◢ 6.0 ◣	149	C LC_105	10.3	0.0	-2.2	-4.4	3.7	2.5	2.5	OK	OK
		◢ 6.0 ◣	149	C LC_146	15.5	0.0	-6.8	4.5	-6.7	3.8	3.8	OK	OK
End plate SLN395 ring	Haunch web SLN395 ring	◢ 6.0 ◣	145	C LC_154	3.3	0.0	-0.3	-0.4	1.8	0.8	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_119	8.8	0.0	-0.8	0.6	-5.0	2.2	0.0	OK	OK
End plate SLN395 ring	Haunch top plate SLN395 ring	◢ 6.0 ◣	149	C LC_146	9.2	0.0	-4.6	-2.7	3.7	2.3	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_146	6.0	0.0	-1.2	3.1	-1.4	1.5	0.0	OK	OK
Haunch top plate SLN395 ring	Haunch web SLN395 ring	◢ 6.0 ◣	342	C LC_146	1.9	0.0	0.5	0.4	1.0	0.5	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_119	1.5	0.0	0.6	-0.6	-0.5	0.4	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Haunch sloping plate SLN395 ring	Haunch web SLN395 ring	◢ 6.0 ◣	271	C LC_119	17.2	0.0	-2.3	-3.8	9.1	4.3	4.3	OK	OK
		◢ 6.0 ◣	271	C LC_121	24.3	0.0	-5.0	3.9	-13.1	6.0	5.9	OK	OK
End plate SLN394 ring	Bottom flange plate SLN394 ring	◢ 6.0 ◣	149	C LC_152	26.2	0.0	-12.8	-12.9	2.9	6.5	5.9	OK	OK
		◢ 6.0 ◣	149	C LC_121	27.0	0.0	-13.6	13.1	-3.1	6.7	5.8	OK	OK
End plate SLN394 ring	Haunch web SLN394 ring	◢ 6.0 ◣	145	C LC_119	8.0	0.0	-1.0	-0.9	4.5	2.0	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_121	3.9	0.0	-0.3	0.4	-2.2	1.0	0.0	OK	OK
End plate SLN394 ring	Haunch top plate SLN394 ring	◢ 6.0 ◣	149	C LC_152	11.1	0.0	-5.4	-5.1	2.4	2.8	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_146	10.0	0.0	-3.9	4.2	-3.2	2.5	0.0	OK	OK
Haunch top plate SLN394 ring	Haunch web SLN394 ring	◢ 6.0 ◣	342	C LC_146	3.7	0.0	1.2	0.9	1.8	0.9	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_121	2.8	0.0	-0.7	1.4	0.8	0.7	0.0	OK	OK
Haunch sloping plate SLN394 ring	Haunch web SLN394 ring	◢ 6.0 ◣	271	C LC_121	21.2	0.0	-2.9	-0.1	12.1	5.2	5.2	OK	OK
		◢ 6.0 ◣	271	C LC_121	23.2	0.0	-4.4	7.3	-11.0	5.7	5.7	OK	OK
End plate column below SLN85 column below	SLN85 column below	◢ 4.2 ◣	521	C LC_135	134.7	0.0	-65.0	-67.3	-10.9	33.3	18.0	OK	OK
		◢ 4.2 ◣	521	C LC_135	147.2	0.0	-74.4	72.3	12.7	36.4	19.9	OK	OK
SLN105+stub radial (ends 200 past the node)-w 1	SLN395 ring-w 1	◢ 7.2	208	C LC_154	57.0	0.0	3.6	-6.3	32.2	14.1	8.8	OK	OK
SLN105+stub radial (ends 200 past the node)-w 1	SLN394 ring-w 1	◢ 7.2	208	C LC_154	56.8	0.0	5.5	6.4	-32.0	14.0	8.9	OK	OK
Haunch sloping plate SLN395 ring	Haunch top plate SLN395 ring	◢ 6.0 ◣	146	C LC_121	3.1	0.0	-1.9	1.4	0.1	0.8	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_146	1.2	0.0	-0.2	0.7	-0.2	0.3	0.0	OK	OK
Haunch sloping plate SLN394 ring	Haunch top plate SLN394 ring	◢ 6.0 ◣	146	C LC_146	4.6	0.0	-4.3	-1.0	-0.1	1.4	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	147	C LC_146	3.3	0.0	-0.6	1.8	-0.5	0.8	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		

Item	Value	Unit	Reference
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5